Homological Algebra

***EYNTKA* Series**

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October 29, 2026

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Part I

Homological Algebra

Throughout this book, the reader has encountered many functors, some notable examples being $\mathrm{Hom}_R(-, N)$, $\mathrm{Hom}_R(M, -)$ and $M \otimes_R -$ among many others. Each of these functors give some information about the objects and the morphisms passing through the functor. As we saw for the Hom and $M \otimes_R -$ functors, certain appropriate objects preserve exactness, while most preserve it only partially. Perhaps a crowning achievement of mid 20th century mathematics is to be able to quantify this lack of exactness, and rather spectacularly use the information given by this quantification to extract a (sometimes copious) amount of information about the objects of study. The great effectiveness with which homological algebra can produce interesting information may seem surprising at first, for example, using homological algebra the notion of *dimension* as it is understood in geometry is readily translatable to algebra, or that we may measure how far away a module is from being a vector-space (details are in K-theory)[1] How this is the case is perhaps best motivated by going through learning the subject to see for oneself, but it may be helpful for some readers to have some of my (the authors) intuition presented to stand on some ground before proceeding.

If we have two objects M_1 and M_2, we may wish to consider a way of constructing a new object M given M_1 and M_2. Ideally, we may form a short exact sequence like so:

$$0 \to M_1 \to M \to M_2 \to 0$$

More generally, we may have a collection of objects $\{M_i\}_{i \in I}$ which we use to construct an object via a long exact sequence. However, it is often not the case that such an exact sequence is too strong, that is, it is not the case that $\mathrm{im}(\varphi_i) = \ker(\varphi_{i+1})$. Even if such a construction exists, we often then want to "transform" this information via a functor which shall then lose us exactness (one can think of a functor as asking some sort of "question" whose output is the answer). In this case, it has been eerily common that the image of φ_i is contained in the kernel of φ_{i+1}:

$$\mathrm{im}(\varphi_i) \subseteq \ker(\varphi_{i+1}) \tag{1}$$

Conceptually, this can be thought as the object "being obstructed" from fully being an extension. This new type of sequence is called a *cochain complex*. The reader with a good memory may recall [8, Exact Chain Complex] where it was given that another name for a long exact sequence is an exact cochain complex. In this way, the condition given by equation 1 is a generalization of the exact condition. The theoretical heart of homological algebra can be thought of how to extract "quantitatively" how a cochain complex *fails* to be an exact cochain complex. For example, in [8, Extension Problem: Prelude to Homological Algebra] it was have eluded that through the use of homology theory, the Ext functor gives us all the ways two modules M_1, M_2 may be extended. In this case, the failure of $\mathrm{Hom}(M, -)$ or $\mathrm{Hom}(-, N)$ preserving injectivity or surjectivity (equivalently, the "obstruction" to these functors being exact) gave rise to the Ext functor and the study of its homology.

The term "algebraic object" has been is a bit ambiguous. What type of algebraic objects can be used to in the theory of homological algebra? As it turns out, not all algebraic objects satisfy the right conditions for the theory of homological algebra (for example, **Grp** and **Fld** will be shown to be inappropriate for a theory of homology). However, there are non-algebraic objects for which a homological theory can be induced (famously, topological spaces and smooth manifolds admit a theory of homological algebra). This spurred the need to generalize the theory of homological algebra to the categorical level, where homological theory is well-defined on *abelian categories*. We thus shall start homological algebra with an overview of abelian categories. For those who are already familiar

[1] This is overviewed in chapter 4

with some homological algebra, they may have studied homological algebra within the category of R-**Mod**, which is a concrete abelian category. Doing so will in fact not loose any generality due to Freyd-Mitchell Theorem (theorem 1.1.15) which states that any [small] abelian category A admits a fully faithful exact functor to a category R-**Mod** for a suitable ring R. Working with the concrete category R-**Mod** brings it benefits as it may be easier to conceptualize the manipulation of elements, and so the reader may ask for the benefit in sticking to working with general abelian categories. As the author, I believe the benefits comes in being comfortable with more advanced category theory. Future algebraists are going to encounter more complicated problems involving category theory (higher order categories, topoi's, and stacks, are all examples the reader may see in the not too distant future). Hence, introducing a categorical perspective will familiarize the reader for more complicated categorical notions in their not too distant future[2].

The reader may wonder why a cochain complex is not called a chain complex. This is because we can work in the dual category, where all arrows are flipped. In the dual category, we shall get chain complexes. Homological algebra in the dual category is theoretically the same, however we shall see that cohomology is often computationally easier to work in, permitting more useful algebraic structures we can use to study. It is for this reason that most mathematicians work with cohomology rather than homology, even though every theorem we shall show for homology translates directly to cohomology[3]. In this book, we shall start with homology, and switch to cohomology when starting to work with more computational parts of the theory to get the reader accustomed to both modes of thinking.

Goal

Chapter 1 shall introduce the theory of abelian categories which shall be the types of categories in which homological algebra is well-defined.

The main result Chapter 2 of the book is trying to convey is what information from R-modules give rise to certain properties in its respective chain complex(es), and conversely what information from it's chain complex(es) gives us information about modules. A (co)chain complex need not be exact, and (co)homology shall be the algebraic object quantifying this failure of exactness. We shall see that given an object M within an abelian category, there are in general an infinite number of ways we may associate a [co]chain complex to it. Thus, there will have to be a way to either associate a canonical [co]chain complex (in a natural functorial manner), or we assign [co]chain complexes to an object up to appropriate notion of isomorphism. The latter shall be the approach in this book, where we shall see that we will want to consider [co]chain complexes up to something called *quasi-isomorphism* of [co]chain complexes. We shall then see what cohomological information we may deduce given different choices of functors and how they transform [co]chain complexes.

After having developed the ground work for extracting cohomological information, in Chapter 4 we shall put this knowledge to good use and apply cohomology theory to G-invariant functors and see what this this can tell us about groups. This shall also finally address what was foreshadowed in [8, Foreshadowing] where it was said the units of a Galois field form a trivial 1st homology group. We shall define the notion of homological dimension which shall line up with the results presented in the section on commutative algebra. Finally, we present an interesting technical result to check

[2] assuming, naturally, that the reader holds great interest in algebra and shall keep pursuing branches within the field

[3] there is even an equivalent algebraic structure for the additional algebraic structure that is induced in cohomology, however it is less useful

flatness of a module over a local ring that builds on the same intuitions the Nakayama lemma ([9, Nakayama's Lemma]) presented.

Finally, this Part concludes with the introduction to spectral sequences in chapter 6, which are the quintessential tool for many concrete computations in homological algebra and any serious work in many disciplines of algebra and geometry. The main idea is that most object of study can be broken down into a filtration of simpler objects whose cohomology is more computed. The spectral sequence will put piece together this cohomological information.

Connection to Singular Homology/Cohomology

For readers familiar with algebraic topology and the theory of (co)homology, the connection between the above abstraction and the homology theory of topological spaces comes from noticing that singular (co)homology on a manifold can be interpreted as taking the projective resolution $C_n(X) = \mathrm{Hom}_{\mathbf{Top}}(\Delta^n, X)$. In particular, given M, we can apply $C_*(-)$ to it and get a projective (even free) resolution from which we extract homological information. This should also motivate the use of $\mathrm{Hom}_R(M, -)$ and $\mathrm{Hom}_R(-, N)$ as the "right" functors to be considering when generalizing classical algebraic topology to the algebraic setting. As the tensor product it adjoint ([8, hom-Tensor Adjoint]) it is another natural candidate.

0.0.1 Foreshadowing: Yoneda's Lemma The Hom is very useful as it is "represented" by an element (either in the domain or codomain). Even more is true: it can be used to represent other functors. In [6, The Yoneda Lemma] this is studied *representable functors* which are functors where $F(-) \Rightarrow \mathrm{Hom}_C(A, -)$ for natural transformation $\Rightarrow$, category C, object $A \in C$, and $F : C \to \mathbf{Set}$. The Yoneda lemma will give a way of representing objects using Hom.

Historical Origins

While modern homological algebra is phrased in the language of categories, its machinery evolved from two distinct 19th-century sources which converged in the mid-20th century.

The algebraic lineage begins around 1890 with David Hilbert's Invariant Theory. To prove the Basis Theorem, Hilbert studied the relations ("syzygies") between generators of an ideal. He constructed a chain of free modules to resolve these relations:

$$0 \to F_n \to \cdots \to F_1 \to F_0 \to M \to 0$$

Hilbert proved the *Syzygy Theorem*, showing that for polynomial rings, this chain of relations eventually terminates in an exact sequence of free modules. However, this theory encountered a significant obstruction when applied to rings with singularities (such as coordinate rings of singular curves). In these non-regular settings, Hilbert's resolutions do not terminate but continue infinitely. This failure shifted the mathematical focus from simply computing invariants to essentially classifying the severity of a singularity by the infinite length of its resolution.

Parallel to this, in the 1930s, number theorists like Emmy Noether, Richard Brauer, and Helmut Hasse were attempting to classify division algebras (generalizations of the quaternions) over number fields. They classified these algebras using "Crossed Products," which relied on complex systems of

functions called "Factor Sets" ($c : G \times G \to K^\times$). While they successfully defined the *Brauer Group* of a field to organize these algebras, the calculations were notoriously opaque. The "Factor Sets" satisfied a messy algebraic identity that seemed arbitrary, acting as a computational blocker that obscured the underlying structure of the fields they were studying.

Simultaneously, from 1895 onwards, the topological roots of the subject were forming as Henri Poincaré classified manifolds. He decomposed spaces into simplices and represented their boundaries using incidence matrices. He defined Betti numbers as the ranks of these matrices. However, Poincaré's numerical approach had a blind spot: rank obliterates "torsion." A twisted shape like the projective plane appeared numerically identical to a trivial one (both have rank 0 in dimension 1).

In 1925, Emmy Noether revolutionized this view. She realized that the matrices topologists were diagonalizing were actually presentations of abelian groups. By shifting focus from the numerical rank to the *Homology Group* itself ($\ker \partial / \operatorname{im} \partial$), she captured the torsion information, effectively "algebraizing" the topology. She showed that the matrix was not just a calculator for a number, but a presentation of a structure.

The convergence of these fields occurred in the 1940s when algebraists attempted to classify *Group Extensions*. Saunders Mac Lane and Samuel Eilenberg attempted to classify these extensions and rediscovered the same "Factor Sets" used by Brauer and Hasse. They discovered that the algebraic identity these functions satisfied was *identical* to the boundary formula for a 2-simplex in topology; essentially they realized they were calculating $H^2(G, N)$ (see [11] for this computation in the EYNTKA series). This was the turning point: it proved that "cohomology" was not just a property of geometric shapes, but a fundamental algebraic structure that arose whenever one tried to "extend" one object by another, whether in topology, group theory, or number theory.

The grand unification came in 1956 when Cartan and Eilenberg realized that the topological concept of "boundaries" and the algebraic concept of "extensions" were instances of the same phenomenon: the *failure of exactness*. They observed that while Hilbert's resolution is exact, applying a functor (like the tensor product or Hom) usually destroys that exactness. They realized that the "homology" groups were simply the mathematical terms required to repair this broken exactness. This birth of the *Derived Functor* (Tor and Ext) transformed the field from the study of static objects to the study of how functors distort structure. To build this machinery, they faced a technical blocker: Hilbert's reliance on "Free Modules." While free modules worked for Homology (Tor), they failed for the dual theory of Cohomology (Ext), as the dual of a free module is rarely free. To achieve the necessary symmetry, they replaced the specific condition of "Freeness" with the categorical properties of *Projectivity* and *Injectivity*. This allowed them to define derived functors for any ring, regardless of whether a basis existed.

With this machinery established, the period from 1955 to 1980 saw these newly developed tools being used to solving the original obstructions in geometry and number theory. In geometry, Jean-Pierre Serre proved that a local ring is regular (non-singular) if and only if it has finite global homological dimension (theorem 4.2.11). This transformed the "problem" of infinite resolutions into a precise detection tool: the non-terminating resolution was effectively the algebraic signature of a geometric singularity. Alexander Grothendieck later introduced *Local Cohomology* ($H^i_{\mathfrak{m}}$) to quantify the depth and severity of these singularities, eventually leading to Intersection Homology which restored Poincaré Duality to singular spaces.

In number theory, the mysterious "Factor Sets" of the 1930s were identified as elements of the second Galois Cohomology group, $H^2(\mathrm{Gal}(L/K), L^\times)$. This realization allowed Emil Artin and John Tate to rewrite Class Field Theory entirely in the language of cohomology. The obscure identities of the past were revealed to be standard cohomological relations, and the *Tate Cohomology* groups ($\hat{H}^i$) were

developed specifically to handle the arithmetic of finite groups, solving the classification problems that had stalled the previous generation (see [18] for the EYNTKA series exposition of modern class field theory).

1

Abelian Categories

This chapter focuses on the category in which we may define "cohomological theories" as not all categories are well-suited for this endeavor

1.1 Definitions and Properties

We have extensively worked with categories which have a multitudes of different properties. Some important terms to recall are the following:

- *zero-object*: an object 0 in a category C $0 \in \text{Obj}(C)$ that is both the initial and final object. Such an object is also called a terminal object. Similarly, a zero morphism is a morphism $f : X \to Y$ such that for any other morphisms $g, h : X \to Y$, $fg = fh$ and $hf = gf$.
- *Products* and *Coproducts*: Given two objects $a, b \in \text{Obj}(C)$, if there exists an object $c \in \text{Obj}(C)$ that is the limit (resp. colimit) of the diagram:

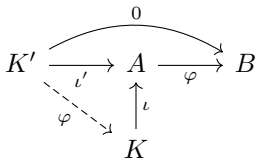
  $$\overset{A}{\bullet} \qquad \overset{B}{\bullet}$$

  then c is the product (resp. coproduct) of a and b, and is usually written as $a \times b$ (resp. $a \amalg b$)
- *kernels* and *cokernels*: Given a morphism $f : a \to b$, the kernel (resp. cokernel) of f is an object k and a morphism $\iota : k \to a$ such that k is the limit (resp. colimit) of the diagarm:

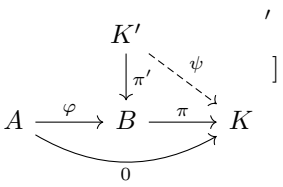
  $$\overset{a}{\bullet} \xrightarrow{f} \overset{b}{\bullet}$$

  with the added restriction that the map $\varphi : k \to y$ must be the zero morphism (namely, by commutativity, $f\iota = 0_{kb}$ must be the zero morphism). In terms of diagrams, we get that that

for any $\iota' : K \to A$ such that $f\pi\iota' = 0$, that there exists a morphism φ such that the following diagram commutes:



and simmilarly for cokernels:



Kernels and cokernels do not necessarily exist in a category, or not all morphisms have kernels (ex. the category of finitely generated R-modules for any ring R need has morphisms without kernels). Furthermore, monomorphism need not be kernels (not all subgroups are normal) and epimorphisms need not be cokernels (recall $\mathbb{Z} \to \mathbb{Q}$ is an epimorphism in **Ring**)

Not all categories we have worked with have these properties. For example, **Fld** does not have products (what would be the characteristic of the product of $\mathbb{F}_p$ and $\mathbb{F}_q$ for $p \neq q$?), some categories do not have zero morphisms (set being an example), some have monomorphisms/epimorphisms that do not correspond to kernels (as showed above). We first limit our attention to categories which have these nice properties:

Definition 1.1.1: Additive Category

Let **A** be a category. Then **A** is said to be *additive* if it has a zero-object, has both finite products and coproducts, and each $\mathrm{Hom}_A(X, Y)$ can be given an abelian group structure such that the composition of maps:

$$\mathrm{Hom}_A(X, Y) \times \mathrm{Hom}_A(Y, Z) \to \mathrm{Hom}_A(X, Z)$$

is bilinear. A functor between additive categories is *additive* if it is a group homomorphism on hom-sets.

A functor between two additive categories is called an *additive functor* if the maps between the hom-sets is a homomorphism.

The term "additive" comes froms the fact that we may interpret the additive category as a monoidal category (a category which we may add a functor $\otimes : C \times C \to C$ that is given the property of a monoid operation, for an additive category this would be the direct product), where the hom-sets are "enriched" to also be abelian groups. Another way of looking at additive categories is in which the algebra of matrices is well-defined.

Example 1.1.2: Additive Categories

1. The category of R-modules is an example of an additive category.

2. **Grp** is not additive, for $\mathrm{Hom}_{\mathbf{Grp}}(A, B)$ does not have an abelian group strucuture satisfying the above condition.
3. The category of matrices over a ring is an additive category.
4. The category of finitely generated R-modules is an additive category

A key realisation is that in the absence of kernels and cokernels as objects monomorphisms and epimorphisms take their place:

Lemma 1.1.3: Additive Categories: Monomorphisms and Kernels I

Let A be an additive category. Then kernels are monomorphisms and cokernels are epimorphisms

***Proof*:**
Let $k\colon K \to A$ be a kernel of $\varphi\colon A \to B$, and let $f, g\colon X \to K$ satisfy $k \circ f = k \circ g$. Because the hom-sets of an additive category are abelian groups and composition is bilinear, this says $k \circ (f - g) = 0$.

Now apply the universal property of the kernel to the zero morphism $0\colon X \to A$. It satisfies $\varphi \circ 0 = 0$, so there is a *unique* morphism $u\colon X \to K$ with $k \circ u = 0$. Both $f - g$ and the zero morphism $X \to K$ have this property, so uniqueness forces $f - g = 0$, that is $f = g$. Hence k is a monomorphism.

The statement for cokernels is the same argument in the opposite category: a cokernel of φ in A is a kernel of φ in A^{op}, an additive category, and a monomorphism there is an epimorphism in A, completing the proof.

Note that kernels and cokernels need not exist in an additive category, hence the converse of each accession need not be the case (in a vacuous way). If a morphism φ has a kernel then we may identify exactly what this kernel is:

Lemma 1.1.4: Additive Categories: Monomorphisms and Kernels II

Let A be an additive category and $\varphi : X \to Y$ a morphism. Then:

1. If φ has a kernel, then φ is a monomorphism if and only if $0 \to X$ is its kernel
2. If φ has a cokernel, then φ is an epimorphism if and only if $Y \to 0$ is its cokernel

***Proof*:**

1. Suppose first that the zero morphism $0 \to X$ is a kernel of φ, and let $f, g\colon Z \to X$ satisfy $\varphi \circ f = \varphi \circ g$. Then $\varphi \circ (f - g) = 0$, so $f - g$ factors through the kernel: there is $u\colon Z \to 0$ with $f - g = 0 \circ u = 0$. Hence $f = g$ and φ is a monomorphism.

   Conversely suppose φ is a monomorphism and let $k\colon K \to X$ be a kernel of φ. From

$\varphi \circ k = 0 = \varphi \circ 0$ and φ monic it follows that $k = 0$. But k is itself a monomorphism by lemma 1.1.3, and $k \circ \mathrm{id}_K = 0 = k \circ 0$, so $\mathrm{id}_K = 0$. An object whose identity is the zero morphism is a zero object, so $K \cong 0$ and the kernel is $0 \to X$.

2. This is the statement of (1) in the opposite category, where kernels and monomorphisms become cokernels and epimorphisms.

This completes the proof.

As mentioned, it is not guaranteed that kernels and cokernels exist. We shall shortly axiomatically add them to our category by adding them into the definition of a new category that supersedes additive categories, but first we show a few more important results about abelian categories. Namely, in an additive category, products and coproducts are equivalent:

Lemma 1.1.5: Additive Categories: Products and Coproducts

Let A be an additive category and $X, Y \in \mathrm{Obj}(A)$. Then if $X \times Y$ exists, so does $X \amalg Y$, and if $X \amalg Y$ exists, so does $X \times Y$. Furthermore,

$$X \times Y \cong X \amalg Y$$

***Proof*:**
Suppose $P = X \times Y$ exists, with projections $p_X \colon P \to X$ and $p_Y \colon P \to Y$. Let $i_X \colon X \to P$ be the unique morphism with $p_X i_X = \mathrm{id}_X$ and $p_Y i_X = 0$, and let $i_Y \colon Y \to P$ be defined symmetrically; both exist by the universal property of the product.

Step 1: The key identity $i_X p_X + i_Y p_Y = \mathrm{id}_P$. Two morphisms out of P into P agree as soon as their composites with p_X and with p_Y agree, again by the universal property of the product. Compute

$$p_X(i_X p_X + i_Y p_Y) = (p_X i_X)p_X + (p_X i_Y)p_Y = \mathrm{id}_X \circ p_X + 0 \circ p_Y = p_X$$

using bilinearity of composition, and symmetrically $p_Y(i_X p_X + i_Y p_Y) = p_Y$. Since id_P has the same two composites, the identity holds.

Step 2: (P, i_X, i_Y) is a coproduct. Let $f \colon X \to Z$ and $g \colon Y \to Z$ be given and set $h = f p_X + g p_Y \colon P \to Z$. Then

$$h i_X = f(p_X i_X) + g(p_Y i_X) = f, \qquad h i_Y = f(p_X i_Y) + g(p_Y i_Y) = g$$

so h is a morphism with the required restrictions. For uniqueness, suppose h' also satisfies $h' i_X = f$ and $h' i_Y = g$. Then by Step 1,

$$h' = h' \mathrm{id}_P = h'(i_X p_X + i_Y p_Y) = (h' i_X)p_X + (h' i_Y)p_Y = f p_X + g p_Y = h$$

Hence P with i_X, i_Y satisfies the universal property of $X \amalg Y$, and in particular $X \times Y \cong X \amalg Y$.

If instead $X \amalg Y$ is assumed to exist, the same argument in the opposite category produces the product, completing the proof.

Hence, additive categories behave very much like abelian groups, however as we pointed out kernels and cokernels need not exist. We thus define a new category that forces these conditions:

Definition 1.1.6: Abelian Category

[Abelian Category] Let A be an additive category. Then A is an *abelian* category if

1. kernels and cokernels exist in A;
2. every monomorphism is the kernel of some morphism, and every epimorphism is the cokernel of some morphism.

The second condition is not a consequence of the first, and it is the one that does the work. An additive category satisfying only (1) is called *pre-abelian*, and there the canonical map from the coimage to the image of a morphism (theorem 1.1.9) need not be an isomorphism; the category of torsion-free abelian groups is pre-abelian but not abelian, since $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ is there both a monomorphism and an epimorphism without being an isomorphism. Condition (2) rules this out, and is what makes lemma 1.1.8 true.

Note that lemma 1.1.4 is a weaker statement, not this one: it characterizes a monomorphism by having *zero* kernel, which says nothing about whether the monomorphism is itself a kernel.

Since each kernel is associated to a monomorphism, we may intuitively think of the association:

$$\text{kernel} \;\Leftrightarrow\; \text{submodule}$$

and similarly:

$$\text{cokernel} \;\Leftrightarrow\; \text{quotient module}$$

In many ways, an abelian category should be thought of as the minimum number of conditions necessary to define exact sequence[1]. To see this, we require the following:

Lemma 1.1.7: Abelian Category: Kernels And Cokernels

Let A be an abelian category. Then every kernel is the kernel of its cokernel, and every cokernel is the cokernel of its kernel.

***Proof*:**
Let $k\colon K \to A$ be a kernel, say of $\varphi\colon A \to B$, and let $c\colon A \to C$ be a cokernel of k. The claim is that k is a kernel of c.

First, $c \circ k = 0$ holds by the definition of a cokernel. Now let $t\colon T \to A$ satisfy $c \circ t = 0$; a unique factorization of t through k is needed. Since $\varphi \circ k = 0$ and c is the cokernel of k, the morphism φ factors as $\varphi = \bar{\varphi} \circ c$ for some $\bar{\varphi}\colon C \to B$. Therefore

$$\varphi \circ t = \bar{\varphi} \circ c \circ t = 0$$

and since k is the kernel of φ, there is a unique $u\colon T \to K$ with $k \circ u = t$. Uniqueness is automatic because k is a monomorphism by lemma 1.1.3. So k satisfies the universal property of $\ker c$.

[1] Those familiar with some algebraic topology may find it insightful that an abelian category is the the category with ninimum number of condition so that the snake lemma (theorem 2.1.20) holds

The dual statement, that every cokernel is the cokernel of its kernel, is this argument in the opposite category, completing the proof.

Lemma 1.1.8: Abelian Categories: Isomorphisms

[Abelian Categories, Isomorphisms] Let A be an abelian category and $\varphi : A \to B$ a morphism in A. Then if φ is a monomorphism and an epimorphism, it is an isomorphism

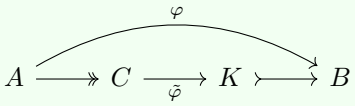
Note that this does not hold in additive categories, nor even in pre-abelian ones: condition (2) of definition 1.1.6 is exactly what the proof below uses.

***Proof*:**
Since φ is an epimorphism, condition (2) of definition 1.1.6 makes it the cokernel of some morphism $\psi\colon C \to A$. By the definition of a cokernel $\varphi \circ \psi = 0$, so ψ factors through $\ker\varphi$ by the universal property of the kernel.

Now φ is also a monomorphism, so $\ker\varphi = 0$ by lemma 1.1.4. The factorization of the previous paragraph then runs through the zero object, so $\psi = 0$ outright and φ is a cokernel of the zero morphism $0 \to A$. But the identity $\mathrm{id}_A\colon A \to A$ is also a cokernel of $0 \to A$: any $f\colon A \to X$ with $f \circ 0 = 0$, which is every f, factors uniquely through id_A. Cokernels are unique up to unique isomorphism, so there is an isomorphism $A \to B$ commuting with φ and id_A, which forces φ to be that isomorphism, as we sought to show.

Theorem 1.1.9: Abelian Category: Canonical Decomposition

[Abelian Category, Canonical Decomposition] Let A be an abelian category and $\varphi : A \to B$ a morphism. Write $C = \mathrm{coker}(\ker\varphi)$ and $K = \ker(\mathrm{coker}\varphi)$. Then φ decomposes in the following cannonical way:



where the first morphism is an epimorphism, the last a monomorphism, and the middle morphism $\tilde{\varphi}$ is an *isomorphism*.

***Proof*:**
Write $k\colon N \to A$ for $\ker\varphi$ and $p\colon A \to C$ for its cokernel, and $q\colon B \to Q$ for $\mathrm{coker}\varphi$ and $\iota\colon K \to B$ for its kernel. By lemma 1.1.3 the morphism p is an epimorphism and ι is a monomorphism.

Step 1: φ factors through ι. Since q is the cokernel of φ, $q \circ \varphi = 0$, and since ι is the kernel of q there is a unique $\varphi'\colon A \to K$ with $\varphi = \iota \circ \varphi'$.

Step 2: φ' factors through p. From $\iota \circ \varphi' \circ k = \varphi \circ k = 0$ and ι monic it follows that $\varphi' \circ k = 0$. As p is the cokernel of k, there is a unique $\tilde{\varphi}\colon C \to K$ with $\varphi' = \tilde{\varphi} \circ p$. Combining, $\varphi = \iota \circ \tilde{\varphi} \circ p$, which is the displayed factorization.

Step 3: ι is the smallest subobject through which φ factors. Let $m\colon V \to B$ be any monomorphism with $\varphi = m \circ \psi$ for some ψ. By condition (2) of definition 1.1.6 the monomorphism m is a kernel, so by lemma 1.1.7 it is the kernel of its own cokernel; write $d\colon B \to D$ for $\operatorname{coker} m$, so that $m = \ker d$. Then

$$d \circ \varphi = d \circ m \circ \psi = 0$$

so d factors through the cokernel of φ, say $d = \bar{d} \circ q$. Since $\iota = \ker q$ gives $q \circ \iota = 0$, it follows that $d \circ \iota = \bar{d} \circ q \circ \iota = 0$, and therefore ι factors through $\ker d = m$.

Step 4: φ' is an epimorphism. Let $c\colon K \to W$ be the cokernel of φ' and $n\colon V \to K$ its kernel. From $c \circ \varphi' = 0$ the morphism φ' factors as $\varphi' = n \circ \psi$, whence $\varphi = \iota \circ n \circ \psi$. The composite $\iota \circ n$ is a monomorphism, being a composite of two (each is a kernel, so lemma 1.1.3 applies), so Step 3 gives a t with $\iota = \iota \circ n \circ t$. As ι is monic this yields $n \circ t = \mathrm{id}_K$, so n is a split epimorphism as well as a monomorphism, hence an isomorphism by lemma 1.1.8. Then $c \circ n = 0$ with n invertible forces $c = 0$; but c is a cokernel, hence an epimorphism by lemma 1.1.3, and $\mathrm{id}_W \circ c = 0 = 0 \circ c$ then gives $\mathrm{id}_W = 0$, so $W \cong 0$. A morphism with zero cokernel is an epimorphism, so φ' is one.

Step 5: $\tilde{\varphi}$ is an isomorphism. By Step 4 the composite $\tilde{\varphi} \circ p = \varphi'$ is an epimorphism, and if $u \circ \tilde{\varphi} = 0$ then $u \circ \tilde{\varphi} \circ p = 0$ and hence $u = 0$; so $\tilde{\varphi}$ is an epimorphism. Dually, running Steps 1 to 4 in the opposite category shows that the induced morphism $C \to B$, which is $\iota \circ \tilde{\varphi}$, is a monomorphism; and a composite $g \circ h$ can be a monomorphism only if h is, so $\tilde{\varphi}$ is a monomorphism. Being both, it is an isomorphism by lemma 1.1.8, completing the proof.

From the above, we may think of $\ker(\operatorname{coker}) : K \to B$ as representing the "image" of φ:

Lemma 1.1.10: Abelian Category: Images

Let $\varphi : A \to B$ be a morphism in an abelian category, and let $\iota : K \to B$ be the kernel of the cokernel of φ. Then:

1. ι is a monomorphism
2. φ factors through ι
3. ι is initial with these properties

***Proof*:**
Part (1) is lemma 1.1.3, since ι is a kernel. Part (2) is Step 1 of the proof of theorem 1.1.9: writing $q = \operatorname{coker}\varphi$, the identity $q \circ \varphi = 0$ makes φ factor uniquely through $\iota = \ker q$.

Part (3) is the minimality established in Step 3 of that proof: if $m\colon V \to B$ is any monomorphism through which φ factors, then ι factors through m. Concretely, m is a kernel by condition (2) of definition 1.1.6, hence the kernel of its own cokernel d by lemma 1.1.7; then $d \circ \varphi = 0$ makes d factor through q, so $d \circ \iota = 0$ and ι factors through $\ker d = m$, as we sought to show.

Definition 1.1.11: Abelian Category: Image

Let $\varphi : A \to B$ be a morphism in an abelian category. Then the *iamge* of φ is $\ker(\text{coker}(\varphi))$ and is denoted $\text{im}(\varphi)$. Similarly, its co-image is $\text{coker}(\ker(\varphi))$ and is dentoed $\text{coim}(\varphi)$.

With our intuition from working with R-modules, given a morphism $\varphi : A \to B$ with image $\iota : K \to B$, we may imagine there is an induced epimorphism $\overline{\varphi} : A \to K$. This is indeed the case:

Proposition 1.1.12: Abelian Category: sub object and quotient objects

Let $\varphi : A \to B$ be a morphism in an abelian category with image $\text{im}\varphi\colon K \to B$ and coimage $\text{coim}\,\varphi\colon A \to C$. Then the induced morphism $A \to K$ is an epimorphism and the induced morphism $C \to B$ is a monomorphism.

***Proof*:**
The induced morphism $A \to K$ is the morphism φ' of theorem 1.1.9, shown to be an epimorphism in Step 4 of that proof. The induced morphism $C \to B$ is the composite $\iota \circ \tilde{\varphi}$, a composite of a monomorphism with an isomorphism, hence a monomorphism, completing the proof.

From the cannonical decomposition, we may say that the image and coimage are "isomorphic" (which is not strictly true since these are morphisms, however by the Freyd-Mitchell Theorem this is a safe way of looking at it).

Definition 1.1.13: Abelian Category: Exactness

Consider a sequence of objects in an abelian category

$$\cdots \to A \xrightarrow{\varphi} B \xrightarrow{\psi} C \to \cdots$$

This sequence is *exact at* B if:

1. $\psi \circ \varphi = 0$
2. $\text{coker}\varphi \circ \ker \psi = 0$

and it is *exact* if it is exact at each of its interior objects.

Example 1.1.14: Categorical Exactness

1. In R-Mod, $\ker \psi$ is the inclusion of $\{b \in B \mid \psi(b) = 0\}$ and $\text{coker}\varphi$ is the projection $B \to B/\text{im}\varphi$. So condition (2) says that every b with $\psi(b) = 0$ lies in $\text{im}\varphi$, condition (1) says $\text{im}\varphi \subseteq \ker \psi$, and together they are the usual condition $\text{im}\varphi = \ker \psi$.
2. The sequence $0 \to A \xrightarrow{\varphi} B \to 0$ is exact if and only if φ is an isomorphism. The cokernel of $0 \to A$ is id_A, so exactness at A says that $\ker \varphi$ is the zero morphism. A kernel is a monomorphism (lemma 1.1.3), so its object is then a zero object, which means that φ is a monomorphism (lemma 1.1.4). In the same way, the kernel of $B \to 0$ is id_B, so exactness at B says that the epimorphism $\text{coker}\varphi$ is zero, which means that φ is an epimorphism. A

morphism that is both is an isomorphism (lemma 1.1.8), and an isomorphism is both.

3. For objects A and B, let $P = A \times B$ with projections p_A, p_B, and let i_A, i_B be as in the proof of lemma 1.1.5, so that $p_A i_A = \mathrm{id}_A$, $p_B i_A = 0$, $p_B i_B = \mathrm{id}_B$ and $i_A p_A + i_B p_B = \mathrm{id}_P$. Then $0 \to A \xrightarrow{i_A} P \xrightarrow{p_B} B \to 0$ is exact. As in (2), exactness at A and at B means that i_A is a monomorphism and p_B an epimorphism, which hold because $p_A i_A = \mathrm{id}_A$ and $p_B i_B = \mathrm{id}_B$. At P, $p_B i_A = 0$, and every t with $p_B t = 0$ satisfies $t = (i_A p_A + i_B p_B)t = i_A(p_A t)$, the factorization being unique as i_A is monic. So i_A is a kernel of p_B, and $\mathrm{coker} i_A \circ \ker p_B = \mathrm{coker} i_A \circ i_A = 0$.

Freyd-Mitchell Theorem

As mentioned, R-mod is an abelian category, and as we shall develop a bit later it the categories R-mod "represent" all abelian categories.

Theorem 1.1.15: Freyd-Mitchell Theorem

Let A be a small abelian category. Then there is a fully faithful exact functor $A \to R\text{-Mod}$ for a suitable ring R. Such a functor reflects exactness: a sequence in A is exact if and only if its image in R-Mod is.

***Proof Key ideas*:**
The proof constructs R as the endomorphism ring of a suitable injective cogenerator in the category of left exact functors $A^{\mathrm{op}} \to \mathbf{Ab}$, and occupies a chapter's worth of category theory that this book does not otherwise need; it is given in full in *An Introduction to Homological Algebra* [23], §1.6. The idea is that A embeds into its category of left exact presheaves by Yoneda, that this larger category is a Grothendieck category with enough injectives, and that a single injective cogenerator there converts the whole category into modules over its endomorphism ring.

The clause about reflecting exactness is immediate from the rest and is what the theorem is used for: a fully faithful exact functor reflects kernels and cokernels, so a diagram chase valid in R-Mod transfers back to A. This is what permits the element-based arguments in lemma 2.1.22 and throughout chapter 6.

1.2 Localization of Categories

The constructions of this book repeatedly produce morphisms that ought to be invertible but are not. The model for what to do about it is already familiar from commutative algebra: a ring R and a multiplicative set S of elements one wishes to invert are replaced by $S^{-1}R$, together with a map $R \to S^{-1}R$ that is universal among ring maps sending S into the units. Nothing in that description mentions elements of a ring rather than morphisms of a category, and the same universal property can be imposed one level up.

The payoff comes in chapter 2, where the class to be inverted is the class of quasi-isomorphisms and

the resulting category is the derived category. The construction is recorded here, before it is needed, because it is a statement about categories and not about complexes.

Definition 1.2.1: Localization of a Category

Let $\mathcal{C}$ be a category and W a class of morphisms of $\mathcal{C}$. A *localization* of $\mathcal{C}$ at W is a category $\mathcal{C}[W^{-1}]$ together with a functor $Q\colon \mathcal{C} \to \mathcal{C}[W^{-1}]$ such that

1. $Q(w)$ is an isomorphism for every $w \in W$;
2. for every category $\mathcal{D}$ and every functor $F\colon \mathcal{C} \to \mathcal{D}$ carrying each element of W to an isomorphism, there is a functor $\overline{F}\colon \mathcal{C}[W^{-1}] \to \mathcal{D}$ with $\overline{F} \circ Q = F$, and $\overline{F}$ is unique with this property.

As with every universal property, the definition determines the answer up to canonical isomorphism as soon as an answer exists: two localizations receive unique comparison functors from one another whose composites must be the identity, by the uniqueness clause applied to Q itself. Existence is the substantive point, and it is settled by a construction that simply writes down the morphisms that formal inversion forces to exist.

Theorem 1.2.2: Existence of the Localization

Let $\mathcal{C}$ be a category and W a class of morphisms. Then a localization $Q\colon \mathcal{C} \to \mathcal{C}[W^{-1}]$ exists.

***Proof*:**

Step 1: The construction. Let $\mathcal{C}[W^{-1}]$ have the same objects as $\mathcal{C}$. For objects X and Y, consider finite *zigzags* from X to Y: alternating strings of morphisms of $\mathcal{C}$

$$X = X_0 \;\; \text{⊢} X_1 \;\; \text{⊢} \;\; \cdots \;\; \text{⊢} X_n = Y$$

in which each link $X_i \text{⊢} X_{i+1}$ is either a morphism $X_i \to X_{i+1}$ of $\mathcal{C}$, called a forward link, or a morphism $X_{i+1} \to X_i$ that belongs to W, called a backward link. Enlarging W to contain all identities and to be closed under composition changes neither the class of functors inverting it nor, therefore, the localization, so assume this without loss of generality; it is what makes the second move below well formed. Impose on the collection of zigzags the smallest equivalence relation generated by: replacing two consecutive forward links by their composite; replacing two consecutive backward links by their composite; deleting a forward or backward link that is an identity; and deleting a pair of adjacent links w forward followed by w backward, or w backward followed by w forward, for $w \in W$. Let $\mathrm{Hom}_{\mathcal{C}[W^{-1}]}(X, Y)$ be the collection of equivalence classes, with composition given by concatenation, which is well defined because the generating relations are local to the links they touch. Concatenation is associative and the empty zigzag is an identity, so this is a category. Let Q send an object to itself and a morphism to the one-link forward zigzag.

Step 2: Q inverts W. For $w \in W$ the backward one-link zigzag on w is a morphism in the opposite direction, and concatenating it with the forward one-link zigzag on w in either order produces a pair of adjacent links which the last generating relation deletes, leaving the empty zigzag. So $Q(w)$ is an isomorphism.

Step 3: The universal property. Let $F\colon \mathcal{C} \to \mathcal{D}$ carry W into isomorphisms. Define $\overline{F}$ to be F on objects, and on a zigzag to be the composite in $\mathcal{D}$ obtained by applying F to each forward link and $F(-)^{-1}$ to each backward link. Each generating relation is respected: composites go to composites by functoriality, identities to identities, and an adjacent pair w, w of opposite orientation goes to $F(w)F(w)^{-1}$ or $F(w)^{-1}F(w)$, both identities. So $\overline{F}$ is well defined on equivalence classes, and it is a functor because concatenation goes to composition. By construction $\overline{F} \circ Q = F$.

For uniqueness, suppose $G \circ Q = F$ for a functor G. Every morphism of $\mathcal{C}[W^{-1}]$ is a concatenation of one-link zigzags, each of which is $Q(f)$ or $Q(w)^{-1}$; a functor is determined by its values on these, and $G(Q(f)) = F(f)$ forces $G(Q(w)^{-1}) = G(Q(w))^{-1} = F(w)^{-1}$. So G agrees with $\overline{F}$ on every morphism, completing the proof.

1.2.3 Warning: Localization and Set Theory The construction above builds each $\mathrm{Hom}_{\mathcal{C}[W^{-1}]}(X, Y)$ out of *all* zigzags, and for a category $\mathcal{C}$ that is not small there is no guarantee that the result is a set rather than a proper class. The property at risk is *local smallness*: the localization of a locally small category need not be locally small.

Two remedies are standard. The first is to assume $\mathcal{C}$ small, which excludes the category of modules over a ring and every category of complexes of modules. The second is to exhibit a calculus of fractions as in theorem 1.2.5, which replaces arbitrary zigzags by roofs. This alone does not bound anything, since the roofs over a fixed object are indexed by a proper class. What is needed in addition is that the W-morphisms into each object admit a *set* that is cofinal among them, and a multiplicative system with that property is what makes the localization locally small.

The zigzag description settles existence and is useless for computation: a morphism is an equivalence class of arbitrarily long strings, and deciding when two strings agree is not something one does by hand. Under a mild hypothesis on W every zigzag collapses to a single roof, and then morphisms become genuinely manipulable.

Definition 1.2.4: Multiplicative System

A class W of morphisms in a category $\mathcal{C}$ is a *multiplicative system* if

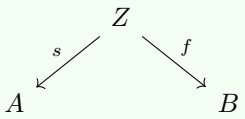
1. W contains every identity and is closed under composition;
2. (Ore condition) for every $f\colon Z \to B$ in $\mathcal{C}$ and every $t\colon V \to B$ in W there are $t'\colon Z' \to Z$ in W and $f'\colon Z' \to V$ in $\mathcal{C}$ with $f \circ t' = t \circ f'$;
3. for every pair $f, g\colon X \to Y$ and every $s\colon Y \to Y'$ in W with $s \circ f = s \circ g$, there is $t\colon X' \to X$ in W with $f \circ t = g \circ t$.

Condition (2) is what lets two roofs be composed, and condition (3) is what makes the resulting notion of equality between roofs transitive. Both are visible in the ring case: for a commutative ring every S is a multiplicative system, and the conditions are the familiar manipulations that put two fractions over a common denominator.

Theorem 1.2.5: Calculus of Fractions

Let W be a multiplicative system in $\mathcal{C}$. Then:

1. every morphism $A \to B$ in $\mathcal{C}[W^{-1}]$ is represented by a *roof*



   with $s \in W$, standing for $Q(f) \circ Q(s)^{-1}$;

2. if two roofs (s, f) and (s', f') are *commonly dominated*, meaning there are $u\colon Z'' \to Z$ and $u'\colon Z'' \to Z'$ with $su = s'u' \in W$ and $fu = f'u'$, then they represent the same morphism;

3. conversely, two roofs representing the same morphism are commonly dominated, so that roofs modulo common domination compute $\mathrm{Hom}_{\mathcal{C}[W^{-1}]}(A, B)$.

***Proof*:**

Step 1: Roofs compose. Given roofs $A \xleftarrow{s} Z \xrightarrow{f} B$ and $B \xleftarrow{t} V \xrightarrow{g} C$, apply the Ore condition to $f\colon Z \to B$ and $t \in W$ to obtain $t'\colon Z' \to Z$ in W and $f'\colon Z' \to V$ with $ft' = tf'$. Then $A \xleftarrow{st'} Z' \xrightarrow{gf'} C$ is a roof, since $st' \in W$ by closure under composition, and in $\mathcal{C}[W^{-1}]$ it represents

$$Q(g)Q(f')Q(t')^{-1}Q(s)^{-1} = Q(g)Q(t)^{-1}Q(f)Q(s)^{-1}$$

the identity following by rearranging $Q(f)Q(t') = Q(t)Q(f')$, both $Q(t)$ and $Q(t')$ being invertible. This is the composite of the two given morphisms.

Step 2: Part (2). Suppose (s, f) and (s', f') are commonly dominated by u, u'. Since $su = s'u'$ lies in W and $Q(s)$ is invertible, $Q(u)$ is invertible too, and then

$$Q(f)Q(s)^{-1} = Q(fu)Q(su)^{-1} = Q(f'u')Q(s'u')^{-1} = Q(f')Q(s')^{-1}$$

so the two roofs represent the same morphism.

Step 3: Common domination is an equivalence relation. Reflexivity and symmetry are immediate. For transitivity, suppose Z_1 dominates (s, f) and (s', f') through u_1, v_1 with $\sigma_1 := su_1 = s'v_1 \in W$, and Z_2 dominates (s', f') and (s'', f'') through u_2, v_2 with $\sigma_2 := s'u_2 = s''v_2 \in W$. Apply the Ore condition to $\sigma_1\colon Z_1 \to A$ and $\sigma_2 \in W$ to get $t'\colon Z_3 \to Z_1$ in W and $w\colon Z_3 \to Z_2$ with $\sigma_1 t' = \sigma_2 w$. Then

$$s'(v_1 t') = \sigma_1 t' = \sigma_2 w = s'(u_2 w)$$

so condition (3) of definition 1.2.4, applied to the parallel pair $v_1t', u_2w\colon Z_3 \rightrightarrows Z'$ and to $s' \in W$, yields $t\colon Z_4 \to Z_3$ in W with $v_1t't = u_2wt$. The roof with middle leg $\sigma_1 t't \in W$ and legs $u_1t't$ and v_2wt then dominates both (s, f) and (s'', f''): the middle leg lies in W by closure, and the two outer compatibilities follow because $fu_1 = f'v_1$ and $f''v_2 = f'u_2$ both factor through f'.

Step 4: Part (1). Every morphism of $\mathcal{C}[W^{-1}]$ is a zigzag, and it suffices to show every zigzag is equivalent to one of the form backward-then-forward. Assign to a zigzag the number of pairs $i < j$ with link i forward and link j backward. Composing two adjacent links of the same orientation strictly decreases the length and does not increase this number; and given an adjacent pair $Z \xrightarrow{f} B \xleftarrow{t} V$ with $t \in W$, the Ore condition gives $Z \xleftarrow{t'} Z' \xrightarrow{f'} V$ with $t' \in W$ and $ft' = tf'$, which represents the same morphism because $Q(t)^{-1}Q(f) = Q(f')Q(t')^{-1}$, and which strictly decreases the count. The count cannot decrease forever, so the process terminates at a zigzag with no forward link preceding a backward one; composing along each run leaves a single roof.

Step 5: Part (3). What remains is the converse of Step 2. The standard route builds the category whose morphisms are roofs modulo common domination, with composition as in Step 1, and checks that this is well defined on classes, associative and unital, and that it satisfies the universal property of definition 1.2.1; comparing with $\mathcal{C}[W^{-1}]$ through that universal property then forces the two to agree, which is exactly the converse. Those verifications are bookkeeping of the same kind as Steps 1 and 3, carrying no further idea, and they are carried out in full in *An Introduction to Homological Algebra* [23]. This completes the proof.

1.2.1 Serre Subcategories and the Quotient

Localizing a category at a class of morphisms is one way to force objects to become isomorphic. There is a second, available when the ambient category is abelian: kill a class of *objects* and keep track of what survives. The result is again abelian, which is what makes it useful, and it is the construction under the localization sequence in K-theory.

Definition 1.2.6: Serre Subcategory

Let $\mathcal{A}$ be an abelian category. A non-empty full subcategory $\mathcal{B} \subseteq \mathcal{A}$ is a *Serre subcategory* if it is closed under subobjects, quotient objects, and extensions: for every short exact sequence

$$0 \to A' \to A \to A'' \to 0$$

in $\mathcal{A}$, the object A lies in $\mathcal{B}$ if and only if both A' and A'' do.

The single biconditional packages all three closure conditions. It is strictly stronger than a two-out-of-three rule: from $A \in \mathcal{B}$ alone it concludes that *both* A' and A'' lie in $\mathcal{B}$, and that direction, which is what gives closure under subobjects and quotients, is the one the proof below leans on most.

Example 1.2.7: Torsion Modules

Let R be a commutative domain. The torsion modules form a Serre subcategory of R-**Mod**: a submodule of a torsion module is torsion, a quotient of one is torsion, and if A' and A'' are torsion in $0 \to A' \to A \to A'' \to 0$ then every $a \in A$ has $ra \in A'$ for some $r \neq 0$ and then $sra = 0$ for some $s \neq 0$, so A is torsion.

By contrast the free R-modules are *not* a Serre subcategory: they are closed under neither quotients (over $R = \mathbb{Z}$, $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2$) nor subobjects in general.

Theorem 1.2.8: The Quotient Abelian Category

Let $\mathcal{A}$ be a small abelian category and $\mathcal{B} \subseteq \mathcal{A}$ a Serre subcategory. Then there is an abelian category $\mathcal{A}/\mathcal{B}$ and an exact functor $\pi\colon \mathcal{A} \to \mathcal{A}/\mathcal{B}$ with the following properties.

1. $\pi(B) = 0$ for every $B \in \mathcal{B}$, and an object A of $\mathcal{A}$ has $\pi(A) = 0$ if and only if $A \in \mathcal{B}$.
2. π is universal: any exact functor $F\colon \mathcal{A} \to \mathcal{D}$ into an abelian category with $F(B) = 0$ for all $B \in \mathcal{B}$ factors as $F = \bar{F} \circ \pi$ for an exact $\bar{F}\colon \mathcal{A}/\mathcal{B} \to \mathcal{D}$, unique up to natural isomorphism.

The category $\mathcal{A}/\mathcal{B}$ is the *Gabriel quotient* of $\mathcal{A}$ by $\mathcal{B}$.

***Proof*:**
The construction is a localization in the sense of definition 1.2.1, taken at the class

$$W = \{\, f \in \mathcal{A} \;:\; \ker f \in \mathcal{B} \text{ and } \operatorname{coker} f \in \mathcal{B} \,\}$$

of morphisms whose kernel and cokernel are both killed, and $\mathcal{A}/\mathcal{B} := \mathcal{A}[W^{-1}]$ with π the canonical functor. Existence is theorem 1.2.2, which needs no hypothesis on W; smallness of $\mathcal{A}$ is what keeps the result locally small, by box 1.2.3.

That W is a multiplicative system is where the Serre condition is consumed, and each clause of definition 1.2.4 uses it differently, so all three are worth doing.

Identities and composition. Every identity has zero kernel and cokernel, and $0 \in \mathcal{B}$ since $\mathcal{B}$ is non-empty and closed under subobjects. For composition, take $f\colon A \to A'$ and $g\colon A' \to A''$ in W. Two elementary exact fragments suffice, and neither needs the full six-term Ker-Coker sequence, which this book does not have at this point. First, $\ker f \subseteq \ker(gf)$ and f carries $\ker(gf)$ into $\ker g$, giving

$$0 \to \ker f \to \ker(gf) \to \ker g$$

so $\ker(gf)$ is an extension of a subobject of $\ker g$ by $\ker f$. Dually,

$$\operatorname{coker} f \to \operatorname{coker}(gf) \to \operatorname{coker} g \to 0$$

exhibits $\operatorname{coker}(gf)$ as an extension of $\operatorname{coker} g$ by a quotient of $\operatorname{coker} f$. Closure under subobjects, quotients and extensions puts both in $\mathcal{B}$, so $gf \in W$.

The Ore condition. Given $t\colon B \to B'$ in W and $f\colon Z \to B'$, form the pullback of t along f. Its projection $t'\colon B \times_{B'} Z \to Z$ has $\ker t' \cong \ker t$ and $\operatorname{coker} t' \hookrightarrow \operatorname{coker} t$. The first is formal; the second holds because $\operatorname{im} t' = f^{-1}(\operatorname{im} t)$, so $Z/f^{-1}(\operatorname{im} t) \hookrightarrow B'/\operatorname{im} t$, an element argument legitimate here because $\mathcal{A}$ is small and theorem 1.1.15 embeds it exactly-and-fully faithfully in a module category. Hence $\ker t' \in \mathcal{B}$ and, by closure under *subobjects*, $\operatorname{coker} t' \in \mathcal{B}$, so $t' \in W$.

The cancellation condition. Suppose $sf = sg$ for $f, g\colon A \to B$ and $s \in W$. Here it is the additive structure of $\mathcal{A}$ that does the work, and no pullback helps. Set $h = f - g$, so $sh = 0$ and hence $\operatorname{im} h \subseteq \ker s$. Since $\ker s \in \mathcal{B}$, closure under subobjects gives $\operatorname{im} h \in \mathcal{B}$. Let $t\colon K \rightarrowtail A$ be the kernel of $A \twoheadrightarrow \operatorname{im} h$. Then $\ker t = 0$ and $\operatorname{coker} t \cong \operatorname{im} h \in \mathcal{B}$, so $t \in W$, and $ht = 0$ gives $ft = gt$.

For (1), $\pi(B) = 0$ because $B \to 0$ lies in W when $B \in \mathcal{B}$. The converse needs an argument, and it is worth giving the right one, since the tempting route is circular: from $\pi(A) = 0$ one would like to say $A \to 0$ is inverted, hence lies in W, hence has kernel A in $\mathcal{B}$. But "inverted $\Rightarrow$ in

> W" is the statement that W is *saturated*, and definition 1.2.1 gives only the inclusion, not the converse.
>
> Argue from clause (3) of theorem 1.2.5 instead. If $\pi(A) = 0$ then $\mathrm{id}_{\pi(A)} = 0$, so the roofs $(\mathrm{id}_A, \mathrm{id}_A)$ and $(\mathrm{id}_A, 0)$ from A to A represent the same morphism, and clause (3) makes them commonly dominated: there is $t\colon Z \to A$ with $t \in W$ and $\mathrm{id}_A \circ t = 0 \circ t$, that is $t = 0$. A zero map has $\mathrm{coker} t = A$, and $t \in W$ puts that cokernel in $\mathcal{B}$. Hence $A \in \mathcal{B}$. Exactness of π and the abelian structure on $\mathcal{A}/\mathcal{B}$ are transported along the calculus of fractions, kernels and cokernels being computed by representative fractions. Clause (2) is clause (2) of definition 1.2.1 together with the observation that an exact F killing $\mathcal{B}$ inverts every element of W: exactness gives $\ker F(f) = F(\ker f) = 0$ and $\mathrm{coker} F(f) = F(\mathrm{coker} f) = 0$, and a map both monic and epic in an abelian category is an isomorphism. That produces a functor $\bar{F}$ with $\bar{F}\pi = F$. That $\bar{F}$ is moreover *exact* then follows in two lines, granting that kernels and cokernels in $\mathcal{A}/\mathcal{B}$ are computed as π-images of those in $\mathcal{A}$: every morphism of $\mathcal{A}/\mathcal{B}$ is $\pi(f)$ composed with an isomorphism, so it is enough to check on π-images, where $\bar{F}(\ker \pi f) = \bar{F}\pi(\ker f) = F(\ker f) = \ker F(f)$. That the transported kernels and cokernels are indeed computed this way is part of the verification recorded as omitted below.
>
> What is omitted is the routine verification that the transported kernels and cokernels satisfy the abelian axioms, which is a diagram chase on representative fractions with no new idea in it. The construction is Gabriel's, as we sought to show.

The quotient is the categorical analogue of localizing a ring: for R commutative and $S \subseteq R$ multiplicative, $S^{-1}R$**-mod** is the quotient of R**-mod** by the S-torsion modules. That example is the one to keep in mind, and it is the case that appears in the localization sequence of algebraic K-theory.

2

Complexes and (co)Homology

Let **A** be a fixed abelian category, which by theorem 1.1.15 may without loss of generality it can be considered ${}_R\text{Mod}$. In this chapter, we shall focus on studying (co)chains of such modules, that is sequence of modules that respect equation (1):

$$\text{im}(d^n) \subseteq \ker(d^{n+1})$$

The collection of all chains shall be denoted $\text{Ch}(\mathbf{A})$. We shall see that $\text{Ch}(\mathbf{A})$ is an abelian category, the notion of extensions (and snakes lemma) is well defined, which will be explored. In the right categories, it is relatively easy to construct a concrete complex given the category has enough objects, for example $M \in \text{Obj}(R\text{-}\mathbf{Mod})$ always has a complex known as a *free resolution* (see [8, Free Resolution]). There will be many different ways of creating complexes given an object, giving us different types of resolutions. We shall see that when a category has enough projective objects (resp. Injective objects), then "up to homotopy equivalence", projective resolutions shall be *initial* (resp. final) in the category of complexes $\text{Ch}(A)$. Thus, we shall take the time to develop the homology theory in categories with enough projective or injective objects. We shall see that under appropriate circumstances, taking (co)homology of a projective resolution of an object can give us some important information about said object.

Homology and cohomology are duals of each other, which means on the theoretical level they are the same. As cohomology is the more computationally useful tool, we shall be focusing on cohomology on this chapter, with the idea that the reader shall be comfortable being able to switch indices when necessary. Taking the cohomology of a complex shall be shown to be an additive functorial process, meaning for a given morphism of complexes $\alpha^\bullet : A^\bullet \to B^\bullet$, we shall get a morphism each cohomology $H^i(\alpha) : H^i(A^\bullet) \to H^i(B^\bullet)$ which will be shown can be to form a new complex $H^\bullet(\alpha) : H^\bullet(A^\bullet) \to H^\bullet(B^\bullet)$.

The goal of this chapter is to show how to extract information given a complexes (co)homology. Given this, it is important to know what information a morphism of complexes preserves in cohomology, that is given a morphism of complexes, $\alpha^\bullet : A^\bullet \to B^\bullet$, when is $H^\bullet(\alpha) : H^\bullet(A^\bullet) \to H^\bullet(B^\bullet)$ an

isomorphism? This shall lead us to the notion of *quasi-isomorphisms*, and their "representation" through mapping cones. As it turns out, quasi-isomorphisms do not behave nicely in general. What we shall then seek out is a stricter concept known as a *homotopy equivalence* that will still be a quasi-isomorphism, but will be invertible in the appropriate category.

2.1 Basic Definition and Properties

Starting with defining the generalization of an exact sequence:

Definition 2.1.1: Chain Complex

A *chain complex* $(C_\bullet, d_\bullet)$ is a sequence of objects and morphisms:

$$\cdots \xleftarrow{d_i} M_i \xleftarrow{d_{i+1}} M_{i+1} \xleftarrow{d_{i+2}} \cdots$$

with the indices running through $\mathbb{Z}$, such that

$$d_i \circ d_{i+1} = 0$$

and each d_i is called a *boundary map*. Similarly, a *cochain complex* $(C^\bullet, d^\bullet)$ is a sequence of objects

$$\cdots \xrightarrow{d^{i-1}} M^i \xrightarrow{d^i} M^{i+1} \xrightarrow{d^{i+1}} \cdots$$

with the indices running through $\mathbb{Z}$, such that

$$d^i \circ d^{i-1} = 0$$

where $d^\bullet$ is called the *differential*[a].

[a] Due to it's origin's in differential geometry and algebraic topology

Chains and cochains can be interchanged by taking

$$M_i = M^{-i}$$

Working with one or the other makes essentially no difference for homological algebra. In certain cases, some structures present themselves more in the cohomological case (ex. a natural ring structure called the *cohomology ring* shall become important to us), and so we opt to work with the cohomology, even though at this stage this can be considered to be an arbitrary choice. The condition that $d^i \circ d^{i-1} = 0$ is the same as $\mathrm{im} d^{i-1} \subseteq \ker d^i$. If the complex is exact, then $\mathrm{im} d^{i-1} = \ker d^i$ for all i. We also often write $C^\bullet$ instead of $(C^\bullet, d^\bullet)$ if the mappings $d^\bullet$ are clear from context or implied. In some textbooks, even the bullet is dropped and M represents the complex. Sometime, for homology, the notation $\partial_\bullet$ shall be used instead of $d_\bullet$ due to it's connection to being the boundary map for topological spaces when creating complexes out of topological spaces[1].The condition that $d^i \circ d^{i-1} = 0$ (resp. $d_i \circ d_{i+1} = 0$) is often abbreviated to $d \circ d = 0$.

Due to historical origins in algebraic topology, the kernel and image of the differential

[1] Similarly, $d^\bullet$ stands for the *differential* due to it's link to the derivative, which would be explored in a class in algebraic topology, or see EYNTKA Algebraic Topology

Definition 2.1.2: Cycles and Boundaries

Let $C_\bullet$ (resp. $C^\bullet$) be a chain complex (resp. cochain complex). Then:

1. $\ker(d_n)$ is called the *n-cycles* of $C_\bullet$ and $\operatorname{im}(d_{n+1})$ is called the n-boundary of $C_\bullet$. The n-cycle is represented as $Z_n = Z_n(C_\bullet)$ and the *n-boundary* is represented as $B_n = B_n(C_\bullet)$
2. $\ker(d^n)$ is called the *n-cocycles* of $C^\bullet$ and $\operatorname{im}(d^{n-1})$ is called the *n-coboundary* of $C^\bullet$. The n-cocycle is represented as $Z^n = Z^n(C^\bullet)$ and the n-coboundary is represented as $B^n = B^n(C^\bullet)$

The name cycles and boundaries has its origin in algebraic topology:

Example 2.1.3: Cycles and Boundaries

Let T be the boundary of a triangle, with vertices v_0, v_1, v_2 and oriented edges $e_{01} = [v_0, v_1]$, $e_{12} = [v_1, v_2]$ and $e_{20} = [v_2, v_0]$. Its simplicial chain complex has $C_1 = \mathbb{Z}e_{01} \oplus \mathbb{Z}e_{12} \oplus \mathbb{Z}e_{20}$, $C_0 = \mathbb{Z}v_0 \oplus \mathbb{Z}v_1 \oplus \mathbb{Z}v_2$ and $C_n = 0$ otherwise, with $d_1[v_i, v_j] = v_j - v_i$. Since $d_1(ae_{01} + be_{12} + ce_{20}) = (c-a)v_0 + (a-b)v_1 + (b-c)v_2$, the 1-cycles are the multiples of $z = e_{01} + e_{12} + e_{20}$, the chain that runs once around the triangle. As $C_2 = 0$, the only 1-boundary is 0, so z is a cycle that is not a boundary, and $Z_1/B_1 \cong \mathbb{Z}$ detects the hole enclosed by T.

The collection of all complexes forms a Category:

Definition 2.1.4: Category Of Complexes

Let $\mathbf{A}$ be an abelian category. Then $\mathrm{Ch}^\bullet(\mathbf{A})$ is a new category called the *category of complexes of A* where

1. $\mathrm{Obj}(\mathrm{Ch}^\bullet(\mathbf{A})) = \{\text{cochain complexes in } \mathbf{A}\}$
2. For two complexes $C^\bullet$ and $N^\bullet$, a morphism $\alpha^\bullet \in \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, N^\bullet)$ consists of morphisms such that the following diagram commute:

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & M^{i-1} & \xrightarrow{d^{i-1}_{C^\bullet}} & M^i & \xrightarrow{d^i_{C^\bullet}} & M^{i+1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle \alpha^{i-1}} & & \downarrow{\scriptstyle \alpha^{i-1}} & & \downarrow{\scriptstyle \alpha^{i-1}} & & \\
\cdots & \longrightarrow & N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1} & \longrightarrow & \cdots
\end{array}$$

Similarly for $\mathrm{Ch}_\bullet(\mathbf{A})$ with chain complexes.

If unambiguous, the bullet is dropped giving $\mathrm{Ch}(\mathbf{A})$. It should be checked that $\mathrm{Ch}(A)$ is indeed a category. There are many variations of $\mathrm{Ch}(A)$ which will be important:

1. $C^+(A)$ is the full subcategory of $\mathrm{Ch}(A)$ of complexes which are bounded from bellow, so that $L^i = 0$ for $i \ll 0$
2. $C^-(A)$ is the full subcategory of $\mathrm{Ch}(A)$ of complexes which are bounded from above, so that $L^i = 0$ for $i \gg 0$
3. $C^{\geq 0}(A)$ (resp. $C^{\leq 0}$) for which $L^i = 0$ for $i \leq 0$ (resp. $i \geq 0$)

4. P, I are full subcategories of A consisting of projective and injective objects in A respectively, and hence we have $\mathrm{Ch}(P)$, $\mathrm{Ch}(I)$, and all its variants

There are also a few simple functors we may immediately define:

1. Take $\iota_r : \mathbf{A} \to \mathrm{Ch}(\mathbf{A})$ that sends an object A form $\mathbf{A}$ to a complex that is zero every and A in the rth degree. We shall commonly identify A with the complex given by ι_0.
2. Take $s_r : \mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{A})$ which shifts over the complex, namely
$$C^\bullet \mapsto M[r]^\bullet$$
where $M[r]^i = M^{i+r}$ and $d^i_{M[r]^\bullet} = (-1)^r d^{i+r}_{C^\bullet}$ (the $(-1)^r$ shall be explained shortly).
3. the canonical truncations $\tau^{\leq n}$ and $\tau^{\geq n}$, defined in definition 2.1.12 once cohomology is available.

More importantly, $\mathrm{Ch}(A)$ is an abelian category, allowing the notion of exactness between complexes to be well-defined:

Lemma 2.1.5: Complex Category Is Abelian

Let A be an abelian category and $\mathrm{Ch}(A)$ the category of complexes of A. Then $\mathrm{Ch}(A)$ is an abelian category, in which kernels and cokernels are computed degreewise. In particular, a sequence of complexes is exact if and only if it is exact in each degree.

***Proof*:**
Write $d^n_C \colon C^n \to C^{n+1}$ for the differentials of a complex $C^\bullet$. Every condition below is checked in each degree n.

Step 1: $\mathrm{Ch}(A)$ *is additive (definition 1.1.1).* For chain maps $\alpha, \beta \colon C^\bullet \to D^\bullet$ put $(\alpha+\beta)^n = \alpha^n + \beta^n$. By bilinearity of composition in A,
$$d^n_D(\alpha^n + \beta^n) = \alpha^{n+1} d^n_C + \beta^{n+1} d^n_C = (\alpha^{n+1} + \beta^{n+1}) d^n_C$$
so $\alpha+\beta$ is a chain map, and the same computation shows that $-\alpha$ and the zero family are chain maps. The group axioms hold in each degree, so $\mathrm{Hom}_{\mathrm{Ch}(A)}(C^\bullet, D^\bullet)$ is an abelian group, and composition, which is computed degreewise, is bilinear because composition in A is. The complex $0^\bullet$ with a zero object in every degree is a zero object of $\mathrm{Ch}(A)$, since the unique morphisms $0 \to C^n$ and $C^n \to 0$ commute with all differentials (every composite through 0 is zero).

For complexes $C^\bullet$ and $D^\bullet$, let $P^n = C^n \times D^n$ with projections p_C, p_D, and let i_C, i_D be the morphisms into P^n with $p_C i_C = \mathrm{id}$, $p_D i_C = 0$, $p_C i_D = 0$ and $p_D i_D = \mathrm{id}$ (the degree is dropped from this notation). Then $i_C p_C + i_D p_D = \mathrm{id}_{P^n}$, since both sides have the same composites with p_C and with p_D. Put $d^n_P = i_C d^n_C p_C + i_D d^n_D p_D$. These identities give
$$d^{n+1}_P d^n_P = i_C d^{n+1}_C d^n_C p_C + i_D d^{n+1}_D d^n_D p_D = 0 \qquad p_C d^n_P = d^n_C p_C \qquad d^n_P i_C = i_C d^n_C$$
and the same identities with D in place of C, so $P^\bullet$ is a complex and p_C, p_D, i_C, i_D are chain maps satisfying the same identities in $\mathrm{Ch}(A)$. If $\alpha \colon T^\bullet \to C^\bullet$ and $\beta \colon T^\bullet \to D^\bullet$ are chain

maps, then $\gamma = i_C\alpha + i_D\beta$ is a chain map with $p_C\gamma = \alpha$ and $p_D\gamma = \beta$, and any γ' with these two properties equals $(i_Cp_C + i_Dp_D)\gamma' = i_C\alpha + i_D\beta$. So $P^\bullet$ is a product of $C^\bullet$ and $D^\bullet$. If $\alpha\colon C^\bullet \to T^\bullet$ and $\beta\colon D^\bullet \to T^\bullet$ are chain maps, then $h = \alpha p_C + \beta p_D$ satisfies $hi_C = \alpha$ and $hi_D = \beta$, and any h' with these two properties equals $h'(i_Cp_C + i_Dp_D) = \alpha p_C + \beta p_D$. So $P^\bullet$ is also a coproduct. Iterating gives all finite products and coproducts, the empty ones being $0^\bullet$, so $\mathrm{Ch}(A)$ is additive.

Step 2: kernels and cokernels, degreewise. Let $f\colon C^\bullet \to D^\bullet$ be a chain map, and suppose $j\colon J^\bullet \to C^\bullet$ is a chain map such that each j^n is a kernel of f^n. Then $fj = 0$. If $g\colon T^\bullet \to C^\bullet$ is a chain map with $fg = 0$, then each $g^n = j^nu^n$ for a unique u^n, and

$$j^{n+1}d_J^nu^n = d_C^nj^nu^n = d_C^ng^n = g^{n+1}d_T^n = j^{n+1}u^{n+1}d_T^n$$

so $d_J^nu^n = u^{n+1}d_T^n$, because j^{n+1} is a monomorphism (lemma 1.1.3). Thus u is the unique chain map with $ju = g$, and j is a kernel of f in $\mathrm{Ch}(A)$. Such a j exists: choose kernels $j^n\colon J^n \to C^n$ of the f^n in A (definition 1.1.6). As $f^{n+1}d_C^nj^n = d_D^nf^nj^n = 0$, there is a unique d_J^n with $j^{n+1}d_J^n = d_C^nj^n$, and $j^{n+2}d_J^{n+1}d_J^n = d_C^{n+1}d_C^nj^n = 0$ gives $d_J^{n+1}d_J^n = 0$, again because j^{n+2} is monic. Kernels are unique up to isomorphism, so every kernel of f in $\mathrm{Ch}(A)$ is a kernel of f^n in each degree.

If $q\colon D^\bullet \to Q^\bullet$ is a chain map such that each q^n is a cokernel of f^n, and $g\colon D^\bullet \to T^\bullet$ is a chain map with $gf = 0$, then each $g^n = v^nq^n$ for a unique v^n, and

$$d_T^nv^nq^n = d_T^ng^n = g^{n+1}d_D^n = v^{n+1}q^{n+1}d_D^n = v^{n+1}d_Q^nq^n$$

gives $d_T^nv^n = v^{n+1}d_Q^n$, because q^n is an epimorphism (lemma 1.1.3). So q is a cokernel of f in $\mathrm{Ch}(A)$. Such a q exists: choose cokernels q^n of the f^n. As $q^{n+1}d_D^nf^n = q^{n+1}f^{n+1}d_C^n = 0$, there is a unique d_Q^n with $d_Q^nq^n = q^{n+1}d_D^n$, and $d_Q^{n+1}d_Q^nq^n = q^{n+2}d_D^{n+1}d_D^n = 0$ gives $d_Q^{n+1}d_Q^n = 0$. As for kernels, every cokernel of f in $\mathrm{Ch}(A)$ is a cokernel of f^n in each degree.

Step 3: monomorphisms and epimorphisms are degreewise. If each f^n is a monomorphism, so is f: from $fg = fh$ we get $f^ng^n = f^nh^n$, hence $g^n = h^n$, for every n. Conversely, let f be a monomorphism of $\mathrm{Ch}(A)$ with kernel j as in Step 2. From $fj = 0 = f \circ 0$ we get $j = 0$, so each $j^n = 0$, and since j^n is a monomorphism, $j^n \circ \mathrm{id}_{J^n} = 0 = j^n \circ 0$ gives $\mathrm{id}_{J^n} = 0$. So J^n is a zero object, $0 \to C^n$ is a kernel of f^n, and f^n is a monomorphism by lemma 1.1.4(1). If each f^n is an epimorphism, then $gf = hf$ gives $g^n = h^n$ for every n, so f is an epimorphism. Conversely, if f is an epimorphism of $\mathrm{Ch}(A)$ with cokernel q as in Step 2, then $qf = 0 = 0 \circ f$ gives $q = 0$, and $\mathrm{id}_{Q^n} \circ q^n = 0 = 0 \circ q^n$ with q^n an epimorphism gives $\mathrm{id}_{Q^n} = 0$. So $D^n \to 0$ is a cokernel of f^n, and f^n is an epimorphism by lemma 1.1.4(2).

Step 4: condition (2) of definition 1.1.6. Let $f\colon C^\bullet \to D^\bullet$ be a monomorphism of $\mathrm{Ch}(A)$ and q a cokernel of f as in Step 2. By Step 3 each f^n is a monomorphism, hence the kernel of some morphism by condition (2) of definition 1.1.6 in A, hence a kernel of its own cokernel q^n by lemma 1.1.7. As f is a chain map, Step 2 applied to q shows that f is a kernel of q in $\mathrm{Ch}(A)$. If instead f is an epimorphism of $\mathrm{Ch}(A)$ and j is a kernel of f as in Step 2, then each f^n is an epimorphism by Step 3, hence a cokernel of its own kernel j^n by condition (2) of definition 1.1.6 and lemma 1.1.7, and Step 2 applied to j shows that f is a cokernel of j. Together with Steps 1 and 2, this verifies definition 1.1.6 for $\mathrm{Ch}(A)$.

Step 5: exactness. By definition 1.1.13, a sequence $C^\bullet \xrightarrow{\varphi} D^\bullet \xrightarrow{\psi} E^\bullet$ is exact at $D^\bullet$ when $\psi\varphi = 0$ and $\operatorname{coker}\varphi \circ \ker\psi = 0$. By Step 2, $\ker\psi$ and $\operatorname{coker}\varphi$ may be taken with components $\ker\psi^n$ and $\operatorname{coker}\varphi^n$, and a chain map is zero exactly when all its components are, so both conditions hold in $\mathrm{Ch}(A)$ exactly when they hold for $C^n \to D^n \to E^n$ for every n. Hence a sequence of complexes is exact if and only if it is exact in each degree, completing the proof.

Since it is an abelian category, we shall be able to consider sequences of complexes, and in particular exact sequences of complexes:

$$\cdots \to L^\bullet \to C^\bullet \to N^\bullet \to \cdots$$

which give a commutative diagram like so:

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
\cdots & \longrightarrow & A_{n+2} & \longrightarrow & A_{n+1} & \longrightarrow & A_n & \longrightarrow & A_{n-1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle \iota} & & \\
\cdots & \longrightarrow & B_{n+2} & \longrightarrow & B_{n+1} & \longrightarrow & B_n & \longrightarrow & B_{n-1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi} & & \\
\cdots & \longrightarrow & C_{n+2} & \longrightarrow & C_{n+1} & \longrightarrow & C_n & \longrightarrow & C_{n-1} & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & & 0 & &
\end{array} \tag{2.1}$$

Example 2.1.6: Complexes

1. Given any object M, we may always create the trivial complex
$$\cdots 0 \to 0 \to M \to 0 \to 0 \to \cdots$$
Verify that this is indeed a complex.
2. As mentioned, In R-**Mod**, the free resolution is an example of a complex.
3. Any short exact sequence can be thought of an (exact) complex if we add 0's that extend arbitrarily to the left and right.
4. Let $C^n = \mathbb{Z}/8\mathbb{Z}$ for $n \geq 0$ and $C^n = 0$ for $n < 0$. Let each d^n be defined as:
$$\overline{x} \mapsto \overline{4x}$$
Show that this is a cochain complex.
5. Let $\{V_i, W_i\}_{i\in\mathbb{Z}}$ be a collection of vector spaces. Define:
$$A_n = V_n \oplus W_n \oplus V_{n-1}$$
Combining the projection and inclusion maps $A_n \to V_{n-1} \to A_{n-1}$, show that $A_\bullet$ forms a chain complex.
6. Let $C^\bullet$ be a cochain complex, choose an object $A \in \mathbf{A}$, and apply $\mathrm{Hom}_A(-, A)$ to each object. Show that this is a cochain complex.

Since Ch($\mathbf{A}$) is an abelian category, kernels and cokernels exist and match with monomorphisms/epimorphisms, hence giving us the following:

Definition 2.1.7: Sub-complexes And Quotient Complexes

Let $C^\bullet$ be a (co)chain complex. Then a (co)chain complex $N^\bullet$ is called a *subcomplex* if each N_n is a sub-object of C_n and the differential on B_n is the restriction of the differential in C_n (categorically, the inclusion map $\iota : N^\bullet \to C^\bullet$ is a morphism of complexes).

Furthermore, C^n/B^n is a *quotient-complex*. Furthermore, by the canonical decomposition of morphisms in abelian categories, for any complex-morphism $\alpha^\bullet$, $\{\ker(\alpha^\bullet)\}$ and $\{\text{coker}(\alpha^\bullet)\}$ is a sub-complex and quotient complex respectively.

2.1.1 Homology and Cohomology

Chain complexes need not be exact. The cohomology of a complex measures how far away a complex is from being exact, that is how far away a complex is form representing a sequence of extensions

Definition 2.1.8: Homology and Cohomology Of A Complex

Let $(C^\bullet, d^\bullet)$ be a complex. Then the ith homology and cohomology of the complex at M^i is, respectively:

$$H_i(C^\bullet) = \frac{\ker d_i}{\text{im} d_{i+1}} \qquad H^i(C^\bullet) := \frac{\ker d^i}{\text{im} d^{i-1}}$$

Though homology and cohomology are similar, it is not necessarily the case that $H^i(C^\bullet) = H_{n-i}(C^\bullet)$ for some appropriate n around which the chain can "pivot". We shall see such examples soon, and later also show under what conditions we may compare homology and cohomology[2]

Example 2.1.9: Cohomology Of Complex

Let's compute the cohomology of sequences in example 2.1.6.

1. The homology and cohomology at the 0th entry is:
$$\frac{\ker(d_0)}{\text{im}(d_1)} = \frac{\ker(d^0)}{\text{im}(d^{-1})} = M$$
which shows that at this position, the chain is "off" by M in being exact.
2. A free resolution $\cdots \to F_1 \to F_0 \to M \to 0$, with M kept as the last term, is exact, meaning $\text{im}(d^{i-1}) = \ker(d^i)$ in every degree, so its homology and cohomology vanish. With M removed the complex $\cdots \to F_1 \to F_0 \to 0$ has homology M in degree 0 and 0 elsewhere.
3. Any exact sequence has trivial homology and cohomology
4. This complex is not exact. Since d^n is multiplication by 4 on $\mathbb{Z}/8\mathbb{Z}$, its kernel is $\{\overline{0}, \overline{2}, \overline{4}, \overline{6}\}$ and its image is $\{\overline{0}, \overline{4}\}$. So $H^0 = \ker d^0 \cong \mathbb{Z}/4\mathbb{Z}$, and $H^n \cong \mathbb{Z}/2\mathbb{Z}$ for every $n \geq 1$.

[2]The universal coefficient theorem and Poincaré duality being two famous examples the reader may have heard of if they took algebraic topology

5. The reader should calculate this cohomology as an exercise
6. This cohomology shall be calculated later in this chapter.

Lemma 2.1.10: Cohomology Functor

For every $i \in \mathbb{Z}$, there exists a functor:

$$H^i : \mathrm{Ch}(A) \to A \qquad C^\bullet \mapsto H^i(C^\bullet)$$

which is additive and covariant.

This shows that taking cohomology is indeed a way of analyzing resolutions, for it preserves enough of the structures of modules and complexes. Notice it is not necessarily an exact functor, hence it will not in general preserve extensions.

***Proof*:**
We compute with elements, as theorem 1.1.15 permits when applied to a small abelian subcategory containing the objects involved. Let $\alpha\colon C^\bullet \to N^\bullet$ be a chain map, so that $d^i_N\alpha^i = \alpha^{i+1}d^i_C$ for every i. If $z \in \ker d^i_C$, then $d^i_N(\alpha^i(z)) = \alpha^{i+1}(d^i_C(z)) = 0$, so α^i maps $\ker d^i_C$ into $\ker d^i_N$. If $z = d^{i-1}_C(w)$, then $\alpha^i(z) = d^{i-1}_N(\alpha^{i-1}(w))$, so α^i maps $\mathrm{im}d^{i-1}_C$ into $\mathrm{im}d^{i-1}_N$. Hence

$$H^i(\alpha)\colon H^i(C^\bullet) \to H^i(N^\bullet) \qquad z + \mathrm{im}d^{i-1}_C \mapsto \alpha^i(z) + \mathrm{im}d^{i-1}_N$$

is a well-defined homomorphism of the quotients of definition 2.1.8. The formula gives $H^i(\mathrm{id}_{C^\bullet}) = \mathrm{id}$ and, since $(\beta\circ\alpha)^i = \beta^i\circ\alpha^i$, also $H^i(\beta\circ\alpha) = H^i(\beta)\circ H^i(\alpha)$, so H^i is a covariant functor. It is additive because $(\alpha+\alpha')^i(z) = \alpha^i(z) + \alpha'^i(z)$ gives $H^i(\alpha+\alpha') = H^i(\alpha) + H^i(\alpha')$, as claimed.

Notice that each H^i takes d^i to the zero map. Hence, we may consider the following functor:

$$H^\bullet : \mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{A})$$

by taking $H^i(C^\bullet)$ for each i and connecting them by the zero-morphism. This is the cohomology of this complex then equals that of $C^\bullet$, namely:

$$H^\bullet(H^\bullet(C^\bullet)) = H^\bullet(C^\bullet)$$

Lemma 2.1.11: Cohomology And Products

Let $\{A^\bullet_\alpha\}$ be a collection of complexes in $\mathbf{A}$, and suppose the coproduct (respectively the product) indexed by α exists and is an exact functor. Then

$$H^n\Big(\bigoplus_\alpha A^\bullet_\alpha\Big) = \bigoplus_\alpha H^n(A^\bullet_\alpha) \qquad H^n\Big(\prod_\alpha A^\bullet_\alpha\Big) = \prod_\alpha H^n(A^\bullet_\alpha)$$

The exactness hypothesis is automatic for *finite* index sets, where the two agree by lemma 1.1.5, and it holds for arbitrary index sets in R-Mod.

***Proof*:**
Treat the coproduct; the product is the same argument in the opposite category. The complex $\bigoplus_\alpha A_\alpha^\bullet$ has n-th term $\bigoplus_\alpha A_\alpha^n$ and differential $\bigoplus_\alpha d_\alpha^n$.

An exact functor preserves kernels and images, so

$$\ker\Big(\bigoplus_\alpha d_\alpha^n\Big) = \bigoplus_\alpha \ker d_\alpha^n, \qquad \operatorname{im}\Big(\bigoplus_\alpha d_\alpha^{n-1}\Big) = \bigoplus_\alpha \operatorname{im} d_\alpha^{n-1}$$

and it preserves the quotient of the second by the first, being exact. Hence

$$H^n\Big(\bigoplus_\alpha A_\alpha^\bullet\Big) = \frac{\bigoplus_\alpha \ker d_\alpha^n}{\bigoplus_\alpha \operatorname{im} d_\alpha^{n-1}} = \bigoplus_\alpha \frac{\ker d_\alpha^n}{\operatorname{im} d_\alpha^{n-1}} = \bigoplus_\alpha H^n(A_\alpha^\bullet)$$

For a finite index set the coproduct is a finite biproduct, which every additive functor preserves, so no hypothesis is needed; in R-Mod both $\bigoplus$ and $\prod$ are exact and kernels and images are computed componentwise, giving the statement for arbitrary index sets. This completes the proof.

A complex can be cut off at a degree n so as to keep exactly its cohomology in degrees at most n, or exactly its cohomology in degrees at least n.

Definition 2.1.12: Canonical Truncation

Let $X \in \mathrm{Ch}(\mathbf{A})$ and $n \in \mathbb{Z}$, and write $Z^n = \ker d^n$ and $B^n = \operatorname{im} d^{n-1}$ as in definition 2.1.2. The *canonical truncations* of X at n are the complexes

$$\begin{aligned} \tau^{\leq n}X\colon &\quad \cdots \to X^{n-2} \to X^{n-1} \to Z^n \to 0 \to \cdots \\ \tau^{\geq n}X\colon &\quad \cdots \to 0 \to X^n/B^n \to X^{n+1} \to X^{n+2} \to \cdots \end{aligned}$$

with Z^n and X^n/B^n in degree n. The map $X^{n-1} \to Z^n$ is d^{n-1}, which lands in Z^n because $d^n d^{n-1} = 0$, and $X^n/B^n \to X^{n+1}$ is induced by d^n, which vanishes on B^n. All other differentials are those of X. The inclusion $Z^n \to X^n$ together with identities in degrees below n is a chain map $i_n\colon \tau^{\leq n}X \to X$, and the quotient $X^n \to X^n/B^n$ together with identities in degrees above n is a chain map $p_n\colon X \to \tau^{\geq n}X$.

Both squares at degree n commute: d^n vanishes on Z^n, and the quotient map kills $\operatorname{im} d^{n-1} = B^n$. A chain map $f\colon X \to Y$ carries $Z^n(X)$ into $Z^n(Y)$ and $B^n(X)$ into $B^n(Y)$, so it induces chain maps $\tau^{\leq n}f$ and $\tau^{\geq n}f$. This makes $\tau^{\leq n}$ and $\tau^{\geq n}$ functors $\mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{A})$, with i_n and p_n natural.

Example 2.1.13: Truncating a Resolution of $\mathbb{Z}/2$

Let X be the complex $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ in degrees -1 and 0. Then $Z^{-1} = \ker(2) = 0$, so $\tau^{\leq -1}X = 0$. Also $B^0 = 2\mathbb{Z}$, so $\tau^{\geq 0}X$ is $\mathbb{Z}/2\mathbb{Z}$ in degree 0, and $p_0\colon X \to \tau^{\geq 0}X$ is the reduction map $\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ in degree 0. Since $B^{-1} = 0$ and $Z^0 = X^0$, both $\tau^{\geq -1}X$ and $\tau^{\leq 0}X$ equal X.

Proposition 2.1.14: Cohomology of the Canonical Truncations

Let $X \in \mathrm{Ch}(\mathbf{A})$ and $n \in \mathbb{Z}$. Then $H^i(i_n)\colon H^i(\tau^{\leq n}X) \to H^i(X)$ is an isomorphism for $i \leq n$, and $H^i(\tau^{\leq n}X) = 0$ for $i > n$. Likewise $H^i(p_n)\colon H^i(X) \to H^i(\tau^{\geq n}X)$ is an isomorphism for $i \geq n$, and $H^i(\tau^{\geq n}X) = 0$ for $i < n$. In particular i_n induces an isomorphism on every H^i when $H^i(X) = 0$ for all $i > n$, and p_n does when $H^i(X) = 0$ for all $i < n$.

***Proof*:**
We compute with elements, as theorem 1.1.15 permits.

Step 1: $\tau^{\leq n}X$. For $i \leq n-2$ the complexes $\tau^{\leq n}X$ and X agree in degrees $i-1$, i and $i+1$, and i_n is the identity there. For $i = n-1$ the kernel of $X^{n-1} \to Z^n$ equals the kernel of d^{n-1}, because $Z^n \to X^n$ is a monomorphism, and the image from degree $n-2$ is unchanged, so $H^{n-1}(i_n)$ is the identity of $H^{n-1}(X)$. In degree n the kernel of $Z^n \to 0$ is Z^n and the image of $X^{n-1} \to Z^n$ is B^n, so $H^n(\tau^{\leq n}X) = Z^n/B^n = H^n(X)$, and $H^n(i_n)$, induced by the inclusion $Z^n \to X^n$, is the identity of Z^n/B^n. Above degree n the complex $\tau^{\leq n}X$ is zero.

Step 2: $\tau^{\geq n}X$. For $i \geq n+2$ the two complexes agree in degrees $i-1$, i, $i+1$. For $i = n+1$ the kernel of d^{n+1} is unchanged, and the image of $X^n/B^n \to X^{n+1}$ equals the image of d^n, because $X^n \to X^n/B^n$ is an epimorphism. In degree n the induced map sends $x + B^n$ to $d^n x$, so its kernel is Z^n/B^n, and nothing maps into degree n. So $H^n(\tau^{\geq n}X) = Z^n/B^n = H^n(X)$, and $H^n(p_n)$ is the identity of Z^n/B^n. Below degree n the complex $\tau^{\geq n}X$ is zero. The last sentence of the statement follows from the two descriptions, completing the proof.

Now, given a morphism of complexes $\alpha^\bullet : C^\bullet \to N^\bullet$, we may ask what (co)homological information such a morphism preserves. In other words, given a morphism $\alpha^\bullet$, when is $H^\bullet(\alpha^\bullet)$ an isomorphism? We give such morphisms a name:

Definition 2.1.15: Quasi-Isomorphism

Let $\alpha^\bullet : C^\bullet \to N^\bullet$ be a morphism of complexes. Then if $H^\bullet(\alpha^\bullet) : H^\bullet(C^\bullet) \to H^\bullet(N^\bullet)$ is an isomorphism, $\alpha^\bullet$ is called a *quasi-isomorphism*

One may wonder if the fact that the (co)homology is isomorphism is strong enough of a condition for the chains to be isomorphic. This turns out to be rather far from the case:

Example 2.1.16: Quasi-Isomorphism

1. The following counter example shall come up with many different minor variations, and so the reader should keep this in mind. Take the following morphism of complexes:

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\cdot 2} & \mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \pi} & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \cdots
\end{array}$$

   This is a quasi-isomorphism, as the reader should check, however it does not have an inverse.

More generally, it is not even an equivalence relation, failing to be symmetric (show there is no morphism the other way around that is a quasi-isomorphism).

2. A quasi-isomorphism between $C^\bullet$ and the complex $H^\bullet(C^\bullet)$ need not exist in a given direction. For $C^\bullet$ as in (1), every chain map $\mathbb{Z}/2 \to C^\bullet$ is zero, since $\mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}) = 0$, so no quasi-isomorphism $H^\bullet(C^\bullet) \to C^\bullet$ exists. What does hold is the case of cohomology in a single degree n: then $i_n \colon \tau^{\leq n} C^\bullet \to C^\bullet$ and $\tau^{\leq n} C^\bullet \to \tau^{\geq n}\tau^{\leq n} C^\bullet = H^n(C^\bullet)[-n]$ are quasi-isomorphisms, by proposition 2.1.14.

3. Let $\iota(A) \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$ be the complex with A at the 0th position and zero's everywhere else. Say that $C^\bullet$ is a resolution such that $C^\bullet \to \iota(A)$ is a quasi-isomorphism. Show that this implies $C^\bullet$ is exact everywhere except possibly at $H^0(M^0)$

The lack of symmetry means that quasi-isomorphisms do not form an equivalence relation, but a *weak equivalence relation* (a relation that is reflective and transitive, but not symmetric). This is rather a pity, since if it did form an equivalence relation, then the equivalence class of morphisms up to weak isomorphism would hold all the necessary (co)homological information. Alas, this was not mean to be. Thus, in the pursuit of finding (co)homological invariants, it productive to ask how (co)homological information can be detected between (co)chain complexes. One idea is to map $\mathrm{Ch}(\mathbf{A})$ into another category $D(\mathbf{A})$ in which quasi-isomorphisms become isomorphisms. If quasi-isomorphisms formed equivalence relationships, we could have formed a congruence category which would satisfy this condition, however as just mentioned this cannot be the case.

Such a functor has already been encountered: the (co)homology functor is already one such example. It is more fruitful to as if this can be done in a *univerasl* way, that is if there is a category $D(\mathbf{A})$ where for any category D for which there exists an additive functor $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to D$ where every quasi-isomorphism in $\mathrm{Ch}(\mathbf{A})$ becomes an isomorphism in D, there exists a unique functor such that the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Ch}(\mathbf{A}) & \xrightarrow{\mathscr{F}} & D \\ \downarrow & \nearrow_{\exists!} & \\ D(\mathbf{A}) & & \end{array} \tag{2.2}$$

The localization of section 1.2 produces this category, called the *derived category* (hence the $D(-)$ notation).

Corollary 2.1.17: Existence of the Derived Category

Let $\mathbf{A}$ be an abelian category and let qis denote the class of quasi-isomorphisms in $\mathrm{Ch}(\mathbf{A})$. Then the localization

$$D(\mathbf{A}) := \mathrm{Ch}(\mathbf{A})[\mathrm{qis}^{-1}]$$

exists, and the canonical functor $Q \colon \mathrm{Ch}(\mathbf{A}) \to D(\mathbf{A})$ is universal among functors carrying quasi-isomorphisms to isomorphisms.

***Proof*:**
Apply theorem 1.2.2 with $\mathcal{C} = \mathrm{Ch}(\mathbf{A})$ and $W = \mathrm{qis}$, which needs no hypothesis on W. The universal property is clause (2) of definition 1.2.1, as we sought to show.

The construction leaves open whether $D(\mathbf{A})$ is locally small.

2.1.18 Warning: Quasi-Isomorphisms of Complexes Are Not a Multiplicative System For $\mathrm{Ch}(\mathbf{A})$ localized at quasi-isomorphisms, the second remedy of box 1.2.3 fails whenever $\mathbf{A}$ has a nonzero object E, and the first fails when $\mathbf{A}$ is a category of modules, since $\mathrm{Ch}(\mathbf{A})$ is then not small. The second fails because the quasi-isomorphisms are *not* a multiplicative system: conditions (2) and (3) of definition 1.2.4 both fail. Let $C^\bullet = (E \xrightarrow{\mathrm{id}} E)$ in degrees $-1, 0$, which is acyclic, so that $0 \to C^\bullet$ and $C^\bullet \to 0$ are quasi-isomorphisms, and let $h\colon E[0] \to C^\bullet$ be the chain map that is the identity in degree 0, where $E[0]$ is E in degree 0. Condition (2) for h and $0 \to C^\bullet$, and condition (3) for the pair $h, 0$ and $C^\bullet \to 0$, each require a quasi-isomorphism $t\colon X^\bullet \to E[0]$ with $h \circ t = 0$. Every chain map $t\colon X^\bullet \to E[0]$ has only a degree-zero component, so $h \circ t = 0$ forces $t = 0$, which is not a quasi-isomorphism since $H^0(E[0]) = E \neq 0$.

The route that repairs this, and the one section 3.2 follows, is to pass first to the homotopy category $K(\mathbf{A})$ of definition 2.1.38. There the quasi-isomorphisms do form a multiplicative system (theorem 3.2.3), theorem 1.2.5 applies, and $D(\mathbf{A}) = K(\mathbf{A})[\mathrm{qis}^{-1}]$.

In some nice categories $\mathbf{A}$, quasi-isomorphisms will line up with the nicer notion of *homotopy-equivalence*, in which case the derived category will be eminently understandable, as section 2.2 shows.

Though quasi-isomorphisms may not be ideal in computing (co)homological invariants, they are still important as the defining concept for (co)homological information. For example, the following shows the relationship between quasi-isomorphism and exactness:

Lemma 2.1.19: Quasi-isomorphisms And Exactness

Let $C^\bullet$ be a cochain complex. Then the following are equivalent:

1. $C^\bullet$ is exact
2. $C^\bullet$ is acyclic, that is $H^\bullet(C^\bullet) = 0$
3. The morphism of complexes $0 \to C^\bullet$ is a quasi-isomorphism

***Proof*:**
(1) $\Leftrightarrow$ (2): exactness of $C^\bullet$ at degree n means $\ker d^n = \mathrm{im} d^{n-1}$, and $H^n(C^\bullet) = \ker d^n / \mathrm{im} d^{n-1}$ by definition 2.1.8. A quotient object is zero exactly when the two are equal, so $C^\bullet$ is exact at every degree if and only if $H^n(C^\bullet) = 0$ for every n.

(2) $\Leftrightarrow$ (3): the zero complex has $H^n(0) = 0$ for every n, and the unique morphism $0 \to C^\bullet$ induces the unique morphism $0 \to H^n(C^\bullet)$ in each degree. That morphism is an isomorphism precisely when $H^n(C^\bullet) = 0$. So $0 \to C^\bullet$ is a quasi-isomorphism, meaning it induces isomorphisms in every degree, if and only if $H^\bullet(C^\bullet) = 0$, completing the proof.

2.1.2 Snake Lemma

We often have multiple ways of constructions a resolution for an object M. We shall often extend resolutions using smaller resolutions. We then get cohomology of this new extended resolution by simply applying the cohomology functor. However, is is this functor exact? More generally, let's say we have a short exact sequence of complexes, then after applying cohomology do we get a short exact sequence of homology complexes? The answer is no: if

$$0 \to L^\bullet \to C^\bullet \to N^\bullet \to 0$$

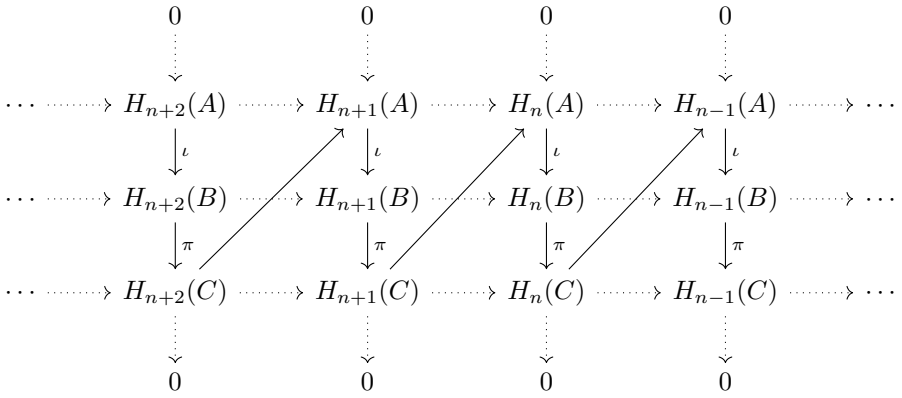
is a short exact sequence, then we get the (not necessarily exact) sequence:

$$H^i(L^\bullet) \to H^i(C^\bullet) \to H^i(N^\bullet)$$

In terms of morphisms, if we have:

$$\alpha^\bullet : C^\bullet \to N^\bullet$$

Then the morphism after applying the cohomology functor $H^\bullet(\alpha^\bullet) : H^\bullet(C^\bullet) \to H^\bullet(N^\bullet)$ need not be exact, i.e. it does not preserve kernels and cokernels. Can we "fix" this in some way by recovering some exact sequence from this and thus allowing our analysis of building more complicated resolutions for M to be independent of cohomology. The answer is yes, Namely, if we take the cohomology of complexes, we may "snake" like in the following diagram:



Theorem 2.1.20: Snake Lemma

Given a short exact sequence of complexes $0 \to A \xrightarrow{i} B \xrightarrow{j} C \to 0$, we may form a long exact sequence of modules using cohomology:

$$\cdots \to H^n(A) \xrightarrow{i^*} H^n(B) \xrightarrow{j^*} H^n(C) \xrightarrow{\partial} H^{n+1}(A) \xrightarrow{i^*} H^{n+1}(B) \xrightarrow{j^*} H^{n+1}(C) \to \cdots$$

Equivalently, in homological indexing, where the differential lowers degree and H_n is written for the homology in degree n, the same statement reads

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{j_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \to \cdots$$

The sequence is natural in the short exact sequence of complexes. Given a commutative diagram of complexes with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{j} & C & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle h} & & \\ 0 & \longrightarrow & A' & \xrightarrow[i']{} & B' & \xrightarrow[j']{} & C' & \longrightarrow & 0 \end{array}$$

the two long exact sequences form a commutative ladder whose vertical morphisms are $H_n(f)$, $H_n(g)$ and $H_n(h)$. Writing ∂' for the connecting morphism of the second row, the square carrying the two connecting morphisms reads $\partial' \circ H_n(h) = H_{n-1}(f) \circ \partial$.

The two forms are the same statement under $H_n = H^{-n}$, and the book uses both. The proof below constructs the homological form, and the homological variant of the proof also appears in [17].

***Proof*:**
We first define the map $\partial : H_n(C) \to H_{n-1}(A)$. We should keep in mind the following diagram:

$$\begin{array}{ccc} & & A_{n-1} \\ & & \downarrow{\scriptstyle i} \\ B_n & \xrightarrow{\partial} & B_{n-1} \\ \downarrow{\scriptstyle j} & & \\ C_n & & \end{array}$$

Let $c \in C_n$ be a cycle so that $c \in \ker \partial_n$. Since j is surjective, $c = j(b)$ for some $b \in B_n$. Then by definition $\partial : B_n \to B_{n-1}$ so $\partial(b) \in B_{n-1}$. Then by commutativity $j(\partial(b)) = \partial(j(b)) = \partial(c) = 0$ (since c is a cycle), thus $\partial b \in \ker(j)$. Since $\ker j = \mathrm{im} i$, $\partial b = i(a)$ for some $a \in A_{n-1}$. Since $i(\partial a) = \partial i(a) = \partial\partial b = 0$ and i is injective, $\partial a = 0$. Thus, define $\partial : H_n(C) \to H_{n-1}(A)$ by $[c] \mapsto [a]$. This is well defined since:

1. The element a is uniquely determined by ∂b since i is injective.
2. For a different choice b' of b, we get $j(b') = j(b)$, so $b' - b \in \ker(j)$, which is equal to $\mathrm{im}(i)$. Thus $b' - b = i(a')$ for some a', hence $b' = b + i(a')$. This replaces a by the homologous

element $a + \partial a'$, since

$$i(a + \partial a') = i(a) + i(\partial a') = \partial b + \partial i(a') = \partial(b + i(a')) = \partial b'$$

3. A different choice of c within its homology class has the form $c + \partial c'$. Since $c' = j(b')$ for some b',
$$c + \partial c' = c + \partial j(b') = c + j(\partial b') = j(b + \partial b')$$
so b is replaced by $b + \partial b'$, which has the same boundary ∂b, and thus a is unchanged.

Finally, $\partial : H_n(C) \to H_{n-1}(A)$ is a homomorphism, for if $\partial[c_1] = [a_1]$ and $\partial[c_2] = [a_2]$ via elements b_1, b_2, then
$$j(b_1 + b_2) = j(b_1) + j(b_2) = c_1 + c_2$$
and
$$i(a_1 + a_2) = i(a_1) + i(a_2) = \partial b_1 + \partial b_2 = \partial(b_1 + b_2)$$
hence
$$\partial([c_1] + [c_2]) = [a_1] + [a_2]$$
so ∂ is a homomorphism. It remains to show the following inclusions.

1. $\mathrm{im} i_* \subseteq \ker j_*$. Since $ji = 0$, $j_* i_* = 0$.

2. $\mathrm{im} j_* \subseteq \ker \partial$. For a cycle $b \in B_n$, the class $[j(b)]$ may be lifted to b itself, and $\partial b = 0 = i(0)$, so $\partial j_* = 0$.

3. $\mathrm{im}\partial \subseteq \ker i_*$. If $\partial[c] = [a]$ through the choice $b \in B_n$, then $i(a) = \partial b$ is a boundary, so $i_*[a] = [\partial b] = 0$.

4. $\ker j_* \subseteq \mathrm{im} i_*$. Let a cycle $b \in B_n$ represent a homology class in $\ker j_*$, so $j(b) = \partial c'$ for some $c' \in C_{n+1}$. Since j is surjective, $c' = j(b')$ for some $b' \in B_{n+1}$. Then since $\partial j(b') = \partial c' = j(b)$,
$$j(b - \partial b') = j(b) - j(\partial b') = j(b) - \partial j(b') = 0$$
Thus, $b - \partial b' = i(a)$ for some $a \in A_n$. Since $i(\partial a) = \partial i(a) = \partial(b - \partial b') = \partial b = 0$ and i is injective, a is a cycle. Thus:
$$i_*[a] = [b - \partial b'] = [b]$$
showing that i_* maps onto $\ker j_*$.

5. $\ker \partial \subseteq \mathrm{im} j_*$. By definition of ∂, if c represents a homology class in $\ker \partial$, then $a = \partial a'$ for some $a' \in A_n$. The element $b - i(a')$ is a cycle since $\partial(b - i(a')) = \partial b - \partial i(a') = \partial b - i(\partial a') = \partial b - i(a) = 0$. Since
$$j(b - i(a')) = j(b) - ji(a') = j(b) = c$$
we get that $j_*[b - i(a')] = [c]$.

6. $\ker i_* \subseteq \mathrm{im}\partial$. Given a cycle $a \in A_{n-1}$ such that $i(a) = \partial b$ for some $b \in B_n$, then $j(b)$ is a cycle since $\partial j(b) = j(\partial b) = ji(a) = 0$, hence $\partial[j(b)] = [a]$.

It remains to prove naturality. Let a commutative diagram as in the statement be given and let $[c] \in H_n(C)$. Running the construction above, $\partial[c] = [a]$, where $b \in B_n$ satisfies $j(b) = c$ and $i(a) = \partial b$. The element $h(c)$ is a cycle, since $\partial h(c) = h(\partial c) = 0$, and it represents $H_n(h)[c]$. Commutativity of the right-hand square gives $j'(g(b)) = h(j(b)) = h(c)$, so $g(b)$ is an admissible choice of preimage for $h(c)$, and commutativity of the left-hand square gives

$$i'(f(a)) = g(i(a)) = g(\partial b) = \partial(g(b))$$

So $f(a)$ is the element of A'_{n-1} that the construction attaches to $h(c)$ through the choice $g(b)$, and since ∂' does not depend on that choice, $\partial'(H_n(h)[c]) = [f(a)] = H_{n-1}(f)(\partial[c])$. The remaining squares of the ladder commute because H_n is a functor and the given diagram commutes, completing the proof.

Example 2.1.21: Uses Of Snake Lemma

1. *The six-term kernel-cokernel sequence.* The classical statement that carries the name "snake" is a special case of theorem 2.1.20. Suppose given a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^0 & \longrightarrow & B^0 & \longrightarrow & C^0 & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle h} & & \\ 0 & \longrightarrow & A^1 & \longrightarrow & B^1 & \longrightarrow & C^1 & \longrightarrow & 0 \end{array}$$

   Read each column as a complex concentrated in degrees 0 and 1, so that $A^\bullet = (A^0 \xrightarrow{f} A^1)$ and likewise for $B^\bullet$ and $C^\bullet$. Exactness of the two rows says precisely that $0 \to A^\bullet \to B^\bullet \to C^\bullet \to 0$ is a short exact sequence of complexes, and for a two-term complex $H^0 = \ker$ and $H^1 = \operatorname{coker}$. Since H^n vanishes outside degrees 0 and 1, the long exact sequence of theorem 2.1.20 has only six nonzero terms:

$$0 \to \ker f \to \ker g \to \ker h \xrightarrow{\partial} \operatorname{coker} f \to \operatorname{coker} g \to \operatorname{coker} h \to 0$$

   The connecting morphism ∂ is the "snake" that crosses from the kernel row to the cokernel row.

2. *A computation.* Apply the six-term sequence to multiplication by a fixed integer $n \geq 1$ on the short exact sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$. Multiplication by n is injective on $\mathbb{Z}$ and bijective on $\mathbb{Q}$, and $\mathbb{Q}/\mathbb{Z}$ is divisible, so

$$\ker f = 0, \quad \ker g = 0, \quad \ker h = \tfrac{1}{n}\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z}$$

$$\operatorname{coker} f = \mathbb{Z}/n\mathbb{Z}, \quad \operatorname{coker} g = 0, \quad \operatorname{coker} h = 0$$

   The six-term sequence collapses to $0 \to \mathbb{Z}/n\mathbb{Z} \xrightarrow{\partial} \mathbb{Z}/n\mathbb{Z} \to 0$, forcing ∂ to be an isomorphism. The explicit description confirms it: an element of $\ker h$ is the class of some $q \in \mathbb{Q}$ with $nq \in \mathbb{Z}$, and ∂ sends it to the class of nq in $\mathbb{Z}/n\mathbb{Z}$; the generator $1/n$ goes to the generator 1.

3. *Where the machine is used later.* Every long exact sequence in this book is an instance of theorem 2.1.20. The long exact sequence of derived functors (theorem 2.3.11) applies it to a short exact sequence of resolutions, the exact triangle of a mapping cone (proposition 2.1.24)

applies it to the cone sequence, and the exact couple that generates a spectral sequence (theorem 6.2.2) is nothing but all the long exact sequences of a filtration assembled at once. The topological applications, the Mayer-Vietoris sequence and the long exact sequence of a pair, are developed where they are used, in *Algebraic Topology* [17].

The most-used consequence of the snake lemma is a criterion for recognizing an isomorphism in the middle of a diagram from isomorphisms on either side. It is stated here because it is needed whenever two exact sequences are compared, and in particular in the comparison theorem for spectral sequences (theorem 6.2.7).

Lemma 2.1.22: Five Lemma

Consider a commutative diagram in an abelian category with exact rows

$$\begin{array}{ccccccccc} A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 & \longrightarrow & A_4 & \longrightarrow & A_5 \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_3} & & \downarrow{\scriptstyle f_4} & & \downarrow{\scriptstyle f_5} \\ B_1 & \longrightarrow & B_2 & \longrightarrow & B_3 & \longrightarrow & B_4 & \longrightarrow & B_5 \end{array}$$

If f_2 and f_4 are isomorphisms, f_1 is an epimorphism, and f_5 is a monomorphism, then f_3 is an isomorphism.
More precisely, f_3 is a monomorphism as soon as f_2 and f_4 are monomorphisms and f_1 is an epimorphism, and f_3 is an epimorphism as soon as f_2 and f_4 are epimorphisms and f_5 is a monomorphism.

***Proof*:**
By theorem 1.1.15 the diagram may be chased with elements. Label the horizontal maps $\alpha_i \colon A_i \to A_{i+1}$ and $\beta_i \colon B_i \to B_{i+1}$.

Step 1: f_3 is a monomorphism. Assume f_2, f_4 monomorphisms and f_1 an epimorphism, and let $a \in A_3$ with $f_3(a) = 0$. Then $f_4(\alpha_3(a)) = \beta_3(f_3(a)) = 0$, and f_4 is a monomorphism, so $\alpha_3(a) = 0$. By exactness of the top row at A_3 there is $b \in A_2$ with $\alpha_2(b) = a$. Now $\beta_2(f_2(b)) = f_3(\alpha_2(b)) = f_3(a) = 0$, so by exactness of the bottom row at B_2 there is $c \in B_1$ with $\beta_1(c) = f_2(b)$. Since f_1 is an epimorphism, $c = f_1(e)$ for some $e \in A_1$. Then

$$f_2(\alpha_1(e)) = \beta_1(f_1(e)) = \beta_1(c) = f_2(b)$$

and f_2 is a monomorphism, so $\alpha_1(e) = b$. Therefore $a = \alpha_2(b) = \alpha_2(\alpha_1(e)) = 0$, the last equality by exactness of the top row at A_2.

Step 2: f_3 is an epimorphism. Assume f_2, f_4 epimorphisms and f_5 a monomorphism, and let $b \in B_3$. Since f_4 is an epimorphism, $\beta_3(b) = f_4(a)$ for some $a \in A_4$. Then

$$f_5(\alpha_4(a)) = \beta_4(f_4(a)) = \beta_4(\beta_3(b)) = 0$$

by exactness of the bottom row at B_4, and f_5 is a monomorphism, so $\alpha_4(a) = 0$. By exactness of the top row at A_4 there is $c \in A_3$ with $\alpha_3(c) = a$. The element $b - f_3(c)$ satisfies

$$\beta_3(b - f_3(c)) = \beta_3(b) - f_4(\alpha_3(c)) = f_4(a) - f_4(a) = 0$$

so by exactness of the bottom row at B_3 there is $e \in B_2$ with $\beta_2(e) = b - f_3(c)$. Since f_2 is an epimorphism, $e = f_2(g)$ for some $g \in A_2$, and then

$$f_3\big(\alpha_2(g) + c\big) = \beta_2(f_2(g)) + f_3(c) = \big(b - f_3(c)\big) + f_3(c) = b$$

so b lies in the image of f_3.

Under the hypotheses of the first statement both steps apply, so f_3 is both a monomorphism and an epimorphism, hence an isomorphism by lemma 1.1.8, completing the proof.

There is another way of looking at chain complexes which shall be important in elucidating triangle categories in section 3.1.

Mapping Cones

Using a particular total complex, we can detect when a quasi-isomorphism is an isomorphism. This result requires the snake lemma to prove, hence the need to put this section after the snake lemma.

Definition 2.1.23: Mapping Cone

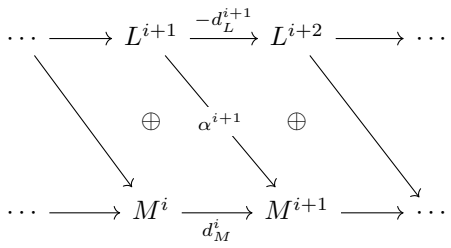
Let $\alpha^\bullet : L^\bullet \to M^\bullet$ be a morphism of complexes. Then the *mapping cone* is the complex denoted by $MC(\alpha^\bullet)$ defined by:

$$MC(\alpha)^i := L[1]^i \oplus M^i = L^{i+1} \oplus M^i$$

where the morphism $d^i_{MC(\alpha)} : MC(\alpha)^i \to MC(\alpha)^{i+1}$ are given by:

$$d^i_{MC(\alpha)}(l, m) := (-d^{i+1}_L(l), \alpha^{i+1}(l) + d^i_M(m))$$

The term mapping cone comes from the topological notion of a mapping cone of a function. Visually, the mapping cone can be seen as the following commutative diagram:



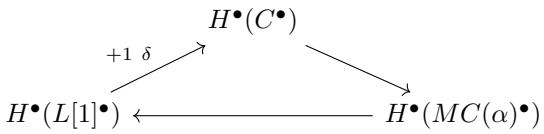
Given $MC(\alpha)^\bullet$, we may define two natural chain maps $C^\bullet \to MC(\alpha)^\bullet$ and $MC(\alpha)^\bullet \to L[1]^\bullet$ induced by the natural exact morphisms $M^i \to L^{i+1} \oplus M^i \to L^{i+1}$ which implies that:

$$0 \to C^\bullet \to MC(\alpha)^\bullet \to L[1]^\bullet \to 0$$

is exact.

Proposition 2.1.24: Mapping Cone And Exact Triangles

There is an exact triangle



where δ is the morphism induced by cohomology

***Proof*:**
The short exact sequence of complexes displayed above,

$$0 \to C^\bullet \to MC(\alpha)^\bullet \to L[1]^\bullet \to 0$$

is exact, so theorem 2.1.20 produces a long exact sequence in cohomology

$$\cdots \to H^n(C^\bullet) \to H^n(MC(\alpha)^\bullet) \to H^n(L[1]^\bullet) \xrightarrow{\delta} H^{n+1}(C^\bullet) \to \cdots$$

Exactness at each of the three kinds of term is precisely exactness of the displayed triangle at each of its three corners, the degree shift $H^n(L[1]^\bullet) = H^{n+1}(L^\bullet)$ being what the label $+1$ records.

The connecting morphism is the one induced by α, with no intervening sign; the projection $MC(\alpha)^\bullet \to L[1]^\bullet$ is a chain map precisely because the shift carries the differential $-d_L$, which is what cancels against the sign in the cone. To see this, unwind definition 2.1.23 on representatives: a class in $H^n(L[1]^\bullet)$ is represented by some $l \in L^{n+1}$ with $d_L l = 0$, and its preimage in $MC(\alpha)^n = L^{n+1} \oplus C^n$ may be taken to be $(l, 0)$. Applying the differential of the cone,

$$d_{MC(\alpha)}(l, 0) = \big(-d_L l,\ \alpha^{n+1}(l)\big) = \big(0,\ \alpha^{n+1}(l)\big)$$

which is the image of $\alpha^{n+1}(l) \in C^{n+1}$ under the inclusion. Hence $\delta[l] = [\alpha^{n+1}(l)]$, that is $\delta = H^{n+1}(\alpha)$, completing the proof.

2.1.25 Remark Not all categories have this triangle property, categories which will have this will be called *triangulated categories*

Proposition 2.1.26: Quasi-isomorphism And Isomorphism

Let $\alpha^\bullet : C^\bullet \to N^\bullet$. Then $H^\bullet(\alpha^\bullet)$ is an isomorphism if and only if $MC(\alpha)^\bullet$ is an exact complex.

***Proof*:**
By proposition 2.1.24, applied with $\alpha\colon C^\bullet \to N^\bullet$, the cone sits in a long exact sequence

$$\cdots \to H^n\big(MC(\alpha)^\bullet\big) \to H^n\big(C[1]^\bullet\big) \xrightarrow{H^{n+1}(\alpha)} H^{n+1}(N^\bullet) \to H^{n+1}\big(MC(\alpha)^\bullet\big) \to \cdots$$

the connecting morphism having been identified there as the map induced by α, and $H^n(C[1]^\bullet) = H^{n+1}(C^\bullet)$.

Suppose $MC(\alpha)^\bullet$ is exact, so that $H^\bullet(MC(\alpha)^\bullet) = 0$ by lemma 2.1.19. Deleting the vanishing terms leaves

$$0 \to H^{n+1}(C^\bullet) \xrightarrow{H^{n+1}(\alpha)} H^{n+1}(N^\bullet) \to 0$$

exact for every n, so each $H^{n+1}(\alpha)$ is an isomorphism.

Conversely suppose every $H^n(\alpha)$ is an isomorphism. Fix n and read the sequence around $H^n(MC(\alpha)^\bullet)$:

$$H^{n-1}(C[1]^\bullet) \xrightarrow{H^n(\alpha)} H^n(N^\bullet) \to H^n\big(MC(\alpha)^\bullet\big) \to H^n(C[1]^\bullet) \xrightarrow{H^{n+1}(\alpha)} H^{n+1}(N^\bullet)$$

The left-hand map is surjective and the right-hand map is injective, both being isomorphisms. Exactness at $H^n(N^\bullet)$ then makes the morphism $H^n(N^\bullet) \to H^n(MC(\alpha)^\bullet)$ zero, and exactness at $H^n(C[1]^\bullet)$ makes the morphism $H^n(MC(\alpha)^\bullet) \to H^n(C[1]^\bullet)$ zero. Exactness at $H^n(MC(\alpha)^\bullet)$ therefore gives

$$H^n\big(MC(\alpha)^\bullet\big) = \ker\big(\text{zero morphism}\big) = \operatorname{im}\big(\text{zero morphism}\big) = 0$$

As n was arbitrary the cone is acyclic, hence exact by lemma 2.1.19, completing the proof.

2.1.3 Chain Homotopies and Homotopic Categories

The notion of homotopies originated in topology as a weaker alternative to continuous functions[3]. The topological notion of homotopy settled on the level of continuous functions, where $f, g : X \to Y$ are homotopic (denoted $f \simeq g$) if there exists a continuous function $F : X \times [0,1] \to Y$ where $F(\cdot, t) = f_t$ is continuous, $f_0 = f$ and $f_1 = g$. If we represent X, Y as one dimensional lines, then we may visualize a homotopy as:

[3] every homeomorphic is a homotopy equivalence, however not every homotopy equivalence is a homeomorphism, hence homotopy can be seen as weaker than continuity

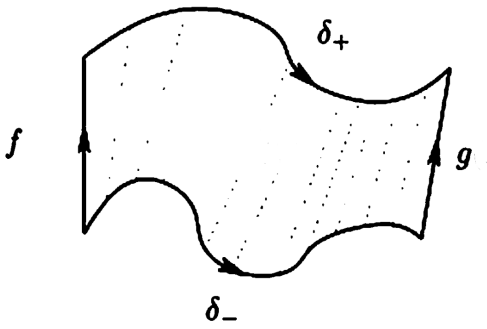


Figure 2.1: homotopy

In the development of homotopy theory, it was found that the (co)homology and homotopy of topological spaces (or their representations as chain complexes) is homology-invariant, hence homotopies are examples of quasi-isomorphisms. The proofs of these results turn out to be independent of the underlying topological structures and is purely algebraic, and hence the notion of homotopy has found a generalization as a purely algebraic notion between two (co)chain complexes.

Due to its origin in topology, the intuitions behind the following definitions will proceed more naturally with a more geometric build-up. Since homotopy may be interpreted purely algebraically, this geometrica build-up will be followed by a more algebraic interpretation. First, the reader may take for granted that given a topological space X, we may induce a chain $C_\bullet$, which is a topological object[4]. Given two continuous maps $f, g : X \to Y$, we may push forward the chain like so:

update image

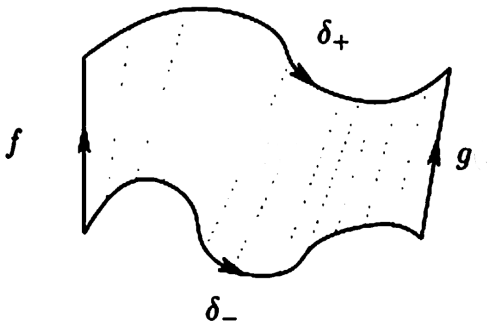


Figure 2.2: homotopy on chain

The space $h(C_\bullet)$ is a topological space, hence we may ask for it's border, $\partial h(C_\bullet)$. By inspecting the image, we see that:

$$\partial h(C_\bullet) = g(C_\bullet) - \delta_+ - f(C_\bullet) + \delta_- = g(C_\bullet) - f(C_\bullet) - h(\partial C_\bullet)$$

re-ordering, we get:

$$g(C_\bullet) - f(C_\bullet) = \partial h(C_\bullet) + h(\partial C_\bullet)$$

[4]The construction of singular chains is in *Algebraic Topology* [17, Singular n-Simplex, Singular Homology Group]. [16] is the standard outside reference.

Abstracting notation, we get:

$$g - f = \partial h + h\partial \tag{2.3}$$

Now, recall that in a chain complex, the image of the function ∂ is called the boundary, and that by definition of homology the boundary becomes trivial. In a topological setting, this vocabulary reflects the concrete of the boundary map ∂: it maps a chain to it's boundary. What this means for us is that when taking homology on both sides, we get:

$$H^k(g|_{C_\bullet} - f|_{C_\bullet}) = \overline{0}$$

meaning the two maps are equivalent up to homology! In other words, this "geometric intuition" shows that a homotopy is an example of a quasi-isomorphism. Even better, homotopies shall form an equivalence relation, hence they are an example of a quasi-isomorphism that shall capture some (co)homological information (and in the case where all quasi-isomorphisms are homotopies, they could fully capture the (co)homological information!)

Though this is the origins of chain homotopies being (co)homology-invariant, since they may be interpreted in pure algebraic terms, it is worthwhile taking a moment to have an algebraic build-up as well. Algebraically, homotopies can be seen as forming equivalence classes of (co)chain complexes up to *obstruction from being split-exact*. Recall that a split exact sequence is a short exact sequence that is also split at each map, that is there exists a section morphism $s^n : C^n \to C^{n-1}$ such that $d^{n-1} \circ s^n \circ d^{n-1} = d$, or if we drop the indices and collect all the section maps:

$$d = dsd$$

In other words, it is possible to "inverse" the map d via the map s in such a way that it "cancels" the effect of d (for we may re-apply the map d after sd and get the same result). The simplest source of split-exact (co)chains are exact (co)chains of vector-spaces, for any vector-space maps split[5]. Given a cochain, we may write:

$$C^n = Z^n \oplus B' \qquad (B^n)' \cong C^n/Z^n = d(C^n) = B^{n+1}$$

$$Z^n = B^n \oplus (H^n)' \qquad (H^n)' = Z^n/B^n = H^n(C)$$

Hence, $C^n \cong Z^n \oplus B^{n+1}$. This allows for the definition of the section map via the composition:

$$C^n \hookrightarrow Z^n \hookrightarrow B^n \cong d(C^n) \cong B^{n+1} \subseteq C^{n+1}$$

We may thus define the maps $d \circ s$ and $s \circ d$ where $d \circ s(C^n) = B^n$ and $s \circ d(C^n)(B^n)' \cong B^{n+1}$. Adding these maps together, we get a map $ds + sd : C^n \to C^n$ whose kernel is the homology:

$$\ker(ds + sd)_n = H^n(C^\bullet)$$

The kernel (and cokernel) of the chain map $ds + sd$ is naturally $H^\bullet(C^\bullet)$. Working over more general R-modules (and more generally in an arbitrary abelian category), we loose the natural section maps, for example $0 \to \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z} \to 0$ does note have a section map for each map. Instead, we may define arbitrary maps $s^n : C^n \to D^{n-1}$ where we may put them in the following (*not* necessarily commutative) diagram:

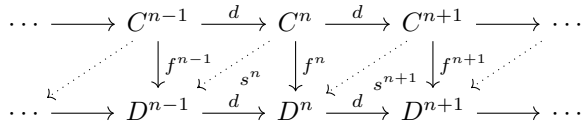
$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & C^{n-1} & \xrightarrow{d} & C^n & \xrightarrow{d} & C^{n+1} & \longrightarrow & \cdots \\
 & & \searrow & & \swarrow^{s^n} & & \swarrow^{s^{n+1}} & & \searrow \\
\cdots & \longrightarrow & D^{n-1} & \xrightarrow{d} & D^n & \xrightarrow{d} & D^{n+1} & \longrightarrow & \cdots
\end{array}$$

[5] More generally, any maps of projective modules split

Given any such collection of morphisms, we may define $f^n = s^{n+1} \circ d^n + d^{n-1} \circ s^n$ (or, if dropping indices and abbreviate, $f = sd + ds$). Notice that even though the maps $s^\bullet$ do not commute, $f^\bullet$ *will* create a commutative diagram, in particular it is a chain map. To see this, notice that (after dropping indices):

$$df = d(ds + sd) = dsd = (ds + sd)d = fd$$

showing it is indeed a chain map:



Thus, given any collection of morphisms $s^\bullet$, we may naturally define a chain map using the differentials. Conversely, is every morphism of complexes representable as $ds + sd$. This is not in general the case; maps being representable like so are given a name:

Definition 2.1.27: Null-homotopic Maps

Let $f^\bullet : C^\bullet \to D^\bullet$ be a morphism of complexes. Then f is called *null-homotopic* if there exists a collection $\{s^i\}_{i\in\mathbb{Z}}$ of morphisms $s^i \colon C^i \to D^{i-1}$ such that

$$f^\bullet = ds + sd$$

The maps $s^\bullet$ are called the *chain contraction* of $f^\bullet$

To show that not all maps are null-homotopic, consider the following lemma:

Lemma 2.1.28: Null-homotopic And Split Exact Complexes

Let $C^\bullet$ be a complex. Then $C^\bullet$ is split exact if and only if the identity map is null-homotopic.

***Proof*:**
Throughout, split exact means exact together with a family $s^n \colon C^n \to C^{n-1}$ satisfying $dsd = d$, and null-homotopic means $\mathrm{id} = ds + sd$ for some such family.

Step 1: Null-homotopic implies split exact. Suppose $\mathrm{id} = dt + td$. For exactness, let $c \in \ker d^n$; then

$$c = dt(c) + td(c) = dt(c)$$

so $c \in \mathrm{im} d^{n-1}$, and the reverse inclusion $\mathrm{im} d^{n-1} \subseteq \ker d^n$ holds because $d \circ d = 0$. For the splitting, compute

$$dtd = d(td) = d(\mathrm{id} - dt) = d - d^2t = d$$

so t is a splitting family.

Step 2: Split exact implies null-homotopic. Suppose $C^\bullet$ is exact and $dsd = d$. Set

$$\varphi = ds + sd$$

Then $d\varphi = dsd = d$ and $\varphi d = dsd = d$, both using $d \circ d = 0$; in particular φ commutes with d.

Put $\psi = \mathrm{id} - \varphi$. From the two identities just proved, $d\psi = d - d = 0$ and $\psi d = d - d = 0$. The first says $\mathrm{im}\psi \subseteq \ker d$, and exactness gives $\ker d = \mathrm{im}d$, so $\mathrm{im}\psi \subseteq \mathrm{im}d$; the second says ψ vanishes on $\mathrm{im}d$. Composing, $\psi \circ \psi = 0$.

Therefore $\varphi = \mathrm{id} - \psi$ is invertible, with inverse $\mathrm{id} + \psi$, and φ^{-1} commutes with d because φ does. Set $t = \varphi^{-1}s$. Then

$$dt + td = d\varphi^{-1}s + \varphi^{-1}sd = \varphi^{-1}(ds + sd) = \varphi^{-1}\varphi = \mathrm{id}$$

so the identity is null-homotopic with contraction t, completing the proof.

The name "null-homotopic" comes from how the (co)homology of such maps are trivial:

Proposition 2.1.29: Null-homotopic And Cohomology

Let $f^\bullet : C^\bullet \to D^\bullet$ be null-homotopic. Then $H^\bullet(f)$ is the zero map

***Proof*:**
We compute with elements, applying theorem 1.1.15 to a small abelian subcategory containing the objects involved. Since f is null-homotopic, there are morphisms $s^i \colon C^i \to D^{i-1}$ with $f^n = d_D^{n-1}s^n + s^{n+1}d_C^n$ for every n (definition 2.1.27). Let $x \in C^n$ be a cocycle representing a class $\overline{x} \in H^n(C^\bullet)$. Since $d_C^n(x) = 0$,

$$f^n(x) = d_D^{n-1}(s^n(x)) + s^{n+1}(d_C^n(x)) = d_D^{n-1}(s^n(x))$$

so $f^n(x)$ is a coboundary of $D^\bullet$. The map $H^n(f)$ of lemma 2.1.10 sends $\overline{x}$ to the class of $f^n(x)$, which is therefore $\overline{0}$ in $H^n(D^\bullet)$, as we sought to show.

Thus, maps that are null-homotopic induce zero (co)homology maps![6] Thus, we may think of maps that are null-homotopic as being the "zero" maps. This leads us to the following definition:

Definition 2.1.30: Chain-homotopy

Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be two morphisms of complexes. Then we say that these two maps are *chain homotopic* or just *homotopic* if there exists maps $h^i : C^i \to D^{i-1}$ such that

$$g^i - f^i = d_{D^\bullet}^{i-1} \circ h^i + h^{i+1} \circ d_{C^\bullet}^i$$

or, dropping indices:

$$g^\bullet - f^\bullet = dh + hd$$

If $f^\bullet, g^\bullet$ are homotopic, we write $f^\bullet \sim g^\bullet$

The s notation has been replaced with h notation to emphasize now the "homotopy" aspect of it rather than the "section" property that was evident in the case of vector spaces. If we were working with chains, then we would have

$$g_\bullet - f_\bullet = \partial h + h\partial$$

[6] TBD How much can we say about the converse, if a map induces the zero morphism, then is it null-homotopic?

which, the reader may recognize as equation (2.3). If we were working with chains, then $h_i : C_i \to D_{i+1}$ would shift up a degree, which would be interpreted in the geometrical setting as moving *up* a dimension, in the same way that homotopies are a "dimension more" than the original spaces X, Y. Furthermore, the reader should verify the "up to homotopy" *is* an equivalence relation, as the reader should verify!

Importantly, two chain-homotopic maps induce the same (co)homology:

Proposition 2.1.31: Chain-Homotopy and Cohomology

Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be chain homotopic. Then

$$H^\bullet(f^\bullet) = H^\bullet(g^\bullet)$$

***Proof*:**
Since $f^\bullet$ and $g^\bullet$ are chain-homotopic, by proposition 2.1.29 for each element in $H^k(C^\bullet)$ (since it is an additive functor):

$$H^k(f^\bullet)(\overline{x}) - H^k(g^\bullet)(\overline{x}) = 0 \qquad \Rightarrow \qquad H^k(f^\bullet)(\overline{x}) = H^k(g^\bullet)(\overline{x})$$

as we sought to show.

The above result shows promise in using chain-homotopies to hold (co)homological information. The following definition is the important connection idea:

Definition 2.1.32: Homotopy Equivalent Chains

Let $f^\bullet : C^\bullet \to D^\bullet$ be a morphism of complexes. Then $f^\bullet$ is a *homotopy equivalence* if there exists a morphisms of complexes $g^\bullet : D^\bullet \to C^\bullet$ such that

$$f^\bullet \circ g^\bullet \sim 1_{D^\bullet} \qquad \text{and} \qquad g^\bullet \circ f^\bullet \sim 1_{C^\bullet}$$

If $f^\bullet : C^\bullet \to D^\bullet$ is a homotopy equivalence, then $C^\bullet$ and $D^\bullet$ are said to be *homotopy equivalent*

Proposition 2.1.33: Homotopy Equivalence Ans Cohomology

Let $f^\bullet : C^\bullet \to D^\bullet$ be a homotopy equivalence. Then $f^\bullet$ is a quasi-isomorphism, that is $H^\bullet(f^\bullet)$ is an isomorphism

***Proof*:**
Since $f^\bullet$ is a homotopy equivalence, we have $g^\bullet$ that is it's homotopic inverse. Then the maps $f^\bullet \circ g^\bullet$ and $g^\bullet \circ f^\bullet$ induce inverse morphisms of cohomology by proposition 2.1.31, and hence give an isomorphism, as we sought to show.

As demonstrated in example 2.1.16 not all quasi-isomorphisms are invertible even up to homotopy equivalence, hence quasi-isomorphisms are more general. As mentioned earlier, in nice enough categories these two concepts shall coincide.

Now that homotopy equivalences is established, can be asked if given an (additive) functor $\mathscr{F} : \mathbf{A} \to \mathbf{B}$, if there are two homotopic complexes in $\mathrm{Ch}(\mathbf{A})$, will the (co)homology be isomorphic in $\mathbf{B}$? In other words, when trying to define the derived category that was mentioned earlier, can this category be defined at least up to homotopy? The answer is yes! The proof of this result essentially follows from the following lemma:

Lemma 2.1.34: Homotopy Equivalence And Additive Functors

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories, $C(\mathscr{F})$ the induced functor on chains. Then if $f^\bullet \sim g^\bullet$ in $\mathrm{Ch}(\mathbf{A})$, $C(\mathscr{F})(f^\bullet) \sim \mathrm{Ch}(\mathbf{B})(g^\bullet)$ in $\mathrm{Ch}(\mathbf{B})$. Furthermore, if $C^\bullet$ and $D^\bullet$ are homotopy equivalence, $C(\mathscr{F})(C^\bullet)$ and $C(\mathscr{F})(D^\bullet)$ are homotopy equivalent

***Proof*:**
Given the first assertion, the second is an immediate consequence, hence we prove the first. Let $f^\bullet \sim g^\bullet$ so that :

$$f^i - g^i = d^{i-1}_{D^\bullet} \circ h^i + h^{i+1} \circ d^i_{C^\bullet}$$

Since $\mathscr{F}$ is an additive functor:

$$\mathscr{F}(f^i) - \mathscr{F}(g^i) = \mathscr{F}\left(d^{i-1}_{D^\bullet}\right) \circ \mathscr{F}(h^i) + \mathscr{F}\left(h^{i+1}\right) \circ \mathscr{F}\left(d^i_{C^\bullet}\right)$$

hence, the morphisms $\mathscr{F}(h^i)$ gives a homotopy between $\mathscr{F}(f^\bullet)$ and $\mathscr{F}(g^\bullet)$, as we sought to show.

Note this does *not* hold for quasi-isomorphisms (exercise 4.15 in Aluffi), needing the stronger condition of exact functors.

Theorem 2.1.35: Derived Category And Homotopy Equivalence

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories. Then if $C^\bullet$ and $D^\bullet$ are homotopy-equivalent, then:

$$H^\bullet(\mathscr{F}(C^\bullet)) \cong H^\bullet(\mathscr{F}(D^\bullet))$$

that is, their cohomology is isomorphic.

***Proof*:**
By lemma 2.1.34, an additive functor carries homotopy-equivalent complexes to homotopy-equivalent complexes, so $\mathscr{F}(C^\bullet)$ and $\mathscr{F}(D^\bullet)$ are homotopy equivalent in $\mathrm{Ch}(\mathbf{B})$. By proposition 2.1.33 a homotopy equivalence is a quasi-isomorphism, so the induced map on cohomology is an isomorphism, giving $H^\bullet(\mathscr{F}(C^\bullet)) \cong H^\bullet(\mathscr{F}(D^\bullet))$, as we sought to show.

Hence, up to homotopy, the choice of cochain shall give the same cohomology information!

Homotopic Categories

With chain homotopies being a nicer sub-class of quasi-isomorphisms, their exploration is worth further pursuing. They are in fact almost the solution to the universal problem given in equation (2.2).

Let us take for granted that a category $D(A)$ exists that satisfies the universal property. What necessary conditions can be deduced about $D(A)$?

Lemma 2.1.36: Derived Category And Additive Functors

Let $\mathbf{D}$ be an additive category, $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to \mathbf{D}$ be an additive functor that takes quasi-isomorphisms to isomorphisms.

1. Let $C^\bullet$ be an exact complex in $\mathrm{Ch}(\mathbf{A})$. Then $\mathscr{F}(C^\bullet) = 0$, that is $\mathscr{F}(C^\bullet)$ is the zero object in $\mathbf{D}$.
2. Let $\alpha^\bullet : C^\bullet \to E^\bullet$ be a morphism of complexes that factors through an exact complex:

$$\begin{array}{ccc} C^\bullet & \xrightarrow{\alpha^\bullet} & E^\bullet \\ & \searrow \quad \nearrow & \\ & D^\bullet & \end{array}$$

   where $D^\bullet$ is exact. Then $\mathscr{F}(\alpha^\bullet)$ is the zero morphism.

***Proof*:**

1. Since $C^\bullet$ is exact, the zero morphism $0^\bullet : C^\bullet \to C^\bullet$ is a quasi-isomorphism, and so $\mathscr{F}(0^\bullet)$ maps to an isomorphism (i.e. an invertible morphism in $\mathbf{D}$). Thus:

$$\mathscr{F}(C^\bullet) \xrightarrow[\mathscr{F}(0^\bullet)]{} \mathscr{F}(C^\bullet) \xrightarrow[\mathscr{F}(0^\bullet)^{-1}]{} \mathscr{F}(C^\bullet), \qquad \mathscr{F}(C^\bullet) \xrightarrow{\mathrm{id}_{\mathscr{F}(C^\bullet)}} \mathscr{F}(C^\bullet)$$

   Since $\mathscr{F}$ is additive functor, in particular a group homomorphism on hom-sets, $\mathscr{F}(0) = 0$. Thus:

$$\mathrm{id}_{\mathscr{F}(C^\bullet)} = 0$$

   It follows that $\mathscr{F}(C^\bullet)$ must be the zero object: an object whose identity morphism is zero admits only the zero endomorphism, so it is isomorphic to 0

2. Applying $\mathscr{F}$ to the diagram in the question, we get the diagram:

$$\begin{array}{ccc} \mathscr{F}(C^\bullet) & \longrightarrow & \mathscr{F}(E^\bullet) \\ & \searrow \quad \nearrow & \\ & \mathscr{F}(D^\bullet) & \end{array}$$

   Since $\mathscr{F}(D^\bullet) = 0$ by the above result, we see that $\mathscr{F}(\alpha^\bullet)$ must be the zero morphisms, completing the proof.

Lemma 2.1.37: Derived Categories And Homotopies

Let $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to \mathbf{D}$ be an additive functor that maps quasi-isomorphisms to isomorphisms. Let $f^\bullet, g^\bullet : C^\bullet \to D^\bullet$ be homotopic morphisms in $\mathrm{Ch}(\mathbf{A})$. Then

$$\mathscr{F}(f^\bullet) = \mathscr{F}(g^\bullet)$$

Note that this was shown explicitly when $\mathscr{F} = H^i(-)$.

***Proof*:**
Since $\mathscr{F}$ is additive, it suffices to show that if $f^\bullet \sim 0$, then $\mathscr{F}(f^\bullet) = 0$. By lemma 2.1.36, it suffices to show $f^\bullet$ factors through an exact complex. The exact complex that shall be used is the mapping cone $MC(\mathrm{id}_{C^\bullet})$ for the identity map $\mathrm{id}_{C^\bullet} : C^\bullet \to C^\bullet$, namely

$$MC(\mathrm{id}_{C^\bullet})^i = C^{i+1} \oplus C^i$$

$$d^i_{MC(\mathrm{id}_{C^\bullet})}(a, b) = (-d^{i+1}_{C^\bullet}(a), a + d^i_{C^\bullet}(b))$$

By proposition 2.1.26, since the identity map $\mathrm{id}_{C^\bullet}$ is (trivially) a quasi-isomorphism, $MC(\mathrm{id}_{C^\bullet})$ is an exact complex.

Now, the goal is to find morphism of complexes where:

$$C^\bullet \to MC(\mathrm{id}_{C^\bullet}) \to D^\bullet$$

the morphism of complexes $C^\bullet \to MC(\mathrm{id}_{C^\bullet})$ is readily available: for each i:

$$b \mapsto (0, b)$$

On the other hand, it is not immediately clear there is a morphism of complexes $MC(\mathrm{id}_{C^\bullet}) \to D^\bullet$. To find such a morphism, we take advantage of the homotopy $f^\bullet \sim 0$, that is there exists maps:

$$h^i : C^i \to D^{i-1}$$

such that $f^\bullet = fh + hd$. With these, define the maps:

$$\pi^i : MC(\mathrm{id}_{C^\bullet})i \to D^i \qquad (a, b) \mapsto h^{i+1}(a) + f^i(b)$$

If this is indeed a morphism of complexes, then the map $f : C^\bullet \to D^\bullet$ factored through an exact complex and hence must be trivial. What must be verified is that:

$$d^{i-1}_{D^\bullet} \circ \pi^{i+1} = \pi^i \circ d^{i-1}_{MC(\mathrm{id}_{C^\bullet})} \tag{2.4}$$

The left hand side of equation (2.4) gives:

$$(a, b) \mapsto h^i(a) + f^{i-1}(b) \mapsto d^{i-1}_{D^\bullet}(h^i(a) + f^{i-1}(b))$$

while the right hand side gives:

$$(a, b) \mapsto (-d^i_{C^\bullet}(a), a + d^{i-1}_{L^\bullet}(b)) \mapsto -h^{i+1}(d^i_{C^\bullet}(a) + f^i(a + d^{i-1}_{C^\bullet}(b))$$

These two sides must be equal. Simplifying the equality, we get:

$$(d^{i-1}_{D^\bullet} \circ h^i + h^{i+1} \circ d^i_{C^\bullet} - f^i)(a) = (f^i \circ d^{i-1}_{C^\bullet} - d^{i-1}_{D^\bullet} \circ f^{-1})(a)$$

for all $a \in C^{i+1}, b \in C^i$. But notice that both sides are zero: the right hand side b $f^\bullet$ is a morphism of complexes, while the left hand side because h is a homotopy and $f^\bullet \sim 0$. But then this gives us our desired result, completing the proof.

Thus, behave well for the desired universal property presented in equation 2.2. We formalize the category in question:

Definition 2.1.38: Homotopy Category

Let $\mathbf{A}$ be an abelian category. Then the *homotopy category* of $\mathbf{A}$, denoted $K(\mathbf{A})$ is the category whose objects ar ethe (co)chain complexes in $\mathbf{A}$ (i.e. $\mathrm{Ch}(\mathbf{A})$), and whose morphisms are:

$$\mathrm{Hom}_{K(\mathbf{A})}(C^\bullet, D^\bullet) = \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, D^\bullet)/\sim$$

where $\sim$ is the homotopy relation, that is, the morphisms are morphisms of complexes up to homotopy.

These are indeed morphisms, namely $\sim$ is transitive, it respects composition (exercise). Naturally, the equivalent bounded variations $K^-(\mathbf{A})$ and $K^+(\mathbf{A})$ are obtained from $\mathrm{Ch}^+(\mathbf{A})$ and $\mathrm{Ch}^-(\mathbf{A})$.

Lemma 2.1.39: Homotopy Category Is Additive

Let $\mathbf{A}$ be an abelian category. Then $K(\mathbf{A})$ is an *additive category*

***Proof*:**
Recall from definition 2.1.38 that $K(\mathbf{A})$ has the same objects as $\mathrm{Ch}(\mathbf{A})$ and $\mathrm{Hom}_{K(\mathbf{A})}(C^\bullet, D^\bullet) = \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, D^\bullet)/\sim$, the quotient by chain homotopy.

Step 1: The hom-sets are abelian groups. The null-homotopic morphisms form a subgroup of $\mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(C^\bullet, D^\bullet)$: if $f = ds + sd$ and $f' = ds' + s'd$ then $f - f' = d(s - s') + (s - s')d$ is null-homotopic. Being a quotient of an abelian group by a subgroup, each hom-set of $K(\mathbf{A})$ is an abelian group, and $f \sim g$ exactly when $f - g$ is null-homotopic.

Step 2: Composition is well defined and bilinear. Suppose $f \sim f'$, say $f - f' = ds + sd$, and let g be a chain map composable on the left. Since g commutes with the differentials,

$$gf - gf' = g(ds + sd) = d(gs) + (gs)d$$

so $gf \sim gf'$; composing on the right with a chain map h gives $fh - f'h = d(sh) + (sh)d$ likewise. Hence composition descends to the quotient, and it is bilinear because it is bilinear in $\mathrm{Ch}(\mathbf{A})$ and the quotient maps are group homomorphisms.

Step 3: Zero object and biproducts. The zero complex is a zero object in $\mathrm{Ch}(\mathbf{A})$ and remains one after the quotient, since its hom-sets are already zero. For two complexes $C^\bullet$ and $D^\bullet$, the degreewise direct sum $C^\bullet \oplus D^\bullet$ carries the four structure morphisms i_C, i_D, p_C, p_D, which are chain maps and satisfy the biproduct identities $p_C i_C = \mathrm{id}$, $p_D i_D = \mathrm{id}$, $p_C i_D = 0$, $p_D i_C = 0$ and $i_C p_C + i_D p_D = \mathrm{id}$ in $\mathrm{Ch}(\mathbf{A})$. Identities between morphisms are preserved by the quotient functor, so they continue to hold in $K(\mathbf{A})$, and an object with such data is a biproduct there as well.

Together these are the axioms of an additive category, completing the proof.

Note that in general homotopic categories are *not* abelian. Indeed, homotopic maps $f^\bullet \sim g^\bullet$ need not have the same kernel/cokernel, giving no natural definition of these concepts. The homotopic category is a type of category that is stronger than additive but weaker than abelian, known as a *triangulated category* (recall box 2.1.25).

In $K(\mathbf{A})$, homotopy equivalences become isomorphisms. This comes straight from the definition, for if $f^\bullet \circ g^\bullet \sim \mathrm{id}$ in $\mathrm{Ch}(\mathbf{A})$, then $f^\bullet \circ g^\bullet = \mathrm{id}$ in $K(\mathbf{A})$, which can be thought of as formally inverting homotopy equivalences. Using this, we have the following almost universal property, or factorization

It looks like a universal property because it is one, though not the one for quasi-isomorphisms. Comparing with definition 1.2.1, the homotopy category is precisely the localization

$$K(\mathbf{A}) \cong \mathrm{Ch}(\mathbf{A})\big[\{\text{homotopy equivalences}\}^{-1}\big]$$

The proposition below is stated with the hypothesis that $\mathscr{F}$ inverts *quasi-isomorphisms*, which is strictly stronger than it needs to be, and that mismatch is what makes it read as an "almost" universal property. Its proof uses only that $\mathscr{F}$ inverts homotopy equivalences. Inverting all quasi-isomorphisms is the stronger requirement that produces $D(\mathbf{A})$ instead (corollary 2.1.17), and the gap between the two is exactly example 2.1.16: a quasi-isomorphism need not be a homotopy equivalence.

Proposition 2.1.40: Homotopy Category Factorization

Let $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to D$ be a functor that maps quasi-isomorphisms to isomorphisms. Then $\mathscr{F}$ factors uniquely through $K(\mathbf{A})$:

$$\begin{array}{ccc} \mathrm{Ch}(\mathbf{A}) & \xrightarrow{\mathscr{F}} & D \\ \downarrow & \nearrow_{\exists!} & \\ K(\mathbf{A}) & & \end{array}$$

***Proof*:**
Since every homotopy equivalence is a quasi-isomorphism (proposition 2.1.33), the hypothesis in particular sends homotopy equivalences to isomorphisms, and that is all the proof uses. Because $\mathscr{F}$ is defined on $\mathrm{Ch}(\mathbf{A})$ and $K(\mathbf{A})$ has the same objects, the only thing to establish is that $f \sim g$ implies $\mathscr{F}(f) = \mathscr{F}(g)$; the factorization and its uniqueness are then forced, since the quotient functor $\mathrm{Ch}(\mathbf{A}) \to K(\mathbf{A})$ is the identity on objects and surjective on morphisms.

Step 1: The cylinder of a complex. For a complex $C^\bullet$ define

$$\mathrm{Cyl}(C)^n = C^n \oplus C^{n+1} \oplus C^n, \qquad d(a,b,c) = (da - b,\ -db,\ dc + b)$$

This squares to zero, since $d^2(a,b,c) = (d(da-b)+db,\ d(db),\ d(dc+b)-db) = (0,0,0)$. Define chain maps

$$i_0(a) = (a,0,0), \qquad i_1(c) = (0,0,c), \qquad p(a,b,c) = a + c$$

Each is compatible with the differentials: $d\,i_0(a) = (da,0,0) = i_0(da)$, similarly for i_1, and $p(d(a,b,c)) = (da-b)+(dc+b) = d(a+c)$. Moreover $pi_0 = \mathrm{id}_C = pi_1$.

Step 2: p is a homotopy equivalence. The composite $i_0 p - \mathrm{id}$ sends (a,b,c) to $(c,-b,-c)$, and setting $s(a,b,c) = (0,-c,0)$ gives

$$ds(a,b,c) = (c,\ dc,\ -c), \qquad sd(a,b,c) = (0,\ -(dc+b),\ 0)$$

whose sum is $(c, -b, -c)$. So $i_0 p \sim \mathrm{id}_{\mathrm{Cyl}(C)}$ while $p i_0 = \mathrm{id}_C$, making i_0 and p mutually inverse homotopy equivalences. The symmetric computation applies to i_1, with the contracting homotopy $s'(a, b, c) = (0, a, 0)$ in place of s, since $i_1 p - \mathrm{id}$ sends (a, b, c) to $(-a, -b, a)$.

Step 3: Homotopies are morphisms out of the cylinder. Let $f, g \colon C^\bullet \to D^\bullet$ with $g - f = ds + sd$. Define $H \colon \mathrm{Cyl}(C) \to D^\bullet$ by $H(a, b, c) = f(a) + s(b) + g(c)$. It is a chain map: expanding both sides, $H(d(a, b, c)) - dH(a, b, c) = (g - f - ds - sd)(b) = 0$. By construction $Hi_0 = f$ and $Hi_1 = g$.

Step 4: Conclusion. Apply $\mathscr{F}$. By Step 2 the morphisms i_0 and i_1 are homotopy equivalences, hence quasi-isomorphisms, so $\mathscr{F}(i_0)$ and $\mathscr{F}(i_1)$ are isomorphisms by hypothesis. From $p i_0 = \mathrm{id} = p i_1$ and functoriality, $\mathscr{F}(p)\mathscr{F}(i_0) = \mathrm{id} = \mathscr{F}(p)\mathscr{F}(i_1)$, and since $\mathscr{F}(i_0)$ is invertible so is $\mathscr{F}(p)$; cancelling it gives $\mathscr{F}(i_0) = \mathscr{F}(p)^{-1} = \mathscr{F}(i_1)$. Therefore

$$\mathscr{F}(f) = \mathscr{F}(Hi_0) = \mathscr{F}(H)\mathscr{F}(i_0) = \mathscr{F}(H)\mathscr{F}(i_1) = \mathscr{F}(Hi_1) = \mathscr{F}(g)$$

so $\mathscr{F}$ identifies homotopic morphisms and factors uniquely through $K(\mathbf{A})$. Note that $\mathscr{F}$ was used only through functoriality and the hypothesis that it inverts quasi-isomorphisms. The direct sums and sums of morphisms above all live in $\mathrm{Ch}(\mathbf{A})$, so no additivity of $\mathscr{F}$ itself is required. This matters in section 3.2, where the functor to be factored is the localization onto $D(\mathbf{A})$, whose additive structure is not available until later. This completes the proof.

Hence, in the case where $K(\mathbf{A})$ is *not* the universal object satisfying equation (2.2), there must be a unique functor from $K(\mathbf{A})$ to $D(\mathbf{A})$.

2.2 Projective and Injective Resolutions

It has been alluded to a few times that for special categories, homotopy equivalence and quasi-isomorphism coincide. This shall now be covered in this section. In particular, this shall linked to the idea of an abelian category $\mathbf{A}$ being bounded from above or bellow, and having *enough projective objects* or *enough injective objects*. These categories shall furthermore be shown to be universal in some appropriate way, making their choice not arbitrary. Recall that in $\mathbf{A} = R\text{-}\mathbf{Mod}$, an R-module M Is called projective if $\mathrm{Hom}_R(M, -)$ is exact, and injective if $\mathrm{Hom}_R(-, M)$ is exact. Recall that projective/injective modules nicely lend themselves to a splitting condition on short exact sequences for which they are one of the modules forming them (recall [8, Equivalencies for Projective Modules] and [8, Injective Modules Equivalences]). This will be important, due to the link of chain homotopies with split exact sequences.

Definition 2.2.1: Projective And Injective Objects

Let $\mathbf{A}$ be an abelian category. Then

1. An object $P \in \mathbf{A}$ is said to be *projective* if the functor $\mathrm{Hom}_A(P, -)$ is exact.
2. An object $Q \in \mathbf{A}$ is said to be *injective* if the functor $\mathrm{Hom}_A(-, Q)$ is exact

[8, Equivalencies for Projective Modules] and [8, Injective Modules Equivalences] hold for abelian

categories more generally (in particular the splitting remarks), as the reader may venture to check. Since the axioms for an abelian category are self-dual, so that $\mathbf{A}^{\mathrm{op}}$ is abelian whenever $\mathbf{A}$ is, the theory on projective objects is dual to that of injective object, and the functor $(-)^{\mathrm{op}}$ maps projective/injectives to injectives/projectives respectively. Thus, many proofs shall simply proof the result for projectives, the result for injectives coming automatically. Note that as we have seen in [8, Extension Problem: Prelude to Homological Algebra] in a fixed [abelian] category $\mathbf{A}$, projective and injective may have very different features.

In what follows, let $\mathbf{P}$ denote the full subcategory of $\mathbf{A}$ determined by the projective objects, and let $\mathbf{I}$ denote the full subcategory of $\mathbf{A}$ determined by the injective objects. Let $K^-(\mathbf{P})$ denote the full subcategory of $K(\mathbf{A})$ consisting of bounded-above complexes of projective objects of $\mathbf{A}$, and $K^+(\mathbf{I})$ be the full subcategory of $K(\mathbf{A})$ consisting of bounded-below complexes of injective objects of $\mathbf{A}$. Note that these are generally not abelian categories, and furthermore projective and injective objects need not exist:

Example 2.2.2: Category With And Without Projective/Injective Resolutions

1. The category of finite abelian groups has no projective or injective, hence no projective/injective resolution.
2. The category of finitely generated abelian groups has projectives, but no injectives, namely since $\mathbb{Z}^n$ for every n are objects this category. Thus, it is possible to create a free, and hence projective, resolution.
3. the dual of the above category has enough injectives but not enough projectives.
4. The category of R-modules has both projective and injective resolutions (projective resolutions comes from free resolutions, while injective resolutions comes from repeated application of [8, Universal Property Of Injective Hulls])

It is thus helpful to add the following condition to an abelian category

Definition 2.2.3: Enough Projective And Enough Injectives

Let $\mathbf{A}$ be an abelian category. Then:

1. we say that $\mathbf{A}$ as *enough projectives* if for every object A of $\mathbf{A}$, there exists a projective object P of $\mathbf{A}$ and an epimorphism $P \twoheadrightarrow A$.
2. we say that $\mathbf{A}$ as *enough injectives* if for every object A of $\mathbf{A}$, there exists a injective object Q of $\mathbf{A}$ and a monomorphism $A \hookrightarrow Q$.

With the important definitions established, the main objective of this section may start going under way. The goal is to show that if we have a quasi-isomorphism between two bounded-above complexes of projectives $f^\bullet : P_0^\bullet \to P_1^\bullet$ (resp. quasi-isomorphism between two bounded-bellow complexes of injectives $f^\bullet : Q_0^\bullet \to Q_1^\bullet$), then $f^\bullet$ is in fact a homotopy-equivalence, from which it can be concluded that in $K^-(\mathbf{P})$ and $K^+(\mathbf{I})$ (homotopy-classes of) quasi-isomorphisms are isomorphisms, which shall give us the desired derived category $D^-(\mathbf{A})$ and $D^+(\mathbf{A})$ in the case where A has enough projectives (resp. enough injectives).

Let us now revisit this splitting condition. Recall that if $C^\bullet$ in $\mathrm{Ch}(\mathbf{A})$ is exact, then the identity

map $\mathrm{id}_{C^\bullet}$ and the trivial map induce the same morphisms in (co)homology. However, there do exist exact complexes that are *not* homotopic to 0. As discussed earlier, this directly links to the splitting condition, namely the identity is homotopic to 0 if the sequence is split exact. Being projective is exactly the condition that causes splitting behavior. With this in mind, let us prove the first lemma:

Lemma 2.2.4: Zero-morphism In Cohomology And Homotopy

Let $\mathbf{A}$ be an abelian category, $P^\bullet \in \mathrm{Ch}^-(\mathbf{P})$ (i.e. $P^i = 0$ for $i > 0$), and let $C^\bullet$ be a complex in $\mathrm{Ch}(\mathbf{A})$ such that $H^i(C^\bullet) = 0$ for $i < 0$ (i.e. has trivial cohomology in negative coefficients). Let $f^\bullet : P^\bullet \to C^\bullet$ be a morphism of complexes that induces the zero-morphism in cohomology: $H^\bullet(f^\bullet) = 0$. Then

$$f^\bullet \sim 0$$

i.e., $f^\bullet$ is null-homotopic.

The injective version of this statement is comes from dualizing the appropriate terms.

***Proof*:**

To show $f^\bullet$ is null-homotopic, we must show there exists maps $h^i : P^i \to C^{i-1}$ such that

$$f^i = d^{i-1}_{C^\bullet} \circ h^i + h^{i+1} \circ d^i_{P^\bullet} \tag{2.5}$$

as a diagram:

$$\begin{array}{ccccccccccccc}
\cdots & \longrightarrow & P^{-2} & \longrightarrow & P^{-1} & \longrightarrow & P^0 & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & \swarrow^{h^{-2}} & \downarrow{\scriptstyle f^{-2}} & \swarrow^{h^{-1}} & \downarrow{\scriptstyle f^{-1}} & \swarrow^{h^0} & \downarrow{\scriptstyle f^0} & \swarrow^{h^1=0} & \downarrow & & \\
\cdots & \longrightarrow & C^{-2} & \longrightarrow & C^{-1} & \longrightarrow & C^0 & \longrightarrow & C^1 & \longrightarrow & \cdots
\end{array}$$

Naturally, $h^i = 0$ for $i > 0$. For the rest, we shall use induction. The base case is $i = 0$. Since the cohomology induced by $f^\bullet$ is zero, f^0 factors through the image of $d^{-1}_{C^\bullet}$:

$$\begin{array}{ccccc}
 & & P^0 & & \\
 & & \downarrow{\scriptstyle f^0} & & \\
C^{-1} & \xrightarrow{d^{-1}_{C^\bullet}} & \mathrm{im} d^{-1}_{C^\bullet} & \longrightarrow & 0
\end{array}$$

Now, since P^0 is projective, there exists a morphism $h^0 : P^0 \to C^{-1}$ such that $f^0 = d^{-1}_{C^\bullet} \circ h^0$, as needed.

Next, assume h^i and h^{i+1} satisfies equation (2.5). The goal is to show h^{i-1} satisfies it too. By equation (2.5) and the fact that $P^\bullet$ is a complex:

$$d^{i-2}_{C^\bullet} \circ h^i \circ d^{i-1}_{P^\bullet} = (f^i - h^{i+1} \circ d^{i-1}_{P^\bullet}) = f^i \circ d^{i-1}_{P^\bullet} - h^{i+1} \circ 0 = f^i \circ d^{i-1}_{P^\bullet}$$

which can be seen by staring at the following diagram:

$$\begin{array}{ccccccccc}
\cdots & \cdots\cdots\rightarrow & P^{i-1} & \xrightarrow{d^{i-1}_{P^\bullet}} & P^i & \xrightarrow{d^i_{P^\bullet}} & P^{i+1} & \cdots\cdots\rightarrow & \\
 & & \vdots & \swarrow^{h^i} & \downarrow{\scriptstyle f^i} & \swarrow^{h^{i+1}} & \vdots & & \\
\cdots & \cdots\cdots\rightarrow & C^{i-1} & \xrightarrow[d^{i-1}_{C^\bullet}]{} & C^i & \cdots\cdots\rightarrow & C^{i+1} & \cdots\cdots\rightarrow &
\end{array}$$

Thus, since f^b is a morphism of cochain complexes:

$$d_{C^\bullet}^{i-1} \circ (f^{i-1} - h^i \circ d_{P^\bullet}^{i-1}) = d_{C^\bullet}^{i-1} \circ f^{i-1} - f^i \circ d_{P^\bullet}^{i-1} = 0$$

Importantly, $(f^{i-1} - h^i \circ d_{P^\bullet}^{i-1})$ factors through $\ker d_{C^\bullet}^{i-1}$. Since $C^\bullet$ is exact at C^{i-1} for $i \leq 0$, $\ker d_{C^\bullet} = \mathrm{im}_{C^\bullet}^{i-2}$. Thus, we get:

$$\begin{array}{ccccc} & & P^{i-1} & & \\ & & \downarrow{\scriptstyle f^{i-1}-h^i\circ d_{P^\bullet}^{i-1}} & & \\ C^{i-2} & \xrightarrow{d_{C^\bullet}^{i-2}} & \mathrm{im}(d_{C^\bullet}^{i-2}) & \longrightarrow & 0 \end{array}$$

Now, since P^{i-1} is projective, there exists a lift $h^{i-1} : P^{i-1} \to C^{i-2}$ such that

$$f^{i-1} - h^i \circ d_{P^\bullet}^{i-1} = d_{C^\bullet}^{i-2} \circ h^{i-1}$$

which, shuffling around, gives us

$$f^{i-1} = d_{C^\bullet}^{i-2} \circ h^{i-1} + h^i \circ d_{P^\bullet}^{i-1}$$

as we sought to show.

Corollary 2.2.5: Exactness Cohomology And Homotopy

Let $\mathbf{A}$ be an abelian category, $P^\bullet \in \mathrm{Ch}^-(\mathbf{P})$ a bounded-above complex of projectives, and $C^\bullet \in \mathrm{Ch}(\mathbf{A})$ an exact complex. Then every morphism $P^\bullet \to C^\bullet$ is homotopic to 0.

***Proof*:**
Every morphism to an exact complex induces the zero morphism in cohomology. After a translation of degrees, which changes neither the hypotheses nor the conclusion, $P^i = 0$ for $i > 0$, and $H^i(C^\bullet) = 0$ for every i, so lemma 2.2.4 applies, completing the proof.

Note the importance of the bounding from above/bellow (depending on whether working with projectives or injective) in the inductive argument, without it there would not be an appropriate base case. Using the above results, the first quasi-isomorphism to homotopy equivalence can be established:

Corollary 2.2.6: Projectives With Quasi-isomorphism And Zero Morphisms

Let $P^\bullet$ (resp. $Q^\bullet$) be a bounded-above exact complex of projectives (resp. a bounded bellow exact complex of injectives). Then $P^\bullet$ (resp. $Q^b b$) is homotopy-equivalent to the zero-complex.

***Proof*:**
Treat the projective case; the injective one is the same argument in the opposite category. A complex is homotopy equivalent to the zero complex exactly when its identity morphism is null-homotopic, since a homotopy inverse to the zero complex forces $\mathrm{id} \sim 0$ and conversely. By lemma 2.1.28 it therefore suffices to show that $P^\bullet$ is split exact.

Write $Z^n = \ker d^n$ and $B^n = \mathrm{im} d^{n-1}$; exactness gives $Z^n = B^n$, so for every n there is a short exact sequence

$$0 \to B^n \to P^n \to B^{n+1} \to 0$$

Since $P^\bullet$ is bounded above there is an N with $P^n = 0$ for $n > N$, whence $B^n = 0$ for $n > N+1$.

Argue by descending induction that each B^n is projective. For $n > N+1$ it is zero. Suppose B^{n+1} is projective. Then the displayed short exact sequence splits, because a surjection onto a projective object admits a section, so $P^n \cong B^n \oplus B^{n+1}$ and B^n is a direct summand of the projective object P^n, hence projective itself.

Every one of the short exact sequences therefore splits. Choosing a section $\sigma^{n+1} \colon B^{n+1} \to P^n$ of d^n for each n and letting $s^{n+1} \colon P^{n+1} \to P^n$ be σ^{n+1} composed with the projection $P^{n+1} \twoheadrightarrow B^{n+1}$ gives $dsd = d$, since $d^n \sigma^{n+1}$ is the identity on B^{n+1} and d factors through B^{n+1}. So $P^\bullet$ is split exact, and lemma 2.1.28 makes $\mathrm{id}_{P^\bullet}$ null-homotopic, completing the proof.

Another important result that will be used in the next section is that quasi-isomorphisms are "non-zero-divisors" up to homotopy wrt morphisms from commplexes of projectives:

Lemma 2.2.7: Quasi-isomorphisms Of Projectives And Zero-divisor

Let $\mathbf{A}$ be an abelian category, $f^\bullet : C^\bullet \to D^\bullet$ be a quasi-isomorphism. Let $P^\bullet$ be a bounded-above complex of projectives, and let $\alpha^\bullet : P^\bullet \to C^\bullet$ be a morphism of cochain complexes such that the composition:

$$P^\bullet \xrightarrow{\alpha^\bullet} C^\bullet \xrightarrow{f^\bullet} D^\bullet$$

is homotopic to the zero-morphism. Then $\alpha^\bullet \sim 0$

***Proof*:**
Let $h^i : P^i \to D^{i-1}$ define the homotopy between $f^\bullet \circ \alpha^\bullet$ and 0, namely:

$$-f^i \circ \alpha^i = d^{i-1}_{C^\bullet} \circ h^i + h^{i+1} \circ d^i_{P^\bullet}$$

Take the mapping cone $MC(f)^\bullet$ of $f^\bullet$ and define the morphisms:

$$\beta^i = (\alpha^, h^i) : P^i \to MC(f)^{i-1}$$

which may be visualized in the following diagram:

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & P^{i-1} & \xrightarrow{d^{i-1}_{P^\bullet}} & P^i & \xrightarrow{d^i_{P^\bullet}} & P^{i+1} & \longrightarrow & \\
 & & \downarrow{\scriptstyle \beta^{i-1}} & & \downarrow{\scriptstyle \beta^i} & & \downarrow{\scriptstyle \beta^{i+1}} & & \\
\cdots & \longrightarrow & C^{i-1} \oplus D^{i-2} & \xrightarrow[d^{i-2}_{MC(f)^\bullet}]{} & C^i \oplus D^{i-1} & \xrightarrow[d^{i-1}_{MC(f)^\bullet}]{} & C^{i+1} \oplus D^i & \longrightarrow &
\end{array}$$

These give a morphism of cochain complexes $P^\bullet \to MC(f)[-1]^\bullet$. Indeed:

$$\beta^{i+1} \circ d^i_{P^\bullet} = (\alpha^{i+1} \circ d^i_{P^\bullet}, h^{i+1} \circ d^i_{P^\bullet})$$
$$-d^{i-1}_{MC(f)} \circ \beta^i = (d^i_{C^\bullet} \circ \alpha^i, -f^i \circ \alpha^i - d^{i-1}_{M^\bullet} \circ h^i)$$

Then by definition of h^i and by the morphism property of $\alpha^\bullet$, these are equal. Now, by hypothesis, $f^\bullet$ is a quasi-isomorphism. Then by proposition 2.1.26 $MC(f)^\bullet$ is an exact complex. Then by corollary 2.2.6 $\beta^\bullet$ is homotopic to 0. Since $\alpha^\bullet$ is the composition:

$$P^\bullet \xrightarrow{\beta^\bullet} MC(f)[-1]^\bullet = L^\bullet \oplus M[-1]^\bullet \to L^\bullet$$

it follows that $\beta^\bullet \sim 0$ implies $\alpha^\bullet \sim 0$, as we sought to show.

Though $\alpha^\bullet$ must be homotopic to 0, it need not be the case that $\alpha^\bullet = 0$. Take for example:

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \cdot 2} & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\cdot 2} & \mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \pi} & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \cdots
\end{array}$$

In this case, $f^\bullet$ is a quasi-isomorphism and $f^\bullet \circ \alpha^\bullet = 0$, and so $\alpha^\bullet \sim 0$, even though it is not equal to 0. With this, one more result is required before the desired goal:

Proposition 2.2.8: projectives, and one-sided inverses

Let $\mathbf{A}$ be an abelian category, $C^\bullet \in \mathrm{Ch}(\mathbf{A})$, $P^\bullet \in \mathrm{Ch}^-(\mathbf{P})$, and $f^\bullet : C^\bullet \to P^\bullet$ a quasi-isomorphism. Then there exists a morphism of complexes $g^\bullet : P^\bullet \to C^\bullet$ such that:

$$f^\bullet \circ g^\bullet \sim \mathrm{id}_{P^\bullet}$$

For injective, $Q^\bullet \in \mathrm{Ch}^+(\mathbf{I})$, $f^\bullet : Q^\bullet \to C^\bullet$ is a quasi-isomorphism, for which then there exists a morphism $g^\bullet : C^\bullet \to Q^\bullet$ such that $g^\bullet \circ f^\bullet \sim \mathrm{id}_{Q^\bullet}$. Notice that this is a one-sided inverse, hence is a slightly more general result than what is being built up to.

***Proof*:**
Since $f^\bullet : C^\bullet \to P^\bullet$ is a quasi-isomorphism, $MC(f)^\bullet = C[1]^\bullet \oplus P^\bullet$ is an exact complex. Define $\rho^\bullet = (0, \mathrm{id}_{P^\bullet})$ to be the morphism:

$$\rho^\bullet : P^\bullet \to MC(f)^\bullet$$

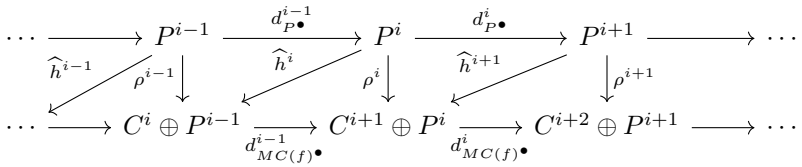
By corollary 2.2.5, since $MC(f)^\bullet$ is exact, $\rho^\bullet$ is homotopic to the zero morphism. Thus, there exists morphisms:

$$\widehat{h}^i : P^i \to MC(f)^{i-1} = C^i \oplus P^{i-1}$$

such that

$$\rho^i = d^{i-1}_{MC(f)^\bullet} \circ \widehat{h}^i + \widehat{h}^{i+1} \circ d^i_{P^\bullet}$$

Visually, we may see this as:



Now, write out $\widehat{h}^i$ out in components so that $\widehat{h}^i = (g^i, h^i)$ where $g^i : P^i \to C^i$ and $h^i : P^i \to P^{i-1}$. We will show that the collection $\{g^i\}_{i\in\mathbb{Z}}$ is a morphism of cochain complexes (i.e. they form commutative squares $g^{i+1} \circ d^i_{P^\bullet} = d^i_{C^\bullet} \circ g^i$) such that $f^\bullet \circ g^\bullet \sim \mathrm{id}_{P^\bullet}$. Indeed, since

$$\begin{aligned}(0, \mathrm{id}_{P^i}) &= \rho^i \\ &= d^{i-1}_{MC(f)^\bullet} \circ \widehat{h}^i + \widehat{h}^{i+1} \circ d^i_{P^\bullet} \\ &= (d_{MC}(f)^\bullet)^{i-1} \circ (g^i, h^i) + (g^{i+1}, h^{i+1}) \circ d^i_{P^\bullet} \\ &= (-d^i_{C^\bullet} \circ g^i, f^i \circ g^i + d^{i-1}_{P^\bullet} \circ h^i) + (g^{i+1} \circ d^i_{P^\bullet}, h^{i+1} \circ d^i_{P^\bullet})\end{aligned}$$

It follows that:

$$\begin{aligned}g^{i+1} \circ d^i_{P^\bullet} - d^i_{C^\bullet} \circ g^i &= 0 \\ \mathrm{id}_{P^i} - f^i \circ g^i &= d^{i-1}_{P^\bullet} \circ h^i + h^{i+1} \circ d^i_{P^\bullet}\end{aligned}$$

which is exactly what we need, hence the morphism of complexes $g^\bullet : P^\bullet - > C^\bullet$ is the needed homotopy right-inverse of $f^\bullet$, as we sought to show.

With this, we can finally prove the desired result:

Theorem 2.2.9: Projective, Quasi-isomorphism Becomes Homotopy

Let $\mathbf{A}$ be an abelian category, and let $\alpha^\bullet : P^\bullet_0 - > P^\bullet_1$ be a quasi-isomorphism between bounded-above complexes of projectives (resp. quasi-isomorphism between two bounded-bellow complexes of injectives $f^\bullet : Q^\bullet_0 \to Q^\bullet_1$). Then $\alpha^b b$ is a homotopy-equivalence.

***Proof*:**
The argument for projectives shall be given, the injective argument following by appropriate dualizing. By proposition 2.2.8, since $P^\bullet_1$ is in $C^-(\mathbf{P})$ and $\alpha^\bullet : P^\bullet_0 \to P^\bullet_1$ is a quasi-isomorphism, there exists a morphism of complexes $\beta^\bullet : P^\bullet_1 \to p^\bullet_0$ such that

$$\alpha^\bullet \circ \beta^\bullet \sim \mathrm{id}^\bullet_{P_1}$$

Then by proposition 2.1.31, $H^\bullet(\alpha^\bullet) \circ H^\bullet(\beta^\bullet) = \mathrm{id}$. Since $H^\bullet(\alpha^\bullet)$ is invertible, so is $H^\bullet(\beta^\bullet)$, and so $\beta^\bullet$ must be a quasi-isomorphism. Since $P^\bullet_0$ is also in $\mathrm{Ch}^-(\mathbf{P})$, we have that there exists a morphism $(\alpha^\bullet)' : P^\bullet_0 \to P^\bullet_1$ such that $\beta^\bullet \circ (\alpha^\bullet)' \sim \mathrm{id}_{P^\bullet_0}$. Since:

$$(\alpha^\bullet)' \sim (\alpha^\bullet \circ \beta^\bullet) \circ (\alpha^\bullet)' = \alpha^\bullet(\beta^\bullet \circ (\alpha^\bullet)') \sim \alpha^\bullet$$

it follows that $\beta^\bullet \sim \alpha^\bullet \sim_{P^\bullet_0}$ (exercise 4.6 in Aluffi). Thus, $\alpha^\bullet$ is indeed a homotopy equivalence, as we sought to show.

Corollary 2.2.10: Projective In Homotopy Category

In $K^-(\mathbf{P})$ and $K^+(\mathbf{I})$, (homotopy classes of) quasi-isomorphissm are isomorphisms.

***Proof*:**
follows immediately form theorem 2.2.9.

2.2.1 Resolutions

With the important equivalence between quasi-isomorphisms and homotopy-equivalence established in the case of complexes of projective (resp. projective) objects, we turn back our attention to the question of finding (co)homological invariants, focusing on categories with enough projective s(resp. Injectives). In particular, given an object A in an abelian category $\mathbf{A}$, it wouldb e desirable to associate to it a (co)chain of projective (resp. injective) objects in some universal way (up to homotopy) from which information can be extracted.

Let us first refine the definition of projective/injective resolutions of an object. Recall that there i a copy of $\mathbf{A}$ in $\mathrm{Ch}(\mathbf{A})$ given by mapping $A \mapsto \iota(A)$.

Definition 2.2.11: Projective And Injective Resolution

Let $A \in \mathrm{Obj}(\mathbf{A})$ be an object in an abelian category A. Then:

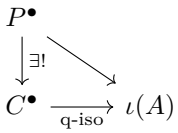
1. A *resolution* of A is a quasi-isomorphism $C^\bullet \to \iota(A)$ for some complex $C^\bullet \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$.
2. A *projective resolution* of A is a quasi-isomorphism $P^\bullet \to \iota(A)$ where $P^\bullet$ is a complex in $\mathrm{Ch}^{\leq 0}(\mathbf{P})$
3. An *injective resolution* of A is a quasi-isomorphism $\iota(A) \to Q^\bullet$ where $Q^\bullet$ is a complex in $\mathrm{Ch}^{\geq 0}(\mathbf{I})$

Often, $P^\bullet$ and $Q^\bullet$ shall be referred to as the projective resolution and injective resolution respectively, omitting the quasi-isomorphism.

2.2.12 Remark There is a point of confusion that may arise. It may seem that the resolutions *themselves* are projective/injective objects in the abelian category $\mathrm{Ch}(\mathbf{A})$. However, this is not the case: a projective object of $\mathrm{Ch}(\mathbf{A})$ must be a complex of projectives that is moreover split exact in the relevant range, which a resolution is not.

Recall further that any resolution $C^\bullet$ will be exact everywhere except possibly at C^0 (exercise, or double-check example 2.1.16). Hence, the non-trivial cohomological information lies in $H^0(C^\bullet)$. Let us now focus on resolutions for a particular object A: consider the category of homotopy classes of quasi-isomorphisms with *target* $\iota(A)$. Then we shall show that, if they exist, projective resolution are

initial in this category.



Analogously, injective resolutions are final in homotopic category of quasi-isomorphisms with *source* $\iota(A)$. Given this universal property, this means that each projective resolution maps uniquely (in $K(\mathbf{A})$) to *every* resolution of A. The following lemma proves this:

Lemma 2.2.13: Projective Resolution As Initial Objects

Let $A \in \mathbf{A}$, let $C^\bullet$ be a resolution of A (so that $C^i = 0$ for $i > 0$, $H^i(C^\bullet) = 0$ for $i < 0$, and $H^0(C^\bullet) = A$), and let $P^\bullet \in \mathrm{Ch}^{\leq 0}(\mathbf{P})$ be a bounded-above complex of projectives. Then any morphism

$$\varphi : H^0(P^\bullet) \to H^0(C^\bullet) \; [= A]$$

is induced by a morphism of complexes $\varphi^\bullet : P^\bullet \to C^\bullet$, and any two such $\varphi^\bullet$ are homotopic.

***Proof*:**

Since both complexes vanish in positive degrees, $H^0(P^\bullet) = \mathrm{coker} d_P^{-1}$ and $H^0(C^\bullet) = \mathrm{coker} d_C^{-1}$; write $\pi_P \colon P^0 \twoheadrightarrow H^0(P^\bullet)$ and $\pi_C \colon C^0 \twoheadrightarrow A$ for the two quotient maps.

Step 1: Degree 0. The composite $\varphi \circ \pi_P \colon P^0 \to A$ is a morphism out of a projective object, and π_C is an epimorphism, so by definition 2.2.1 it lifts: there is $\varphi^0 \colon P^0 \to C^0$ with $\pi_C \circ \varphi^0 = \varphi \circ \pi_P$.

Step 2: Degree -1. Compute

$$\pi_C \circ \varphi^0 \circ d_P^{-1} = \varphi \circ \pi_P \circ d_P^{-1} = 0$$

since $\pi_P \circ d_P^{-1} = 0$. Hence $\varphi^0 \circ d_P^{-1}$ factors through $\ker \pi_C = \mathrm{im} d_C^{-1}$. As P^{-1} is projective and d_C^{-1} is an epimorphism onto its image, the map $P^{-1} \to \mathrm{im} d_C^{-1}$ lifts to $\varphi^{-1} \colon P^{-1} \to C^{-1}$ with $d_C^{-1} \circ \varphi^{-1} = \varphi^0 \circ d_P^{-1}$.

Step 3: The induction. Suppose φ^{-i} has been constructed for $i < n$ with every square commuting. Then

$$d_C^{-n+1} \circ \left(\varphi^{-n+1} \circ d_P^{-n}\right) = \varphi^{-n+2} \circ d_P^{-n+1} \circ d_P^{-n} = 0$$

so $\varphi^{-n+1} \circ d_P^{-n}$ lands in $\ker d_C^{-n+1}$, which for $n \geq 2$ equals $\mathrm{im} d_C^{-n}$ because $C^\bullet$ is exact in degree $-n+1 < 0$; the case $n = 1$ is Step 2, where $\ker \pi_C$ plays that role. Projectivity of P^{-n} again lifts it along the epimorphism $C^{-n} \twoheadrightarrow \mathrm{im} d_C^{-n}$, producing φ^{-n} with $d_C^{-n} \circ \varphi^{-n} = \varphi^{-n+1} \circ d_P^{-n}$. This constructs $\varphi^\bullet$ in every degree, and Step 1 arranges that it induces φ on H^0.

Step 4: Uniqueness up to homotopy. Let $\varphi^\bullet$ and $\psi^\bullet$ both induce φ, and set $f^\bullet = \varphi^\bullet - \psi^\bullet$. On H^0 it induces $\varphi - \varphi = 0$, and in every negative degree $H^i(C^\bullet) = 0$, so $H^i(f^\bullet) = 0$ there as well; in positive degrees both complexes vanish. Hence $H^\bullet(f^\bullet) = 0$. The hypotheses of lemma 2.2.4 are met, $P^\bullet$ being a bounded-above complex of projectives and $C^\bullet$ having vanishing cohomology in negative degrees, so $f^\bullet \sim 0$, that is $\varphi^\bullet \sim \psi^\bullet$, completing the proof.

Since these are initial (resp. final) objects in the category, we have a unique isomorphism beetween any two solutions to the universal problem. In this case, this may be stated as followed:

Proposition 2.2.14: Unique Projection Resolution

Let $\mathbf{A}$ be an abelian category. Then any two projective (resp. injective) resolutions of an object A in $\mathbf{A}$ are homotopy equivalent

***Proof*:**
Let $P^\bullet$ and $Q^\bullet$ be two projective resolutions of A, so that each is a bounded-above complex of projectives with $H^0 = A$. Apply lemma 2.2.13 twice to the morphism id_A: once with source $P^\bullet$ and target $Q^\bullet$, giving $f^\bullet\colon P^\bullet \to Q^\bullet$, and once the other way, giving $g^\bullet\colon Q^\bullet \to P^\bullet$. Each is unique up to homotopy.

The composite $g^\bullet \circ f^\bullet\colon P^\bullet \to P^\bullet$ induces id_A on H^0, and so does $\mathrm{id}_{P^\bullet}$. Both are therefore lifts of the same morphism id_A, and the uniqueness clause of lemma 2.2.13 gives $g^\bullet \circ f^\bullet \sim \mathrm{id}_{P^\bullet}$. Symmetrically $f^\bullet \circ g^\bullet \sim \mathrm{id}_{Q^\bullet}$. Hence $f^\bullet$ is a homotopy equivalence with homotopy inverse $g^\bullet$, as we sought to show. The injective case is dual.

A final important observation is that this lift is functorial!

Proposition 2.2.15: Projective Resolution Functoriality

Let $\mathbf{A}$ be an abelian category, and let A_0, A_1 be objects of $\mathbf{A}$. Let $P_0^\bullet$ and $P_1^\bullet$ be projective resolutions for A_0 and A_1 respectively. Then every morphism $\varphi : A_0 \to A_1$ in $\mathbf{A}$ induces a morphism $\alpha^\bullet : P_0^\bullet \to P_1^\bullet$ that is unique up to homotopy

***Proof*:**
Notice that φ can be interpreted as a morphism $\varphi : H^0(P_0^\bullet) \to H^0(P_1^\bullet)$. The complex $P_0^\bullet$ consists of projectives, and $P1^\bullet$ is a resolution of A_1. Thus, by lemma 2.2.13, there exists a list α^b that is unique up to homotopy, as we sought to show.

Corollary 2.2.16: Full Subcategory Of Homotopic Category

Let $\mathbf{A}$ be an abelian category with enough projectives (resp. enough injectives). Then $\mathbf{A}$ is equivalent to a full subcategory of $K^-(\mathbf{P})$ (resp. $K^+(\mathbf{I})$).

***Proof*:**
Treat the projective case. Choose for each object A of $\mathbf{A}$ a projective resolution $P_A^\bullet$, possible because $\mathbf{A}$ has enough projectives, and send $A \mapsto P_A^\bullet$. By proposition 2.2.15 every $\varphi\colon A_0 \to A_1$ lifts to a morphism of resolutions, well defined up to homotopy, so this is a functor $R\colon \mathbf{A} \to K^-(\mathbf{P})$: composites and identities are preserved because a lift of a composite and the composite of lifts both lift the same morphism and so agree up to homotopy, which is equality in $K^-(\mathbf{P})$.

R is faithful. If $R(\varphi) = 0$ then a lift $\varphi^\bullet$ of φ is null-homotopic, so it induces the zero morphism

on cohomology by proposition 2.1.29. Since $H^0(\varphi^\bullet) = \varphi$ under the identifications $H^0(P^\bullet_{A_i}) = A_i$, this gives $\varphi = 0$.

R is full. Let $f^\bullet \colon P^\bullet_{A_0} \to P^\bullet_{A_1}$ be any morphism in $K^-(\mathbf{P})$, represented by a chain map. It induces $\varphi := H^0(f^\bullet) \colon A_0 \to A_1$, and $f^\bullet$ is then a lift of φ. By the uniqueness clause of lemma 2.2.13 any two lifts of φ are homotopic, so $R(\varphi) = [f^\bullet] = f^\bullet$ in $K^-(\mathbf{P})$.

A fully faithful functor is an equivalence onto its essential image, which is a full subcategory of $K^-(\mathbf{P})$. Hence $\mathbf{A}$ is equivalent to a full subcategory of $K^-(\mathbf{P})$, and dually to one of $K^+(\mathbf{I})$ when $\mathbf{A}$ has enough injectives, completing the proof.

2.2.2 Derived Category with Enough Projectives

Let us finally confirm that the solution to the universal problem in equation 2.2. Recall that the derived category is a category $D(\mathbf{A})$ where for any category D for which there exists an additive functor $\mathscr{F} : \mathrm{Ch}(\mathbf{A}) \to D$ where every quasi-isomorphism in $\mathrm{Ch}(\mathbf{A})$ becomes an isomorphism in D, there exists a unique functor such that the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Ch}(\mathbf{A}) & \xrightarrow{\mathscr{F}} & D \\ \downarrow & \nearrow_{\exists!} & \\ D(\mathbf{A}) & & \end{array}$$

In the case where $\mathbf{A}$ has enough projectives (resp. enough injectives) and is bounded from above (resp. from bellow), then $K^-(\mathbf{P})$ (resp. $K^+(\mathbf{I})$) is the solution to the above universal problem. In particular, there is an equivalence of categories between $K^-(\mathbf{P})$ and $D^-(\mathbf{A})$. The formal definition of the derived category shall be given in section 3.2, hence this equivalence shall not be proven, however the universal property is readily provable, and once the derived category is formally defined it will be the equivalence essentially come to proper manipulation of universal properties.

To start, a general procedure for finding projective/injective resolutions for any (co)chain must be explored. The discussion so far has been for projective/injective resolution, that is quasi-isomorphisms $P^\bullet \to \iota(A)$ or $\iota(A) \to Q^\bullet$. Is it possible to replace $\iota(A)$ with any bounded complex? The answer is yes, though rather technical. The importance of this will be the existence of a functor p that maps $\mathrm{Ch}^-(\mathbf{P})$ into $K^-(\mathbf{P})$, which shall be important for solving the universal problem above, as well as finding functors between derived categories. Such functors shall also be key in solving the "extension problem" that was mentioned all the way back in [8, Extension Problem: Prelude to Homological Algebra].

Theorem 2.2.17: Resolution And Complexes

Let $\mathbf{A}$ be an abelian category with enough projectives and $C^\bullet$ a complex in $C^-(\mathbf{A})$. Then there exists a bounded-above complex of projective $P^\bullet$ and a quasi-isomorphism $P^\bullet \to C^\bullet$, where $P^\bullet$ is uniquely defined up to homotopy equivalence. Furthermore, every morphism $\alpha^\bullet$ of complexes in $C^-(\mathbf{A})$ lifts to a morphism of the corresponding projective resolutions in $K^-(\mathbf{P})$, which is uniquely defined up to homotopy.

***Proof Key ideas*:**
The uniqueness and lifting halves are already available; only the existence half needs a construction, and that construction is a descending induction whose bookkeeping is long without being conceptually informative, so it is given in outline here and in full in *An Introduction to Homological Algebra* [23].

Existence. Let $C^\bullet$ be bounded above, say $C^n = 0$ for $n > N$. Build P^n by descending induction on n, starting with $P^n = 0$ for $n > N$. At each stage one has already constructed the terms above degree n together with the maps to $C^\bullet$, and one chooses a projective object surjecting onto the pullback of C^n against the cycles constructed one degree up. That single choice simultaneously arranges surjectivity on cohomology in degree n and injectivity in degree $n+1$, which is what makes the resulting map a quasi-isomorphism; enough projectives is exactly what gives the surjection at each stage.

Uniqueness up to homotopy equivalence. Two such projective complexes over $C^\bullet$ are homotopy equivalent by the argument of proposition 2.2.14, with its object-level input lemma 2.2.13 replaced by the lifting of morphisms in the next paragraph, whose existence comes from corollary 2.2.5 and whose uniqueness up to homotopy comes from lemma 2.2.7. Applied to $\mathrm{id}_{C^\bullet}$, it gives a map each way over $C^\bullet$, and both composites lift the identity, hence are homotopic to it by that uniqueness.

Lifting of morphisms. Given $\alpha^\bullet \colon C^\bullet \to D^\bullet$ and projective complexes $P^\bullet \to C^\bullet$, $Q^\bullet \to D^\bullet$, the composite $P^\bullet \to C^\bullet \to D^\bullet$ is a morphism from a bounded-above complex of projectives into a complex quasi-isomorphic to $Q^\bullet$. Existence of a factorization through $Q^\bullet$ comes from corollary 2.2.5, applied to the cone of $Q^\bullet \to D^\bullet$, and its uniqueness up to homotopy from the cancellation property lemma 2.2.7. Together they give a well-defined morphism in $K^-(\mathbf{P})$.

Thus, given an abelian category $\mathbf{A}$ with enough projective, there is a functor

$$\mathscr{P} : C^-(\mathbf{A}) \to K^-(\mathbf{P})$$

associating each (bounded-above) complex $C^\bullet$ to a projective resolution $\mathscr{P}(C^\bullet)$ in a functorial manner (so that morphisms $f^\bullet$, $g^\bullet$ are lifted and $\mathscr{P}(g^\bullet \circ f^\bullet) = \mathscr{P}(g^\bullet) \circ \mathscr{P}(f^\bullet)$ since both sides are classes of homotopy lifts of $g^\bullet \circ f^\bullet$). The uniqueness of $\mathscr{P}$ is up to natural transformation, that is if another functor $\mathscr{P}'$ is chosen that assigns a different choice of projective resolution, then by theorem 2.2.17 there is a homotopy equivalence between $\mathscr{P}(C^\bullet)$ and $\mathscr{P}'(C^\bullet)$, hence an isomorphism $\varphi_{C^\bullet} : \mathscr{P}(C^\bullet) \to \mathscr{P}'(C^\bullet)$ in $K^-(\mathbf{P})$. Since the identity on $C^\bullet$ lifts to a unique map up to homotopy, the isomorphism $\varphi_{C^\bullet}$ is unique. Furthermore, given a morphism $f^\bullet : C^\bullet \to D^\bullet$, the diagram:

$$\begin{array}{ccc} \mathscr{P}(C^\bullet) & \xrightarrow{\mathscr{P}(f^\bullet)} & \mathscr{P}(D^\bullet) \\ {\scriptstyle \varphi_{C^\bullet}}\downarrow{\scriptstyle \sim} & & {\scriptstyle \sim}\downarrow{\scriptstyle \varphi_{D^\bullet}} \\ \mathscr{P}'(C^\bullet) & \xrightarrow[\mathscr{P}'(f^\bullet)]{} & \mathscr{P}'(D^\bullet) \end{array}$$

commutes (again by the uniqueness of lifts up to homotopy, can you see why)? Thus, we have the unique natural isomorphism for any two choices of functors giving a projective resolution.

Theorem 2.2.18: Universal Property Of Derived Category With Enough Projectives

Let $\mathbf{A}$ be an abelian category with enough projectives, and $\mathscr{P}$ a functor sending bounded-above cochains to projective resolutions. Then:

1. If $f^\bullet$ is a quasi-isomorphism in $C^-(\mathbf{A})$, then $\mathscr{P}(f^\bullet)$ is an isomorphism in $K^-(\mathbf{P})$
2. If $\mathscr{F} : C^-(\mathbf{A}) \to D$ is an additive functor that takes quasi-isomorphisms to isomorphisms, then there exists a unique functor $\widetilde{\mathscr{F}} : K^-(\mathbf{P}) \to D$ unique up to natural isomorphism such that the following diagram commutes up to natural isomorphism:

$$\begin{array}{ccc} C^-(\mathbf{A}) & \xrightarrow{\mathscr{F}} & D \\ {\scriptstyle\mathscr{P}}\downarrow & \nearrow_{\widetilde{\mathscr{F}}} & \\ K^-(\mathbf{P}) & & \end{array}$$

where commuting up to natural transformation means there exists a natural transformation $\nu : \widetilde{\mathscr{F}} \circ \mathscr{P} \rightsquigarrow \mathscr{F}$ such that:

$$\nu_{C^\bullet} : \widetilde{\mathscr{F}} \circ \mathscr{P}(C^\bullet) \xrightarrow{\sim} \mathscr{F}(C^\bullet)$$

is an isomorphism for every complex $C^\bullet$ in $C^-(\mathbf{A})$.

The fact that the commutative diagram commutes up to natural isomorphism reflects that the homotopic category is categorically equivalent to the derived category. If $\mathbf{A}$ has enough injective, then we may instead define the functor $\mathscr{I} : C^+(\mathbf{A}) \to K^+(\mathbf{I})$ that shall satisfy the dual universal property.

***Proof*:**

1. Let $f^\bullet : C^\bullet \to D^\bullet$ be any morphism in $C^-(\mathbf{A})$. By theorem 2.2.17, $f^\bullet$ lifts to a morphism of resolutions which commutes up to homotopy in the following diagram:

$$\begin{array}{ccc} \mathscr{P}(C^\bullet) & \xrightarrow{\mathscr{P}(f^\bullet)} & \mathscr{P}(C^\bullet) \\ {\scriptstyle\mu_{C^\bullet}}\downarrow & & \downarrow{\scriptstyle\mu_{D^\bullet}} \\ C^\bullet & \xrightarrow{f^\bullet} & D^\bullet \end{array}$$

where $\mu_{C^\bullet}, \mu_{D^\bullet}$ are the quasi-isomorphisms of the projective resolutions. Since the diagram commute up to homootpy, it commutes after taking cohomology, thus since $f^\bullet$ is a quasi-isomorphism, so is $\mathscr{P}(f^\bullet)$. Then by corollary 2.2.10, $\mathscr{P}(f^\bullet)$ is an isomorphism.

2. Define $\widetilde{\mathscr{F}} : K^-(\mathbf{P}) \to D$ via:

$$\widetilde{\mathscr{F}}(P^\bullet) := \mathscr{F}(P^\bullet) \qquad \widetilde{\mathscr{F}}(f^\bullet) := \mathscr{F}(f^\bullet)$$

where $\widetilde{\mathscr{F}}(f^\bullet)$ can be any representative of $f^\bullet$. This map is well-defined, since homotopic morphisms have the same image in D (lemma 2.1.37). Naturally, $\widetilde{\mathscr{F}}$ is a functor. We need

to check the commutativity up to morphism of the diagram and the uniqueness of $\widetilde{\mathscr{F}}$ up to natural isomorphism.

First, let $C^\bullet \in C^-(\mathbf{A})$. Then (by construction of $\mathscr{P}$, there exists a quasi-isomorphism

$$\mu_{C^\bullet} : \mathscr{P}(C^\bullet) \to C^\bullet$$

Since $\mathscr{F}$ maps quasi-isomorphisms to isomorphism, the induced morphism $\nu_{C^\bullet} := \mathscr{F}(\mu_{C^\bullet})$:

$$\nu_{C^\bullet} : \widetilde{\mathscr{F}}(\mathscr{P}(C^\bullet)) = \mathscr{F}(\mathscr{P}(C^\bullet)) \to \mathscr{F}(C^\bullet)$$

is an isomorphism. What's left to check is that $\nu_{C^\bullet}$ defines a natural transformation. Let $f^\bullet : C^\bullet \to D^\bullet$ in $C^-(\mathbf{A})$. We want to show the following diagram commutes:

$$\begin{array}{ccc} \widetilde{\mathscr{F}}(\mathscr{P}(C^\bullet)) & \xrightarrow{\widetilde{\mathscr{F}}(\mathscr{P}(f^\bullet))} & \widetilde{\mathscr{F}}(\mathscr{P}(D^\bullet)) \\ {\scriptstyle \nu_{C^\bullet}}\downarrow & & \downarrow{\scriptstyle \nu_{D^\bullet}} \\ \mathscr{F}(C^\bullet) & \xrightarrow{\mathscr{F}(f^\bullet)} & \mathscr{F}(D^\bullet) \end{array}$$

This diagram is obtained by applying $\mathscr{F}$ to the diagram discussed above the theorem:

$$\begin{array}{ccc} \mathscr{P}(C^\bullet) & \xrightarrow{\mathscr{P}(f^\bullet)} & \mathscr{P}(D^\bullet) \\ {\scriptstyle \varphi_{C^\bullet}}\downarrow{\scriptstyle \sim} & & {\scriptstyle \sim}\downarrow{\scriptstyle \varphi_{D^\bullet}} \\ \mathscr{P}'(C^\bullet) & \xrightarrow[\mathscr{P}'(f^\bullet)]{} & \mathscr{P}'(D^\bullet) \end{array}$$

This diagram commutes up to homotopy, hence the other diagram commutes directly (lemma 2.1.37).

For uniqueness up to isomorphism, say $\overline{\mathscr{F}} : K^-(\mathbf{P}) \to D$ is another functor satisfying the properties. Then there exists a natural isomorphism:

$$\overline{\nu} : \overline{\mathscr{F}} \circ \mathscr{P} \rightsquigarrow \mathscr{F}$$

Now, for any $P^\bullet \in K^-(\mathbf{P})$, take the composition:

$$\overline{\mathscr{F}}(P^\bullet) \xrightarrow[\sim]{\overline{\mathscr{F}}(\mu_{P^\bullet})^{-1}} \overline{\mathscr{F}}(\mathscr{P}((P^\bullet)) \xrightarrow[\sim]{\overline{\nu}_{P^\bullet}} = \widetilde{\mathscr{F}}(P^\bullet)$$

If φ is this composition, then it is easy to verify that $\varphi : \overline{\mathscr{F}} \to \widetilde{\mathscr{F}}$ is a natural isomorphism, completing the proof.

2.3 Derived Functors: Ext and Tor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor. Often, we study information about $\mathbf{A}$ by passing it through $\mathbf{B}$ (think of $- \otimes N$ or $\mathrm{Hom}_R(N, -)$). As alluded to in [8, Ext and Tor], such information may be the number of extensions two objects may have, the amount of torsion an object may have. In [8, Ext and Tor], it was claimed that this comes from "fixing" the lack of exactness of a left- or right-exact

functor using homological algebra. We are finally in a position to make this precise. In this section, we shall show how to take a (not-necessarily left/right exact) additive functor between two abelian categories with $\mathbf{A}$ having enough projectives (resp. injectives) $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ and get a *derived functor*:

$$D^-(\mathscr{F}) : D^-(\mathbf{A}) \to D^-(\mathbf{B}) \qquad (\text{resp. } D^-(\mathscr{F}) : D^-(\mathbf{A}) \to D^-(\mathbf{B}))$$

This functor shall be used to develop a long-exact sequence in $\mathbf{B}$, given a short exact sequence of objects in $\mathbf{A}$.

2.3.1 Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories. Then $\mathscr{F}$ extends to a functor:

$$\mathrm{Ch}(\mathscr{F}) : \mathrm{Ch}(\mathbf{A}) \to \mathrm{Ch}(\mathbf{B})$$

namely, if $C^\bullet$ is a complex:

$$\mathscr{F}(d^{i+1}_{C^\bullet}) \circ \mathscr{F}(d^i_{C^\bullet}) = \mathscr{F}(d^{i+1}_{C^\bullet} \circ d^i_{C^\bullet}) = \mathscr{F}(0) = 0$$

hence, $\mathrm{Ch}(\mathscr{F})(C^\bullet)$ is indeed a complex. Then since homotopic morphisms are sent to homotopic morphisms (lemma 2.1.34), $\mathrm{Ch}(\mathscr{F})$ induces a morphism $K(\mathscr{F})$:

$$K(\mathscr{F}) : K(\mathbf{A}) \to K(\mathbf{B})$$

From the results we have proved, we have the following commutative diagram:

$$\begin{array}{ccc} \mathbf{A} & \xrightarrow{\mathscr{F}} & \mathbf{B} \\ {\scriptstyle\iota}\downarrow & & \downarrow{\scriptstyle\iota} \\ K(\mathbf{A}) & \xrightarrow{K(\mathscr{F})} & K(\mathbf{B}) \end{array}$$

Furthermore, if the complexes are bounded, then certainly $\mathrm{Ch}(\mathscr{F})$ sends bounded complexes to bounded complexes, and hence the same holds for $K(\mathscr{F})$. Now, if $\mathbf{A}$ has enough projectives so that we may consider $K^-(\mathbf{P})$, can we restrict $K(\mathbf{A})$ to $K(P(\mathbf{A}))$, and map to $K(P(\mathbf{B}))$ (where $P(\mathbf{A})$ and $P(\mathbf{B})$ are the projectives of $\mathbf{A}$ and $\mathbf{B}$ respectively)? This is not the case in general; for example we saw that $- \otimes N$ in R-**Mod** does not preserve projectives. If $\mathbf{B}$ has enough projectives, then we may choose a projective resolution functor $\mathscr{P}_B : \mathrm{Ch}^-(B) \to K^-(\mathbf{P}^{-1})$ mapping each complex to a projective resolution in a functorial manner. Since $\mathscr{P}_B$ sends homotopic morphisms to homotopic morphisms (lemma 2.2.7), $\mathscr{P}_B$ descends to a functor $K(\mathscr{P}_B)$. Combining these facts, we may define the following functor:

Definition 2.3.1: Left-Derived (right-Derived) Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor. Then the *left-derived functor* of $\mathscr{F}$ is the composition:

$$K^-(P(\mathbf{A})) \xrightarrow[\iota_A]{} K^-(\mathbf{A}) \xrightarrow[K(\mathscr{F})]{} K^-(\mathbf{B}) \xrightarrow[K(\mathscr{P}_B))]{} K^-(P(\mathbf{B})) \qquad \mathbf{L}\mathscr{F} : K^-(\mathbf{A}) \to K^-(P(\mathbf{B}))$$

where ι_A is the inclusion of $K^-(P(\mathbf{A}))$ as a full subcategory of $K^-(\mathbf{A})$. The resulting functor is denoted $\mathbf{L}\mathscr{F}$.

It may seem that the natural notation would be $D^-(\mathscr{F})$, however the $\mathbf{L}\mathscr{F}$ notation was established before the abstraction given above, and has stuck. Similarly, if $\mathbf{A}$, $\mathbf{B}$ has enough injectives, we may define the right-derived functor $\mathbf{R}\mathscr{F}$.

It is natural to want to characterize $\mathbf{L}\mathscr{F}$ and $\mathbf{R}\mathscr{F}$ in terms of some universal property. Naturally, this universal property would be defined up to natural isomorphism, for $\mathscr{P}_A$ and $\mathscr{P}_B$ are chosen up to natural isomorphism. This turns to be a bit subtle. For example, there is no natural functor which shall always make the following diagram commute:

$$\begin{array}{ccc} \hat{\mathbf{A}} & \dashrightarrow^{??} & \hat{\mathbf{B}} \\ \downarrow & & \downarrow \\ K^-(P(\mathbf{A})) & \xrightarrow{\mathbf{L}\mathscr{F}} & K^-(P(\mathbf{B})) \end{array}$$

The reason is, it is not necessarily the case that a complex whose cohomology is concentrated at 0 would map via $\mathbf{L}\mathscr{F}$ to a complex with the same property. In fact, the reader may expect this, for the image of a functor $\mathscr{F}$ may present some non-trivial cohomological information, (think of $\mathscr{F} := \mathrm{Hom}_R(N, -)$ where N is non-projective). If the functor $\mathscr{F}$ is *exact*, then then there is essentially no "change in cohomology". This is captured in the following proposition:

Proposition 2.3.2: Derived Functors And Exactness

Let $\mathscr{F}$ be an additive *exact* functor. Then $\hat{\mathbf{A}}$ is in fact sent to $\hat{\mathbf{B}}$ via $\mathbf{L}\mathscr{F}$

***Proof*:**
Let $P^\bullet \in \hat{\mathbf{A}}$ so that $H^i(P^\bullet) = 0$ for all $i \neq 0$. Then $\mathbf{L}\mathscr{F}(P^\bullet)$ is given by choosing a projective resolution $P^\bullet_{\mathscr{F}(P^\bullet)}$ of $\mathscr{F}(P^\bullet)$. Since quasi-isomorphissm preserve cohomology and (key!) exact functors *commute* with cohomology, an exact functor carrying kernels to kernels and images to images and hence subquotients to subquotients:

$$H^i(\mathbf{L}\mathscr{F}(P^\bullet)) = H^i(P^\bullet_{\mathscr{F}(P^\bullet)}) \cong H^i(\mathscr{F}(P^\bullet)) \cong \mathscr{F}(H^i(P^\bullet)) = \mathscr{F}(0) \overset{i\neq 0}{=} 0$$

Thus, $\mathbf{L}\mathscr{F}(P^\bullet)$ does indeed map to an object of $\hat{\mathbf{B}}$.

Even more is true if $\mathscr{F}$ is exact: $H^i(\mathbf{L}\mathscr{F}(P^\bullet)) \cong \mathscr{F}(H^0(P^\bullet))$, saying that the restriction of $\mathbf{L}\mathscr{F}$ agrees with $\mathscr{F}$ (modulo equivalence of categories). Hence, in general we shall be interested in non-exact functors, usually functors that are only left or right exact.

Returning to finding an appropriate universal property for derived functors, another natural candidate would be given by taking a "step back" from the above diagram and consider instead if the following diagram commutes (up to natural isomorphism)

$$\begin{array}{ccc} K^-(\mathbf{A}) & \xrightarrow{K(\mathscr{F})} & K^-(\mathbf{B}) \\ \downarrow{\scriptstyle \mathscr{P}_A} & & \downarrow{\scriptstyle \mathscr{P}_B} \\ K^-(P(\mathbf{A})) & \xrightarrow[\mathbf{L}\mathscr{F}]{} & K^-(P(\mathbf{B})) \end{array}$$

in particular, that there may be a natural isomorphism:

$$\mathbf{L}\mathscr{F} \circ \mathscr{P}_A \overset{\sim}{\rightsquigarrow} \mathscr{P}_B \circ K(\mathscr{F})$$

Unfortunately, this is not in general the case

Example 2.3.3: Lacking Natural Isomorphism
(exercise 7.3 in Aluffi, should I put solution here?)

The following is the best we can hope for:

Proposition 2.3.4: Universal Property of (left-)Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between abelian categories. Then $\mathbf{L}\mathscr{F}$ satisfies the following universal property:

1. There is a natural transformation:
$$\mathbf{L}\mathscr{F} \circ \mathscr{P}_A \overset{\sim}{\rightsquigarrow} \mathscr{P}_B \circ K(\mathscr{F})$$
2. for every functor $\mathscr{G} : K^-(P(\mathbf{A})) \to K^-(P(\mathbf{B}))$ and every natural tarnsformation $\gamma : \mathscr{G} \circ \mathscr{P}_A \rightsquigarrow \mathscr{P}_B \circ K(\mathscr{F})$, there is a unique (up to natural isomorphism) natural transformation $\mathscr{G} \rightsquigarrow \mathbf{L}\mathscr{F}$ inducign a factorization:
$$\gamma : \mathscr{G} \circ \mathscr{P}_A \rightsquigarrow \mathbf{L}\mathscr{F} \circ \mathscr{P}_A \rightsquigarrow \mathscr{P}_B \circ K(\mathscr{F})$$

A similar result holds for right-derived functors.

***Proof Key ideas*:**
The argument is the usual one for a Kan-extension-style universal property, and it is stated here rather than carried out because the ambient apparatus is not yet in place: the definition of $D(\mathbf{A})$ used below rests on the localization machinery that section 1.2 defers. The full treatment is in *An Introduction to Homological Algebra* [23].

For (1), the natural transformation is the comparison map induced by functoriality of projective resolutions: $\mathscr{P}_A$ picks a resolution, $K(\mathscr{F})$ applies $\mathscr{F}$ termwise, and proposition 2.2.15 makes the resulting square commute up to homotopy, which is commutativity in K^-.

For (2), given $\mathscr{G}$ and γ, the factorization is forced on objects because a projective resolution is initial among complexes of projectives mapping to the object (lemma 2.2.13), so any transformation out of $\mathscr{G} \circ \mathscr{P}_A$ is determined by its value on resolutions; uniqueness up to natural isomorphism is then the uniqueness up to homotopy of those comparison maps (proposition 2.2.14).

(a word on composition)

2.3.2 Long Exact Sequences of Derived Functors

With the derived functor established and being the best candidate for holding (co)homology information, we proceed with extracting this informaiton: we may take $\mathbf{L}_i\mathscr{F} := H^{-i} \circ \mathbf{L}\mathscr{F}$. The reason for taking H^{-i} instead of i is since the indexing works out better. This produced a functor

$\mathbf{L}_i\mathscr{F} : D^-(\mathbf{A}) \to B$. Notice that B does not need to have enough projectives for:

$$\mathbf{L}_i\mathscr{F}(P^\bullet) = H^{-i}(\mathscr{P}_B(\mathrm{Ch}(\mathscr{F})(P^\bullet))) \cong H^{-i}(\mathrm{Ch}(\mathscr{F})(P^\bullet))$$

where the last isomorphism comes from $\mathscr{P}_B(\mathrm{Ch}(\mathscr{F})(P^\bullet)))$ being quasi-isomorphic to $\mathrm{Ch}(\mathscr{F})(P^\bullet)$. Though this abstract approach is useful, the following more concrete version of $\mathbf{L}_i\mathscr{F}$ shall prove more useful for computations.

Definition 2.3.5: ith Left-Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between two abelian categories where $\mathbf{A}$ has enough projectives. Then the *ith left-derived functor* $\mathbf{L}_i\mathscr{F} : A \to B$ is given by:

$$\mathbf{L}_i\mathscr{F} := H^{-i} \circ \mathbf{L}\mathscr{F} \circ \mathscr{P}_A \circ \iota_A$$

The complex in $\mathrm{Ch}(\mathbf{B})$ with $\mathbf{L}_i\mathscr{F}(M)$ in the $-i$th degree with vanishing differentials is denoted $\mathbf{L}_\bullet\mathscr{F}(M)$.

To be precise, if $M \in \mathbf{A}$, then $\mathbf{L}_i(M)$ is found by

1. finding any projective resolution $P^\bullet_M$ of M
2. Applying $\mathrm{Ch}(\mathscr{F})$ to $P^\bullet_M$, $\mathrm{Ch}(\mathscr{F})(P^\bullet_M)$
3. take the $(-i)$th cohomology of this complex, $H^{-i}(\mathrm{Ch}(\mathscr{F})(P^\bullet_M))$

With all the build-up done so far, the resulting cohomology is independent of choice, for:

1. Any two projective resolutions are homotopy-equivalent
2. the image of homotopy-equivalent complexes via additive functors have isomorphic cohomology

Naturally, the *ith right-derived functor* $\mathbf{R}^i\mathscr{F}$ of an additive functor $\mathscr{F}$ and the assocaited complex $\mathbf{R}^\bullet\mathscr{F}$ is defined simplify (given $\mathbf{A}$ has enough injectives), namely

$$\mathbf{R}^i\mathscr{F} := H^i \circ \mathbf{R}\mathscr{F} \circ \mathscr{Q}_A \circ \iota_A$$

A note should be said about *contravriant functors*: since they can always be viewed as *covariant* functor $\mathbf{A}^{\mathrm{op}} \to B$, $\mathbf{A}$ would instead have to have the appropriate injectives or projectives swapped. The right-derived functor of an additive *contravariant functor* would need $\mathbf{A}$ to have enough *projectives*, since $\mathbf{A}^{\mathrm{op}}$ would have to have enough injectives. The $\mathbf{L}_i$ and $\mathbf{R}_i$ notation can be thought of as "looking" at the cohomology provided by $\mathscr{F}$. As mentioned earlier, if $\mathscr{F}$ is exact, then the derived functors are *trivial* in terms of cohomology, hence the derived functors are only interesting if $\mathscr{F}$ is not exact, at which point these functors give you interesting information.

Example 2.3.6: ith Derived Functors

It is finally time to bring back [8, Ext and Tor]

1. Let $\mathbf{A} = R$-**Mod** for a commutative ring R, pick an R-module N, and define

$$- \otimes_R N$$

The left-derived functor of $- \otimes_R N$ is labeled

$$- \overset{\mathbf{L}}{\otimes}_R N : D^-(R\text{-Mod}) \to D^-(R\text{-Mod})$$

The ith derived left-functor of $- \otimes_R N$ is denoted:

$$\operatorname{Tor}_i^R(- \otimes_R N) = H^{-i}((-)_\bullet \otimes N)$$

where the right hand side of the equality comes from the result elaborated in [8, Ext and Tor]. It was mentioned that a flat resolution may be used instead of a projective resolution, however this was does not currently fit within the theory. This shall be addressed in the next section.

2. Similarly, $\operatorname{Hom}_R(-, N)$ and $\operatorname{Hom}_R(N, -)$ admits a right derived functors

$$\operatorname{Ext}_R^i(M, N) = H^i(\operatorname{Hom}_R(M, N^\bullet))$$

For $\operatorname{Hom}_R(-, N)$, do not forget that one must take the opposite category and a projective resolution.

Commutativity of R is used here only so that the values are again R-modules. Over an arbitrary ring R the same definitions apply to $- \otimes_R N \colon \text{Mod-}R \to \mathbf{Ab}$ for a left R-module N, and to Hom_R between left R-modules, with values in $\mathbf{Ab}$.

The most important part of left-/right-derived functors is their use in fixing exactness after applying a not necessarily exact functor $\mathscr{F}$. On a high level, since the derived category there is not an abelian category (hence not necessarily containing kernels and cokernels, nor is it really feasible to investigate such categories), the notion of *triangulated categories* is useful for categories where given a "short exact sequence", we can create a long exact sequence (that is, a category in which snakes lemma can be applied). We shall not investigate the idea of triangulated categories, but the following is a concrete results needed for any rigorous exploration of triangulated categories.

Assuming the whole that that our category $\mathbf{A}$ has enough projectives, take a short exact sequence:

$$0 \to L \to M \to N \to 0$$

Since there is enough projectives, can the projective resolutions be arranged to create a short exact sequence? The answer is yes, and given the following apt name:

Lemma 2.3.7: Horseshoe Lemma

Let

$$0 \to L \to M \to N \to 0$$

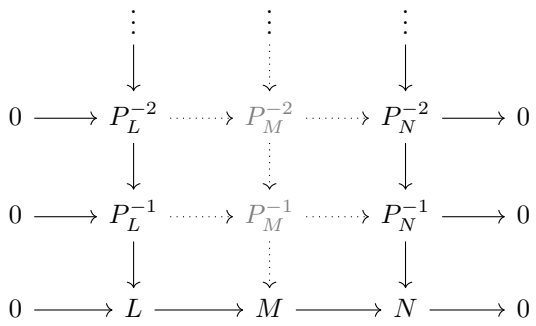
be an exact sequence in an abelian category $\mathbf{A}$ with enough projectives, and let $P_L^\bullet, P_N^\bullet$ be projective resolutions of L, N respectively. Then there exists an exact sequence:

$$0 \to P_L^\bullet \to P_M^\bullet \to P_N^\bullet \to 0$$

where $P_M^\bullet$ is a projective resolution of M, inducing the original sequence in cohomology (i.e. applying H^0). Moreover the sequence is *degreewise split*: the construction produces $P_M^i = P_L^i \oplus P_N^i$ in each degree, with the two displayed maps the inclusion and the projection.

The dual statement holds in an abelian category with enough injectives: given injective resolutions $L \to I_L^\bullet$ and $N \to I_N^\bullet$, there is a degreewise split exact sequence $0 \to I_L^\bullet \to I_M^\bullet \to I_N^\bullet \to 0$ with $I_M^\bullet$ an injective resolution of M. It is this form that theorem 6.4.2 uses, and it follows by applying the projective statement in the opposite category.

***Proof*:**
Visually, we want to find objects and maps to fill in this commutative diagram:



where the dotted lines and the P_M^i are the goal (the name of the lemma comes from this horseshoe shaped diagram). I claim that $P_M^i = P_L^i \oplus P_N^i$ (which the reader should recall or reprove is projective) with the natural inclusion and projection maps shall work. The argument is done via induction. Since P_N^0 is projective and $M \to N$ is an epimorphism, the morphism $P_N^0 \to N$ lifts to a isomorphism m $\beta : P_N^0 \to M$. On the other hand, P_L^0 maps to M via $\alpha : P_L^0 \to L \to M$. Thus, there is a morphism:

$$\pi_M := \alpha \oplus \beta : P_L^0 \oplus P_N^0 \to M$$

diving us the diagram

$$\begin{array}{ccccccccc}
 & & \vdots & & \vdots & & \vdots & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & P_L^{-2} & \cdots\cdots\rightarrow & P_M^{-2} & \cdots\cdots\rightarrow & P_N^{-2} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & P_L^{-1} & \cdots\cdots\rightarrow & P_M^{-1} & \cdots\cdots\rightarrow & P_N^{-1} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & P_L^{0} & \longrightarrow & P_L^0 \oplus P_N^0 & \longrightarrow & P_N^0 & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & L & \longrightarrow & M & \longrightarrow & N & \longrightarrow & 0
\end{array}$$

The snake lemma and π_L being an epimorphism gives that the kernels of the vertical maps form an exact sequence, and hence we have an epimorphism from P_L^{-1} and P_N^{-1} to the respective kernels since the column of the original diagram are exact:

$$\begin{array}{ccccccccc}
 & & P_L^{-1} & & & & P_N^{-1} & & \\
 & & \downarrow{\scriptstyle \pi_L^{-1}} & & & & \downarrow{\scriptstyle \pi_N^{-1}} & & \\
0 & \longrightarrow & \ker \pi_L & \longrightarrow & \ker \pi_M & \longrightarrow & \ker \pi_N & \longrightarrow & 0
\end{array}$$

Now, repeat as before, defining $P_M^{-1} = P_L^{-1} \oplus P_N^{-1}$ and define $\pi_M^{-1} : P_L^{-1} \oplus P_N^{-1} \to \ker \pi_M \to P_L^0 \oplus P_N^0$ giving the diagram:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & P_L^{-1} & \longrightarrow & P_L^{-1} \oplus P_N^{-1} & \longrightarrow & P_N^{-1} & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \pi_L^{-1}} & & \downarrow{\scriptstyle \pi_M^{-1}} & & \downarrow{\scriptstyle \pi_N^{-1}} & & \\
0 & \longrightarrow & P_L^{0} & \longrightarrow & P_L^0 \oplus P_N^0 & \longrightarrow & P_N^0 & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \pi_L} & & \downarrow{\scriptstyle \pi_M} & & \downarrow{\scriptstyle \pi_N} & & \\
0 & \longrightarrow & L & \longrightarrow & M & \longrightarrow & N & \longrightarrow & 0
\end{array}$$

By then by the snake lemma, $P_L^{-1} \oplus P_N^{-1} \to \ker \pi_M$ is an epimorphism, and hence we get exactness at $P_L^0 \oplus P_N^0$. Continuing inductively constructs the rest of the diagram.

It is tempting to say from this result that the mapping $\mathbf{A} \to K(\mathbf{A})$ is exact, however $K(\mathbf{A})$ is not abelian and hence this is simply undefined. The above is should instead be interpreted as follows: $P_M^\bullet$ is the mapping cone of a morphism: $\rho^\bullet : P_N[-1]^\bullet \to P_L^\bullet$, giving the distinguished triangle:

$$\begin{array}{ccc}
 & P_L^\bullet & \\
{\scriptstyle +1}\nearrow & & \searrow \\
P_N^\bullet & \longleftarrow & MC(\rho)^\bullet [= P_M^\bullet]
\end{array}$$

This is as close as we can get to preserving exactness.

The horseshoe construction is functorial in the short exact sequence it resolves. Once lifts of the two outer resolutions have been chosen, compatibility with the splitting leaves only one entry of the middle lift to be built, and that entry is what the following lemma constructs.

Lemma 2.3.8: Functoriality of the Horseshoe Construction

Consider a morphism of short exact sequences in an abelian category with enough injectives,

$$\begin{array}{ccccccccc} 0 & \longrightarrow & L & \longrightarrow & M & \longrightarrow & N & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \lambda} & & \downarrow{\scriptstyle \mu} & & \downarrow{\scriptstyle \nu} & & \\ 0 & \longrightarrow & L' & \longrightarrow & M' & \longrightarrow & N' & \longrightarrow & 0 \end{array}$$

and injective resolutions $I^\bullet_L, I^\bullet_N, I^\bullet_{L'}, I^\bullet_{N'}$ of the four outer objects, and injective resolutions $I^\bullet_M$, $I^\bullet_{M'}$ of M and M' sitting in *degreewise split* short exact sequences $0 \to I^\bullet_L \to I^\bullet_M \to I^\bullet_N \to 0$ and $0 \to I^\bullet_{L'} \to I^\bullet_{M'} \to I^\bullet_{N'} \to 0$ lifting the two given rows. (This is what lemma 2.3.7 produces, but the argument uses only the splitting, which is automatic whenever the outer terms are injective.) Given lifts $\lambda^\bullet \colon I^\bullet_L \to I^\bullet_{L'}$ and $\nu^\bullet \colon I^\bullet_N \to I^\bullet_{N'}$ of λ and ν, there is a lift $\mu^\bullet \colon I^\bullet_M \to I^\bullet_{M'}$ of μ compatible with both, that is making the induced diagram of resolutions commute.

The dual statement holds in an abelian category with enough projectives: given a morphism of short exact sequences of the same shape, projective resolutions of the four outer objects, degreewise split short exact sequences $0 \to P^\bullet_L \to P^\bullet_M \to P^\bullet_N \to 0$ and $0 \to P^\bullet_{L'} \to P^\bullet_{M'} \to P^\bullet_{N'} \to 0$ lifting the two rows, and lifts $\lambda^\bullet$ of λ and $\nu^\bullet$ of ν, there is a lift $\mu^\bullet \colon P^\bullet_M \to P^\bullet_{M'}$ of μ compatible with both. It follows by applying the injective statement in the opposite category, and it is this form that theorem 2.3.11 uses.

***Proof*:**

By lemma 2.3.7 the middle resolutions are $I^q_M = I^q_L \oplus I^q_N$ and $I^q_{M'} = I^q_{L'} \oplus I^q_{N'}$ in each degree, with the displayed maps the inclusion and the projection. Compatibility with those two maps forces $\mu^\bullet$ to have the shape

$$\mu^q = \begin{pmatrix} \lambda^q & \sigma^q \\ 0 & \nu^q \end{pmatrix}$$

so the only freedom is the family $\sigma^q \colon I^q_N \to I^q_{L'}$, and the content of the lemma is that it can be chosen so that $\mu^\bullet$ is a chain map lifting μ.

Write the differential of $I^\bullet_M$ as $\begin{pmatrix} d_L & \tau \\ 0 & d_N \end{pmatrix}$ and that of $I^\bullet_{M'}$ as $\begin{pmatrix} d_{L'} & \tau' \\ 0 & d_{N'} \end{pmatrix}$, where τ and τ' are the off-diagonal terms the horseshoe construction produces. The chain-map condition $\mu^{q+1} d = d \mu^q$ reads, in the only entry that is not automatic,

$$\lambda^{q+1}\tau^q + \sigma^{q+1} d^q_N = d^q_{L'} \sigma^q + \tau'^q \nu^q$$

so, writing $\theta^q := d^q_{L'}\sigma^q + \tau'^q\nu^q - \lambda^{q+1}\tau^q$, the family σ must satisfy $\sigma^{q+1} \circ d^q_N = \theta^q$ for every q.

The base case. Write the augmentation of $I^\bullet_M$ through the splitting as $\varepsilon_M = (\varphi,\ \varepsilon_N \pi)^{\mathsf{T}}$, where

$\pi\colon M \to N$ is the given epimorphism and $\varphi\colon M \to I^0_L$ restricts to ε_L on L; similarly for M'. The morphism σ^0 is not given by the hypothesis, it has to be built. Set $\theta^{-1} := \varphi'\mu - \lambda^0\varphi\colon M \to I^0_{L'}$. On the subobject L this is $\varepsilon_{L'}\lambda - \varepsilon_{L'}\lambda = 0$, so θ^{-1} kills $L = \ker\pi$ and factors as $\bar\theta \circ \pi$ for a unique $\bar\theta\colon N \to I^0_{L'}$. Since $\varepsilon_N\colon N \to I^0_N$ is a monomorphism and $I^0_{L'}$ is injective, $\bar\theta$ extends along it to $\sigma^0\colon I^0_N \to I^0_{L'}$ with $\sigma^0\varepsilon_N\pi = \theta^{-1}$, which is the degree-zero instance of the required identity and is the only place the compatibility of μ with λ and ν is used.

The inductive step. Suppose $\sigma^{\le q}$ has been chosen. To extend θ^q along d^q_N it suffices that θ^q vanish on $\ker d^q_N = \mathrm{im} d^{q-1}_N$, and this is a direct computation. Squaring the differential of $I^\bullet_M$ gives $d_L\tau + \tau d_N = 0$, and likewise $d_{L'}\tau' + \tau' d_{N'} = 0$; using these, the inductive hypothesis $\sigma^q d^{q-1}_N = \theta^{q-1}$, the chain-map identities for λ and ν, and $d \circ d = 0$,

$$\theta^q \circ d^{q-1}_N = \left(\lambda^{q+1} d^q_L - d^q_{L'}\lambda^q\right) \circ \tau^{q-1} = 0$$

the last equality because $\lambda^\bullet$ is a chain map. For $q = 0$ the same conclusion holds with $\ker d^0_N = \mathrm{im}\varepsilon_N$ in place of $\mathrm{im} d^{-1}_N$: applying θ^0 to $\varepsilon_N\pi$ and using $d^0_L\varphi + \tau^0\varepsilon_N\pi = 0$, which is the first component of $d^0_M\varepsilon_M = 0$, together with the base case, gives $\theta^0\varepsilon_N\pi = (\lambda^1 d^0_L - d^0_{L'}\lambda^0)\varphi = 0$, and π is an epimorphism. Hence in every degree θ^q factors through $\mathrm{im} d^q_N$, and since $\mathrm{im} d^q_N \hookrightarrow I^{q+1}_N$ is a monomorphism and $I^{q+1}_{L'}$ is injective, definition 2.2.1 extends the factored morphism to $\sigma^{q+1}\colon I^{q+1}_N \to I^{q+1}_{L'}$ with $\sigma^{q+1} d^q_N = \theta^q$, as required.

The resulting $\mu^\bullet$ is a chain map by construction, is compatible with the inclusion and projection by its shape, and lifts μ because it does so in degree -1, that is on the augmentations, completing the proof.

Corollary 2.3.9: Resolution Of Projective Resolutions

Let $M^\bullet$ be a complex bounded from above in a category $\mathbf{A}$ with enough projectives. Then there exists a complex $P^b_{M^\bullet} b$:

$$\cdots \to P^\bullet_{M^{-3}} \to P^\bullet_{M^{-2}} \to P^\bullet_{M^{-1}} \to\!\!\to P^\bullet_{M^0} \to 0$$

where each $P^\bullet_{M^i}$ is a projection resolution of M^i and inducing $M^\bullet$ in cohomology. If $M^\bullet$ is exact, then $P^\bullet_{M^\bullet}$ can be chosen to be exact.

***Proof*:**
Since $M^\bullet$ is a complex, it can be broken down into two short exact sequences:

$$0 \to K^i \to M^i \to I^{i+1} \to 0 \qquad 0 \to I^i \to K^i \to H^i \to 0$$

where K^i is the kernel of $M^i \to M^{i+1}$, I^{i+1} is its image, and H^i the cohomology at M^i. Chose arbitrary projective resolutions for each I^i and H^i for all i. Then by the horseshoe lemma, we get a projective resolution for K^i that fits within the short exact sequence:

$$0 \to P^\bullet_{I^i} \to P^\bullet_{K^i} \to P^\bullet_{H^i} \to 0$$

Applying the horseshoe lemma gain, we get a short exact sequence:

$$0 \to P^\bullet_{K^i} \to P^\bullet_{M^i} \to P^\bullet_{I^{i+1}} \to 0$$

Using these, the $P^\bullet_{M^i}$ can be used to construct a cochain complex, the differentials obtains by composition. If $M^\bullet$ is exact, then $I^i = K^i$, and so the construction gives $P^\bullet_{I^i} = P^\bullet_{K^i}$, so that $P^\bullet_{M^\bullet}$ is also exact.

Lemma 2.3.10: Projective Resolution and Functors

Let $\mathbf{A}$ be an abelian category and let

$$0 \to L^\bullet \to M^\bullet \to P^\bullet \to 0$$

be an exact sequence of complexes in $\mathbf{A}$, where P^i is projective for all i. Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor between abelian categories. Then the sequence:

$$0 \to \mathscr{F}(L^\bullet) \to \mathscr{F}(M^\bullet) \to \mathscr{F}(P^\bullet) \to 0$$

is exact

Note that there is nothing about $\mathscr{F}$ being an exact functor! This can be thought of as the generalization of the properties of projective modules which preserve exactness!!

***Proof*:**

Fix i. By lemma 2.1.5 the sequence $0 \to L^i \xrightarrow{\iota} M^i \xrightarrow{\pi} P^i \to 0$ is exact. The cokernel of $0 \to L^i$ is id_{L^i}, so exactness at L^i makes $\ker\iota$ the zero morphism. A kernel is a monomorphism (lemma 1.1.3), and a zero monomorphism has a zero object as its source, so ι is a monomorphism (lemma 1.1.4). Dually, exactness at P^i makes π an epimorphism. Exactness at M^i gives $\pi\iota = 0$ and $\mathrm{coker}\iota \circ \ker\pi = 0$. The monomorphism ι is the kernel of some morphism (condition (2) of definition 1.1.6), hence the kernel of $\mathrm{coker}\iota$ (lemma 1.1.7). So $\ker\pi = \iota a$ for some a, and $\pi\iota = 0$ gives $\iota = (\ker\pi)\, b$ for some b. Then $\iota = \iota ab$ and $\ker\pi = (\ker\pi)\, ba$, and as both are monomorphisms, ab and ba are identities. So a is an isomorphism and ι is a kernel of π. Applying the exact functor $\mathrm{Hom}_{\mathbf{A}}(P^i, -)$ (definition 2.2.1) to it shows that $s \mapsto \pi s$ maps $\mathrm{Hom}_{\mathbf{A}}(P^i, M^i)$ onto $\mathrm{Hom}_{\mathbf{A}}(P^i, P^i)$, so there is $s\colon P^i \to M^i$ with $\pi s = \mathrm{id}_{P^i}$. Then $\pi(\mathrm{id}_{M^i} - s\pi) = \pi - \pi = 0$, so $\mathrm{id}_{M^i} - s\pi = \iota r$ for a unique $r\colon M^i \to L^i$. From $\iota r\iota = \iota - s\pi\iota = \iota$ and ι monic, $r\iota = \mathrm{id}_{L^i}$. So

$$\pi\iota = 0 \qquad r\iota = \mathrm{id}_{L^i} \qquad \pi s = \mathrm{id}_{P^i} \qquad \iota r + s\pi = \mathrm{id}_{M^i}$$

An additive functor preserves composites, identities, zero morphisms and sums of morphisms, so the same four identities hold for $\mathscr{F}(\iota)$, $\mathscr{F}(\pi)$, $\mathscr{F}(r)$ and $\mathscr{F}(s)$ in $\mathbf{B}$. They make $0 \to \mathscr{F}(L^i) \to \mathscr{F}(M^i) \to \mathscr{F}(P^i) \to 0$ exact: $\mathscr{F}(\iota)$ has the left inverse $\mathscr{F}(r)$, so it is a monomorphism, $\mathscr{F}(\pi)$ has the right inverse $\mathscr{F}(s)$, so it is an epimorphism, and $\mathscr{F}(\iota)$ is a kernel of $\mathscr{F}(\pi)$, because $\mathscr{F}(\pi)\mathscr{F}(\iota) = 0$ and every t with $\mathscr{F}(\pi)t = 0$ satisfies

$$t = \big(\mathscr{F}(\iota)\mathscr{F}(r) + \mathscr{F}(s)\mathscr{F}(\pi)\big)t = \mathscr{F}(\iota)\big(\mathscr{F}(r)t\big)$$

with the factorization unique as $\mathscr{F}(\iota)$ is monic. This holds in every degree i, so lemma 2.1.5 gives the exactness of the sequence of complexes, completing the proof.

Theorem 2.3.11: Long Exact Sequence From Derived Functor

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be an additive functor of abelian categories, and assume $\mathbf{A}$ has enough projective. Then every short exact sequence in $\mathbf{A}$

$$0 \to L \xrightarrow{\alpha} M \xrightarrow{\beta} N \to 0$$

[covariantly] functorially induces a long exact sequence in $\mathbf{B}$, whose unlabelled horizontal arrows are $L_i\mathscr{F}(\alpha)$ and $L_i\mathscr{F}(\beta)$:

$$\begin{array}{l} \cdots \longrightarrow L_2\mathscr{F}(L) \longrightarrow L_2\mathscr{F}(M) \longrightarrow L_2\mathscr{F}(N) \\ \qquad\qquad \xrightarrow{\delta_2} \\ \quad L_1\mathscr{F}(L) \longrightarrow L_1\mathscr{F}(M) \longrightarrow L_1\mathscr{F}(N) \\ \qquad\qquad \xrightarrow{\delta_1} \\ \quad L_0\mathscr{F}(L) \longrightarrow L_0\mathscr{F}(M) \longrightarrow L_0\mathscr{F}(N) \longrightarrow 0 \end{array}$$

The assignment is natural in the short exact sequence. Given a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & L & \longrightarrow & M & \longrightarrow & N & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \lambda} & & \downarrow{\scriptstyle \mu} & & \downarrow{\scriptstyle \nu} & & \\ 0 & \longrightarrow & L' & \longrightarrow & M' & \longrightarrow & N' & \longrightarrow & 0 \end{array}$$

the two long exact sequences form a commutative ladder whose vertical morphisms are $L_i\mathscr{F}(\lambda)$, $L_i\mathscr{F}(\mu)$ and $L_i\mathscr{F}(\nu)$. Writing δ_i' for the connecting morphisms of the second row, the squares carrying the connecting morphisms read $\delta_i' \circ L_i\mathscr{F}(\nu) = L_{i-1}\mathscr{F}(\lambda) \circ \delta_i$.

***Proof*:**
Choose projective resolutions $P_L^\bullet$ and $P_N^\bullet$. By lemma 2.3.7 the given short exact sequence lifts to a degreewise split short exact sequence $0 \to P_L^\bullet \to P_M^\bullet \to P_N^\bullet \to 0$ of projective resolutions, which by lemma 2.3.10 creates a short exact sequence:

$$0 \to \mathscr{F}(P_L^\bullet) \to \mathscr{F}(P_M^\bullet) \to \mathscr{F}(P_N^\bullet) \to 0$$

Since the cohomology of these complexes is precisely the left-derived functors of $\mathscr{F}$ (definition 2.3.5), the long exact cohomology sequence is given by theorem 2.1.20.

Naturality. Resolve the second row the same way, as $0 \to P_{L'}^\bullet \to P_{M'}^\bullet \to P_{N'}^\bullet \to 0$. By proposition 2.2.15 the morphisms λ and ν lift to morphisms of resolutions $\lambda^\bullet \colon P_L^\bullet \to P_{L'}^\bullet$ and $\nu^\bullet \colon P_N^\bullet \to P_{N'}^\bullet$, and the projective form of lemma 2.3.8 then produces a lift $\mu^\bullet \colon P_M^\bullet \to P_{M'}^\bullet$ of μ compatible with the inclusion and the projection. The three lifts form a morphism of short exact sequences of complexes, and $\mathscr{F}$, being a functor, carries the two commuting squares to

commuting squares:

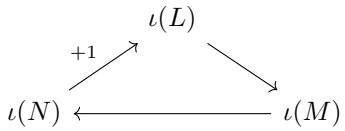
$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathscr{F}(P_L^\bullet) & \longrightarrow & \mathscr{F}(P_M^\bullet) & \longrightarrow & \mathscr{F}(P_N^\bullet) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \mathscr{F}(\lambda^\bullet)} & & \downarrow{\scriptstyle \mathscr{F}(\mu^\bullet)} & & \downarrow{\scriptstyle \mathscr{F}(\nu^\bullet)} & & \\ 0 & \longrightarrow & \mathscr{F}(P_{L'}^\bullet) & \longrightarrow & \mathscr{F}(P_{M'}^\bullet) & \longrightarrow & \mathscr{F}(P_{N'}^\bullet) & \longrightarrow & 0 \end{array}$$

Both rows are exact by lemma 2.3.10, so the naturality clause of theorem 2.1.20, read through $H_n = H^{-n}$, gives a commutative ladder between the two long exact sequences, the squares carrying the connecting morphisms included.

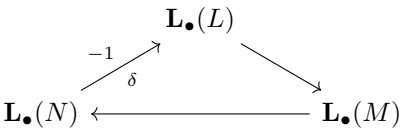
The vertical morphisms of that ladder are the asserted ones. By definition 2.3.5 the value of $L_i\mathscr{F}$ on a morphism is H^{-i} of the image under $\mathrm{Ch}(\mathscr{F})$ of a lift of that morphism to projective resolutions, and the value does not depend on the lift chosen: two lifts are homotopic by proposition 2.2.15, $\mathrm{Ch}(\mathscr{F})$ carries homotopic morphisms to homotopic morphisms by lemma 2.1.34, and homotopic morphisms agree on cohomology by proposition 2.1.31. Since $\lambda^\bullet$, $\mu^\bullet$ and $\nu^\bullet$ are such lifts, the vertical morphisms are $L_i\mathscr{F}(\lambda)$, $L_i\mathscr{F}(\mu)$ and $L_i\mathscr{F}(\nu)$.

The same argument identifies the horizontal arrows, which is what makes the sequence a long exact sequence *of the derived functors of* α *and* β rather than of unnamed induced maps. The horseshoe sequence $0 \to P_L^\bullet \to P_M^\bullet \to P_N^\bullet \to 0$ of lemma 2.3.7 induces the original short exact sequence on H^0, so its degreewise inclusion and projection are lifts of α and β to the chosen projective resolutions. Applying $\mathrm{Ch}(\mathscr{F})$ and H^{-i} to them therefore gives $L_i\mathscr{F}(\alpha)$ and $L_i\mathscr{F}(\beta)$, by the same independence of the lift, completing the proof.

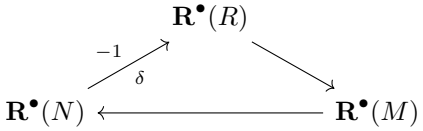
The change from short exact sequence to long exact sequence can be unified under the notion of triangles, namely this theorem can be seen as the triangle:



inducing the triangle



where the vertices of this triangle are complexes in $C^{\leq 0}(\mathbf{B})$. Naturally, the dual situation happens for right derived functors for categories that have enough injectives, iin which case the above triangle induces a triangle:



where the vertices of this triangle are complexes in $C^{\geq 0}(\mathbf{B})$.

The dual statement is what the right-derived functors satisfy, and it is recorded separately because Ext in its second variable is computed that way.

Corollary 2.3.12: Long Exact Sequence From a Right-Derived Functor

Let $\mathscr{F}\colon \mathbf{A} \to \mathbf{B}$ be an additive functor of abelian categories with $\mathbf{A}$ having enough injectives. Then every short exact sequence $0 \to L \xrightarrow{\alpha} M \xrightarrow{\beta} N \to 0$ in $\mathbf{A}$ induces a long exact sequence

$$\begin{aligned} 0 &\to R^0\mathscr{F}(L) \\ &\to R^0\mathscr{F}(M) \\ &\to R^0\mathscr{F}(N) \xrightarrow{\delta^0} R^1\mathscr{F}(L) \\ &\to R^1\mathscr{F}(M) \\ &\to R^1\mathscr{F}(N) \xrightarrow{\delta^1} R^2\mathscr{F}(L) \\ &\to \cdots \end{aligned}$$

whose unlabelled arrows are $R^i\mathscr{F}(\alpha)$ and $R^i\mathscr{F}(\beta)$, and which is functorial in the short exact sequence: a morphism from it to a second short exact sequence $0 \to L' \to M' \to N' \to 0$, with components $\lambda\colon L \to L'$, $\mu\colon M \to M'$ and $\nu\colon N \to N'$, induces a commutative ladder between the two long exact sequences whose vertical morphisms are $R^i\mathscr{F}(\lambda)$, $R^i\mathscr{F}(\mu)$ and $R^i\mathscr{F}(\nu)$, the squares carrying the connecting morphisms included.

***Proof*:**
Apply theorem 2.3.11 in the opposite categories. An injective object of $\mathbf{A}$ is a projective object of $\mathbf{A}^{op}$, so $\mathbf{A}$ having enough injectives is $\mathbf{A}^{op}$ having enough projectives; the short exact sequence reverses to $0 \to N \to M \to L \to 0$ in $\mathbf{A}^{op}$; and the left-derived functors of $\mathscr{F}^{op}$ are the right-derived functors of $\mathscr{F}$ with the arrows reversed, by definition 2.3.1. Transporting the conclusion back gives the displayed sequence.

The naturality clause transports with it. A morphism of short exact sequences in $\mathbf{A}$ with components λ, μ, ν is, in $\mathbf{A}^{op}$, a morphism from the reversed second sequence to the reversed first with components $\nu^{op}, \mu^{op}, \lambda^{op}$, so theorem 2.3.11 produces a commutative ladder in $\mathbf{B}^{op}$ whose vertical morphisms are the left-derived functors of $\mathscr{F}^{op}$ applied to those three. Read in $\mathbf{B}$, that ladder is the asserted one, completing the proof.

These consideration were for general additive functors. In most cases we saw, there is usually a left-exact or right-exact functor (indeed, the nature of the left-derived and right-derived series almost force us to choose "a side"). If a functor is given to be either left- or right-exact, then it can be directly recovered from their derived functors, as we shall now show.

Proposition 2.3.13: Right-Exactness And Derived Functors

Let $\mathbf{A}$ be an abelian category with enough projectives and $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be a right-exact additive functor. Then

1. $\mathbf{L}_i\mathscr{F} = 0$ for $i < 0$
2. $L_0\mathscr{F}$ is naturally isomorphic to $\mathscr{F}$

The dual result holds for the category with enough injectives. Do note that the *right*-exact functor is associated to the *left* derived functors, and vice-versa for left-exact functors which would have the associated *right* derived functors. This is because the derived functors are "fixing" the exactness on the other side.

***Proof*:**

Since projective resolution $P^\bullet$ of an object M in $\mathbf{A}$ is in $C^{\leq 0}(\mathbf{A})$, $C(\mathscr{F})(P^\bullet)$ is 0 in positive degree, and hence so is its cohomology. Since $H_i = H^{-i}$, the first claim is immediate.

For the second, an exact sequence:

$$P^{-1} \to P^0 \to M \to 0$$

will map to an exact sequence by $\mathscr{F}$ (due to right-exactness)

$$\mathscr{F}(P^{-1}) \to \mathscr{F}(P^0) \to \mathscr{F}(M) \to 0$$

This induces an isomorphism: $\nu_M : \mathbf{L}_0\mathscr{F}(M) = H^0(C(\mathscr{F})(P^\bullet)) \xrightarrow{\sim} \mathscr{F}(M)$. If $\varphi : M_0 \to M_1$ is a morphism, then there exists a morphism of projective resolutions $\alpha^\bullet : P_0^\bullet \to P_1^\bullet$:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & P_0^{-1} & \longrightarrow & P_0^0 & \longrightarrow & M_0 & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha^{-1}} & & \downarrow{\scriptstyle \alpha^0} & & \downarrow{\scriptstyle \varphi} & & \\ \cdots & \longrightarrow & P_1^{-1} & \longrightarrow & P_1^0 & \longrightarrow & M_1 & \longrightarrow & 0 \end{array}$$

induced by φ where $H^0(\alpha^\bullet)$ agrees with φ. Applying $\mathscr{F}$ and truncating to $\alpha^\bullet$, we get the commutative diagram:

$$\begin{array}{ccccccc} \mathscr{F}(P_0^{-1}) & \longrightarrow & \mathscr{F}(P_0^0) & \longrightarrow & \mathscr{F}(M_0) & \longrightarrow & 0 \\ \downarrow{\scriptstyle \mathscr{F}(\alpha^{-1})} & & \downarrow{\scriptstyle \mathscr{F}(\alpha^0)} & & \downarrow{\scriptstyle \mathscr{F}(\varphi)} & & \\ \mathscr{F}(P_1^{-1}) & \longrightarrow & \mathscr{F}(P_1^0) & \longrightarrow & \mathscr{F}(M_1) & \longrightarrow & 0 \end{array}$$

with exact rows since $\mathscr{F}$ is right-exact. But then the following diagram commutes:

$$\begin{array}{ccc} \mathbf{L}_0\mathscr{F}(M_0) & \xrightarrow{\mathbf{L}_0\mathscr{F}(\varphi)} & \mathbf{L}_0\mathscr{F}(M_1) \\ \downarrow{\scriptstyle \nu_{M_0}} & & \downarrow{\scriptstyle \nu_{M_1}} \\ \mathscr{F}(M_0) & \xrightarrow[\mathscr{F}(\varphi)]{} & \mathscr{F}(M_1) \end{array}$$

The square commutes because $\mathbf{L}_0\mathscr{F}(\varphi) = H^0(C(\mathscr{F})(\alpha^\bullet))$ is the map that $\mathscr{F}(\alpha^0)$ induces on the cokernels of the maps $\mathscr{F}(P_i^{-1}) \to \mathscr{F}(P_i^0)$, each ν_{M_i} is the map that $\mathscr{F}(P_i^0) \to \mathscr{F}(M_i)$ induces on the same cokernel, and the right-hand square of the previous diagram commutes. This completes the proof.

This finally gives the desired recovery of exactness promised way back. Given a short exact sequence

$$0 \to L \to M \to N \to 0$$

then applying a right-exact functor

$$\mathscr{F}(L) \to \mathscr{F}(M) \to \mathscr{F}(N) \to 0$$

will give an exact sequence, which can be stretched out to form a long exact sequence using the derived functors:

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & L_2\mathscr{F}(L) & \longrightarrow & L_2\mathscr{F}(M) & \longrightarrow & L_2\mathscr{F}(N) \\
& \xrightarrow{\delta_2} & L_1\mathscr{F}(L) & \longrightarrow & L_1\mathscr{F}(M) & \longrightarrow & L_1\mathscr{F}(N) \\
& \xrightarrow{\delta_1} & \mathscr{F}(L) & \longrightarrow & \mathscr{F}(M) & \longrightarrow & \mathscr{F}(N) \longrightarrow 0
\end{array}$$

In this way, $\mathbf{L}_1\mathscr{F}$ measures the extent to which $\mathscr{F}$ fails to be left-exact, and each progressive $\mathbf{L}_i\mathscr{F}$ gives further measurement, further information, about this failure. When in a geometrical setting, this failure is often thought of as some form of obstruction (famously, in algebraic topology, this obstruction is interpreted as "'holes", in K-theory).

Similarly, $R^0\mathscr{F}$ is naturally isomorphic to $\mathscr{F}$ if $\mathscr{F}$ is left-exact and the right-derived functors fit into the diagram:

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathscr{F}(L) & \longrightarrow & \mathscr{F}(M) & \longrightarrow & \mathscr{F}(N) \\
& \xrightarrow{\delta^0} & R^1\mathscr{F}(L) & \longrightarrow & R^1\mathscr{F}(M) & \longrightarrow & R^1\mathscr{F}(N) \\
& \xrightarrow{\delta^1} & R^2\mathscr{F}(L) & \longrightarrow & R^2\mathscr{F}(M) & \longrightarrow & R^2\mathscr{F}(N) \longrightarrow \cdots
\end{array}$$

Thus, we have successfully captured the cohomological information that a functor may provide!

Acyclic Functors

There still remain some questions to be answered, one that was mentioned was that instead of using a projective resolution for Tor, a flat resolution may be used instead and give the same answer.

Definition 2.3.14: Acyclic Functor

Let $\mathscr{F}$ be an additive functor. An object M of $\mathbf{A}$ is *$\mathscr{F}$-acyclic* (with respect to left-derived functors) if $\mathbf{L}_i\mathscr{F}(M) = 0$ for all $i \neq 0$.

A left exact functor such as $\mathrm{Hom}_R(N, -)$ is derived through injective resolutions, and the objects on which its right-derived functors vanish can take the place of injectives in that computation (theorem 2.3.18 below).

Definition 2.3.15: Acyclic for a Left-Exact Functor

Let $\mathscr{F}: \mathbf{A} \to \mathbf{B}$ be a left exact additive functor between abelian categories, with $\mathbf{A}$ having enough injectives. An object M of $\mathbf{A}$ is *$\mathscr{F}$-acyclic* if $R^i\mathscr{F}(M) = 0$ for all $i \geq 1$. Injective objects are $\mathscr{F}$-acyclic for every left exact $\mathscr{F}$, since an injective I is its own injective resolution, with I in degree 0 and zero elsewhere.

Projectives are acyclic for every right-exact functor in the sense of definition 2.3.14, and injectives for every left-exact functor in the sense of definition 2.3.15, but such a functor may admit other acyclic objects.

Example 2.3.16: Functor-acyclic Objects

Let R be a commutative ring. Recall that an R-module M is *flat* if $- \otimes_R M$ is exact (equivalently, by symmetry of $\otimes$ that $M \otimes_R -$ is exact). Then flat modules are *acyclic* with respect to tensor products, namely if N is flat:

$$\mathrm{Tor}_R^i(M, N) = 0 \qquad \forall i \neq 0,\ \forall M$$

and symmetrically if M is flat. Thus, flat modules are $(- \otimes_R N)$-acyclic for all modules N.

What shall be built up to now is that $\mathscr{F}$-acyclic resolutions can be used to compute $\mathbf{L}_i\mathscr{F}$.

Theorem 2.3.17: Derived Functors And Acyclic Functors

Let $\mathscr{F} : \mathbf{A} \to \mathbf{B}$ be a right-exact functor of abelian categories where $\mathbf{A}$ has enough projectives. Let $A^\bullet$ be a resolution of M where each A^i is $\mathscr{F}$-acyclic. Then for all i:

$$\mathbf{L}_i\mathscr{F}(M) \cong H^{-i}(\mathrm{Ch}(\mathscr{F})(A^\bullet))$$

Note that if each A^i is projective, the statement follows by definition.

***Proof*:**

The 0th and -1th position will be computed directly, and each successive position comes from induction. Take the exact sequence

$$A^{-1} \to A^0 \to M \to 0$$

which creates another exact sequence in $\mathbf{B}$ after applying the right-exact functor $\mathscr{F}$:

$$\mathscr{F}(A^{-1}) \to \mathscr{F}(A^0) \to \mathscr{F}(M) \to 0$$

Then:

$$H^0(\mathrm{Ch}(\mathscr{F})(A^\bullet)) \cong \mathscr{F}(M) \cong \mathbf{L}_0\mathscr{F}(M)$$

Now, if K is the kernel of $A^0 \to M$, then we have a short exact sequence:

$$0 \to K \to A^0 \to M \to 0$$

Thus applying the left-derived functors we form the long exact sequence:

$$\mathbf{L}_1\mathscr{F}(A^0) \to \mathbf{L}_1\mathscr{F}(M) \to \mathscr{F}(K) \to \mathscr{F}(A^0) \to \mathscr{F}(M) \to 0$$

Since A^0 is acyclic, $L_1\mathscr{F}(A^0) = 0$, hence by exactness

$$\mathbf{L}_1\mathscr{F}(M) \cong \ker(\mathscr{F}(K) \to \mathscr{F}(A^0)) \cong H^{-1}(\mathrm{Ch}(\mathscr{F})(A^\bullet))$$

where the last isomorphism holds because K is the cokernel of $A^{-2} \to A^{-1}$, so that right exactness makes $\mathscr{F}(K)$ the cokernel of $\mathscr{F}(A^{-2}) \to \mathscr{F}(A^{-1})$, and $\mathscr{F}(A^{-1}) \to \mathscr{F}(A^0)$ factors through it. For the rest, we do induction on $i \geq 2$, assuming the statement for every object of $\mathbf{A}$, every $\mathscr{F}$-acyclic resolution of it, and every index below i. Truncating and shifting $A^\bullet$ so that A^{-1} is in the 0 degree:

$$\cdots \to A^{-3} \to A^{-2} \to A^{-1} \to 0 \to \cdots$$

we get an $\mathscr{F}$-acyclic resolution of K, and so by the induction hypothesis $\mathbf{L}_{i-1}\mathscr{F}(K)$ can be computed by applying $\mathrm{Ch}(\mathscr{F})$ to the complex and taking cohomology, namely:

$$\mathbf{L}_{i-1}\mathscr{F}(K) \cong H^{-i}(\mathrm{Ch}(\mathscr{F})(A^\bullet))$$

On the other hand, the other terms in the long exact sequence obtained using the left derived functors gives:

$$\cdots \to \mathbf{L}_i\mathscr{F}(A^0) \to L_i\mathscr{F}(M) \to \mathbf{L}_{i-1}\mathscr{F}(K) \to \mathbf{L}_{i-1}\mathscr{F}(A^0) \to \cdots$$

since A^0 is $\mathscr{F}$-acyclic and $i > 1$, we get:

$$\mathbf{L}_i\mathscr{F}(M) \cong \mathbf{L}_{i-1}\mathscr{F}(K)$$

giving the desired result.

Thus, for the Tor functor, a flat resolution computes the left-derived functors. Since $\mathscr{F}$-acyclic objects suffice in the computation, one can imagine that there is no need for $\mathbf{A}$ to have enough projectives to compute it. This is indeed the case, given that $\mathbf{A}$ has enough $\mathscr{F}$-acyclic objects (this is an exercise in Aluffi, 8.14, with bifunctors).

Sheaf cohomology is the right-derived functor of global sections, and it is computed in practice from resolutions by flasque sheaves, or by fine sheaves on a paracompact space, which are acyclic for global sections and need not be injective. The dimension shift of the last proof, run with injective resolutions, justifies this for every left exact functor.

Theorem 2.3.18: Acyclic Resolutions Compute Right-Derived Functors

Let $\mathscr{F}\colon \mathbf{A} \to \mathbf{B}$ be a left exact additive functor between abelian categories, with $\mathbf{A}$ having enough injectives. Let M be an object of $\mathbf{A}$ and

$$0 \to M \xrightarrow{\eta} A^0 \xrightarrow{d^0} A^1 \xrightarrow{d^1} A^2 \to \cdots$$

an exact sequence in which every A^i is $\mathscr{F}$-acyclic, and write $A^\bullet$ for the complex $A^0 \to A^1 \to \cdots$ with A^0 in degree 0. Then for every $i \geq 0$

$$R^i\mathscr{F}(M) \cong H^i\big(\mathscr{F}(A^\bullet)\big)$$

The isomorphism is natural: for a morphism of two such sequences, with component $\varphi\colon M \to M'$ on the resolved objects, it carries $R^i\mathscr{F}(\varphi)$ to the map induced on $H^i(\mathscr{F}(A^\bullet))$.

***Proof*:**

Step 1: Degree zero. Left exactness of $\mathscr{F}$, applied to the exact sequence $0 \to M \to A^0 \to A^1$, makes $0 \to \mathscr{F}(M) \to \mathscr{F}(A^0) \to \mathscr{F}(A^1)$ exact, so $\mathscr{F}(\eta)$ identifies $\mathscr{F}(M)$ with $H^0(\mathscr{F}(A^\bullet))$. There is a natural isomorphism $R^0\mathscr{F} \cong \mathscr{F}$, by proposition 2.3.13 applied to $\mathscr{F}^{op}$ as in the proof of corollary 2.3.12. This is the case $i = 0$.

Step 2: The dimension shift. Let K be the cokernel of η, which d^0 identifies with $\ker(d^1)$ by exactness. Corollary 2.3.12, applied to $0 \to M \to A^0 \to K \to 0$, gives an exact sequence

$$\mathscr{F}(A^0) \to \mathscr{F}(K) \xrightarrow{\delta^0} R^1\mathscr{F}(M) \to R^1\mathscr{F}(A^0) = 0$$

after the identifications $R^0\mathscr{F} \cong \mathscr{F}$ of Step 1. For $i \geq 2$ the same long exact sequence places $\delta^{i-1}\colon R^{i-1}\mathscr{F}(K) \to R^i\mathscr{F}(M)$ between $R^{i-1}\mathscr{F}(A^0) = 0$ and $R^i\mathscr{F}(A^0) = 0$, so δ^{i-1} is an isomorphism.

Step 3: Degree one. The sequence $0 \to K \to A^1 \to A^2$ is exact, so left exactness identifies $\mathscr{F}(K)$ with $\ker(\mathscr{F}(d^1))$, and $\mathscr{F}(d^0)$ factors as $\mathscr{F}(A^0) \to \mathscr{F}(K) \to \mathscr{F}(A^1)$. Hence the cokernel of $\mathscr{F}(A^0) \to \mathscr{F}(K)$ is $\ker(\mathscr{F}(d^1))/\mathrm{im}(\mathscr{F}(d^0)) = H^1(\mathscr{F}(A^\bullet))$, and Step 2 gives $R^1\mathscr{F}(M) \cong H^1(\mathscr{F}(A^\bullet))$.

Step 4: Induction. The sequence $0 \to K \to A^1 \to A^2 \to \cdots$ is an exact sequence of the same kind for K, with the complex $A^{\bullet+1}$ whose degree-j term is A^{j+1}. Let $i \geq 2$ and assume the theorem in degree $i-1$ for every object and every such sequence. Then $R^{i-1}\mathscr{F}(K) \cong H^{i-1}(\mathscr{F}(A^{\bullet+1})) = H^i(\mathscr{F}(A^\bullet))$, the equality because, for $i - 1 \geq 1$, both are computed from the maps $\mathscr{F}(A^{i-1}) \to \mathscr{F}(A^i) \to \mathscr{F}(A^{i+1})$. Composing with the isomorphism δ^{i-1} of Step 2 gives $R^i\mathscr{F}(M) \cong H^i(\mathscr{F}(A^\bullet))$.

Step 5: Naturality. A morphism of two such sequences induces a morphism between the short exact sequences of Step 2 and a morphism between the sequences for K and K'. The identifications of Step 1, the connecting morphisms, by the functoriality clause of corollary 2.3.12, and the identifications of kernels and cokernels in Step 3 all commute with the induced maps. So the induction carries naturality from each degree to the next, completing the proof.

2.4 δ-Functors and Grothendieck's Effaceability Criterion

The right-derived functors of section 2.3 are built from an injective resolution, and the resolution is a choice. The construction does not depend on that choice, but two constructions that look nothing alike cannot be compared by inspecting resolutions, and comparisons of that kind are what the theory is used for. Grothendieck's answer, in the 1957 Tôhoku paper, is to axiomatize what the derived functors do rather than how they are made. What they do is corollary 2.3.12: a family of functors $(T^i)_{i\geq 0}$ turns each short exact sequence into a long exact sequence, compatibly with morphisms of short exact sequences. A family with that structure is a δ-functor.

The two axioms do not determine the higher terms from the degree-zero one, as example 2.4.2 below shows. What determines them is a universal property inside the category of δ-functors, and a universal δ-functor is determined by its degree-zero term alone. That property cannot be checked by inspection, and this is where the criterion of the section enters: a δ-functor whose positive-degree terms are killed by embedding each object into a larger one is universal. The hypothesis, effaceability, asks for one embedding per object, and it is what makes two theories agreeing in degree zero isomorphic in every degree.

Definition 2.4.1: δ-Functor

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. A *δ-functor* T from $\mathcal{A}$ to $\mathcal{B}$ consists of a family $(T^i)_{i\geq 0}$ of additive functors $T^i\colon \mathcal{A}\to\mathcal{B}$ together with, for every short exact sequence

$$\xi\colon \qquad 0\to A'\xrightarrow{\alpha} A\xrightarrow{\beta} A''\to 0$$

in $\mathcal{A}$ and every $i\geq 0$, a morphism $\delta^i_\xi\colon T^i(A'')\to T^{i+1}(A')$ in $\mathcal{B}$, subject to two conditions.

1. (Exactness.) For every such ξ the sequence

$$\begin{aligned}0\to T^0(A')\to T^0(A)\to T^0(A'')\xrightarrow{\delta^0_\xi} T^1(A')\to T^1(A)\to T^1(A'')\\ \xrightarrow{\delta^1_\xi} T^2(A')\to T^2(A)\to\cdots\end{aligned}$$

   is exact, the unlabelled arrows being $T^i(\alpha)$ and $T^i(\beta)$.

2. (Naturality of δ.) For every commutative diagram with exact rows

$$\begin{array}{cccccccccc}\xi\colon & 0 & \longrightarrow & A' & \xrightarrow{\alpha} & A & \xrightarrow{\beta} & A'' & \longrightarrow & 0\\ & & & \downarrow{\scriptstyle g'} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle g''} & & \\ \eta\colon & 0 & \longrightarrow & B' & \longrightarrow & B & \longrightarrow & B'' & \longrightarrow & 0\end{array}$$

   and every $i\geq 0$, the square

$$\begin{array}{ccc}T^i(A'') & \xrightarrow{\delta^i_\xi} & T^{i+1}(A')\\ \downarrow{\scriptstyle T^i(g'')} & & \downarrow{\scriptstyle T^{i+1}(g')}\\ T^i(B'') & \xrightarrow[\delta^i_\eta]{} & T^{i+1}(B')\end{array}$$

   commutes.

Condition (1) has an immediate consequence: its opening segment $0\to T^0(A')\to T^0(A)\to T^0(A'')$ says that T^0 is left exact, in the sense fixed by [8, Left Exactness of $\mathrm{Hom}_R(D,-)$](2): a short exact sequence is carried to an exact sequence of that three-term shape. A δ-functor is therefore a left exact functor together with a chosen way of continuing its exact sequence to the right, and the question the section answers is how much freedom that continuation has. The indexing above is the cohomological one, matching corollary 2.3.12. The homological variant, in which the connecting morphisms lower the index, is this notion transported to the opposite categories, and only the cohomological form is written out below.

Example 2.4.2: The Constant δ-Functor on Ab

Take $\mathcal{A}=\mathcal{B}=\mathbf{Ab}$, set $T^i=\mathrm{id}_{\mathbf{Ab}}$ for every $i\geq 0$, and let every δ^i_ξ be the zero morphism. This is a δ-functor. Each T^i is additive, and for $\xi\colon 0\to A'\to A\to A''\to 0$ the sequence of condition (1) reads

$$0\to A'\to A\to A''\xrightarrow{0} A'\to A\to A''\xrightarrow{0} A'\to\cdots$$

Exactness at the first A' is injectivity of $A' \to A$, and at each copy of A it is exactness of ξ. At each copy of A'' the image of $A \to A''$ is all of A'' by surjectivity, and the kernel of the zero map out of A'' is also all of A''. At each copy of A' after the first, the image of the zero map is 0, which is the kernel of the monomorphism $A' \to A$. Condition (2) holds because both composites in its square are zero.

The example uses no resolutions and no derived functors, and it shows that conditions (1) and (2) do not determine $(T^i)_{i>0}$ from T^0. Indeed the family with $T^0 = \mathrm{id}_{\mathbf{Ab}}$, with $T^i = 0$ for $i > 0$ and with all connecting morphisms zero, is a second δ-functor on $\mathbf{Ab}$ with the same degree-zero term: its long exact sequence is $0 \to A' \to A \to A'' \to 0 \to 0 \to \cdots$, exact at A'' because $A \to A''$ is an epimorphism. Pinning the higher terms down needs a comparison between δ-functors, so morphisms come first.

Definition 2.4.3: Morphism of δ-Functors

Let S and T be δ-functors from $\mathcal{A}$ to $\mathcal{B}$, with connecting morphisms written δ_S and δ_T. A *morphism of δ-functors* $f\colon T \to S$ is a family $(f^i)_{i\geq 0}$ of natural transformations $f^i\colon T^i \Rightarrow S^i$ such that for every short exact sequence $\xi\colon 0 \to A' \to A \to A'' \to 0$ and every $i \geq 0$ the square

$$\begin{array}{ccc} T^i(A'') & \xrightarrow{\delta^i_{T,\xi}} & T^{i+1}(A') \\ \downarrow{\scriptstyle f^i_{A''}} & & \downarrow{\scriptstyle f^{i+1}_{A'}} \\ S^i(A'') & \xrightarrow[\delta^i_{S,\xi}]{} & S^{i+1}(A') \end{array}$$

commutes.

Degreewise composition of two morphisms of δ-functors is again one, since the two commuting squares stack, and the identity natural transformations form a morphism, so the δ-functors from $\mathcal{A}$ to $\mathcal{B}$ form a category. The universal property below is initiality in that category among the δ-functors with a prescribed degree-zero term.

Definition 2.4.4: Universal δ-Functor

A δ-functor T from $\mathcal{A}$ to $\mathcal{B}$ is *universal* if for every δ-functor S from $\mathcal{A}$ to $\mathcal{B}$ and every natural transformation $\varphi\colon T^0 \Rightarrow S^0$ there is exactly one morphism of δ-functors $f\colon T \to S$ with $f^0 = \varphi$.

The quantifier runs over all δ-functors S and all φ, and the morphism produced goes out of T. So universality of T says that any competing family carrying long exact sequences receives a canonical comparison from T, determined by its degree-zero component alone.

Example 2.4.2 is not universal, and the failure is in uniqueness rather than existence. Take $S = T$ there and $\varphi = \mathrm{id}_{T^0}$. The identity morphism of T extends φ, and so does the family that is the identity in degree zero and multiplication by 2 in every positive degree: multiplication by 2 is a natural transformation $\mathrm{id}_{\mathbf{Ab}} \Rightarrow \mathrm{id}_{\mathbf{Ab}}$ because every homomorphism of abelian groups commutes with it, and the square of definition 2.4.3 commutes because all the connecting morphisms are zero. The two extensions differ on $T^1(\mathbb{Z}) = \mathbb{Z}$.

Definition 2.4.5: Effaceable Functor

An additive functor $T\colon \mathcal{A} \to \mathcal{B}$ is *effaceable* if for every object A of $\mathcal{A}$ there is a monomorphism $u\colon A \hookrightarrow M$ in $\mathcal{A}$ with $T(u) = 0$. Such a u is said to *efface* T at A. A δ-functor $(T^i)_{i\geq 0}$ is *effaceable in positive degrees* if T^i is effaceable for every $i > 0$.

The condition asks that the morphism $T^i(u)$ vanish, not that the object $T^i(M)$ vanish. The second is strictly stronger, and it is what happens whenever M can be taken injective, but the proof below uses only the first. Nothing in the definition asks for a functorial choice of u either: one monomorphism per object is enough, and different objects may be effaced by embeddings with no relation to each other.

The criterion has a hypothesis that is checked object by object and a conclusion about every δ-functor at once.

Theorem 2.4.6: An Effaceable δ-Functor Is Universal

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories and let $T = (T^i)_{i\geq 0}$ be a δ-functor from $\mathcal{A}$ to $\mathcal{B}$ that is effaceable in positive degrees. Then T is universal.

***Proof*:**
Let S be a δ-functor from $\mathcal{A}$ to $\mathcal{B}$ and $\varphi\colon T^0 \Rightarrow S^0$ a natural transformation. Set $f^0 := \varphi$ and construct f^{n+1} from f^n by induction on $n \geq 0$, the induction hypothesis being that $f^n\colon T^n \Rightarrow S^n$ is a natural transformation. Two facts about abelian categories are used repeatedly, both read off theorem 1.1.9: applied to an epimorphism ϵ, that theorem factors ϵ through $\operatorname{coker}(\ker \epsilon)$ by an isomorphism, so ϵ is a cokernel of its kernel; applied to a monomorphism μ, it factors μ through $\ker(\operatorname{coker}\mu)$ by an isomorphism, so μ is a kernel of its cokernel.

Step 1: Effacing short exact sequences. Fix $n \geq 0$ and an object A of $\mathcal{A}$, and let $u\colon A \to M$ be a monomorphism with $T^{n+1}(u) = 0$, which exists because T^{n+1} is effaceable. Put $N := \operatorname{coker} u$, so that

$$\xi_u\colon \qquad 0 \to A \xrightarrow{u} M \xrightarrow{v} N \to 0$$

is a short exact sequence. The relevant segment of the long exact sequence of T at ξ_u is

$$T^n(M) \xrightarrow{T^n(v)} T^n(N) \xrightarrow{\delta^n_{T,\xi_u}} T^{n+1}(A) \xrightarrow{T^{n+1}(u)} T^{n+1}(M)$$

The last morphism is zero, so exactness at $T^{n+1}(A)$ makes the image of δ^n_{T,ξ_u} all of $T^{n+1}(A)$, that is, δ^n_{T,ξ_u} is an epimorphism. Exactness at $T^n(N)$ identifies its kernel with the image of $T^n(v)$.

Step 2: The construction. Keep u and ξ_u, and consider $\delta^n_{S,\xi_u} \circ f^n_N\colon T^n(N) \to S^{n+1}(A)$. Naturality of f^n applied to v gives $f^n_N \circ T^n(v) = S^n(v) \circ f^n_M$, and $\delta^n_{S,\xi_u} \circ S^n(v) = 0$ because those two morphisms are consecutive in the long exact sequence of S at ξ_u. Hence $\delta^n_{S,\xi_u} \circ f^n_N \circ T^n(v) = 0$. Write $k\colon K \to T^n(N)$ for the kernel of δ^n_{T,ξ_u}, which by Step 1 is the image of $T^n(v)$. The induced morphism $T^n(M) \to K$ is an epimorphism (proposition 1.1.12), so the displayed vanishing gives $\delta^n_{S,\xi_u} \circ f^n_N \circ k = 0$. As δ^n_{T,ξ_u} is an epimorphism it is a cokernel of k, so there is exactly one

morphism

$$f^{n+1}_{A,u}\colon T^{n+1}(A) \longrightarrow S^{n+1}(A) \qquad \text{with} \qquad f^{n+1}_{A,u} \circ \delta^n_{T,\xi_u} = \delta^n_{S,\xi_u} \circ f^n_N$$

Step 3: Comparison of two effacements. Suppose given a commutative diagram with exact rows

$$\begin{array}{lccccccccc} \xi_u\colon & 0 & \longrightarrow & A & \xrightarrow{u} & M & \xrightarrow{v} & N & \longrightarrow & 0 \\ & & & \downarrow{\scriptstyle a} & & \downarrow & & \downarrow{\scriptstyle c} & & \\ \xi_w\colon & 0 & \longrightarrow & B & \xrightarrow{w} & P & \longrightarrow & Q & \longrightarrow & 0 \end{array}$$

in which $T^{n+1}(u) = 0$ and $T^{n+1}(w) = 0$. Then $S^{n+1}(a) \circ f^{n+1}_{A,u} = f^{n+1}_{B,w} \circ T^{n+1}(a)$. Since δ^n_{T,ξ_u} is an epimorphism, it is enough to check that the two sides agree after composing with it. For the first, Step 2, naturality of δ_S for the displayed morphism, and naturality of f^n applied to c give

$$S^{n+1}(a) f^{n+1}_{A,u} \delta^n_{T,\xi_u} = S^{n+1}(a) \delta^n_{S,\xi_u} f^n_N = \delta^n_{S,\xi_w} S^n(c) f^n_N = \delta^n_{S,\xi_w} f^n_Q T^n(c)$$

For the second, naturality of δ_T for the same morphism, followed by Step 2 for w, gives

$$f^{n+1}_{B,w} T^{n+1}(a) \delta^n_{T,\xi_u} = f^{n+1}_{B,w} \delta^n_{T,\xi_w} T^n(c) = \delta^n_{S,\xi_w} f^n_Q T^n(c)$$

The two agree.

Step 4: Independence of the effacement. Let $u\colon A \to M$ and $u'\colon A \to M'$ both efface T^{n+1} at A. The morphism $(u, u')\colon A \to M \oplus M'$ is a monomorphism, because composing it with the projection to M gives u. Writing ι_M and $\iota_{M'}$ for the two inclusions, $(u, u') = \iota_M u + \iota_{M'} u'$, so additivity of T^{n+1} gives

$$T^{n+1}\big((u, u')\big) = T^{n+1}(\iota_M) T^{n+1}(u) + T^{n+1}(\iota_{M'}) T^{n+1}(u') = 0$$

and (u, u') effaces T^{n+1} at A as well. The projection $M \oplus M' \to M$ carries (u, u') to u, hence induces a morphism from the effacing short exact sequence of (u, u') to that of u whose left-hand vertical map is id_A. Step 3 with $a = \mathrm{id}_A$ then gives $f^{n+1}_{A,(u,u')} = f^{n+1}_{A,u}$, and the other projection gives $f^{n+1}_{A,(u,u')} = f^{n+1}_{A,u'}$ in the same way. So the morphism of Step 2 does not depend on the effacement, and it is written f^{n+1}_A from now on.

Step 5: Naturality. Let $g\colon A \to B$ be a morphism of $\mathcal{A}$, and choose effacements $u\colon A \to M$ of T^{n+1} at A and $w\colon B \to P$ at B. The morphism $(u, wg)\colon A \to M \oplus P$ is a monomorphism, again because its composite with the projection to M is u, and

$$T^{n+1}\big((u, wg)\big) = T^{n+1}(\iota_M) T^{n+1}(u) + T^{n+1}(\iota_P) T^{n+1}(w) T^{n+1}(g) = 0$$

since $T^{n+1}(u) = 0$ and $T^{n+1}(w) = 0$. So (u, wg) effaces T^{n+1} at A. The projection $M \oplus P \to P$ carries (u, wg) to $wg = w \circ g$, so it induces a morphism from the effacing short exact sequence of (u, wg) to that of w whose left-hand vertical map is g. Step 3 gives $S^{n+1}(g) \circ f^{n+1}_{A,(u,wg)} = f^{n+1}_{B,w} \circ T^{n+1}(g)$, and by Step 4 the two subscripted morphisms are f^{n+1}_A and f^{n+1}_B. Hence f^{n+1} is a natural transformation $T^{n+1} \Rightarrow S^{n+1}$, which is the induction hypothesis one degree up.

Step 6: Compatibility with every connecting morphism. Let $\xi\colon 0 \to A \xrightarrow{a} M \xrightarrow{b} N \to 0$ be an arbitrary short exact sequence, choose an effacement $u\colon A \to E$ of T^{n+1} at A, and form the pushout of a along u,

$$P := \operatorname{coker}\Big(A \xrightarrow{(a,-u)} M \oplus E\Big)$$

with structure morphisms $m\colon M \to P$ and $e\colon E \to P$, so that $ma = eu$.

The morphism m is a monomorphism. Let $k\colon K \to M$ satisfy $mk = 0$. Then the composite of $(k,0)\colon K \to M \oplus E$ with the cokernel morphism $\pi\colon M \oplus E \to P$ vanishes, so $(k,0)$ factors through $\ker \pi$. Now $(a,-u)$ is a monomorphism, because its composite with the projection to M is a, and π is its cokernel, so $(a,-u)$ is a kernel of π. Hence there is $h\colon K \to A$ with $ah = k$ and $-uh = 0$. As u is a monomorphism $h = 0$, and therefore $k = 0$. So $0 \to M$ is a kernel of m, and lemma 1.1.4(1) makes m a monomorphism.

Consequently $a_1 := ma = eu$ is a monomorphism, being a composite of two, and $T^{n+1}(a_1) = T^{n+1}(e)T^{n+1}(u) = 0$. So

$$\xi_1\colon \qquad 0 \to A \xrightarrow{a_1} P \to \operatorname{coker}(a_1) \to 0$$

effaces T^{n+1} at A, and the identity of A together with m and the induced morphism $\gamma\colon N \to \operatorname{coker}(a_1)$ on cokernels is a morphism of short exact sequences $\xi \to \xi_1$. Steps 2 and 4 give $f_A^{n+1} \circ \delta^n_{T,\xi_1} = \delta^n_{S,\xi_1} \circ f^n_{\operatorname{coker}(a_1)}$, while naturality of δ_T and of δ_S for that morphism gives $\delta^n_{T,\xi_1} \circ T^n(\gamma) = \delta^n_{T,\xi}$ and $\delta^n_{S,\xi_1} \circ S^n(\gamma) = \delta^n_{S,\xi}$, since that morphism is the identity on A. Therefore

$$f_A^{n+1}\delta^n_{T,\xi} = f_A^{n+1}\delta^n_{T,\xi_1}T^n(\gamma) = \delta^n_{S,\xi_1}f^n_{\operatorname{coker}(a_1)}T^n(\gamma) = \delta^n_{S,\xi_1}S^n(\gamma)f^n_N = \delta^n_{S,\xi}f^n_N$$

the third equality being naturality of f^n applied to γ. The induction therefore produces a family $f = (f^i)_{i\geq 0}$ of natural transformations satisfying the condition of definition 2.4.3, that is, a morphism of δ-functors with $f^0 = \varphi$.

Step 7: Uniqueness. Let f and g be morphisms of δ-functors $T \to S$ with $f^0 = g^0 = \varphi$, and suppose $f^j = g^j$ for all $j \leq n$. Given an object A, choose an effacement $u\colon A \to M$ of T^{n+1} at A and form ξ_u as in Step 1. Both f_A^{n+1} and g_A^{n+1} satisfy $h \circ \delta^n_{T,\xi_u} = \delta^n_{S,\xi_u} \circ f^n_N$, by definition 2.4.3 and the induction hypothesis, and δ^n_{T,ξ_u} is an epimorphism by Step 1, so $f_A^{n+1} = g_A^{n+1}$. Hence $f = g$, and T is universal, as we sought to show.

The hypothesis is one embedding per object and per positive degree, and the conclusion quantifies over every δ-functor at once. Step 1 is what makes that exchange possible: an effacing embedding makes δ^n an epimorphism, so a morphism out of $T^{n+1}(A)$ is pinned down by its effect on T^n of the quotient, one degree lower. Without effaceability the induction has nothing to descend on, and example 2.4.2 is the failing case: there $T^1 = \mathrm{id}_{\mathbf{Ab}}$, so a monomorphism $u\colon A \to M$ has $T^1(u) = u$, which vanishes only when $A = 0$.

The use of universality is that it turns a comparison in degree zero into a comparison in every degree.

Corollary 2.4.7: A Universal δ-Functor Is Determined by Its Degree-Zero Part

Let S and T be universal δ-functors from $\mathcal{A}$ to $\mathcal{B}$ and let $\varphi\colon T^0 \Rightarrow S^0$ be a natural isomorphism. Then the unique morphism of δ-functors $f\colon T \to S$ with $f^0 = \varphi$ has every f^i a natural isomorphism. Equivalently, *any* morphism of δ-functors between universal δ-functors whose degree-zero component is a natural isomorphism is itself an isomorphism. In particular, a universal δ-functor with a prescribed degree-zero term is unique up to a unique isomorphism of δ-functors.

***Proof*:**
Universality of T gives a unique morphism $f\colon T \to S$ with $f^0 = \varphi$, and universality of S a unique morphism $g\colon S \to T$ with $g^0 = \varphi^{-1}$. The degreewise composite $g \circ f\colon T \to T$ is a morphism of δ-functors with $(g \circ f)^0 = \varphi^{-1} \circ \varphi = \mathrm{id}_{T^0}$, and id_T is another one. Applying definition 2.4.4 to T with S taken to be T itself and φ taken to be id_{T^0}, the two coincide, so $g \circ f = \mathrm{id}_T$. The same argument in S gives $f \circ g = \mathrm{id}_S$. Hence each f^i is a natural isomorphism with inverse g^i. For the second sentence of the statement, a morphism of δ-functors $h\colon T \to S$ whose component h^0 is a natural isomorphism is, by universality of T, the unique morphism extending h^0, so it is the f just treated with $\varphi = h^0$, completing the proof.

The corollary is what makes universality usable: to identify two families of functors in all degrees it is enough to identify them in degree zero, provided both are universal. The criterion of theorem 2.4.6 is what verifies that proviso, and the family it applies to most often is the one built in section 2.3.

Corollary 2.4.8: The Right-Derived Functors Form a Universal δ-Functor

Let $\mathcal{A}$ be an abelian category with enough injectives, let $\mathcal{B}$ be an abelian category, and let $F\colon \mathcal{A} \to \mathcal{B}$ be an additive functor. Then:

1. the family $(R^iF)_{i\geq 0}$, equipped with the connecting morphisms of corollary 2.3.12, is a δ-functor from $\mathcal{A}$ to $\mathcal{B}$;
2. $R^iF(I) = 0$ for every injective object I of $\mathcal{A}$ and every $i > 0$; consequently $(R^iF)_{i\geq 0}$ is effaceable in positive degrees and therefore universal;
3. if F is left exact then R^0F is naturally isomorphic to F, and $(R^iF)_{i\geq 0}$ is then, up to a unique isomorphism of δ-functors, the only universal δ-functor from $\mathcal{A}$ to $\mathcal{B}$ whose degree-zero term is naturally isomorphic to F.

***Proof*:**

1. Each R^iF is additive. Let $\varphi, \psi\colon A \to B$, and choose injective resolutions $A \to I^\bullet$ and $B \to J^\bullet$. Applying proposition 2.2.15 in the opposite category, φ and ψ lift to morphisms of resolutions $\varphi^\bullet, \psi^\bullet\colon I^\bullet \to J^\bullet$, unique up to homotopy, and $\varphi^\bullet + \psi^\bullet$ is then a lift of $\varphi + \psi$. Since F is additive, $\mathrm{Ch}(F)(\varphi^\bullet + \psi^\bullet) = \mathrm{Ch}(F)(\varphi^\bullet) + \mathrm{Ch}(F)(\psi^\bullet)$ degreewise, and H^i carries a sum of morphisms of complexes to the sum of the induced morphisms, so $R^iF(\varphi+\psi) = R^iF(\varphi)+R^iF(\psi)$. Condition (1) of definition 2.4.1 is the exactness assertion of corollary 2.3.12 together with its identification of the unlabelled arrows as $R^iF(\alpha)$ and $R^iF(\beta)$, and condition (2) is its functoriality in the short exact sequence.

2. Let I be injective. The complex with I in degree 0 and 0 elsewhere, augmented by id_I, is an injective resolution of I, so $R^iF(I) = H^i(F(I) \to 0 \to \cdots)$, which is 0 for $i > 0$. Given an object A and $i > 0$, enough injectives gives a monomorphism $A \hookrightarrow I$ with I injective (definition 2.2.3(2)), and the induced morphism $R^iF(A) \to R^iF(I) = 0$ is zero because its target is. So R^iF is effaceable for every $i > 0$, and theorem 2.4.6 applies.

3. Let $0 \to A \xrightarrow{\varepsilon_A} I^0 \xrightarrow{d^0} I^1 \to \cdots$ be an injective resolution. Left exactness is a hypothesis about short exact sequences, and $0 \to A \to I^0 \xrightarrow{d^0} I^1$ is not one, so the passage to $\ker F(d^0)$ takes two steps. First, F preserves monomorphisms: a monomorphism $\nu\colon X \to Y$ is a kernel of its cokernel (theorem 1.1.9), so $0 \to X \xrightarrow{\nu} Y \to \operatorname{coker}\nu \to 0$ is short exact and left exactness makes $F(\nu)$ a monomorphism. Second, factor $d^0 = \mu \circ p$ with $p\colon I^0 \to \operatorname{im}d^0$ an epimorphism and $\mu\colon \operatorname{im}d^0 \to I^1$ a monomorphism (theorem 1.1.9). Exactness of the resolution at A and at I^0 makes ε_A a kernel of d^0, hence of p since μ is a monomorphism, so $0 \to A \xrightarrow{\varepsilon_A} I^0 \xrightarrow{p} \operatorname{im}d^0 \to 0$ is short exact and $0 \to F(A) \to F(I^0) \xrightarrow{F(p)} F(\operatorname{im}d^0)$ is exact. Since $F(\mu)$ is a monomorphism, $\ker F(d^0) = \ker F(p)$, and that kernel is the image of the monomorphism $F(\varepsilon_A)$. Hence $F(A) \cong \ker F(d^0) = H^0\big(\mathrm{Ch}(F)(I^\bullet)\big) = R^0F(A)$, the isomorphism being induced by $F(\varepsilon_A)$. For a morphism $\varphi\colon A \to B$ and a lift $\varphi^\bullet$ of φ to the chosen resolutions, the augmentation square $\varepsilon_B \circ \varphi = \varphi^0 \circ \varepsilon_A$ gives $F(\varepsilon_B)F(\varphi) = F(\varphi^0)F(\varepsilon_A)$, and $R^0F(\varphi)$ is the restriction of $F(\varphi^0)$ to the two kernels, so the isomorphism is natural in A. For the final clause, $(R^iF)_{i\geq 0}$ is universal by (2) and its degree-zero term is F up to the isomorphism just produced, so corollary 2.4.7 gives the uniqueness, completing the proof.

Part (3) is the form in which the corollary is applied: a family of functors carrying long exact sequences, agreeing with F in degree zero and effaceable in positive degrees, *is* the family of right-derived functors of F, by a canonical isomorphism compatible with every connecting morphism. The converse question, whether universality forces effaceability, has an affirmative answer as soon as enough injectives are available.

Corollary 2.4.9: Universality and Effaceability Agree When There Are Enough Injectives

Let $\mathcal{A}$ be an abelian category with enough injectives, let $\mathcal{B}$ be an abelian category, and let T be a δ-functor from $\mathcal{A}$ to $\mathcal{B}$. Then the following are equivalent:

1. T is universal;
2. $T^i(I) = 0$ for every injective object I of $\mathcal{A}$ and every $i > 0$;
3. T^i is effaceable for every $i > 0$.

***Proof*:**
That (3) implies (1) is theorem 2.4.6. For (1) implies (2), let T be universal. Condition (1) of definition 2.4.1 makes T^0 left exact, so by corollary 2.4.8(2) and (3) the family $(R^iT^0)_{i\geq 0}$ is a universal δ-functor with R^0T^0 naturally isomorphic to T^0. Both T and $(R^iT^0)_{i\geq 0}$ are universal and their degree-zero terms are naturally isomorphic, so corollary 2.4.7 gives an isomorphism of δ-functors $T \cong (R^iT^0)_{i\geq 0}$. For I injective and $i > 0$ this gives $T^i(I) \cong R^iT^0(I) = 0$ by corollary 2.4.8(2). For (2) implies (3), let $i > 0$ and let A be an object. Enough injectives gives

a monomorphism $A \hookrightarrow I$ with I injective, and $T^i(A) \to T^i(I) = 0$ is zero because its target is. Hence T^i is effaceable, completing the proof.

A worked case shows the two corollaries, and it can be carried out to the end because the functor below has an injective resolution of length one available at every object.

Example 2.4.10: The δ-Functor $\operatorname{Ext}^\bullet_{\mathbb{Z}}(\mathbb{Z}/n, -)$

Fix an integer $n \geq 1$ and let $F = \operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n, -)\colon \mathbf{Ab} \to \mathbf{Ab}$, which is left exact. The category $\mathbf{Ab}$ is $\mathbb{Z}$-$\mathbf{Mod}$ and has enough injectives by example 2.2.2(4), and $R^iF = \operatorname{Ext}^i_{\mathbb{Z}}(\mathbb{Z}/n, -)$ by example 2.3.6(2), so corollary 2.4.8 makes $\big(\operatorname{Ext}^i_{\mathbb{Z}}(\mathbb{Z}/n, -)\big)_{i\geq 0}$ a universal δ-functor whose degree-zero term is naturally isomorphic to F.

Its values can be written down. By [8, Baer's Criterion](2), applied to the principal ideal domain $\mathbb{Z}$, an abelian group is injective if and only if it is divisible. Every abelian group embeds in a divisible one: write $A = L/K$ with L the free abelian group on the underlying set of A, note that $L \subseteq L \otimes_{\mathbb{Z}} \mathbb{Q}$ and that $K \subseteq L$, and observe that the kernel of $A = L/K \to (L \otimes_{\mathbb{Z}} \mathbb{Q})/K$ consists of the classes of elements of L that lie in K, hence is zero, while the target is a quotient of the divisible group $L \otimes_{\mathbb{Z}} \mathbb{Q}$ and is therefore divisible. Fix such an embedding $A \hookrightarrow D$ with D divisible. The quotient D/A is divisible as well, so

$$0 \to A \to D \to D/A \to 0$$

is an injective resolution of A of length one, and $\operatorname{Ext}^i_{\mathbb{Z}}(\mathbb{Z}/n, A) = 0$ for $i \geq 2$.

For any abelian group B a homomorphism $\mathbb{Z}/n \to B$ is determined by the image of 1, which may be any element killed by n, so $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/n, B) = B[n] := \{b \in B \,\big|\, nb = 0\}$. Applying F to the resolution leaves the two-term complex $D[n] \to (D/A)[n]$, whence

$$\operatorname{Ext}^0_{\mathbb{Z}}(\mathbb{Z}/n, A) = \ker\big(D[n] \to (D/A)[n]\big) = D[n] \cap A = A[n]$$

which agrees with left exactness, and $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n, A) = \operatorname{coker}\big(D[n] \to (D/A)[n]\big)$. That cokernel is A/nA. Send the class of $d + A$, where $nd \in A$, to $nd + nA$. This is well defined, since replacing d by $d + a$ with $a \in A$ changes nd by $na \in nA$, and it kills the image of $D[n]$, on which $nd = 0$. It is surjective because D is divisible: given $a \in A$ choose $d \in D$ with $nd = a$, and then $d + A$ lies in $(D/A)[n]$ and maps to $a + nA$. It is injective because $nd \in nA$, say $nd = na$ with $a \in A$, forces $n(d - a) = 0$, so that $d + A = (d - a) + A$ lies in the image of $D[n]$.

The universal δ-functor extending $A \mapsto A[n]$ is therefore

$$\operatorname{Ext}^0_{\mathbb{Z}}(\mathbb{Z}/n, A) = A[n], \qquad \operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n, A) = A/nA, \qquad \operatorname{Ext}^i_{\mathbb{Z}}(\mathbb{Z}/n, A) = 0 \;\; (i \geq 2)$$

and the long exact sequence attached to $0 \to A' \to A \to A'' \to 0$ has only six nonzero terms,

$$0 \to A'[n] \to A[n] \to A''[n] \xrightarrow{\delta^0} A'/nA' \to A/nA \to A''/nA'' \to 0$$

Effaceability is visible in this display: for D divisible $D/nD = 0$, so the morphism $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n, A) \to \operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n, D)$ induced by $A \hookrightarrow D$ is zero, and in degrees $i \geq 2$ both groups vanish.

Applications of the criterion follow one pattern. Two families of functors are each shown to carry long exact sequences functorial in the short exact sequence; each is shown effaceable in positive degrees, possibly by different classes of effacing objects; and the two degree-zero terms are matched by a natural isomorphism. Theorem 2.4.6 makes both universal, and corollary 2.4.7 then upgrades the degree-zero match to an isomorphism in every degree, compatible with every connecting morphism. The two classes of effacing objects need not be related, and neither has to consist of injective objects, since definition 2.4.5 asks only that the induced morphism vanish.

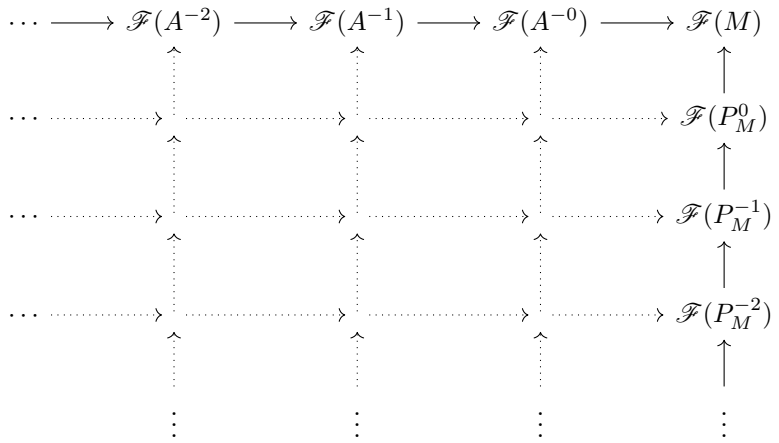
Exercise 2.4.1

1. Let $G\colon \mathcal{A} \to \mathcal{B}$ be an exact additive functor. Show that setting $T^i := G$ for every $i \geq 0$, with all connecting morphisms zero, defines a δ-functor from $\mathcal{A}$ to $\mathcal{B}$, and that this δ-functor is universal if and only if G is the zero functor.
2. Show that a universal δ-functor T with $T^0 = 0$ has $T^i = 0$ for every $i \geq 0$.
3. Let $\mathcal{A}$ be an abelian category with enough injectives and let $F, G\colon \mathcal{A} \to \mathcal{B}$ be left exact additive functors admitting a natural isomorphism $\eta\colon F \Rightarrow G$. Show that there is exactly one isomorphism of δ-functors $(R^iF)_{i\geq 0} \to (R^iG)_{i\geq 0}$ whose degree-zero component corresponds to η, and give the argument without choosing a single resolution.
4. Fix integers $m, n \geq 1$ and put $d = \gcd(m, n)$. Write out the six-term exact sequence of example 2.4.10 for the short exact sequence $0 \to \mathbb{Z} \xrightarrow{m} \mathbb{Z} \to \mathbb{Z}/m \to 0$, computing each of the six groups, and verify the exactness of the resulting sequence directly.
5. Let $\mathcal{A}$ be an abelian category with enough injectives, let $F\colon \mathcal{A} \to \mathcal{B}$ be additive, and let $0 \to A \to I \to C \to 0$ be exact with I injective. Show that $R^iF(C) \cong R^{i+1}F(A)$ for every $i \geq 1$ and that $R^1F(A) \cong \operatorname{coker}\big(R^0F(I) \to R^0F(C)\big)$.
6. Let T be a δ-functor from $\mathcal{A}$ to $\mathcal{B}$ and let $E\colon \mathcal{B} \to \mathcal{C}$ be an exact additive functor. Show that $(E \circ T^i)_{i\geq 0}$, with connecting morphisms $E(\delta^i_\xi)$, is a δ-functor from $\mathcal{A}$ to $\mathcal{C}$, and that it is universal whenever T is effaceable in positive degrees.
7. Let T be a universal δ-functor from $\mathcal{A}$ to $\mathcal{B}$. Show that $f \mapsto f^0$ is a bijection from the set of morphisms of δ-functors $T \to T$ onto the set of natural transformations $T^0 \Rightarrow T^0$, and that it carries composition to composition.

2.5 Double Complexes

This final section shall be the basis for being able to compute many complexes. It was mentioned in box 2.2.12 that $\operatorname{Ext}^i_R(M, N)$ may be computed by using either a projective resolution of M or an injective resolution on N, both give the same answer. More generally, we may compute the resolution of either M or N and get the same value. This section will now prove this result. This will make many computations much simpler, for usually one object in is readily understandable while the other is being analyzed.

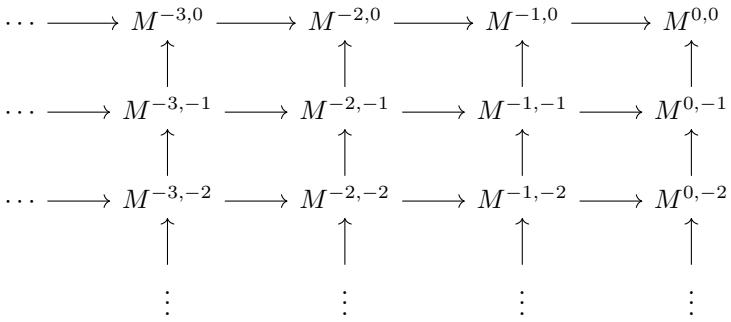
To motivate this, switching from projective to $\mathscr{F}$-acyclic resolutions can be viewed in the following important alternative way. Consider the following diagram where the top row and right column are cochain:



in some way, there was a reason to go from the top chain to the right chain, with the intermediate values "interpolating" the result. The natural candidate for this would be a complex of complex. Needing to bound it in one direction, say we are looking at $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$. Such an object would be of the form:

$$\cdots \to M^{-3,\bullet} \xrightarrow{d_h^{-3,\bullet}} M^{-2,\bullet} \xrightarrow{d_h^{-2,\bullet}} M^{-1,\bullet} \xrightarrow{d_h^{-1,\bullet}} M^{0,\bullet} \to 0 \to \cdots$$

The h in the subscript is to represent the fact we are going *horizontally*, and the respective morphism of each $d_h^{-i,\bullet}$ will be going vertically and are denoted $d_v^{i,j}$, or simply d_v to simply show the function's direction:



Given a complex of complex, it would be nice to reduce this to being a complex, a more familiar setting. One strategy the reader may come up with is to take:

$$\bigoplus_{i+j=n} M^{i,j}$$

with the natural morphisms between them. However these morphisms *almost* complexes: instead of commutativity for each square, they are *anti-commutative*. To fix this, create a new type of object that will satisfy this condition:

Definition 2.5.1: Double Complex

Let $A = \{A_{p,q}\}$ be a family of objects from $\mathbf{A}$. Then A together with maps

$$d^h : A_{p,q} \to A_{p-1,q} \qquad \delta^v : A_{p,q} \to A_{p,q-1}$$

satisfying $d^h \circ d^h = 0$, $\delta^v \circ \delta^v = 0$ and $d^h \circ \delta^v + \delta^v \circ d^h = 0$ is called a *double complex* or *bicomplex*.
Both differentials *lower* an index; this homological convention is the one used throughout, and in particular in chapter 6. A cochain double complex, whose differentials raise indices, is the same thing read through $A_{p,q} = A^{-p,-q}$.

We may visualize a double complex like so, with d^h running to the right and δ^v running downwards, each lowering its index:

$$\begin{array}{ccccccccc}
 & & \vdots & & \vdots & & \vdots & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
\cdots & \longrightarrow & A_{p+1,q+1} & \longrightarrow & A_{p,q+1} & \longrightarrow & A_{p-1,q+1} & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
\cdots & \longrightarrow & A_{p+1,q} & \longrightarrow & A_{p,q} & \longrightarrow & A_{p-1,q} & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
\cdots & \longrightarrow & A_{p+1,q-1} & \longrightarrow & A_{p,q-1} & \longrightarrow & A_{p-1,q-1} & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & \vdots & & \vdots & & \vdots & &
\end{array} \tag{2.6}$$

Note this is not a commutative diagram but an *anti-commutative* one. That is a convention, not a restriction, and it is worth pinning down once because both forms occur in practice. A grid whose squares *commute*, which is what a complex of complexes literally gives, is converted into a double complex in the sense above by replacing the vertical differential with

$$\delta^v_{p,q} := (-1)^p \, \tilde{\delta}^v_{p,q}$$

where $\tilde{\delta}^v$ is the commuting one. The sign makes the squares anticommute, which is exactly what makes $d^h + \delta^v$ square to zero on the total complex. Every later construction, here and in chapter 6, assumes the anticommuting form; when a source of double complexes produces commuting squares, this twist is understood to have been applied.

As mentioned, we can form a total complex from this:

Proposition 2.5.2: Total Complex

Let $A = \{A^{p,q}\}$ be a double complex. Then define the total complex $\mathrm{Tot}^{\Pi}(A)$ and $\mathrm{Tot}^{\oplus}(A)$ by:

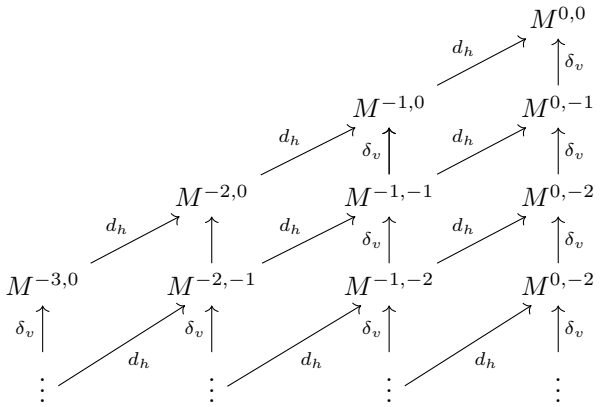
$$\mathrm{Tot}^{\Pi}(A)_n = \prod_{p+q=n} A_{p,q} \qquad \mathrm{Tot}^{\oplus}(A)_n = \bigoplus_{p+q=n} A_{p,q}$$

with $d^{\mathrm{tot}} = d^h + \delta^v$, which squares to zero precisely because the two differentials anticommute. The two agree whenever only finitely many $A_{p,q}$ are nonzero on each diagonal $p+q=n$, as for bounded-above tensor and Hom grids.

***Proof*:**
Computing:

$$d_{tot} \circ d_{tot} = (d_h + \delta_v) \circ (d_h + \delta_v) = d_h \circ d_h + d_h \circ \delta_v + \delta_v \circ d_h + \delta_v \circ \delta_v = 0$$

Note that the total complex does not exist in all categories, namely it may not exist in categories which are not closed under infinite products and coproducts (ex. in the category of finite abelian groups). A category which is closed for infinite products is called *complete* and *cocomplete* if it is closed for infinite sums. We can visualize a total complex by taking the diagram in equation (2.6) and moving it like so:



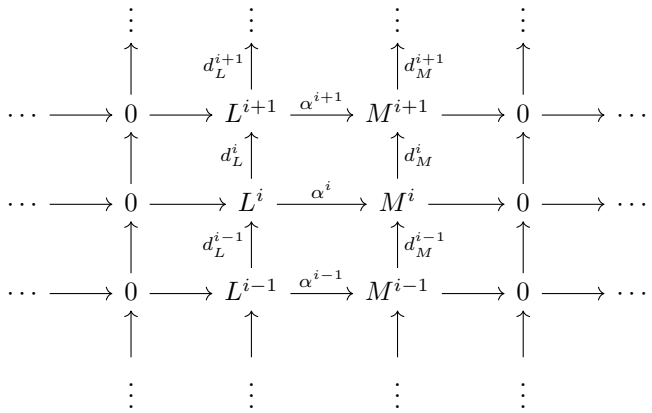
It is then easy to visualize how we get a total complex form this.

Example 2.5.3: Total Complexs

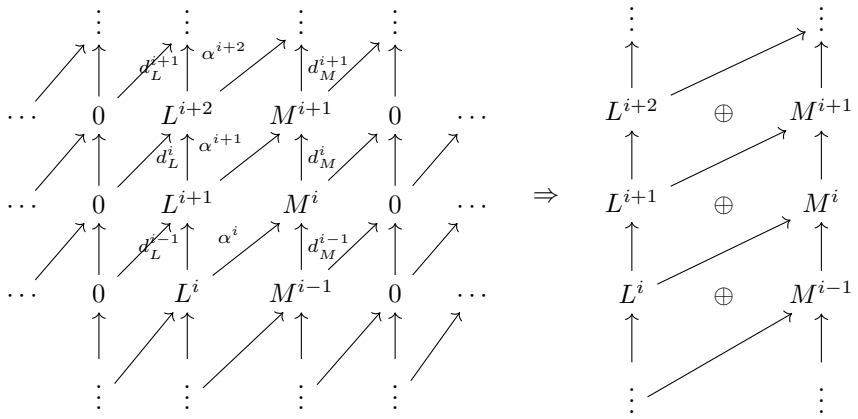
1. Consider a double complex:

$$\cdots \to 0 \to L^\bullet \xrightarrow{\alpha^\bullet} M^\bullet \to 0 \to \cdots$$

which we may represent as:



we shall get the following:



which is exactly the definition of a mapping cone!

2. The operators $\mathrm{Hom}_A(-,-)$ and $-\otimes-$ both give rise to double complexes. Focusing on the $\mathrm{Hom}_A(-,-)$ example, consider a chain in $L^\bullet \in \mathrm{Ch}^{\leq 0}(\mathbf{A})$ and $M^\bullet \in \mathrm{Ch}^{\geq 0}(\mathbf{A})$. These can be used to form a double complex, or isomorphically a complex in $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$:

$$\cdots \to \mathrm{Hom}_A(L^\bullet, M^0) \to \mathrm{Hom}_A(L^\bullet, M^1) \to \mathrm{Hom}_A(L^\bullet, M^2) \to \cdots$$

which creates the following upper-quadrant grid:

$$\begin{array}{ccccccc}
\vdots & & \vdots & & \vdots & & \\
\uparrow & & \uparrow & & \uparrow & & \\
\mathrm{Hom}_A(L^{-2}, M^0) & \longrightarrow & \mathrm{Hom}_A(L^{-2}, M^1) & \longrightarrow & \mathrm{Hom}_A(L^{-2}, M^2) & \longrightarrow & \cdots \\
\uparrow & & \uparrow & & \uparrow & & \\
\mathrm{Hom}_A(L^{-1}, M^0) & \longrightarrow & \mathrm{Hom}_A(L^{-1}, M^1) & \longrightarrow & \mathrm{Hom}_A(L^{-1}, M^2) & \longrightarrow & \cdots \\
\uparrow & & \uparrow & & \uparrow & & \\
\mathrm{Hom}_A(L^0, M^0) & \longrightarrow & \mathrm{Hom}_A(L^0, M^1) & \longrightarrow & \mathrm{Hom}_A(L^0, M^2) & \longrightarrow & \cdots
\end{array}$$

Flipping the above diagram across the main diagonal, we would get the double complex:

$$\cdots \to \mathrm{Hom}_A(L^0, M^\bullet) \to \mathrm{Hom}_A(L^{-1}, M^\bullet) \to \mathrm{Hom}_A(L^{-2}, M^\bullet) \to \cdots$$

These two double complexes are isomorphic (as the reader should check). Due to this, by slight abuse of notation the total complex of both can be denoted $TC(\mathrm{Hom}_A(L, M))^\bullet$, where the degree i term is:

$$\mathrm{Hom}_A(L^{-i}, M^0) \oplus \mathrm{Hom}_A(L^{-i+1}, M^1) \oplus \cdots \oplus 0\mathrm{Hom}_A(L^{-1}, M^{i-1}) \oplus 0\mathrm{Hom}_A(L^0, M^i)$$

The reader should verify that:

(a) $\ker d^i_{tot}$ is equal to morphisms of cochains $L^\bullet \to M[i]^\bullet$

(b) The image $\mathrm{im} d^{i-1}_{tit}$ consists of morphisms that are homotopy equivalent to 0

(c) Thus, the cohomology of $TC(\mathrm{Hom}_A(M, N))^\bullet$ "paramaterizes" the morphisms in the homotopic category

To ground this, consider $TC(\mathrm{Hom}_A(M, N))^0 = \mathrm{Hom}_A(L^0, M^0)$. Then $d_{tot}(\alpha) = 0$ means that both $\alpha \circ d^{-1}_{L^\bullet} \in \mathrm{Hom}_A(L^{-1}, M^0)$ and $d^0_{M^\bullet} \in \mathrm{Hom}_A(L^0, M^1)$ both vanish. The reader should draw out this diagram, and notice that this will make α have the condition to be a morphism of cochain complexes $L^\bullet \to M^\bullet$!

$$H^0(TC(\mathrm{Hom}_A(M, N))^\bullet) = \mathrm{Hom}_{\mathrm{Ch}(\mathbf{A})}(L^\bullet, M^\bullet) \overset{!}{=} \mathrm{Hom}_{K(\mathbf{A})}(L^\bullet, M^\bullet)$$

where the $\overset{!}{=}$ equality comes from the fact that there is no nontrivial homotopies in this case! For $H^i(TC(\mathrm{Hom}_A(M, N))^\bullet)$, with some care for signs, it can be identified with $\mathrm{Hom}_{K(\mathbf{A})}(L^\bullet, M[i]^\bullet)$

The cohomology of double complexes requires some book-keeping through the use of *spectral sequences*, which shall be explored in greater depth in chapter 6. For the current goal, it is enough to consider when a double-complex is exact:

Proposition 2.5.4: Exactness Of Complex Of Complex

Let $\mathbf{A}$ be an abelian category and let

$$\cdots \to M^{-3,\bullet} \to M^{-2,\bullet} \to M^{-1,\bullet} \to M^{0,\bullet} \to 0$$

be a complex in $\mathrm{Ch}^{\leq 0}(\mathrm{Ch}^{\leq 0}(\mathbf{A}))$, regarded as a double complex $M^{\bullet,\bullet}$ with $M^{p,q} = 0$ unless $p \leq 0$ and $q \leq 0$, and let $T^\bullet$ denote its total complex. Then $T^\bullet$ is exact if *either* of the following holds:

1. the displayed complex is exact in $\mathrm{Ch}^{\leq 0}(\mathbf{A})$, equivalently every row $M^{\bullet,q}$ is exact;
2. every column $M^{p,\bullet}$ is exact.

Neither hypothesis implies the other, and either one alone suffices; this is the form in which the result is used below and in chapter 6. The same statement holds in $\mathrm{Ch}^{\geq 0}(\mathrm{Ch}^{\geq 0}(\mathbf{A}))$, by reversing the direction of every index. Note further that only one of the two directions need be bounded: case (2) uses only $p \leq 0$ and case (1) only $q \leq 0$, the bound being what makes the following staircase terminate; with only one bound the diagonals are infinite, so there the total complex must be taken to be $\mathrm{Tot}^{\oplus}$, since the staircase needs a least index in the support of a given element. The result is known as the *acyclic assembly lemma*; see *An Introduction to Homological Algebra* [23] for the variants in the other quadrants and for the distinction between Tot^{Π} and $\mathrm{Tot}^{\oplus}$, which does not arise here because each diagonal of $M^{\bullet,\bullet}$ carries only finitely many nonzero terms.

***Proof*:**

Step 1: Reduction to case (2). Transposing the double complex, that is exchanging the two indices and the two differentials, converts rows into columns and leaves the total complex unchanged up to the sign of the differential, which does not affect exactness. So case (1) follows from case (2) applied to the transpose, and only the latter needs proof.

Step 2: The staircase. Assume every column is exact, write d_h and d_v for the two differentials and $d = d_h + d_v$ for the differential of $T^\bullet$. Let $x \in T^n$ satisfy $dx = 0$; the claim is that $x = dy$ for some $y \in T^{n-1}$. Write $x = \sum_{p+q=n} x^{p,q}$, a finite sum.

Let p_1 be least with $x^{p_1,q_1} \neq 0$, where $q_1 = n - p_1$; if there is no such p_1 then $x = 0$ and there is nothing to prove. Consider the component of the equation $dx = 0$ in bidegree $(p_1, q_1 + 1)$. Only two terms can contribute to it, namely $d_v x^{p_1,q_1}$ and $d_h x^{p_1-1,q_1+1}$, and the second vanishes because $p_1 - 1 < p_1$ and p_1 was least. Hence

$$d_v x^{p_1,q_1} = 0$$

so x^{p_1,q_1} is a cocycle in the column $M^{p_1,\bullet}$. That column is exact, so there is $y^{p_1,q_1-1} \in M^{p_1,q_1-1}$ with $d_v y^{p_1,q_1-1} = x^{p_1,q_1}$.

Step 3: Termination. Replace x by

$$x' = x - d\big(y^{p_1,q_1-1}\big) = x - d_v y^{p_1,q_1-1} - d_h y^{p_1,q_1-1}$$

This is again a cocycle, since it differs from x by a coboundary. Its component in bidegree (p_1, q_1) is $x^{p_1,q_1} - d_v y^{p_1,q_1-1} = 0$, and $d_h y^{p_1,q_1-1}$ lives in bidegree $(p_1 + 1, q_1 - 1)$, so no component

with $p < p_1$ is disturbed and all of those were already zero. The least index with a nonzero component has therefore increased by at least one.

Because $M^{p,q} = 0$ for $p > 0$, this least index cannot exceed 0, so after at most $|p_1| + 1$ repetitions every component vanishes. Summing the finitely many y's produced along the way gives $y \in T^{n-1}$ with $dy = x$, so $T^\bullet$ is exact at degree n. As n was arbitrary, this completes the proof.

Example 2.5.5: Exactness Of Double Complex

Let $P^\bullet \in C^{\leq 0}(\mathbf{A})$ be a complex and $L^\bullet \in C^{\geq 0}(\mathbf{A})$ an exact complex. Since each P^i is projective, $\mathrm{Hom}_{\mathbf{A}}(P^i, L^\bullet)$ for each i. Hence, the complex:

$$0 \to \mathrm{Hom}_{\mathbf{A}}(P^0, L^\bullet) \to \mathrm{Hom}_{\mathbf{A}}(P^{-1}, L^\bullet) \to \mathrm{Hom}_{\mathbf{A}}(P^{-2}, L^\bullet) \to \cdots$$

is exact. Using example 2.5.3, we see that:

$$\mathrm{Hom}_{K(\mathbf{A})}(P^\bullet, L[i]^\bullet) = 0 \qquad \forall i \in \mathbb{Z}$$

This implies that every cochain morphism from a bounded-above projective complex to a bounded-bellow exact complex is homotopic to 0. This just proved corollary 2.2.5 (with an extra boundedness condition)! In some textbooks, due to the speed at which some results can be proved using double-complexes, they are introduced earlier. However, the more "down-to-earth" introduction has is favored in this chapter since it demonstrated key homological concepts.

Next, we look at resolutions of complexes.. As we saw in definition 2.2.11, a (projective) resolution is a quasi-isomorphism $M^\bullet \to \iota(A)$ for some complex $M^\bullet \in \mathrm{Obj}(\mathrm{Ch}(\mathbf{A}))$, and more generally a projective resolution for a general complex is a quasi-isomorphism $P^\bullet \to C^\bullet$ for some projective complex $P^\bullet$. Generalizing the definition of resolution of the first kind, $\iota(A)$, and say that resolution of a complex $N^\bullet \in \mathrm{Ch}^\bullet(A) = A'$ is a complex of complex $M^{\bullet,\bullet} \in \mathrm{Ch}^{\leq 0}(A')$ and a quasi-isomorphism

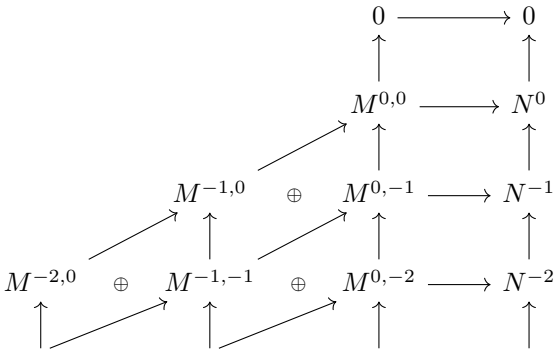
$$\begin{array}{ccccccccc} \cdots & \longrightarrow & M^{-2,\bullet} & \xrightarrow{d_h} & M^{-2,\bullet} & \xrightarrow{d_h} & M^{-2,\bullet} & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & N^\bullet & \longrightarrow & 0 & \longrightarrow & \cdots \end{array} \tag{2.7}$$

where the diagram commutes. We shall show that this definition of a resolution induces a resolution as given in theorem 2.2.17. To see this, notice that in equation (2.7), $\alpha^\bullet \circ d_h = 0$, hence we can collapse the diagram like so:

$$\cdots \to M^{-2,\bullet} \xrightarrow{-d_h} M^{-1,\bullet} \xrightarrow{d_h} M^{0,\bullet} \xrightarrow{\alpha^\bullet} N^\bullet \to 0 \cdots$$

Call this complex $M_N^{\bullet,\bullet}$. Two important diagrams can be derived here. The first is by taking the

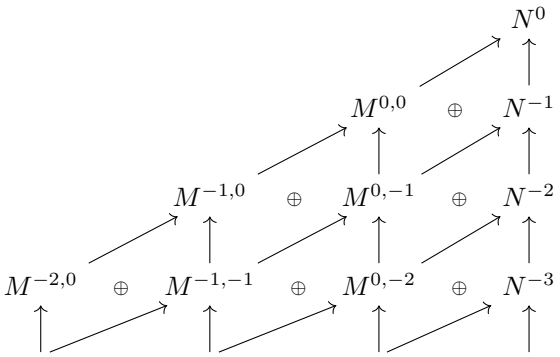
total complex of $M^{\bullet,\bullet}$ and taking a morphism from this to $N^\bullet$:



This gives a morphism:

$$TC(\alpha^\bullet) : TC(M^{\bullet,\bullet}) \to N^\bullet$$

where the morphism is defined since $TC(-)$ is a functorial. Another diagram we can create is by considering the total complex of $M_N^{\bullet,\bullet}$:



These two are related in the following way:

Proposition 2.5.6: Total Complex Of Resolution

Let $M^{\bullet,\bullet}$ be a resolution of $N^\bullet$ and let $M_N^{\bullet,\bullet}$ be defined as above. Then

$$TC(M_N)^\bullet \cong MC\big(TC(\alpha)\big)[-1]^\bullet$$

canonically, equivalently $TC(M_N)[1]^\bullet \cong MC(TC(\alpha))^\bullet$.

***Proof*:**
Both sides are computed degreewise, so it suffices to match the terms and the differentials.

Step 1: The terms. The complex $M_N^{\bullet,\bullet}$ is $M^{\bullet,\bullet}$ with $N^\bullet$ appended in the column $p = 1$, to the right of $M^{0,\bullet}$ and joined to it by the augmentation $\alpha^\bullet$. A term N^j therefore sits in total degree

$1+j$, so its contribution to total degree n is N^{n-1} and

$$TC(M_N)^n = \Big(\bigoplus_{i+j=n} M^{i,j}\Big) \oplus N^{n-1} = TC(M)^n \oplus N^{n-1}$$

which is what the diagram above the proposition displays, $M^{0,0}$ being grouped with N^{-1}. On the other side $TC(\alpha)\colon TC(M)^\bullet \to N^\bullet$ is the totalized augmentation, and since definition 2.1.23 puts the *source* of a morphism in the shifted slot,

$$MC\big(TC(\alpha)\big)^n = TC(M)^{n+1} \oplus N^n$$

Comparing the two, $TC(M_N)^n = MC(TC(\alpha))^{n-1}$, which is the shift asserted in the statement.

Step 2: The differentials. On $TC(M_N)^\bullet$ the differential is the total differential of the augmented double complex: on the M part it is $d_h + d_v$, the component carrying $M^{0,\bullet}$ into the appended column is $\alpha^\bullet$, and on the N part it is $-d_N$, the sign being the $(-1)^p$ twist at $p=1$ that makes the augmented grid anticommute. So $d_{TC(M_N)}(m,n) = (d_{TC(M)}m,\ \alpha(m) - d_N n)$. By definition 2.1.23 the cone differential is $d_{MC}(l,n) = (-d_{TC(M)}l,\ \alpha(l) + d_N n)$, so under the identification of Step 1 the two differ by an overall sign.

The shift convention $d_{[r]} = (-1)^r d$ gives $d_{MC(TC(\alpha))[-1]}(m,n) = -\big(-d_{TC(M)}m,\ \alpha(m) + d_N n\big) = (d_{TC(M)}m,\ -\alpha(m) - d_N n)$, so the two differentials agree in the $TC(M)$ component and differ exactly in the sign of the α-component. They are therefore not equal on the nose, but the map

$$\varphi(m,n) = (m,-n)$$

is an isomorphism of complexes intertwining them: $\varphi\big(d_{TC(M_N)}(m,n)\big) = (d_{TC(M)}m,\ -\alpha(m) + d_N n) = d_{MC(TC(\alpha))[-1]}\big(\varphi(m,n)\big)$. Hence $TC(M_N)^\bullet \cong MC(TC(\alpha))[-1]^\bullet$, canonically, completing the proof.

Using this, we can see how the resolution in $\mathrm{Ch}^{\leq 0}(A')$ leads to a resolution as defined in theorem 2.2.17:

Theorem 2.5.7: Total Complexes And Resolution Of Complexes

Let $\mathbf{A}$ be an abelian category, $\mathbf{A}' = \mathrm{Ch}^{\leq 0}(\mathbf{A})$, and $N^\bullet \in A'$. Let $M^{\bullet,\bullet} \in \mathrm{Ch}^{\leq 0}(\mathbf{A}')$ with total complex $T^\bullet$.

1. Assume that $M^{\bullet,\bullet}$ is a resolution of $N^\bullet$ in $\mathrm{Ch}^{\leq 0}(\mathbf{A}')$. Then $T^\bullet$ is quasi-isomorphic to $N^\bullet$ in $\mathbf{A}'$.
2. Assuming the cohomology of $M^{i,\bullet}$ is concentrated at 0. Then $T^\bullet$ is quasi-isomorphic to the complex of cohomology objects induced by the complex $M^{\bullet,\bullet}$:

$$\cdots \to H^0(M^{-3,\bullet}) \to H^0(M^{-2,\bullet}) \to H^0(M^{-1,\bullet}) \to H^0(M^{0,\bullet}) \to 0 \cdots$$

Notice that this theorem is a direct extensions of proposition 2.5.4. This is also a special case of results we shall see in chapter 6

***Proof*:**
Notice that (2) follows from (1) by flipping the corresponding double-complex across the diagonal, hence we prove the first. For the first, if $M_N^{\bullet,\bullet}$ is quasi-isomorphic, then $TC(M_N)^\bullet$ is exact by proposition 2.5.4, which by proposition 2.5.6 is the mapping cone of the morphism $T^\bullet \to N^\bullet$, which by proposition 2.1.26 shows that $\alpha^\bullet : T^\bullet \to N^\bullet$ is a quasi-isomorphism, as we sought to show.

Using this theorem, a simple alternative to resolving complexes (theorem 2.2.17). By the horseshoe lemma, every bounded-above complex $N^\bullet$ in an abelian category with enough projectives may be viewed as a complex induced by H^0 from a complex:

$$\cdots \to P^{-3,\bullet} \to P^{-2,\bullet} \to P^{-1,\bullet} \to P^{0,\bullet} \to 0$$

where $P^{i,\bullet}$ are projective resolutions of N^i. Then by theorem 2.5.7(2) the total complex of $P^{\bullet,\bullet}$ is a projective resolution of $N^\bullet$. This then can give theorem 2.2.17. It is still worth presenting theorem 2.2.17 to firmly grasp the fundamentals of homological algebra.

2.5.1 Applications to Computations of Ext and Tor

Let us now show that we may compute $\mathrm{Ext}(M,N)$ or $\mathrm{Tor}(M,N)$ using a resolution of either term. Going back to the $\mathscr{F}$-acyclic resolution of an object M:

$$\cdots \to A^{-2} \to A^{-1} \to A^0 \to M \to 0 \to \cdots$$

We may create a complex of projective resolution by corollary 2.3.9:

$$\cdots \to P^{-2,\bullet} \to P^{-1,\bullet} \to P^{0,\bullet} \to P_M^\bullet \to 0 \to \cdots$$

where $P^{i,\bullet}$ is a projective resolution of A^{-i}. Since the resolution by projective resolution's is exact, the rows of the double-complex corresponding to $P^{\bullet,\bullet}$ are projective resolutions of P_M^i. Now, applying the functor $\mathscr{F}$, we get the diagram:

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & \mathscr{F}(A^{-2}) & \longrightarrow & \mathscr{F}(A^{-1}) & \longrightarrow & \mathscr{F}(A^0) & \longrightarrow & \mathscr{F}(M) \\
 & & \uparrow & & \uparrow & & \uparrow & & \uparrow \\
\cdots & \longrightarrow & \mathscr{F}(P_M^{-2,0}) & \longrightarrow & \mathscr{F}(P_M^{-1,0}) & \longrightarrow & \mathscr{F}(P_M^{0,0}) & \longrightarrow & \mathscr{F}(P_M^0) \\
 & & \uparrow & & \uparrow & & \uparrow & & \uparrow \\
\cdots & \longrightarrow & \mathscr{F}(P_M^{-2,-1}) & \longrightarrow & \mathscr{F}(P_M^{-1,-1}) & \longrightarrow & \mathscr{F}(P_M^{0,-1}) & \longrightarrow & \mathscr{F}(P_M^{-1}) \\
 & & \uparrow & & \uparrow & & \uparrow & & \uparrow \\
\cdots & \longrightarrow & \mathscr{F}(P_M^{-2,-2}) & \longrightarrow & \mathscr{F}(P_M^{-1,-2}) & \longrightarrow & \mathscr{F}(P_M^{0,-2}) & \longrightarrow & \mathscr{F}(P_M^{-2}) \\
 & & \uparrow & & \uparrow & & \uparrow & & \uparrow \\
 & & \vdots & & \vdots & & \vdots & & \vdots
\end{array}$$

Since each row and column of this diagram (excluding the top row and right column) is exact, by theorem 2.5.7 the total complex of:

$$\cdots \mathscr{F}(P^{-2,\bullet}) \to \mathscr{F}(P^{-1,\bullet}) \to \mathscr{F}(P^{0,\bullet}) \to 0 \to \cdots$$

is quasi-isomorphic to both $\mathrm{Ch}(\mathscr{F})(A^\bullet)$ and $\mathrm{Ch}(\mathscr{F})(P^\bullet)$, that is:

$$TC(\mathscr{F}(P))^\bullet \sim \mathrm{Ch}(\mathscr{F})(A^\bullet) \sim \mathrm{Ch}(\mathscr{F})(P^\bullet)$$

Thus, by definition of quasi-isomorphism

$$\mathbf{L}_i\mathscr{F}(M) \cong H^{-i}(\mathrm{Ch}(\mathscr{F})(A^\bullet)) \cong H^{-i}(\mathrm{Ch}(\mathscr{F})(P^\bullet))$$

Applying these results to Ext and Tor will finally give us the following results:

Theorem 2.5.8: Computation Of Tor and Ext

Let R be a ring. If M is a right and N a left R-module, with projective resolutions $P^\bullet_M$ and $P^\bullet_N$, then

$$\mathrm{Tor}^R_i(M,N) \cong H^{-i}(P^\bullet_M \otimes_R N) \cong H^{-i}(M \otimes_R P^\bullet_N)$$

If M and N are left R-modules, $P^\bullet_M$ is a projective resolution of M and $Q^\bullet_N$ an injective resolution of N, then

$$\mathrm{Ext}^i_R(M,N) \cong H^i(\mathrm{Hom}_R(P^\bullet_M, N)) \cong H^i(\mathrm{Hom}_R(M, Q^\bullet_N))$$

The index for Tor is $-i$ because a projective resolution sits in nonpositive degrees (definition 2.2.11), so that $P^{-i}_M \otimes_R N$ is the term of degree $-i$.

***Proof*:**
We first treat Tor. Take the complex:

$$\cdots \to P^{-2}_M \otimes_R P^\bullet_N \to P^{-1}_M \otimes_R P^\bullet_N \to P^0_M \otimes_R P^\bullet_N \to 0 \cdots$$

Each P^j_N is projective, hence flat ([8, Projective Modules Are Flat]), so $- \otimes_R P^j_N$ is exact, and the cohomology of the complex is concentrated at 0 and equals $M \otimes_R P^\bullet_N$. By theorem 2.5.7 we have that :

$$TC(P^\bullet_M \otimes_R P^\bullet_N)^\bullet \qquad \text{is quasi-isomorphic to} \qquad M \otimes_R P^\bullet_N$$

Similarly, since each P^i_M is projective, hence flat, the complex $P^i_M \otimes_R P^\bullet_N$ is a resolution of $P^i_M \otimes N$, i.e. the cohomology of each $P^i_M \otimes_R P^\bullet_N$ is concentrated at degree 0 and equals $P^i_M \otimes_R N$. By theorem 2.5.7 we have that :

$$TC(P^\bullet_M \otimes_R P^\bullet_N)^\bullet \qquad \text{is quasi-isomorphic to} \qquad P^\bullet_M \otimes_R N$$

but then, the cohomologies are identical.

For Ext the double complex is $\mathrm{Hom}_R(P^{-p}_M, Q^q_N)$. Each $\mathrm{Hom}_R(-, Q^q_N)$ is exact, Q^q_N being injective, so the complex $\mathrm{Hom}_R(P^\bullet_M, Q^q_N)$ is a resolution of $\mathrm{Hom}_R(M, Q^q_N)$. Each $\mathrm{Hom}_R(P^{-p}_M, -)$ is exact, P^{-p}_M being projective, so $\mathrm{Hom}_R(P^{-p}_M, Q^\bullet_N)$ is a resolution of $\mathrm{Hom}_R(P^{-p}_M, N)$. The same two applications of theorem 2.5.7, with every index reversed as after proposition 2.5.4, identify both $\mathrm{Hom}_R(M, Q^\bullet_N)$ and $\mathrm{Hom}_R(P^\bullet_M, N)$ up to quasi-isomorphism with the total complex, as we sought to show.

3

The Derived Category

Every derived functor in chapter 2 was computed by choosing a resolution and then proving the answer independent of the choice. That pattern is a symptom. The object of the theory is not the module A but the complex that resolves it, and not the complex but its class up to quasi-isomorphism. The resolutions are representatives, and the independence proofs are what working with representatives costs. The derived category is the category in which that thing is an object. Its construction was already carried out in section 1.2 and recorded as corollary 2.1.17: it is the localization $D(\mathbf{A}) = \mathrm{Ch}(\mathbf{A})[\mathrm{qis}^{-1}]$, and it needs no hypothesis on projectives or injectives.

What that construction does not say is what kind of category results. In general it is not abelian, and the failure is not a technicality: kernels and cokernels are the first casualties of inverting quasi-isomorphisms. Something weaker survives, strong enough to carry every long exact sequence the subject uses, and it has a name and a short list of axioms. This chapter states them, builds $D(\mathbf{A})$ inside that framework, and then recovers Ext, Tor and the Grothendieck spectral sequence as statements about it. Its notation follows Weibel and Gelfand-Manin. section 3.1 flags the one place where the two disagree.

3.1 Triangulated Categories

A homotopy category is in general not abelian. The kernel and cokernel of a homotopy class of maps depend on the representative, so neither is well defined on the class. Yet a derived functor $\mathrm{D}(\mathscr{F})$ still carries a long exact sequence, so some structure survives. Locating it starts from a short exact sequence of complexes

$$0 \to L^\bullet \to M^\bullet \to N^\bullet \to 0$$

The snake lemma (theorem 2.1.20) relates $N^\bullet$ and $L^\bullet$ after passing to cohomology, through the connecting morphism. Before passing to cohomology there is no map between them at all. Adjoining

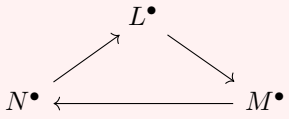
the zero morphism $N^\bullet \xrightarrow{0} L^\bullet$ closes the sequence into a cycle, and the one short exact sequence becomes three:

$$N^\bullet \xrightarrow{0} L^\bullet \to M^\bullet$$
$$L^\bullet \to M^\bullet \to N^\bullet$$
$$M^\bullet \to N^\bullet \xrightarrow{0} L^\bullet$$

These carry the same information, and folding any one of them gives a triangle.

Definition 3.1.1: Exact Triangle

Let $L^\bullet, M^\bullet, N^\bullet \in \mathrm{Ch}(\mathbf{A})$ be three complexes. If there exist morphisms such that the following diagram is exact (at each vertex):

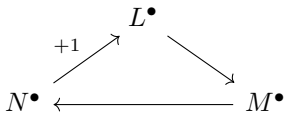


Then the diagram is called an *exact triangle* of $\mathrm{Ch}(\mathbf{A})$. This diagram is also sometimes denoted

$$L^\bullet \to M^\bullet \to N^\bullet \to$$

if it is easier to write it out flat.

The particular triangle obtained by folding the three sequences above is



where the $+1$ morphism is the zero map. The label anticipates what that map becomes once the construction is done properly: not zero, but a map $N^\bullet \to L[1]^\bullet$ raising cohomological degree by one. Stretched out, the triangle is a single long sequence

$$\cdots \to N^{i-1} \xrightarrow{0} L^i \to M^i \to N^i \xrightarrow{0} L^{i+1} \to \cdots$$

Applying the cohomology functor and replacing the zero maps by the connecting morphism gives

$$\cdots \to H^{i-1}(N^\bullet) \xrightarrow{\delta^{i-1}} H^i(L^\bullet) \to H^i(M^\bullet) \to H^i(N^\bullet) \xrightarrow{\delta^i} H^{i+1}(L^\bullet) \to \cdots$$

The maps δ^i do not come from the cohomology functor, which is the precise sense in which $\mathrm{Ch}(\mathbf{A})$ is the wrong category to apply it in. The axioms below name the structure that carries δ as part of its data.

The translation functor on $\mathrm{Ch}(\mathbf{A})$, $\mathrm{K}(\mathbf{A})$ and $D(\mathbf{A})$ is the shift of definition 2.1.23. For $n \in \mathbb{Z}$, a complex X and a chain map f it is

$$X[n]^i := X^{i+n}, \qquad d^i_{X[n]} := (-1)^n d^{i+n}_X, \qquad f[n]^i := f^{i+n} \tag{3.1}$$

so that $H^i(X[n]^\bullet) = H^{i+n}(X^\bullet)$. For $n = 1$ this is the shift definition 2.1.23 uses: its projection $MC(\alpha) \to L[1]$ is a chain map because $d_{L[1]} = -d_L$. It is an automorphism of each of these categories, with inverse $X \mapsto X[-1]$.

3.1.2 Warning: Two Conventions for the Translation Weibel writes the translation functor T and reserves the bracket for the opposite shift, so that his T is his $[-1]$ and his identification reads $H^i(A[-1]) \cong H^{i+1}(A)$. This book writes the translation as $[1]$ throughout. *The functors agree*: both raise cohomological degree by one, and $[1]$ here is T there. Only the bracket differs. A reader checking an axiom against Weibel §10.2 should translate $T \rightsquigarrow [1]$ and leave the signs unchanged.

With the translation fixed, the axioms can be stated. A *triangle* on an ordered triple (A, B, C) of objects of an additive category $\mathcal{K}$ equipped with an automorphism $[1]$ is a triple of morphisms $u\colon A \to B$, $v\colon B \to C$, $w\colon C \to A[1]$, displayed

$$A \xrightarrow{u} B \xrightarrow{v} C \xrightarrow{w} A[1]$$

A *morphism of triangles* (f, g, h) from (u, v, w) on (A, B, C) to (u', v', w') on (A', B', C') is a triple of morphisms making the three squares of

$$\begin{array}{ccccccc} A & \xrightarrow{u} & B & \xrightarrow{v} & C & \xrightarrow{w} & A[1] \\ {\scriptstyle f}\downarrow & & {\scriptstyle g}\downarrow & & {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle f[1]} \\ A' & \xrightarrow[u']{} & B' & \xrightarrow[v']{} & C' & \xrightarrow[w']{} & A'[1] \end{array}$$

commute.

Definition 3.1.3: Triangulated Category

A *triangulated category* is an additive category $\mathcal{K}$ together with an automorphism $[1]\colon \mathcal{K} \to \mathcal{K}$ (the *translation*) and a class of triangles called *distinguished*, satisfying:

(TR1) Every morphism $u\colon A \to B$ embeds in a distinguished triangle (u, v, w). For every object A the triangle $(\mathrm{id}_A, 0, 0)$ on $(A, A, 0)$ is distinguished. Any triangle isomorphic to a distinguished triangle is distinguished.

(TR2) *(Rotation.)* If (u, v, w) is distinguished on (A, B, C), then so are

$$\big(v,\ w,\ -u[1]\big) \text{ on } (B, C, A[1]) \qquad \text{and} \qquad \big(-w[-1],\ u,\ v\big) \text{ on } (C[-1], A, B)$$

(TR3) *(Morphisms.)* Given distinguished triangles (u, v, w) on (A, B, C) and (u', v', w') on (A', B', C') and morphisms $f\colon A \to A'$, $g\colon B \to B'$ with $gu = u'f$, there exists $h\colon C \to C'$ making (f, g, h) a morphism of triangles.

(TR4) *(The octahedral axiom.)* Let $u\colon A \to B$ and $v\colon B \to C$ be morphisms, and suppose given distinguished triangles

$$(u,\ j,\ \delta) \text{ on } (A, B, C'), \quad (v,\ x,\ i) \text{ on } (B, C, A'), \quad (vu,\ y,\ \varepsilon) \text{ on } (A, C, B')$$

Then there is a distinguished triangle $\big(f,\ g,\ (j[1])\, i\big)$ on (C', B', A') such that

$$\delta = \varepsilon f, \qquad x = gy, \qquad yv = fj, \qquad u[1]\,\varepsilon = ig$$

The four axioms are Verdier's. A triangulated category is *not* an abelian category, and the distinguished triangles are what stand in for its short exact sequences.

Three comments on the axioms, in the order a reader meets the difficulties.

The sign in (TR2) is not decoration, and what it does is easy to misstate. Rotating three times returns $(-u[1], -v[1], -w[1])$, and $(-\mathrm{id}, \mathrm{id}, -\mathrm{id})$ is an isomorphism of triangles from that to $(u[1], v[1], -w[1])$: the translate carrying a single sign. That sign does not in general come off. For a diagonal $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ with $\varepsilon_i = \pm\mathrm{id}$ landing on $(u[1], v[1], w[1])$, the three squares read

$$(\varepsilon_1 + \varepsilon_2)\, u[1] = 0, \qquad (\varepsilon_2 + \varepsilon_3)\, v[1] = 0, \qquad (\varepsilon_3 + \varepsilon_1)\, w[1] = 0$$

Where an edge is not annihilated by 2, its square forces the corresponding $\varepsilon_k = -\varepsilon_{k+1}$, and the three relations together give $2\varepsilon_1 = 0$, which no diagonal satisfies. Where an edge *is* killed by 2 the constraint is vacuous and a diagonal can exist: on the triangle $(\mathrm{id}_A, 0, 0)$ of (TR1), whose second and third edges vanish, $(\mathrm{id}, -\mathrm{id}, \mathrm{id})$ is such an isomorphism. In particular $(-\mathrm{id}, -\mathrm{id}, -\mathrm{id})$ is not such an isomorphism: conjugating every vertex by $-\mathrm{id}$ fixes every edge, so it is an automorphism of a triangle and not a map to the sign-flipped one. What forces the sign in the first place is the model, exercise 3.1.1 (1) producing $-\alpha[1]$ from the cone construction, so an axiom without it would fail for $\mathrm{K}(\mathbf{A})$.

(TR3) asserts existence of the filler h and not uniqueness, and the failure of uniqueness is the standing defect of triangulated categories, the reason the axioms hand back no functorial cone and the reason the theory does not by itself glue. What (TR3) does give is proposition 3.1.10, recorded below once the tools to prove it are in place.

The octahedral axiom (TR4) is the one that resists reading. Its content is a compatibility between the three cones attached to a composable pair $A \xrightarrow{u} B \xrightarrow{v} C$: the cones of u, of v, and of vu are not independent, and (TR4) says the first and third fit into a distinguished triangle whose third vertex is the second. In the abelian setting the corresponding statement is the third isomorphism theorem, $(C/A)/(B/A) \cong C/B$, and (TR4) is what survives of it. The four displayed identities say that the octahedron built from the four triangles has its remaining four faces commuting.

3.1.4 Note: What the Axioms Do Not Give The axioms give no limits or colimits beyond finite direct sums, no functorial cone, and no notion of exactness for a general diagram. A particular triangulated category may still have some of this. $\mathrm{K}(R\text{-Mod})$ has all small products and coproducts, formed degreewise, since a family of null-homotopies assembles componentwise. But nothing in definition 3.1.3 produces them, so no argument from the axioms may use them. Restoring the missing structure canonically is what the enhancements of a triangulated category (differential graded, or stable ∞-categorical, [12]) exist for.

Two kinds of functor are relevant, one preserving the triangulated structure and one converting it into exactness in an abelian category.

Definition 3.1.5: Triangulated Functor

Let $\mathcal{K}$ and $\mathcal{K}'$ be triangulated categories. A *triangulated functor* (or *exact functor* of triangulated categories) is an additive functor $F\colon \mathcal{K} \to \mathcal{K}'$ together with a natural isomorphism $F(A[1]) \cong F(A)[1]$, carrying distinguished triangles to distinguished triangles.

Definition 3.1.6: Cohomological Functor

Let $\mathcal{K}$ be a triangulated category and $\mathbf{A}$ an abelian category. An additive functor $H\colon \mathcal{K} \to \mathbf{A}$ is *cohomological* if for every distinguished triangle (u, v, w) on (A, B, C) the sequence

$$H(A) \xrightarrow{H(u)} H(B) \xrightarrow{H(v)} H(C)$$

is exact at $H(B)$.

The definition asks for exactness at one spot, and rotation delivers the rest: applying H to every rotate of a distinguished triangle and splicing gives the long exact sequence

$$\cdots \to H(A[n]) \to H(B[n]) \to H(C[n]) \to H(A[n+1]) \to \cdots$$

This is the mechanism promised at the start of the section. The connecting morphism δ is no longer constructed by a diagram chase. It is the image under H of the third edge w, which the triangle carried all along.

Lemma 3.1.7: Consecutive Maps of a Distinguished Triangle Compose to Zero

Let (u, v, w) be a distinguished triangle on (A, B, C) in a triangulated category $\mathcal{K}$. Then $vu = 0$, $wv = 0$ and $u[1]\, w = 0$.

***Proof*:**
By (TR1) the triangle $(\mathrm{id}_A, 0, 0)$ on $(A, A, 0)$ is distinguished. Take $f = \mathrm{id}_A$ and $g = u$ in (TR3), against the two distinguished triangles

$$A \xrightarrow{\mathrm{id}_A} A \xrightarrow{0} 0 \xrightarrow{0} A[1] \qquad \text{and} \qquad A \xrightarrow{u} B \xrightarrow{v} C \xrightarrow{w} A[1]$$

The required square commutes, since $u \circ \mathrm{id}_A = u = u \circ \mathrm{id}_A$. So (TR3) gives $h\colon 0 \to C$ making the triple a morphism of triangles, and commutativity of the middle square reads $v \circ u = h \circ 0 = 0$. This is $vu = 0$.

Rotating the triangle once by (TR2) gives the distinguished triangle $(v, w, -u[1])$ on $(B, C, A[1])$. Its first two maps compose to zero by the case already proved, so $wv = 0$. Rotating once more gives $(w, -u[1], -v[1])$, whence $(-u[1])w = 0$ and so $u[1]\, w = 0$.

Proposition 3.1.8: Hom Is Cohomological

Let $\mathcal{K}$ be a triangulated category and $X \in \mathcal{K}$. Then $\mathrm{Hom}_{\mathcal{K}}(X, -)\colon \mathcal{K} \to \mathbf{Ab}$ is cohomological, and $\mathrm{Hom}_{\mathcal{K}}(-, X)\colon \mathcal{K}^{\mathrm{op}} \to \mathbf{Ab}$ is cohomological.

***Proof*:**
Let (u, v, w) be distinguished on (A, B, C) and let $X \in \mathcal{K}$. Write $u_* = \mathrm{Hom}(X, u)$ and similarly for v.

Step 1: $v_* u_* = 0$. By lemma 3.1.7 the composite vu is zero, so $v_* u_*(\varphi) = vu\varphi = 0$ for every $\varphi \in \mathrm{Hom}(X, A)$. Hence $\mathrm{im} u_* \subseteq \ker v_*$.

Step 2: $\ker v_* \subseteq \mathrm{im} u_*$. Let $\psi \in \mathrm{Hom}(X, B)$ with $v\psi = 0$. Consider the distinguished triangle $(\mathrm{id}_X, 0, 0)$ on $(X, X, 0)$ from (TR1) and the triangle (u, v, w), and rotate both once by (TR2) to get distinguished triangles on $(X, 0, X[1])$ and $(B, C, A[1])$. The square

$$\begin{array}{ccc} X & \longrightarrow & 0 \\ {\scriptstyle\psi}\downarrow & & \downarrow \\ B & \xrightarrow[v]{} & C \end{array}$$

commutes precisely because $v\psi = 0$, so (TR3) gives a morphism $X[1] \to A[1]$ completing it to a morphism of triangles. Since $[1]$ is an automorphism, that morphism is $\varphi[1]$ for a unique $\varphi\colon X \to A$, and commutativity of the remaining square gives $u\varphi = \psi$. Hence $\psi \in \mathrm{im} u_*$.

The two inclusions give exactness at $\mathrm{Hom}(X, B)$, which is the definition. The contravariant statement is the same argument applied in $\mathcal{K}^{\mathrm{op}}$, which is triangulated with translation $[-1]$.

Two consequences follow at once, and the second is the substitute for functoriality promised above.

Corollary 3.1.9: The Five Lemma for Distinguished Triangles

Let (f, g, h) be a morphism of distinguished triangles in $\mathcal{K}$, from (u, v, w) on (A, B, C) to (u', v', w') on (A', B', C'). If f and g are isomorphisms, so is h.

***Proof*:**
Fix $X \in \mathcal{K}$ and apply $\mathrm{Hom}_{\mathcal{K}}(X, -)$. By proposition 3.1.8 and rotation, each triangle gives a long exact sequence of abelian groups, and (f, g, h) induces a morphism between them. In the five consecutive terms

$$\mathrm{Hom}(X, A) \to \mathrm{Hom}(X, B) \to \mathrm{Hom}(X, C) \to \mathrm{Hom}(X, A[1]) \to \mathrm{Hom}(X, B[1])$$

the maps induced by f, g, $f[1]$ and $g[1]$ are isomorphisms, $[1]$ being an automorphism of $\mathcal{K}$. The five lemma (lemma 2.1.22) in $\mathbf{Ab}$ gives that $h_* \colon \mathrm{Hom}(X, C) \to \mathrm{Hom}(X, C')$ is an isomorphism.

It remains to produce an inverse for h itself, which needs no Yoneda argument. Take $X = C'$: surjectivity of h_* gives $k \colon C' \to C$ with $hk = \mathrm{id}_{C'}$. Take $X = C$: both kh and id_C are carried by h_* to h, since $h(kh) = (hk)h = h$, so injectivity of h_* forces $kh = \mathrm{id}_C$. Thus k is a two-sided inverse and h is an isomorphism.

Proposition 3.1.10: A Distinguished Triangle Is Determined by Any One of Its Maps

Let (u, v, w) on (A, B, C) and (u, v', w') on (A, B, C') be distinguished triangles sharing the morphism $u \colon A \to B$. Then there is an isomorphism $h \colon C \to C'$ with $(\mathrm{id}_A, \mathrm{id}_B, h)$ a morphism of triangles.

***Proof*:**
Apply (TR3) with $f = \mathrm{id}_A$ and $g = \mathrm{id}_B$. The required square commutes, both composites being u, so there is $h \colon C \to C'$ making $(\mathrm{id}_A, \mathrm{id}_B, h)$ a morphism of triangles. Its two outer maps are identities, hence isomorphisms, so corollary 3.1.9 gives that h is an isomorphism.

The cone of a morphism is therefore unique up to isomorphism, and that is all: the isomorphism h is produced by (TR3), which does not claim it is unique, so the axioms provide no functorial choice of cone. Everything awkward about triangulated categories descends from this one gap.

One fact the verification needs is used twice more in this chapter and is stated nowhere earlier, so it is recorded here.

Lemma 3.1.11: The Cone on an Identity Is Null-Homotopic

Let $L^\bullet \in \mathrm{Ch}(\mathbf{A})$. Then $MC(\mathrm{id}_{L^\bullet})$ is null-homotopic, hence exact, and is isomorphic to 0 in $\mathrm{K}(\mathbf{A})$.

***Proof*:**
By definition 2.1.23, $MC(\mathrm{id}_L)^i = L^{i+1} \oplus L^i$ with

$$d^i(a,b) = \big(-d_L^{i+1}(a),\ a + d_L^i(b)\big)$$

Define $s^i \colon MC(\mathrm{id}_L)^i \to MC(\mathrm{id}_L)^{i-1} = L^i \oplus L^{i-1}$ by $s^i(a,b) := (b,0)$. Then

$$\begin{aligned} d^{i-1}s^i(a,b) &= d^{i-1}(b,0) = \big(-d_L^i(b),\ b\big), \\ s^{i+1}d^i(a,b) &= s^{i+1}\big(-d_L^{i+1}(a),\ a + d_L^i(b)\big) = \big(a + d_L^i(b),\ 0\big) \end{aligned}$$

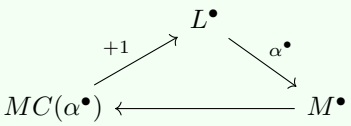
Adding, $\big(ds + sd\big)(a,b) = \big(-d_L^i(b) + a + d_L^i(b),\ b\big) = (a,b)$, so $ds + sd = \mathrm{id}$ and s is a contracting homotopy for the identity.

Exactness follows: by proposition 2.1.29 a null-homotopic map induces the zero map on cohomology, so $H^i(\mathrm{id}) = 0$ for every i. But H^i is a functor, so $H^i(\mathrm{id}) = \mathrm{id}_{H^i(MC(\mathrm{id}_L))}$. An identity that is also the zero map forces $H^i\big(MC(\mathrm{id}_L)\big) = 0$. In $\mathrm{K}(\mathbf{A})$ the identity of $MC(\mathrm{id}_L)$ and the zero morphism are the same homotopy class, which is what it means for the object to be isomorphic to 0 there.

The example the chapter is built on is $\mathrm{K}(\mathbf{A})$, the homotopy category of definition 2.1.38. Its translation is the shift, and its distinguished triangles are declared by the mapping cone.

Theorem 3.1.12: The Homotopy Category Is Triangulated

Let $\mathbf{A}$ be an abelian category. Then $\mathrm{K}(\mathbf{A})$ is a triangulated category, with translation the shift [1] and with distinguished triangles those isomorphic in $\mathrm{K}(\mathbf{A})$ to a *mapping cone triangle*



for some chain map $\alpha^\bullet \colon L^\bullet \to M^\bullet$. The same holds for the bounded variants $\mathrm{K}^+(\mathbf{A})$, $\mathrm{K}^-(\mathbf{A})$ and $\mathrm{K}^b(\mathbf{A})$.

***Proof*:**
That $\mathrm{K}(\mathbf{A})$ is additive is lemma 2.1.39, and the shift is an automorphism of it because it is an automorphism of $\mathrm{Ch}(\mathbf{A})$ carrying chain homotopies to chain homotopies.

(TR1). Every $\alpha^\bullet$ embeds in its own mapping cone triangle, which is distinguished by definition. The triangle $(\mathrm{id}_A, 0, 0)$ is distinguished because $MC(\mathrm{id}_A)$ is null-homotopic (lemma 3.1.11), hence zero in $\mathrm{K}(\mathbf{A})$, so the cone triangle on id_A is isomorphic to it. Closure under isomorphism holds by fiat, the class being defined as the isomorphism closure.

(TR2), (TR3), (TR4). Neither (TR2) nor (TR4) holds in $\mathrm{Ch}(\mathbf{A})$ on the nose. The rotate of a cone triangle, and the octahedron's fourth triangle, are cone triangles only up to an explicit homotopy

equivalence between cones, the underlying graded objects being different in general. (TR3) is not like that. Its filler is $h(l,m) = (f(l), g(m))$, a chain map exactly because the hypothesis square $gu = u'f$ closes the second component, so (TR3) holds at the chain level whenever that square commutes strictly. What changes in $\mathrm{K}(\mathbf{A})$ is that the square commutes only up to homotopy, and the filler must absorb the chosen homotopy. The constructions are in Weibel [23], rotation at Example 10.1.6 in §10.1 and (TR3) with the octahedral axiom at Proposition 10.2.4, whose closing homotopy check that book also leaves to its reader. Exercise 3.1.1 (1) asks for the rotation case here, whose sign is the one worth seeing produced.

The theorem also settles a question the mapping cone raises. $\mathrm{Ch}(\mathbf{A})$ itself is *not* triangulated, and the obstruction is visible in one line: (TR1) forces $(\mathrm{id}_{L^\bullet}, 0, 0)$ to be distinguished, so by rotation $L^\bullet$ must be the cone of $0 \to L^\bullet$ up to isomorphism, which it is, but (TR1) applied to $\mathrm{id}_{L^\bullet}$, together with proposition 3.1.10 comparing the two distinguished triangles that share $\mathrm{id}_{L^\bullet}$, forces $MC(\mathrm{id}_{L^\bullet})$ to be isomorphic to 0. The cone of the identity is not the zero complex. It is only null-homotopic, so the isomorphism exists in $\mathrm{K}(\mathbf{A})$ and not in $\mathrm{Ch}(\mathbf{A})$. Passing to homotopy classes is exactly what is needed, and no less.

What remains is that $D(\mathbf{A})$ is triangulated too, and that in $D(\mathbf{A})$, unlike in $\mathrm{K}(\mathbf{A})$, every short exact sequence of complexes yields a distinguished triangle. Both are taken up in section 3.2.

Exercise 3.1.1

1. Verify (TR2) for $\mathrm{K}(\mathbf{A})$ in the rotation direction. Given a chain map $\alpha^\bullet \colon L^\bullet \to M^\bullet$, construct a homotopy equivalence $MC\big(M^\bullet \to MC(\alpha^\bullet)\big) \to L[1]^\bullet$, and check that under it the third map of the rotated triangle is $-\alpha[1]^\bullet$ and not $\alpha[1]^\bullet$. Where in the construction does the sign enter?

2. Rotate a distinguished triangle (u, v, w) three times using (TR2) and write down the result. Show that it is $(-u[1], -v[1], -w[1])$ on $(A[1], B[1], C[1])$, and check that $(-\mathrm{id}, \mathrm{id}, -\mathrm{id})$ is an isomorphism of triangles from it to $(u[1], v[1], -w[1])$. Then read off the three squares as conditions $(\varepsilon_k + \varepsilon_{k+1})e_k = 0$ on the edges e_k, and show that no diagonal $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ with $\varepsilon_i = \pm\mathrm{id}$ is an isomorphism onto $(u[1], v[1], w[1])$ *provided no edge is annihilated by* 2. Then check that the diagonal $(\mathrm{id}, -\mathrm{id}, \mathrm{id})$ given above for $(\mathrm{id}_A, 0, 0)$ satisfies all three squares, and say why it does not contradict the general case. Where would the argument break if (TR2) carried no sign?

3. Let (u, v, w) be a distinguished triangle on (A, B, C) with $C = 0$. Show that u is an isomorphism. (Use proposition 3.1.8 and the argument that closes corollary 3.1.9.)

4. Show that a distinguished triangle (u, v, w) with $w = 0$ splits, in the sense that v admits a section $s \colon C \to B$ and $B \cong A \oplus C$ compatibly with u and v. Deduce that in $\mathrm{K}(\mathbf{A})$ a short exact sequence of complexes that is degreewise split gives a distinguished triangle, and find a short exact sequence of complexes that does not.

5. Let $A \subseteq B \subseteq C$ be subobjects in an abelian category, and consider the three quotient maps. Identify the objects C', A' and B' of (TR4) in this situation, and check that the fourth distinguished triangle it produces is

$$B/A \longrightarrow C/A \longrightarrow C/B$$

This is the third isomorphism theorem, and it is what (TR4) abstracts. Verify the four identities of (TR4) against it.

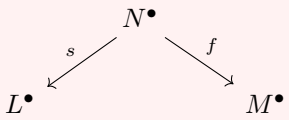
3.2 The Derived Category as a Triangulated Category

The derived category was produced in section 1.2 as a localization and recorded as corollary 2.1.17. That construction is universal and says nothing about how a morphism of $D(\mathbf{A})$ looks. This section gives the concrete description, shows that $D(\mathbf{A})$ inherits the structure of definition 3.1.3, and proves the property that motivated the whole apparatus: in $D(\mathbf{A})$, and not in $\mathrm{K}(\mathbf{A})$, every short exact sequence of complexes becomes a distinguished triangle. The treatment follows Aluffi's *Algebra: Chapter 0* [1] and Weibel [23] §10.4.

The objects of $D(\mathbf{A})$ are those of $\mathrm{Ch}(\mathbf{A})$. What changes is the morphisms, because the quasi-isomorphisms must become invertible. A first attempt would localize $\mathrm{Ch}(\mathbf{A})$ directly. It fails, and for a reason already recorded in box 2.1.18: the quasi-isomorphisms of $\mathrm{Ch}(\mathbf{A})$ do not form a multiplicative system, so theorem 1.2.5 does not apply and the localization is produced only as a category whose Hom-classes need not be sets. In $\mathrm{K}(\mathbf{A})$ they do form a multiplicative system, and the calculus of fractions then presents every morphism concretely.

Definition 3.2.1: Roof

Let $L^\bullet, M^\bullet \in \mathrm{K}(\mathbf{A})$. A *roof* from $L^\bullet$ to $M^\bullet$ is a diagram



in $\mathrm{K}(\mathbf{A})$ in which s is a quasi-isomorphism. Two roofs $(N^\bullet, s, f)$ and $(N'^\bullet, s', f')$ are *equivalent* if there is a third roof $(N''^\bullet, t, g)$ and morphisms $a\colon N''^\bullet \to N^\bullet$ and $b\colon N''^\bullet \to N'^\bullet$ with $sa = t = s'b$ and $fa = g = f'b$ in $\mathrm{K}(\mathbf{A})$. The maps a and b are then automatically quasi-isomorphisms, by two-out-of-three applied to $sa = t$ and $s'b = t$, so nothing is lost by not requiring it.

The roof is the fraction $f \circ s^{-1}$ written before s^{-1} exists. Composition is where the multiplicative system earns its place: to compose fs^{-1} with gt^{-1} one must rewrite $s^{-1}g$ as $g'(s')^{-1}$ for some new roof, and the Ore condition inside theorem 1.2.5 is exactly the hypothesis that this is always possible.

Everything in this section rests on one theorem, and it is the reason the passage from $\mathrm{Ch}(\mathbf{A})$ to $\mathrm{K}(\mathbf{A})$ is made at all. In $\mathrm{Ch}(\mathbf{A})$ the quasi-isomorphisms are not a multiplicative system, as box 2.1.18 shows by example. In $\mathrm{K}(\mathbf{A})$ they are, and the proof is where the triangulated structure of section 3.1 first does work.

One preliminary is needed. Proposition 2.1.24 gives a long exact cohomology sequence for a mapping cone triangle, and since every distinguished triangle of $\mathrm{K}(\mathbf{A})$ is isomorphic to one, that is a statement about all of them.

Lemma 3.2.2: Cohomology Is Cohomological on the Homotopy Category

For each n the functor $H^n\colon \mathrm{K}(\mathbf{A}) \to \mathbf{A}$ is cohomological in the sense of definition 3.1.6. In particular, if (u, v, w) is a distinguished triangle of $\mathrm{K}(\mathbf{A})$ on (A, B, C) with C exact, then u is a quasi-isomorphism.

***Proof*:**
H^n is well defined on $\mathrm{K}(\mathbf{A})$ because homotopic maps induce the same map on cohomology (proposition 2.1.31).

Let (u, v, w) be distinguished. By theorem 3.1.12 it is isomorphic to a mapping cone triangle, for which proposition 2.1.24 gives the long exact cohomology sequence, exact in particular at the middle term. An isomorphism of triangles induces isomorphisms on each H^n, and exactness transports along them, so the sequence $H^n(A) \to H^n(B) \to H^n(C)$ is exact for (u, v, w) as well.

For the consequence, rotate by (TR2) and splice as after definition 3.1.6 to obtain

$$H^{n-1}(C) \to H^n(A) \xrightarrow{H^n(u)} H^n(B) \to H^n(C)$$

exact for every n. If C is exact the outer terms vanish, so each $H^n(u)$ is an isomorphism.

The theorem itself is Weibel [23] Proposition 10.4.1(1). Both substantial conditions follow the same pattern: replace a quasi-isomorphism by the triangle it sits in, note that the third vertex is exact, and use the exactness of Hom on that triangle to factor the morphism that must be killed.

Theorem 3.2.3: Quasi-Isomorphisms Form a Multiplicative System

Let $\mathbf{A}$ be an abelian category. The class of quasi-isomorphisms in $\mathrm{K}(\mathbf{A})$ is a multiplicative system in the sense of definition 1.2.4.

***Proof*:**
Throughout, if t is a quasi-isomorphism then any distinguished triangle on t has exact third vertex: the mapping cone triangle on t has third vertex $MC(t)$, exact by proposition 2.1.26, and any other is isomorphic to it by proposition 3.1.10, an isomorphism of $\mathrm{K}(\mathbf{A})$ being a homotopy equivalence and so preserving cohomology (proposition 2.1.33).

Condition (1). Each $H^n(\mathrm{id})$ is an identity, and $H^n(gf) = H^n(g)H^n(f)$, so identities are quasi-isomorphisms and a composite of two is one.

Condition (2), the Ore condition. Let $f\colon Z \to B$ be any morphism and $t\colon V \to B$ a quasi-isomorphism. Embed t in a distinguished triangle

$$V \xrightarrow{t} B \xrightarrow{g} C \longrightarrow V[1]$$

by (TR1). By the paragraph above, C is exact. Apply (TR1) to $gf\colon Z \to C$ and rotate backwards by (TR2) to obtain a distinguished triangle

$$Z' \xrightarrow{t'} Z \xrightarrow{gf} C \longrightarrow Z'[1]$$

Its third vertex C is exact, so t' is a quasi-isomorphism by lemma 3.2.2.

It remains to factor ft' through t. By lemma 3.1.7 applied to the second triangle, $(gf)t' = 0$,

that is $g(ft') = 0$. Apply $\mathrm{Hom}_{\mathrm{K}(\mathbf{A})}(Z', -)$ to the first triangle: by proposition 3.1.8 the sequence

$$\mathrm{Hom}(Z', V) \xrightarrow{t_*} \mathrm{Hom}(Z', B) \xrightarrow{g_*} \mathrm{Hom}(Z', C)$$

is exact at the middle term. Since ft' lies in $\ker g_*$, there is $f' \colon Z' \to V$ with $tf' = ft'$, which is condition (2).

Condition (3). Let $f, g \colon X \to Y$ and let $s \colon Y \to Y'$ be a quasi-isomorphism with $sf = sg$. Put $h := f - g$, using that $\mathrm{K}(\mathbf{A})$ is additive (lemma 2.1.39). Then $sh = 0$, and it suffices to find a quasi-isomorphism t with $ht = 0$.

Embed s in a distinguished triangle and rotate backwards to put s in the middle:

$$W \xrightarrow{r} Y \xrightarrow{s} Y' \longrightarrow W[1]$$

The third vertex of a triangle on s is exact, and W is its shift, so W is exact as well. Applying $\mathrm{Hom}_{\mathrm{K}(\mathbf{A})}(X, -)$ and proposition 3.1.8 gives exactness of

$$\mathrm{Hom}(X, W) \xrightarrow{r_*} \mathrm{Hom}(X, Y) \xrightarrow{s_*} \mathrm{Hom}(X, Y')$$

at the middle term, and $h \in \ker s_*$, so $h = rh'$ for some $h' \colon X \to W$.

Now apply (TR1) to h' and rotate backwards, giving a distinguished triangle

$$X' \xrightarrow{t} X \xrightarrow{h'} W \longrightarrow X'[1]$$

whose third vertex W is exact, so t is a quasi-isomorphism by lemma 3.2.2. By lemma 3.1.7, $h't = 0$, whence

$$ht = rh't = 0$$

that is, $ft = gt$. This is condition (3), and the three conditions together are the definition.

The theorem discharges an obligation the book has carried since corollary 2.1.17. Box 2.1.18 observed that the quasi-isomorphisms of $\mathrm{Ch}(\mathbf{A})$ are not a multiplicative system and pointed here for the proof that those of $\mathrm{K}(\mathbf{A})$ are. That makes roofs available. Local smallness needs in addition the cofinal set of box 1.2.3, taken up after the next theorem.

Theorem 3.2.4: Roofs Compute the Derived Category

Let $\mathbf{A}$ be an abelian category. Then every morphism $L^\bullet \to M^\bullet$ of $D(\mathbf{A})$ is represented by a roof, two roofs represent the same morphism exactly when they are equivalent, composition is induced by the Ore condition, and the localization functor $Q\colon \mathrm{K}(\mathbf{A}) \to D(\mathbf{A})$ sends f to the class of the roof $(L^\bullet, \mathrm{id}, f)$.
The equivalence classes form a *set* when $\mathbf{A}$ is such that the quasi-isomorphisms into each complex admit a cofinal set. Without that hypothesis they are not guaranteed to be sets. The hypothesis is sufficient and is not necessary for a particular Hom-class to be a set: $\mathrm{Hom}_{D(\mathbf{A})}(0, M^\bullet)$ is a singleton for every $\mathbf{A}$, since any two roofs out of 0 are equivalent through the roof $(0, \mathrm{id}_0, 0)$.

***Proof*:**
By theorem 3.2.3 the quasi-isomorphisms of $\mathrm{K}(\mathbf{A})$ form a multiplicative system, so theorem 1.2.5 applies and gives this description of $\mathrm{K}(\mathbf{A})[\mathrm{qis}^{-1}]$.

It remains to identify that localization with $D(\mathbf{A})$, which corollary 2.1.17 defines as $\mathrm{Ch}(\mathbf{A})[\mathrm{qis}^{-1}]$ and so does not do on its own. The bridge is proposition 2.1.40: *any* functor out of $\mathrm{Ch}(\mathbf{A})$ inverting quasi-isomorphisms factors uniquely through $\mathrm{K}(\mathbf{A})$, and the factored functor inverts the quasi-isomorphisms there, those being the images of quasi-isomorphisms. So a functor out of $\mathrm{Ch}(\mathbf{A})$ inverting quasi-isomorphisms is the same thing as a functor out of $\mathrm{K}(\mathbf{A})$ doing so, compatibly with composition, and the two localizations satisfy the same universal property of definition 1.2.1 over *all* functors. They are therefore canonically isomorphic.

The quantifier matters. Restricting the comparison to *additive* functors would be circular, since the additive structure on $D(\mathbf{A})$ is derived in theorem 3.2.7 from the roof description being established here.

Inverting a single morphism is the case $M^\bullet = N^\bullet$, $f = \mathrm{id}$: the roof $(N^\bullet, s, \mathrm{id})$ is the inverse of $Q(s)$.

The set-hood clause is not a formality, and box 1.2.3 is where the difficulty was recorded: a calculus of fractions replaces zigzags by roofs without bounding anything, the roofs over a fixed object being indexed by a proper class. What bounds them is a *set* of quasi-isomorphisms cofinal among those into each complex. Every construction in this chapter is stated for $D(\mathbf{A})$ as a category. Only the statements quantifying over its Hom-*sets* need the hypothesis, and they say so. For a Grothendieck abelian category, which includes R-Mod and the categories of sheaves of abelian groups on a site (example 5.1.5), every Hom-class of $D(\mathbf{A})$ is a set by corollary 5.1.21, whose proof uses K-injective resolutions and no cofinality hypothesis.

3.2.5 Note: What the Bounded Variants Are For Writing $D^+(\mathbf{A})$, $D^-(\mathbf{A})$ and $D^b(\mathbf{A})$ for the full subcategories of $D(\mathbf{A})$ on the complexes with cohomology bounded below, bounded above, and bounded respectively, each is *equivalent* to the localization of the corresponding $\mathrm{K}^{\pm}(\mathbf{A})$ at its quasi-isomorphisms. Equivalent and not equal: the localization of $\mathrm{K}^b(\mathbf{A})$ has only bounded complexes as objects, whereas $D^b(\mathbf{A})$ as defined here contains every complex with bounded cohomology, such as the exact 2-periodic complex $\cdots \to \mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4 \to \cdots$. Truncation gives essential surjectivity: if $H^i(X) = 0$ for $i < m$, then $p_m \colon X \to \tau^{\geq m} X$ of definition 2.1.12 is a quasi-isomorphism onto a bounded-below complex by proposition 2.1.14, and dually $i_m \colon \tau^{\leq m} X \to X$ when $H^i(X) = 0$ for $i > m$. For the bounded case, if $H^i(X) = 0$ outside $[m, M]$ then $i_M \colon \tau^{\leq M} X \to X$ is a quasi-isomorphism, and so is $p_m \colon \tau^{\leq M} X \to \tau^{\geq m} \tau^{\leq M} X$, since $H^i(\tau^{\leq M} X) \cong H^i(X) = 0$ for $i < m$ by proposition 2.1.14, and $\tau^{\geq m} \tau^{\leq M} X$ is bounded. The equivalence is Weibel [23] 10.3.15. The bound is what makes derived functors constructible: a resolution of a bounded-above complex by projectives is again bounded above, and section 1.2's machinery has something to run on. Chapter 3 states results for $D(\mathbf{A})$ where no bound is needed and says so explicitly where one is.

When $\mathbf{A}$ has enough projectives the bounded-above derived category has a second, entirely computational, description, and the book already has it.

Proposition 3.2.6: Enough Projectives Computes D^-

Let $\mathbf{A}$ be an abelian category with enough projectives and let $\mathbf{P}$ be its full subcategory of projectives. Then $\mathrm{K}^-(\mathbf{P})$ satisfies the universal property of $D^-(\mathbf{A})$: every additive functor out of $C^-(\mathbf{A})$ carrying quasi-isomorphisms to isomorphisms factors through $\mathrm{K}^-(\mathbf{P})$, uniquely up to natural isomorphism.

***Proof*:**
This is theorem 2.2.18, whose part (1) gives that a quasi-isomorphism of $C^-(\mathbf{A})$ becomes an isomorphism in $\mathrm{K}^-(\mathbf{P})$ and whose part (2) is the factorization. That quasi-isomorphisms are already isomorphisms in $\mathrm{K}^-(\mathbf{P})$, so that no further localization is needed, is corollary 2.2.10.

The point is worth stating plainly because it is easy to misremember in a form that is false. In general $\mathrm{K}(\mathbf{A})$ is *not* the localization of $\mathrm{Ch}(\mathbf{A})$ at the quasi-isomorphisms, with or without enough projectives. It inverts the homotopy equivalences, and for $\mathbf{A} = \mathbf{Ab}$ those are strictly fewer: example 2.1.16 exhibits a quasi-isomorphism with no inverse in $\mathrm{K}(\mathbf{Ab})$. What enough projectives gives is that the homotopy category of *projective* complexes, $\mathrm{K}^-(\mathbf{P})$, already has the quasi-isomorphisms inverted, and that is a statement about $\mathbf{P}$ and not about $\mathbf{A}$.

The hedge *in general* is doing work. Over a semisimple $\mathbf{A}$, say Vect_k, which has enough projectives, every complex splits as the direct sum of its cohomology and contractible pieces, so every quasi-isomorphism is already a homotopy equivalence and the two classes coincide. There $\mathrm{K}(\mathbf{A})$ *is* the localization, so over such an $\mathbf{A}$ the claim above is false there. That is why the general statement is given with a cited witness.

With the morphisms described, the structure of section 3.1 transfers.

Theorem 3.2.7: The Derived Category Is Triangulated

Let $\mathbf{A}$ be an abelian category. Then $D(\mathbf{A})$ is a triangulated category, with translation the shift [1] and with distinguished triangles those isomorphic in $D(\mathbf{A})$ to the image under Q of a distinguished triangle of $\mathrm{K}(\mathbf{A})$. The localization functor $Q\colon \mathrm{K}(\mathbf{A}) \to D(\mathbf{A})$ is a triangulated functor in the sense of definition 3.1.5. The same holds for D^{+}, D^{-} and D^{b}.

***Proof*:**
The shift commutes with localization, since it carries quasi-isomorphisms to quasi-isomorphisms, and so descends to an automorphism of $D(\mathbf{A})$. $D(\mathbf{A})$ is additive because $\mathrm{K}(\mathbf{A})$ is (lemma 2.1.39) and the localization of an additive category at a multiplicative system is again additive, sums of roofs being formed over a common denominator and biproducts surviving. Theorem 1.2.5 is a statement about categories with no additive content, and the fact used here is Weibel [23] 10.3.11.

(TR1). A morphism of $D(\mathbf{A})$ is a roof fs^{-1}, and $Q(s)$ is invertible, so it is enough to embed f. Theorem 3.1.12 does that in $\mathrm{K}(\mathbf{A})$, and applying Q and composing with the isomorphism $Q(s)$ gives a distinguished triangle on fs^{-1}. The triangle $(\mathrm{id}, 0, 0)$ is the Q-image of the corresponding triangle of $\mathrm{K}(\mathbf{A})$, and closure under isomorphism holds because the class was defined as an isomorphism closure.

(TR2), (TR3), (TR4). These are Weibel [23] Proposition 10.4.1(2), and the deferral is deliberate. The obstacle is that a distinguished triangle of $D(\mathbf{A})$ is only *isomorphic* to a Q-image, and that (TR3)'s hypothesis square commutes in $D(\mathbf{A})$ and so only up to homotopy after lifting. Verdier's argument straightens the data with the Ore condition, rebuilds the triangles over the refined vertices using (TR1), (TR3) and corollary 3.1.9, applies the axiom in $\mathrm{K}(\mathbf{A})$, and conjugates back. Weibel calls the verification straightforward but lengthy, and it needs two-sided fractions, which definition 1.2.4 does not carry. No argument in this book depends on reproducing it.

Q carries distinguished triangles to distinguished triangles by the definition given, and commutes with the shift by the first paragraph, so it is a triangulated functor. The bounded cases are the same argument applied to $\mathrm{K}^{\pm}(\mathbf{A})$.

Proposition 3.2.8: Cohomology Is Cohomological on $D(\mathbf{A})$

For each n the functor $H^{n}\colon D(\mathbf{A}) \to \mathbf{A}$ is well defined and cohomological in the sense of definition 3.1.6.

***Proof*:**
H^{n} is well defined on $D(\mathbf{A})$ because a quasi-isomorphism induces an isomorphism on H^{n} by definition, so H^{n} inverts the class being localized and factors through Q by the universal property of theorem 1.2.2.

For the cohomological property, let (u, v, w) be distinguished in $D(\mathbf{A})$. By definition it is

isomorphic to Q of a distinguished triangle of $\mathrm{K}(\mathbf{A})$, which by theorem 3.1.12 is isomorphic to a mapping cone triangle. For a mapping cone triangle the required exactness at the middle term is proposition 2.1.24, which produces exactly the long exact cohomology sequence of the cone. Exactness is preserved along the isomorphisms, so it holds for (u, v, w).

The failure that section 3.1 opened with can now be repaired. A short exact sequence of complexes has a connecting morphism on cohomology but, in $\mathrm{K}(\mathbf{A})$, need not have any morphism $N^\bullet \to L[1]^\bullet$ at all. The cone gives a candidate, and in $D(\mathbf{A})$ the candidate is an isomorphism.

Lemma 3.2.9: The Cone of an Injection Computes the Quotient

Let

$$0 \to L^\bullet \xrightarrow{u} M^\bullet \xrightarrow{v} N^\bullet \to 0$$

be a short exact sequence in $\mathrm{Ch}(\mathbf{A})$, meaning exact in each degree. Define $\varphi\colon MC(u)^\bullet \to N^\bullet$ by $\varphi^i(l, m) := v^i(m)$, where $MC(u)^i = L^{i+1} \oplus M^i$ as in definition 2.1.23. Then φ is a chain map and a quasi-isomorphism.

***Proof*:**

Step 1: φ is a chain map. With the differential of definition 2.1.23,

$$\varphi^{i+1}\big(d^i_{MC(u)}(l, m)\big) = \varphi^{i+1}\big(-d^{i+1}_L(l),\ u^{i+1}(l) + d^i_M(m)\big) = v^{i+1}u^{i+1}(l) + v^{i+1}d^i_M(m)$$

The first term vanishes because $vu = 0$, the sequence being exact at $M^\bullet$. The second is $d^i_N v^i(m)$ because v is a chain map. So $\varphi^{i+1}d^i_{MC(u)} = d^i_N\varphi^i$.

Step 2: the kernel of φ. In degree i, $\ker\varphi^i = \{(l, m) : v^i(m) = 0\} = L^{i+1} \oplus \ker v^i = L^{i+1} \oplus u^i(L^i)$, using exactness at $M^\bullet$. Since u is a monomorphism, $u^i(L^i) \cong L^i$, so $\ker\varphi^\bullet$ is the complex with $(\ker\varphi)^i = L^{i+1} \oplus L^i$ and differential $(l, l') \mapsto (-d^{i+1}_L(l),\ l + d^i_L(l'))$ read through u. This is exactly $MC(\mathrm{id}_{L^\bullet})$.

Step 3: φ is surjective in each degree, since v is. So

$$0 \to MC(\mathrm{id}_{L^\bullet}) \to MC(u)^\bullet \xrightarrow{\varphi} N^\bullet \to 0$$

is a short exact sequence of complexes. By lemma 3.1.11 the cone on an identity map is null-homotopic, hence exact, so $H^i\big(MC(\mathrm{id}_{L^\bullet})\big) = 0$ for every i. The long exact cohomology sequence of the displayed short exact sequence (theorem 2.1.20) then has zeroes flanking every $H^i(\varphi)$, forcing each $H^i(\varphi)$ to be an isomorphism. Thus φ is a quasi-isomorphism.

Theorem 3.2.10: Every Short Exact Sequence Gives a Distinguished Triangle in $D(\mathbf{A})$

Let $0 \to L^\bullet \xrightarrow{u} M^\bullet \xrightarrow{v} N^\bullet \to 0$ be a short exact sequence in $\mathrm{Ch}(\mathbf{A})$. Then there is a morphism $w\colon N^\bullet \to L[1]^\bullet$ in $D(\mathbf{A})$ for which

$$L^\bullet \xrightarrow{u} M^\bullet \xrightarrow{v} N^\bullet \xrightarrow{w} L[1]^\bullet$$

is a distinguished triangle of $D(\mathbf{A})$, and $H^\bullet(w) = -\delta$, where δ is the connecting morphism of theorem 2.1.20.

***Proof*:**
Let $\iota\colon M^\bullet \to MC(u)^\bullet$ and $\pi\colon MC(u)^\bullet \to L[1]^\bullet$ be the two natural chain maps of definition 2.1.23, so that (u, ι, π) is the mapping cone triangle on u and is distinguished in $\mathrm{K}(\mathbf{A})$ by theorem 3.1.12. Let $\varphi\colon MC(u)^\bullet \to N^\bullet$ be the quasi-isomorphism of lemma 3.2.9.

Step 1: φ is invertible in $D(\boldsymbol{A})$. It is a quasi-isomorphism, and Q inverts exactly those.

Step 2: the triangle. Set $w := Q(\pi) \circ Q(\varphi)^{-1}\colon N^\bullet \to L[1]^\bullet$. The triple $(\mathrm{id}_{L^\bullet}, \mathrm{id}_{M^\bullet}, Q(\varphi))$ is an isomorphism of triangles from $Q(u, \iota, \pi)$ to (u, v, w), provided the middle square commutes, that is $\varphi \circ \iota = v$. In degree i, $\iota^i(m) = (0, m)$ and $\varphi^i(0, m) = v^i(m)$, so the square commutes on the nose in $\mathrm{Ch}(\mathbf{A})$.

The third square commutes by the definition of w. Since $Q(u, \iota, \pi)$ is distinguished in $D(\mathbf{A})$ by theorem 3.2.7, and distinguished triangles are closed under isomorphism by (TR1), the triangle (u, v, w) is distinguished.

Step 3: the sign. Let $x \in N^n$ be a cocycle. Choose $m \in M^n$ with $v(m) = x$. Then $v(d_M m) = d_N x = 0$, so $d_M m = u(\lambda)$ for a unique $\lambda \in L^{n+1}$, and $\delta[x] = [\lambda]$ by the construction in theorem 2.1.20. Now compute $H^n(w)[x]$ directly. A cocycle of $MC(u)^n$ lying over x is a pair (l, m) with $\varphi(l, m) = v(m) = x$ and

$$0 = d_{MC(u)}(l, m) = \big(-d_L(l),\ u(l) + d_M(m)\big)$$

The second component forces $u(l) = -d_M m = -u(\lambda)$, and u is a monomorphism, so $l = -\lambda$. Such a pair does exist: the first component asks $d_L(\lambda) = 0$, which holds because $u(d_L\lambda) = d_M(u\lambda) = d_M d_M m = 0$ and u is a monomorphism. Since $w = Q(\pi) \circ Q(\varphi)^{-1}$ and $\pi(l, m) = l$ carries no sign in definition 2.1.23,

$$H^n(w)[x] = [l] = [-\lambda] = -\delta[x]$$

so $H^\bullet(w) = -\delta$.

The sign is not an artifact of this book's conventions, which is the natural thing to assume and is wrong. Definition 2.1.23 puts the sign on d_L and none on α or on the projection π. Weibel flips both: his cone differential carries $-\alpha$ and his projection carries a minus. Under his conventions the second component forces $u(l) = +d_M m$, so $l = +\lambda$, and the signed projection returns $-\lambda$

again. The two flips cancel each other, and $-\delta$ is what both conventions produce. A mixed convention, flipping exactly one of the two, is what would give $+\delta$.

Nor is the sign available to be absorbed into w. The diagonal comparison of exercise 3.1.1 (2) fails for $(u, v, -w)$ against (u, v, w), so $(u, v, -w)$ is not known to be distinguished.

The contrast with $\mathrm{K}(\mathbf{A})$ is not a matter of the argument being harder there. It is false there.

Example 3.2.11: A Short Exact Sequence That Is No Triangle in $\mathrm{K}(\mathbf{Ab})$

Regard $\mathbb{Z}/2$ and $\mathbb{Z}/4$ as complexes concentrated in degree 0, and take the short exact sequence of abelian groups

$$0 \to \mathbb{Z}/2 \xrightarrow{\cdot 2} \mathbb{Z}/4 \to \mathbb{Z}/2 \to 0$$

There is no morphism w in $\mathrm{K}(\mathbf{Ab})$ making $(\cdot 2, \pi, w)$ a distinguished triangle on $(\mathbb{Z}/2, \mathbb{Z}/4, \mathbb{Z}/2)$.

Suppose there were, and apply $\mathrm{Hom}_{\mathrm{K}(\mathbf{Ab})}(\mathbb{Z}/2, -)$, which is cohomological by proposition 3.1.8. Every complex in sight is concentrated in degree 0, so $\mathbb{Z}/2[1]$ and $\mathbb{Z}/2[-1]$ are concentrated in degrees -1 and 1, and a chain map between complexes concentrated in different degrees is zero. Both outer terms of the long exact sequence therefore vanish, leaving

$$0 \to \mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}/2) \to \mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}/4) \to \mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}/2) \to 0$$

exact in $\mathbf{Ab}$, homotopies between degree-0 complexes being zero. Counting: the two outer groups have order 2, so the middle must have order 4. But $\mathrm{Hom}_{\mathbf{Ab}}(\mathbb{Z}/2, \mathbb{Z}/4)$ consists of the maps sending 1 to an element killed by 2, namely 0 and 2, so it has order 2. The contradiction shows no such w exists.

The argument used nothing about w itself, which is the point. The obstruction is not that the third map is hard to write down. It is that no *morphism* closes the triangle at all, whatever is placed at the third edge. The object $\mathbb{Z}/2$ does sit at that vertex in other distinguished triangles, and $\cdot 2$ does embed in one, namely its own cone triangle with third vertex $MC(\cdot 2)$.

In $D(\mathbf{Ab})$ the triangle exists, by theorem 3.2.10. What differs is not the category's objects but its morphisms: w is a roof, invisible from $\mathrm{K}(\mathbf{Ab})$.

Two criteria fall out of the roof description and are used constantly.

Proposition 3.2.12: Recognizing Zero in the Derived Category

Let $\mathbf{A}$ be an abelian category.

1. A complex $X^\bullet$ is isomorphic to 0 in $D(\mathbf{A})$ if and only if $X^\bullet$ is exact.
2. A chain map $f\colon X^\bullet \to Y^\bullet$ becomes the zero morphism in $D(\mathbf{A})$ if and only if there is a quasi-isomorphism $t\colon X''^\bullet \to X^\bullet$ with ft null-homotopic.

***Proof*:**

(1) If $X^\bullet$ is exact then $X^\bullet \to 0$ is a quasi-isomorphism, which Q inverts, so $X^\bullet \cong 0$ in $D(\mathbf{A})$. Conversely, if $X^\bullet \cong 0$ there, then $H^i(X^\bullet) \cong H^i(0) = 0$ for every i, since H^i is a functor on $D(\mathbf{A})$ by proposition 3.2.8. So $X^\bullet$ is exact.

(2) By theorem 3.2.4 the morphism $Q(f)$ is the class of the roof $(X^\bullet, \mathrm{id}, f)$ and the zero morphism is the class of $(X^\bullet, \mathrm{id}, 0)$. Two roofs with the same left leg represent the same morphism exactly when their right legs agree after precomposition with a quasi-isomorphism, which is the equivalence of definition 3.2.1. So $Q(f) = 0$ exactly when there is a quasi-isomorphism t with $ft = 0$ in $\mathrm{K}(\mathbf{A})$, and equality in $\mathrm{K}(\mathbf{A})$ is null-homotopy in $\mathrm{Ch}(\mathbf{A})$.

The criterion is stated with t on the *source* because that is the side definition 1.2.4 controls: its condition (3) cancels a quasi-isomorphism on the target and returns one on the source, and not conversely. The dual form, a quasi-isomorphism s on the target with sf null-homotopic, is equivalent, but deriving it needs the opposite cancellation condition, which the definition used here does not carry. Weibel [23] states the axioms in self-dual form for exactly this reason.

3.2.13 Note: Recovering A from $D(\mathbf{A})$ Distinct abelian categories can have equivalent derived categories, so $\mathbf{A}$ is not recovered from $D(\mathbf{A})$ on its own. What recovers it is an extra piece of data, a *t-structure*, whose heart is an abelian category. Section 5.3 constructs the standard t-structure on $D(\mathbf{A})$, whose heart is equivalent to $\mathbf{A}$, and section 5.4 develops the general theory. The middle-perversity t-structure, whose heart is the category of perverse sheaves, is the subject of the *Perverse Sheaves* volume [13].

3.2.1 Stable Module Theory

Two maps $f, g : M \to N$ of R-modules are *projectively homotopic* when $f - g$ factors through a projective module, $f - g : M \to P \to N$. This is the module-level form of the chain homotopy of section 2.1, and it recovers the geometric intuition directly: two modules can be homotopy equivalent without either one being resolved first. The associated equivalence on objects is *stable equivalence*: modules M and N are stably equivalent when

$$M \oplus P \cong N \oplus P$$

for some projective P. This relation is the starting point of K_0, developed in [5].

Exercise 3.2.1

1. Let $s\colon N^\bullet \to L^\bullet$ be a quasi-isomorphism. Show that the roof $(N^\bullet, s, \mathrm{id}_{N^\bullet})$ represents a two-sided inverse to $Q(s)$ in $D(\mathbf{A})$, and that the equivalence relation of definition 3.2.1 is what makes the two composites equal to the identity roofs.

2. Give a complex of abelian groups that is exact, nonzero in infinitely many degrees, and therefore isomorphic to 0 in $D(\mathbf{Ab})$ while being isomorphic to 0 in neither $\mathrm{Ch}(\mathbf{Ab})$ nor $\mathrm{K}(\mathbf{Ab})$. (For the last point, exhibit the failure of a contracting homotopy, or cite exercise 3.1.1 (4).)

3. Lemma 3.1.11 shows $MC(\mathrm{id}_{L^\bullet})$ is null-homotopic, hence isomorphic to 0 in $\mathrm{K}(\mathbf{A})$. Show that it is nevertheless *not* the zero complex in $\mathrm{Ch}(\mathbf{A})$ whenever $L^\bullet \neq 0$. Then explain why lemma 3.2.9 concludes only that φ is a *quasi-isomorphism*, and not that it is an isomorphism of complexes, even though its kernel is isomorphic to 0 in $\mathrm{K}(\mathbf{A})$.

4. Find chain maps f and g of abelian groups such that f is zero in $D(\mathbf{Ab})$ without being null-homotopic, and g induces the zero map on every cohomology group without being zero in

$D(\mathbf{Ab})$. (For the second, complexes of length two and a map built from multiplication by an integer will do.)

5. Show that $Q\colon \mathrm{K}(\mathbf{Ab}) \to D(\mathbf{Ab})$ is not full by producing a morphism of $D(\mathbf{Ab})$ that is not Q of any chain map. (Example 3.2.11 contains one.)

3.3 Adapted Classes and Total Derived Functors

Section 2.3 built $R^n\mathscr{F}$ by choosing a resolution, applying $\mathscr{F}$ term by term, taking cohomology, and then proving the answer independent of the choice. The derived category was introduced to make that last step unnecessary, so the natural thing to want now is a single functor $R\mathscr{F}$ on $D^+(\mathbf{A})$ whose cohomology returns the $R^n\mathscr{F}$. The obstacle is immediate: $\mathscr{F}$ applied term by term does *not* descend to $D^+(\mathbf{A})$, because a left exact functor need not carry quasi-isomorphisms to quasi-isomorphisms. If it did, all its higher derived functors would vanish.

Example 3.3.1: Termwise Application Does Not Descend
Let $\mathscr{F} = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2, -)$ on $\mathbf{Ab}$, which is left exact. The chain map

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{2} & \mathbb{Z} \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbb{Z}/2 \end{array}$$

from the complex $(\mathbb{Z} \xrightarrow{2} \mathbb{Z})$ in degrees $-1, 0$ to $\mathbb{Z}/2$ in degree 0 is a quasi-isomorphism. Both complexes have cohomology only in degree 0, and there the induced map is the reduction $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2$, an isomorphism. Apply $\mathscr{F}$ term by term. The upper complex becomes $\mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}) \to \mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z})$, which is $0 \to 0$. The lower becomes $\mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}/2) = \mathbb{Z}/2$ in degree 0. The induced map on H^0 is $0 \to \mathbb{Z}/2$, not an isomorphism.

So the termwise functor does not send this quasi-isomorphism to an isomorphism, and by the universal property of definition 1.2.1 it does not factor through $D^+(\mathbf{Ab})$.

The upper complex is a *projective* resolution of $\mathbb{Z}/2$, and that is the diagnosis. Projective resolutions compute the derived functors of right exact functors. But $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2, -)$ is left exact, and the complexes on which it behaves are the injective ones. What follows makes that choice of complexes part of the construction.

The repair is to apply $\mathscr{F}$ only to complexes drawn from a class of objects on which it behaves, and then to check that no information is lost by the restriction. Both halves are packaged in the following definition, which is Gelfand-Manin [15] III.6.3.

Definition 3.3.2: Adapted Class

Let $\mathscr{F}\colon \mathbf{A} \to \mathbf{B}$ be a left exact additive functor. A class $\mathcal{R} \subseteq \operatorname{Ob}\mathbf{A}$, stable under finite direct sums, is *adapted* to $\mathscr{F}$ if

1. $\mathscr{F}$ carries every acyclic complex in $\mathrm{K}^+(\mathcal{R})$ to an acyclic complex, and
2. every object of $\mathbf{A}$ is a subobject of an object of $\mathcal{R}$.

For a right exact $\mathscr{F}$ the two conditions are dualized: $\mathscr{F}$ carries acyclic complexes in $\mathrm{K}^-(\mathcal{R})$ to acyclic complexes, and every object of $\mathbf{A}$ is a quotient of an object of $\mathcal{R}$.

Condition (2) is what makes the restriction harmless, since it lets every bounded-below complex be replaced by one built from $\mathcal{R}$. Condition (1) is what makes it useful.

Example 3.3.3: Adapted Classes

1. If $\mathbf{A}$ has enough injectives, the class of injective objects is adapted to every left exact $\mathscr{F}$. It is stable under finite direct sums, since $\operatorname{Hom}(-, I_1 \oplus I_2) \cong \operatorname{Hom}(-, I_1) \oplus \operatorname{Hom}(-, I_2)$ is exact when both summands are. Condition (2) is the definition of enough injectives. For condition (1), an acyclic bounded-below complex I of injectives is homotopy equivalent to 0 by the injective case of corollary 2.2.6, so id_I is null-homotopic. The additive functor $\mathscr{F}$ carries that null-homotopy to one of $\mathrm{id}_{\mathscr{F}(I)}$ (lemma 2.1.34), and a complex whose identity is null-homotopic is exact by proposition 2.1.29.

2. Dually, with enough projectives the projectives are adapted to every right exact $\mathscr{F}$.

3. Let R be a ring and N a left R-module. The flat right R-modules form a class adapted to $- \otimes_R N\colon \text{Mod-}R \to \mathbf{Ab}$. The class is stable under finite direct sums, since $(F_1 \oplus F_2) \otimes_R -$ is the direct sum of the exact functors $F_1 \otimes_R -$ and $F_2 \otimes_R -$. Every module is a quotient of a free module, and free modules are flat by [8, Projective Modules Are Flat], which is the quotient form of condition (2).

   For condition (1), let K be an acyclic bounded-above complex of flat right modules, with $K^i = 0$ for $i > a$, let $\pi\colon P \to N$ be a projective resolution with $P^j = 0$ for $j > 0$, and let C be the exact complex $\cdots \to P^{-1} \to P^0 \xrightarrow{\pi} N$ with N in degree 1. Let T be the total complex of the double complex $K^i \otimes_R C^j$, with differentials $d_K \otimes 1$ and $(-1)^i(1 \otimes d_C)$. Each column $K^i \otimes_R C$ is exact because K^i is flat, so T is exact by proposition 2.5.4, applied after re-indexing so that K and C vanish in positive degrees. After this re-indexing the twist of section 2.5 is $(-1)^{i-a}$, which differs from the sign $(-1)^i$ above by the constant $(-1)^a$, and $x \otimes y \mapsto (-1)^{aj}\, x \otimes y$ for $y \in C^j$ is an isomorphism between the two total complexes, so exactness transfers. The same holds for the quotient below. The terms with $j = 1$ form a subcomplex S with $S^n = K^{n-1} \otimes_R N$ and differential $d_K \otimes 1$, because d_C vanishes on C^1. The quotient T/S is the total complex of $K^i \otimes_R P^j$, whose rows $K \otimes_R P^j$ are exact because each P^j is projective, hence flat. So T/S is exact by proposition 2.5.4 again, and the long exact sequence of theorem 2.1.20 for $0 \to S \to T \to T/S \to 0$ makes S exact, which says that $K \otimes_R N$ is exact.

   This class can be strictly larger than the class of projectives: over $\mathbb{Z}$ the module $\mathbb{Q}$ is flat

and not projective ([8, Flat Modules]). That is the point of allowing a class: a flat resolution, projective or not, computes $L(-\otimes_R N)$.

An adapted class computes the derived category, in the following precise sense. Write $S_{\mathcal{R}}$ for the class of quasi-isomorphisms in $\mathrm{K}^+(\mathcal{R})$.

Theorem 3.3.4: An Adapted Class Computes D^+

Let $\mathcal{R}$ be adapted to a left exact $\mathscr{F}\colon \mathbf{A}\to\mathbf{B}$. Then $S_{\mathcal{R}}$ is a multiplicative system in $\mathrm{K}^+(\mathcal{R})$, and the canonical functor

$$\mathrm{K}^+(\mathcal{R})\big[S_{\mathcal{R}}^{-1}\big]\longrightarrow D^+(\mathbf{A})$$

is an equivalence of categories. The corresponding statement with K^- and D^- holds for a right exact $\mathscr{F}$.

The proof is Gelfand-Manin [15] III.6.4. It repeats their construction of injective resolutions of complexes (III.5.22) with $\mathcal{R}$ in place of the injectives, and its input is the single fact about $\mathcal{R}$ that condition (2) is for: every bounded-below complex K admits a quasi-isomorphism $K\to L$ with L in $\mathrm{K}^+(\mathcal{R})$ (property $(*)$ in that proof). Full faithfulness follows through Gelfand-Manin's criterion III.2.10, condition b2: for every quasi-isomorphism $s\colon L'\to K$ with L' in $\mathrm{K}^+(\mathcal{R})$ there is a quasi-isomorphism $t\colon K\to L$ with ts in $S_{\mathcal{R}}$, which $(*)$ provides. That criterion uses fractions with the quasi-isomorphism on the target, the form theorem 3.2.3 takes when applied to $\mathbf{A}^{\mathrm{op}}$. Essential surjectivity follows from $(*)$ once an object of $D^+(\mathbf{A})$ has been replaced by a bounded-below complex, which is the truncation step of box 3.2.5, by the canonical truncation $\tau^{\geq m}$ of definition 2.1.12 (Weibel [23] 10.3.15).

With the equivalence in hand the construction is short.

Definition 3.3.5: Total Derived Functor

Let $\mathscr{F}\colon \mathbf{A}\to\mathbf{B}$ be left exact and $\mathcal{R}$ a class adapted to it. Define $R\mathscr{F}$ on $\mathrm{K}^+(\mathcal{R})$ term by term,

$$\big(R\mathscr{F}\big)(K^\bullet)^i := \mathscr{F}(K^i)$$

Condition (1) of definition 3.3.2 makes $R\mathscr{F}$ carry the quasi-isomorphisms of $\mathrm{K}^+(\mathcal{R})$ to isomorphisms, so it descends to $\mathrm{K}^+(\mathcal{R})[S_{\mathcal{R}}^{-1}]\to D^+(\mathbf{B})$. Composing with an inverse $\Phi\colon D^+(\mathbf{A})\to\mathrm{K}^+(\mathcal{R})[S_{\mathcal{R}}^{-1}]$ of the equivalence of theorem 3.3.4 gives the *total right derived functor*

$$R\mathscr{F}\colon D^+(\mathbf{A})\longrightarrow D^+(\mathbf{B})$$

The total left derived functor $L\mathscr{F}$ of a right exact $\mathscr{F}$ is defined dually, on D^-.

That $R\mathscr{F}$ carries quasi-isomorphisms to isomorphisms is where condition (1) enters. A morphism $u\colon K\to L$ of $\mathrm{K}^+(\mathcal{R})$ is a quasi-isomorphism exactly when its cone is acyclic (proposition 2.1.26). The cone has terms $K^{n+1}\oplus L^n$, which lie in $\mathcal{R}$ because $\mathcal{R}$ is stable under finite direct sums, so it is a complex of $\mathrm{K}^+(\mathcal{R})$ and condition (1) makes its image under $\mathscr{F}$ acyclic. Being additive, $\mathscr{F}$ commutes with the formation of cones, so $\mathscr{F}(u)$ has acyclic cone and is a quasi-isomorphism.

The definition depends on two choices, $\mathcal{R}$ and Φ, and neither is canonical. What makes $R\mathscr{F}$ well defined regardless is a universal property. To state it, write $\Psi\colon \mathrm{K}^+(\mathcal{R})[S_{\mathcal{R}}^{-1}]\to D^+(\mathbf{A})$ for the equivalence of theorem 3.3.4, $\beta\colon \mathrm{id}\to\Psi\Phi$ for the natural isomorphism that makes Φ an inverse of Ψ,

$Q_{\mathcal{R}}$ for the localization functor of $\mathrm{K}^+(\mathcal{R})$, and $\bar{\mathscr{F}}\colon \mathrm{K}^+(\mathcal{R})[S_{\mathcal{R}}^{-1}] \to D^+(\mathbf{B})$ for the functor induced by $\mathscr{F}$ term by term, so that $R\mathscr{F} = \bar{\mathscr{F}}\Phi$.

Theorem 3.3.6: The Total Derived Functor Is Universal

Let $\mathscr{F}\colon \mathbf{A} \to \mathbf{B}$ be left exact, let $\mathcal{R}$ be adapted to $\mathscr{F}$, and let $R\mathscr{F}$ be built from $\mathcal{R}$ and Φ as in definition 3.3.5. Write $Q_{\mathbf{A}}$ and $Q_{\mathbf{B}}$ for the localization functors of $\mathrm{K}^+(\mathbf{A})$ and $\mathrm{K}^+(\mathbf{B})$. For $X \in \mathrm{K}^+(\mathbf{A})$ put $Y := \Phi(Q_{\mathbf{A}}X)$, choose a quasi-isomorphism $s\colon X \to Z$ with Z in $\mathrm{K}^+(\mathcal{R})$, let $\gamma\colon Y \to Z$ be the isomorphism of $\mathrm{K}^+(\mathcal{R})[S_{\mathcal{R}}^{-1}]$ with $\Psi(\gamma) = Q_{\mathbf{A}}(s)\,\beta_X^{-1}$, and set

$$\varepsilon_{\mathscr{F}}(X) := \bar{\mathscr{F}}(\gamma)^{-1} \circ Q_{\mathbf{B}}\big(\mathscr{F}(s)\big)\colon \mathscr{F}(X) \longrightarrow \bar{\mathscr{F}}(Y) = R\mathscr{F}(Q_{\mathbf{A}}X)$$

1. $R\mathscr{F}$ is a triangulated functor. The morphism $\varepsilon_{\mathscr{F}}(X)$ does not depend on the choice of s, and these morphisms form a natural transformation $\varepsilon_{\mathscr{F}}\colon Q_{\mathbf{B}} \circ \mathrm{K}^+(\mathscr{F}) \to R\mathscr{F} \circ Q_{\mathbf{A}}$.
2. The pair $(R\mathscr{F}, \varepsilon_{\mathscr{F}})$ is initial: for every triangulated $\mathcal{G}\colon D^+(\mathbf{A}) \to D^+(\mathbf{B})$ and every natural transformation $\varepsilon\colon Q_{\mathbf{B}} \circ \mathrm{K}^+(\mathscr{F}) \to \mathcal{G} \circ Q_{\mathbf{A}}$ there is a unique natural transformation $\eta\colon R\mathscr{F} \to \mathcal{G}$ with $\varepsilon = (\eta * Q_{\mathbf{A}}) \circ \varepsilon_{\mathscr{F}}$.
3. Consequently $R\mathscr{F}$ is independent of $\mathcal{R}$ and Φ, up to a unique natural isomorphism compatible with the transformations $\varepsilon_{\mathscr{F}}$.
4. Dually, let $\mathscr{F}$ be right exact and $\mathcal{R}$ adapted to it. The total left derived functor $L\mathscr{F}$ carries a natural transformation $\varepsilon_{\mathscr{F}}\colon L\mathscr{F} \circ Q_{\mathbf{A}} \to Q_{\mathbf{B}} \circ \mathrm{K}^-(\mathscr{F})$, defined dually, and the pair is terminal: for every triangulated $\mathcal{G}\colon D^-(\mathbf{A}) \to D^-(\mathbf{B})$ and every natural transformation $\varepsilon\colon \mathcal{G} \circ Q_{\mathbf{A}} \to Q_{\mathbf{B}} \circ \mathrm{K}^-(\mathscr{F})$ there is a unique natural transformation $\eta\colon \mathcal{G} \to L\mathscr{F}$ with $\varepsilon = \varepsilon_{\mathscr{F}} \circ (\eta * Q_{\mathbf{A}})$. In particular $L\mathscr{F}$ is independent of the choices up to a unique compatible natural isomorphism.

The morphism γ exists because Z and Y are objects of $\mathrm{K}^+(\mathcal{R})$ and Ψ is fully faithful, and it is an isomorphism because $\Psi(\gamma)$ is one. The quasi-isomorphism s exists by property $(*)$ of the proof of theorem 3.3.4. When γ is the class of a quasi-isomorphism $t\colon Y \to Z$ of $\mathrm{K}^+(\mathcal{R})$, so that $X \xrightarrow{s} Z \xleftarrow{t} Y$ represents β_X, the definition reads $\varepsilon_{\mathscr{F}}(X) = Q_{\mathbf{B}}(\mathscr{F}(t))^{-1} \circ Q_{\mathbf{B}}(\mathscr{F}(s))$, with $\mathscr{F}(t)$ a quasi-isomorphism by condition (1) of definition 3.3.2. That is the formula of Gelfand-Manin [15] III.6.10, whose text writes $\mathrm{K}^+(F)(s)$ at the point where t is meant.

The general case reduces to this one. By Gelfand-Manin III.6.4 the class $S_{\mathcal{R}}$ is localizing in their sense (III.2.6), which includes the Ore condition with the quasi-isomorphism on the target, so by their Remark III.2.9, the right-roof variant of Lemma III.2.8, the isomorphism γ is $Q_{\mathcal{R}}(u)^{-1}Q_{\mathcal{R}}(t)$ for some morphisms $t\colon Y \to W$ and $u\colon Z \to W$ of $\mathrm{K}^+(\mathcal{R})$ with u in $S_{\mathcal{R}}$. Then $Q_{\mathbf{A}}(t) = \Psi(Q_{\mathcal{R}}(t)) = Q_{\mathbf{A}}(u)\,\Psi(\gamma)$ is invertible, so t is a quasi-isomorphism too (proposition 3.2.8), and

$$Q_{\mathbf{A}}(t) = Q_{\mathbf{A}}(u)\,Q_{\mathbf{A}}(s)\,\beta_X^{-1} = Q_{\mathbf{A}}(us)\,\beta_X^{-1}$$

So $X \xrightarrow{us} W \xleftarrow{t} Y$ represents β_X as in Gelfand-Manin III.6.10, with W in $\mathrm{K}^+(\mathcal{R})$. Since $\bar{\mathscr{F}} \circ Q_{\mathcal{R}} = Q_{\mathbf{B}} \circ \mathscr{F}$ on $\mathrm{K}^+(\mathcal{R})$,

$$\bar{\mathscr{F}}(\gamma)^{-1}\,Q_{\mathbf{B}}(\mathscr{F}(s)) = Q_{\mathbf{B}}(\mathscr{F}(t))^{-1}\,Q_{\mathbf{B}}(\mathscr{F}(u))\,Q_{\mathbf{B}}(\mathscr{F}(s)) = Q_{\mathbf{B}}(\mathscr{F}(t))^{-1}\,Q_{\mathbf{B}}(\mathscr{F}(us))$$

which is their $\varepsilon_F(X)$ for that diagram. So $\varepsilon_{\mathscr{F}}$ as defined here is Gelfand-Manin's, and III.6.10, which proves that theirs does not depend on the diagram and is natural in X, gives the second and third assertions of (1).

The statement is Gelfand-Manin's Definition III.6.6, with the right exact form (4) stated after it. That the construction of definition 3.3.5 satisfies it is their Theorem III.6.8, proved in III.6.8 to III.6.11: $R\mathscr{F}$ is triangulated by III.6.8(a) and Lemma III.6.9, $\varepsilon_{\mathscr{F}}$ is constructed and shown to be well defined and natural in III.6.10, and the universal property is verified in III.6.11. The proof is deferred to Gelfand-Manin: it is four pages of manipulation of roofs and introduces no idea used later. Independence of the choices is III.6.7: two constructions, each initial, have unique comparison maps between them, and these are mutually inverse.

The reconciliation with section 2.3 is a computation with the definitions.

Theorem 3.3.7: Classical Derived Functors Are the Cohomology of the Total One

Let $\mathbf{A}$ have enough injectives and let $\mathscr{F}$ be left exact. For every object A of $\mathbf{A}$, regarded as a complex concentrated in degree 0,

$$H^n\big(R\mathscr{F}(A)\big) \cong R^n\mathscr{F}(A)$$

natural in A, where $R^n\mathscr{F}$ is the nth right derived functor of section 2.3.

***Proof*:**

Step 1: the class. Take $\mathcal{R}$ to be the class of injectives, adapted to $\mathscr{F}$ by example 3.3.3(1). By theorem 3.3.6(3) this changes $R\mathscr{F}$ only by a canonical natural isomorphism, so it suffices to prove the theorem for this $\mathcal{R}$.

Step 2: the object. Let $\iota_A\colon A \to I_A$ be an injective resolution, which exists by hypothesis. It is a quasi-isomorphism from the complex A in degree 0 to $I_A \in \mathrm{K}^+(\mathcal{R})$, so $Q_{\mathbf{A}}(\iota_A)\,\beta_A^{-1}\colon \Psi\Phi(A) \to \Psi(I_A)$ is an isomorphism of $D^+(\mathbf{A})$. As Ψ is fully faithful, it is $\Psi(c_A)$ for a unique isomorphism $c_A\colon \Phi(A) \to I_A$ of $\mathrm{K}^+(\mathcal{R})[S_{\mathcal{R}}^{-1}]$. Applying $\bar{\mathscr{F}}$ gives an isomorphism $R\mathscr{F}(A) = \bar{\mathscr{F}}\Phi(A) \cong \bar{\mathscr{F}}(I_A) = \mathscr{F}(I_A)$, and so

$$H^n\big(R\mathscr{F}(A)\big) \cong H^n\big(\mathscr{F}(I_A)\big)$$

The right-hand side is canonically isomorphic to $R^n\mathscr{F}(A)$ as defined in section 2.3, the two injective resolutions of A involved being homotopy equivalent (proposition 2.2.14, injective case) and $\mathscr{F}$ carrying homotopy equivalences to homotopy equivalences (lemma 2.1.34).

Step 3: naturality. Let $\varphi\colon A \to A'$ be a morphism and $\tilde{\varphi}\colon I_A \to I_{A'}$ a lift of it, with $\tilde{\varphi}\,\iota_A$ homotopic to $\iota_{A'}\varphi$, which exists by the injective case of proposition 2.2.15. Then $Q_{\mathbf{A}}(\tilde{\varphi})\,Q_{\mathbf{A}}(\iota_A) = Q_{\mathbf{A}}(\iota_{A'})\,\varphi$ in $D^+(\mathbf{A})$, and with the naturality of β,

$$\begin{aligned}\Psi\big(c_{A'}\,\Phi(\varphi)\big) &= Q_{\mathbf{A}}(\iota_{A'})\,\beta_{A'}^{-1}\,\Psi\Phi(\varphi) = Q_{\mathbf{A}}(\iota_{A'})\,\varphi\,\beta_A^{-1}\\ &= Q_{\mathbf{A}}(\tilde{\varphi})\,Q_{\mathbf{A}}(\iota_A)\,\beta_A^{-1} = \Psi\big(Q_{\mathcal{R}}(\tilde{\varphi})\big)\,\Psi(c_A)\end{aligned}$$

the last step because Ψ agrees with $Q_{\mathbf{A}}$ on morphisms of $\mathrm{K}^+(\mathcal{R})$. As Ψ is faithful, $c_{A'}\,\Phi(\varphi) = Q_{\mathcal{R}}(\tilde{\varphi})\,c_A$. Applying $\bar{\mathscr{F}}$ and then H^n shows that the isomorphism of Step 2 carries $H^n\big(R\mathscr{F}(\varphi)\big)$ to $H^n\big(\mathscr{F}(\tilde{\varphi})\big)$, which is $R^n\mathscr{F}(\varphi)$ by the definition in section 2.3, completing the proof.

So nothing computed earlier is displaced. What is gained is that $R\mathscr{F}(A)$ exists before its cohomology is taken, and that the gain shows up immediately when two functors are composed.

Theorem 3.3.8: Derived Functor of a Composite

Let $\mathbf{A}, \mathbf{B}, \mathcal{C}$ be abelian categories with $\mathbf{A}$ and $\mathbf{B}$ having enough injectives, and let

$$\mathscr{F}\colon \mathbf{A} \to \mathbf{B}, \qquad \mathcal{G}\colon \mathbf{B} \to \mathcal{C}$$

be left exact additive functors. There is always a natural transformation

$$\zeta\colon R(\mathcal{G}\circ\mathscr{F}) \longrightarrow R\mathcal{G}\circ R\mathscr{F}$$

and it is an isomorphism whenever $\mathscr{F}$ carries injective objects of $\mathbf{A}$ to $\mathcal{G}$-acyclic objects of $\mathbf{B}$, in the sense of definition 2.3.15.

The natural transformation comes from the universal property of theorem 3.3.6(2) applied to $R(\mathcal{G}\circ\mathscr{F})$. The isomorphism comes from the observation that for A a bounded-below complex of injectives the three complexes involved already agree. $R(\mathcal{G}\circ\mathscr{F})(A)$ and $R\mathscr{F}(A)$ are computed term by term, and so is $R\mathcal{G}(\mathscr{F}(A))$, because $\mathscr{F}(A)$ is a bounded-below complex of $\mathcal{G}$-acyclic objects and the $\mathcal{G}$-acyclic objects form a class adapted to $\mathcal{G}$ (Gelfand-Manin [15] III.6.16(b)). Every object of $D^+(\mathbf{A})$ is isomorphic to such a complex. The transformation ζ is Weibel [23] 10.8.2(1), and the isomorphism under this hypothesis is his Corollary 10.8.3. His letters are the opposite of the ones used here: he writes the composite as FG with G applied first, so his $R(FG) \cong RF \circ RG$ is the statement above with the names exchanged. The convention here matches theorem 6.5.1, which is what this theorem feeds.

3.3.9 Note: This Is the Grothendieck Spectral Sequence, Before Indices Theorem 6.5.1 carries the same hypothesis, that $\mathscr{F}$ send injectives to $\mathcal{G}$-acyclics, and concludes with a spectral sequence

$$E_2^{p,q} = R^p\mathcal{G}\big(R^q\mathscr{F}(A)\big) \implies R^{p+q}(\mathcal{G}\circ\mathscr{F})(A)$$

The relation between the two is that theorem 3.3.8 is the statement at the level of the derived category, and the spectral sequence is what remains of it after cohomology is taken. Filtering $R\mathscr{F}(A)$ by its canonical truncations (definition 2.1.12) and applying $R\mathcal{G}$ gives, after renumbering the pages (the canonical filtration's E_1 is the E_2 below), a spectral sequence with $E_2^{p,q} = R^p\mathcal{G}\big(R^q\mathscr{F}(A)\big)$. In the proof of theorem 6.5.1 the same page comes from the filtration by resolution degree of a Cartan-Eilenberg resolution of $\mathscr{F}(I)$, for I an injective resolution of A, which is the second spectral sequence of theorem 6.4.5 for $\mathcal{G}$. The isomorphism ζ is what says the abutment is $R^{p+q}(\mathcal{G}\circ\mathscr{F})(A)$ and not something else.

This is worth seeing in both directions. The spectral sequence is the computational form and is what chapter 6 uses. The derived-category form carries strictly more information, since a spectral sequence loses the extension data that $R\mathcal{G}\circ R\mathscr{F}$ retains as an object.

Exercise 3.3.1

1. Exhibit two different classes adapted to $-\otimes_R N$ over a commutative ring R, and check directly that the two resulting constructions of $L(-\otimes_R N)$ agree on a module with a short flat resolution. Which part of theorem 3.3.6 makes the agreement automatic?

2. Prove directly that an acyclic bounded-below complex of injectives is split exact, the fact behind example 3.3.3(1), and deduce condition (1) of definition 3.3.2 for the class of injectives.

(Build the splittings by induction up the complex, using injectivity at each stage.)

3. Show that if $\mathscr{F}$ is exact then every class stable under finite direct sums and satisfying condition (2) of definition 3.3.2 is adapted to $\mathscr{F}$, and conclude that $R\mathscr{F}$ is the functor induced by $\mathscr{F}$ term by term, with $R^n\mathscr{F} = 0$ for $n \neq 0$. Show also that the stability hypothesis cannot be omitted: in **Ab**, the groups not isomorphic to $\mathbb{Z} \oplus \mathbb{Z}$ form a class satisfying condition (2) that is not stable under finite direct sums.

4. Give left exact $\mathscr{F}$ and $\mathcal{G}$ for which $\mathscr{F}$ does *not* carry injectives to $\mathcal{G}$-acyclics, and show that the conclusion of theorem 3.3.8 fails for them, so the hypothesis is not decorative. (One source of examples: take $\mathcal{G}$ to be a functor with a nonvanishing R^1 on an object in the image of an injective under $\mathscr{F}$.)

5. Definition 3.3.5 produces $R\mathscr{F}$ on $D^+(\mathbf{A})$ and not on $D(\mathbf{A})$. Identify the step in the construction that uses the bound, and explain what would go wrong for an unbounded complex.

3.4 RHom and the Derived Tensor Product

The functors Ext and Tor of section 2.3 have two variables, and section 2.5.1 computed each by resolving either one of them. Section 3.3 derives a functor of a single variable. This section derives Hom and $\otimes$ in both variables, to bifunctors RHom and $\otimes^L$ on the bounded derived categories with

$$H^n R\mathrm{Hom}(A,B) \cong \mathrm{Ext}^n(A,B), \qquad H^{-n}\left(A \otimes^L_R B\right) \cong \mathrm{Tor}^R_n(A,B)$$

for objects A and B. The agreement of the two computations, resolving the first variable or the second, becomes an isomorphism in the derived category, and $\mathrm{Ext}^n(A,B)$ becomes the group of morphisms $A \to B[n]$ of $D(\mathcal{A})$, a description with no resolution in it.

The treatment follows Weibel [23] 10.6 and 10.7, with the sign conventions of Gelfand-Manin [15] III.6.14. By box 3.2.5 every object of $D^+(\mathcal{A})$ is isomorphic to a bounded-below complex, and $D^+(\mathcal{A})$ is equivalent to the localization of $\mathrm{K}^+(\mathcal{A})$ at its quasi-isomorphisms. Objects of $D^+(\mathcal{A})$ are therefore taken to be bounded-below complexes, and objects of $D^-(\mathcal{A})$ bounded-above ones. Complexes with no bound are treated in chapter 5.

The morphisms of $\mathrm{K}(\mathcal{A})$ from a complex X to the shifts of a complex Y are the cohomology groups of one complex of abelian groups.

Definition 3.4.1: Hom Complex

Let $X, Y \in \mathrm{Ch}(\mathcal{A})$. The *Hom complex* $\mathrm{Hom}^\bullet(X,Y)$ is the complex of abelian groups with

$$\mathrm{Hom}^n(X,Y) := \prod_{i\in\mathbb{Z}} \mathrm{Hom}_{\mathcal{A}}\left(X^i, Y^{i+n}\right)$$

whose elements are families $\varphi = (\varphi^i)_{i\in\mathbb{Z}}$, and with differential

$$D\varphi := d_Y \circ \varphi - (-1)^n \varphi \circ d_X, \qquad \text{that is} \qquad (D\varphi)^i = d_Y^{i+n}\varphi^i - (-1)^n \varphi^{i+1} d_X^i$$

for $\varphi \in \mathrm{Hom}^n(X,Y)$.

The product is taken in **Ab**, where it exists whatever $\mathcal{A}$ is. For φ of degree n,

$$D(D\varphi) = d_Y d_Y \varphi - (-1)^n d_Y \varphi d_X + (-1)^n d_Y \varphi d_X - \varphi d_X d_X = 0$$

so $\mathrm{Hom}^\bullet(X,Y)$ is a complex. The sign matches the shift convention (3.1), under which the differential of $Y[n]$ is $(-1)^n d_Y$.

Example 3.4.2: The Hom Complex of a Resolution of $\mathbb{Z}/2$

Let P be the complex $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ in degrees -1 and 0, with basis e of P^{-1} and f of P^0, so that $de = 2f$. It is a projective resolution of $\mathbb{Z}/2$. The group $\mathrm{Hom}^n(P,P)$ vanishes unless $-1 \le n \le 1$, and

$$\mathrm{Hom}^{-1}(P,P) = \mathrm{Hom}(P^0, P^{-1}) \cong \mathbb{Z}, \qquad \mathrm{Hom}^1(P,P) = \mathrm{Hom}(P^{-1}, P^0) \cong \mathbb{Z},$$
$$\mathrm{Hom}^0(P,P) = \mathrm{Hom}(P^{-1}, P^{-1}) \oplus \mathrm{Hom}(P^0, P^0) \cong \mathbb{Z}^2$$

where $h \in \mathbb{Z}$ stands for $f \mapsto he$, the pair (a,b) for $e \mapsto ae$ and $f \mapsto bf$, and c for $e \mapsto cf$. In degree -1 the differential is $Dh = dh + hd$, which sends e to $h(2f) = 2he$ and f to $d(he) = 2hf$, so $Dh = (2h, 2h)$. In degree 0 it is $D(a,b) = d(a,b) - (a,b)d$, which sends e to $2af - 2bf$. So $\mathrm{Hom}^\bullet(P,P)$ is

$$0 \to \mathbb{Z} \xrightarrow{h \mapsto (2h,2h)} \mathbb{Z}^2 \xrightarrow{(a,b) \mapsto 2a-2b} \mathbb{Z} \to 0$$

Its cohomology is 0 in degree -1, $\{(a,a)\}/\{(2h,2h)\} \cong \mathbb{Z}/2$ in degree 0, generated by the identity $(1,1)$, and $\mathbb{Z}/2\mathbb{Z}$ in degree 1, generated by the map φ with $\varphi(e) = f$.

Lemma 3.4.3: The Hom Complex Computes the Homotopy Category

Let $X, Y \in \mathrm{Ch}(\mathcal{A})$ and $n \in \mathbb{Z}$.

1. The n-cocycles of $\mathrm{Hom}^\bullet(X,Y)$ are the chain maps $X \to Y[n]$, with the shift of (3.1), the n-coboundaries are the null-homotopic ones, and so
$$H^n \mathrm{Hom}^\bullet(X,Y) = \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}\big(X, Y[n]\big)$$
2. Chain maps $f\colon X' \to X$ and $g\colon Y \to Y'$ induce chain maps $f^*\colon \varphi \mapsto \varphi f$ and $g_*\colon \varphi \mapsto g\varphi$, and homotopic chain maps induce homotopic ones. Hence $(X,Y) \mapsto \mathrm{Hom}^\bullet(X,Y)$ is a functor $\mathrm{K}(\mathcal{A})^{\mathrm{op}} \times \mathrm{K}(\mathcal{A}) \to \mathrm{K}(\mathbf{Ab})$.
3. Under the identification of (1), $H^n(f^*)$ is precomposition with f and $H^n(g_*)$ is post-composition with $g[n]$.

***Proof*:**
Step 1: cocycles and coboundaries. For $\varphi \in \mathrm{Hom}^n(X,Y)$ the equation $D\varphi = 0$ reads $(-1)^n d_Y \varphi = \varphi d_X$. Since $d_{Y[n]} = (-1)^n d_Y$ and $Y[n]^i = Y^{i+n}$ by (3.1), this says that the family $\varphi^i\colon X^i \to Y[n]^i$ is a chain map $X \to Y[n]$. Next, $\varphi = D\psi$ for $\psi \in \mathrm{Hom}^{n-1}(X,Y)$ means $\varphi = d_Y\psi + (-1)^n \psi d_X$. Putting $s := (-1)^n \psi$, so that $s^i\colon X^i \to Y[n]^{i-1}$, this is

$$\varphi = d_{Y[n]}\, s + s\, d_X$$

which by definition 2.1.27 says that φ is null-homotopic as a map $X \to Y[n]$. Conversely a

null-homotopy s gives $\varphi = D\big((-1)^n s\big)$. Two chain maps are equal in $\mathrm{K}(\mathcal{A})$ exactly when their difference is null-homotopic (definition 2.1.38), which gives the displayed equality.

Step 2: functoriality. Since $f d_{X'} = d_X f$,

$$D(\varphi f) = d_Y \varphi f - (-1)^n \varphi f d_{X'} = \big(d_Y \varphi - (-1)^n \varphi d_X\big) f = (D\varphi) f$$

and similarly $D(g\varphi) = g\, D\varphi$, so f^* and g_* are chain maps. If $g - g' = d_{Y'} k + k d_Y$, then $H(\varphi) := k\varphi$ satisfies

$$D(k\varphi) + k\, D\varphi = d_{Y'} k\varphi + (-1)^n k\varphi d_X + k d_Y \varphi - (-1)^n k \varphi d_X = (g - g')\varphi$$

so $g_* - g'_* = DH + HD$. If $f - f' = d_X k + k d_{X'}$, then $H(\varphi) := (-1)^n \varphi k$ satisfies

$$D\big((-1)^n \varphi k\big) + (-1)^{n+1} (D\varphi) k = (-1)^n d_Y \varphi k + \varphi k d_{X'} + (-1)^{n+1} d_Y \varphi k + \varphi d_X k = \varphi (f - f')$$

so $f^* - f'^* = DH + HD$. As $(g\varphi) f = g(\varphi f)$, the two kinds of induced map commute, and $\mathrm{Hom}^\bullet$ is a functor on the product of homotopy categories.

Step 3: the induced maps. $H^n(f^*)$ sends the class of a chain map $\varphi\colon X \to Y[n]$ to the class of φf, and $H^n(g_*)$ sends it to the class of $g\varphi$, which as a map $X \to Y'[n]$ is $g[n] \circ \varphi$, the shift $g[n]$ having the same components as g by (3.1), completing the proof.

For the complex P of example 3.4.2 the lemma gives $\mathrm{Hom}_{\mathrm{K}(\mathbf{Ab})}(P, P) \cong \mathbb{Z}/2$, generated by id_P, and $\mathrm{Hom}_{\mathrm{K}(\mathbf{Ab})}(P, P[1]) \cong \mathbb{Z}/2$, generated by the chain map φ with $\varphi(e) = f$, which is therefore not null-homotopic.

On a bounded-below complex of injectives, and out of a bounded-above complex of projectives, the homotopy category already computes the morphisms of the derived category. This is Weibel [23] Corollary 10.4.7.

Lemma 3.4.4: Complexes of Projectives and Injectives Compute $D(\mathcal{A})$

Let $\mathcal{A}$ be an abelian category and $X \in \mathrm{Ch}(\mathcal{A})$.

1. If P is a bounded-above complex of projectives, then $Q\colon \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(P, X) \to \mathrm{Hom}_{D(\mathcal{A})}(P, X)$ is bijective.
2. If I is a bounded-below complex of injectives, then $Q\colon \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(X, I) \to \mathrm{Hom}_{D(\mathcal{A})}(X, I)$ is bijective.

***Proof*:**
Step 1: (1), surjectivity. By theorem 3.2.4 a morphism $P \to X$ of $D(\mathcal{A})$ is the class of a roof (N, s, f) with $s\colon N \to P$ a quasi-isomorphism, and that class is $Q(f)\, Q(s)^{-1}$. As P is a bounded-above complex of projectives and s is a quasi-isomorphism into it, proposition 2.2.8 gives a chain map $g\colon P \to N$ with sg homotopic to id_P. (Its proof runs through lemma 2.2.4, stated for complexes vanishing in positive degrees. A translation of degrees puts P in that form and changes neither hypothesis nor conclusion.) Applying Q gives $Q(s)Q(g) = \mathrm{id}_P$, and as $Q(s)$ is invertible, $Q(g) = Q(s)^{-1}$. The given morphism is therefore $Q(f)Q(g) = Q(fg)$.

Step 2: (1), injectivity. The functor Q is additive by theorem 3.2.7, so it suffices to show that $Q(h) = 0$ forces h to be null-homotopic, for a chain map $h\colon P \to X$. By proposition 3.2.12(2) there is a quasi-isomorphism $t\colon N \to P$ with ht null-homotopic, and proposition 2.2.8 gives $g\colon P \to N$ with tg homotopic to id_P. Composition is well defined on homotopy classes, so h is homotopic to htg, which is homotopic to 0.

Step 3: (2). Reindexing $X^n \mapsto X^{-n}$ identifies $\mathrm{Ch}(\mathcal{A}^{\mathrm{op}})$ with $\mathrm{Ch}(\mathcal{A})^{\mathrm{op}}$. It carries quasi-isomorphisms to quasi-isomorphisms and null-homotopies to null-homotopies, so it identifies $\mathrm{K}(\mathcal{A}^{\mathrm{op}})$ with $\mathrm{K}(\mathcal{A})^{\mathrm{op}}$ and, the universal property of definition 1.2.1 being self-dual, $D(\mathcal{A}^{\mathrm{op}})$ with $D(\mathcal{A})^{\mathrm{op}}$, compatibly with Q. An injective object of $\mathcal{A}$ is a projective object of $\mathcal{A}^{\mathrm{op}}$, so I becomes a bounded-above complex of projectives of $\mathcal{A}^{\mathrm{op}}$. Statement (1) for the abelian category $\mathcal{A}^{\mathrm{op}}$ is then statement (2) for $\mathcal{A}$, completing the proof.

Corollary 3.4.5: The Hom Complex Preserves Quasi-Isomorphisms

Let $s\colon X \to X'$ and $t\colon Y \to Y'$ be quasi-isomorphisms in $\mathrm{Ch}(\mathcal{A})$.

1. If I is a bounded-below complex of injectives, then $s^*\colon \mathrm{Hom}^\bullet(X', I) \to \mathrm{Hom}^\bullet(X, I)$ is a quasi-isomorphism.
2. If P is a bounded-above complex of projectives, then $t_*\colon \mathrm{Hom}^\bullet(P, Y) \to \mathrm{Hom}^\bullet(P, Y')$ is a quasi-isomorphism.

***Proof*:**
(1) Fix n. The shift $I[n]$ is again a bounded-below complex of injectives. By lemma 3.4.3(1) and (3), $H^n(s^*)$ is the map $\mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(X', I[n]) \to \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(X, I[n])$, $\psi \mapsto \psi s$. The bijections Q of lemma 3.4.4(2) carry it to $\psi \mapsto \psi \circ Q(s)$ from $\mathrm{Hom}_{D(\mathcal{A})}(X', I[n])$ to $\mathrm{Hom}_{D(\mathcal{A})}(X, I[n])$, because Q is a functor, and that map is bijective with inverse $\psi \mapsto \psi \circ Q(s)^{-1}$.

(2) The same argument applies with lemma 3.4.4(1): $H^n(t_*)$ becomes postcomposition with $Q(t[n])$ on $\mathrm{Hom}_{D(\mathcal{A})}(P, Y[n])$, which is bijective because $t[n]$ is a quasi-isomorphism, $H^i(t[n])$ being $H^{i+n}(t)$, completing the proof.

When $\mathcal{A}$ has enough injectives, corollary 3.4.5(1) shows that resolving the second variable by injectives gives a functor of the first variable that inverts quasi-isomorphisms.

Definition 3.4.6: *R*Hom

Let $\mathcal{A}$ have enough injectives. For every bounded-below complex B fix an *injective resolution* $\iota_B\colon B \to I_B$, that is, a quasi-isomorphism into a bounded-below complex of injectives. For $A \in \mathrm{Ch}(\mathcal{A})$ set

$$R\mathrm{Hom}(A, B) := \mathrm{Hom}^\bullet(A, I_B)$$

regarded as an object of $D(\mathbf{Ab})$.

Injective resolutions exist: theorem 2.2.17, applied to $\mathcal{A}^{\mathrm{op}}$, which has enough projectives because $\mathcal{A}$ has enough injectives, gives one for every bounded-below complex. When A is an object of $\mathcal{A}$

placed in degree 0, the group $\mathrm{Hom}^n(A, I_B)$ has the single factor $\mathrm{Hom}_{\mathcal{A}}(A, I_B^n)$ and $D\varphi = d_{I_B}\varphi$. So $R\mathrm{Hom}(A, B)$ is the complex $\mathrm{Hom}_{\mathcal{A}}(A, I_B)$, the left exact functor $\mathrm{Hom}_{\mathcal{A}}(A, -)$ applied term by term to I_B, which is the value at B of the total derived functor of definition 3.3.5 for the functor $\mathrm{Hom}_{\mathcal{A}}(A, -)$ and the class of injectives, adapted to it by example 3.3.3(1).

Example 3.4.7: $R\mathrm{Hom}(\mathbb{Z}/n, \mathbb{Z})$

In **Ab** take $B = \mathbb{Z}$ and the injective resolution $\mathbb{Z} \to I_{\mathbb{Z}}$, where $I_{\mathbb{Z}}$ is $\mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ in degrees 0 and 1. The groups $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are divisible, hence injective, as recorded in example 2.4.10, and $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ is exact, so the inclusion is a quasi-isomorphism. For $n \geq 1$,

$$\mathrm{Hom}(\mathbb{Z}/n, \mathbb{Q}) = 0, \qquad \mathrm{Hom}(\mathbb{Z}/n, \mathbb{Q}/\mathbb{Z}) = \{x \in \mathbb{Q}/\mathbb{Z} : nx = 0\} \cong \mathbb{Z}/n$$

the first because $\mathbb{Q}$ has no nonzero element of finite order, the second generated by the class of $1/n$. So $R\mathrm{Hom}(\mathbb{Z}/n, \mathbb{Z})$ is the complex $0 \to \mathbb{Z}/n$ with $\mathbb{Z}/n$ in degree 1, which is $(\mathbb{Z}/n)[-1]$. Its cohomology is $\mathrm{Hom}(\mathbb{Z}/n, \mathbb{Z}) = 0$ in degree 0 and $\mathbb{Z}/n$ in degree 1, the values of $\mathrm{Ext}^0_{\mathbb{Z}}(\mathbb{Z}/n, \mathbb{Z})$ and $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n, \mathbb{Z})$ found in example 2.4.10.

Up to a canonical isomorphism $R\mathrm{Hom}(A, B)$ does not depend on the chosen resolutions, and it is a functor of both variables at once. The statement is Weibel [23] Lemma 10.7.3 and Theorem 10.7.4.

Theorem 3.4.8: $R\mathrm{Hom}$ **Is a Bifunctor**

Let $\mathcal{A}$ have enough injectives.

1. For every morphism $\beta\colon B \to B'$ of $D^+(\mathcal{A})$ there is a unique morphism $\tilde{\beta}\colon I_B \to I_{B'}$ of $\mathrm{K}(\mathcal{A})$ with $Q(\tilde{\beta})\,Q(\iota_B) = Q(\iota_{B'})\,\beta$, and $\beta \mapsto \tilde{\beta}$ is a functor $D^+(\mathcal{A}) \to \mathrm{K}(\mathcal{A})$.
2. $R\mathrm{Hom}$ is a bifunctor
$$R\mathrm{Hom}\colon D(\mathcal{A})^{\mathrm{op}} \times D^+(\mathcal{A}) \longrightarrow D(\mathbf{Ab})$$
whose value on $Q(f)$, for a chain map $f\colon A' \to A$, and on β is $Q\big(\mathrm{Hom}^\bullet(f, \tilde{\beta})\big)$.
3. Replacing the injective resolutions by another choice $\iota'_B\colon B \to I'_B$ changes $R\mathrm{Hom}$ by the natural isomorphism $\mathrm{Hom}^\bullet(A, c_B)$, where $c_B\colon I_B \to I'_B$ is the unique morphism of $\mathrm{K}(\mathcal{A})$ with $Q(c_B)\,Q(\iota_B) = Q(\iota'_B)$.
4. If A is bounded above, then $R\mathrm{Hom}(A, B)$ is bounded below. Hence $R\mathrm{Hom}$ restricts to a bifunctor $D^-(\mathcal{A})^{\mathrm{op}} \times D^+(\mathcal{A}) \to D^+(\mathbf{Ab})$.

***Proof*:**

Step 1: the second variable. By lemma 3.4.4(2), with $X = I_B$ and the bounded-below complex of injectives $I_{B'}$, the map $Q\colon \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(I_B, I_{B'}) \to \mathrm{Hom}_{D(\mathcal{A})}(I_B, I_{B'})$ is bijective. As $Q(\iota_B)$ is invertible, $Q(\iota_{B'})\,\beta\,Q(\iota_B)^{-1}$ lies in the target, so it is $Q(\tilde{\beta})$ for exactly one $\tilde{\beta}$, and that is the stated condition. For $\beta'\colon B' \to B''$,

$$Q(\tilde{\beta}'\tilde{\beta})\,Q(\iota_B) = Q(\tilde{\beta}')\,Q(\iota_{B'})\,\beta = Q(\iota_{B''})\,\beta'\beta$$

so $\tilde{\beta}'\tilde{\beta}$ satisfies the condition defining $\widetilde{\beta'\beta}$ and the two are equal. Likewise $\widetilde{\mathrm{id}_B} = \mathrm{id}_{I_B}$. This is (1).

Step 2: the first variable. Fix B. By lemma 3.4.3(2), $A \mapsto \mathrm{Hom}^\bullet(A, I_B)$ is a functor $\mathrm{K}(\mathcal{A})^{\mathrm{op}} \to \mathrm{K}(\mathbf{Ab})$, and by corollary 3.4.5(1) it carries quasi-isomorphisms to quasi-isomorphisms. Followed by $Q\colon \mathrm{K}(\mathbf{Ab}) \to D(\mathbf{Ab})$ it carries the quasi-isomorphisms of $\mathrm{K}(\mathcal{A})$ to isomorphisms. By theorem 3.2.4, $D(\mathcal{A})$ is the localization of $\mathrm{K}(\mathcal{A})$ at its quasi-isomorphisms, so $D(\mathcal{A})^{\mathrm{op}}$ is that of $\mathrm{K}(\mathcal{A})^{\mathrm{op}}$, and the universal property of definition 1.2.1 factors the functor uniquely through $D(\mathcal{A})^{\mathrm{op}}$. The factorization is $R\mathrm{Hom}(-, B)$. Its value on $Q(f)$ is $Q(f^*)$, and its value on $Q(s)^{-1}$, for a quasi-isomorphism s, is $Q(s^*)^{-1}$.

Step 3: both variables. Put $R\mathrm{Hom}(A, \beta) := Q\big(\mathrm{Hom}^\bullet(A, \tilde{\beta})\big)$, a functor of β by Step 1 and lemma 3.4.3(2). For $\alpha\colon A' \to A$ in $D(\mathcal{A})$ and $\beta\colon B \to B'$ consider the square

$$\begin{array}{ccc} R\mathrm{Hom}(A, B) & \xrightarrow{R\mathrm{Hom}(A,\beta)} & R\mathrm{Hom}(A, B') \\ {\scriptstyle R\mathrm{Hom}(\alpha,B)}\downarrow & & \downarrow{\scriptstyle R\mathrm{Hom}(\alpha,B')} \\ R\mathrm{Hom}(A', B) & \xrightarrow[R\mathrm{Hom}(A',\beta)]{} & R\mathrm{Hom}(A', B') \end{array}$$

If $\alpha = Q(f)$ for a chain map f, both composites are $Q\big(\mathrm{Hom}^\bullet(f, \tilde{\beta})\big)$, because f^* and $\tilde{\beta}_*$ commute (lemma 3.4.3(2)). If $\alpha = Q(s)^{-1}$ for a quasi-isomorphism $s\colon A \to A'$, the square is the square for $Q(s)$ with its two vertical isomorphisms inverted, so it commutes. Squares of this shape paste along composites in α, and every morphism of $D(\mathcal{A})$ is $Q(f)Q(s)^{-1}$ by theorem 3.2.4, so the square commutes for every α. Setting $R\mathrm{Hom}(\alpha, \beta) := R\mathrm{Hom}(\alpha, B') \circ R\mathrm{Hom}(A, \beta)$, the commutativity of these squares is exactly what makes $R\mathrm{Hom}$ preserve composites in the pair (α, β). This is (2).

Step 4: independence of the resolutions. As in Step 1, lemma 3.4.4(2) gives unique $c_B\colon I_B \to I'_B$ and $c'_B\colon I'_B \to I_B$ in $\mathrm{K}(\mathcal{A})$ with $Q(c_B)Q(\iota_B) = Q(\iota'_B)$ and $Q(c'_B)Q(\iota'_B) = Q(\iota_B)$. Both $c'_B c_B$ and id_{I_B} satisfy the condition defining $\widetilde{\mathrm{id}_B}$, so $c'_B c_B = \mathrm{id}_{I_B}$, and symmetrically $c_B c'_B = \mathrm{id}_{I'_B}$. So c_B is an isomorphism of $\mathrm{K}(\mathcal{A})$ and $\mathrm{Hom}^\bullet(A, c_B)$ one of $\mathrm{K}(\mathbf{Ab})$. Writing $\tilde{\beta}'$ for the morphism of Step 1 built from the primed resolutions, $c_{B'}\tilde{\beta}$ and $\tilde{\beta}' c_B$ both satisfy $Q(-)\,Q(\iota_B) = Q(\iota'_{B'})\,\beta$, so they are equal by the same uniqueness. With lemma 3.4.3(2), and Step 3 for the passage from chain maps to all morphisms of $D(\mathcal{A})$, the isomorphisms $\mathrm{Hom}^\bullet(A, c_B)$ are natural in both variables. This is (3).

Step 5: bounds. Suppose $A^i = 0$ for $i > a$ and $I_B^j = 0$ for $j < b$. A factor $\mathrm{Hom}(A^i, I_B^{i+n})$ of $\mathrm{Hom}^n(A, I_B)$ is nonzero only if $i \le a$ and $i + n \ge b$, which forces $n \ge b - a$. So $\mathrm{Hom}^n(A, I_B) = 0$ for $n < b - a$, completing the proof.

Dually, when $\mathcal{A}$ has enough projectives, let $\pi_A\colon P_A \to A$ be a *projective resolution* of a bounded-above complex A, that is, a quasi-isomorphism from a bounded-above complex of projectives. Then $\mathrm{Hom}^\bullet(P_A, B)$ defines a bifunctor $D^-(\mathcal{A})^{\mathrm{op}} \times D(\mathcal{A}) \to D(\mathbf{Ab})$ (Weibel [23] Theorem 10.7.4). When both kinds of resolution exist, the two constructions agree, which is Weibel's Exercise 10.7.1.

Proposition 3.4.9: $R\mathrm{Hom}$ May Be Computed by Resolving Either Variable

Let $\mathcal{A}$ have enough injectives and enough projectives, let A be a bounded-above complex with a projective resolution $\pi_A\colon P_A \to A$, and let B be a bounded-below complex. Then both chain maps in

$$R\mathrm{Hom}(A,B) = \mathrm{Hom}^\bullet(A, I_B) \xrightarrow{\pi_A^*} \mathrm{Hom}^\bullet(P_A, I_B) \xleftarrow{(\iota_B)_*} \mathrm{Hom}^\bullet(P_A, B)$$

are quasi-isomorphisms. In particular $R\mathrm{Hom}(A,B) \cong \mathrm{Hom}^\bullet(P_A, B)$ in $D(\mathbf{Ab})$.

***Proof*:**
The map π_A^* is a quasi-isomorphism by corollary 3.4.5(1), with $I = I_B$ and $s = \pi_A$. The map $(\iota_B)_*$ is one by corollary 3.4.5(2), with $P = P_A$ and $t = \iota_B$. Projective resolutions exist by theorem 2.2.17, completing the proof.

In example 3.4.7, the projective resolution $\mathbb{Z} \xrightarrow{n} \mathbb{Z}$ of $\mathbb{Z}/n$ in degrees -1 and 0 gives $\mathrm{Hom}^\bullet(P,\mathbb{Z}) = \big(\mathbb{Z} \xrightarrow{-n} \mathbb{Z}\big)$ in degrees 0 and 1, the sign being the $-(-1)^0\varphi d_P$ of definition 3.4.1. Its cohomology is again $\mathbb{Z}/n$ in degree 1 and 0 elsewhere.

Theorem 3.4.10: Ext as Morphisms in the Derived Category

Let $\mathcal{A}$ have enough injectives.

1. For every complex A, every bounded-below complex B and every $n \in \mathbb{Z}$ there is an isomorphism
$$\mathrm{Hom}_{D(\mathcal{A})}\big(A, B[n]\big) \cong H^n R\mathrm{Hom}(A,B)$$
natural in A and B.

2. For objects A and B of $\mathcal{A}$, regarded as complexes in degree 0, and every $n \in \mathbb{Z}$, there are isomorphisms natural in A and B
$$\mathrm{Ext}^n_{\mathcal{A}}(A,B) \cong \mathrm{Hom}_{D(\mathcal{A})}\big(A, B[n]\big) \cong H^n R\mathrm{Hom}(A,B)$$
where $\mathrm{Ext}^n_{\mathcal{A}}(A,-) = R^n\mathrm{Hom}_{\mathcal{A}}(A,-)$ is the right derived functor of section 2.3, as in example 2.3.6(2). In particular $\mathrm{Hom}_{D(\mathcal{A})}(A, B[n]) = 0$ for $n < 0$, and $Q\colon \mathrm{Hom}_{\mathcal{A}}(A,B) \to \mathrm{Hom}_{D(\mathcal{A})}(A,B)$ is bijective.

***Proof*:**
Step 1: (1). The shift of $Q(\iota_B)$ is an isomorphism $B[n] \to I_B[n]$ of $D(\mathcal{A})$, and composing with it gives $\mathrm{Hom}_{D(\mathcal{A})}(A, B[n]) \cong \mathrm{Hom}_{D(\mathcal{A})}(A, I_B[n])$. Since $I_B[n]$ is a bounded-below complex of injectives, lemma 3.4.4(2) identifies this with $\mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(A, I_B[n])$, and lemma 3.4.3(1) identifies that with $H^n\mathrm{Hom}^\bullet(A, I_B) = H^n R\mathrm{Hom}(A,B)$. Each identification is natural in chain maps of A, by lemma 3.4.3(3) and because Q is a functor, hence in all morphisms of $D(\mathcal{A})$, each being $Q(f)Q(s)^{-1}$. Each is natural in B because $Q(\tilde{\beta}[n])\,Q(\iota_B[n]) = Q(\iota_{B'}[n])\,\beta[n]$ by theorem 3.4.8(1).

Step 2: (2). By theorem 3.4.8(3) the choice of ι_B does not matter, so take ι_B to be an injective

resolution of B in the sense of definition 2.2.11, with $I_B^n = 0$ for $n < 0$. As observed after definition 3.4.6, $R\mathrm{Hom}(A, B)$ is then the complex $\mathrm{Hom}_{\mathcal{A}}(A, I_B)$, whose nth cohomology is $R^n\mathrm{Hom}_{\mathcal{A}}(A, -)(B) = \mathrm{Ext}^n_{\mathcal{A}}(A, B)$ by the definition of right derived functors in section 2.3. Together with (1) this gives the displayed isomorphisms, natural in A because a morphism $A' \to A$ acts on both sides by precomposition. For naturality in B, let $b\colon B \to B'$ be a morphism of $\mathcal{A}$. The map $\mathrm{Ext}^n_{\mathcal{A}}(A, b)$ of section 2.3 is induced by a comparison lift $\hat{b}\colon I_B \to I_{B'}$ with $\hat{b}\,\iota_B$ homotopic to $\iota_{B'} b$ (the injective case of proposition 2.2.15). Then $Q(\hat{b})\,Q(\iota_B) = Q(\iota_{B'})\,Q(b)$, so $\hat{b} = \tilde{b}$ in $\mathrm{K}(\mathcal{A})$ by the uniqueness in theorem 3.4.8(1), with $\tilde{b}$ the morphism there for $Q(b)$. Hence $\mathrm{Ext}^n_{\mathcal{A}}(A, b) = H^n\mathrm{Hom}_{\mathcal{A}}(A, \hat{b})$ is $H^n R\mathrm{Hom}(A, Q(b))$, and with (1) the displayed isomorphisms are natural in B. For $n < 0$ the group $\mathrm{Hom}_{\mathcal{A}}(A, I_B^n)$ is zero.

Step 3: $n = 0$. The sequence $0 \to B \xrightarrow{\iota_B} I_B^0 \to I_B^1$ is exact, and $\mathrm{Hom}_{\mathcal{A}}(A, -)$ is left exact, so $H^0\mathrm{Hom}_{\mathcal{A}}(A, I_B) = \ker\big(\mathrm{Hom}(A, I_B^0) \to \mathrm{Hom}(A, I_B^1)\big)$ consists of the maps $\iota_B\varphi$ for $\varphi \in \mathrm{Hom}_{\mathcal{A}}(A, B)$. Follow φ through Step 1: $Q(\varphi)$ goes to $Q(\iota_B)Q(\varphi) = Q(\iota_B\varphi)$, then to the homotopy class of the chain map $\iota_B\varphi\colon A \to I_B$, then to the cocycle $\iota_B\varphi \in \mathrm{Hom}(A, I_B^0)$. So the composite of Q with the bijection $\mathrm{Hom}_{D(\mathcal{A})}(A, B) \to H^0\mathrm{Hom}_{\mathcal{A}}(A, I_B) \cong \mathrm{Hom}_{\mathcal{A}}(A, B)$ is the identity of $\mathrm{Hom}_{\mathcal{A}}(A, B)$, and Q is bijective, completing the proof.

Part (2) is Weibel [23] Theorem 10.7.4 and Corollary 10.7.5, and Gelfand-Manin [15] III.6.14. Gelfand-Manin take $\mathrm{Hom}_{D(\mathcal{A})}(A, B[n])$ as the definition of $\mathrm{Ext}^n_{\mathcal{A}}(A, B)$ (III.5.3), which needs no injectives and no projectives, and prove that Q is fully faithful on objects of $\mathcal{A}$ with no hypothesis on $\mathcal{A}$ (III.5.2). When $\mathcal{A}$ also has enough projectives, proposition 3.4.9 computes the same groups from a projective resolution of A.

Example 3.4.11: The Extension $\mathbb{Z}/4$ as a Morphism of $D(\mathbf{Ab})$

Let $u\colon \mathbb{Z}/2 \to \mathbb{Z}/4$ be multiplication by 2 and $v\colon \mathbb{Z}/4 \to \mathbb{Z}/2$ reduction modulo 2. Theorem 3.2.10 completes $0 \to \mathbb{Z}/2 \xrightarrow{u} \mathbb{Z}/4 \xrightarrow{v} \mathbb{Z}/2 \to 0$ to a distinguished triangle (u, v, w) of $D(\mathbf{Ab})$ with $w\colon \mathbb{Z}/2 \to \mathbb{Z}/2[1]$, and example 3.2.11 shows that no morphism of $\mathrm{K}(\mathbf{Ab})$ completes it to a distinguished triangle there.

By theorem 3.4.10(2) and example 2.4.10, $\mathrm{Hom}_{D(\mathbf{Ab})}(\mathbb{Z}/2, \mathbb{Z}/2[1]) \cong \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) \cong \mathbb{Z}/2$, and w is its nonzero element. To see that $w \neq 0$, rotate the triangle to $(v, w, -u[1])$ and apply $\mathrm{Hom}_{D(\mathbf{Ab})}(\mathbb{Z}/2, -)$, which is cohomological by proposition 3.1.8, $D(\mathbf{Ab})$ being triangulated by theorem 3.2.7. Exactness of

$$\mathrm{Hom}_{D(\mathbf{Ab})}(\mathbb{Z}/2, \mathbb{Z}/4) \xrightarrow{v_*} \mathrm{Hom}_{D(\mathbf{Ab})}(\mathbb{Z}/2, \mathbb{Z}/2) \xrightarrow{w_*} \mathrm{Hom}_{D(\mathbf{Ab})}(\mathbb{Z}/2, \mathbb{Z}/2[1])$$

shows that $w = 0$ would make v_* surjective. By the last clause of theorem 3.4.10(2) the first two groups are $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/4)$ and $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2)$, with v_* composition with v. The two homomorphisms $\mathbb{Z}/2 \to \mathbb{Z}/4$ send 1 to 0 and to 2, and $v(2) = 0$, so v_* is zero and misses $\mathrm{id}_{\mathbb{Z}/2}$. Hence $w \neq 0$.

So the short exact sequence determines, through its triangle, the nonzero element of $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2)$. Through the resolution P of example 3.4.2, $\mathrm{Hom}_{D(\mathbf{Ab})}(\mathbb{Z}/2, \mathbb{Z}/2[1]) \cong \mathrm{Hom}_{D(\mathbf{Ab})}(P, P[1]) \cong \mathrm{Hom}_{\mathrm{K}(\mathbf{Ab})}(P, P[1])$ by lemma 3.4.4(1), and w corresponds to the chain map φ with $\varphi(e) = f$. No chain map $\mathbb{Z}/2 \to \mathbb{Z}/2[1]$ represents w, the two complexes being concentrated in the different degrees 0 and -1, so that $\mathrm{Hom}_{\mathrm{K}(\mathbf{Ab})}(\mathbb{Z}/2, \mathbb{Z}/2[1]) = 0$.

The tensor product is derived in the same way, by resolving with projective or flat modules. For a ring R, write Mod-R and R-Mod for the categories of right and left R-modules. Both have enough projectives, free modules being projective (example 2.2.2(4)).

Definition 3.4.12: Total Tensor Complex

Let X be a complex of right R-modules and Y a complex of left R-modules. The *total tensor complex* $\mathrm{Tot}(X \otimes_R Y)$ is the complex of abelian groups with

$$\mathrm{Tot}(X \otimes_R Y)^n := \bigoplus_{i+j=n} X^i \otimes_R Y^j, \qquad d(x \otimes y) := d_X x \otimes y + (-1)^i x \otimes d_Y y$$

for $x \in X^i$ and $y \in Y^j$, the sum being a direct sum in **Ab**. For chain maps $f\colon X \to X'$ and $g\colon Y \to Y'$, the chain map $\mathrm{Tot}(f \otimes g)$ is $x \otimes y \mapsto f(x) \otimes g(y)$.

The differential is defined on the whole direct sum, since it carries the summand $X^i \otimes_R Y^j$ into the two summands $X^{i+1} \otimes_R Y^j$ and $X^i \otimes_R Y^{j+1}$. This is the direct-sum total complex $\mathrm{Tot}^{\oplus}$ of proposition 2.5.2. When X and Y are bounded above each sum is finite: if $X^i = 0$ for $i > a$ and $Y^j = 0$ for $j > b$, then $i + j = n$ forces $n - b \le i \le a$, and $\mathrm{Tot}(X \otimes_R Y)^n = 0$ for $n > a + b$. The sign is the twist $(-1)^p$ of section 2.5, which makes the squares of the grid $X^i \otimes_R Y^j$ anticommute, and

$$d\big(d(x \otimes y)\big) = d_X d_X x \otimes y + (-1)^{i+1} d_X x \otimes d_Y y + (-1)^i d_X x \otimes d_Y y + x \otimes d_Y d_Y y = 0$$

The map $\mathrm{Tot}(f \otimes g)$ commutes with d because f and g commute with the differentials and preserve degrees.

Example 3.4.13: Two Total Tensor Complexes for $\mathbb{Z}/2$

Let $R = \mathbb{Z}$ and let P be the resolution $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ of example 3.4.2, with $de = 2f$. For $Y = \mathbb{Z}/2$ in degree 0, $\mathrm{Tot}(P \otimes \mathbb{Z}/2)$ is $\mathbb{Z}/2 \xrightarrow{0} \mathbb{Z}/2$ in degrees -1 and 0, with cohomology $\mathbb{Z}/2$ in both degrees. For $Y = P$, the complex $\mathrm{Tot}(P \otimes P)$ has bases $\{e \otimes e\}$, $\{e \otimes f,\ f \otimes e\}$ and $\{f \otimes f\}$ in degrees -2, -1 and 0, and

$$d(e \otimes e) = 2f \otimes e - 2e \otimes f, \qquad d(e \otimes f) = d(f \otimes e) = 2f \otimes f$$

the minus sign being $(-1)^{-1}$. The kernel of d in degree -1 is spanned by $e \otimes f - f \otimes e$, and the image of d in that degree is spanned by twice it, so the cohomology is 0 in degree -2, $\mathbb{Z}/2$ in degree -1 and $\mathbb{Z}/2$ in degree 0, the same as for $\mathrm{Tot}(P \otimes \mathbb{Z}/2)$. Without the sign, $d(e \otimes e)$ would be $2f \otimes e + 2e \otimes f$, and $d \circ d$ would send $e \otimes e$ to $8f \otimes f \neq 0$.

Lemma 3.4.14: Tensoring with a Bounded-Above Flat Complex Preserves Quasi-Isomorphisms

Let R be a ring. All complexes below are bounded above, of right R-modules on the left of $\otimes_R$ and of left R-modules on the right.

1. If f and f' are homotopic, so are $\mathrm{Tot}(f\otimes g)$ and $\mathrm{Tot}(f'\otimes g)$, and likewise in the second variable. Hence $\mathrm{Tot}(-\otimes_R -)$ is a functor on the product of the homotopy categories, with values in $\mathrm{K}^-(\mathbf{Ab})$.
2. If X is exact and every Y^j is flat, or Y is exact and every X^i is flat, then $\mathrm{Tot}(X\otimes_R Y)$ is exact.
3. If every Y^j is flat and $s\colon X\to X'$ is a quasi-isomorphism, then $\mathrm{Tot}(s\otimes Y)$ is a quasi-isomorphism. If every X^i is flat and $t\colon Y\to Y'$ is a quasi-isomorphism, then $\mathrm{Tot}(X\otimes t)$ is a quasi-isomorphism.

***Proof*:**
Step 1: homotopies. Here $\mathrm{Tot}(f\otimes g)$ is bilinear in (f,g). If $f-f'=dh+hd$, then $H(x\otimes y):=h(x)\otimes g(y)$ satisfies, for $x\in X^i$,

$$(dH+Hd)(x\otimes y)=dhx\otimes gy+(-1)^{i-1}hx\otimes dgy+hdx\otimes gy+(-1)^i hx\otimes gdy=(dh+hd)x\otimes gy$$

using $gd=dg$, so $\mathrm{Tot}\big((f-f')\otimes g\big)=dH+Hd$. If $g-g'=dk+kd$, then $H(x\otimes y):=(-1)^i f(x)\otimes k(y)$ satisfies

$$(dH+Hd)(x\otimes y)=(-1)^i dfx\otimes ky+fx\otimes dky+(-1)^{i+1}fdx\otimes ky+fx\otimes kdy=fx\otimes(dk+kd)y$$

using $fd=df$. This is (1), the degree bound having been checked after definition 3.4.12.

Step 2: exactness. The groups $X^i\otimes_R Y^j$ with the maps $d_X\otimes 1$ and $(-1)^i(1\otimes d_Y)$ form a double complex with anticommuting squares whose total complex is $\mathrm{Tot}(X\otimes_R Y)$. Suppose X exact and every Y^j flat. For fixed j, the row $X\otimes_R Y^j$ with differential $d_X\otimes 1$ is the image of X under $-\otimes_R Y^j$, which is exact by [8, Flat Module]. An exact functor preserves kernels and images, so it carries the exact complex X to an exact complex, and every row is exact. In the other case $X^i\otimes_R -$ is exact for every i, and every column $X^i\otimes_R Y$, with differential $(-1)^i(1\otimes d_Y)$, is exact. If $X^i=0$ for $i>a$ and $Y^j=0$ for $j>b$, re-indexing by $p=i-a$ and $q=j-b$ places the double complex in the bidegrees $p,q\le 0$ without changing any object or map. The sign $(-1)^i$ differs from the twist $(-1)^p=(-1)^{i-a}$ of section 2.5 by the constant $(-1)^a$, and $x\otimes y\mapsto(-1)^{aj}x\otimes y$, for $y\in Y^j$, is an isomorphism between the two total complexes. So proposition 2.5.4 makes the total complex exact.

Step 3: the first variable of (3). Let every Y^j be flat and $s\colon X\to X'$ a quasi-isomorphism. As graded groups,

$$\begin{aligned} MC\big(\mathrm{Tot}(s\otimes Y)\big)^n &= \mathrm{Tot}(X\otimes_R Y)^{n+1}\oplus\mathrm{Tot}(X'\otimes_R Y)^n \\ &= \bigoplus_{i+j=n}\big(X^{i+1}\oplus X'^i\big)\otimes_R Y^j = \mathrm{Tot}\big(MC(s)\otimes_R Y\big)^n \end{aligned}$$

and the differentials agree. For $x \in X^{i+1}$, $x' \in X'^{i}$ and $y \in Y^{j}$, the differential of $\mathrm{Tot}(MC(s) \otimes_R Y)$ sends $(x, x') \otimes y$ to

$$\big(-dx,\ sx + dx'\big) \otimes y + (-1)^i (x, x') \otimes dy$$

since (x, x') has degree i in $MC(s)$. By definition 2.1.23, the differential of the cone sends $(x \otimes y,\ x' \otimes y)$ to

$$\big(-d(x \otimes y),\ sx \otimes y + d(x' \otimes y)\big) = \big(-dx \otimes y + (-1)^i x \otimes dy,\ sx \otimes y + dx' \otimes y + (-1)^i x' \otimes dy\big)$$

the sign $(-1)^{i+1}$ of $x \in X^{i+1}$ in $d(x \otimes y)$ being negated. These are the same element. The cone $MC(s)$ is exact by proposition 2.1.26 and bounded above, so Step 2 makes $\mathrm{Tot}(MC(s) \otimes_R Y) = MC(\mathrm{Tot}(s \otimes Y))$ exact, and proposition 2.1.26 makes $\mathrm{Tot}(s \otimes Y)$ a quasi-isomorphism.

Step 4: the second variable of (3). Here the identification needs a sign. Define $\theta\colon \mathrm{Tot}\big(X \otimes_R MC(t)\big) \to MC\big(\mathrm{Tot}(X \otimes t)\big)$ by $\theta\big(x \otimes (y, y')\big) := \big((-1)^i x \otimes y,\ x \otimes y'\big)$ for $x \in X^i$, $y \in Y^{j+1}$ and $y' \in Y'^{j}$, a bijection in each degree. Applying θ to $d\big(x \otimes (y, y')\big) = dx \otimes (y, y') + (-1)^i x \otimes (-dy,\ ty + dy')$ gives

$$\big((-1)^{i+1} dx \otimes y - x \otimes dy,\ dx \otimes y' + (-1)^i x \otimes ty + (-1)^i x \otimes dy'\big)$$

and the cone differential applied to $\theta\big(x \otimes (y, y')\big)$ gives $\big(-(-1)^i d(x \otimes y),\ (-1)^i x \otimes ty + d(x \otimes y')\big)$, the same element. So θ is an isomorphism of complexes. With every X^i flat and $MC(t)$ exact, Step 2 and proposition 2.1.26 finish as in Step 3, completing the proof.

Definition 3.4.15: Derived Tensor Product

Let R be a ring. For every bounded-above complex B of left R-modules fix a projective resolution $\pi_B\colon P_B \to B$. For a bounded-above complex A of right R-modules, the *derived tensor product* of A and B is

$$A \otimes^L_R B := \mathrm{Tot}(A \otimes_R P_B)$$

regarded as an object of $D^-(\mathbf{Ab})$.

Projective resolutions exist by theorem 2.2.17. For a fixed right module M, $B \mapsto M \otimes^L_R B$ is the complex $M \otimes_R P_B$, the right exact functor $M \otimes_R -$ applied term by term, which is the total left derived functor $L(M \otimes_R -)$ of definition 3.3.5 with the projectives, adapted by example 3.3.3(2).

Example 3.4.16: $\mathbb{Z}/m \otimes^L \mathbb{Z}/n$

Let $R = \mathbb{Z}$, let $m, n \geq 1$ and let $d = \gcd(m, n)$. The complex $\mathbb{Z} \xrightarrow{n} \mathbb{Z}$ in degrees -1 and 0 is a projective resolution of $\mathbb{Z}/n$, so $\mathbb{Z}/m \otimes^L_{\mathbb{Z}} \mathbb{Z}/n$ is

$$\mathbb{Z}/m \xrightarrow{n} \mathbb{Z}/m$$

in degrees -1 and 0. Its cohomology is $\{x \in \mathbb{Z}/m : nx = 0\} \cong \mathbb{Z}/d$ in degree -1, $\mathbb{Z}/(m\mathbb{Z} + n\mathbb{Z}) = \mathbb{Z}/d$ in degree 0, and 0 elsewhere. For $m = n = 2$ it is the complex $\mathrm{Tot}(P \otimes \mathbb{Z}/2)$ of example 3.4.13 with the two factors exchanged.

Theorem 3.4.17: The Derived Tensor Product Is a Balanced Bifunctor

Let R be a ring.

1. $\otimes_R^L$ is a bifunctor $D^-(\text{Mod-}R) \times D^-(R\text{-Mod}) \to D^-(\mathbf{Ab})$, and another choice of projective resolutions changes it by a natural isomorphism, as in theorem 3.4.8(3).
2. If $\varphi\colon F \to A$ is a quasi-isomorphism from a bounded-above complex of flat right modules, then both chain maps in
$$A \otimes_R^L B = \operatorname{Tot}(A \otimes_R P_B) \xleftarrow{\varphi\otimes 1} \operatorname{Tot}(F \otimes_R P_B) \xrightarrow{1\otimes \pi_B} \operatorname{Tot}(F \otimes_R B)$$
are quasi-isomorphisms, so $A \otimes_R^L B \cong \operatorname{Tot}(F \otimes_R B)$ in $D(\mathbf{Ab})$. When $F = P_A$ is a projective resolution of A, this isomorphism is natural in chain maps of A and of B.
3. If $\psi\colon G \to B$ is a quasi-isomorphism from a bounded-above complex of flat left modules, then $A \otimes_R^L B \cong \operatorname{Tot}(A \otimes_R G)$ in $D(\mathbf{Ab})$.

***Proof*:**

Step 1: the first variable. Fix B. Every P_B^j is projective, hence flat by [8, Projective Modules Are Flat]. By lemma 3.4.14(1), $A \mapsto \operatorname{Tot}(A \otimes_R P_B)$ is a functor $\mathrm{K}^-(\text{Mod-}R) \to \mathrm{K}^-(\mathbf{Ab})$, and by lemma 3.4.14(3) it carries quasi-isomorphisms to quasi-isomorphisms. Followed by Q, it inverts the quasi-isomorphisms of $\mathrm{K}^-(\text{Mod-}R)$, so it factors uniquely through the localization of $\mathrm{K}^-(\text{Mod-}R)$ at them. By box 3.2.5 the canonical functor from that localization to $D^-(\text{Mod-}R)$ is an equivalence, so it is fully faithful. With the objects of $D^-(\text{Mod-}R)$ taken to be the bounded-above complexes it is also bijective on objects, hence an isomorphism of categories, and the factorization is a functor on $D^-(\text{Mod-}R)$.

Step 2: the second variable. By lemma 3.4.4(1), with the bounded-above complex of projectives P_B, each morphism $\beta\colon B \to B'$ of $D^-(R\text{-Mod})$ determines a unique $\tilde{\beta}\colon P_B \to P_{B'}$ in $\mathrm{K}(R\text{-Mod})$ with $Q(\pi_{B'})\,Q(\tilde{\beta}) = \beta\,Q(\pi_B)$. The uniqueness argument of Step 1 of the proof of theorem 3.4.8 makes $\beta \mapsto \tilde{\beta}$ a functor. Set $A \otimes_R^L \beta := Q\big(\operatorname{Tot}(A \otimes \tilde{\beta})\big)$.

Step 3: (1). Steps 3 and 4 of the proof of theorem 3.4.8 apply with $\operatorname{Tot}(- \otimes_R -)$ in place of $\mathrm{Hom}^\bullet$, lemma 3.4.14(1) in place of lemma 3.4.3(2), and lemma 3.4.4(1) in place of lemma 3.4.4(2), with one change. Those steps pass from chain maps to all morphisms by writing a morphism as a roof $Q(f)Q(s)^{-1}$ (theorem 3.2.4), and the apex of a roof between bounded-above complexes need not be bounded above, so it lies outside $\mathrm{K}^-(\text{Mod-}R)$, the domain of the functor of Step 1, and lemma 3.4.14 is proved only for bounded-above complexes. In place of roofs, use that every morphism of the localization of $\mathrm{K}^-(\text{Mod-}R)$, which is $D^-(\text{Mod-}R)$ by Step 1, is a composite of morphisms $Q(f)$ and $Q(s)^{-1}$ with f and s chain maps of bounded-above complexes and s a quasi-isomorphism. The construction in the proof of theorem 1.2.2 exhibits every morphism in this form, and any two localizations are identified by an isomorphism commuting with Q. The interchange square in Step 3 of the proof of theorem 3.4.8 commutes for each factor $Q(f)$ and $Q(s)^{-1}$ by the argument given there, hence for every morphism, and the same factorization gives the naturality in Step 4 of that proof. The values are bounded above by the degree bound after definition 3.4.12.

Step 4: (2). The map $\varphi \otimes 1$ is a quasi-isomorphism by the first half of lemma 3.4.14(3), every P_B^j being flat, and $1 \otimes \pi_B$ is one by the second half, every F^i being flat. For naturality when $F = P_A$, let $a\colon A \to A'$ and $b\colon B \to B'$ be chain maps, let $\tilde{a}\colon P_A \to P_{A'}$ be a lift with $\pi_{A'}\tilde{a}$ homotopic to $a\pi_A$ (theorem 2.2.17), and let $\tilde{b}$ be the morphism of Step 2 for $Q(b)$, so that $\pi_{B'}\tilde{b}$ and $b\pi_B$ are homotopic by the injectivity in lemma 3.4.4(1). Then

$$(\pi_{A'} \otimes 1)(\tilde{a} \otimes \tilde{b}) = \pi_{A'}\tilde{a} \otimes \tilde{b}, \qquad (1 \otimes \pi_{B'})(\tilde{a} \otimes \tilde{b}) = \tilde{a} \otimes \pi_{B'}\tilde{b}$$

are homotopic to $(a \otimes \tilde{b})(\pi_A \otimes 1)$ and $(\tilde{a} \otimes b)(1 \otimes \pi_B)$ by lemma 3.4.14(1), so both squares commute in $\mathrm{K}(\mathbf{Ab})$.

Step 5: (3). Let $\pi_A\colon P_A \to A$ be a projective resolution. By (2), $A \otimes_R^L B \cong \mathrm{Tot}(P_A \otimes_R B)$. By lemma 3.4.14(3), the maps $1 \otimes \psi\colon \mathrm{Tot}(P_A \otimes_R G) \to \mathrm{Tot}(P_A \otimes_R B)$, every P_A^i being flat, and $\pi_A \otimes 1\colon \mathrm{Tot}(P_A \otimes_R G) \to \mathrm{Tot}(A \otimes_R G)$, every G^j being flat, are quasi-isomorphisms. So $\mathrm{Tot}(P_A \otimes_R B) \cong \mathrm{Tot}(A \otimes_R G)$ in $D(\mathbf{Ab})$, completing the proof.

Weibel [23] Definition 10.6.1 and Theorem 10.6.3 construct $\otimes_R^L$ by resolving the second variable, as here, and (2) is Weibel's Exercise 10.6.1. By (2), for a fixed left module N the functor $A \mapsto A \otimes_R^L N \cong P_A \otimes_R N$ is the total left derived functor $L(- \otimes_R N)$ of definition 3.3.5.

Theorem 3.4.18: Tor **as the Cohomology of the Derived Tensor Product**

Let R be a ring, A a right R-module and B a left R-module, regarded as complexes in degree 0. For every $n \in \mathbb{Z}$ there is an isomorphism natural in A and B

$$\mathrm{Tor}_n^R(A, B) \cong H^{-n}\big(A \otimes_R^L B\big)$$

where $\mathrm{Tor}_n^R(-, B) = L_n(- \otimes_R B)$ is the left derived functor of definition 2.3.5, as in example 2.3.6(1). In particular $H^{-n}(A \otimes_R^L B) = 0$ for $n < 0$, $H^0(A \otimes_R^L B) \cong A \otimes_R B$, and $\mathrm{Tor}_n^R(A, B) \cong H^{-n}(A \otimes_R P_B)$ for a projective resolution P_B of B.

***Proof*:**
Let $\pi_A\colon P_A \to A$ be a projective resolution in the sense of definition 2.2.11, with $P_A^n = 0$ for $n > 0$. By theorem 3.4.17(2) there is an isomorphism $A \otimes_R^L B \cong \mathrm{Tot}(P_A \otimes_R B)$ natural in A and B, and since B sits in degree 0 this is the complex $P_A \otimes_R B$ with differential $d \otimes 1$. Its $(-n)$th cohomology is $L_n(- \otimes_R B)(A) = \mathrm{Tor}_n^R(A, B)$ by definition 2.3.5, whose construction, stated in example 2.3.6(1) for a commutative ring, uses no commutativity. For $n < 0$ the term P_A^{-n} is zero. In degree 0, the sequence $P_A^{-1} \to P_A^0 \to A \to 0$ is exact and $- \otimes_R B$ is right exact, so $H^0(P_A \otimes_R B) = \mathrm{coker}\big(P_A^{-1} \otimes_R B \to P_A^0 \otimes_R B\big) \cong A \otimes_R B$. The last statement is definition 3.4.15 itself, $\mathrm{Tot}(A \otimes_R P_B)$ being $A \otimes_R P_B$ when A sits in degree 0, completing the proof.

The theorem is Weibel [23] Theorem 10.6.3 and Corollary 10.6.4, and Gelfand-Manin [15] III.6.15. Its last statement is the balance of Tor, here over an arbitrary ring.

Over a commutative ring R a module is both a left and a right module, and each $X^i \otimes_R Y^j$ and each $\mathrm{Hom}_R(X^i, Y^j)$ is an R-module. Then $\mathrm{Tot}(X \otimes_R Y)$ and the Hom complex $\mathrm{Hom}_R^\bullet(X, Y)$ of definition 3.4.1, formed in R-Mod, are complexes of R-modules. A chain map of complexes of R-modules is a quasi-isomorphism exactly when it is one of complexes of abelian groups, cohomology

being computed on the underlying groups, so the constructions and proofs above run inside R-Mod. Writing $D(R)$ for $D(R\text{-Mod})$, they give

$$\otimes_R^L \colon D^-(R) \times D^-(R) \to D^-(R), \qquad R\mathrm{Hom}_R \colon D(R)^{\mathrm{op}} \times D^+(R) \to D(R)$$

with $R\mathrm{Hom}_R(B,C) = \mathrm{Hom}_R^\bullet(B, I_C)$. The forgetful functor R-Mod $\to$ **Ab** is exact, so it induces a functor $D(R) \to D(\mathbf{Ab})$, which carries these two bifunctors to those of theorem 3.4.17 and theorem 3.4.8.

Theorem 3.4.19: Adjunction Between $\otimes^L$ and RHom

Let R be a commutative ring, let $A, B \in D^-(R)$ and let $C \in D^+(R)$. There is an isomorphism

$$\mathrm{Hom}_{D(R)}\big(A \otimes_R^L B,\ C\big) \cong \mathrm{Hom}_{D(R)}\big(A,\ R\mathrm{Hom}_R(B,C)\big)$$

natural in A and C.

***Proof*:**

Step 1: the adjunction for complexes. Let X and Y be bounded-above complexes and Z any complex of R-modules, and define

$$\Phi \colon \mathrm{Hom}_R^\bullet\big(\mathrm{Tot}(X \otimes_R Y), Z\big) \to \mathrm{Hom}_R^\bullet\big(X, \mathrm{Hom}_R^\bullet(Y,Z)\big), \qquad \Phi(\varphi)(x)(y) := \varphi(x \otimes y)$$

In degree n the source is the product over pairs (i,j) of $\mathrm{Hom}_R(X^i \otimes_R Y^j, Z^{i+j+n})$, a homomorphism out of a finite direct sum being a family of homomorphisms, and the target is the product over the same pairs of $\mathrm{Hom}_R\big(X^i, \mathrm{Hom}_R(Y^j, Z^{i+j+n})\big)$, a homomorphism into a product being a family. On each factor Φ is the bijection of [8, Hom-Tensor Adjunction], with both rings equal to R. For the differentials, let φ have degree n, $x \in X^i$ and $y \in Y^j$. By definition 3.4.1 and definition 3.4.12,

$$(D\varphi)(x \otimes y) = d_Z\varphi(x \otimes y) - (-1)^n \varphi(dx \otimes y) - (-1)^{n+i}\varphi(x \otimes dy)$$

On the other side $\Phi(\varphi)(x)$ has degree $i+n$ in $\mathrm{Hom}_R^\bullet(Y,Z)$, where D is $d_Z \circ - - (-1)^{i+n}(-) \circ d_Y$, so

$$\begin{aligned}\big(D\Phi(\varphi)\big)(x)(y) &= \big(D(\Phi(\varphi)(x))\big)(y) - (-1)^n \Phi(\varphi)(dx)(y) \\ &= d_Z\varphi(x \otimes y) - (-1)^{i+n}\varphi(x \otimes dy) - (-1)^n \varphi(dx \otimes y)\end{aligned}$$

The two agree, so Φ is an isomorphism of complexes.

Step 2: resolutions. Let $\pi_A \colon P_A \to A$ and $\pi_B \colon P_B \to B$ be the projective resolutions and $\iota_C \colon C \to I_C$ the injective resolution. By theorem 3.4.17(2) with $F = P_A$, the chain map $\pi_A \otimes 1 \colon T := \mathrm{Tot}(P_A \otimes_R P_B) \to \mathrm{Tot}(A \otimes_R P_B) = A \otimes_R^L B$ is a quasi-isomorphism. By corollary 3.4.5(1), $\pi_B^* \colon \mathrm{Hom}_R^\bullet(B, I_C) \to \mathrm{Hom}_R^\bullet(P_B, I_C) =: H$ is one, so $R\mathrm{Hom}_R(B,C) \cong H$ in $D(R)$.

Step 3: the chain of isomorphisms. Using Step 2 and $Q(\iota_C)$, then lemma 3.4.4(2), lemma 3.4.3(1),

Step 1, lemma 3.4.3(1), lemma 3.4.4(1), and finally Step 2 with $Q(\pi_A)$,

$$\begin{aligned}\mathrm{Hom}_{D(R)}(A \otimes^L_R B, C) &\cong \mathrm{Hom}_{D(R)}(T, I_C) \cong \mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(T, I_C) = H^0\mathrm{Hom}^\bullet_R(T, I_C)\\ &\cong H^0\mathrm{Hom}^\bullet_R(P_A, H) = \mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(P_A, H) \cong \mathrm{Hom}_{D(R)}(P_A, H)\\ &\cong \mathrm{Hom}_{D(R)}\big(A, R\mathrm{Hom}_R(B, C)\big)\end{aligned}$$

Step 4: naturality. By lemma 3.4.4(1), a morphism $\alpha\colon A \to A'$ of $D^-(R)$ determines a unique $\tilde{\alpha}\colon P_A \to P_{A'}$ in $\mathrm{K}(R\text{-Mod})$ with $Q(\pi_{A'})\,Q(\tilde{\alpha}) = \alpha\,Q(\pi_A)$, and under the isomorphisms of Step 2 it acts as $Q(\mathrm{Tot}(\tilde{\alpha} \otimes 1))$ on T, by the factorization in Step 1 of the proof of theorem 3.4.17, and as $Q(\tilde{\alpha})$ on P_A. By theorem 3.4.8(1), a morphism $\gamma\colon C \to C'$ acts on I_C as $Q(\tilde{\gamma})$. Each isomorphism of Step 3 commutes with precomposition by these and postcomposition by $\tilde{\gamma}$: the two uses of lemma 3.4.3(1) by its part (3), the two uses of lemma 3.4.4 because Q is a functor, and Φ because $\Phi\big(\varphi \circ \mathrm{Tot}(\tilde{\alpha} \otimes 1)\big) = \Phi(\varphi) \circ \tilde{\alpha}$ and $\Phi(\tilde{\gamma} \circ \varphi) = \tilde{\gamma}_* \circ \Phi(\varphi)$ by the formula for Φ. So the composite isomorphism is natural in A and C, completing the proof.

This is Weibel [23] Theorem 10.8.7, which states it as an adjunction between $-\otimes^L_R B\colon D^-(R) \to D^-(R)$ and $R\mathrm{Hom}_R(B, -)\colon D^+(R) \to D^+(R)$, for fixed $B \in D^-(R)$. The two functors have different domains, so the precise content is the bijection of the theorem, natural in $A \in D^-(R)$ and $C \in D^+(R)$. That $R\mathrm{Hom}_R(B, C)$ lies in $D^+(R)$ is theorem 3.4.8(4).

3.4.20 Note: Where the Bounds Enter The bounds enter at two places. Lemma 3.4.4 needs P bounded above and I bounded below, through proposition 2.2.8, and lemma 3.4.14(2) needs the flat factor bounded above, through the acyclic assembly lemma proposition 2.5.4. The bound on the other factor only makes the sums in definition 3.4.12 finite. With the direct sums of that definition, the exactness in (2) needs only the bound on the flat factor, by the remark after proposition 2.5.4. The first bound cannot be dropped: example 5.1.7 gives an exact complex of injective $\mathbb{Z}/4$-modules, unbounded in both directions, for which the conclusion of lemma 3.4.4(2) fails, and exercise 3.4.1 (1) asks for a second one. Defining $R\mathrm{Hom}$ and $\otimes^L$ on unbounded complexes, with the adjunction of theorem 3.4.19, requires K-injective and K-flat resolutions, which chapter 5 constructs.

Exercise 3.4.1

1. Let k be a field, $\Lambda = k[x]/(x^2)$, and I the complex of Λ-modules with $I^n = \Lambda$ for every $n \in \mathbb{Z}$ and every differential multiplication by x. Show that Λ is an injective Λ-module, that I is exact, and that id_I is not null-homotopic. Conclude that $\mathrm{Hom}_{\mathrm{K}(\Lambda\text{-Mod})}(I, I) \neq 0 = \mathrm{Hom}_{D(\Lambda\text{-Mod})}(I, I)$, so that the bound in lemma 3.4.4(2) cannot be dropped. (For the null-homotopy, use lemma 2.1.28.)

2. Let $a, b \geq 1$ and $d = \gcd(a, b)$. Compute $H^n R\mathrm{Hom}(\mathbb{Z}/a, \mathbb{Z}/b)$ for every n twice: from the injective resolution $\mathbb{Z}/b \to \mathbb{Q}/\mathbb{Z} \xrightarrow{b} \mathbb{Q}/\mathbb{Z}$, whose first map sends 1 to the class of $1/b$, and, through proposition 3.4.9, from the projective resolution $\mathbb{Z} \xrightarrow{a} \mathbb{Z}$ of $\mathbb{Z}/a$. Check that both give $\mathbb{Z}/d$ for $n = 0, 1$ and 0 otherwise, in agreement with example 2.4.10.

3. Let $\mathcal{A}$ have enough injectives, let A be an object and let $0 \to B' \xrightarrow{u} B \xrightarrow{v} B'' \to 0$ be a short exact sequence in $\mathcal{A}$. Using theorem 3.2.10, proposition 3.1.8 and theorem 3.4.10, derive a

long exact sequence

$$0 \to \mathrm{Hom}(A, B') \to \mathrm{Hom}(A, B) \to \mathrm{Hom}(A, B'') \to \mathrm{Ext}^1(A, B') \to \cdots$$
$$\cdots \to \mathrm{Ext}^n(A, B) \to \mathrm{Ext}^n(A, B'') \to \mathrm{Ext}^{n+1}(A, B') \to \cdots$$

in which $\mathrm{Ext}^n(A, B') \to \mathrm{Ext}^n(A, B)$ is $\mathrm{Ext}^n(A, u)$. State and derive the corresponding sequence in the first variable for a short exact sequence $0 \to A' \to A \to A'' \to 0$.

4. Let A be the complex of abelian groups with $A^k = \mathbb{Z}$ for $k \geq 0$, $A^k = 0$ for $k < 0$, and every differential zero. Show that $H^{-k}R\mathrm{Hom}(A, \mathbb{Z}) \cong \mathbb{Z}$ for every $k \geq 0$, so that $R\mathrm{Hom}(A, \mathbb{Z})$ does not lie in $D^+(\mathbf{Ab})$ and the hypothesis of theorem 3.4.8(4) cannot be dropped. (Use the resolution $I_{\mathbb{Z}}$ of example 3.4.7 and lemma 2.1.11.)

5. Let $s\colon P \to \mathbb{Z}/2$ be the projective resolution of example 3.4.2. Show that $\mathrm{Tot}(s \otimes \mathbb{Z}/2)$ is not a quasi-isomorphism. Which hypothesis of lemma 3.4.14(3) fails, and which complex in Step 2 of its proof fails to be exact? Compare example 3.3.1, which tests the same quasi-isomorphism against $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2, -)$.

6. Let R be a commutative ring. Show that $R \otimes^L_R B \cong B$ in $D(R)$ for every $B \in D^-(R)$, and that $\mathrm{Hom}_{D(R)}(R, X) \cong H^0(X)$ for every complex X of R-modules. Combine these with theorem 3.4.19 for $A = R$ to obtain $\mathrm{Hom}_{D(R)}(B, C) \cong H^0R\mathrm{Hom}_R(B, C)$ for $B \in D^-(R)$ and $C \in D^+(R)$, and compare with theorem 3.4.10(1).

7. Let R be a commutative ring and X, Y bounded-above complexes of R-modules. Show that $\mathrm{sw}(x \otimes y) := (-1)^{ij}\, y \otimes x$, for $x \in X^i$ and $y \in Y^j$, is an isomorphism of complexes $\mathrm{Tot}(X \otimes_R Y) \to \mathrm{Tot}(Y \otimes_R X)$, and that the same formula without the sign is not a chain map when $X = Y = P$ is the complex of example 3.4.13. Deduce that $A \otimes^L_R B \cong B \otimes^L_R A$ in $D(R)$ for $A, B \in D^-(R)$, and that $\mathrm{Tor}^R_n(A, B) \cong \mathrm{Tor}^R_n(B, A)$ for R-modules A and B.

4

Applications of (co)Homology

With cohomology theory more securely under our belt, it is time to use it! All the work in this chapter can be learnt without the need for any algebraic topology background, as it is the goal of the book to minimize the need for external material. However, the proceeding ideas are abstractions of ideas in algebraic topology, and ignoring the source of inspiration for these ideas is itself a disservice to the pedagogy of the material. Therefore, the section will be written in such a way that a background in homology/cohomology will be advantageous. Namely, it is best if the reader is familiar with simplicial and singular homology, how cohomology is derived from $\mathrm{Hom}(C_\bullet(X), G) = C^\bullet(X; G)$ and the relation of homology to cohomology (ex. the universal coefficient theorem for cohomology). The reader may choose to consult [17] or [16] for the details.

This chapter does *not* present an in-depth coverage of any of the material, in particular these subjects are a large source of modern development in mathematics (as of the early 21st century). The material presented is to demonstrate to the reader the power of cohomological theories so that the reader has an easier time when reading the more dense books containing this information. For a book in the EYNTKA series covering group and Galois cohomology, see [11], and for K-theory see [19].

In this chapter we:

1. Shall use the above theory to find cohomological invariants of groups
2. Shall show some ways of interpreting Ext and Tor
3. Shall show that the notion of dimension from the geometric setting can be brought over to the algebraic!
4. Some K-theory will certainly interest the reader

We shall not do sheaf cohomology, as (naturally) it required sheaf theory that was not covered, and it is an exploration for EYNTKA Algebraic Geometry [4].

4.1 Group Cohomology

We start with no link to algebraic topology, and shall include the relation to singular cohomology before the examples.

Given a (not necessarily abelian) group G, what would it mean to extract cohomological information from it? A (co)chain would have to naturally be formed, and a functor must be found to derive. Groups themselves do not form an abelian category, but it was shown that representations, that is $k[G]$-modules, do form them. This is perhaps a first place in which we may think we can study groups, however this category is too nice (think of (co)homology with $\mathbb{Z}$ vs. $\mathbb{R}/\mathbb{C}$ coefficients). What is more useful is it's generalization:

Definition 4.1.1: G-Modules

Let G be an arbitrary group, and M an abelian group. Then M is a *G-module* if there exists a [left] group-action on M. Equivalently, if there exists a $\mathbb{Z}[G]$-module structure on M.

The reader is free to parse through the technicalities of the group action. All $k[G]$-modules presented in [9, Representation of Finite Groups] are $\mathbb{Z}[G]$-modules with the action restricted. An example of a G-module that is not a representation is the action of the group $\mathrm{Gal}_K(L)$ on the module L/K (a galois extension of L over K). All finite groups G have a non-trivial G-module given by the action of G on the group-ring $k[G]$ (note the addition is abelian). From a geometric perspective, modules represent sheaves (or in special cases vector bundles) over some schemes,

As we have seen many times at this point, the object acts on an abelian group can give lots of information on the object acting on it and the object being acted on. Given a G-module M, we can define the following:

Definition 4.1.2: G-Invariant subgroup

Let M be a G-module. Then the *G-invariant* set of M is the set:

$$M^G := \{m \in M \mid g \cdot m = m,\ \forall g \in G\}$$

which is the largest trivial G-submodule of M.

It is easy to see that $(-)^G$ is a functor. The following will help in computing cohomology

Corollary 4.1.3: Hom Representation of G-invariant

$$M^G \cong \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, M^G) \cong \mathrm{Hom}_G(\mathbb{Z}, M)$$

***Proof*:**
Give $\mathbb{Z}$ the trivial G-action, which is the action the statement requires.

For any abelian group N, evaluation at 1 is an isomorphism $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, N) \to N$, with inverse sending x to the map $m \mapsto mx$. Taking $N = M^G$ gives the first isomorphism.

For the second, let $\varphi\colon \mathbb{Z} \to M$ be $\mathbb{Z}[G]$-linear and put $m = \varphi(1)$. Equivariance gives $\varphi(g \cdot 1) = g\varphi(1)$, and $g \cdot 1 = 1$ because the action on $\mathbb{Z}$ is trivial, so $m = gm$ for every $g \in G$, that is $m \in M^G$. Conversely, any $m \in M^G$ defines a $\mathbb{Z}[G]$-linear map $\mathbb{Z} \to M$, $k \mapsto km$, since $g(km) = k(gm) = km$. These assignments are mutually inverse and additive, so $\mathrm{Hom}_G(\mathbb{Z}, M) \cong M^G$, completing the proof.

Similarly, we can define:

Definition 4.1.4: G-Coinvariant subgroup

Let M be a G-module. Then the *G-coinvariant* of M is the set:

$$M_G := \frac{M}{\langle (gm - m) \mid g \in G, m \in M \rangle}$$

and is the maximal invariant quotient module with the trivial action.

Just like $(-)^G$, $(-)_G$ is also a functor. The reader should further verify that

Proposition 4.1.5: Tensor Representation of Coinvariant Module

$$M_G \cong_G \mathbb{Z} \otimes_{\mathbb{Z}[G]} M$$

Proposition 4.1.6: Invariants Is Left-exact

The functor $(-)^G$ is left-exact, the functor $(-)_G$ is right exact.

***Proof*:**
First, prove that $(-)^G$ is right-adjoint to the trivial-action functor.

$$\mathrm{Hom}(T(M), N) \cong \mathrm{Hom}(M, N^G)$$

And similarly for $(-)_G$; it is left-adjoint to the trivial-action functor. As $T(-)$ is an exact functor, $(-)^G$ and $(-)_G$ are left-exact and right-exact respectively, completing the proof.

Both of these functors are not exact:

Example 4.1.7: Lack Of Exactness's
Let $G = C_2$ be the cyclic group of order 2 and:

$$0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where $\epsilon(1 \cdot a + \sigma \cdot b) = a + b$ and I_G is the kernel. This sequence is exact. Take $(-)^G$:

$$0 \to (I_G)^G \to (\mathbb{Z}[G])^G \xrightarrow{\epsilon^G} (\mathbb{Z})^G \to 0$$

i.e.:

$$0 \to 0 \to \Delta \xrightarrow{\epsilon^G} \mathbb{Z} \to 0$$

where $\Delta = \{(a,a) \mid a \in \mathbb{Z}\}$. Then ϵ^G is no longer surjective as $\epsilon^G(a,a) = 2a$. The above example can also be used to show $(-)_G$ is not left-exact. More generally, if $0 \to M_2^G \to M_2 \to M_3 \to 0$ is exact, taking $(-)^G$ does not change $M - 2^G$, but now we get an isomorphism between the 1st and second terms, so it cannot be surjective.

More generally, let G act trivially on a coset $[m]$ of a quotient M/L. Can this action always be lifted so that G acts trivially on a representative m?

Thus, it is fruitful to take the ith right-derived functor of $(-)^G$ (similarly the left-derived functors of $(-)_G$):

$$\mathbf{R}^i(-)^G := H^i(G,-)$$

which would give us the long exact sequence of abelian groups:

$$0 \to L^G \to M^G \to N^G \to H^1(G,L) \to H^1(G,M) \to \cdots$$

Since $M^G = \mathrm{Hom}_G(\mathbb{Z}, M)$ by taking an injective resolution of M, or a projective resolution of $\mathbb{Z}$, we can write:

$$H^i(G,M) = \mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

Note that $H^0(G,M) = M^G$, hence the higher order cohomology groups represent some failure for points to be fixed. Let us compute some examples to show the power of the machinery developed in chapter 2 before concretely demonstrating how the cohomology groups represent the obstruction of points being fixed.

Example 4.1.8: Group Cohomologies – Homological Algebra Computations

1. The trivial group has trivial cohomology
2. (Integers) Let $G = \mathbb{Z}$. Then $\mathbb{Z}[G] \cong \mathbb{Z}[x, x^{-1}]$. Then since $\mathbb{Z}[x,x^{-1}]/(1-x) \cong_G \mathbb{Z}$ with the trivial action, define the following free (hence projective) resolution:

   $$\cdots \to 0 \to \mathbb{Z}[x,x^{-1}] \xrightarrow{\cdot(1-x)} \mathbb{Z}[x,x^{-1}] \to 0 \to \cdots$$

   Applying the (contravariant) $\mathrm{Hom}_{\mathbb{Z}[x,x^{-1}]}(-,M)$, we can compute the cohomology of

   $$\cdots \to 0 \to M \xrightarrow{\cdot(1-x)} M \to 0 \to \cdots$$

   Then we can conclude by direct computation:

   $$H^0(\mathbb{Z},M) \cong \ker(M \xrightarrow{\cdot(1-x)} M) = M^{\mathbb{Z}}$$

   and:

   $$H^1(\mathbb{Z},M) = \frac{M}{(1-x)M}$$

   and furthermore $H^i(\mathbb{Z},M) = 0$ for $i \notin \{0,1\}$
3. Let $G = C_n$ be a finite cyclic group of over n with generator σ. Let us form a free resolution of $\mathbb{Z}[G]$ by taking advantage of the fact that $\sigma - 1$ and $N = \sum_{i=0}^{n-1} \sigma^i$ are zero divisors:

   $$N \cdot (\sigma - 1) = (1 + \sigma + \cdots + \sigma^{n-1})(\sigma - 1) = \sigma^n - 1 = 1 - 1 = 0$$

This gives a resolution:

$$\cdots \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where

$$\epsilon\left(\sum_g a_g g\right) = \sum_g a_g$$

is the usual augmentation map. Then:

$$H^k(C_n, M) = \begin{cases} \frac{M^G}{NM} & k \text{ even}, k \geq 2 \\ \frac{{}_N M}{(\sigma-1)M} & k \text{ odd}, k \geq 1 \end{cases} \qquad \text{where} \qquad {}_N M = \{m \in M \mid N \cdot m = 0\}$$

For example, if $M = \mathbb{Z}$, then:

$$H^k(C_n, \mathbb{Z}) = \begin{cases} \mathbb{Z}/m\mathbb{Z} & k \text{ even}, k \geq 2 \\ 0 & \text{otherwise} \end{cases}$$

4. Let $G = *_{s\in S}\mathbb{Z}$ be the free product of S copies of $\mathbb{Z}$. This has a simple free resolution, namely take:
$$\epsilon : \mathbb{Z}[G] \to \mathbb{Z}$$
where the kernel is:
$$I_S = \sum_{s\in S} \mathbb{Z}[G] \cdot (s-1)$$
It should be verified that I_S is a free G-module, hence we have a free-resoluion:
$$0 \to I_S \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$
and cohomology:
$$H^k(*_{s\in S}\mathbb{Z}, \mathbb{Z}^G) = \begin{cases} \mathbb{Z} & k = 0 \\ \bigoplus_{s\in S} \mathbb{Z} & k = 1 \\ 0 & k \geq 2 \end{cases}$$

5. (If you know some algebraic geometry) *Galois Cohomology*: Take the category of varieties W over k such that when extending scalars to $\overline{k}$ we have $V/\overline{k} \cong W/\overline{k}$. Let's label it $\mathbf{Var}_{k\to\overline{k}}$, and say that the objects are:
$$\text{Obj}(\mathbf{Var}_{k\to\overline{k}}) = \left\{W/k \;\middle|\; V/\overline{k} \cong W/\overline{k}\right\}$$
and the morphism are regular morphisms. Then I claim that:
$$\mathbf{Var}_{k\to\overline{k}} \cong H^1(\text{Gal}_k(\overline{k}), \text{Aut}(V/\overline{k}))$$
To see this, first note that an element $\sigma \in \text{Gal}_k(\overline{k})$ acts on a $\overline{k}$-variety $W/\overline{k} = W_k \times_{\text{Spec}\,(k)} \text{Spec}\,(\overline{k})$ via:
$$\text{id}_{W_k} \times \text{Spec}\,(\sigma^{-1}) : W/\overline{k} \to W/\overline{k}$$
Or, for those less comfortable with fibered produced from algebraic geometry, they can think of this map as the "equivalent":
$$\text{id}_{W_k} \otimes_k \sigma^{-1} : k[W] \otimes_k \overline{k}$$

With the action defined, let us look at the correspondence. Take $W_k \in \mathbf{Var}_{k\to\overline{k}}$ so that $\varphi : W_{\overline{k}} \xrightarrow{\sim} V_{\overline{k}}$ is an isomorphism. Then notice that this isomorphism induces a map $\varphi \circ \sigma^{-1} \circ \varphi^{-1}$, which is an automorphism on $W_{\overline{k}}$. The collection of all these maps as σ varies can uniquely define W_k (k is the fixed subfield over all σ). This is true up to a choice of $\mathrm{Aut}_k(V)$, and so we quotient out by it. Then we can see this creates the correspondence between these two categories, and shows how far away the category $\mathbf{Var}_{k\to\overline{k}}$ is from being invariant to the group-action of the galois group.

6. (if you know elliptic curves) An excellent example of where galois cohomology is immediately useful is in the theory of elliptic curves. On a high level, it is relatively easy to find the solution of an elliptic curve over $\mathbb{C}$, but it becomes much more difficult to solve it over finite extensions of $\mathbb{Q}$, or $\mathbb{Q}$ itself. If the reader has read (or is reading) [10], then the have encountered the short exact sequence of $\mathrm{Gal}_K(\overline{K})$-modules:

$$0 \to E[m] \to E(\overline{K}) \xrightarrow{[m]} E(\overline{K}) \to 0$$

where $E[m]$ are the m-torsion points of an elliptic curve ($m \geq 2$), K is a number field, $E(\overline{K})$ are the points of the elliptic curve over $\overline{K}$. Let $G = \mathrm{Gal}_K(\overline{K})$. Then when considering solutions over K, the short exact sequence sequence becomes a long exact sequences:

$$0 \to E(K)[m] \to E(K) \xrightarrow{[m]} E(K) \xrightarrow{\delta} H^1(G, E[m]) \\ \to H^1(G, E(\overline{K})) \xrightarrow{[m]} H^1(G, E(\overline{K}) \to \cdots$$

This is the foundation on which the torsion-free points of E/K will be computed!

7. If the reader knows about the classifying space BG, then:

$$H^n(BG, \mathbb{Z}) \cong H^n(G, \mathbb{Z})$$

where he left hand side is singular cohomology and the right hand side is group cohomology.

What obstruction information does this capture? Since there is a failure in surjectivity $\varphi : M \twoheadrightarrow N$ when applying $(-)^G$, this means that a fixed element $n \in N^G$ does *not* necessarily come form a fixed element $m \in M^G$. Thus, even though $\varphi(m) = n$ exists, it need not be the case that there exists an $m' \in M^G$ such that $\varphi^G(m') = n$. To measure this failure, take the short exact sequence

$$0 \to L \xrightarrow{\iota} M \xrightarrow{\varphi} N \to 0$$

and let $n \in N^G$. As $N^G \subseteq N$, there exists an $m \in M$ such that $\varphi(m) = n$. Consider any $g \in G$ and take $g \cdot m - m$. Then applying the G-homomorphism:

$$\varphi(g \cdot m - m) = g \cdot \varphi(m) - \varphi(m) = g \cdot n - n \overset{!}{=} n - n = 0$$

where $\overset{!}{=}$ comes from $n \in N^G$. By exactness $\ker\varphi = \mathrm{im}\iota$, so there must exist an element $\lambda_m(g) \in L$ such that

$$\iota(\lambda_m(g)) = g \cdot m - m$$

Now, $n \in N^G$ can be lifted if and only if $g \cdot m - m = 0$ for all $g \in G$, since then $m \in M^G$. In terms of elements of L:

$$\forall g \in G,\ \iota(\lambda_m(g)) = 0$$

by injectivity, it must be that $\lambda_m(g) = 0$ for all $g \in G$. Thus, the function $\lambda_m : G \to L$ is non-trivial if the lift does not exist. Let us explore the properties of λ_m. First, if $g_1, g_2 \in G$ then:

$$\begin{aligned}\iota(\lambda_m(g_1g_2)) &= g_1g_2 \cdot m - m \\ &= g_1g_2 \cdot m - g_1 \cdot m + g_1 \cdot m - m \\ &= g_1 \cdot (g_2 \cdot m - m) + (g_1 \cdot m - m) \\ &= g_1\iota(\lambda_m(g_2)) + \iota(\lambda_m(g_1)) \\ &= \iota(g_1\lambda_m(g_2)) + \iota(\lambda_m(g_1)) \\ &= \iota(g_1\lambda_m(g_2) + \lambda_m(g_1))\end{aligned}$$

Thus, as f is injective (more specifically a monomorphism) we get:

$$\lambda_m(g_1g_2) = \lambda_m(g_1) + g_1\lambda_m(g_2)$$

A function with this property is given a name:

Definition 4.1.9: 1-Cocycle

A function $f : G \to M$ is a called a *1-cocycle*, *1-derivation*, or *crossed homomorphism* if:

$$f(gh) = f(g) + gf(h)$$

The set of all derivations is denoted $Z^1(G, M)$. If it is clear from context, the "1-" is dropped.

Now, the above λ_m was dependent on a choice of m. But $M \hookrightarrow N$ is surjective, and hence there may be another m' such that $\varphi(m') = n$. As φ is a G-homomorphism

$$\varphi(m - m') = c - c = 0$$

and so $m - m' \in \ker\varphi = \mathrm{im}\iota$. Thus, there is an $\ell \in L$ such that

$$m - m' = \iota(\ell) \qquad \text{or} \qquad m' = m + \iota(\ell)$$

Thus, let us relate $\lambda_{m'}$ to λ_m:

$$\begin{aligned}\iota(\lambda_{m'}(g)) &= g \cdot m' - m' \\ &= g \cdot (m + \iota(\ell)) - m - \iota(\ell) \\ &= g \cdot m + g \cdot \iota(\ell)) - m - \iota(\ell) \\ &= (g \cdot m - m) + g \cdot \iota(\ell) - \iota(\ell) \\ &= \iota(\lambda_m(g)) + \iota(g \cdot \ell) - \iota(\ell) \\ &= \iota(\lambda_m(g) + g \cdot \ell - \iota(\ell))\end{aligned}$$

Thus, as ι is a monomorphism:

$$\lambda_{m'}(g) = \lambda_m(g) + g \cdot \ell - \ell$$

Thus, the term $g \cdot \ell - \ell$ can be seen as the "trivial" part of the obstruction, as it does not change the difficulty in finding an obstruction. This is given a name:

Definition 4.1.10: 1-Coboundary

A function $f_m : G \to M$ is a called a *1-coboundary* or *principal derivation* if:

$$f_m(g) = g \cdot m - m$$

The set of all boundary maps is denoted $B^1(G, M)$.If it is clear from context, the "1-" is dropped.

Note that a coboundary is a cocycle:

$$f_m(gh) = gh \cdot m - m = gh \cdot m - h \cdot m + h \cdot m - m = f_m(g) + gf_m(h)$$

Hence $B^1(G, M) \subseteq Z^1(G, M)$. It shall be shown in a moment that:

$$H^1(G, M) \cong \frac{Z^1(G, M)}{B^1(G, M)}$$

First, let us show there is a map $L^G \to H^1(G, M)$. Indeed there is a well-defined G-module homomorphism:

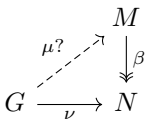
$$\ell \mapsto [g \mapsto g \cdot m - m]$$

where m is in the fiber of ℓ.

Let us next motivate the 2nd order cohomology group. Let us assume the cocycle-representation and the derived functor representation are isomorphic for the next part. Let us start with the long exact sequence we know is given:

$$0 \to L^G \xrightarrow{\alpha} M^G \xrightarrow{\beta} N^G \xrightarrow{\delta^0} H^1(G, L) \xrightarrow{\alpha_*} H^1(G, M) \xrightarrow{\beta_*} H^1(G, N) \xrightarrow{\delta^1} H^2(G, L) \cdots$$

Consider $[\nu] \in H^1(G, N)$; let us call it a 1-obstruction. Then can it be lifted some $[\mu] \in H^1(G, M)$, namely does there exist $[\mu]$ such that $\beta_*([\mu]) = [\nu]$. To find out, first take some representative cocycle $\nu : G \to N$ (notice we are now working with a map, not an element). As $M \hookrightarrow N$ is surjective, we have a diagram:



We can certainly lift each point *locally*, namely given a point $\nu(g) = n \in N$ there is a point (or points) m such that $\beta(m) = n$[1]. But this lifting can fail *globally*, namely such a lift is not guaranteed to be a 1-cocycle. This mirrors the general pattern in cohomology where having local data of a function does not imply the existence of a global function that restricts to all the local data. Let us measure this failure of lifting. If the lifted μ is a cocycle, it would satisfy:

$$\mu(g_1 g_2) - \mu(g_1) - g_1 \mu(g_2) = 0$$

Thus, define the function:

$$\psi : G \times G \to M, \qquad \psi(g_1, g_2) = \mu(g_1 g_2) - \mu(g_1) - g_1 \mu(g_2)$$

[1]Technically, the axiom of choice is invoked here for making a potentially uncountable number of choices

Let us show that for any $(g_1, g_2) \in G \times G$, $\psi(g_1, g_2) \in \ker \beta$. Indeed:

$$\begin{aligned}
\beta(\psi(g_1, g_2)) &= \beta(\mu(g_1 g_2) - \mu(g_1) - g_1\mu(g_2) \\
&= \beta(\mu(g_1 g_2)) - \beta(\mu(g_1)) - \beta(g_1\mu(g_2)) \\
&\overset{!}{=} \nu(g_1 g_2)) - \nu(g_1) - g_1\nu(g_2)) \\
&= 0 && \nu \text{ is a cocycle}
\end{aligned}$$

where $\overset{!}{=}$ comes from μ being defined as a lift of ν and hence $\beta \circ \mu = \nu$. Thus, by exactness $\ker \beta = \text{im}\alpha$ and so for each pair $(g_1, g_2) \in G \times G$, there exists a unique element in L that maps to $\psi(g_1, g_2)$, namely there exists a function $\Omega : G \times G \to L$ such that

$$\alpha(\Omega(g_1, g_2)) = \psi(g_1, g_2)$$

This Ω would now be our new 2-cocycle and are denoted $Z^2(G, M)$. Doing a similar but longer computations such as the one for the 1-cycle, we can find that:

$$g_1\Omega(g_2, g_3) - \Omega(g_1 g_2, g_3) + \Omega(g_1, g_2 g_3) - \Omega(g_1, g_2) = 0$$

Similarly, an explicit computation would fine that 2-boundaries (given a 1-chain ψ) are of the form

$$\Omega(g_1, g_2) = g_1 \cdot \psi(g_2) - \psi(g_1 g_2) + \psi(g_1)$$

These are label $B^2(G, M)$ and:

$$H^2(G, M) \cong \frac{Z^2(G, M)}{B^2(G, M)}$$

The same process continues for higher order cohomology. For the reader who knows some algebraic topology, the following informal discussion grounds the higher order group-cohomology as the singular (or simplicial) cohomology of BG, the classifying space of G with the property that $\pi_1(BG) = G$ and $\pi_n(BG) = \{0\}$ for all $n > 1$ (see [17, Existence of the Universal Bundle, Cohomology of BG is Group Cohomology]). In particular, regarding a G-module M as a local coefficient system on BG,

$$H^n(BG; M) \cong H^n(G, M)$$

where the left side is cohomology with local coefficients. When G acts trivially on M it is the singular cohomology $H^n_{\text{Sing}}(BG; M)$ with constant coefficients. Using the nerve or bar construction on BG and considering the usual covering $EG \to BG$ we get the following correspondence:

Geometric Object in EG	Correspondence in BG	Algebraic equivalent
0-Simplices (Vertices)	A single 0-simplex, base point $*$.	elements of G
1-Simplices (Edges)	Directed edge for each element $g \in G$. An arrow from $*$ to $*$ labeled by g.	Functions on G
2-Simplices (Triangles)	Oriented triangle for each pair $(g_1, g_2) \in G \times G$. Edges are the 1-simplices for g_1, g_2, and the product $g_1 g_2$.	Functions on $G \times G$
3-Simplices (Tetrahedral)	Oriented tetrahedron for each triple $(g_1, g_2, g_3) \in G \times G \times G$. Faces are the triangles for (g_2, g_3), $(g_1, g_2 g_3)$, etc.	Functions on $G \times G \times G$

n-Simplices	n-tuple $(g_1, \dots, g_n) \in G^n$.	Functions on G^n

Where "functions" here means set-function (to take into account the "failed" lifting), and the transition maps d_i captures the G-module homomorphism information. Geometrically, the G-module homomorphisms correspond to G-equivariant maps $f : G^n \to M$.

Using this intuition, we can construct a standard way of defining a projective resolution of $\mathbb{Z}$ by G-modules. First, let $\mathbb{Z}[G^n]$ be a G-module with the diagonal action given by multiplying on the left, or more specifically:

$$g \cdot (g_1, ..., g_n) = (gg_1, ..., gg_n)$$

Note that this diagonal map corresponds to the natural action of left multiplication by g on EG. These modules are free G-modules; clearly $\mathbb{Z}[G]$ is a $\mathbb{Z}$-module with basis given by the elements G. Then for $n \geq 0$:

$$\mathbb{Z}[G^{n+1}] \cong \bigoplus_{(g_1,...,g_n)\in G^n} \mathbb{Z}[G] \cdot (1, g_1, ..., g_n)$$

which shows that these are free. From this, we can consider the following projective (even free) resolution:

Definition 4.1.11: Standard Resolution

Let G be a group. Then the *standard resolution* of $\mathbb{Z}$ by G-modules is the exact sequence of G-module homomorphism:

$$\cdots \to \mathbb{Z}[G^{n+1}] \xrightarrow{d_n} \mathbb{Z}[G^n] \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where $\mathbb{Z}[G^{n+1}]$ carries the diagonal G-action and, on the basis G^{n+1},

$$d_n(g_0, ..., g_n) = \sum_{i=0}^{n} (-1)^i (g_0, ..., \hat{g}_i, ..., g_n)$$

the term $\hat{g}_i$ being omitted from the tuple, while $\epsilon(g) = 1$ for every $g \in G$. The map ϵ is also called the *augmentation map*.

Note that each $\mathbb{Z}[G^{n+1}]$ is a free $\mathbb{Z}[G]$-module, the diagonal action permuting a basis freely. We claimed this is an exact sequence, however this ought to be verified.

Lemma 4.1.12: Standard Resolution is Exact

The standard resolution of $\mathbb{Z}$ by G-modules is exact

This shows that the nomenclature of cocycles and coboundaries above is used correctly.

***Proof*:**
Exactness is a statement about the underlying abelian groups, so it is enough to produce a contracting homotopy by $\mathbb{Z}$-linear maps; these need not be G-equivariant, and indeed the ones below are not.

Step 1: The maps. Write $e \in G$ for the identity element and define

$$s_{-1}\colon \mathbb{Z} \to \mathbb{Z}[G], \quad 1 \mapsto (e), \qquad s_n\colon \mathbb{Z}[G^{n+1}] \to \mathbb{Z}[G^{n+2}], \quad (g_0, \dots, g_n) \mapsto (e, g_0, \dots, g_n)$$

extended $\mathbb{Z}$-linearly.

Step 2: The contracting identity in positive degrees. Fix $n \geq 1$ and compute $d_{n+1}s_n(g_0, \dots, g_n) = d_{n+1}(e, g_0, \dots, g_n)$. Omitting the entry in position 0 gives $(g_0, \dots, g_n)$ with sign $+1$; omitting the entry in position $j \geq 1$, which is g_{j-1}, gives $(e, g_0, \dots, \hat{g}_{j-1}, \dots, g_n)$ with sign $(-1)^j$. Reindexing by $i = j - 1$ and using $(-1)^{i+1} = -(-1)^i$,

$$d_{n+1}s_n(g_0, \dots, g_n) = (g_0, \dots, g_n) - \sum_{i=0}^{n}(-1)^i(e, g_0, \dots, \hat{g}_i, \dots, g_n)$$

The subtracted sum is exactly $s_{n-1}d_n(g_0, \dots, g_n)$, since s_{n-1} prepends e to each term of $d_n(g_0, \dots, g_n) = \sum_i (-1)^i(g_0, \dots, \hat{g}_i, \dots, g_n)$. Hence

$$d_{n+1}s_n + s_{n-1}d_n = \mathrm{id}$$

Step 3: The bottom of the complex. At the augmentation, $\epsilon s_{-1}(1) = \epsilon(e) = 1$, so $\epsilon s_{-1} = \mathrm{id}_{\mathbb{Z}}$. In degree zero,

$$d_1 s_0(g_0) = d_1(e, g_0) = (g_0) - (e), \qquad s_{-1}\epsilon(g_0) = s_{-1}(1) = (e)$$

so $d_1 s_0 + s_{-1}\epsilon = \mathrm{id}$ as well.

Step 4: Conclusion. The augmented complex therefore has its identity morphism null-homotopic as a complex of abelian groups. By Step 1 of the proof of lemma 2.1.28, a complex whose identity is null-homotopic is exact. Hence the standard resolution is exact, completing the proof.

Hence, after taking the contravariant functor $\mathrm{Hom}_{\mathbb{Z}[G]}(-, M)$, we get the cochain complex:

$$0 \to \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G], M) \to \cdots \to \mathrm{Hom}_{\mathbb{Z}[G^n]}(-, M) \xrightarrow{d_n^*} \mathrm{Hom}_{\mathbb{Z}[G^{n+1}]}(-, M) \to \cdots$$

where d_n^* maps $\varphi \mapsto \varphi \circ d_n$. Similarly, we can take the covariant functor

$$\cdots \to \mathbb{Z}[G^3] \otimes_{\mathbb{Z}[G]} A \xrightarrow{d_{2*}} \mathbb{Z}[G^2] \otimes_{\mathbb{Z}[G]} A \xrightarrow{d_{1*}} \mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} A \to 0$$

where d_{n*} is given by $(g_0, ..., g_n) \otimes a \mapsto d_n(g_0, ..., g_n) \otimes a$, the map ϵ is the augmentation map given by $\sum a_g g \mapsto \sum a_g$, and we view the modules as *right* $\mathbb{Z}[G]$-modules with the action

$$(g_1, ..., g_n) \cdot g = (g_1 g, ..., g_n g)$$

Let us now better understand these hom-sets and relate to the above discussion. Let

Definition 4.1.13: coChain

Let

$$C^n(G,M) = (\mathrm{Hom}_{\mathbf{Set}}(G^n, M), +)$$

with group operation defined on the maps given by point-wise addition. This group is called the *n-cochain group*.

From this, we can define a boundary map:

$$\begin{aligned} d^n &: C^n(G,M) \to C^{n+1}(G,M) \\ d^n(f)(g_0,...,g_n) &= g_0 f(g_1,...,g_n) - f(g_0g_1, g_2, ..., g_n) \\ &\quad + f(g_0, g_1g_2, ..., g_n) - \cdots + (-1)^n f(g_0,...,g_{n-2}, g_{n-1}g_n) + (-1)^{n+1} f(g_0,...,g_{n-1}) \end{aligned}$$

Then $d^{n+1} \circ d^n = 0$, giving us a cochain complex. We can then have:

Definition 4.1.14: Cohomology Group – using Cocycles and Coboundaries

$$H^n(G,M) = \frac{\ker d^n}{\mathrm{im} d^{n-1}} = \frac{Z^n(G,M)}{B^n(G,M)}$$

We sum up many of the results that we have shown before using the new notation:

1. $C^0(G,M) = M$, and $H^0(G,M) \cong M^G$
2. $d^0 : C^0(G,M) \to C^1(G,M)$ is given by $d^0(a)(g) = ga - a$
3. $H^0(G,M) \cong \ker d^0 = M^G$
4. $\mathrm{im} d^0 = \{f : G \to M \mid \exists a \in M \text{ such that } f(g) = ga - a, \ \forall g \in G\}$ (principal crossed homomorphism)
5. $d^1 : C^1(G,M) \to C^2(G,M)$ given by $d^1(f)(g,h) = gf(h) - f(gh) + f(g)$
6. $\ker d^1 = \{f : G \to M \mid f(gh) = f(g) + gf(h)\}$ (crossed homomorphisms)
7. If $0 \to A \to B \to C \to 0$ is exact, we get an exact sequence of cochains for every n:

$$0 \to C^n(G,A) \to C^n(G,B) \to C^n(G,C) \to 0$$

We shall show that the collection of complexes given by the above is isomorphic to those given by the standard resolution

Proposition 4.1.15: Characterizing Hom of Resolution

Let M be a G-module. Then for every $n \geq 0$ there is an isomorphism:

$$\Phi^n : \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^{n+1}], M) \xrightarrow{\cong} C^n(G, M)$$

where the map $\varphi : \mathbb{Z}[G^{n+1}] \to M$ goes to $f : G^n \to M$ given by:

$$f(g_1, ..., g_n) = \varphi(1, g_1, g_1g_2, g_1g_2g_3, ..., g_1g_2g_3 \cdots g_n)$$

The isomorphism Φ^n commutes with boundary maps, that is

$$\Phi^{n+1} \circ d^*_{n+1} = d^n \circ \Phi^n$$

Hence the two cochain complexes agree: the one built from the standard resolution and the one built from cocycles and coboundaries are isomorphic, so the two descriptions of $H^n(G, M)$ coincide.

***Proof*:**

Step 1: Φ^n is a bijection. The group G acts diagonally on the basis G^{n+1} of $\mathbb{Z}[G^{n+1}]$, and this action is free, since $g \cdot (h_0, \ldots, h_n) = (h_0, \ldots, h_n)$ forces $gh_0 = h_0$ and hence $g = e$. Each orbit contains exactly one tuple whose first entry is e, namely $h_0^{-1} \cdot (h_0, \ldots, h_n)$. A $\mathbb{Z}[G]$-linear map out of $\mathbb{Z}[G^{n+1}]$ is therefore determined freely and without constraint by its values on the tuples $(e, k_1, \ldots, k_n)$, so restriction to those tuples is a bijection onto $\mathrm{Hom}_{\mathbf{Set}}(G^n, M) = C^n(G, M)$.

The substitution in the statement is the reparametrization $k_i = g_1g_2 \cdots g_i$ of that indexing. It is a bijection $G^n \to G^n$, with inverse $g_1 = k_1$ and $g_i = k_{i-1}^{-1}k_i$ for $i \geq 2$. Composing the two bijections gives Φ^n, and it is additive because both steps are, so Φ^n is an isomorphism of abelian groups.

Step 2: Compatibility with the differentials. Let $\varphi \in \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^{n+1}], M)$ and set $f = \Phi^n(\varphi)$. Write $h_i = g_1 \cdots g_i$, so that $f(g_1, \ldots, g_n) = \varphi(e, h_1, \ldots, h_n)$. Applying Φ^{n+1} to $d^*_{n+1}\varphi = \varphi \circ d_{n+1}$ and evaluating at $(g_0, \ldots, g_n)$ gives $\varphi\big(d_{n+1}(e, k_1, \ldots, k_{n+1})\big)$ where $k_i = g_0g_1 \cdots g_{i-1}$.

Expand d_{n+1} by definition 4.1.11. The term omitting position 0 is $(k_1, \ldots, k_{n+1})$; pulling out the common left factor $k_1 = g_0$ by equivariance of φ turns it into $g_0 \cdot \varphi(e, g_0^{-1}k_2, \ldots) = g_0 f(g_1, \ldots, g_n)$. The term omitting position j for $1 \leq j \leq n$ merges g_{j-1} and g_j, contributing $(-1)^j f(g_0, \ldots, g_{j-1}g_j, \ldots, g_n)$, and the term omitting the last position drops g_n, contributing $(-1)^{n+1} f(g_0, \ldots, g_{n-1})$. Summing,

$$\Phi^{n+1}(d^*_{n+1}\varphi)(g_0, \ldots, g_n) = g_0 f(g_1, \ldots, g_n) + \sum_{j=1}^{n} (-1)^j f(g_0, \ldots, g_{j-1}g_j, \ldots, g_n) + (-1)^{n+1} f(g_0, \ldots, g_{n-1})$$

which is precisely $d^n(f)(g_0, \ldots, g_n)$ as defined above. Hence $\Phi^{n+1} \circ d^*_{n+1} = d^n \circ \Phi^n$.

The maps $\Phi^\bullet$ are therefore isomorphisms commuting with the differentials, that is an isomorphism

of cochain complexes, completing the proof.

Due to the relevance of the standard representation, we have the following definition:

Definition 4.1.16: Induced and Coinduced G-module

Let G be a group and A an abelian group. Then:

$$\mathrm{CoInd}^G(A) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], A)$$

with the G-action defined as $(g\varphi)(z) = \varphi(zg)$ is called the *coinduced G-module* associated to A. Similarly

$$\mathrm{Ind}^G(A) = \mathbb{Z}[G] \otimes_{\mathbb{Z}} A$$

with G-action defined by $g(z \otimes a) = (gz) \otimes a$ is the is the induced G-module associated to A.

A result that is common in many Cohomologies theories is some version of the following:

Lemma 4.1.17: Induced and Coinduced Equivalence

Let G be a finite group and A an abelian group. Then

$$\mathrm{Ind}^G(A) \cong \mathrm{CoInd}^G(A)$$

As a consequence, if $B = \mathrm{Ind}^G(A)$ or $B = \mathrm{CoInd}^G(A)$, then

$$H_0(G, B) \cong H^0(G, B) \cong A$$

and they are both zero for $n \geq 1$.

***Proof*:**

Define the map: $\alpha : \mathrm{CoInd}^G(A) \to \mathrm{Ind}^G(A)$ given by:

$$\varphi \mapsto \sum_{g \in G} g^{-1} \otimes \varphi(g)$$

$$(g^{-1} \mapsto a) \leftarrow\!\shortmid\, g \otimes a$$

with $(g^{-1} \mapsto a)(h) = 0$ for $h \in G \setminus \{g^{-1}\}$. These are certainly inverses, and easily verifiable to be G-module homomorphisms, a technique also used in the proof of Maschke's Theorem (see [9, Maschke's Theorem]).

For the last part, write $B = \mathrm{CoInd}^G(A) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], A)$, which the first part identifies with $\mathrm{Ind}^G(A)$. Let $P_\bullet \to \mathbb{Z}$ be a resolution by free $\mathbb{Z}[G]$-modules; each P_i is then also free as an abelian group. The adjunction

$$\mathrm{Hom}_{\mathbb{Z}[G]}\big(M, \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], A)\big) \cong \mathrm{Hom}_{\mathbb{Z}}(M, A)$$

of [8, Hom-Tensor Adjunction], natural in M, turns the complex computing $H^\bullet(G, B)$ into $\mathrm{Hom}_{\mathbb{Z}}(P_\bullet, A)$. Its cohomology is $\mathrm{Ext}^\bullet_{\mathbb{Z}}(\mathbb{Z}, A)$, which is A in degree 0 and vanishes in positive degrees because $\mathbb{Z}$ is projective over $\mathbb{Z}$. Dually $P_\bullet \otimes_{\mathbb{Z}[G]} (\mathbb{Z}[G] \otimes_{\mathbb{Z}} A) \cong P_\bullet \otimes_{\mathbb{Z}} A$ computes $\mathrm{Tor}^{\mathbb{Z}}_\bullet(\mathbb{Z}, A)$, which is A in degree 0 and zero above. So $H^0(G, B) \cong H_0(G, B) \cong A$ and both vanish for $n \geq 1$, as we sought to show.

Proposition 4.1.18: Torsion of H^1

If G has order n and is finite, then every element of $H^1(G, M)$ has order dividing n.

***Proof*:**
By proposition 4.1.15, $H^1(G, M) = Z^1(G, M)/B^1(G, M)$. Let $\delta \in Z^1(G, M)$, so that $\delta(gh) = \delta(g) + g\delta(h)$ for all $g, h \in G$, and put $S = \sum_{h \in G} \delta(h)$. Fix g and sum the cocycle identity over $h \in G$. As $h \mapsto gh$ permutes G, the left side sums to S, so

$$S = n\,\delta(g) + gS$$

that is, $n\,\delta(g) = g(-S) - (-S)$ for every $g \in G$. So $n\delta$ is the coboundary f_{-S} of definition 4.1.10, and n kills the class of δ in $H^1(G, M)$. Hence the order of every class divides n, as claimed.

Proposition 4.1.19: M finitely generated, then H1 finite

Let M be a finitely generated G-module and G finite. Then $H^1(G, M)$ is finite.

***Proof*:**
Let $m_1, \ldots, m_r$ generate M as a G-module. Since G is finite, the finitely many elements gm_i generate M as an abelian group, so the group $C^1(G, M)$ of all functions $G \to M$, which is the product of $|G|$ copies of M, is a finitely generated abelian group, a quotient of some $\mathbb{Z}^k$. The preimage in $\mathbb{Z}^k$ of the subgroup $Z^1(G, M)$ is free of finite rank by [8, Submodule Of Free Module Over PID], as $\mathbb{Z}$ is a PID, so $Z^1(G, M)$ is finitely generated, and so is its quotient $H^1(G, M) = Z^1(G, M)/B^1(G, M)$ (proposition 4.1.15). If s elements generate $H^1(G, M)$, each has order dividing $|G|$ by proposition 4.1.18, so $H^1(G, M)$ has at most $|G|^s$ elements, completing the proof.

Finally, let us prove the foreshadowing in [8, Foreshadowing]. Let L/K be a cyclic extension so that $G = \mathrm{Gal}_K(L) \cong \mathbb{Z}/m\mathbb{Z}$. The G acts on $L^\times$ in the natural way, and hence we find it's group cohomology. First:

Lemma 4.1.20: Cocycles using Norm and Trace

Let L/K be a cyclic extension $G = \mathrm{Gal}_K(L) \cong \mathbb{Z}/m\mathbb{Z}$ with generator σ, and $Z^1(G, L^\times)$ the 1-cocycles. Let $U_{L/K} = \left\{ l \in L \;\middle|\; \mathrm{Nm}_{L/K}(l) = 1 \right\}$ (sometimes this is denoted L^1). Then as groups:

$$Z^1(G, L^\times) \cong U_{L/K}$$
$$c \mapsto c(\sigma)$$

Similarly:

$$Z^1(G, L) \cong T_{L/K}$$

where $T_{L/K}$ is the traceless elements[a]

[a] There is less common notation, sometimes it is denoted L^0

***Proof*:**
Let M be $L^\times$ or L, with its group law written additively as in definition 4.1.9, so that the cocycle condition reads $c(gh) = c(g) + g(c(h))$ and the neutral element 0_M is 1 when $M = L^\times$ and 0 when $M = L$. For $\beta \in M$ and $k \geq 0$ put

$$s_k(\beta) = \sum_{i=0}^{k-1} \sigma^i(\beta)$$

with $s_0(\beta) = 0_M$. For $M = L^\times$ and $k \geq 1$, $s_k(\beta)$ is the product $\beta\,\sigma(\beta)\cdots\sigma^{k-1}(\beta)$. Splitting the sum at $i = j$ gives, for all $j, k \geq 0$,

$$s_{j+k}(\beta) = s_j(\beta) + \sigma^j\big(s_k(\beta)\big)$$

The extension L/K is Galois with group $G = \{\sigma^i : 0 \leq i < m\}$, so the norm and the trace are the product and the sum of the conjugates $\sigma^i(\beta)$ [20, Chapter I, Proposition 2.6]: $s_m(\beta)$ is $\mathrm{Nm}_{L/K}(\beta)$ for $M = L^\times$ and $\mathrm{tr}_{L/K}(\beta)$ for $M = L$. As $\mathrm{Nm}_{L/K}(0) = 0$, the set $U_{L/K}$ lies in $L^\times$, so in both cases the claimed target is $\{\beta \in M : s_m(\beta) = 0_M\}$.

Step 1: $c \mapsto c(\sigma)$ is a homomorphism into the target. The group law of $Z^1(G, M)$ is pointwise, so the map is a homomorphism. Let $c \in Z^1(G, M)$ and $\beta = c(\sigma)$. The cocycle condition at $g = h = e$ gives $c(e) = c(e) + c(e)$, so $c(e) = 0_M$. At $g = \sigma$ and $h = \sigma^k$ it gives $c(\sigma^{k+1}) = \beta + \sigma(c(\sigma^k))$, so by induction on k, using the splitting formula with $j = 1$, $c(\sigma^k) = s_k(\beta)$ for every $k \geq 0$. For $k = m$ this reads $s_m(\beta) = c(\sigma^m) = c(e) = 0_M$, so β lies in the target.

Step 2: injectivity. By Step 1, $c(\sigma^k) = s_k(c(\sigma))$ for all k, so c is determined by $c(\sigma)$.

Step 3: surjectivity. Let $\beta \in M$ with $s_m(\beta) = 0_M$. The splitting formula with $k = m$ gives $s_{j+m}(\beta) = s_j(\beta) + \sigma^j(0_M) = s_j(\beta)$, so $s_k(\beta)$ depends only on the class of k modulo m, and $c(\sigma^k) = s_k(\beta)$ defines a function $c\colon G \to M$. For $g = \sigma^j$ and $h = \sigma^k$ with $j, k \geq 0$ the splitting formula reads $c(gh) = c(g) + g(c(h))$, so $c \in Z^1(G, M)$, and $c(\sigma) = s_1(\beta) = \beta$. Hence $c \mapsto c(\sigma)$ is an isomorphism from $Z^1(G, L^\times)$ onto $U_{L/K}$ and from $Z^1(G, L)$ onto $T_{L/K}$, as claimed.

Theorem 4.1.21: Hilbert's 90th Theorem: Cohomology Upgrade

Using the above lemma's notation:

$$H^1(G, L^\times) \cong \{1\}$$

***Proof*:**
By proposition 4.1.15, $H^1(G, L^\times) = Z^1(G, L^\times)/B^1(G, L^\times)$, so it suffices to show that every cocycle c is a coboundary. Written multiplicatively, the coboundaries of definition 4.1.10 are the functions $g \mapsto g(\gamma)/\gamma$ with $\gamma \in L^\times$. By lemma 4.1.20, $\beta = c(\sigma)$ satisfies $\mathrm{Nm}_{L/K}(\beta) = 1$, and a cocycle is determined by its value at σ, so it suffices to find $\gamma \in L^\times$ with $\beta = \sigma(\gamma)/\gamma$.

Put $s_0 = 1$ and $s_k = \beta\,\sigma(\beta)\cdots\sigma^{k-1}(\beta)$ for $k \geq 1$, so that $s_{k+1} = \beta\,\sigma(s_k)$ and, as in the proof of lemma 4.1.20, $s_m = \mathrm{Nm}_{L/K}(\beta) = 1$. The automorphisms $\sigma^0, \ldots, \sigma^{m-1}$ of L are distinct, hence linearly independent over L ([8, linear independence of Automorphisms (Dedekind's Lemma)]), and the coefficients $s_0, \ldots, s_{m-1}$ are nonzero, so some $\theta \in L$ gives

$$\alpha = \sum_{k=0}^{m-1} s_k\,\sigma^k(\theta) \neq 0$$

Applying σ and multiplying by β,

$$\beta\,\sigma(\alpha) = \sum_{k=0}^{m-1} \beta\,\sigma(s_k)\,\sigma^{k+1}(\theta) = \sum_{k=1}^{m} s_k\,\sigma^k(\theta) = \alpha - \theta + s_m\,\sigma^m(\theta) = \alpha$$

because $s_m = 1$ and $\sigma^m = \mathrm{id}$. So $\beta = \alpha/\sigma(\alpha) = \sigma(\gamma)/\gamma$ with $\gamma = \alpha^{-1}$, and c is the coboundary of γ, completing the proof.

Note this theorem generalizes to arbitrary finite Galois extensions; see [11]. This result should now make intuitive sense: the elements of K are *exactly* the $\mathrm{Gal}_K(L)$-invariant element. Note that if there were more complicated groups instead of $L^\times$, there can be more interesting cohomology groups, see [11] for more details.

Exercise 4.1.1

1. Let A, B be any G-modules. Show that $H^n(G, A \oplus B) \cong H^n(G, A) \oplus H^n(G, B)$
2. Show that $H^0(G, \mathrm{CoInd}^G(A)) \cong A$ and $H^n(G, \mathrm{CoInd}^G(A)) \cong 0$ for all $n \geq 1$ (hint: consider $\zeta : \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^n], \mathrm{CoInd}^G(A)) \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G^n], A)$ given by $\varphi \mapsto (z \mapsto \varphi(z)(1))$. What is the inverse of this map?)
3. Show that $H_0(G, \mathrm{Ind}^G(A)) \cong A$ and $H_n(G, \mathrm{Ind}^G(A)) \cong 0$ for all $n \geq 1$.

4.2 Homological Dimension

We now turn to measuring the "complexity" of modules and rings using the length of their resolutions. Before defining these measures rigorously, it is worth stepping back to 1890 to understand why mathematicians began counting the length of resolutions in the first place.

Historical Motivation: Invariant Theory

In the late 19th century, algebra was dominated by *Invariant Theory*. Imagine a group G acting on a polynomial ring $S = \mathbb{C}[x_1, \dots, x_n]$ (for example, by linear substitutions of variables). Mathematicians were looking for *invariant ring* S^G:

$$S^G = \{f \in S \big| g \cdot f = f \text{ for all } g \in G\}$$

The central question was if S^G finitely generated? Can we find a finite set of basic invariant polynomials $f_1, \dots, f_m$ such that every other invariant is just a polynomial combination of these?

Hilbert showed we can map a polynomial ring onto the invariants:

$$\varphi : \mathbb{C}[y_1, \dots, y_m] \twoheadrightarrow S^G \qquad y_i \mapsto f_i$$

and then the kernel $I = \ker(\varphi)$ represents the *relations* between the invariants. Hence, the study of invariants reduced to the study of the kernel. To find these invariants, 19th-century computationalists (like Paul Gordan) asked tried computing them and found the following problem:

1. **First Syzygies:** The ideal I describes the relations. By the Basis Theorem, I is finitely generated.
2. **Second Syzygies:** Are these relations independent? Or are there "relations between the relations"? If we have a relation $Ar_1 + Br_2 = 0$, that is a second syzygy.
3. **Higher Syzygies:** We can keep going. We can form a module of second syzygies and ask for its generators, then ask for the relations between *those* generators.

Mathematicians feared this chain of relations might continue forever, becoming infinitely complex. Hilbert's *Syzygy Theorem* proved that if you are working with a polynomial ring in n variables, this process *must stop*. Specifically, after at most n steps, the module of relations becomes free—there are no more "hidden" relations.

This was part of David Hilbert's seminal paper laying the foundation of commutative geometry. In particular, he proved three theorems: The *Basis Theorem*, the *Nullstellensatz*, and the *Syzygy Theorem*.

4.2.1 Etymology The word *Syzygy* comes from the Greek *syzygia*, meaning "yoked together". In astronomy, it refers to the alignment of three celestial bodies (like the Sun, Earth, and Moon). In algebra, it refers to a relation that "yokes" generators together to zero.

4.2.1 Projective and Global Dimension

We can now formalize Hilbert's discovery using the language of resolutions.

Definition 4.2.2: Projective Dimension

Let M be an R-module. The *projective dimension* of M, denoted $\mathrm{pd}_R(M)$, is the minimal length n of a projective resolution of M:

$$0 \to P_n \to \cdots \to P_1 \to P_0 \to M \to 0$$

If no finite resolution exists, we say $\mathrm{pd}_R(M) = \infty$.

Definition 4.2.3: Global Dimension

The *global dimension* of a ring R, denoted $\mathrm{gldim}(R)$, is the supremum of the projective dimensions of all R-modules:

$$\mathrm{gldim}(R) = \sup\{\mathrm{pd}_R(M) \big| M \in R\text{-}\mathbf{Mod}\}$$

Lemma 4.2.4: Detecting Projective Dimension with Ext

Let R be a ring, M an R-module and $m \geq 0$ an integer. The following are equivalent:

1. $\mathrm{pd}_R(M) \leq m$
2. $\mathrm{Ext}^i_R(M, N) = 0$ for all R-modules N and all $i > m$.
3. $\mathrm{Ext}^{m+1}_R(M, N) = 0$ for all R-modules N.

***Proof*:**

For a projective resolution $\cdots \to P_1 \to P_0 \to M \to 0$, theorem 2.5.8 computes $\mathrm{Ext}^i_R(M, N)$ as $H^i(\mathrm{Hom}_R(P_\bullet, N))$, where the cochain complex $\mathrm{Hom}_R(P_\bullet, N)$ has $\mathrm{Hom}_R(P_i, N)$ in degree i.

Step 1: (1) implies (2), and (2) implies (3). If $\mathrm{pd}_R(M) \leq m$, then definition 4.2.2 gives a projective resolution of M with $P_i = 0$ for every $i > m$. For $i > m$ the group $\mathrm{Hom}_R(P_i, N)$ is zero, and so is its cohomology $\mathrm{Ext}^i_R(M, N)$. Condition (3) is the case $i = m + 1$ of (2).

Step 2: Dimension shifting. Fix a projective resolution

$$\cdots \to P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} M \to 0$$

of M, which exists by example 2.2.2(4). Put $K_j = \mathrm{im}(d_j)$ for $j \geq 0$. Then $K_0 = M$, and $K_j = \ker(d_{j-1})$ for $j \geq 1$ by exactness. Let $\pi \colon P_m \to K_m$ be the map d_m with codomain K_m. The complex

$$\cdots \to P_{m+2} \xrightarrow{d_{m+2}} P_{m+1} \xrightarrow{d_{m+1}} P_m \xrightarrow{\pi} K_m \to 0$$

is exact: π is onto, $\ker(\pi) = \ker(d_m) = \mathrm{im}(d_{m+1})$, and exactness at the remaining terms is that of the resolution of M. So $Q_j = P_{m+j}$ is a projective resolution of K_m. For $j \geq 1$ the

cohomology of $\mathrm{Hom}_R(Q_\bullet, N)$ in degree j is computed from

$$\mathrm{Hom}_R(P_{m+j-1}, N) \xrightarrow{d^*_{m+j}} \mathrm{Hom}_R(P_{m+j}, N) \xrightarrow{d^*_{m+j+1}} \mathrm{Hom}_R(P_{m+j+1}, N)$$

and the same two maps compute the cohomology of $\mathrm{Hom}_R(P_\bullet, N)$ in degree $m+j$. By theorem 2.5.8, applied to both resolutions,

$$\mathrm{Ext}^j_R(K_m, N) \cong \mathrm{Ext}^{m+j}_R(M, N) \qquad (j \geq 1)$$

Step 3: A module K with $\mathrm{Ext}^1_R(K, N) = 0$ for every R-module N is projective. By example 2.2.2(4), K has a resolution

$$\cdots \to F_2 \xrightarrow{e_2} F_1 \xrightarrow{e_1} F_0 \xrightarrow{e_0} K \to 0$$

by free modules, and free modules are projective ([8, Free Modules Are Projective]). Let $L = \ker(e_0)$, with inclusion $\iota\colon L \to F_0$, and let $q\colon F_1 \to L$ be the map e_1 with codomain $\mathrm{im}(e_1) = L$, so that $e_1 = \iota \circ q$ and q is onto. Since $\iota \circ q \circ e_2 = e_1 \circ e_2 = 0$ and ι is injective, $q \circ e_2 = 0$, so q is a cocycle of degree 1 in $\mathrm{Hom}_R(F_\bullet, L)$. The cohomology of that complex in degree 1 is $\mathrm{Ext}^1_R(K, L) = 0$ (theorem 2.5.8), so q is a coboundary: $q = g \circ e_1$ for some $g\colon F_0 \to L$. Then $(g \circ \iota) \circ q = g \circ e_1 = q$, and as q is onto, $g \circ \iota = \mathrm{id}_L$.

Every $p \in F_0$ equals $\iota(g(p)) + \big(p - \iota(g(p))\big)$, and the second summand lies in $\ker(g)$. Also $\iota(L) \cap \ker(g) = 0$, because $g \circ \iota = \mathrm{id}_L$. So $F_0 = \iota(L) \oplus \ker(g)$. The restriction of e_0 to $\ker(g)$ is injective, since $\ker(e_0) \cap \ker(g) = \iota(L) \cap \ker(g) = 0$, and it is onto K, since e_0 is onto and vanishes on $\iota(L)$. Hence $K \cong \ker(g)$ is isomorphic to a direct summand of the free module F_0, and K is projective by [8, Equivalencies for Projective Modules].

Step 4: (3) implies (1). Take the resolution and the modules K_j of Step 2. By (3) and Step 2, $\mathrm{Ext}^1_R(K_m, N) \cong \mathrm{Ext}^{m+1}_R(M, N) = 0$ for every R-module N, so K_m is projective by Step 3. If $m = 0$, then $M = K_0$ is projective, and $0 \to M \xrightarrow{\mathrm{id}} M \to 0$ is a projective resolution of length 0. If $m \geq 1$, then $K_m = \ker(d_{m-1})$, and

$$0 \to K_m \to P_{m-1} \to \cdots \to P_1 \to P_0 \to M \to 0$$

is exact, a projective resolution of length m. In both cases $\mathrm{pd}_R(M) \leq m$ by definition 4.2.2, completing the proof.

Theorem 4.2.5: Global Dimension of Polynomial Rings

Let R be a nonzero commutative ring. Then

$$\mathrm{gldim}(R[x]) = \mathrm{gldim}(R) + 1$$

***Proof*:**

Step 1: The characteristic exact sequence. Let M be any $R[x]$-module. Restricting scalars makes M an R-module, and $R[x] \otimes_R M$ is an $R[x]$-module.

Define a sequence of $R[x]$-modules:

$$0 \to R[x] \otimes_R M \xrightarrow{\alpha} R[x] \otimes_R M \xrightarrow{\beta} M \to 0$$

where the maps are defined by:

- $\alpha(f(x) \otimes m) = xf(x) \otimes m - f(x) \otimes xm$
- $\beta(f(x) \otimes m) = f(x)m$

Both formulas are R-bilinear in $(f(x), m)$, because x acts R-linearly on M, so they define maps on $R[x] \otimes_R M$, and both maps are $R[x]$-linear.

- *Surjectivity of β:* For any $m \in M$, $\beta(1 \otimes m) = m$, so β is surjective.
- *Exactness at the middle:* The elements $1 \otimes m$ generate $R[x] \otimes_R M$ as an $R[x]$-module and $\beta \circ \alpha$ is $R[x]$-linear, so the computation $\beta(\alpha(1 \otimes m)) = \beta(x \otimes m - 1 \otimes xm) = xm - xm = 0$ gives $\mathrm{im}(\alpha) \subseteq \ker(\beta)$. Conversely, any element in $R[x] \otimes_R M$ can be written as $\sum_{i=0}^{d} x^i \otimes m_i$. If it lies in $\ker(\beta)$, then $\sum_{i=0}^{d} x^i m_i = 0$. We can rewrite our original element as:

  $$\sum_{i=0}^{d} x^i \otimes m_i = \sum_{i=0}^{d} (x^i \otimes m_i - 1 \otimes x^i m_i)$$

  Since $x^i \otimes m - 1 \otimes x^i m = \alpha\left(\sum_{j=0}^{i-1} x^{i-1-j} \otimes x^j m\right)$ for every $m \in M$, the element is in the image of α.
- *Injectivity of α:* Suppose $\sum_{i=0}^{d} x^i \otimes m_i \in \ker(\alpha)$. Then:

  $$\sum_{i=0}^{d} x^{i+1} \otimes m_i - \sum_{i=0}^{d} x^i \otimes xm_i = 0$$

  Because $R[x]$ is free over R, we can compare the coefficients of $x^k \otimes -$ degree by degree. For degree 0, we have $-1 \otimes xm_0 = 0 \implies xm_0 = 0$. For degree $k > 0$, we have $m_{k-1} - xm_k = 0 \implies m_{k-1} = xm_k$. For degree $d+1$, we have $m_d = 0$. Working backward from $m_d = 0$, we get $m_{d-1} = xm_d = 0$, and by induction, $m_i = 0$ for all i. Hence, α is injective.

Step 2: Extending scalars along $R \to R[x]$. Let M be an R-module and $P_\bullet \to M$ a projective resolution over R. Because $R[x]$ is free over R, it is flat ([8, Projective Modules Are Flat]), so tensoring the resolution with $R[x]$ preserves exactness. Each $R[x] \otimes_R P_i$ is a projective $R[x]$-module by [8, Equivalencies for Projective Modules]: if $P_i \oplus Q \cong R^{(I)}$ for an R-module Q and a set I, then $(R[x] \otimes_R P_i) \oplus (R[x] \otimes_R Q) \cong R[x]^{(I)}$, because $R[x] \otimes_R -$ commutes with direct sums ([8, Arbitrary Direct Sums Commute with Tensor Products]) and carries $R^{(I)}$ to $R[x]^{(I)}$ ([8, Extension of Scalars of a Free Module]). Therefore $R[x] \otimes_R P_\bullet$ is a projective resolution of $R[x] \otimes_R M$ over $R[x]$. Taking $P_\bullet$ of length $\mathrm{pd}_R(M)$ when that is finite gives

$$\mathrm{pd}_{R[x]}(R[x] \otimes_R M) \leq \mathrm{pd}_R(M)$$

Let N be an $R[x]$-module. Regard $\mathrm{Hom}_{R[x]}(-,N)$ as a functor on the opposite category of $R[x]$-modules. Its right derived functors, computed from projective resolutions as in example 2.3.6, are the functors $\mathrm{Ext}^i_{R[x]}(-,N)$ by theorem 2.5.8, and the opposite category has enough injectives, because $R[x]$-modules have projective resolutions (example 2.2.2(4)). For an $R[x]$-module M, corollary 2.3.12 therefore turns the sequence of Step 1, regarded as a short exact sequence in the opposite category, into the long exact sequence

$$\cdots \to \mathrm{Ext}^i_{R[x]}(R[x]\otimes_R M,N) \xrightarrow{\alpha^*} \mathrm{Ext}^i_{R[x]}(R[x]\otimes_R M,N) \to \mathrm{Ext}^{i+1}_{R[x]}(M,N)$$
$$\to \mathrm{Ext}^{i+1}_{R[x]}(R[x]\otimes_R M,N) \xrightarrow{\alpha^*} \mathrm{Ext}^{i+1}_{R[x]}(R[x]\otimes_R M,N) \to \cdots$$

in which α^* is the map induced by α.

Step 3: The upper bound $gldim(R[x]) \le gldim(R)+1$. If $\mathrm{gldim}(R)=\infty$, the right-hand side is infinite. Otherwise put $n=\mathrm{gldim}(R)$, and let M be an $R[x]$-module. As an R-module, M has $\mathrm{pd}_R(M)\le n$ by definition 4.2.3, so $\mathrm{pd}_{R[x]}(R[x]\otimes_R M)\le n$ by Step 2, and by lemma 4.2.4 the groups $\mathrm{Ext}^i_{R[x]}(R[x]\otimes_R M,N)$ vanish for every $R[x]$-module N and every $i\ge n+1$. In the long exact sequence of Step 2 with $i=n+1$, the terms on either side of $\mathrm{Ext}^{n+2}_{R[x]}(M,N)$ are therefore zero, and exactness reads

$$0\to \mathrm{Ext}^{n+2}_{R[x]}(M,N)\to 0$$

So $\mathrm{Ext}^{n+2}_{R[x]}(M,N)=0$ for every $R[x]$-module N, and lemma 4.2.4 gives $\mathrm{pd}_{R[x]}(M)\le n+1$. Because this holds for all $R[x]$-modules, $\mathrm{gldim}(R[x])\le n+1$.

Step 4: Let M and K be R-modules, regarded as $R[x]$-modules on which x acts as 0. If $\mathrm{Ext}^i_R(M,K)\ne 0$ for some $i\ge 0$, then $\mathrm{Ext}^{i+1}_{R[x]}(M,K)\ne 0$. Choose a projective resolution $P_\bullet\to M$ over R. By Step 2, $R[x]\otimes_R P_\bullet$ is a projective resolution of $R[x]\otimes_R M$ over $R[x]$. Restriction along $p\mapsto 1\otimes p$ gives isomorphisms

$$\mathrm{Hom}_{R[x]}(R[x]\otimes_R P_j,K)\cong \mathrm{Hom}_R(P_j,K)$$

by [8, Extension of Scalars Is Left Adjoint to Restriction], and they commute with the differentials, since $(1\otimes d)(1\otimes p)=1\otimes d(p)$. Taking cohomology, theorem 2.5.8 on both sides gives

$$\mathrm{Ext}^i_{R[x]}(R[x]\otimes_R M,K)\cong \mathrm{Ext}^i_R(M,K)\quad \text{for all } i\ge 0$$

Since $xM=0$, the map α of Step 1 is $\alpha(f(x)\otimes m)=xf(x)\otimes m$, which is multiplication by x on $R[x]\otimes_R M$. Write μ for multiplication by x on each $R[x]\otimes_R P_j$. It commutes with the $R[x]$-linear differentials and augmentation, so it is a morphism of resolutions lifting α. By proposition 2.2.15 every lift of α is homotopic to μ, and homotopic morphisms induce the same map after applying $\mathrm{Hom}_{R[x]}(-,K)$ and taking cohomology (lemma 2.1.34, proposition 2.1.31). So α^* on $\mathrm{Ext}^i_{R[x]}(R[x]\otimes_R M,K)$ is induced by the cochain map $\psi\mapsto\psi\circ\mu$ on $\mathrm{Hom}_{R[x]}(R[x]\otimes_R P_\bullet,K)$. That cochain map is zero: $(\psi\circ\mu)(z)=\psi(xz)=x\psi(z)=0$ for every z, because ψ is $R[x]$-linear and $xK=0$. Hence $\alpha^*=0$ in every degree.

The long exact sequence of Step 2 with $N=K$ therefore breaks into short exact sequences, one for each $i\ge 0$, which read

$$0\to \mathrm{Ext}^i_R(M,K)\to \mathrm{Ext}^{i+1}_{R[x]}(M,K)\to \mathrm{Ext}^{i+1}_R(M,K)\to 0$$

after the isomorphism above. The injection on the left gives the claim.

Step 5: The lower bound gldim$(R[x]) \geq$ gldim$(R)+1$. Let $m \geq 0$ be an integer with $m \leq \text{gldim}(R)$. Some R-modules M and K satisfy $\text{Ext}^m_R(M,K) \neq 0$. For $m = 0$ take $M = K = R$: computing with the projective resolution $0 \to R \xrightarrow{\text{id}} R \to 0$, theorem 2.5.8 gives $\text{Ext}^0_R(R,R) \cong \text{Hom}_R(R,R)$, which contains $\text{id}_R \neq 0$ because $R \neq 0$. For $m \geq 1$, $\text{gldim}(R) > m-1$, so by definition 4.2.3 some R-module M has $\text{pd}_R(M) > m-1$, and lemma 4.2.4, applied with $m-1$ in place of m, gives an R-module K with $\text{Ext}^m_R(M,K) \neq 0$.

Regard M and K as $R[x]$-modules on which x acts as 0. By Step 4, $\text{Ext}^{m+1}_{R[x]}(M,K) \neq 0$, so $\text{pd}_{R[x]}(M) > m$ by lemma 4.2.4, and $\text{gldim}(R[x]) \geq m+1$. If $\text{gldim}(R) = n$ is finite, the case $m = n$ gives $\text{gldim}(R[x]) \geq n+1$. If $\text{gldim}(R) = \infty$, every $m \geq 0$ is allowed, and $\text{gldim}(R[x]) = \infty$.

Combining the upper and lower bounds gives $\text{gldim}(R[x]) = \text{gldim}(R) + 1$, completing the proof.

Hilbert's theorem is effectively the calculation of the global dimension of polynomial rings.

Theorem 4.2.6: Hilbert's Syzygy Theorem

Let k be a field and $R = k[x_1, \ldots, x_n]$ be the polynomial ring in n variables. Then every finitely generated R-module has finite projective dimension, and:

$$\text{gldim}(R) = n$$

***Proof*:**
For $n = 0$, $R = k$ is a field, over which every module is free, so $\text{gldim}(k) = 0$. For $n \geq 1$, write $R = S[x_n]$ with $S = k[x_1, \ldots, x_{n-1}]$. By induction on n, $\text{gldim}(S) = n-1$, and theorem 4.2.5, applied to the nonzero ring S, gives $\text{gldim}(R) = \text{gldim}(S) + 1 = n$. By definition 4.2.3, every R-module, finitely generated or not, therefore has projective dimension at most n.

The residue field $k \cong R/(x_1, \ldots, x_n)$ attains this bound. Let V be a k-vector space of dimension n with basis $e_1, \ldots, e_n$. The *Koszul complex* of $x_1, \ldots, x_n$ is $\bigwedge^\bullet V \otimes_k R$, augmented to k:

$$0 \to \bigwedge^n V \otimes R \xrightarrow{d_n} \cdots \xrightarrow{d_2} \bigwedge^1 V \otimes R \xrightarrow{d_1} R \xrightarrow{\epsilon} k \to 0$$

where d_i sends $e_{j_1} \wedge \cdots \wedge e_{j_i} \otimes 1$ to $\sum_{s=1}^{i} (-1)^{s+1} x_{j_s}\, e_{j_1} \wedge \cdots \wedge \widehat{e_{j_s}} \wedge \cdots \wedge e_{j_i} \otimes 1$, so that $d_1(e_i \otimes 1) = x_i$. Because the variables $x_1, \ldots, x_n$ form a regular sequence, this complex is exact ([8, The Koszul Complex of the Variables]), so it is a free resolution of k of length n and $\text{pd}_R(k) \leq n$. For the reverse inequality, compute $\text{Tor}^R_n(k,k)$ with this resolution. Every d_i has its matrix entries in $(x_1, \ldots, x_n)$, so after applying $- \otimes_R k$ every differential vanishes, and $\text{Tor}^R_n(k,k) \cong \bigwedge^n V \cong k \neq 0$. A projective resolution of k of length at most $n-1$ would compute $\text{Tor}^R_n(k,k) = 0$, so $\text{pd}_R(k) = n$, completing the proof.

This theorem has implications beyond just counting relations:

1. *Degree of the Hilbert Polynomial:* Hilbert originally used this theorem to prove that the Hilbert Function $H_M(t) = \dim_k(M_t)$ of a graded module eventually becomes a polynomial. The finite resolution allows us to calculate $H_M(t)$ as an alternating sum of the Hilbert functions of free modules, which are essentially binomial coefficients $\binom{t+k}{k}$.

2. *Structure of Projective Modules:* A celebrated problem posed by Serre (related to this theorem) asked if every projective module over $k[x_1, \dots, x_n]$ is free. This is the algebraic analogue of "vector bundles over affine space are trivial". This was solved affirmatively in 1976 (the *Quillen-Suslin Theorem*). This implies that over polynomial rings, "projective resolution" and "free resolution" are synonymous.

4.2.2 Dimension of Local Rings

Computing the global dimension for general rings seems daunting, as one must check *every* module. However, we can simplify this task using derived functors.

Proposition 4.2.7: Detecting Projective Dimension with Tor

Let R be a ring, M a right R-module and $n \geq 0$ an integer, and consider the conditions

1. $\mathrm{pd}_R(M) \leq n$
2. $\mathrm{Tor}_i^R(M, N) = 0$ for all left R-modules N and all $i > n$.
3. $\mathrm{Tor}_{n+1}^R(M, N) = 0$ for all left R-modules N.

Then (1) implies (2), and (2) implies (3). If R is commutative and Noetherian and M is finitely generated, then (3) implies (1), so the three conditions are equivalent.

***Proof*:**

For a projective resolution $\cdots \to P_1 \to P_0 \to M \to 0$, theorem 2.5.8 computes $\mathrm{Tor}_i^R(M, N)$ as $H_i(P_\bullet \otimes_R N)$, the homology at the term $P_i \otimes_R N$.

Step 1: (1) implies (2), and (2) implies (3). If $\mathrm{pd}_R(M) \leq n$, then definition 4.2.2 gives a projective resolution of M with $P_i = 0$ for every $i > n$. For $i > n$ the module $P_i \otimes_R N$ is zero, and so is $\mathrm{Tor}_i^R(M, N)$. Condition (3) is the case $i = n + 1$ of (2).

Step 2: A resolution by finitely generated free modules. From here on R is commutative and Noetherian, M is finitely generated, and (3) holds. Then M has a projective resolution

$$\cdots \to P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} M \to 0$$

in which every P_j is a finitely generated free module. To build it, write M as a quotient of a finitely generated free module P_0. Every submodule of a finitely generated module over the Noetherian ring R is finitely generated [2, Props. 6.2 and 6.5], so $\ker(d_0)$ is a quotient of a finitely generated free module P_1. The same argument applied to $\ker(d_j)$ produces P_{j+1} and d_{j+1} for every $j \geq 1$.

Step 3: Dimension shifting. Put $K_n = \mathrm{im}(d_n)$, and let $\pi \colon P_n \to K_n$ be the map d_n with codomain K_n. The complex

$$\cdots \to P_{n+2} \xrightarrow{d_{n+2}} P_{n+1} \xrightarrow{d_{n+1}} P_n \xrightarrow{\pi} K_n \to 0$$

is exact: π is onto, $\ker(\pi) = \ker(d_n) = \mathrm{im}(d_{n+1})$, and exactness at the remaining terms is that of the resolution of M. So $Q_j = P_{n+j}$ is a projective resolution of K_n. For $j \geq 1$ the homology

of $Q_\bullet \otimes_R N$ at $Q_j \otimes_R N$ is computed from

$$P_{n+j+1} \otimes_R N \xrightarrow{d_{n+j+1}\otimes 1} P_{n+j} \otimes_R N \xrightarrow{d_{n+j}\otimes 1} P_{n+j-1} \otimes_R N$$

and the same two maps compute the homology of $P_\bullet \otimes_R N$ at $P_{n+j} \otimes_R N$. By theorem 2.5.8, applied to both resolutions, and by (3),

$$\operatorname{Tor}_1^R(K_n, N) \cong \operatorname{Tor}_{n+1}^R(M, N) = 0$$

for every R-module N.

Step 4: K_n is flat. Let $\psi\colon N' \to N$ be an injective homomorphism of R-modules with cokernel N'', and put $T = K_n \otimes_R -$. Since R-modules have projective resolutions (example 2.2.2(4)), theorem 2.3.11 applies to T and to the short exact sequence $0 \to N' \to N \to N'' \to 0$, and gives an exact sequence

$$L_1T(N'') \to L_0T(N') \xrightarrow{L_0T(\psi)} L_0T(N)$$

By definition 2.3.5, $L_1T(N'')$ is $H_1(K_n \otimes_R P''_\bullet)$ for a projective resolution $P''_\bullet$ of N'', which is $\operatorname{Tor}_1^R(K_n, N'') = 0$ by theorem 2.5.8 and Step 3. So $L_0T(\psi)$ is injective. The functor T is right exact ([8, $D \otimes_R -$ Is Right Exact]), so L_0T is naturally isomorphic to T by proposition 2.3.13, and $T(\psi) = 1 \otimes \psi$ is injective as well. As ψ was an arbitrary injection, K_n is flat ([8, Conditions for $A \otimes_R -$ to Be Exact]).

Step 5: (3) implies (1). The exact sequence $P_{n+1} \to P_n \to K_n \to 0$ of Step 3, with P_{n+1} and P_n finitely generated free, shows that K_n is finitely presented. A finitely presented flat module is projective [23, Thm. 3.2.7], so K_n is projective by Step 4. If $n = 0$, then $M = K_0$ is projective and $\operatorname{pd}_R(M) = 0$. If $n \geq 1$, then $K_n = \ker(d_{n-1})$, and

$$0 \to K_n \to P_{n-1} \to \cdots \to P_1 \to P_0 \to M \to 0$$

is a projective resolution of length n. In both cases $\operatorname{pd}_R(M) \leq n$ by definition 4.2.2, completing the proof.

Without the hypotheses on R and M, (3) does not imply (1). The $\mathbb{Z}$-module $\mathbb{Q}$ is torsion-free, hence flat ([8, Flat Equals Torsion-Free Over a PID]), so the exact functor $\mathbb{Q} \otimes_{\mathbb{Z}} -$ keeps a projective resolution $P_\bullet \to N$ of any abelian group N exact, and $\operatorname{Tor}_1^{\mathbb{Z}}(\mathbb{Q}, N) \cong H_1(\mathbb{Q} \otimes_{\mathbb{Z}} P_\bullet) = 0$ by theorem 2.5.8. This is (3) for $n = 0$. Yet $\mathbb{Q}$ is not projective: over the principal ideal domain $\mathbb{Z}$ projective modules are free ([8, Projective Modules Over a PID]), every element of $\mathbb{Q}$ is twice an element of $\mathbb{Q}$, and a basis element e of a nonzero free abelian group is not twice any element, since the coefficient of e in $2v$ is even. The ring $\mathbb{Z}$ is Noetherian, so finite generation of M is the hypothesis that fails.

It is more common that we consider rings that already have certain properties and verify how these conditions affect the global dimension. For example, given a local ring, the work is significantly reduced. In the local case, we can verify condition (3) by checking a *single* module: the residue field.

Lemma 4.2.8: Projective Dimension over Local Rings

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring and M a finitely generated R-module. Then:

$$\mathrm{pd}_R(M) \leq n \iff \mathrm{Tor}_{n+1}^R(M, k) = 0$$

Consequently, $\mathrm{pd}_R(M)$ is the smallest n such that $\mathrm{Tor}_i^R(M, k) = 0$ for all $i > n$.

***Proof*:**
The forward direction is immediate from proposition 4.2.7. For the converse, let $\cdots \to P_1 \to P_0 \to M \to 0$ be a *minimal* free resolution, meaning that every P_i is free of finite rank β_i and the matrices of the maps $P_{i+1} \to P_i$ have entries in $\mathfrak{m}$.

Such a resolution exists. Choose $m_1, \ldots, m_\beta \in M$ whose images form a k-basis of $M/\mathfrak{m}M$. They generate M by [7, Nakayama's Lemma III], so $e_i \mapsto m_i$ defines a surjection $R^\beta \to M$. Its kernel lies in $\mathfrak{m}R^\beta$, since a relation $\sum_i a_i m_i = 0$ reduces modulo $\mathfrak{m}$ to a relation among the basis vectors of $M/\mathfrak{m}M$, which forces every a_i into $\mathfrak{m}$. As a submodule of a finitely generated module over the Noetherian ring R, the kernel is finitely generated [2, Props. 6.2 and 6.5], so the same construction applies to it in place of M. Continuing in this way produces a free resolution in which the image of each map $P_{i+1} \to P_i$ is the kernel of $P_i \to P_{i-1}$ (of $P_0 \to M$ when $i = 0$), which lies in $\mathfrak{m}P_i$. So every matrix entry lies in $\mathfrak{m}$.

The differentials in the complex $P_\bullet \otimes_R k$ are therefore all zero maps, and by theorem 2.5.8

$$\mathrm{Tor}_i^R(M, k) \cong H_i(P_\bullet \otimes k) \cong P_i \otimes k \cong k^{\beta_i}$$

If $\mathrm{Tor}_{n+1}^R(M, k) = 0$, then $k^{\beta_{n+1}} = 0$, so $\beta_{n+1} = 0$ and $P_{n+1} = 0$. The map $P_n \to P_{n-1}$ (the map $P_0 \to M$ when $n = 0$) then has kernel equal to the image of $P_{n+1} = 0$, so $0 \to P_n \to \cdots \to P_0 \to M \to 0$ is a projective resolution of length n, and $\mathrm{pd}_R(M) \leq n$ by definition 4.2.2. For the consequence, the forward direction applied with $i - 1$ in place of n gives $\mathrm{Tor}_i^R(M, k) = 0$ for every $i > \mathrm{pd}_R(M)$, and if $\mathrm{Tor}_i^R(M, k) = 0$ for all $i > n$, then the converse, applied to $\mathrm{Tor}_{n+1}^R(M, k) = 0$, gives $\mathrm{pd}_R(M) \leq n$.

A regular local ring of dimension m has its maximal ideal generated by m elements. These elements cut the ring down one dimension at a time without meeting a zero-divisor, and that is what makes the Koszul complex on them a resolution of the residue field.

Definition 4.2.9: Regular Sequence

Let R be a ring and M an R-module. A sequence of elements $x_1, \ldots, x_n \in R$ is called an *M-regular sequence* if:

1. $(x_1, \ldots, x_n)M \neq M$
2. For $i = 1, \ldots, n$, the element x_i is not a zero-divisor on the quotient module $M/(x_1, \ldots, x_{i-1})M$.

Lemma 4.2.10: The Maximal Ideal of a Regular Local Ring Is Generated by a Regular Sequence

Let $(R, \mathfrak{m})$ be a Noetherian regular local ring of dimension m, and let $x_1, \dots, x_m$ generate $\mathfrak{m}$. Then $x_1, \dots, x_m$ is an R-regular sequence.

***Proof*:**
Induct on m. For $m = 0$ the sequence is empty. For $m > 0$, a regular local ring is a domain ([7, A Regular Local Ring Is a Domain]), and $x_1 \neq 0$: regularity makes $\dim_k \mathfrak{m}/\mathfrak{m}^2 = m$, where $k = R/\mathfrak{m}$, so an m-element generating set of $\mathfrak{m}$ is minimal and none of its members can vanish. Hence x_1 is a nonzerodivisor, and $\dim R/(x_1) = m - 1$ by [7, One Equation Cuts Dimension by One], applicable because R is local. The maximal ideal of $R/(x_1)$ is generated by the $m - 1$ images of $x_2, \dots, x_m$, so $R/(x_1)$ is regular local of dimension $m - 1$, and by induction those images form an $R/(x_1)$-regular sequence. Since $(x_1, \dots, x_m) = \mathfrak{m} \neq R$, prefixing x_1 gives an R-regular sequence, completing the proof.

This leads to the fundamental equality attributed to Auslander, Buchsbaum, and Serre, which connects the homological algebra back to the geometry of the ring.

Theorem 4.2.11: Auslander-Buchsbaum-Serre Theorem

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring. Then the global dimension is determined by the residue field:

$$\operatorname{gldim}(R) = \operatorname{pd}_R(k)$$

Furthermore, $\operatorname{gldim}(R)$ is finite if and only if R is a *regular local ring*, that is, if and only if $\mathfrak{m}$ can be generated by $\dim R$ elements ([7, Regular Local Ring]).

This theorem is the bridge between algebra and geometry. Regularity is a condition on the number of generators of $\mathfrak{m}$, equivalently on the associated graded ring ([7, Characterizations of a Regular Local Ring]), while global dimension is a homological condition on resolutions. Geometrically the regular local rings are the local rings at the nonsingular points of a variety, which is the reading [14] develops.

***Proof*:**
By definition 4.2.3, $\operatorname{gldim}(R) \geq \operatorname{pd}_R(k)$, which settles the case $\operatorname{pd}_R(k) = \infty$. Otherwise let $n = \operatorname{pd}_R(k)$, and let M be any finitely generated R-module. By lemma 4.2.8, $\operatorname{pd}_R(M) \leq n$ holds once $\operatorname{Tor}^R_{n+1}(M, k) = 0$. Since $\operatorname{pd}_R(k) = n$, definition 4.2.2 gives a projective resolution $Q_\bullet$ of k with $Q_i = 0$ for every $i > n$. Computing with this resolution of the second variable, theorem 2.5.8 gives

$$\operatorname{Tor}^R_{n+1}(M, k) \cong H_{n+1}(M \otimes_R Q_\bullet) = 0$$

because the term $M \otimes_R Q_{n+1}$ is zero. Hence $\operatorname{pd}_R(M) \leq n$. This holds for every finitely generated M, in particular for every cyclic module R/I. By the Global Dimension Theorem [23, Thm. 4.1.2], $\operatorname{gldim}(R)$ is the supremum of $\operatorname{pd}_R(R/I)$ over the ideals I of R, so $\operatorname{gldim}(R) \leq n$, and hence $\operatorname{gldim}(R) = n$.

For the second statement, let R be regular of dimension m and let $x_1, \dots, x_m$ generate $\mathfrak{m}$. By lemma 4.2.10 they form a regular sequence, so by [8, Exactness of the Koszul Complex]

the Koszul complex on $x_1, \dots, x_m$ is a finite free resolution of $R/\mathfrak{m} = k$ of length m. Hence $\mathrm{pd}_R(k) \le m$, and $\mathrm{gldim}(R) = \mathrm{pd}_R(k) \le m$ is finite. The converse, that finite global dimension forces R to be regular, is Serre's theorem [23, Main Thm. 4.4.16].

Definition 4.2.12: Regular Ring

A Noetherian ring A is *regular* if $A_\mathfrak{p}$ is a regular local ring for every prime ideal $\mathfrak{p}$ of A.

Lemma 4.2.13: Localization Does Not Raise Global Dimension

Let A be a commutative ring and $S \subseteq A$ a multiplicative set. Every $S^{-1}A$-module M, regarded as an A-module through $A \to S^{-1}A$, satisfies $\mathrm{pd}_{S^{-1}A}(M) \le \mathrm{pd}_A(M)$. Consequently

$$\mathrm{gldim}(S^{-1}A) \le \mathrm{gldim}(A)$$

***Proof*:**
Let M be an $S^{-1}A$-module. If $\mathrm{pd}_A(M) = \infty$ there is nothing to prove. Otherwise put $n = \mathrm{pd}_A(M)$ and choose, by definition 4.2.2, a projective resolution

$$0 \to P_n \to \cdots \to P_1 \to P_0 \to M \to 0$$

over A. Localization at S preserves short exact sequences ([7, Localization Is Exact]), hence every exact sequence, since an exact sequence is spliced from short exact sequences through its images. So

$$0 \to S^{-1}P_n \to \cdots \to S^{-1}P_1 \to S^{-1}P_0 \to S^{-1}M \to 0$$

is an exact sequence of $S^{-1}A$-modules. Each $S^{-1}P_i$ is projective over $S^{-1}A$: if $P_i \oplus Q \cong A^{(I)}$ for an A-module Q and a set I, then $S^{-1}P_i \oplus S^{-1}Q \cong (S^{-1}A)^{(I)}$, because localization commutes with direct sums.

The localized sequence ends in $S^{-1}M$, and the map $\varphi\colon M \to S^{-1}M$, $\varphi(m) = m/1$, is an isomorphism of $S^{-1}A$-modules. An element $s \in S$ acts on M as the unit $s/1$ of $S^{-1}A$. For $a \in A$ and $s \in S$, the element $m' = (a/s)m$ of M satisfies $sm' = am$, so

$$\frac{a}{s} \cdot \varphi(m) = \frac{am}{s} = \frac{sm'}{s} = \frac{m'}{1} = \varphi\Big(\frac{a}{s}\, m\Big)$$

and φ is $S^{-1}A$-linear. It is injective, since $m/1 = 0$ means $um = 0$ for some $u \in S$, and u acts on M invertibly. It is surjective, since the display with $a = 1$ reads $m/s = \varphi\big((1/s)m\big)$. Composing the augmentation with φ^{-1} turns the localized sequence into a projective resolution of M over $S^{-1}A$ of length n, so $\mathrm{pd}_{S^{-1}A}(M) \le n$.

By definition 4.2.3, $\mathrm{gldim}(S^{-1}A)$ is the supremum of $\mathrm{pd}_{S^{-1}A}(M)$ over all $S^{-1}A$-modules M, and each of these is at most $\mathrm{pd}_A(M) \le \mathrm{gldim}(A)$, completing the proof.

Combined with theorem 4.2.11, the lemma shows that a Noetherian ring A of finite global dimension is regular, since each $A_\mathfrak{p}$ is then a Noetherian local ring of global dimension at most $\mathrm{gldim}(A)$.

Corollary 4.2.14: Polynomial Rings over Regular Rings Are Regular

Let A be a regular ring. Then the polynomial ring $A[t_1, \dots, t_n]$ is a regular ring for every $n \geq 0$.

***Proof*:**
Since $A[t_1, \dots, t_n] = A[t_1, \dots, t_{n-1}][t_n]$, induction on n reduces the claim to $n = 1$. Write t for the variable. The ring $A[t]$ is Noetherian by Hilbert's Basis Theorem ([7, Hilbert's Basis Theorem (HBT)]).

Let P be a prime ideal of $A[t]$ and $\mathfrak{p} = P \cap A$, a prime ideal of A. The multiplicative set $A \setminus \mathfrak{p}$ is contained in $A[t] \setminus P$, and inverting it coefficientwise gives $(A \setminus \mathfrak{p})^{-1}A[t] = A_{\mathfrak{p}}[t]$. Let T be the image of $A[t] \setminus P$ in $A_{\mathfrak{p}}[t]$. By the universal property of localization ([7, Universal Property of Localization]), both $T^{-1}(A_{\mathfrak{p}}[t])$ and $A[t]_P$ are initial among $A[t]$-algebras in which every element of $A[t] \setminus P$ becomes a unit. Hence $A[t]_P \cong T^{-1}(A_{\mathfrak{p}}[t])$ is a localization of $A_{\mathfrak{p}}[t]$.

The ring $A_{\mathfrak{p}}$ is a regular local ring by definition 4.2.12, in particular Noetherian and local, so the implication from regularity to finite global dimension in theorem 4.2.11 gives $\operatorname{gldim}(A_{\mathfrak{p}}) < \infty$. Then lemma 4.2.13 and theorem 4.2.5 give

$$\operatorname{gldim}(A[t]_P) \leq \operatorname{gldim}(A_{\mathfrak{p}}[t]) = \operatorname{gldim}(A_{\mathfrak{p}}) + 1 < \infty$$

The ring $A[t]_P$ is local, and it is Noetherian as a localization of the Noetherian ring $A[t]$ ([7, Ideals of Localization]). The implication from finite global dimension to regularity in theorem 4.2.11 therefore makes $A[t]_P$ a regular local ring. As P was an arbitrary prime of $A[t]$, the Noetherian ring $A[t]$ is regular.

Geometrically, $\operatorname{Spec} A[t_1, \dots, t_n] = \operatorname{Spec} A \times_{\operatorname{Spec} \mathbb{Z}} \mathbb{A}^n_{\mathbb{Z}}$, so the product of a regular affine scheme with affine n-space is regular.

4.2.3 Beyond Regularity: The Hierarchy of Singularities

The Auslander-Buchsbaum-Serre theorem (theorem 4.2.11) presents a stark dichotomy: a local ring either has finite global dimension (and is geometrically smooth), or its global dimension is completely infinite.

For decades, algebraists sought to understand the "infinite" case. If a ring is singular, just *how* singular is it? How chaotic do the infinite free resolutions become? This inquiry led to a classification of local Noetherian rings, a hierarchy that measures the severity of singularities based on homological behavior.

The complexity of a ring that fails to be regular is measured with the regular sequences of definition 4.2.9. Intuitively, a regular sequence is a sequence of elements that "cut down" the dimension of the module as cleanly and efficiently as possible, without creating any hidden nilpotent, and the maximal length of one is a new invariant.

Definition 4.2.15: Depth

Let $(R, \mathfrak{m})$ be a local Noetherian ring and M a finitely generated R-module. The *depth* of M, denoted $\operatorname{depth}(M)$, is the maximum length of an M-regular sequence contained in the maximal ideal $\mathfrak{m}$.

Depth is always bounded above by the Krull dimension ($\operatorname{depth}(R) \leq \operatorname{kdim}(R)$, proved as corollary 4.2.22 below). The gap between a ring's depth and its dimension is one of the primary measures of how "bad" a singularity is.

As defined, depth is a maximum over regular sequences, which makes it awkward to compute and awkward to use: knowing that *some* regular sequence has length n says nothing about whether a *given* regular element extends to one. Rees's theorem removes both difficulties at once by identifying depth with a vanishing degree for Ext, from which it follows that every maximal regular sequence has the same length.

Theorem 4.2.16: Depth via Ext (Rees)

Let $(R, \mathfrak{m}, k)$ be a Noetherian local ring and $M \neq 0$ a finitely generated R-module. If $x_1, \dots, x_n \in \mathfrak{m}$ is an M-regular sequence, then

$$\operatorname{Ext}^i_R(k, M) = 0 \text{ for } i < n, \qquad \operatorname{Ext}^n_R(k, M) \cong \operatorname{Hom}_R\big(k,\, M/(x_1, \dots, x_n)M\big)$$

Consequently every maximal M-regular sequence in $\mathfrak{m}$ has length $\min\big\{i \;\big|\; \operatorname{Ext}^i_R(k, M) \neq 0\big\}$, and in particular they all have the same length, namely $\operatorname{depth}(M)$.

***Proof*:**

Step 1: $\mathfrak{m}$ acts as zero on $\operatorname{Ext}^i_R(k, M)$. For $x \in \mathfrak{m}$, multiplication by x on $\operatorname{Ext}^i_R(k, M)$ is induced by multiplication by x on the first argument, which is the zero map $k \to k$; functoriality makes the induced map zero.

Step 2: the displayed formulas, by induction on n. For $n = 0$ there is nothing to prove, $\operatorname{Ext}^0_R(k, M) \cong \operatorname{Hom}_R(k, M)$ by proposition 2.3.13. Let $n \geq 1$ and put $M' = M/x_1M$, which is nonzero by Nakayama's lemma. As x_1 is a non-zero divisor on M, there is a short exact sequence

$$0 \to M \xrightarrow{x_1} M \to M' \to 0$$

and corollary 2.3.12, applied to $\operatorname{Hom}_R(k, -)$, gives a long exact sequence in which every map induced by x_1 vanishes by Step 1. It therefore breaks into short exact sequences

$$0 \to \operatorname{Ext}^i_R(k, M) \to \operatorname{Ext}^i_R(k, M') \to \operatorname{Ext}^{i+1}_R(k, M) \to 0$$

for every i. Now $x_2, \dots, x_n$ is an M'-regular sequence of length $n-1$, so by induction $\operatorname{Ext}^i_R(k, M') = 0$ for $i < n-1$ and $\operatorname{Ext}^{n-1}_R(k, M') \cong \operatorname{Hom}_R(k, M/(x_1, \dots, x_n)M)$. Reading the displayed sequence for $i < n-1$ forces $\operatorname{Ext}^i_R(k, M) = 0$ and $\operatorname{Ext}^{i+1}_R(k, M) = 0$, so $\operatorname{Ext}^j_R(k, M) = 0$ for all $j < n$. Reading it at $i = n-1$, the left-hand term vanishes, so $\operatorname{Ext}^n_R(k, M) \cong \operatorname{Ext}^{n-1}_R(k, M')$, which is the claimed Hom.

Step 3: maximality. A regular sequence $x_1, \dots, x_n$ is maximal exactly when no element of $\mathfrak{m}$ is a non-zero divisor on $N = M/(x_1, \dots, x_n)M$, that is when $\mathfrak{m}$ consists of zero-divisors on N.

The zero-divisors on N are the union of its associated primes ([7, The Zero-Divisors Are the Union of the Associated Primes]), of which there are finitely many ([7, Finiteness of Associated Primes]), so prime avoidance ([7, Prime Avoidance]) makes this equivalent to $\mathfrak{m} \in \operatorname{Ass}(N)$, that is to $\operatorname{Hom}_R(k, N) \neq 0$. By Step 2 that says $\operatorname{Ext}^n_R(k, M) \neq 0$, while $\operatorname{Ext}^i_R(k, M) = 0$ for $i < n$. So n is the least degree in which $\operatorname{Ext}^\bullet_R(k, M)$ is nonzero, a quantity independent of the sequence chosen, completing the proof.

Two consequences are immediate and are used throughout what follows: a given M-regular sequence in $\mathfrak{m}$ extends to one of length $\operatorname{depth}(M)$, since any maximal extension of it has that length; and $\operatorname{depth}(M) = 0$ exactly when $\mathfrak{m} \in \operatorname{Ass}(M)$.

Using this, we can make the Auslander-Buchsbaum-Serre theorem more refined by looking at modules over R:

Theorem 4.2.17: Auslander-Buchsbaum Formula

Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring, and let M be a non-zero finitely generated R-module with finite projective dimension. Then:

$$\operatorname{pd}_R(M) + \operatorname{depth}(M) = \operatorname{depth}(R)$$

***Proof Key ideas*:**
The proof is an induction on $\operatorname{depth}(M)$ whose steps are standard commutative algebra rather than homological algebra; it is given in full in *An Introduction to Homological Algebra* [23, Thm. 4.4.15].

The base case is $\operatorname{depth}(M) = 0$, where $\mathfrak{m}$ is an associated prime of M and one shows $\operatorname{pd}_R(M) = \operatorname{depth}(R)$ directly from a minimal free resolution: the length of that resolution is detected by $\operatorname{Tor}^R_i(M, k)$ against the residue field (lemma 4.2.8), and the depth of R bounds where those groups can survive.

The inductive step reduces the depth of M by one. If $\operatorname{depth}(M) > 0$ and $\operatorname{depth}(R) > 0$, prime avoidance applied to $\operatorname{Ass}(R) \cup \operatorname{Ass}(M)$ produces an $x \in \mathfrak{m}$ regular on both M and R; passing to M/xM over $R/(x)$ drops both $\operatorname{depth}(M)$ and $\operatorname{depth}(R)$ by one while leaving pd unchanged, so the formula for the quotient gives the formula for M. The remaining case is $\operatorname{depth}(R) = 0$, handled separately: there finite projective dimension forces M to be free, so both sides of the formula vanish.

The two theorems combine into the numerical statement that the applications actually use: for a regular local ring the global dimension is not merely finite, it is the Krull dimension.

Corollary 4.2.18: Global Dimension of a Regular Local Ring

Let $(R, \mathfrak{m}, k)$ be a Noetherian regular local ring of dimension m. Then

$$\operatorname{gldim}(R) \;=\; \operatorname{depth}(R) \;=\; \dim R \;=\; m$$

In particular every finitely generated R-module has projective dimension at most $\dim R$.

***Proof*:**
Step 1: $\operatorname{depth}(R) \geq m$. Choose $x_1, \ldots, x_m$ generating $\mathfrak{m}$, possible by regularity. By lemma 4.2.10 they form a regular sequence, which lies in $\mathfrak{m}$ and has length m, so $\operatorname{depth}(R) \geq m$.

Step 2: $\operatorname{pd}_R(k) \leq m$. By Step 1 and [8, Exactness of the Koszul Complex], the Koszul complex on $x_1, \ldots, x_m$ is a finite free resolution of $R/\mathfrak{m} = k$ of length m. So $\operatorname{pd}_R(k) \leq m$.

Step 3: assembly. Theorem 4.2.11 gives $\operatorname{gldim}(R) = \operatorname{pd}_R(k)$, finite by Step 2, so the Auslander-Buchsbaum formula (theorem 4.2.17) applies to $M = k$. Every element of $\mathfrak{m}$ annihilates the nonzero module k and is therefore a zerodivisor on it, so no element of $\mathfrak{m}$ is k-regular and $\operatorname{depth}(k) = 0$. The formula reads $\operatorname{pd}_R(k) = \operatorname{depth}(R)$. Combining the three steps,

$$m \leq \operatorname{depth}(R) = \operatorname{pd}_R(k) = \operatorname{gldim}(R) \leq m$$

so $\operatorname{depth}(R)$, $\operatorname{pd}_R(k)$ and $\operatorname{gldim}(R)$ all equal $m = \dim R$, completing the proof.

Using depth and other structural properties, local rings can be classified into a strict chain of implications:

$$\text{Regular} \implies \text{Complete Intersection} \implies \text{Gorenstein} \implies \text{Cohen-Macaulay}$$

The chain stops at Cohen-Macaulay: it does *not* continue to normal or reduced. The hypersurface $k[[x, y]]/(xy)$ is a complete intersection, hence Gorenstein and Cohen-Macaulay, and it is reduced, but it is not normal, being not even a domain. By Serre's criterion (theorem 4.3.2) normality is the conjunction of R_1 and S_2; Cohen-Macaulay gives S_2 and says nothing about R_1, which is where the implication fails. The correct companion chain runs

$$\text{Regular} \implies \text{Normal} \implies \text{Reduced}$$

While these definitions are strictly for *local* rings, we say a Noetherian ring has one of these properties if its localization at every prime ideal has it (for regularity this is definition 4.2.12).

Complete Intersections: These are the "nicest" singularities. Geometrically, they are spaces cut out by exactly the correct number of equations (e.g., a 1D curve in $\mathbb{C}^3$ defined by exactly two equations). Algebraically, R is the quotient of a regular local ring by a regular sequence. While modules over these rings can have infinite projective dimension, their free resolutions are highly predictable, eventually becoming periodic.

Gorenstein Rings: These rings represent spaces with "symmetric" singularities. They are exactly the spaces where a generalized form of Poincaré Duality still holds. Algebraically, a local ring is Gorenstein if it has finite injective dimension over itself. The free resolutions over Gorenstein rings exhibit a mirror symmetry when the functor $\operatorname{Hom}_R(-, R)$ is applied.

Cohen-Macaulay Rings: This is perhaps the most important class of singular rings in algebraic geometry, and the one the subsection below treats properly. A local ring is Cohen-Macaulay if its depth equals its Krull dimension (definition 4.2.19). Geometrically, these spaces are "equidimensional" and lack hidden, embedded fuzzy components. Here, the Auslander-Buchsbaum formula perfectly aligns with the geometric dimension.

Normal Rings: A ring is normal if it is integrally closed in its total ring of fractions. Geometrically, this means the space is "smooth in codimension 1". For example, a 2D normal surface can have

an isolated singular point (like the tip of a cone), but it cannot have a singular crease or folded line running through it.

Reduced Rings: A ring is reduced if it contains no non-zero nilpotents ($x^n = 0 \implies x = 0$). Geometrically, this simply means the space has no "thickness" or "fuzz"; it is a classical geometric space uniquely determined by the values of functions at its points.

4.2.4 Cohen-Macaulay Rings

Of the classes just listed, Cohen-Macaulay is the one downstream geometry actually uses, and it deserves more than a description. Two facts are wanted: that a system of parameters in such a ring is automatically a regular sequence, so that the naive count of equations is the honest one; and that the ring carries no embedded associated primes, so that a subscheme cut out this way has no hidden components. Both follow from a single inequality relating depth to the dimensions of the associated primes. Associated primes, prime avoidance, and the identification of the zero-divisors with the union of the associated primes are those of [7, Associated Prime, Prime Avoidance, Zero-Divisors and Annihilators].

Definition 4.2.19: Cohen-Macaulay Ring

A Noetherian local ring $(R, \mathfrak{m})$ is *Cohen-Macaulay* if $\operatorname{depth}(R) = \operatorname{kdim}(R)$. A Noetherian ring is Cohen-Macaulay if its localization at every prime is. More generally a finitely generated module $M \neq 0$ over a Noetherian local ring is *Cohen-Macaulay* if $\operatorname{depth}(M) = \operatorname{kdim}(M)$, the dimension of $R/\operatorname{Ann}(M)$.

Lemma 4.2.20: Depth Drops by One Modulo a Non-Zero Divisor

Let $(R, \mathfrak{m})$ be Noetherian local, $M \neq 0$ finitely generated, and $x \in \mathfrak{m}$ a non-zero divisor on M. Then $\operatorname{depth}(M/xM) = \operatorname{depth}(M) - 1$.

***Proof*:**
Prepending x to any M/xM-regular sequence in $\mathfrak{m}$ yields an M-regular sequence, so $\operatorname{depth}(M) \geq \operatorname{depth}(M/xM) + 1$ directly from definition 4.2.15.

For the reverse, extend x to a maximal M-regular sequence $x, y_1, \ldots, y_m$ in $\mathfrak{m}$; this is possible, and the result has length $\operatorname{depth}(M)$, by theorem 4.2.16. Then $y_1, \ldots, y_m$ is an M/xM-regular sequence, so $\operatorname{depth}(M/xM) \geq m = \operatorname{depth}(M) - 1$. Combining the two, both are equalities, as we sought to show.

Proposition 4.2.21: Depth Is Bounded by the Dimension of Every Associated Prime

Let $(R, \mathfrak{m})$ be Noetherian local, $M \neq 0$ finitely generated, and $\mathfrak{p} \in \operatorname{Ass}(M)$. Then

$$\operatorname{depth}(M) \leq \operatorname{kdim}(R/\mathfrak{p})$$

***Proof*:**
Induct on $n = \operatorname{depth}(M)$. For $n = 0$ there is nothing to prove. Let $n > 0$ and let $x \in \mathfrak{m}$ be a non-zero divisor on M, which exists because the depth is positive. Since $\mathfrak{p}$ consists of zero-divisors on M, [7, The Zero-Divisors Are the Union of the Associated Primes] gives $x \notin \mathfrak{p}$.

Write $\mathfrak{p} = \operatorname{Ann}(m)$ with $m \neq 0$. We first arrange $m \notin xM$. If $m = xm_1$ then $\operatorname{Ann}(m_1) = \mathfrak{p}$: an r killing m_1 kills m, and conversely $rm = 0$ gives $rxm_1 = 0$, whence $rm_1 = 0$ as x is a non-zero divisor. Iterating produces $m = x^k m_k$ with $\operatorname{Ann}(m_k) = \mathfrak{p}$ for each k, and the submodules $Rm \subseteq Rm_1 \subseteq Rm_2 \subseteq \cdots$ ascend. As M is Noetherian the chain stabilizes, say $Rm_k = Rm_{k+1}$, so $m_{k+1} = rm_k = rx\, m_{k+1}$ for some r; then $(1 - rx)m_{k+1} = 0$, and $1 - rx$ is a unit because $x \in \mathfrak{m}$, forcing $m_{k+1} = 0$ and hence $m = 0$. That contradiction means the iteration halts, so we may take $m \notin xM$.

Let $N = Rm$ and let $\overline{N}$ be its image in M/xM, nonzero by the previous paragraph. Every element of $\mathfrak{p}$ kills N, hence $\overline{N}$; and x kills $\overline{N}$ too, since $xN \subseteq xM$. So $\operatorname{Ann}(\overline{N}) \supseteq \mathfrak{p} + (x)$. Choose $\mathfrak{q} \in \operatorname{Ass}(\overline{N}) \subseteq \operatorname{Ass}(M/xM)$, nonempty by [7, Zero-Divisors and Annihilators]. Then $\mathfrak{q} \supseteq \mathfrak{p} + (x)$, and $\mathfrak{q} \neq \mathfrak{p}$ because $x \notin \mathfrak{p}$, so $\mathfrak{q} \supsetneq \mathfrak{p}$ and $\operatorname{kdim}(R/\mathfrak{q}) \leq \operatorname{kdim}(R/\mathfrak{p}) - 1$. By lemma 4.2.20 and the inductive hypothesis applied to M/xM and $\mathfrak{q}$,

$$n - 1 \;=\; \operatorname{depth}(M/xM) \;\leq\; \operatorname{kdim}(R/\mathfrak{q}) \;\leq\; \operatorname{kdim}(R/\mathfrak{p}) - 1$$

which is the assertion.

Corollary 4.2.22: Depth Is at Most the Dimension

For $M \neq 0$ finitely generated over a Noetherian local ring, $\operatorname{depth}(M) \leq \operatorname{kdim}(M)$. In particular $\operatorname{depth}(R) \leq \operatorname{kdim}(R)$, so Cohen-Macaulay is the extremal case.

***Proof*:**
$\operatorname{Ass}(M)$ is nonempty ([7, Zero-Divisors and Annihilators]); pick $\mathfrak{p}$ in it. Then $\mathfrak{p} \supseteq \operatorname{Ann}(M)$, so $\operatorname{kdim}(R/\mathfrak{p}) \leq \operatorname{kdim}(R/\operatorname{Ann}(M)) = \operatorname{kdim}(M)$, and proposition 4.2.21 bounds the depth by the left-hand side.

This is the inequality asserted informally above, and a Cohen-Macaulay ring is one in which it is an equality (definition 4.2.19).

Corollary 4.2.23: A Cohen-Macaulay Ring Has No Embedded Associated Primes

Let $(R, \mathfrak{m})$ be a Cohen-Macaulay local ring of dimension d. Then every $\mathfrak{p} \in \operatorname{Ass}(R)$ satisfies $\operatorname{kdim}(R/\mathfrak{p}) = d$, and every such $\mathfrak{p}$ is a minimal prime of R. In particular R has no embedded associated primes, and R is equidimensional.

***Proof*:**
For $\mathfrak{p} \in \operatorname{Ass}(R)$, proposition 4.2.21 gives $d = \operatorname{depth}(R) \leq \operatorname{kdim}(R/\mathfrak{p}) \leq d$, so $\operatorname{kdim}(R/\mathfrak{p}) = d$. If $\mathfrak{p}$ were not minimal it would strictly contain a prime $\mathfrak{p}_0$, and then $\operatorname{kdim}(R/\mathfrak{p}) < \operatorname{kdim}(R/\mathfrak{p}_0) \leq d$, a contradiction. An associated prime that is minimal is by definition not embedded.

Theorem 4.2.24: A System of Parameters in a Cohen-Macaulay Local Ring Is a Regular Sequence

Let $(R, \mathfrak{m})$ be a Cohen-Macaulay local ring of dimension d and let $x_1, \dots, x_d \in \mathfrak{m}$ be a system of parameters, that is d elements generating an $\mathfrak{m}$-primary ideal. Then $x_1, \dots, x_d$ is a regular sequence, and $R/(x_1, \dots, x_d)$ is Cohen-Macaulay of dimension zero.

***Proof*:**
Induct on d, the case $d = 0$ being the empty sequence. Let $d > 0$. Since $x_1, \dots, x_d$ generate an $\mathfrak{m}$-primary ideal, the images of $x_2, \dots, x_d$ generate an $\mathfrak{m}$-primary ideal of $R/(x_1)$, so $\operatorname{kdim} R/(x_1) \leq d-1$ by the dimension theorem of [7, Dimension Theorem]; and $\operatorname{kdim} R/(x_1) \geq d-1$ by Krull's height theorem there. So $\operatorname{kdim} R/(x_1) = d-1$.

Now x_1 is a non-zero divisor. Suppose $x_1 \in \mathfrak{p}$ for some $\mathfrak{p} \in \operatorname{Ass}(R)$. By corollary 4.2.23, $\operatorname{kdim}(R/\mathfrak{p}) = d$; but $\mathfrak{p} \supseteq (x_1)$ gives a surjection $R/(x_1) \twoheadrightarrow R/\mathfrak{p}$ and hence $\operatorname{kdim} R/(x_1) \geq d$, contradicting the previous paragraph. So x_1 lies outside every associated prime and is a non-zero divisor by [7, The Zero-Divisors Are the Union of the Associated Primes].

By lemma 4.2.20, $\operatorname{depth} R/(x_1) = d - 1 = \operatorname{kdim} R/(x_1)$, so $R/(x_1)$ is Cohen-Macaulay of dimension $d-1$, and the images of $x_2, \dots, x_d$ are a system of parameters for it. The inductive hypothesis makes them a regular sequence there, which is exactly the statement that $x_1, \dots, x_d$ is a regular sequence on R and that the final quotient is Cohen-Macaulay of dimension zero.

The form geometry reaches for is slightly more general: the elements need not be a full system of parameters, only as many as the height of the ideal they generate.

Corollary 4.2.25: Height-Many Elements in a Cohen-Macaulay Ring Form a Regular Sequence

Let R be a Cohen-Macaulay local ring and let $x_1, \dots, x_h \in R$ generate an ideal $I = (x_1, \dots, x_h)$ of height h. Then $x_1, \dots, x_h$ is a regular sequence.

***Proof*:**
Let $\mathfrak{p}$ be a prime minimal over I with $\operatorname{ht} \mathfrak{p} = h$, which exists because the height of I is the minimum of $\operatorname{ht} \mathfrak{q}$ over primes $\mathfrak{q}$ minimal over I. Localizing, $R_{\mathfrak{p}}$ is Cohen-Macaulay of dimension h (localization of a Cohen-Macaulay local ring at a prime is Cohen-Macaulay, and $\dim R_{\mathfrak{p}} = \operatorname{ht} \mathfrak{p} = h$), and the images of $x_1, \dots, x_h$ generate an ideal primary to the maximal ideal of $R_{\mathfrak{p}}$, since $\mathfrak{p}$ is minimal over I. They are therefore a system of parameters of $R_{\mathfrak{p}}$, hence a regular sequence there by theorem 4.2.24.

To descend, induct on i: suppose $x_1, \dots, x_{i-1}$ is a regular sequence on R and that x_i is a zero divisor on $M = R/(x_1, \dots, x_{i-1})$. Then x_i lies in an associated prime $\mathfrak{q}$ of M ([7, Zero-Divisors and Annihilators]), and M is Cohen-Macaulay of dimension $\dim R - i + 1$ by theorem 4.2.24 applied step by step, so $\mathfrak{q}$ has no embedded components by corollary 4.2.23 and $\dim R/\mathfrak{q} = \dim R - i + 1$. But $\mathfrak{q} \supseteq (x_1, \dots, x_i)$ forces $\operatorname{ht} \mathfrak{q} \geq$ the height of that ideal, and comparing with the localized statement above bounds $\dim R/\mathfrak{q}$ strictly below $\dim R - i + 1$, a contradiction. So each x_i is a non-zero divisor on the preceding quotient, as we sought to show.

4.3 Serre's Criterion for Normality

Depth and regularity are conditions on a local ring; imposing them at every prime, with a strength that grows with the height, produces the two families of conditions Serre introduced. Their point is the theorem below: normality, which is a statement about integral closure and has no homological content on its face, is exactly the conjunction of a regularity condition in codimension one and a depth condition everywhere.

Definition 4.3.1: Serre's Conditions

Let R be a Noetherian ring and $k \geq 0$ an integer. Then R satisfies

- (R_k) if $R_{\mathfrak{p}}$ is a regular local ring for every prime $\mathfrak{p}$ with $\operatorname{ht}\mathfrak{p} \leq k$;
- (S_k) if $\operatorname{depth}(R_{\mathfrak{p}}) \geq \min\left(k,\ \operatorname{ht}\mathfrak{p}\right)$ for every prime $\mathfrak{p}$.

Both conditions weaken as attention moves to primes of large height: (R_k) says nothing about primes of height above k, and (S_k) asks only for depth k once the height reaches k. A regular ring satisfies (R_k) for every k, and a Cohen-Macaulay ring satisfies (S_k) for every k, since there $\operatorname{depth} R_{\mathfrak{p}} = \operatorname{kdim} R_{\mathfrak{p}} = \operatorname{ht}\mathfrak{p}$.

Theorem 4.3.2: Serre's Criterion for Normality

Let R be a Noetherian domain. Then R is integrally closed in its fraction field if and only if it satisfies (R_1) and (S_2).

***Proof*:**

Necessity. Suppose R is normal. For (R_1), a prime of height zero is (0) and $R_{(0)}$ is a field, which is regular; and for $\operatorname{ht}\mathfrak{p} = 1$ the ring $R_{\mathfrak{p}}$ is a discrete valuation ring by [7, A Normal Domain Is the Intersection of Its Height-One Localizations], hence regular.

For (S_2) let $\mathfrak{p}$ be a prime. If $\operatorname{ht}\mathfrak{p} = 0$ there is nothing to prove. If $\operatorname{ht}\mathfrak{p} = 1$ then $R_{\mathfrak{p}}$ is a discrete valuation ring, whose maximal ideal is generated by a non-zero divisor, so its depth is at least $1 = \min(2, 1)$. Suppose $\operatorname{ht}\mathfrak{p} \geq 2$, so that $\min(2, \operatorname{ht}\mathfrak{p}) = 2$. The ring $R_{\mathfrak{p}}$ is a Noetherian normal domain, so any nonzero $b \in \mathfrak{p}R_{\mathfrak{p}}$ is a non-zero divisor and $\operatorname{depth}(R_{\mathfrak{p}}) \geq 1$. By [7, Associated Primes of a Principal Ideal Have Height One], every associated prime of $bR_{\mathfrak{p}}$ has height one, so $\mathfrak{p}R_{\mathfrak{p}}$, of height at least two, is not one of them; hence $\operatorname{depth}(R_{\mathfrak{p}}/bR_{\mathfrak{p}}) \neq 0$ by the second consequence of theorem 4.2.16. Then lemma 4.2.20 gives $\operatorname{depth}(R_{\mathfrak{p}}) = \operatorname{depth}(R_{\mathfrak{p}}/bR_{\mathfrak{p}}) + 1 \geq 2$.

Sufficiency. Suppose R satisfies (R_1) and (S_2), and write K for its fraction field. Every prime of height one has $R_{\mathfrak{p}}$ regular local of dimension one by (R_1), hence a discrete valuation ring, hence integrally closed. An intersection of integrally closed subrings of K each with fraction field K is integrally closed, so it suffices to prove

$$R = \bigcap_{\operatorname{ht}\mathfrak{p}=1} R_{\mathfrak{p}}$$

One inclusion is immediate. For the other, let x lie in every such $R_{\mathfrak{p}}$, write $x = a/b$ with $b \neq 0$, and suppose $a \notin (b)$. Then $\operatorname{Ann}(\overline{a}) = (b : a)$ is a proper ideal consisting of zero-divisors on $R/(b)$, so as in [7, The Zero-Divisors Are the Union of the Associated Primes] together with prime

avoidance it lies in a single $\mathfrak{q} \in \operatorname{Ass}(R/(b))$.

The height of $\mathfrak{q}$ is one. It is at least one because R is a domain and $b \neq 0$. If it were at least two, then (S_2) would give $\operatorname{depth}(R_\mathfrak{q}) \geq 2$, and b is a non-zero divisor in $\mathfrak{q}R_\mathfrak{q}$, so lemma 4.2.20 would give $\operatorname{depth}(R_\mathfrak{q}/bR_\mathfrak{q}) \geq 1$, whence $\mathfrak{q}R_\mathfrak{q} \notin \operatorname{Ass}(R_\mathfrak{q}/bR_\mathfrak{q})$ by theorem 4.2.16. But associated primes localize, so $\mathfrak{q} \in \operatorname{Ass}(R/(b))$ makes $\mathfrak{q}R_\mathfrak{q}$ associated to $R_\mathfrak{q}/bR_\mathfrak{q}$, a contradiction.

So $\operatorname{ht}\mathfrak{q} = 1$ and $x \in R_\mathfrak{q}$ by hypothesis; that produces $s \notin \mathfrak{q}$ with $sa \in bR$, so $s \in (b : a) \subseteq \mathfrak{q}$, a contradiction. Hence $a \in (b)$ and $x \in R$, completing the proof.

Geometrically, (R_1) says a normal variety is smooth away from a closed set of codimension at least two, and (S_2) says a function defined outside such a set extends across it. That reading, and the divisor theory it supports, is [14].

4.4 The Local Criterion for Flatness

One of the most powerful applications of Nakayama's Lemma and homological algebra is the Local Criterion for Flatness. Checking whether a module is flat directly from the definition (that tensoring preserves exactness) can be extremely difficult. The local criterion reduces this to checking exactness against a single module (the residue field), or checking that regular sequences map to regular sequences.

Theorem 4.4.1: Local Criterion for Flatness

Let $\varphi : (R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a local homomorphism of Noetherian local rings, and let M be a finitely generated S-module. Then the following are equivalent:

1. M is flat over R.
2. $\operatorname{Tor}_1^R(M, R/\mathfrak{m}) = 0$.
3. The natural multiplication map $\mathfrak{m} \otimes_R M \to \mathfrak{m}M$ is an isomorphism.

***Proof*:**

The implication (1) $\Rightarrow$ (2) is trivial since M is flat implies $\operatorname{Tor}_i^R(M, -) = 0$ for all $i > 0$. The equivalence (2) $\iff$ (3) follows from applying $- \otimes_R M$ to the short exact sequence $0 \to \mathfrak{m} \to R \to R/\mathfrak{m} \to 0$, which gives the long exact sequence in Tor:

$$0 = \operatorname{Tor}_1^R(R, M) \to \operatorname{Tor}_1^R(R/\mathfrak{m}, M) \to \mathfrak{m} \otimes_R M \to R \otimes_R M \to R/\mathfrak{m} \otimes_R M \to 0$$

Since $R \otimes_R M \cong M$, the kernel of $M \to M/\mathfrak{m}M$ is precisely $\mathfrak{m}M$. Thus $\operatorname{Tor}_1^R(R/\mathfrak{m}, M) = 0$ if and only if $\mathfrak{m} \otimes_R M \to \mathfrak{m}M$ is injective (and it is always surjective).

For (2) $\Rightarrow$ (1), we must show that $\operatorname{Tor}_1^R(M, N) = 0$ for all finitely generated R-modules N. We proceed by induction on the length of N. By Nakayama's Lemma ([7, Nakayama's Lemma I]), if $N \neq 0$, we can find a non-zero quotient $N \to R/\mathfrak{m}$. Let K be the kernel, so we have $0 \to K \to N \to R/\mathfrak{m} \to 0$. The long exact sequence in Tor gives:

$$\operatorname{Tor}_1^R(K, M) \to \operatorname{Tor}_1^R(N, M) \to \operatorname{Tor}_1^R(R/\mathfrak{m}, M) = 0$$

Thus, $\operatorname{Tor}_1^R(K,M)$ surjects onto $\operatorname{Tor}_1^R(N,M)$. By iterating this process (using the Artin-Rees lemma and the $\mathfrak{m}$-adic filtration on N; [7, Artin-Rees Lemma]), one can show that $\operatorname{Tor}_1^R(N,M) \subseteq \mathfrak{m}\operatorname{Tor}_1^R(N,M)$. Since M is a finitely generated S-module and S is Noetherian, $\operatorname{Tor}_1^R(N,M)$ is a finitely generated S-module. By Nakayama's Lemma (applied over S, since $\mathfrak{m}S \subseteq \mathfrak{n}$), we conclude $\operatorname{Tor}_1^R(N,M) = 0$.

In algebraic geometry, we often deal with regular local rings and regular sequences. The Local Criterion gives a corollary that tests flatness with regular sequences. Together with theorem 4.2.24, it is the main input to the miracle flatness theorem of *Algebraic Geometry*:[2] for a local homomorphism $(A,\mathfrak{m}) \to (B,\mathfrak{n})$ of Noetherian local rings with A regular and B Cohen-Macaulay, B is flat over A if and only if $\dim B = \dim A + \dim B/\mathfrak{m}B$.

Corollary 4.4.2: Regular Sequence Criterion for Flatness

Let $\varphi : (R,\mathfrak{m}) \to (S,\mathfrak{n})$ be a local homomorphism of Noetherian local rings, and M a finitely generated S-module. Let $x \in \mathfrak{m}$ be a non-zero divisor on R. Then M is flat over R if and only if both of the following hold if and only if:

1. x is a non-zero divisor on M, and
2. M/xM is flat over $R/(x)$.

By induction, if $x_1,\dots,x_d \in \mathfrak{m}$ is a regular sequence in R, then M is flat over R if and only if $x_1,\dots,x_d$ forms a regular sequence on M and $M/(x_1,\dots,x_d)M$ is flat over $R/(x_1,\dots,x_d)$.

***Proof*:**

($\Rightarrow$): If M is flat over R, applying $- \otimes_R M$ to the exact sequence $0 \to R \xrightarrow{\cdot x} R \to R/(x) \to 0$ yields the exact sequence $0 \to M \xrightarrow{\cdot x} M \to M/xM \to 0$. Thus x is a non-zero divisor on M. Furthermore, base change preserves flatness, so $M/xM \cong M \otimes_R R/(x)$ is flat over $R/(x)$.

($\Leftarrow$): Assume x is regular on M and M/xM is flat over $R/(x)$. We want to show M is flat. Since M/xM is flat over $R/(x)$, $\operatorname{Tor}_1^{R/(x)}(R/\mathfrak{m}, M/xM) = 0$. Combined with the fact that x is regular on M, this gives $\operatorname{Tor}_1^R(R/\mathfrak{m}, M) = 0$: taking a free resolution $F_\bullet \to M$ over R, regularity of x on both R and M makes $F_\bullet \otimes_R R/(x)$ a free resolution of M/xM over $R/(x)$, so $\operatorname{Tor}_i^{R/(x)}(N, M/xM) \cong \operatorname{Tor}_i^R(N,M)$ for every $R/(x)$-module N, and $N = R/\mathfrak{m}$ is one. By the Local Criterion for Flatness (theorem 4.4.1), M is flat over R.

The variant that geometry actually asks for reverses the roles: rather than a regular sequence downstairs, one is handed an element that is a nonzerodivisor *on the fibre*, and wants to conclude that it was a nonzerodivisor upstairs all along. This is how flatness of a family cut out by equations gets verified one equation at a time.

[2] [4, Miracle Flatness for Local Rings].

Corollary 4.4.3: Fibrewise Nonzerodivisor Criterion for Flatness

Let $\varphi : (R, \mathfrak{m}) \to (S, \mathfrak{n})$ be a local homomorphism of Noetherian local rings with $\kappa = R/\mathfrak{m}$, let M be a finitely generated S-module that is flat over R, and let $u \in \mathfrak{n}$. If u is a non-zero divisor on $M/\mathfrak{m}M = M \otimes_R \kappa$, then u is a non-zero divisor on M, and M/uM is flat over R.

***Proof*:**
Write $K = \ker(u : M \to M)$ and $M' = M/K$, so that multiplication by u factors as a surjection $M \twoheadrightarrow M'$ followed by an injection $M' \hookrightarrow M$, and $N := M/uM$ is the cokernel of the latter. Both sequences

$$0 \to K \to M \to M' \to 0, \qquad 0 \to M' \xrightarrow{u} M \to N \to 0$$

are exact by construction.

Step 1: N is flat. Apply $- \otimes_R \kappa$ to the second sequence. Flatness of M gives $\operatorname{Tor}_1^R(\kappa, M) = 0$, so the long exact sequence begins

$$0 \to \operatorname{Tor}_1^R(\kappa, N) \to M' \otimes_R \kappa \to M \otimes_R \kappa \to N \otimes_R \kappa \to 0$$

Now right-exactness applied to the first sequence makes $M \otimes_R \kappa \to M' \otimes_R \kappa$ surjective, and composing it with $M' \otimes_R \kappa \to M \otimes_R \kappa$ recovers multiplication by u on $M \otimes_R \kappa$, which is injective by hypothesis. A surjection whose composite with a further map is injective is itself injective, so $M \otimes_R \kappa \to M' \otimes_R \kappa$ is an isomorphism, and therefore $M' \otimes_R \kappa \to M \otimes_R \kappa$ is injective as well. Hence $\operatorname{Tor}_1^R(\kappa, N) = 0$, and N is flat over R by theorem 4.4.1.

Step 2: $K = 0$. Since M and N are both flat over R, the second sequence forces M' to be flat too: for any R-module T the long exact sequence reads $0 = \operatorname{Tor}_2^R(T, N) \to \operatorname{Tor}_1^R(T, M') \to \operatorname{Tor}_1^R(T, M) = 0$. In particular $\operatorname{Tor}_1^R(\kappa, M') = 0$, so applying $- \otimes_R \kappa$ to the first sequence leaves $K \otimes_R \kappa \to M \otimes_R \kappa$ injective. But Step 1 showed $M \otimes_R \kappa \to M' \otimes_R \kappa$ is an isomorphism, so exactness at $M \otimes_R \kappa$ makes the image of $K \otimes_R \kappa$ zero. An injective map with zero image is zero, so $K/\mathfrak{m}K = 0$. As K is a submodule of the finitely generated module M over the Noetherian ring S, it is finitely generated, and $\mathfrak{m}S \subseteq \mathfrak{n}$, so Nakayama gives $K = 0$.

A third consequence is the one deformation arguments in geometry lean on: a resolution of a flat object stays a resolution after restriction to a fibre. Without it a resolution built on a total space says nothing about the special fibres it was built to compare.

Proposition 4.4.4: Restriction of a Resolution to a Fibre

Let R be a ring, A an R-algebra, and M an A-module that is flat over R. Let

$$0 \to G_n \to G_{n-1} \to \cdots \to G_0 \to M \to 0$$

be an exact sequence of A-modules with every G_i flat over R. Then for every R-algebra $R \to k$, applying $- \otimes_R k$ leaves the sequence exact:

$$0 \to G_n \otimes_R k \to \cdots \to G_0 \otimes_R k \to M \otimes_R k \to 0$$

In particular, if $X \to S$ is a morphism of schemes, $\mathscr{F}$ a quasi-coherent sheaf on X flat over S, and $G_\bullet \to \mathscr{F}$ a resolution by quasi-coherent sheaves flat over S, then the restriction of $G_\bullet \to \mathscr{F}$ to any fibre X_s is a resolution of $\mathscr{F}|_{X_s}$.

***Proof*:**
Break the sequence into short exact pieces $0 \to Z_{i+1} \to G_i \to Z_i \to 0$ with $Z_0 = M$ and $Z_{n+1} = 0$. We show by induction on i that every Z_i is flat over R, the base case $Z_0 = M$ being the hypothesis. Given that Z_i is R-flat, the sequence $0 \to Z_{i+1} \to G_i \to Z_i \to 0$ has flat right-hand and middle terms, so the long exact sequence of $\operatorname{Tor}^R(-, N)$ gives $0 = \operatorname{Tor}_2^R(Z_i, N) \to \operatorname{Tor}_1^R(Z_{i+1}, N) \to \operatorname{Tor}_1^R(G_i, N) = 0$ for every N, whence Z_{i+1} is flat. The induction therefore gives flatness of every syzygy.

Now tensor each short exact piece with k over R. Flatness of Z_i makes $\operatorname{Tor}_1^R(Z_i, k) = 0$, so $0 \to Z_{i+1} \otimes_R k \to G_i \otimes_R k \to Z_i \otimes_R k \to 0$ is exact. Splicing the pieces back together reassembles the long sequence, still exact. The sheaf statement is this one applied on affine opens, flatness and exactness both being local, as we sought to show.

Exercise 4.4.1

1. Let $R = k[x, y]$. Find a free resolution for the module $M = R/(x^2, xy, y^2)$. What is the length of this resolution?
2. Show that over $R = \mathbb{Z}$, the module $M = \mathbb{Q}$ has projective dimension 1, but does not have a finite *free* resolution if we require finitely generated free modules (which is impossible since $\mathbb{Q}$ is not finitely generated).
3. Why does the Syzygy theorem fail for $R = k[x_1, x_2, \dots]$ (infinite variables)? Find a module with infinite projective dimension.

4.5 The Limit of a Tower and Its First Derived Functor

The functors derived so far in this chapter have been Hom and $\otimes$, and the objects they were applied to were modules over a ring. The functor of this section is neither, and its source category is not a category of modules: it is the limit, applied to a sequence of abelian groups indexed by $\mathbb{N}$ with maps running backwards. The limit of such a sequence is left exact and not exact, so it has derived functors like any other left-exact functor. What is special is how few of them there are. Exactly one

is nonzero, it admits a description with no resolutions in it at all, and it vanishes under a hypothesis satisfied in most of the situations where the limit is taken.

That last point is why the section exists. A cohomology theory evaluated on an infinite complex, the cohomology of an infinite-dimensional classifying space, the completion of a ring: each is computed as a limit over a sequence of finite approximations, and in each the answer is the naive limit *provided* a single obstruction group vanishes. Naming that group is what allows the provision to be checked rather than assumed.[3]

Definition 4.5.1: Tower of Abelian Groups

A *tower* of abelian groups is a sequence $A_\bullet = (A_n)_{n\geq 0}$ of abelian groups together with homomorphisms

$$A_0 \xleftarrow{f_1} A_1 \xleftarrow{f_2} A_2 \xleftarrow{f_3} \cdots$$

called the *transition maps*. For $m \geq n$ write $f_{n,m}\colon A_m \to A_n$ for the composite $f_{n+1}\circ\cdots\circ f_m$, with $f_{n,n} = \mathrm{id}_{A_n}$.
A *morphism of towers* $u\colon A_\bullet \to B_\bullet$ is a family of homomorphisms $u_n\colon A_n \to B_n$ with $u_n \circ f_{n+1} = g_{n+1}\circ u_{n+1}$ for every n, where $g_\bullet$ denotes the transition maps of $B_\bullet$. Towers and their morphisms form an abelian category, in which kernels, cokernels, images and exactness are all computed degreewise.

That the category is abelian with everything computed degreewise is what makes the section possible: a sequence of towers is exact exactly when each of its levels is exact, so the hypotheses of every result below can be checked one group at a time. The limit itself is the group of coherent families, and the observation that turns it into something derivable is that this group is the kernel of a single explicit map.

Definition 4.5.2: The Limit and $\varprojlim^1$ of a Tower

Let $A_\bullet$ be a tower of abelian groups. Define

$$d^A\colon \prod_{n\geq 0} A_n \to \prod_{n\geq 0} A_n, \qquad d^A\big((a_n)_n\big) = \big(a_n - f_{n+1}(a_{n+1})\big)_n$$

The *limit* of the tower is $\varprojlim A_\bullet := \ker d^A$, and the *first derived functor of the limit* is

$$\varprojlim{}^1 A_\bullet := \operatorname{coker} d^A$$

Both are functors from towers to abelian groups: a morphism $u\colon A_\bullet \to B_\bullet$ induces $\prod u_n$ on products, and $d^B \circ \prod u_n = \prod u_n \circ d^A$ because $u_n f_{n+1} = g_{n+1}u_{n+1}$, so $\prod u_n$ carries $\ker d^A$ into $\ker d^B$ and descends to cokernels.

The kernel is the usual object under a different description. An element $(a_n) \in \prod A_n$ lies in $\ker d^A$ precisely when $a_n = f_{n+1}(a_{n+1})$ for every n, which is the condition that the family be compatible with the transition maps; so $\ker d^A$ is the group of coherent sequences, with the projections $\varprojlim A_\bullet \to A_n$

[3]The older literature writes this group $\varprojlim^1$ and calls it the first derived functor of the inverse limit. This series says *limit* and names the shape of the diagram separately, so the notation here is $\varprojlim^1$ and the phrase is "the limit of a tower".

obtained by restricting the product projections. The cokernel measures how far d^A is from being surjective, and the content of the next theorem is that this single group is the entire obstruction to the limit being exact.

Theorem 4.5.3: The Six-Term Sequence of the Limit

Let $0 \to A_\bullet \to B_\bullet \to C_\bullet \to 0$ be a short exact sequence of towers of abelian groups. Then there is an exact sequence

$$0 \to \varprojlim A_\bullet \to \varprojlim B_\bullet \to \varprojlim C_\bullet \xrightarrow{\partial} \varprojlim{}^1 A_\bullet \to \varprojlim{}^1 B_\bullet \to \varprojlim{}^1 C_\bullet \to 0$$

natural in the short exact sequence.

The sequence says two things worth separating. Reading the left half, the limit is left exact but its failure to preserve surjections is entirely accounted for by the connecting map ∂ into $\varprojlim^1 A_\bullet$: if that group vanishes, $\varprojlim B_\bullet \to \varprojlim C_\bullet$ is onto. Reading the right half, the sequence *stops*. There is no seven-term continuation, and proposition 4.5.4 explains why: the higher derived functors are zero, so no further obstruction can appear.

***Proof*:**

Write $P(A) = \prod_{n\geq 0} A_n$ and likewise for B and C, and let $u\colon A_\bullet \to B_\bullet$, $v\colon B_\bullet \to C_\bullet$ be the given morphisms.

Step 1: The two rows are short exact. By definition 4.5.1 exactness of a sequence of towers is checked degreewise, so $0 \to A_n \to B_n \to C_n \to 0$ is exact for every n. A product of short exact sequences of abelian groups is short exact, since kernels, images and surjectivity of a family of maps into a product are all tested coordinatewise (this is the exactness of arbitrary products in R-Mod recorded in lemma 2.1.11). Hence

$$0 \to P(A) \xrightarrow{\prod u_n} P(B) \xrightarrow{\prod v_n} P(C) \to 0$$

is exact, and the same sequence serves as both rows of the diagram below.

Step 2: The squares commute. Take the vertical maps to be d^A, d^B, d^C:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & P(A) & \xrightarrow{\prod u_n} & P(B) & \xrightarrow{\prod v_n} & P(C) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle d^A} & & \downarrow{\scriptstyle d^B} & & \downarrow{\scriptstyle d^C} & & \\ 0 & \longrightarrow & P(A) & \xrightarrow{\prod u_n} & P(B) & \xrightarrow{\prod v_n} & P(C) & \longrightarrow & 0 \end{array}$$

Commutativity of the left square is the computation already made in definition 4.5.2: for $(a_n) \in P(A)$,

$$d^B\big((u_n(a_n))_n\big) = \big(u_n(a_n) - g_{n+1}(u_{n+1}(a_{n+1}))\big)_n = \big(u_n(a_n) - u_n(f_{n+1}(a_{n+1}))\big)_n$$

using $g_{n+1}u_{n+1} = u_n f_{n+1}$, and the right-hand side is $\prod u_n$ applied to $d^A((a_n))$. The right square is the same computation for v.

Step 3: Apply the snake lemma and identify the terms. Read each of the three vertical maps as a complex concentrated in degrees 0 and 1, so that the diagram of Step 2 is a short exact

sequence of two-term complexes. This is exactly the reading carried out in example 2.1.21(1): for a two-term complex $H^0 = \ker$ and $H^1 = \operatorname{coker}$, and $H^n = 0$ otherwise, so the long exact sequence of theorem 2.1.20 has only six nonzero terms,

$$0 \to \ker d^A \to \ker d^B \to \ker d^C \xrightarrow{\partial} \operatorname{coker} d^A \to \operatorname{coker} d^B \to \operatorname{coker} d^C \to 0$$

The connecting map is the one built in the proof of theorem 2.1.20, by the recipe: given $c \in \ker d^C$, choose $b \in P(B)$ with $\prod v_n(b) = c$, note that $d^B(b)$ has zero image in $P(C)$ and so equals $\prod u_n(a)$ for a unique $a \in P(A)$, and set $\partial(c) := [a] \in \operatorname{coker} d^A$. By definition 4.5.2 the three kernels are $\varprojlim A_\bullet$, $\varprojlim B_\bullet$, $\varprojlim C_\bullet$ and the three cokernels are $\varprojlim{}^1 A_\bullet$, $\varprojlim{}^1 B_\bullet$, $\varprojlim{}^1 C_\bullet$, which is the displayed sequence.

Step 4: Naturality. Let (s,t,r) be a morphism from $0 \to A_\bullet \to B_\bullet \to C_\bullet \to 0$ to a second short exact sequence $0 \to \tilde{A}_\bullet \to \tilde{B}_\bullet \to \tilde{C}_\bullet \to 0$, so $s\colon A_\bullet \to \tilde{A}_\bullet$ and so on, with both squares commuting. Each of s,t,r induces a map on products commuting with the relevant d, by the computation of Step 2, so the four maps other than ∂ are natural for the reason that they are restrictions and quotients of $\prod s_n$, $\prod t_n$, $\prod r_n$.

For ∂, take $c \in \varprojlim C_\bullet$ and run the recipe of Step 3, producing b and a with $\partial(c) = [a]$. Applying s gives $\left[\prod s_n(a)\right]$. Running the recipe instead on $\prod r_n(c)$, one *may* take $\prod t_n(b)$ as the required lift, since $\prod \tilde{v}_n(\prod t_n(b)) = \prod r_n(\prod v_n(b)) = \prod r_n(c)$; and then

$$d^{\tilde{B}}\Big(\prod t_n(b)\Big) = \prod t_n\big(d^B(b)\big) = \prod t_n\Big(\prod u_n(a)\Big) = \prod \tilde{u}_n\Big(\prod s_n(a)\Big)$$

so the unique preimage is $\prod s_n(a)$ and $\tilde{\partial}(\prod r_n(c)) = \left[\prod s_n(a)\right]$ as well. The two agree, and the value does not depend on the lift chosen, that being the well-definedness already established for $\tilde{\partial}$ in the target sequence. Hence the square through ∂ commutes, completing the proof.

The name given to $\operatorname{coker} d^A$ in definition 4.5.2 has so far been unearned: nothing above identifies it with a derived functor in the sense of definition 2.3.5 and the right-derived construction recorded alongside it. The identification is genuine, and it is what permits both the notation and the assertion made after theorem 4.5.3 that the sequence terminates.

Proposition 4.5.4: $\varprojlim{}^1$ Is the First Derived Functor of the Limit

The category of towers of abelian groups has enough injectives, and the functor $\varprojlim$ is left exact. Writing $R^n \varprojlim$ for its right derived functors,

$$R^0 \varprojlim \cong \varprojlim, \qquad R^1 \varprojlim \cong \varprojlim{}^1, \qquad R^n \varprojlim = 0 \text{ for } n \geq 2$$

***Proof Key ideas*:**
Two ingredients are used here and not proved: that the category of towers has enough injectives, and that the right derived functors of $\varprojlim$ are computed by the two-term complex

$$P(A) \xrightarrow{d^A} P(A)$$

placed in degrees 0 and 1. Both are established in [23, §3.5]. The deferral is deliberate rather than incidental: constructing the injectives is long, and nothing in this book or in the books citing

this section uses more than the two-term description, which definition 4.5.2 and theorem 4.5.3 give on their own.

Granting the two ingredients, the rest is a reading-off. Left exactness of $\varprojlim$ is the left half of theorem 4.5.3, proved above without any of this. The displayed complex has $H^0 = \ker d^A = \varprojlim A_\bullet$ and $H^1 = \operatorname{coker} d^A = \varprojlim{}^1 A_\bullet$ by definition 4.5.2, and it is concentrated in degrees 0 and 1, so $H^n = 0$ for $n \geq 2$. These three groups are the values of $R^0\varprojlim$, $R^1\varprojlim$ and $R^n\varprojlim$ for $n \geq 2$, which is the statement.

With the sequence in hand the practical question is when the obstruction vanishes. The condition is not that the transition maps be surjective, which would be too strong to be useful, but that the images they produce inside any fixed level eventually stop shrinking.

Definition 4.5.5: Mittag-Leffler Condition

A tower $A_\bullet$ satisfies the *Mittag-Leffler condition* if for every n the decreasing chain of subgroups

$$A_n = \operatorname{im} f_{n,n} \supseteq \operatorname{im} f_{n,n+1} \supseteq \operatorname{im} f_{n,n+2} \supseteq \cdots$$

is eventually constant: there is an $m_0 \geq n$ with $\operatorname{im} f_{n,m} = \operatorname{im} f_{n,m_0}$ for all $m \geq m_0$. The common value $A'_n := \bigcap_{m \geq n} \operatorname{im} f_{n,m}$ is the *stable image* in level n.

The condition is deliberately weak. It permits the transition maps to be far from surjective, and it permits the stable image to be a proper subgroup of A_n; it forbids only an infinite strictly decreasing chain of images inside a single level. A tower whose transition maps are all surjective satisfies it with $A'_n = A_n$, and a tower whose transition maps are eventually zero in each level satisfies it with $A'_n = 0$.

Theorem 4.5.6: Mittag-Leffler Vanishing

If a tower $A_\bullet$ of abelian groups satisfies the Mittag-Leffler condition, then $\varprojlim{}^1 A_\bullet = 0$.

***Proof*:**

The claim is that d^A is surjective. Two special cases are treated first, and the general case reduces to them.

Step 1: Surjective transition maps. Suppose every f_{n+1} is surjective, and let $(b_n) \in P(A)$ be given. Set $a_0 := 0$, and having chosen a_n, use surjectivity of f_{n+1} to choose $a_{n+1} \in A_{n+1}$ with $f_{n+1}(a_{n+1}) = a_n - b_n$. Then

$$a_n - f_{n+1}(a_{n+1}) = a_n - (a_n - b_n) = b_n$$

for every n, so $d^A((a_n)) = (b_n)$ and d^A is surjective.

Step 2: Transition maps eventually zero in each level. Suppose instead that for each n there is an $m_1 \geq n$ with $f_{n,m} = 0$ for all $m \geq m_1$. Given $(b_n) \in P(A)$, the sum

$$a_n := \sum_{k \geq n} f_{n,k}(b_k)$$

has only finitely many nonzero terms and so defines an element of A_n. Since $f_{n+1} \circ f_{n+1,k} = f_{n,k}$

for $k \geq n+1$,

$$f_{n+1}(a_{n+1}) = \sum_{k \geq n+1} f_{n,k}(b_k) = a_n - f_{n,n}(b_n) = a_n - b_n$$

so again $a_n - f_{n+1}(a_{n+1}) = b_n$ and d^A is surjective.

Step 3: The general Mittag-Leffler case. Let A'_n be the stable image of definition 4.5.5, and let $m_0(n)$ be an index with $\mathrm{im} f_{n,m} = A'_n$ for all $m \geq m_0(n)$. The subgroups A'_n form a subtower: for $m \geq n+1$ one has $f_{n+1}(\mathrm{im} f_{n+1,m}) = \mathrm{im} f_{n,m}$, so $f_{n+1}(A'_{n+1}) \subseteq \bigcap_{m \geq n+1} \mathrm{im} f_{n,m} = A'_n$.

That subtower has surjective transition maps. Let $x \in A'_n$ and choose $m \geq \max\{m_0(n), m_0(n+1)\}$. Since $x \in \mathrm{im} f_{n,m}$ there is $z \in A_m$ with $f_{n,m}(z) = x$; put $y := f_{n+1,m}(z)$. Then $y \in \mathrm{im} f_{n+1,m} = A'_{n+1}$ because $m \geq m_0(n+1)$, and $f_{n+1}(y) = f_{n+1}(f_{n+1,m}(z)) = f_{n,m}(z) = x$. So $f_{n+1} \colon A'_{n+1} \to A'_n$ is onto, and Step 1 gives $\varprojlim{}^1 A'_\bullet = 0$.

The quotient tower $(A_n/A'_n)_n$ has transition maps that are eventually zero in each level: for $m \geq m_0(n)$ the image of $A_m/A'_m \to A_n/A'_n$ is $(\mathrm{im} f_{n,m} + A'_n)/A'_n = A'_n/A'_n = 0$. So Step 2 gives $\varprojlim{}^1(A_\bullet/A'_\bullet) = 0$.

Now apply theorem 4.5.3 to the short exact sequence of towers $0 \to A'_\bullet \to A_\bullet \to A_\bullet/A'_\bullet \to 0$, which is exact degreewise by construction. Its last three terms read

$$\varprojlim{}^1 A'_\bullet \to \varprojlim{}^1 A_\bullet \to \varprojlim{}^1 (A_\bullet/A'_\bullet) \to 0$$

with the two outer groups zero, so $\varprojlim{}^1 A_\bullet = 0$, completing the proof.

Corollary 4.5.7: Surjective Towers Have Vanishing $\varprojlim^1$

If every transition map of a tower $A_\bullet$ is surjective, then $\varprojlim{}^1 A_\bullet = 0$. Consequently, for a short exact sequence of towers $0 \to A_\bullet \to B_\bullet \to C_\bullet \to 0$ in which $A_\bullet$ has surjective transition maps, the sequence of limits

$$0 \to \varprojlim A_\bullet \to \varprojlim B_\bullet \to \varprojlim C_\bullet \to 0$$

is exact.

***Proof*:**
A surjective tower satisfies the Mittag-Leffler condition with $A'_n = A_n$, so theorem 4.5.6 applies; equivalently this is Step 1 of that proof on its own. Substituting $\varprojlim{}^1 A_\bullet = 0$ into theorem 4.5.3 makes the connecting map ∂ land in the zero group, so the first four terms of the six-term sequence give the displayed short exact sequence, as we sought to show.

The corollary is the form in which the result is usually met, and it is the statement proved directly, without the surrounding theory, in [7, Limits and Exactness for Inverse Systems], where it is what makes completion along a surjective system an exact operation. The theory developed here names the group that appears once the surjectivity hypothesis is dropped: the cokernels left unlabelled in that argument are the $\varprojlim{}^1$ terms.

Example 4.5.8: The Tower of Residue Rings $\mathbb{Z}/p^n\mathbb{Z}$

Fix a prime p and let $T_\bullet$ be the tower $T_n = \mathbb{Z}/p^n\mathbb{Z}$ with transition maps the natural reductions $\mathbb{Z}/p^{n+1}\mathbb{Z} \twoheadrightarrow \mathbb{Z}/p^n\mathbb{Z}$ (so $T_0 = 0$). A coherent family is a sequence of residues agreeing under reduction, which is precisely a p-adic integer, so

$$\varprojlim T_\bullet = \mathbb{Z}_p$$

Every transition map is surjective, so corollary 4.5.7 gives $\varprojlim^1 T_\bullet = 0$. This is the typical situation: the obstruction is absent and the limit behaves as expected.

The next example shows that the obstruction is not always absent, and that it can be large. It is worth working through, because a reader who meets only towers with surjective transition maps will reasonably conclude that $\varprojlim^1$ is a formality.

Example 4.5.9: A Tower With Nonvanishing $\varprojlim^1$

Keep p fixed and let $S_\bullet$ be the tower of subgroups $S_n = p^n\mathbb{Z} \subseteq \mathbb{Z}$, with transition maps the inclusions $p^{n+1}\mathbb{Z} \hookrightarrow p^n\mathbb{Z}$. Rescaling by $p^n\mathbb{Z} \cong \mathbb{Z}$ identifies $S_\bullet$ with the tower

$$\mathbb{Z} \xleftarrow{p} \mathbb{Z} \xleftarrow{p} \mathbb{Z} \xleftarrow{p} \cdots$$

Mittag-Leffler fails: inside $S_n = p^n\mathbb{Z}$ the image $\mathrm{im} f_{n,m}$ is $p^m\mathbb{Z}$, and these subgroups shrink strictly as m grows, so the chain in definition 4.5.5 is never constant. The limit vanishes, since $\varprojlim S_\bullet \subseteq \bigcap_{n\geq 0} p^n\mathbb{Z} = 0$.

To compute $\varprojlim^1 S_\bullet$, let $\underline{\mathbb{Z}}$ denote the constant tower with $\underline{\mathbb{Z}}_n = \mathbb{Z}$ and all transition maps the identity, and let $T_\bullet$ be the tower of example 4.5.8. The quotient maps $\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$ assemble into a short exact sequence of towers

$$0 \to S_\bullet \to \underline{\mathbb{Z}} \to T_\bullet \to 0$$

exact in each level because $p^n\mathbb{Z}$ is the kernel of $\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$, and compatible with the transition maps by construction. The identity transition maps of $\underline{\mathbb{Z}}$ are surjective, so $\varprojlim \underline{\mathbb{Z}} = \mathbb{Z}$ and $\varprojlim^1 \underline{\mathbb{Z}} = 0$ by corollary 4.5.7; and $\varprojlim T_\bullet = \mathbb{Z}_p$ with $\varprojlim^1 T_\bullet = 0$ by example 4.5.8. Feeding all of this into theorem 4.5.3 leaves

$$0 \to 0 \to \mathbb{Z} \to \mathbb{Z}_p \xrightarrow{\partial} \varprojlim{}^1 S_\bullet \to 0 \to 0 \to 0$$

where the map $\mathbb{Z} \to \mathbb{Z}_p$ is the canonical inclusion. Exactness identifies ∂ as a surjection whose kernel is the image of $\mathbb{Z}$, so

$$\varprojlim{}^1 S_\bullet \cong \mathbb{Z}_p/\mathbb{Z}$$

This is nonzero, and not by a small margin: $\mathbb{Z}_p$ is uncountable while $\mathbb{Z}$ is countable, so the quotient is uncountable. The failure of $\varprojlim$ to be exact on this sequence is exactly the failure of the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}_p$ to be onto.

Exercise 4.5.1

1. Let A be an abelian group and $\underline{A}$ the constant tower with every level A and every transition map the identity. Compute $\varprojlim \underline{A}$ and $\varprojlim^1 \underline{A}$ directly from definition 4.5.2, without invoking corollary 4.5.7, by exhibiting a solution of $d^{\underline{A}}((a_n)) = (b_n)$ for an arbitrary (b_n).

2. Show that a tower in which every A_n is a finite group satisfies the Mittag-Leffler condition, and conclude that $\varprojlim^1 A_\bullet = 0$. Where does the argument fail for a tower of finitely generated abelian groups?

3. Give a morphism of towers $u\colon B_\bullet \to C_\bullet$ that is surjective in every level but for which $\varprojlim B_\bullet \to \varprojlim C_\bullet$ is not surjective. Name the group in theorem 4.5.3 that receives the resulting obstruction, and say which tower it is computed from.

4. Let $A_\bullet$ be a tower and let $A^{(2)}_\bullet$ be the tower $A_0 \leftarrow A_2 \leftarrow A_4 \leftarrow \cdots$ obtained by keeping the even levels and composing transition maps in pairs. Show that $\varprojlim A_\bullet \cong \varprojlim A^{(2)}_\bullet$ and that $A_\bullet$ satisfies the Mittag-Leffler condition if and only if $A^{(2)}_\bullet$ does.

5. Let $A_\bullet$ be the tower of subgroups $A_n = (n!)\mathbb{Z} \subseteq \mathbb{Z}$ with the inclusions as transition maps, so that rescaling identifies it with $\mathbb{Z} \xleftarrow{1} \mathbb{Z} \xleftarrow{2} \mathbb{Z} \xleftarrow{3} \cdots$, the n-th transition map being multiplication by n. Show that Mittag-Leffler fails and that $\varprojlim A_\bullet = 0$, then run the argument of example 4.5.9 with $\mathbb{Z}/(n!)\mathbb{Z}$ in place of $\mathbb{Z}/p^n\mathbb{Z}$ to compute $\varprojlim^1 A_\bullet$. The limit of the quotient tower is the profinite completion $\widehat{\mathbb{Z}}$ rather than $\mathbb{Z}_p$; say which property of the sequence $(n!)$ is responsible, and give $\varprojlim^1 A_\bullet$ itself.

6. In the situation of example 4.5.9, make the connecting map $\partial\colon \mathbb{Z}_p \to \varprojlim^1 S_\bullet$ explicit by running the construction of ∂ from theorem 4.5.3: given a p-adic integer presented as a coherent family of residues, produce a representative in $\prod_n S_n$ of the class it maps to, and check that the class does not depend on the lifts chosen.

5

Unbounded Derived Categories and t-Structures

The derived functors of chapter 3 are defined on complexes bounded on one side, bounded below for RF and for the second variable of $R\mathrm{Hom}$ and bounded above for LF and $\otimes^L$, because the injective and projective resolutions that compute them are bounded on that side. For a bounded-below complex I of injectives the map $Q\colon \mathrm{Hom}_{\mathrm{K}}(X, I) \to \mathrm{Hom}_D(X, I)$ is bijective for every complex X (lemma 3.4.4(2)), and without the bound this fails. The complex of $\mathbb{Z}/4$-modules with $\mathbb{Z}/4$ in every degree and every differential multiplication by 2 consists of injectives and is exact, yet its identity is not null-homotopic (example 5.1.7). The K-injective and K-flat complexes of Spaltenstein [21] are resolutions that need no bound. With them the right derived functor of an additive functor out of a Grothendieck abelian category, and $R\mathrm{Hom}$ and $\otimes^L$ over a commutative ring, extend to unbounded complexes, and the tensor-Hom adjunction of theorem 3.4.19 holds for the extended functors.

The second half answers the question of box 3.2.13, what in $D(\mathcal{A})$ recovers $\mathcal{A}$. The canonical truncations of definition 2.1.12 descend to $D(\mathcal{A})$. There they cut out the full subcategories of complexes with cohomology in degrees at most 0 and at least 0, and every complex sits in a distinguished triangle whose first vertex lies in the first subcategory and whose third vertex has cohomology in degrees at least 1. The complexes in both subcategories, those with cohomology concentrated in degree 0, form a full subcategory equivalent to $\mathcal{A}$. The axioms satisfied by the two halves define the t-structures of Beilinson, Bernstein and Deligne [3], and the heart of every t-structure is an abelian category.

5.1 Grothendieck Abelian Categories and K-Injective Resolutions

A bounded-below complex of injectives I computes the morphisms into it in the derived category: by lemma 3.4.4(2) the map $Q\colon \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(X, I) \to \mathrm{Hom}_{D(\mathcal{A})}(X, I)$ is bijective for every complex X. Without the bound this fails already for a complex of injective $\mathbb{Z}/4$-modules (example 5.1.7), and the same terms $\mathbb{Z}/4$ carry a second complex for which it holds (example 5.1.10). The replacement for the bound is therefore a condition on the complex as a whole, K-injectivity [21]. Over a Grothendieck abelian category, such as the category of modules over a ring or of sheaves of modules on a ringed site, every complex has a K-injective resolution. With these resolutions the Hom-classes of $D(\mathcal{A})$ are sets, and every additive functor has a right derived functor on all of $D(\mathcal{A})$, which for a left exact functor agrees on bounded-below complexes with the total derived functor of section 3.3.

Grothendieck's construction of injective objects, and its extension to K-injective complexes, produce them as colimits of transfinite sequences of monomorphisms, and both arguments need colimits over directed index sets to preserve exactness ([22, Tag 079H], [22, Tag 079P]).

Definition 5.1.1: Directed System and Its Colimit

A *directed poset* is a nonempty partially ordered set I in which any two elements have a common upper bound. Let $\mathcal{C}$ be a category. A *directed system* over I in $\mathcal{C}$ is a family of objects $(A_i)_{i\in I}$ with morphisms $\varphi_{ij}\colon A_i \to A_j$ for $i \le j$, such that $\varphi_{ii} = \mathrm{id}$ and $\varphi_{jk}\varphi_{ij} = \varphi_{ik}$ whenever $i \le j \le k$. A *morphism* $f\colon (A_i, \varphi_{ij}) \to (B_i, \psi_{ij})$ of directed systems over I is a family of morphisms $f_i\colon A_i \to B_i$ with $f_j\varphi_{ij} = \psi_{ij}f_i$ for $i \le j$.

A *colimit* of (A_i, φ_{ij}) is an object $\varinjlim_i A_i$ with morphisms $\lambda_i\colon A_i \to \varinjlim_i A_i$ satisfying $\lambda_j\varphi_{ij} = \lambda_i$, such that for every object T and every family $t_i\colon A_i \to T$ with $t_j\varphi_{ij} = t_i$ there is a unique $t\colon \varinjlim_i A_i \to T$ with $t\lambda_i = t_i$ for all i.

The simplest directed systems are increasing unions, and for them the colimit is the union.

Example 5.1.2: An Increasing Union of Submodules

Let M be an R-module and $(M_i)_{i\in I}$ a family of submodules indexed by a directed poset, with $M_i \subseteq M_j$ for $i \le j$ and $\bigcup_i M_i = M$. With the inclusions as φ_{ij} this is a directed system, and M with the inclusions $\lambda_i\colon M_i \to M$ is its colimit. Given a compatible family $t_i\colon M_i \to T$, define $t(x) := t_i(x)$ for any i with $x \in M_i$. This does not depend on i: if $x \in M_i \cap M_j$, choose $k \ge i, j$, and compatibility gives $t_i(x) = t_k(x) = t_j(x)$. Two elements x, y lie in a common M_k, so $t(rx + y) = t_k(rx + y) = rt(x) + t(y)$ and t is a homomorphism, with $t\lambda_i = t_i$. It is the only one, since every element of M lies in some M_i.

For instance $\mathbb{Q} = \bigcup_{n\ge 0} \frac{1}{n!}\mathbb{Z}$, an increasing union since $\frac{1}{n!}\mathbb{Z} \subseteq \frac{1}{(n+1)!}\mathbb{Z}$, and every p/q with $q \ge 1$ equals $p(q-1)!/q!$. Multiplication by $n!$ identifies $\frac{1}{n!}\mathbb{Z}$ with $\mathbb{Z}$, and under these identifications the inclusion $\frac{1}{n!}\mathbb{Z} \subseteq \frac{1}{(n+1)!}\mathbb{Z}$ becomes multiplication by $n+1$ on $\mathbb{Z}$. So $\mathbb{Q}$ is the colimit of $\mathbb{Z} \xrightarrow{1} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{3} \cdots$, with $\lambda_n(a) = a/n!$.

A morphism f of directed systems induces $\varinjlim f\colon \varinjlim_i A_i \to \varinjlim_i B_i$, the unique morphism with

$(\varinjlim f)\lambda_i^A = \lambda_i^B f_i$. It exists because the family $\lambda_i^B f_i$ is compatible: $\lambda_j^B f_j \varphi_{ij} = \lambda_j^B \psi_{ij} f_i = \lambda_i^B f_i$. In an abelian category with all set-indexed direct sums every directed system has a colimit, the cokernel of the morphism

$$\bigoplus_{i \le j} A_i \longrightarrow \bigoplus_{i \in I} A_i$$

whose restriction to the summand indexed by the pair (i, j) is $e_i - e_j\varphi_{ij}$, where e_i is the inclusion of the summand A_i. A morphism $t\colon \bigoplus_i A_i \to T$ vanishes on this morphism exactly when $te_i = te_j\varphi_{ij}$ for all $i \le j$, that is, when $(te_i)_i$ is a compatible family. So the universal properties of the direct sum and of the cokernel combine into that of the colimit, with λ_i the composite of e_i and the projection onto the cokernel.

Proposition 5.1.3: Directed Colimits of Modules Are Exact

Let R be a ring, I a directed poset and (M_i, φ_{ij}) a directed system of R-modules over I.

1. The colimit $\varinjlim_i M_i$ exists. Every element of it is $\lambda_i(x)$ for some $i \in I$ and $x \in M_i$, and $\lambda_i(x) = 0$ if and only if $\varphi_{ik}(x) = 0$ for some $k \ge i$.

2. Let $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ be morphisms of directed systems of R-modules over I such that $0 \to A_i \to B_i \to C_i \to 0$ is exact for every i. Then the sequence

$$0 \to \varinjlim_i A_i \xrightarrow{\varinjlim f} \varinjlim_i B_i \xrightarrow{\varinjlim g} \varinjlim_i C_i \to 0$$

   is exact.

***Proof*:**

1. The category of R-modules has direct sums, so the colimit is the quotient $(\bigoplus_i M_i)/N$, where N is the submodule generated by the elements $e_i(x) - e_j(\varphi_{ij}x)$ for $i \le j$ and $x \in M_i$. An element of the quotient is a finite sum $\sum_{r=1}^{s} \lambda_{i_r}(x_r)$. As I is directed, induction on s gives $k \in I$ with $k \ge i_1, \dots, i_s$, and then $\lambda_{i_r}(x_r) = \lambda_k\varphi_{i_r k}(x_r)$, so the sum is $\lambda_k\big(\sum_r \varphi_{i_r k}(x_r)\big)$. If $\varphi_{ik}(x) = 0$ then $\lambda_i(x) = \lambda_k\varphi_{ik}(x) = 0$.

   Conversely let $\lambda_i(x) = 0$, that is, $e_i(x) \in N$. The generators are closed under multiplication by scalars, since $r\big(e_a(y) - e_b(\varphi_{ab}y)\big) = e_a(ry) - e_b(\varphi_{ab}(ry))$, so

$$e_i(x) = \sum_{r=1}^{s} \big(e_{a_r}(y_r) - e_{b_r}(\varphi_{a_r b_r} y_r)\big)$$

   with $a_r \le b_r$. Let J be the finite set $\{i, a_1, b_1, \dots, a_s, b_s\}$ and choose an upper bound k of J. Both sides of the equation lie in the summand $\bigoplus_{j \in J} M_j$, on which $\psi(e_j(z)) := \varphi_{jk}(z)$ defines a homomorphism ψ to M_k. For $a \le b$ in J, $\psi\big(e_a(y) - e_b(\varphi_{ab}y)\big) = \varphi_{ak}(y) - \varphi_{bk}\varphi_{ab}(y) = 0$, so applying ψ to the equation gives $\varphi_{ik}(x) = 0$.

2. Write φ, ψ, χ for the transition maps of A, B, C, and use (1) to write every element of a colimit as $\lambda_i(x)$ for some i and x. If $(\varinjlim f)(\lambda_i a) = \lambda_i(f_i a) = 0$, then (1) gives $k \ge i$ with $0 = \psi_{ik}f_i(a) = f_k\varphi_{ik}(a)$. As f_k is injective, $\varphi_{ik}(a) = 0$ and $\lambda_i(a) = 0$, so $\varinjlim f$ is injective.

Every $\lambda_i(c)$ is $\lambda_i(g_i b) = (\varinjlim g)(\lambda_i b)$ for some $b \in B_i$, so $\varinjlim g$ is surjective. The composite $(\varinjlim g)(\varinjlim f)$ satisfies $(\varinjlim g)(\varinjlim f)\lambda_i = \lambda_i g_i f_i = 0$ for every i, so it is 0 by the uniqueness in the universal property. Finally let $(\varinjlim g)(\lambda_i b) = \lambda_i(g_i b) = 0$. By (1) there is $k \geq i$ with $0 = \chi_{ik} g_i(b) = g_k \psi_{ik}(b)$, so $\psi_{ik}(b) = f_k(a)$ for some $a \in A_k$, by exactness at B_k. Then $\lambda_i(b) = \lambda_k \psi_{ik}(b) = \lambda_k f_k(a) = (\varinjlim f)(\lambda_k a)$, which is exactness at $\varinjlim_i B_i$ and completes the proof.

Directedness enters (2) through (1): a common upper bound k places two elements of a colimit in a single M_k, where they can be compared. The three properties of R-Mod that Grothendieck's axioms abstract are arbitrary direct sums, the exactness of proposition 5.1.3(2), and an object detecting nonzero morphisms. For the last, a homomorphism $f\colon M \to N$ is nonzero exactly when $f(m) \neq 0$ for some m, that is, when $f \circ u \neq 0$ for the map $u\colon R \to M$, $r \mapsto rm$.

Definition 5.1.4: Grothendieck Abelian Category

An abelian category $\mathcal{A}$ is a *Grothendieck abelian category* if it satisfies the following.

(AB3) Every family $(A_t)_{t \in T}$ of objects indexed by a set has a direct sum $\bigoplus_t A_t$.

(AB5) Colimits over directed posets are exact: for every directed poset I and all morphisms of directed systems $0 \to A \to B \to C \to 0$ over I such that $0 \to A_i \to B_i \to C_i \to 0$ is exact for every i, the sequence $0 \to \varinjlim_i A_i \to \varinjlim_i B_i \to \varinjlim_i C_i \to 0$ is exact.

(G) There is a *generator*: an object U such that for every nonzero morphism $f\colon X \to Y$ there is a morphism $u\colon U \to X$ with $fu \neq 0$.

The colimits in (AB5) exist by (AB3), as the cokernel of the morphism $\bigoplus_{i \leq j} A_i \to \bigoplus_{i \in I} A_i$ with components $e_i - e_j \varphi_{ij}$. This is the definition of [22, Tag 079B], whose discussion of (AB5) states the exactness condition for systems over a directed set, as here. The generator there is required to satisfy the following: for every subobject $N \subseteq M$ with $N \neq M$ some morphism $U \to M$ does not factor through N. The two forms agree. Given such N, apply (G) to the nonzero quotient $f\colon M \to M/N$, and note that $u\colon U \to M$ factors through $N = \ker f$ exactly when $fu = 0$. Given a nonzero $f\colon X \to Y$, apply the subobject form to $\ker f \neq X$.

Example 5.1.5: Module and Sheaf Categories

1. For a ring R the category R-Mod is a Grothendieck abelian category. Direct sums give (AB3), proposition 5.1.3(2) is (AB5), and R is a generator: if $f\colon M \to N$ is nonzero, choose m with $f(m) \neq 0$ and let $u\colon R \to M$ be $r \mapsto rm$. Then $fu(1) = f(m) \neq 0$.

2. For a ringed site $(\mathcal{C}, \mathcal{O})$, the category of $\mathcal{O}$-modules is a Grothendieck abelian category [22, Tag 07A5]. For the constant sheaf of rings $\mathbb{Z}$ its objects are the sheaves of abelian groups on $\mathcal{C}$, and for $\mathcal{C}$ the open sets of a topological space they are the sheaves of abelian groups on the space.

The axioms of definition 5.1.4 are Grothendieck's, from the paper in which he proved that injective objects exist in this generality [22, Tag 079A]. In the proof the generator plays the role that the ring plays in Baer's criterion ([8, Baer's Criterion]), and its subobjects the role of the ideals.

Theorem 5.1.6: Grothendieck Abelian Categories Have Enough Injectives

Let $\mathcal{A}$ be a Grothendieck abelian category. There are a functor $E\colon \mathcal{A} \to \mathcal{A}$ and a natural transformation $\upsilon\colon \mathrm{id} \to E$ such that for every object M the morphism $\upsilon_M\colon M \to E(M)$ is a monomorphism and $E(M)$ is injective. In particular $\mathcal{A}$ has enough injectives (definition 2.2.3).

The proof is a transfinite construction that uses nothing from homological algebra beyond (AB5) and the generator, and it is deferred to [22, Tag 079H].

***Proof Key ideas*:**
Fix a generator U. The replacement for Baer's criterion is [22, Tag 079G]: an object is injective if every morphism into it from a subobject $N \subseteq U$ extends to U. For an object M, form the pushout $\mathbf{M}(M)$ of $M \leftarrow \bigoplus N \to \bigoplus U$, the sums running over all pairs of a subobject $N \subseteq U$ and a morphism $\varphi\colon N \to M$. Then every such φ extends to $U \to \mathbf{M}(M)$, and $M \to \mathbf{M}(M)$ is a monomorphism. Iterate transfinitely, taking colimits at limit ordinals, up to an ordinal whose cofinality exceeds the cardinality of the set of subobjects of U. A morphism from a subobject of U into the final stage then factors through an earlier stage, by (AB5) and that cardinality bound [22, Tag 079F], and so extends to U. The criterion makes the final stage injective, and every step is functorial in M, which gives E and υ as claimed.

For R-Mod, through example 5.1.5(1), the theorem gives enough injectives over an arbitrary ring, together with injective embeddings compatible with homomorphisms.

When the category is fixed, $\mathrm{Hom}_{\mathrm{K}}$ and Hom_D abbreviate $\mathrm{Hom}_{\mathrm{K}(\mathcal{A})}$ and $\mathrm{Hom}_{D(\mathcal{A})}$. Recall from lemma 2.1.19 that a complex is acyclic, meaning that its cohomology vanishes, exactly when it is exact.

Example 5.1.7: An Exact Complex of Injectives That Is Not Contractible
Let I be the complex of $\mathbb{Z}/4$-modules with $I^n = \mathbb{Z}/4$ for every $n \in \mathbb{Z}$ and every differential multiplication by 2.

1. $\mathbb{Z}/4$ is an injective $\mathbb{Z}/4$-module. By [8, Baer's Criterion](1) it suffices to extend homomorphisms from the ideals 0, $2\mathbb{Z}/4$ and $\mathbb{Z}/4$ to $\mathbb{Z}/4$, and only $2\mathbb{Z}/4 = \{0, 2\}$ needs an argument. A homomorphism $g\colon 2\mathbb{Z}/4 \to \mathbb{Z}/4$ satisfies $2g(2) = g(4) = g(0) = 0$, so $g(2) \in \{0, 2\}$. If $g(2) = 0$ then g extends by 0, and if $g(2) = 2$ it extends by the identity of $\mathbb{Z}/4$.
2. I is exact: the kernel and the image of multiplication by 2 on $\mathbb{Z}/4$ are both $\{0, 2\}$.
3. id_I is not null-homotopic. Every $\mathbb{Z}/4$-linear endomorphism of $\mathbb{Z}/4$ is multiplication by its value at 1, so a family $s^n\colon I^n \to I^{n-1}$ consists of multiplications by elements $a_n \in \mathbb{Z}/4$. Then $d^{n-1}s^n + s^{n+1}d^n$ is multiplication by $2a_n + 2a_{n+1}$, which sends 1 into $\{0, 2\}$ and so is never the identity.

So id_I is a nonzero element of $\mathrm{Hom}_{\mathrm{K}(\mathbb{Z}/4\text{-Mod})}(I, I)$. But I is exact, hence isomorphic to 0 in $D(\mathbb{Z}/4\text{-Mod})$ by proposition 3.2.12(1), and $\mathrm{Hom}_{D(\mathbb{Z}/4\text{-Mod})}(I, I) = 0$. The map $Q\colon \mathrm{Hom}_{\mathrm{K}}(I, I) \to \mathrm{Hom}_D(I, I)$ is not injective, and the conclusion of lemma 3.4.4(2) fails for this complex of injectives.

The defect in the example is a nonzero homotopy class of maps from an acyclic complex into I, namely id_I.

Definition 5.1.8: K-Injective Complex

Let $\mathcal{A}$ be an abelian category. A complex $I \in \mathrm{Ch}(\mathcal{A})$ is *K-injective* if $\mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(X, I) = 0$ for every acyclic complex X. A *K-injective resolution* of a complex X is a quasi-isomorphism $X \to I$ with I K-injective.

This is [22, Tag 070H]. The complex of example 5.1.7 is not K-injective, since id_I is a nonzero element of $\mathrm{Hom}_{\mathrm{K}}(I, I)$ with I acyclic.

Lemma 5.1.9: Shifts of K-Injective Complexes

Let $\mathcal{A}$ be an abelian category and $I \in \mathrm{Ch}(\mathcal{A})$ a K-injective complex. Then $I[n]$ is K-injective for every $n \in \mathbb{Z}$.

***Proof*:**
Let X be an acyclic complex. By (3.1) the shift $[-n]$ is an automorphism of $\mathrm{K}(\mathcal{A})$ inverse to $[n]$, so $f \mapsto f[-n]$ is a bijection from $\mathrm{Hom}_{\mathrm{K}}(X, I[n])$ to $\mathrm{Hom}_{\mathrm{K}}(X[-n], I)$. The complex $X[-n]$ is acyclic, since $H^i(X[-n]) = H^{i-n}(X) = 0$ by (3.1), and I is K-injective, so $\mathrm{Hom}_{\mathrm{K}}(X[-n], I) = 0$. Hence $\mathrm{Hom}_{\mathrm{K}}(X, I[n]) = 0$, as we sought to show.

By lemma 3.4.3(1), $H^n\mathrm{Hom}^\bullet(X, I) = \mathrm{Hom}_{\mathrm{K}}(X, I[n])$, so with the lemma I is K-injective exactly when the Hom complex $\mathrm{Hom}^\bullet(X, I)$ is acyclic for every acyclic X. This is the definition of [21, p. 1.1], which credits it to J. Bernstein.

Example 5.1.10: Complexes of Injectives With Zero Differential
Let $(E_n)_{n\in\mathbb{Z}}$ be injective objects of $\mathcal{A}$ and E the complex with $E^n = E_n$ and every differential zero. For every complex X,

$$\mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(X, E) \cong \prod_{n\in\mathbb{Z}} \mathrm{Hom}_{\mathcal{A}}\big(H^n(X), E_n\big)$$

and in particular E is K-injective.

Write $Z^n = \ker d_X^n$ and $B^n = \mathrm{im} d_X^{n-1}$. A chain map $f\colon X \to E$ is a family $f^n\colon X^n \to E_n$ with $f^{n+1}d_X^n = 0 \circ f^n = 0$, so each f^n vanishes on B^n, and the chain maps are $\prod_n \mathrm{Hom}(X^n/B^n, E_n)$. A null-homotopy s has $f^n = 0 \circ s^n + s^{n+1}d_X^n = s^{n+1}d_X^n$ with $s^{n+1}\colon X^{n+1} \to E_n$. The differential d_X^n factors as $X^n \to X^n/Z^n \cong B^{n+1} \hookrightarrow X^{n+1}$, and since E_n is injective, $\mathrm{Hom}(X^{n+1}, E_n) \to \mathrm{Hom}(B^{n+1}, E_n)$ is surjective (definition 2.2.1). So the maps of the form $s^{n+1}d_X^n$ are exactly those factoring through $X^n \to X^n/Z^n$, that is, $\mathrm{Hom}(X^n/Z^n, E_n)$. Applying the exact functor $\mathrm{Hom}(-, E_n)$ to

$$0 \to H^n(X) = Z^n/B^n \to X^n/B^n \to X^n/Z^n \to 0$$

identifies $\mathrm{Hom}(X^n/B^n, E_n)/\mathrm{Hom}(X^n/Z^n, E_n)$ with $\mathrm{Hom}(H^n(X), E_n)$, by restriction. Over $\mathbb{Z}/4$ with every $E_n = \mathbb{Z}/4$, injective by example 5.1.7(1), this complex is K-injective, while the complex of example 5.1.7 has the same terms and differential 2 and is not.

Proposition 5.1.11: K-Injective Complexes Compute Morphisms in $D(\mathcal{A})$

Let $\mathcal{A}$ be an abelian category and $I \in \mathrm{Ch}(\mathcal{A})$. The following are equivalent.

1. I is K-injective.
2. For every quasi-isomorphism $s\colon X' \to X$ the map $s^*\colon \mathrm{Hom}_{\mathrm{K}}(X, I) \to \mathrm{Hom}_{\mathrm{K}}(X', I)$, $f \mapsto fs$, is bijective.
3. For every complex X the map $Q\colon \mathrm{Hom}_{\mathrm{K}}(X, I) \to \mathrm{Hom}_{D}(X, I)$ is bijective.

***Proof*:**

(1) implies (2). By theorem 3.1.12 the mapping cone triangle $X' \xrightarrow{s} X \xrightarrow{v} MC(s) \xrightarrow{w} X'[1]$ is distinguished in $\mathrm{K}(\mathcal{A})$, and by (TR2) so is $\big(-w[-1], s, v\big)$ on $(MC(s)[-1], X', X)$. The complex $MC(s)$ is acyclic by proposition 2.1.26, and so is $MC(s)[-1]$. The functor $\mathrm{Hom}_{\mathrm{K}}(-, I)$ is cohomological by proposition 3.1.8, and applied to the two triangles it gives sequences

$$\mathrm{Hom}_{\mathrm{K}}(MC(s), I) \xrightarrow{v^*} \mathrm{Hom}_{\mathrm{K}}(X, I) \xrightarrow{s^*} \mathrm{Hom}_{\mathrm{K}}(X', I) \xrightarrow{(-w[-1])^*} \mathrm{Hom}_{\mathrm{K}}(MC(s)[-1], I)$$

exact at $\mathrm{Hom}_{\mathrm{K}}(X, I)$ and at $\mathrm{Hom}_{\mathrm{K}}(X', I)$. The two outer groups vanish because I is K-injective, so s^* is injective and surjective.

(2) implies (3). By theorem 3.2.4 a morphism $X \to I$ of $D(\mathcal{A})$ is $Q(f)Q(s)^{-1}$ for a roof (X', s, f), with $s\colon X' \to X$ a quasi-isomorphism. By (2), $f = gs$ in $\mathrm{K}(\mathcal{A})$ for some $g\colon X \to I$, and then $Q(f)Q(s)^{-1} = Q(g)$, so Q is surjective. The functor Q is additive by theorem 3.2.7. If $Q(h) = 0$, then proposition 3.2.12(2) gives a quasi-isomorphism $t\colon X'' \to X$ with ht null-homotopic, that is, $t^*(h) = 0$, and the injectivity of t^* gives $h = 0$ in $\mathrm{K}(\mathcal{A})$.

(3) implies (1). An acyclic X is isomorphic to 0 in $D(\mathcal{A})$ by proposition 3.2.12(1), so $\mathrm{Hom}_{D}(X, I) = 0$, and the bijection of (3) gives $\mathrm{Hom}_{\mathrm{K}}(X, I) = 0$, completing the proof.

The equivalence is [22, Tag 070I]. Condition (3) says that the morphisms into a K-injective complex are the same in $D(\mathcal{A})$ as in $\mathrm{K}(\mathcal{A})$, so they need no roofs. For a bounded-below complex of injectives, condition (3) is lemma 3.4.4(2).

Corollary 5.1.12: Bounded-Below Complexes of Injectives Are K-Injective

Let $\mathcal{A}$ be an abelian category. Every bounded-below complex of injective objects of $\mathcal{A}$ is K-injective.

***Proof*:**

For such a complex I, lemma 3.4.4(2) is condition (3) of proposition 5.1.11, as required.

So the injective resolutions of bounded-below complexes used in chapter 3, for instance the resolutions ι_B of definition 3.4.6, are K-injective resolutions. A quasi-isomorphism between bounded-below complexes of injectives is a homotopy equivalence by theorem 2.2.9, and the same holds with the bound replaced by K-injectivity.

Corollary 5.1.13: Quasi-Isomorphisms Between K-Injective Complexes

Let $\mathcal{A}$ be an abelian category.

1. A quasi-isomorphism $s\colon I \to J$ between K-injective complexes is a homotopy equivalence.
2. If $a\colon X \to I$ and $a'\colon X \to I'$ are K-injective resolutions, there is a homotopy equivalence $c\colon I \to I'$ with $ca = a'$ in $\mathrm{K}(\mathcal{A})$, and c is unique up to homotopy with this property.

***Proof*:**

1. By proposition 5.1.11(2) for I, the map $s^*\colon \mathrm{Hom}_{\mathrm{K}}(J, I) \to \mathrm{Hom}_{\mathrm{K}}(I, I)$ is bijective, so there is $t\colon J \to I$ with $ts = \mathrm{id}_I$ in $\mathrm{K}(\mathcal{A})$. Then $(st)s = s = \mathrm{id}_J s$, and $s^*\colon \mathrm{Hom}_{\mathrm{K}}(J, J) \to \mathrm{Hom}_{\mathrm{K}}(I, J)$ is injective by the same proposition for J, so $st = \mathrm{id}_J$ in $\mathrm{K}(\mathcal{A})$ and s is a homotopy equivalence.
2. By proposition 5.1.11(2) for I' and the quasi-isomorphism a, there is a unique class $c \in \mathrm{Hom}_{\mathrm{K}}(I, I')$ with $ca = a'$. Then $H^n(c)H^n(a) = H^n(a')$ with $H^n(a)$ and $H^n(a')$ isomorphisms, so c is a quasi-isomorphism and, by (1), a homotopy equivalence, as claimed.

For a bounded-below complex of injectives I, every quasi-isomorphism $s\colon X \to X'$ induces a quasi-isomorphism $s^*\colon \mathrm{Hom}^\bullet(X', I) \to \mathrm{Hom}^\bullet(X, I)$ of Hom complexes by corollary 3.4.5(1).

Corollary 5.1.14: Hom into a K-Injective Complex Preserves Quasi-Isomorphisms

Let $\mathcal{A}$ be an abelian category, $I \in \mathrm{Ch}(\mathcal{A})$ a K-injective complex and $s\colon X \to X'$ a quasi-isomorphism. Then $s^*\colon \mathrm{Hom}^\bullet(X', I) \to \mathrm{Hom}^\bullet(X, I)$ is a quasi-isomorphism.

***Proof*:**
Fix $n \in \mathbb{Z}$. By lemma 3.4.3(1) and (3), $H^n(s^*)$ is the map $\mathrm{Hom}_{\mathrm{K}}(X', I[n]) \to \mathrm{Hom}_{\mathrm{K}}(X, I[n])$, $\psi \mapsto \psi s$. The complex $I[n]$ is K-injective by lemma 5.1.9, so this map is bijective by proposition 5.1.11(2), as we sought to show.

The existence of K-injective resolutions for modules is Spaltenstein's theorem [21, Proposition 3.11], and its proof runs through the limit of a tower and its first derived functor from section 4.5. A complex X is the limit of the tower of its truncations $\tau^{\geq -n}X$ (example 5.1.16). Each truncation is bounded below and so has a bounded-below injective resolution, and the resolutions can be chosen to form a tower with split surjective transition maps. The limit of that tower is a K-injective resolution of X.

For a tower $A_\bullet$ of R-modules the map d^A of definition 4.5.2 is R-linear, so $\varprojlim A_\bullet$ and $\varprojlim{}^1 A_\bullet$ are R-modules. A homomorphism $M \to \varprojlim A_\bullet$ is the same as a family of homomorphisms $M \to A_n$ compatible with the transition maps, since $\varprojlim A_\bullet = \ker d^A$ is the module of coherent sequences.

Definition 5.1.15: Tower of Complexes and Its Limit

Let R be a ring. A *tower of complexes* of R-modules $C_\bullet = (C_n)_{n\geq 0}$ is a family of complexes C_n with chain maps $q_{n+1}\colon C_{n+1} \to C_n$. For each i the differentials $d^i\colon C_n^i \to C_n^{i+1}$ form a morphism of towers, the q_{n+1} being chain maps, so they induce $\varprojlim_n C_n^i \to \varprojlim_n C_n^{i+1}$ by definition 4.5.2, with square zero. The *limit* $\varprojlim C_\bullet$ is the resulting complex, with $(\varprojlim C_\bullet)^i = \varprojlim_n C_n^i$.

Example 5.1.16: The Tower of Truncations of a Complex

Let X be a complex of R-modules and put $X_n := \tau^{\geq -n}X$ for $n \geq 0$ (definition 2.1.12). Let $r_{n+1}\colon X_{n+1} \to X_n$ be zero in degrees below $-n$, the quotient $X^{-n} \to X^{-n}/B^{-n}$ in degree $-n$, and the identity in degrees above $-n$. It is a chain map: in degree $-n-1$ the differential of X_{n+1} lands in B^{-n}, which the quotient kills, and in degree $-n$ the induced differential of X_n composed with the quotient is d^{-n}. So $X_\bullet$ is a tower of complexes. Comparing degree by degree, the quotient maps $p_{-n}\colon X \to X_n$ satisfy $r_{n+1}p_{-n-1} = p_{-n}$, so they induce a chain map $X \to \varprojlim X_\bullet$. In degree i the maps p^i_{-n} and r^i_{n+1} are identities once $n > -i$, and a coherent sequence in a tower whose transition maps are identities from some index N on is determined by its term of index N. So $X \to \varprojlim X_\bullet$ is an isomorphism in every degree, and X is the limit of its truncations.

The projections $\varprojlim C_\bullet \to C_n$ are chain maps. For each m the groups $H^m(C_n)$ with the maps $H^m(q_{n+1})$ form a tower $H^m(C_\bullet)$.

Lemma 5.1.17: The Cohomology of a Limit of Complexes

Let R be a ring and $(C_n)_{n\geq 0}$ a tower of complexes of R-modules whose transition maps $q^i_{n+1}\colon C^i_{n+1} \to C^i_n$ are surjective for every i and n. Then for every $m \in \mathbb{Z}$ there is an exact sequence

$$0 \to \varprojlim{}^1 H^{m-1}(C_\bullet) \to H^m\big(\varprojlim C_\bullet\big) \to \varprojlim H^m(C_\bullet) \to 0$$

in which the second map is induced by the projections $\varprojlim C_\bullet \to C_n$.

***Proof*:**

Let $P := \prod_n C_n$, the product taken degreewise with the product differential, and let $\delta\colon P \to P$ be $(c_n)_n \mapsto \big(c_n - q_{n+1}(c_{n+1})\big)_n$, the map d^A of definition 4.5.2 for the tower $A_\bullet = C^i_\bullet$ in each degree i. It is a chain map, since the q_{n+1} are, and by definition $\varprojlim C_\bullet = \ker\delta$. The cokernel of δ in degree i is $\varprojlim{}^1 C^i_\bullet$, which vanishes by corollary 4.5.7. So

$$0 \to \varprojlim C_\bullet \to P \xrightarrow{\delta} P \to 0$$

is a short exact sequence of complexes, and theorem 2.1.20 gives the exact sequence

$$H^{m-1}(P) \xrightarrow{H^{m-1}(\delta)} H^{m-1}(P) \xrightarrow{\partial} H^m\big(\varprojlim C_\bullet\big) \to H^m(P) \xrightarrow{H^m(\delta)} H^m(P)$$

Products are exact in R-Mod, so $H^m(P) = \prod_n H^m(C_n)$ by lemma 2.1.11, and under this identification $H^m(\delta)$ is

$$([c_n])_n \longmapsto \big([c_n] - H^m(q_{n+1})[c_{n+1}]\big)_n$$

That is the map d^A of definition 4.5.2 for the tower $H^m(C_\bullet)$, so $\ker H^m(\delta) = \varprojlim H^m(C_\bullet)$, and likewise $\operatorname{coker} H^{m-1}(\delta) = \varprojlim{}^1 H^{m-1}(C_\bullet)$. Exactness at the three middle terms makes ∂ induce an injection of $\operatorname{coker} H^{m-1}(\delta)$ onto the kernel of $H^m(\varprojlim C_\bullet) \to H^m(P)$, and makes the image of that map $\ker H^m(\delta)$. The components of $H^m(\varprojlim C_\bullet) \to \prod_n H^m(C_n)$ are induced by the projections, as claimed.

This is Weibel [23] 3.5.8, in its form for cochain complexes, where it is proved under the Mittag-Leffler condition. Exercise 5.1.1 (8) recovers example 4.5.9 from it.

Lemma 5.1.18: A Limit of K-Injective Complexes Along Split Surjections Is K-Injective

Let R be a ring and $(I_n)_{n\geq 0}$ a tower of complexes of R-modules, with transition maps $q_{n+1}\colon I_{n+1} \to I_n$, such that every I_n is K-injective and every q^i_{n+1} is a split surjection. Then $\varprojlim I_\bullet$ is K-injective.

***Proof*:**
Let M be an acyclic complex of R-modules and write $I = \varprojlim I_\bullet$. In degree k,

$$\operatorname{Hom}^k(M, I) = \prod_i \operatorname{Hom}\big(M^i, \varprojlim_n I_n^{i+k}\big) = \prod_i \varprojlim_n \operatorname{Hom}\big(M^i, I_n^{i+k}\big) = \varprojlim_n \operatorname{Hom}^k(M, I_n)$$

by the description of homomorphisms into a limit, and because a product of limits is the limit of the products, the maps d^A being formed coordinatewise. The differential of definition 3.4.1 is formed by composing with differentials, so this is an isomorphism of complexes $\operatorname{Hom}^\bullet(M, I) \cong \varprojlim_n \operatorname{Hom}^\bullet(M, I_n)$, the transition maps being the chain maps $(q_{n+1})_*$ of lemma 3.4.3(2). These are surjective in each degree: if σ is a section of q^{i+k}_{n+1}, then $\sigma\psi$ is a preimage of $\psi\colon M^i \to I_n^{i+k}$. Each $\operatorname{Hom}^\bullet(M, I_n)$ is acyclic, since by lemma 3.4.3(1)

$$H^j \operatorname{Hom}^\bullet(M, I_n) = \operatorname{Hom}_{\mathrm{K}}(M, I_n[j]) = 0$$

with M acyclic and $I_n[j]$ K-injective by lemma 5.1.9. Lemma 5.1.17 for the tower $\operatorname{Hom}^\bullet(M, I_\bullet)$ of complexes of abelian groups places $H^0\operatorname{Hom}^\bullet(M, I)$ between $\varprojlim{}^1$ and $\varprojlim$ of towers of zero groups, so $\operatorname{Hom}_{\mathrm{K}}(M, I) = H^0\operatorname{Hom}^\bullet(M, I) = 0$, as we sought to show.

The lemma is [22, Tag 070L].

Theorem 5.1.19: K-Injective Resolutions Exist

Let $\mathcal{A}$ be a Grothendieck abelian category and $X \in \mathrm{Ch}(\mathcal{A})$. There is a quasi-isomorphism $\iota_X\colon X \to I_X$ such that I_X is K-injective, every I_X^n is an injective object and every ι_X^n is a monomorphism. Moreover the choice can be made functorial: there are a functor $\mathbf{I}\colon \mathrm{Ch}(\mathcal{A}) \to \mathrm{Ch}(\mathcal{A})$ and a natural transformation $\iota\colon \mathrm{id} \to \mathbf{I}$ such that $I_X = \mathbf{I}(X)$ and ι_X have these properties for every X.

For $\mathcal{A} = R$-Mod the first statement is proved below by Spaltenstein's argument. The same strategy works in every abelian category with enough injectives and exact countable products [22, Tag 090Y]. For a general Grothendieck abelian category, and for the functorial form, the proof is a transfinite

construction in the style of the proof of theorem 5.1.6, built from its functorial injective embeddings and from a set of representatives of the bounded-above acyclic complexes of bounded size. It is due to Serpé, generalizing Spaltenstein, and it is long and set-theoretic, so it is deferred to [22, Tag 079P], with the attribution at [22, Tag 079I].

***Proof*:**
Let $\mathcal{A} = R\text{-Mod}$. By theorem 5.1.6 and example 5.1.5(1) it has enough injectives.

Step 1: the tower of truncations. Let $X_\bullet$ be the tower of truncations of example 5.1.16, with $X_n = \tau^{\geq -n}X$, transition maps r_{n+1} and quotient maps $p_{-n}\colon X \to X_n$ satisfying $r_{n+1}p_{-n-1} = p_{-n}$. By proposition 2.1.14, $H^i(p_{-n})$ is an isomorphism for $i \geq -n$ and $H^i(X_n) = 0$ for $i < -n$.

Step 2: a tower of resolutions. We construct bounded-below complexes of injectives I_n, quasi-isomorphisms $\alpha_n\colon X_n \to I_n$ and chain maps $q_{n+1}\colon I_{n+1} \to I_n$, each q^i_{n+1} split surjective, with

$$q_{n+1}\alpha_{n+1} = \alpha_n r_{n+1}$$

as chain maps. Bounded-below complexes have injective resolutions by theorem 2.2.17 applied to the opposite category, as after definition 3.4.6, and this gives $\alpha_0\colon X_0 \to I_0$. Given α_n, choose likewise a quasi-isomorphism $\beta\colon X_{n+1} \to J$ with J a bounded-below complex of injectives. The complex I_n is K-injective by corollary 5.1.12, so proposition 5.1.11(2) for β gives a chain map $g\colon J \to I_n$ and a homotopy h, with $h^i\colon X^i_{n+1} \to I_n^{i-1}$, such that

$$u := \alpha_n r_{n+1} - g\beta = dh + hd$$

Let P be the complex with $P^i := I^i_n \oplus I^{i-1}_n$ and $d(a,b) := (da, -a - db)$. Then $d(da, -a - db) = (0, -da + da + ddb) = 0$, so P is a complex, and it is acyclic: if $d(a,b) = 0$ then $a = -db$, and $(a,b) = d(-b, 0)$. Its terms are injective and it is bounded below. The projection $\pi(a,b) := a$ is a chain map $P \to I_n$, split surjective in each degree with section $a \mapsto (a, 0)$. Define

$$I_{n+1} := J \oplus P, \qquad q_{n+1} := (g, \pi), \qquad \alpha_{n+1}(x) := \big(\beta x, (ux, -hx)\big)$$

The map $k(x) := (ux, -hx)$ is a chain map $X_{n+1} \to P$: the first coordinates of $dk(x)$ and $k(dx)$ are dux and udx, equal because u is a chain map, and the second ones are $-ux + dhx$ and $-hdx$, equal because $u = dh + hd$. So $\alpha_{n+1} = (\beta, k)$ is a chain map, and $q_{n+1}\alpha_{n+1} = g\beta + \pi k = g\beta + u = \alpha_n r_{n+1}$. As $H^i(P) = 0$, the map $H^i(\alpha_{n+1})$ is $H^i(\beta)$ followed by the inclusion of $H^i(J)$ as $H^i(J \oplus P) = H^i(J) \oplus 0$, so α_{n+1} is a quasi-isomorphism. Finally q_{n+1} is split surjective in each degree, with section $a \mapsto (0, (a, 0))$.

Step 3: the limit. Let $I := \varprojlim I_\bullet$. It is K-injective by lemma 5.1.18, since each I_n is K-injective by corollary 5.1.12. Its terms are injective. Choosing sections of the q^i_{n+1}, each I^i_{n+1} decomposes as $I^i_n \oplus \ker q^i_{n+1}$ with q^i_{n+1} the projection, so by induction $I^i_n \cong I^i_0 \oplus K_1 \oplus \cdots \oplus K_n$ with $K_m := \ker q^i_m$ and transition maps forgetting the last summand. Hence $I^i \cong I^i_0 \times \prod_{m \geq 1} K_m$. Each K_m is injective as a direct summand of an injective module, and a product of injective modules is injective, since $\mathrm{Hom}(-, \prod_t E_t) = \prod_t \mathrm{Hom}(-, E_t)$ and products of exact sequences of abelian groups are exact. The chain maps $\alpha_n p_{-n}\colon X \to I_n$ are compatible, as $q_{n+1}\alpha_{n+1}p_{-n-1} = \alpha_n r_{n+1}p_{-n-1} = \alpha_n p_{-n}$, so they define a chain map $\alpha\colon X \to I$.

Step 4: α is a quasi-isomorphism. Fix m. For $n \geq -m$ the map $H^m(\alpha_n p_{-n})\colon H^m(X) \to$

$H^m(I_n)$ is an isomorphism by Step 1, α_n being a quasi-isomorphism. For $n \geq -m$, the relation $H^m(q_{n+1})H^m(\alpha_{n+1}p_{-n-1}) = H^m(\alpha_n p_{-n})$ then makes $H^m(q_{n+1})$ an isomorphism, and for $n < -m$ its target $H^m(I_n) \cong H^m(X_n)$ is 0. So every transition map of the tower $H^m(I_\bullet)$ is surjective, for every m, and $\varprojlim{}^1 H^{m-1}(I_\bullet) = 0$ by corollary 4.5.7. By lemma 5.1.17, $H^m(I) \to \varprojlim H^m(I_\bullet)$ is an isomorphism. Let $N := \max(0, -m)$. A coherent sequence in a tower whose transition maps are isomorphisms from N on is determined by its term of index N, so the projection $\varprojlim H^m(I_\bullet) \to H^m(I_N)$ is an isomorphism. The composite $H^m(X) \to H^m(I) \to \varprojlim H^m(I_\bullet) \to H^m(I_N)$ is $H^m(\alpha_N p_{-N})$, an isomorphism, so $H^m(\alpha)$ is an isomorphism.

Step 5: a monomorphic resolution. Choose monomorphisms $e^i \colon X^i \to E^i$ into injective modules. Let P' be the complex with $P'^i := E^i \oplus E^{i+1}$ and $d(a,b) := (b,0)$, so that $d^2 = 0$. The maps $s(a,b) := (0,a)$, from P'^i to $P'^{i-1} = E^{i-1} \oplus E^i$, satisfy $ds + sd = \mathrm{id}$, since $(ds+sd)(a,b) = (a,0) + (0,b)$. So $\mathrm{id}_{P'}$ is null-homotopic, P' is acyclic by proposition 2.1.29, and every chain map f into P' is null-homotopic, being homotopic to $0 \circ f$. Thus P' is K-injective. The map $x \mapsto (e^i x, e^{i+1} dx)$ is a chain map $X \to P'$, as both d of it and its value on dx are $(e^{i+1}dx, 0)$, and it is a monomorphism in each degree because e^i is. Put $I_X := I \oplus P'$ and

$$\iota_X := \big(\alpha, x \mapsto (e^i x, e^{i+1} dx)\big) \colon X \longrightarrow I_X$$

a monomorphism in each degree. It is a quasi-isomorphism, since $H^i(I_X) = H^i(I) \oplus 0$ with $H^i(\iota_X)$ equal to $H^i(\alpha)$. The complex I_X has injective terms, and it is K-injective: for acyclic M, $\mathrm{Hom}_{\mathrm{K}}(M, I_X) \cong \mathrm{Hom}_{\mathrm{K}}(M, I) \oplus \mathrm{Hom}_{\mathrm{K}}(M, P') = 0$, since $\mathrm{K}(R\text{-Mod})$ is additive (lemma 2.1.39). This completes the proof for modules.

In Step 4 the towers $H^m(I_\bullet)$ are eventually constant, so their $\varprojlim{}^1$ vanishes and the limit has the cohomology of X. Without that, lemma 5.1.17 would leave a $\varprojlim{}^1$ term in $H^m(I)$. The auxiliary complex P of Step 2 turns commutativity up to homotopy into strict commutativity. The maps produced by proposition 5.1.11(2) commute only up to homotopy, while a map into a limit needs a strictly compatible family, and a chain map into P is a chain map into I_n together with a null-homotopy of it.

Example 5.1.20: A K-Injective Resolution of the Complex $\mathbb{Z}$ in Every Degree
Let X be the complex of abelian groups with $X^n = \mathbb{Z}$ for every n and every differential zero. For each n let J_n be the complex $\mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$, the quotient map, in degrees n and $n+1$. The groups $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are injective (example 3.4.7). Let I be the complex with $I^n = \mathbb{Q} \oplus \mathbb{Q}/\mathbb{Z}$, the factor $\mathbb{Q}$ coming from J_n and the factor $\mathbb{Q}/\mathbb{Z}$ from J_{n-1}, and differential $d(x,t) = (0,[x])$, where $[x] \in \mathbb{Q}/\mathbb{Z}$ lies in I^{n+1}. Then $d^2 = 0$, and I is the degreewise product $\prod_n J_n$, each degree having only two nonzero factors.

The map $\nu \colon X \to I$, $a \mapsto (a,0)$, is a chain map, since $d\nu(a) = (0,[a]) = 0$ for $a \in \mathbb{Z}$. It is a quasi-isomorphism: the cocycles of I^n are $\mathbb{Z} \oplus \mathbb{Q}/\mathbb{Z}$, the coboundaries are $0 \oplus \mathbb{Q}/\mathbb{Z}$, and $H^n(I) \cong \mathbb{Z}$ is generated by $(1,0) = \nu(1)$. And I is K-injective. A chain map $M \to \prod_n J_n$ is a family of chain maps $M \to J_n$, and the same holds for null-homotopies, so $\mathrm{Hom}_{\mathrm{K}}(M, I) = \prod_n \mathrm{Hom}_{\mathrm{K}}(M, J_n)$. For acyclic M every factor vanishes by corollary 5.1.12. So ν is a K-injective resolution of a complex unbounded in both directions, and ν is a monomorphism in each degree.

Theorem 3.2.4 describes the morphisms of $D(\mathcal{A})$ by roofs, which range over all quasi-isomorphisms into a complex, and it guarantees that the classes of roofs form sets only when the quasi-isomorphisms

into each complex admit a cofinal set. K-injective resolutions give a set-theoretic bound with no such hypothesis.

Corollary 5.1.21: The Derived Category Has Small Hom-Sets

Let $\mathcal{A}$ be a Grothendieck abelian category, $X, Y \in \mathrm{Ch}(\mathcal{A})$, and $\iota_Y \colon Y \to I_Y$ a K-injective resolution. Then

$$\mathrm{Hom}_{\mathrm{K}}(X, I_Y) \longrightarrow \mathrm{Hom}_D(X, Y), \qquad f \longmapsto Q(\iota_Y)^{-1} Q(f)$$

is bijective. In particular $\mathrm{Hom}_{D(\mathcal{A})}(X, Y)$ is a set.

***Proof*:**
The map is the composite of the bijection $Q \colon \mathrm{Hom}_{\mathrm{K}}(X, I_Y) \to \mathrm{Hom}_D(X, I_Y)$ of proposition 5.1.11(3) and of composition with the isomorphism $Q(\iota_Y)^{-1}$. The resolution exists by theorem 5.1.19. The source is a quotient of the set of chain maps $X \to I_Y$, which is a subset of the set $\prod_n \mathrm{Hom}_{\mathcal{A}}(X^n, I_Y^n)$, completing the proof.

The corollary gives the conclusion that theorem 3.2.4 states under its cofinality hypothesis, without that hypothesis, for every Grothendieck abelian category, R-Mod included. In definition 3.3.5 the total derived functor of a left exact F was obtained by applying F term by term to a resolution in an adapted class, with bounded-below complexes throughout. K-injective resolutions do the same with no bound and with no exactness assumption on F. For an additive $F \colon \mathcal{A} \to \mathcal{B}$, applying F term by term carries null-homotopies to null-homotopies (lemma 2.1.34), so it defines a functor $\mathrm{K}(F) \colon \mathrm{K}(\mathcal{A}) \to \mathrm{K}(\mathcal{B})$. Write $Q_{\mathcal{A}}$ and $Q_{\mathcal{B}}$ for the localization functors, and R^+F for the total derived functor of definition 3.3.5, with its transformation ε_F of theorem 3.3.6.

Corollary 5.1.22: Right Derived Functors on Unbounded Complexes

Let $\mathcal{A}$ be a Grothendieck abelian category, $\mathcal{B}$ an abelian category and $F \colon \mathcal{A} \to \mathcal{B}$ an additive functor. For every complex X fix a K-injective resolution $\iota_X \colon X \to I_X$ (theorem 5.1.19).

1. For every morphism $\varphi \colon X \to Y$ of $D(\mathcal{A})$ there is a unique $I_\varphi \in \mathrm{Hom}_{\mathrm{K}}(I_X, I_Y)$ with $Q_{\mathcal{A}}(I_\varphi)\, Q_{\mathcal{A}}(\iota_X) = Q_{\mathcal{A}}(\iota_Y)\, \varphi$, and $X \mapsto I_X$, $\varphi \mapsto I_\varphi$ is a functor $I_{(-)} \colon D(\mathcal{A}) \to \mathrm{K}(\mathcal{A})$.

2. Put $RF := Q_{\mathcal{B}} \circ \mathrm{K}(F) \circ I_{(-)}$ and $\varepsilon_X := Q_{\mathcal{B}}\big(F(\iota_X)\big) \colon F(X) \to RF(X)$. Then $\varepsilon \colon Q_{\mathcal{B}} \circ \mathrm{K}(F) \to RF \circ Q_{\mathcal{A}}$ is natural, and (RF, ε) is initial: for every functor $G \colon D(\mathcal{A}) \to D(\mathcal{B})$ and every natural transformation $\eta \colon Q_{\mathcal{B}} \circ \mathrm{K}(F) \to G \circ Q_{\mathcal{A}}$ there is a unique natural transformation $\theta \colon RF \to G$ with $\eta = (\theta * Q_{\mathcal{A}}) \circ \varepsilon$. Hence RF does not depend on the chosen resolutions, up to a unique natural isomorphism compatible with ε.

3. RF is a triangulated functor.

4. If F is left exact, then for every bounded-below complex X there is an isomorphism $\theta_X \colon RF(X) \to R^+F(X)$ in $D(\mathcal{B})$, natural in X for the morphisms of $D^+(\mathcal{A})$, with $\theta_X \varepsilon_X = \varepsilon_F(X)$. In particular $H^n\big(RF(A)\big) \cong R^nF(A)$ for every object A of $\mathcal{A}$, natural in A.

***Proof*:**
Write Q for $Q_{\mathcal{A}}$ where no confusion arises. Since I_Y is K-injective, $Q\colon \mathrm{Hom}_{\mathrm{K}}(Z, I_Y) \to \mathrm{Hom}_D(Z, I_Y)$ is bijective for every Z (proposition 5.1.11(3)), and $Q(\iota_X)$ is invertible.

(1). The morphism $Q(\iota_Y)\varphi Q(\iota_X)^{-1}$ of $\mathrm{Hom}_D(I_X, I_Y)$ has a unique preimage under Q, which is I_φ. Uniqueness gives $I_{\mathrm{id}} = \mathrm{id}$, $I_{\varphi'\varphi} = I_{\varphi'} I_\varphi$ and $I_{\varphi+\varphi'} = I_\varphi + I_{\varphi'}$, since the right-hand sides satisfy the defining equations. For a chain map $f\colon X \to Y$, the classes $I_{Qf}\,\iota_X$ and $\iota_Y f$ in $\mathrm{Hom}_{\mathrm{K}}(X, I_Y)$ have the same image under Q, so $I_{Qf}\,\iota_X = \iota_Y f$ in $\mathrm{K}(\mathcal{A})$.

(2). Naturality of ε: for a chain map $f\colon X \to Y$,

$$RF(Qf)\,\varepsilon_X = Q_{\mathcal{B}}F\big(I_{Qf}\,\iota_X\big) = Q_{\mathcal{B}}F(\iota_Y f) = \varepsilon_Y\, Q_{\mathcal{B}}F(f)$$

by (1). If W is K-injective, then $\iota_W\colon W \to I_W$ is a quasi-isomorphism between K-injective complexes, hence a homotopy equivalence (corollary 5.1.13(1)), so $F(\iota_W)$ is a homotopy equivalence by lemma 2.1.34 and ε_W is an isomorphism. Given (G, η), put $\mu_W := \eta_W \varepsilon_W^{-1}\colon RF(W) \to G(W)$ for K-injective W. For a morphism $\psi\colon W \to W'$ of $\mathrm{K}(\mathcal{A})$ between K-injective complexes, naturality of ε and η along ψ gives $\mu_{W'} RF(Q\psi) = \eta_{W'} Q_{\mathcal{B}}F(\psi)\varepsilon_W^{-1} = G(Q\psi)\mu_W$.

If θ satisfies $\theta_X \varepsilon_X = \eta_X$ for every X, then $\theta_W = \mu_W$ for K-injective W, and naturality along $Q(\iota_X)$ forces

$$\theta_X = G(Q\iota_X)^{-1}\,\mu_{I_X}\,RF(Q\iota_X)$$

so θ is unique. Conversely define θ_X by this formula. Naturality of ε along ι_X and of η along ι_X give

$$\theta_X\varepsilon_X = G(Q\iota_X)^{-1}\,\eta_{I_X}\,\varepsilon_{I_X}^{-1}\,\varepsilon_{I_X}\,Q_{\mathcal{B}}F(\iota_X) = G(Q\iota_X)^{-1}\,G(Q\iota_X)\,\eta_X = \eta_X$$

For $\varphi\colon X \to Y$ in $D(\mathcal{A})$ put $\psi := I_\varphi$, so that $Q(\iota_Y)\varphi = Q(\psi)Q(\iota_X)$. Applying RF and G to this equation gives $RF(Q\iota_Y)RF(\varphi) = RF(Q\psi)RF(Q\iota_X)$ and $G(Q\iota_Y)^{-1}G(Q\psi) = G(\varphi)G(Q\iota_X)^{-1}$, so, with the naturality of μ along ψ,

$$\theta_Y RF(\varphi) = G(Q\iota_Y)^{-1}\mu_{I_Y} RF(Q\psi)RF(Q\iota_X) = G(Q\iota_Y)^{-1}G(Q\psi)\mu_{I_X} RF(Q\iota_X) = G(\varphi)\,\theta_X$$

So θ is natural. Two choices of resolutions give two initial pairs, and the comparison transformations between them compose to identities by the uniqueness proved above.

(3). RF is additive, as $I_{(-)}$, $\mathrm{K}(F)$ and $Q_{\mathcal{B}}$ are. *Shift.* The complex $I_X[1]$ is K-injective by lemma 5.1.9, and $\iota_X[1]$ is a quasi-isomorphism since $H^n(\iota_X[1]) = H^{n+1}(\iota_X)$. As in (1) there is a unique $\omega_X \in \mathrm{Hom}_{\mathrm{K}}(I_{X[1]}, I_X[1])$ with $Q(\omega_X)Q(\iota_{X[1]}) = Q(\iota_X[1])$. Applying the functor H^n on $D(\mathcal{A})$ (proposition 3.2.8) gives $H^n(\omega_X)H^n(\iota_{X[1]}) = H^n(\iota_X[1])$ with both $H^n(\iota_{X[1]})$ and $H^n(\iota_X[1])$ isomorphisms, so ω_X is a quasi-isomorphism between K-injective complexes and hence an isomorphism of $\mathrm{K}(\mathcal{A})$ by corollary 5.1.13(1). For $\varphi\colon X \to Y$ the classes $\omega_Y I_{\varphi[1]}$ and $I_\varphi[1]\omega_X$ both satisfy the equation $Q(-)Q(\iota_{X[1]}) = Q(\iota_Y[1])\varphi[1]$, so they are equal and ω is natural. Since F is additive, $F(Z[1]) = F(Z)[1]$ for every complex Z, the differential $-d$ going to $-F(d)$. So $\zeta_X := Q_{\mathcal{B}}F(\omega_X)\colon RF(X[1]) \to RF(X)[1]$ is a natural isomorphism.

Triangles. Let $u\colon X \to Y$ be a chain map with mapping cone triangle $X \xrightarrow{u} Y \xrightarrow{v} MC(u) \xrightarrow{w} X[1]$, and put $\tilde{u} := I_{Qu}$, so that $\tilde{u}\,\iota_X = \iota_Y u$ in $\mathrm{K}(\mathcal{A})$ by (1). Let $I_Y \xrightarrow{\tilde{v}} MC(\tilde{u}) \xrightarrow{\tilde{w}} I_X[1]$ complete

the cone triangle of $\tilde{u}$. The cone $MC(\tilde{u})$ is K-injective: for acyclic M, the functor $\mathrm{Hom}_{\mathrm{K}}(M,-)$ is cohomological (proposition 3.1.8), and on the rotation $(\tilde{v},\tilde{w},-\tilde{u}[1])$ of the cone triangle it gives an exact sequence $\mathrm{Hom}_{\mathrm{K}}(M,I_Y)\to\mathrm{Hom}_{\mathrm{K}}(M,MC(\tilde{u}))\to\mathrm{Hom}_{\mathrm{K}}(M,I_X[1])$ with vanishing outer terms. By (TR3) in $\mathrm{K}(\mathcal{A})$ there is $c\colon MC(u)\to MC(\tilde{u})$ making (ι_X,ι_Y,c) a morphism of triangles, so $cv=\tilde{v}\,\iota_Y$ and $\tilde{w}c=\iota_X[1]\,w$. Its image under Q is a morphism of distinguished triangles of $D(\mathcal{A})$ (theorem 3.2.7) whose first two components are isomorphisms, so $Q(c)$ is an isomorphism by corollary 3.1.9, and c is a quasi-isomorphism. Let $\kappa\in\mathrm{Hom}_{\mathrm{K}}(I_{MC(u)},MC(\tilde{u}))$ be the unique class with $Q(\kappa)Q(\iota_{MC(u)})=Q(c)$, an isomorphism of $\mathrm{K}(\mathcal{A})$ by the argument used for ω_X.

We claim that $(\mathrm{id},\mathrm{id},\kappa)$ is an isomorphism of triangles from $\big(\tilde{u},I_{Qv},\omega_X I_{Qw}\big)$ to $(\tilde{u},\tilde{v},\tilde{w})$. The targets $MC(\tilde{u})$ and $I_X[1]$ of the remaining two squares are K-injective, so it suffices to compare their two sides after applying Q and composing with the isomorphisms $Q(\iota_Y)$ and $Q(\iota_{MC(u)})$ respectively. Using (1),

$$Q\big(\kappa\, I_{Qv}\big)Q(\iota_Y)=Q(\kappa)Q(\iota_{MC(u)})Q(v)=Q(cv)=Q(\tilde{v})Q(\iota_Y)$$
$$Q(\tilde{w}\kappa)Q(\iota_{MC(u)})=Q(\tilde{w}c)=Q(\iota_X[1])Q(w)=Q(\omega_X)Q(\iota_{X[1]})Q(w)=Q\big(\omega_X I_{Qw}\big)Q(\iota_{MC(u)})$$

Cancelling the isomorphisms $Q(\iota_Y)$ and $Q(\iota_{MC(u)})$ and using proposition 5.1.11(3) for the targets gives $\kappa\, I_{Qv}=\tilde{v}$ and $\tilde{w}\kappa=\omega_X I_{Qw}$ in $\mathrm{K}(\mathcal{A})$, which proves the claim. So the triangle $\big(\tilde{u},I_{Qv},\omega_X I_{Qw}\big)$ is distinguished in $\mathrm{K}(\mathcal{A})$, by (TR1). By definition 2.1.23, $F(MC(\tilde{u}))$ is $MC(F(\tilde{u}))$ term by term, with F applied to the inclusion and the projection, so $\mathrm{K}(F)$ carries cone triangles, and with (TR1) all distinguished triangles, to distinguished triangles of $\mathrm{K}(\mathcal{B})$, and $Q_{\mathcal{B}}$ carries those to distinguished triangles of $D(\mathcal{B})$. The image of our triangle is $\big(RF(Qu),RF(Qv),\zeta_X RF(Qw)\big)$, which is therefore distinguished. Every distinguished triangle of $D(\mathcal{A})$ is isomorphic to the image under Q of a mapping cone triangle (theorem 3.2.7 and theorem 3.1.12). The functor RF, with the natural ζ, carries isomorphic triangles to isomorphic triangles, and the distinguished triangles of $D(\mathcal{B})$ are closed under isomorphism, which proves (3).

(4). By the last assertion of (2), a second choice of resolutions changes RF and ε by a natural isomorphism compatible with ε, and composing with it carries an isomorphism θ as in (4) for one choice to one for the other. So we may take ι_X to be a quasi-isomorphism into a bounded-below complex of injectives whenever X is bounded below, as in Step 2 of the proof of theorem 5.1.19. Such ι_X is a K-injective resolution by corollary 5.1.12, and then I_φ is a morphism of $\mathrm{K}^+(\mathcal{A})$ for every morphism φ of $D^+(\mathcal{A})$ between bounded-below complexes. By theorem 3.3.6(3) we may compute R^+F with the class of injectives, adapted to F by example 3.3.3(1), $\mathcal{A}$ having enough injectives by theorem 5.1.6. Write $\mathcal{R}$ for this class, $\Psi\colon \mathrm{K}^+(\mathcal{R})[S_{\mathcal{R}}^{-1}]\to D^+(\mathcal{A})$ for the equivalence of theorem 3.3.4, Φ for the inverse of Ψ from which R^+F is built, $\beta\colon \mathrm{id}\to\Psi\Phi$ for the natural isomorphism exhibiting Φ as an inverse of Ψ, and $\bar{F}\colon \mathrm{K}^+(\mathcal{R})[S_{\mathcal{R}}^{-1}]\to D^+(\mathcal{B})$ for the functor induced by F term by term, so that $R^+F=\bar{F}\Phi$. Let W be a bounded-below complex of injectives. It lies in $\mathrm{K}^+(\mathcal{R})$, so the quasi-isomorphism $s\colon W\to Z$ in the definition of $\varepsilon_F(W)$ in theorem 3.3.6 may be taken to be id_W, the result not depending on s by part (1) of that theorem. That definition then reads $\varepsilon_F(W)=\bar{F}(\gamma)^{-1}Q_{\mathcal{B}}\big(F(\mathrm{id}_W)\big)=\bar{F}(\gamma)^{-1}$, where $\gamma\colon \Phi(Q_{\mathcal{A}}W)\to W$ is the morphism of $\mathrm{K}^+(\mathcal{R})[S_{\mathcal{R}}^{-1}]$ with $\Psi(\gamma)=Q_{\mathcal{A}}(\mathrm{id}_W)\beta_W^{-1}=\beta_W^{-1}$. As Ψ is an equivalence and β_W^{-1} is an isomorphism, γ is an isomorphism, hence so is $\bar{F}(\gamma)$, and $\varepsilon_F(W)$ is an isomorphism. Set

$$\theta_X:=R^+F(Q\iota_X)^{-1}\,\varepsilon_F(I_X)\colon RF(X)=Q_{\mathcal{B}}F(I_X)\longrightarrow R^+F(X)$$

an isomorphism. Naturality of ε_F along ι_X gives $\theta_X \varepsilon_X = R^+F(Q\iota_X)^{-1}\,\varepsilon_F(I_X)\,Q_{\mathcal{B}}F(\iota_X) = \varepsilon_F(X)$. For a morphism $\varphi\colon X \to Y$ of $D^+(\mathcal{A})$ between bounded-below complexes, naturality of ε_F along I_φ and the relation $Q(\iota_Y)^{-1}Q(I_\varphi) = \varphi\, Q(\iota_X)^{-1}$ give

$$\theta_Y RF(\varphi) = R^+F(Q\iota_Y)^{-1}R^+F\big(Q(I_\varphi)\big)\varepsilon_F(I_X) = R^+F(\varphi)\,R^+F(Q\iota_X)^{-1}\varepsilon_F(I_X) = R^+F(\varphi)\,\theta_X$$

For an object A of $\mathcal{A}$, applying H^n to θ_A and using theorem 3.3.7 gives $H^n(RF(A)) \cong R^nF(A)$, natural in A, completing the proof.

Part (2) is a stronger universal property than the one in theorem 3.3.6(2), since it quantifies over all functors G and not only triangulated ones, and by (3) the pair (RF, ε) belongs to the triangulated class. As in the bounded case, $RF(X)$ is F applied term by term to a resolution of X, now a K-injective one, and for X itself K-injective, ε_X identifies $RF(X)$ with $F(X)$. Applied term by term to a complex of injectives that is not K-injective, F need not carry quasi-isomorphisms to quasi-isomorphisms, as in example 3.3.1, where the complex was a projective resolution.

Example 5.1.23: Applying a Functor Term by Term to a Complex of Injectives
Let I be the complex of example 5.1.7 and $F := \mathrm{Hom}_{\mathbb{Z}/4}(\mathbb{Z}/2, -)\colon \mathbb{Z}/4\text{-Mod} \to \mathbf{Ab}$, where $\mathbb{Z}/2 = \mathbb{Z}/4/(2)$. A homomorphism $\varphi\colon \mathbb{Z}/2 \to \mathbb{Z}/4$ is determined by $\varphi(1)$, subject to $2\varphi(1) = \varphi(2) = 0$, so $F(\mathbb{Z}/4) \cong \{0, 2\} \cong \mathbb{Z}/2$. The differential of I induces $\varphi \mapsto 2\varphi$, which is zero since $2 \cdot 2 = 0$ in $\mathbb{Z}/4$. So $F(I)$ is the complex with $\mathbb{Z}/2$ in every degree and zero differential, and $H^n(F(I)) \cong \mathbb{Z}/2$ for every n. On the other hand I is acyclic, so $I \to 0$ is a quasi-isomorphism into the K-injective complex 0, a K-injective resolution. By the last assertion of corollary 5.1.22(2), $RF(I)$ may be computed with this resolution, so $RF(I) \cong F(0) = 0$, while term-by-term application of F to the injective terms of I has nonzero cohomology in every degree. In particular the term-by-term functor does not carry the quasi-isomorphism $I \to 0$ to a quasi-isomorphism.

Exercise 5.1.1

1. Let R be a ring and $(C_i)_{i\in I}$ a directed system of complexes of R-modules, that is, a directed system in $\mathrm{Ch}(R\text{-Mod})$, and let C be the complex with $C^n = \varinjlim_i C_i^n$ and the differential induced by those of the C_i. Using proposition 5.1.3, show that the maps $H^n(C_i) \to H^n(C)$ induce an isomorphism $\varinjlim_i H^n(C_i) \cong H^n(C)$ for every n.

2. Let $\mathcal{A}$ be an abelian category satisfying (AB3) with a generator U in the sense of (G). Show that every object X is a quotient of the direct sum $\bigoplus_{u\in \mathrm{Hom}(U,X)} U$, by applying (G) to the cokernel of the morphism $\bigoplus_u U \to X$ that restricts to u on the summand indexed by u.

3. Show that the category of finitely generated abelian groups is abelian and is not a Grothendieck abelian category. (If a countable family of copies of $\mathbb{Z}$ had a direct sum S in this category, then $\mathrm{Hom}(S, \mathbb{Z}/2)$ would be in bijection with $\prod_{n\in\mathbb{N}} \mathrm{Hom}(\mathbb{Z}, \mathbb{Z}/2)$. Compare cardinalities.)

4. Let $\mathcal{A}$ be abelian. Show that $I \oplus J$ is K-injective if and only if I and J are, and more generally that if J is a retract of a K-injective complex I in $\mathrm{K}(\mathcal{A})$, meaning that there are $a\colon J \to I$ and $b\colon I \to J$ with $ba = \mathrm{id}_J$ in $\mathrm{K}(\mathcal{A})$, then J is K-injective.

5. Show that a complex I is acyclic and K-injective if and only if id_I is null-homotopic. Deduce that if X is acyclic and $\iota_X\colon X \to I_X$ is a K-injective resolution, then id_{I_X} is null-homotopic.

6. Let I be the complex of example 5.1.7 and regard $\mathbb{Z}/2 = \mathbb{Z}/4/(2)$ and $\mathbb{Z}/4$ as complexes

concentrated in degree 0. Show that $\mathrm{Hom}_{\mathrm{K}(\mathbb{Z}/4\text{-Mod})}(\mathbb{Z}/4, I) = 0$ and $\mathrm{Hom}_{\mathrm{K}(\mathbb{Z}/4\text{-Mod})}(\mathbb{Z}/2, I) \cong \mathbb{Z}/2$. Conclude that $Q\colon \mathrm{Hom}_{\mathrm{K}}(\mathbb{Z}/2, I) \to \mathrm{Hom}_{D}(\mathbb{Z}/2, I)$ is not injective, so that condition (3) of proposition 5.1.11 fails for I already at a complex concentrated in one degree.

7. Let $\mathcal{A}$ be abelian and let $A \xrightarrow{u} B \xrightarrow{v} C \xrightarrow{w} A[1]$ be a distinguished triangle of $\mathrm{K}(\mathcal{A})$. Show that if two of A, B, C are K-injective, then so is the third. (For an acyclic M, apply $\mathrm{Hom}_{\mathrm{K}}(M, -)$ to the triangle and to its two rotations in (TR2).)

8. Fix a prime p. For $n \geq 0$ let C_n be the complex $\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$, the reduction map, in degrees 0 and 1, and let $C_{n+1} \to C_n$ be the identity in degree 0 and the reduction in degree 1. Compute $H^0(C_n)$, $H^1(C_n)$ and $\varprojlim C_\bullet$, and use lemma 5.1.17 with example 4.5.8 to show that $\varprojlim^1$ of the tower $(p^n\mathbb{Z})_n$, with the inclusions as transition maps, is isomorphic to $\mathbb{Z}_p/\mathbb{Z}$, recovering example 4.5.9.

5.2 K-Flat Complexes and the Unbounded Tensor-Hom Adjunction

Section 5.1 resolves every complex of a Grothendieck abelian category on the right by a K-injective complex, and corollary 5.1.22 uses these resolutions to derive additive functors on unbounded complexes. The tensor product is derived by resolving on the left, and box 3.4.20 records where the left resolutions of section 3.4 needed a bound: lemma 3.4.4(1) needs a bounded-above complex of projectives, and lemma 3.4.14(2) a bounded-above complex of flat modules. This section replaces both by conditions on a complex as a whole, K-projectivity and K-flatness, in the form given by Spaltenstein [21], and constructs a resolution of both kinds for every complex of modules. The derived tensor product and $R\mathrm{Hom}$ then become bifunctors on all of $D(R)$, adjoint to one another as in theorem 3.4.19, and they agree with the constructions of section 3.4 on bounded complexes. Over a principal ideal domain every complex of torsion-free modules is K-flat, so that $F \otimes^L_{\mathbb{Z}} \mathbb{Z}/n$ is F/nF for every complex F of torsion-free abelian groups.

Throughout, R is a commutative ring and $D(R) = D(R\text{-Mod})$. As explained before theorem 3.4.19, the total tensor complex $\mathrm{Tot}(X \otimes_R Y)$ of definition 3.4.12 and the Hom complex $\mathrm{Hom}^\bullet(X, Y)$ of definition 3.4.1, formed in R-Mod, are then complexes of R-modules. Neither definition requires a bound on X or Y: Tot is formed with direct sums and $\mathrm{Hom}^\bullet$ with products.

Conditions on the individual terms do not control an unbounded complex. Example 5.2.2(3) and example 5.2.6 below give an acyclic complex of free $\mathbb{Z}/4$-modules for which the conclusions of lemma 3.4.4(1) and lemma 3.4.14(3) fail. The conditions that replace the bounds are tested against every acyclic complex at once.

Definition 5.2.1: K-Projective and K-Flat Complexes

Let X be a complex of R-modules.

1. X is *K-projective* if $\mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(X, A) = 0$ for every acyclic complex A of R-modules.
2. X is *K-flat* if $\mathrm{Tot}(X \otimes_R A)$ is acyclic for every acyclic complex A of R-modules.

A *K-projective resolution* of a complex Y is a quasi-isomorphism $P \to Y$ with P K-projective, and a *K-flat resolution* is a quasi-isomorphism $F \to Y$ with F K-flat.

If A is acyclic, so is every shift $A[n]$, and $H^n\mathrm{Hom}^\bullet(X,A) = \mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(X, A[n])$ by lemma 3.4.3(1). So X is K-projective exactly when $\mathrm{Hom}^\bullet(X,A)$ is acyclic for every acyclic complex A, which is the definition of [21, p. 1.1], where it is attributed to J. Bernstein. K-flatness is [21, p. 5.1], stated there for complexes of sheaves of modules on a commutative ringed space, of which complexes of R-modules are the case of a one-point space with structure ring R. Both conditions say that a functor of complexes built from X carries acyclic complexes to acyclic complexes: $\mathrm{Hom}^\bullet(X,-)$ in the first case, $\mathrm{Tot}(X \otimes_R -)$ in the second.

Example 5.2.2: Bounded-Above Projective Complexes, Single Modules, and a Free Complex over $\mathbb{Z}/4$

1. Let P be a bounded-above complex of projective R-modules and A an acyclic complex. By proposition 3.2.12(1), $A \cong 0$ in $D(R)$, so $\mathrm{Hom}_{D(R)}(P,A) = 0$, and lemma 3.4.4(1) identifies $\mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(P,A)$ with this group. So P is K-projective.

2. Let M be an R-module, regarded as a complex in degree 0. Then $\mathrm{Tot}(M \otimes_R A)^n = M \otimes_R A^n$ with differential $1 \otimes d_A$, and $\mathrm{Hom}^n(M,A) = \mathrm{Hom}_R(M, A^n)$ with differential $\varphi \mapsto d_A\varphi$. If M is flat, the exact functor $M \otimes_R -$ of [8, Flat Module] preserves kernels and images, so it carries acyclic complexes to acyclic ones and M is K-flat. If M is projective, it is K-projective by (1). Conversely, let $0 \to N' \to N \to N'' \to 0$ be a short exact sequence, placed in degrees -1, 0, 1 as an acyclic complex A. If M is K-flat, then $M \otimes_R A$ is acyclic, which says that $M \otimes_R -$ is exact on this sequence, so M is flat. If M is K-projective, then $\mathrm{Hom}^\bullet(M,A)$ is acyclic, and its vanishing cohomology in degree 1 says that $\mathrm{Hom}_R(M,N) \to \mathrm{Hom}_R(M,N'')$ is surjective. Every surjection $N \to N''$ arises this way, so M is projective by [8, Equivalencies for Projective Modules](2).

3. Let I be the complex of example 5.1.7, with $I^n = \mathbb{Z}/4$ for every $n \in \mathbb{Z}$ and every differential multiplication by 2. Its terms are free $\mathbb{Z}/4$-modules. By example 5.1.7(2) and (3), I is acyclic and id_I is a nonzero element of $\mathrm{Hom}_{\mathrm{K}(\mathbb{Z}/4\text{-Mod})}(I,I)$, so I is not K-projective. The same example shows that $Q\colon \mathrm{Hom}_{\mathrm{K}}(I,I) \to \mathrm{Hom}_D(I,I)$ is not injective, so the conclusion of lemma 3.4.4(1) fails for this complex of projective modules.

Together with (1), item (3) shows that a complex of projective modules is K-projective when it is bounded above and need not be otherwise. Item (2) shows that both notions extend the module-level ones.

Write $\mathcal{A} = R\text{-Mod}$. Reindexing $X^n \mapsto X^{-n}$ identifies $\mathrm{Ch}(\mathcal{A}^{\mathrm{op}})$ with $\mathrm{Ch}(\mathcal{A})^{\mathrm{op}}$, and by Step 3 of the proof of lemma 3.4.4 it carries quasi-isomorphisms to quasi-isomorphisms and null-homotopies to null-homotopies, identifies $\mathrm{K}(\mathcal{A}^{\mathrm{op}})$ with $\mathrm{K}(\mathcal{A})^{\mathrm{op}}$, and identifies $D(\mathcal{A}^{\mathrm{op}})$ with $D(\mathcal{A})^{\mathrm{op}}$ compatibly with Q. Write X° for the complex of $\mathcal{A}^{\mathrm{op}}$ corresponding to a complex X of R-modules. By lemma 2.1.19, X is acyclic exactly when $0 \to X$ is a quasi-isomorphism, and this chain map corresponds to $X^\circ \to 0$, so X is acyclic exactly when X° is. As $\mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(P,A) = \mathrm{Hom}_{\mathrm{K}(\mathcal{A}^{\mathrm{op}})}(A^\circ, P^\circ)$ for all complexes P and A, a complex P of R-modules is K-projective exactly when P° is a K-injective complex of the abelian category $\mathcal{A}^{\mathrm{op}}$.

Lemma 5.2.3: K-Projective Complexes Compute Morphisms in $D(R)$

Let P be a K-projective complex of R-modules.

1. For every quasi-isomorphism $t\colon Y \to Y'$, the chain map $t_* := \mathrm{Hom}^\bullet(P,t)\colon \mathrm{Hom}^\bullet(P,Y) \to \mathrm{Hom}^\bullet(P,Y')$ is a quasi-isomorphism.
2. For every complex X, the map $Q\colon \mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(P,X) \to \mathrm{Hom}_{D(R)}(P,X)$ is bijective.
3. A quasi-isomorphism $u\colon P' \to P$ whose source P' is also K-projective is a homotopy equivalence.

***Proof*:**
(2) and (3). Under the correspondence $X \mapsto X^\circ$ above, P° is a K-injective complex of $\mathcal{A}^{\mathrm{op}}$. For a complex X, proposition 5.1.11(3) for $\mathcal{A}^{\mathrm{op}}$ makes $Q\colon \mathrm{Hom}_{\mathrm{K}(\mathcal{A}^{\mathrm{op}})}(X^\circ,P^\circ) \to \mathrm{Hom}_{D(\mathcal{A}^{\mathrm{op}})}(X^\circ,P^\circ)$ bijective, and under the identifications of $\mathrm{K}(\mathcal{A}^{\mathrm{op}})$ with $\mathrm{K}(\mathcal{A})^{\mathrm{op}}$ and of $D(\mathcal{A}^{\mathrm{op}})$ with $D(\mathcal{A})^{\mathrm{op}}$, compatible with Q, this map is $Q\colon \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(P,X) \to \mathrm{Hom}_{D(\mathcal{A})}(P,X)$. This is (2). For (3), the chain map $u^\circ\colon P^\circ \to P'^\circ$ is a quasi-isomorphism between K-injective complexes of $\mathcal{A}^{\mathrm{op}}$, so by corollary 5.1.13(1) there is $v^\circ\colon P'^\circ \to P^\circ$ with $v^\circ u^\circ$ and $u^\circ v^\circ$ homotopic to the identities. The reindexing carries null-homotopies to null-homotopies, so the corresponding $v\colon P \to P'$ has uv and vu homotopic to the identities, and u is a homotopy equivalence.

(1). Fix n. By lemma 3.4.3(1) and (3), $H^n(t_*)$ is the map

$$\mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(P,Y[n]) \longrightarrow \mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(P,Y'[n]), \qquad \psi \longmapsto t[n]\circ\psi$$

The bijections Q of (2) carry it to $\psi \mapsto Q(t[n])\circ\psi$ on morphisms of $D(R)$, because Q is a functor, and that map is bijective with inverse $\psi \mapsto Q(t[n])^{-1}\circ\psi$, since $t[n]$ is a quasi-isomorphism, $H^i(t[n])$ being $H^{i+n}(t)$, completing the proof.

K-flatness is defined so that $\mathrm{Tot}(X \otimes_R -)$ carries quasi-isomorphisms to quasi-isomorphisms. The identities in the proof of lemma 3.4.14 that relate cones and homotopies to total tensor complexes are identities between elementary tensors, and the bound in that lemma enters only through proposition 2.5.4 in its Step 2. Over a commutative ring the two factors can also be exchanged.

Lemma 5.2.4: The Total Tensor Complex of Unbounded Complexes

Let X and Y be complexes of R-modules.

1. The map $\mathrm{sw}_{X,Y}(x \otimes y) := (-1)^{ij}\, y \otimes x$, for $x \in X^i$ and $y \in Y^j$, is an isomorphism of complexes $\mathrm{Tot}(X \otimes_R Y) \to \mathrm{Tot}(Y \otimes_R X)$. In particular X is K-flat if and only if $\mathrm{Tot}(A \otimes_R X)$ is acyclic for every acyclic complex A.
2. If $f, f' \colon X \to X'$ are homotopic, so are $\mathrm{Tot}(f \otimes g)$ and $\mathrm{Tot}(f' \otimes g)$ for every chain map g, and likewise in the second variable. Hence $\mathrm{Tot}(- \otimes_R -)$ is a functor $\mathrm{K}(R\text{-Mod}) \times \mathrm{K}(R\text{-Mod}) \to \mathrm{K}(R\text{-Mod})$.
3. If X is K-flat and $t \colon Y \to Y'$ is a quasi-isomorphism, then $\mathrm{Tot}(X \otimes t)$ and $\mathrm{Tot}(t \otimes X)$ are quasi-isomorphisms.
4. If X is K-flat and $u \colon X \to X'$ is a homotopy equivalence, then X' is K-flat.

***Proof*:**

1. As R is commutative, $(x, y) \mapsto (-1)^{ij} y \otimes x$ is R-bilinear on $X^i \times Y^j$, so $\mathrm{sw}_{X,Y}$ is defined summand by summand, and $\mathrm{sw}_{Y,X}$ is its inverse, since $(-1)^{ij}(-1)^{ji} = 1$. For $x \in X^i$ and $y \in Y^j$, definition 3.4.12 gives
$$\mathrm{sw}\big(d(x \otimes y)\big) = \mathrm{sw}\big(dx \otimes y + (-1)^i x \otimes dy\big) = (-1)^{(i+1)j}\, y \otimes dx + (-1)^{i+i(j+1)}\, dy \otimes x$$
and
$$d\big(\mathrm{sw}(x \otimes y)\big) = (-1)^{ij}\big(dy \otimes x + (-1)^j y \otimes dx\big)$$
The exponents agree, $(i+1)j \equiv ij + j$ and $i + i(j+1) \equiv ij \pmod 2$, so $\mathrm{sw}_{X,Y}$ is a chain map. Applied with $Y = A$ it shows that $\mathrm{Tot}(X \otimes_R A)$ and $\mathrm{Tot}(A \otimes_R X)$ are isomorphic complexes.
2. Step 1 of the proof of lemma 3.4.14 writes down, for a homotopy h from f to f', a map $H(x \otimes y) = h(x) \otimes g(y)$ with $\mathrm{Tot}\big((f - f') \otimes g\big) = dH + Hd$, and the analogous map in the second variable. Both are defined and verified on elementary tensors, and an element of $\mathrm{Tot}(X \otimes_R Y)^n$ is a finite sum of elementary tensors, so the verification holds with no bound on X or Y. Composition of chain maps is preserved because $\mathrm{Tot}\big((f_1 f_2) \otimes (g_1 g_2)\big) = \mathrm{Tot}(f_1 \otimes g_1)\mathrm{Tot}(f_2 \otimes g_2)$.
3. Step 4 of the proof of lemma 3.4.14 defines an isomorphism of complexes $\theta \colon \mathrm{Tot}\big(X \otimes_R MC(t)\big) \to MC\big(\mathrm{Tot}(X \otimes t)\big)$ by a formula on elementary tensors, and the verification there uses no bound. As t is a quasi-isomorphism, $MC(t)$ is acyclic by proposition 2.1.26, so $\mathrm{Tot}(X \otimes_R MC(t))$ is acyclic because X is K-flat, and $\mathrm{Tot}(X \otimes t)$ is a quasi-isomorphism by proposition 2.1.26. Step 3 of the same proof identifies $MC\big(\mathrm{Tot}(t \otimes X)\big)$ with $\mathrm{Tot}\big(MC(t) \otimes_R X\big)$, again by a computation on elementary tensors, and this complex is acyclic by (1). So $\mathrm{Tot}(t \otimes X)$ is a quasi-isomorphism.
4. Let v be a homotopy inverse of u and A an acyclic complex. By (2), $\mathrm{Tot}(u \otimes \mathrm{id}_A)$ is a homotopy equivalence with homotopy inverse $\mathrm{Tot}(v \otimes \mathrm{id}_A)$, so it induces isomorphisms on cohomology by proposition 2.1.33. Since $\mathrm{Tot}(X \otimes_R A)$ is acyclic, so is $\mathrm{Tot}(X' \otimes_R A)$, completing the proof.

For a module M and $p \in \mathbb{Z}$, write $M[-p]$ for the complex with M in degree p and 0 elsewhere, as in (3.1).

Proposition 5.2.5: Unions and Bounded-Above Complexes of Flat Modules Are K-Flat

1. Let X be a complex of R-modules and $X_0 \subset X_1 \subset \cdots$ subcomplexes with $X = \bigcup_m X_m$. Suppose that every X_m is K-flat, and that $X_m^i \otimes_R N \to X^i \otimes_R N$ is injective for all m and i and every R-module N. Then X is K-flat.
2. Every bounded-above complex of flat R-modules is K-flat.

***Proof*:**

1. Let A be acyclic. The chain map $\mathrm{Tot}(X_m \otimes_R A) \to \mathrm{Tot}(X \otimes_R A)$ induced by the inclusion is, in each degree, a direct sum of the maps $X_m^i \otimes_R A^j \to X^i \otimes_R A^j$, which are injective by hypothesis. Identify $\mathrm{Tot}(X_m \otimes_R A)$ with its image T_m, a subcomplex. Then $T_0 \subset T_1 \subset \cdots$, and every element of $\mathrm{Tot}(X \otimes_R A)^n$ lies in some T_m: it is a finite sum of elementary tensors $x \otimes a$, and each x lies in some X_m. Let z be a cocycle of $\mathrm{Tot}(X \otimes_R A)$ of degree n, and choose m with $z \in T_m^n$. Then z is a cocycle of T_m, which is acyclic because X_m is K-flat, so $z = dw$ for some $w \in T_m^{n-1}$, and z is a coboundary of $\mathrm{Tot}(X \otimes_R A)$.
2. Let F be a bounded-above complex of flat modules and A an acyclic complex.

   Step 1: one flat module. For a flat module M in degree p, $\mathrm{Tot}(M[-p] \otimes_R A)^n = M \otimes_R A^{n-p}$, with differential $(-1)^p(1 \otimes d_A)$ by definition 3.4.12. The exact functor $M \otimes_R -$ carries the acyclic complex A to an acyclic complex, and the sign changes no kernel or image, so $\mathrm{Tot}(M[-p] \otimes_R A)$ is acyclic.

   Step 2: bounded complexes. Suppose $F^i = 0$ outside $a \le i \le b$, and induct on $b - a$, the case $b = a$ being Step 1. As $F^{b+1} = 0$, the top term $F^b[-b]$ is a subcomplex of F, and the quotient F'' has the terms F^i for $a \le i \le b-1$. In each degree $0 \to F^b[-b] \to F \to F'' \to 0$ is split, so it remains split exact after applying $- \otimes_R A^j$, and

   $$0 \to \mathrm{Tot}\big(F^b[-b] \otimes_R A\big) \to \mathrm{Tot}(F \otimes_R A) \to \mathrm{Tot}(F'' \otimes_R A) \to 0$$

   is a short exact sequence of complexes. The outer complexes are acyclic by Step 1 and by induction, so the long exact sequence of theorem 2.1.20 makes the middle one acyclic.

   Step 3: bounded-above complexes. Let $X_m \subset F$ be the subcomplex with $X_m^i = F^i$ for $i \ge -m$ and $X_m^i = 0$ for $i < -m$. It is a subcomplex because the differential raises degree, it is bounded because F is bounded above, and its terms are flat, so it is K-flat by Step 2. The X_m increase with union F, and each $X_m^i \to F^i$ is an identity or the inclusion of 0, so it stays injective after $- \otimes_R N$. Part (1) makes F K-flat, completing the proof.

Part (2) is [22, Tag 064K]. Part (1) is a special case of [22, Tag 06Y3], which shows, with no injectivity hypothesis, that every filtered colimit of K-flat complexes is K-flat, using that filtered colimits of

modules are exact and commute with tensor products. The injectivity hypothesis in (1) makes each $\mathrm{Tot}(X_m \otimes_R A)$ a subcomplex of $\mathrm{Tot}(X \otimes_R A)$, so that a cocycle can be treated inside one of them.

Example 5.2.6: An Acyclic Free Complex over $\mathbb{Z}/4$ That Is Not K-Flat

Let I be the complex of example 5.1.7, as in example 5.2.2(3), and let P be the complex

$$\cdots \xrightarrow{2} \mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4 \to 0$$

with the last $\mathbb{Z}/4$ in degree 0, and $\pi\colon P \to \mathbb{Z}/2$ the reduction map in degree 0. The kernel of the reduction is $2\mathbb{Z}/4$, the image of the last differential, and in every negative degree the kernel and the image of multiplication by 2 are $2\mathbb{Z}/4$, so π is a quasi-isomorphism. Now P is a bounded-above complex of free, hence flat, modules ([8, Projective Modules Are Flat]), so it is K-flat by proposition 5.2.5(2). As I is acyclic, $\mathrm{Tot}(P \otimes I)$ is acyclic, and so is $\mathrm{Tot}(I \otimes P)$ by lemma 5.2.4(1). On the other hand $\mathrm{Tot}(I \otimes \mathbb{Z}/2)$ has $\mathbb{Z}/4 \otimes \mathbb{Z}/2 \cong \mathbb{Z}/2$ in every degree with differential multiplication by 2, which is zero on $\mathbb{Z}/2$, so its cohomology is $\mathbb{Z}/2$ in every degree. Hence $\mathrm{Tot}(I \otimes \pi)$ is not a quasi-isomorphism, and I is not K-flat by lemma 5.2.4(3).

The resolutions are built from the canonical truncations $\tau^{\leq n}X$ of definition 2.1.12. Each is bounded above, so it has a projective resolution by theorem 2.2.17, and the resolutions can be chosen so that each is a degreewise direct summand of the next. K-projectivity of their union then reduces to the acyclicity of the limit of a tower of acyclic complexes in the sense of definition 5.1.15, which lemma 5.1.17 gives.

Theorem 5.2.7: Every Complex of Modules Has a K-Projective Resolution

Let R be a commutative ring.

1. Every complex X of R-modules has a K-projective resolution $\pi\colon P \to X$ in which every P^i is projective and P is also K-flat.
2. Every K-projective complex of R-modules is K-flat.

***Proof*:**

1. *Step 1: extending a resolution along a chain map.* Let $f\colon P' \to B$ be a chain map between bounded-above complexes, with every P'^i projective. The claim is that there are a bounded-above complex S of projectives, a complex P'' with $P''^i = S^{i+1} \oplus P'^i$ containing P' as the subcomplex of elements $(0, y)$, and a quasi-isomorphism $h\colon P'' \to B$ with $h(0, y) = f(y)$. The cone $MC(f)$ is bounded above, so theorem 2.2.17 gives a quasi-isomorphism $g\colon S \to MC(f)[-1]$ with S a bounded-above complex of projectives. By definition 2.1.23 and (3.1), $MC(f)[-1]^i = P'^i \oplus B^{i-1}$ with differential $(p, b) \mapsto (dp,\ -fp - db)$. So $g = (g', g'')$ with $g'^i\colon S^i \to P'^i$ and $g''^i\colon S^i \to B^{i-1}$, and $gd = dg$ reads

$$g'd = dg', \qquad g''d + dg'' = -fg' \tag{5.1}$$

Thus $g'\colon S \to P'$ is a chain map. Let $P'' := MC(-g')$, with $P''^i = S^{i+1} \oplus P'^i$ and $d(x, y) = (-dx,\ -g'x + dy)$, and set $h(x, y) := g''x + fy$. By (5.1),

$$h\big(d(x, y)\big) = -g''dx - fg'x + fdy = dg''x + fdy = d\,h(x, y)$$

so h is a chain map, and $h(0, y) = f(y)$. The cone $MC(h)$ has terms $S^{i+2} \oplus P'^{i+1} \oplus B^i$, the same as $MC(g)[1]$, and the two differentials are

$$d_{MC(h)}(x, y, b) = \big(dx,\ g'x - dy,\ g''x + fy + db\big)$$
$$d_{MC(g)[1]}(x, y, b) = \big(dx,\ -g'x - dy,\ -g''x + fy + db\big)$$

so $(x, y, b) \mapsto (-x, y, b)$ is an isomorphism of complexes $MC(h) \to MC(g)[1]$. As g is a quasi-isomorphism, $MC(g)$ is acyclic by proposition 2.1.26, hence so are $MC(g)[1]$ and $MC(h)$, and h is a quasi-isomorphism by proposition 2.1.26.

Step 2: resolving the truncations. Write $X_n := \tau^{\leq n} X$ for $n \geq 0$, so that $X_0 \subset X_1 \subset \cdots$ are bounded-above subcomplexes of X by definition 2.1.12. Let $f_0 \colon P_0 \to X_0$ be a projective resolution, from theorem 2.2.17. Given a quasi-isomorphism $f_{n-1} \colon P_{n-1} \to X_{n-1}$ from a bounded-above complex of projectives, apply Step 1 to f_{n-1} followed by the inclusion $X_{n-1} \subset X_n$. This gives a bounded-above complex S_n of projectives, a complex P_n with $P_n^i = S_n^{i+1} \oplus P_{n-1}^i$ containing P_{n-1}, and a quasi-isomorphism $f_n \colon P_n \to X_n$ extending f_{n-1}.

Step 3: the union. Let $P := \bigcup_n P_n$, the complex with $P^i = P_0^i \oplus \bigoplus_{n \geq 1} S_n^{i+1}$ whose differential restricts to that of each P_n. A homomorphism out of a direct sum is a family of homomorphisms, and each member lifts along a surjection, so P^i is projective by [8, Equivalencies for Projective Modules](2). The f_n are compatible, so they define a chain map $\pi \colon P \to X$. Fix i. For $n \geq i$, $H^i(P_n) \to H^i(X_n)$ is bijective, and so is $H^i(X_n) \to H^i(X)$ by proposition 2.1.14. A cocycle of P of degree i lies in some P_n with $n \geq i$, so $H^i(\pi)$ is surjective, since $H^i(P_n) \to H^i(X)$ is, and injective: if $\pi(z)$ is a coboundary, then $[z] = 0$ in $H^i(P_n)$, so z is a coboundary in $P_n \subset P$. So π is a quasi-isomorphism.

Step 4: P is K-projective. Let A be acyclic. The complexes $\mathrm{Hom}^\bullet(P_n, A)$ for $n \geq 0$, with the restriction maps $\mathrm{Hom}^\bullet(P_{n+1}, A) \to \mathrm{Hom}^\bullet(P_n, A)$, which are chain maps by lemma 3.4.3(2), form a tower of complexes of R-modules $\mathrm{Hom}^\bullet(P_\bullet, A)$ in the sense of definition 5.1.15. A homomorphism $P^i \to A^{i+k}$ is the same as a family of homomorphisms $P_n^i \to A^{i+k}$ compatible with the restrictions, since $P^i = \bigcup_n P_n^i$, and a product of limits is the limit of the products, so the restrictions $\mathrm{Hom}^\bullet(P, A) \to \mathrm{Hom}^\bullet(P_n, A)$ induce an isomorphism of complexes $\mathrm{Hom}^\bullet(P, A) \cong \varprojlim \mathrm{Hom}^\bullet(P_\bullet, A)$. Each restriction $\mathrm{Hom}^k(P_{n+1}, A) \to \mathrm{Hom}^k(P_n, A)$ is surjective, since a homomorphism on P_n^i extends by 0 on the summand S_{n+1}^{i+1} of P_{n+1}^i. Each P_n is a bounded-above complex of projectives, so it is K-projective by example 5.2.2(1), and $\mathrm{Hom}^\bullet(P_n, A)$ is acyclic. In the exact sequence of lemma 5.1.17 for this tower, the outer terms are $\varprojlim{}^1$ and $\varprojlim$ of towers of zero modules, so they vanish, and $H^m \mathrm{Hom}^\bullet(P, A) = 0$ for every m. In particular $\mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(P, A) = H^0 \mathrm{Hom}^\bullet(P, A) = 0$.

Step 5: P is K-flat. Each P_n is a bounded-above complex of projective, hence flat, modules ([8, Projective Modules Are Flat]), so it is K-flat by proposition 5.2.5(2). Each P_n^i is a direct summand of P^i, so $P_n^i \otimes_R N \to P^i \otimes_R N$ is split injective for every module N, and proposition 5.2.5(1) makes P K-flat.

2. Let P be K-projective, and let $\pi \colon P' \to P$ be the resolution of (1), with P' K-projective and K-flat. By lemma 5.2.3(3), π is a homotopy equivalence, and P is K-flat by lemma 5.2.4(4), completing the proof.

Part (1) is the K-projective half of [21, Theorem C], and Step 1 of its proof is the inductive step in the proof of [21, Lemma 3.3], written out at the level of chain maps. [22, Tag 06XX] constructs a variant whose resolutions are also surjective in each degree. Part (2) is [21, Proposition 5.8], here for modules over a commutative ring. Two K-projective resolutions $\pi\colon P \to Y$ and $\pi'\colon P' \to Y$ are related by a unique morphism $c\colon P \to P'$ of $\mathrm{K}(R\text{-Mod})$ with $Q(\pi')Q(c) = Q(\pi)$, by lemma 5.2.3(2), and the same uniqueness in the other direction makes c an isomorphism of $\mathrm{K}(R\text{-Mod})$, that is a homotopy equivalence.

Example 5.2.8: A K-Projective Resolution of $\mathbb{Z}/2$ in Every Degree

Let $R = \mathbb{Z}$ and let V be the complex with $V^k = \mathbb{Z}/2$ for every $k \in \mathbb{Z}$ and every differential zero. For each k, let E_k be the complex $\mathbb{Z}s_k \xrightarrow{2} \mathbb{Z}t_k$ with s_k in degree $k-1$, t_k in degree k and $ds_k = 2t_k$, and let $P := \bigoplus_k E_k$. Then

$$P^i = \mathbb{Z}t_i \oplus \mathbb{Z}s_{i+1}, \qquad dt_i = 0, \qquad ds_{i+1} = 2t_{i+1}$$

Let $\pi\colon P \to V$ send t_i to 1 and s_{i+1} to 0. It is a chain map, because $\pi(ds_{i+1}) = \pi(2t_{i+1}) = 0$. The cocycles of P in degree i are $\mathbb{Z}t_i$, the coboundaries are $2\mathbb{Z}t_i$, so $H^i(P) \cong \mathbb{Z}/2$ generated by the class of t_i, which π sends to the generator of V^i. So π is a quasi-isomorphism. For an acyclic A, a family of maps out of P^i is a family of maps out of the summands, so $\mathrm{Hom}^\bullet(P, A) = \prod_k \mathrm{Hom}^\bullet(E_k, A)$ as complexes. Each E_k is K-projective by example 5.2.2(1), so each factor is acyclic, and the product is acyclic by lemma 2.1.11. So P is a K-projective resolution of V, and it is K-flat by theorem 5.2.7(2).

Definition 5.2.9: Derived Tensor Product of Unbounded Complexes

For every complex Y of R-modules fix a K-projective resolution $\pi_Y\colon P_Y \to Y$. For a complex X of R-modules, the *derived tensor product* of X and Y is

$$X \otimes^L_R Y := \mathrm{Tot}(X \otimes_R P_Y)$$

regarded as an object of $D(R)$.

The resolutions exist by theorem 5.2.7(1), and every P_Y is K-flat by theorem 5.2.7(2). The formula is that of definition 3.4.15 with the bounded-above projective resolution replaced by a K-projective one, and proposition 5.2.21 identifies the two constructions on bounded-above complexes.

Example 5.2.10: $\mathbb{Z}/2 \otimes^L V$ for $\mathbb{Z}/2$ in Every Degree

Let V and P be as in example 5.2.8, and take $P_V := P$. Then $\mathbb{Z}/2 \otimes^L_{\mathbb{Z}} V = \mathrm{Tot}(\mathbb{Z}/2 \otimes_{\mathbb{Z}} P)$ has $\mathbb{Z}/2\,t_i \oplus \mathbb{Z}/2\,s_{i+1}$ in degree i, and every differential is 0, because $2 = 0$ in $\mathbb{Z}/2$. So

$$H^i\big(\mathbb{Z}/2 \otimes^L_{\mathbb{Z}} V\big) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2 \qquad \text{for every } i \in \mathbb{Z}$$

whereas $\mathrm{Tot}(\mathbb{Z}/2 \otimes_{\mathbb{Z}} V) = V$ has $\mathbb{Z}/2$ in each degree. The class of s_{i+1} spans the cohomology in degree i of the summand $\mathbb{Z}/2 \otimes_{\mathbb{Z}} E_{i+1}$, which is $\mathrm{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/2, \mathbb{Z}/2) \cong \mathbb{Z}/2$ for the summand $\mathbb{Z}/2$ of V in degree $i+1$, placed in degree i as in theorem 3.4.18.

Proposition 5.2.11: The Unbounded Derived Tensor Product Is a Balanced Bifunctor

Let R be a commutative ring.

1. $\otimes^L_R$ is a bifunctor $D(R) \times D(R) \to D(R)$, and another choice of the K-projective resolutions changes it by a natural isomorphism.

2. If $\varphi\colon F \to X$ is a quasi-isomorphism with F K-flat, then both chain maps in

$$X \otimes^L_R Y = \operatorname{Tot}(X \otimes_R P_Y) \xleftarrow{\varphi\otimes 1} \operatorname{Tot}(F \otimes_R P_Y) \xrightarrow{1\otimes \pi_Y} \operatorname{Tot}(F \otimes_R Y)$$

are quasi-isomorphisms, so $X \otimes^L_R Y \cong \operatorname{Tot}(F \otimes_R Y)$ in $D(R)$. If $\psi\colon G \to Y$ is a quasi-isomorphism with G K-flat, then $X \otimes^L_R Y \cong \operatorname{Tot}(X \otimes_R G)$ in $D(R)$. In particular $X \otimes^L_R Y \cong \operatorname{Tot}(X \otimes_R Y)$ whenever X or Y is K-flat.

***Proof*:**

1. *Step 1: the first variable.* Fix Y. By lemma 5.2.4(2), $X \mapsto \operatorname{Tot}(X \otimes_R P_Y)$ is a functor $\mathrm{K}(R\text{-Mod}) \to \mathrm{K}(R\text{-Mod})$, and by lemma 5.2.4(3), P_Y being K-flat, it carries quasi-isomorphisms to quasi-isomorphisms. Followed by Q it inverts them, and $D(R)$ is the localization of $\mathrm{K}(R\text{-Mod})$ at the quasi-isomorphisms by theorem 3.2.4, so it factors uniquely through $D(R)$ by the universal property of definition 1.2.1.

   Step 2: the second variable. Let $\beta\colon Y \to Y'$ be a morphism of $D(R)$. By lemma 5.2.3(2) for the K-projective complex P_Y, there is a unique $\tilde{\beta}\colon P_Y \to P_{Y'}$ in $\mathrm{K}(R\text{-Mod})$ with $Q(\pi_{Y'})\,Q(\tilde{\beta}) = \beta\,Q(\pi_Y)$, namely the one with $Q(\tilde{\beta}) = Q(\pi_{Y'})^{-1}\beta\,Q(\pi_Y)$. For $\beta'\colon Y' \to Y''$, both $\tilde{\beta}'\tilde{\beta}$ and $\widetilde{\beta'\beta}$ satisfy the condition defining $\widetilde{\beta'\beta}$, so they are equal, and $\widetilde{\mathrm{id}_Y} = \mathrm{id}$. Set $X \otimes^L_R \beta := Q\big(\operatorname{Tot}(X \otimes \tilde{\beta})\big)$, a functor of β by lemma 5.2.4(2).

   Step 3: both variables. For $\alpha\colon X \to X'$ in $D(R)$, the square formed by $\alpha \otimes^L_R Y$, $\alpha \otimes^L_R Y'$, $X \otimes^L_R \beta$ and $X' \otimes^L_R \beta$ commutes. For $\alpha = Q(f)$ with f a chain map, both composites are $Q\big(\operatorname{Tot}(f \otimes \tilde{\beta})\big)$. For $\alpha = Q(s)^{-1}$ with s a quasi-isomorphism, the square is that of $Q(s)$ with two parallel sides inverted. Every morphism of $D(R)$ is $Q(f)\,Q(s)^{-1}$ by theorem 3.2.4, and squares of this shape paste along composites. Setting $\alpha \otimes^L_R \beta := (\alpha \otimes^L_R Y') \circ (X \otimes^L_R \beta)$, the commuting squares make $\otimes^L_R$ preserve composites in the pair (α, β), as in Step 3 of the proof of theorem 3.4.8.

   Step 4: other resolutions. Let $\pi'_Y\colon P'_Y \to Y$ be other K-projective resolutions. By lemma 5.2.3(2) there are unique morphisms $c_Y\colon P_Y \to P'_Y$ and $c'_Y\colon P'_Y \to P_Y$ of $\mathrm{K}(R\text{-Mod})$ with $Q(\pi'_Y)Q(c_Y) = Q(\pi_Y)$ and $Q(\pi_Y)Q(c'_Y) = Q(\pi'_Y)$. Both $c'_Y c_Y$ and id_{P_Y} satisfy the condition defining $\widetilde{\mathrm{id}_Y}$, so they are equal, and symmetrically $c_Y c'_Y = \mathrm{id}$. So $\operatorname{Tot}(X \otimes c_Y)$ is an isomorphism of $\mathrm{K}(R\text{-Mod})$. It is natural in X because Tot is a bifunctor, and in Y because $c_{Y'}\tilde{\beta}$ and $\tilde{\beta}'c_Y$, with $\tilde{\beta}'$ built from the primed resolutions, both satisfy $Q(\pi'_{Y'})\,Q(-) = \beta\,Q(\pi_Y)$ and are therefore equal.

2. The two chain maps in the display of (2) are quasi-isomorphisms by lemma 5.2.4(3), the

first because P_Y is K-flat and the second because F is, and inverting the first in $D(R)$ gives $X \otimes^L_R Y \cong \mathrm{Tot}(F \otimes_R Y)$. For the second statement apply the first with $F = P_X$, which is K-flat by theorem 5.2.7(2), to get $X \otimes^L_R Y \cong \mathrm{Tot}(P_X \otimes_R Y)$. The chain maps

$$\mathrm{Tot}(P_X \otimes_R Y) \xleftarrow{1\otimes\psi} \mathrm{Tot}(P_X \otimes_R G) \xrightarrow{\pi_X \otimes 1} \mathrm{Tot}(X \otimes_R G)$$

are quasi-isomorphisms by lemma 5.2.4(3), because P_X and G are K-flat. The last statement is the case $F = X$, $\varphi = \mathrm{id}$, or $G = Y$, $\psi = \mathrm{id}$, completing the proof.

By (2), a K-flat resolution of either variable computes $\otimes^L_R$, so a complex known to be K-flat needs no further resolution. A complex of flat modules need not be K-flat: the complex I of example 5.2.6 has free terms.

Theorem 5.2.12: Complexes of Flat Modules Are K-Flat When Submodules of Flat Modules Are Flat

Let R be a commutative ring in which every submodule of a flat R-module is flat. Then every complex of flat R-modules is K-flat.

***Proof*:**

Step 1: the truncations. Let F be a complex of flat modules, and write Z^m and B^{m+1} for the kernel and the image of $d^m \colon F^m \to F^{m+1}$. Both are submodules of flat modules, hence flat. The canonical truncation $\tau^{\leq m}F$ of definition 2.1.12 has the terms F^i for $i < m$, Z^m in degree m, and 0 above, so it is a bounded-above complex of flat modules, and it is K-flat by proposition 5.2.5(2).

Step 2: the union. The subcomplexes $\tau^{\leq 0}F \subset \tau^{\leq 1}F \subset \cdots$ of F have union F, since F^i is the term of $\tau^{\leq m}F$ in degree i once $m > i$. Fix m, i and a module N. For $i < m$ the map $(\tau^{\leq m}F)^i \otimes_R N \to F^i \otimes_R N$ is the identity, and for $i > m$ its source is 0. For $i = m$ it is the map $Z^m \otimes_R N \to F^m \otimes_R N$ induced by the inclusion. The functor $- \otimes_R N$ is right exact by [8, $D \otimes_R -$ Is Right Exact], the tensor product being symmetric over the commutative ring R, and its first left derived functor is $\mathrm{Tor}^R_1(-, N)$ by example 2.3.6(1). By theorem 2.3.11 and proposition 2.3.13(2), the short exact sequence $0 \to Z^m \to F^m \xrightarrow{d^m} B^{m+1} \to 0$ gives an exact sequence

$$\mathrm{Tor}^R_1(B^{m+1}, N) \to Z^m \otimes_R N \to F^m \otimes_R N$$

By theorem 2.5.8, $\mathrm{Tor}^R_1(B^{m+1}, N) \cong H^{-1}(B^{m+1} \otimes_R P_N)$ for a projective resolution P_N of N. The complex P_N is exact in degree -1, and the functor $B^{m+1} \otimes_R -$ is exact because B^{m+1} is flat, so it preserves kernels and images and $H^{-1}(B^{m+1} \otimes_R P_N) = 0$. Hence $Z^m \otimes_R N \to F^m \otimes_R N$ is injective, and proposition 5.2.5(1) makes F K-flat, completing the proof.

The hypothesis on R is used once, to make the cycle modules Z^m and the boundary modules B^{m+1} of F flat. Over $\mathbb{Z}/4$ it fails at the cycle modules $Z^m = 2\mathbb{Z}/4 \cong \mathbb{Z}/2$ of the complex I of example 5.2.6, which are submodules of the free module $\mathbb{Z}/4$. The module $\mathbb{Z}/2$ is not flat over $\mathbb{Z}/4$: tensoring the injection $\mathbb{Z}/2 \to \mathbb{Z}/4$, $1 \mapsto 2$, with $\mathbb{Z}/2$ gives the map $\mathbb{Z}/2 \to \mathbb{Z}/2$ sending 1 to $2 = 0$. Correspondingly, $Z^m \otimes \mathbb{Z}/2 \to I^m \otimes \mathbb{Z}/2$ is this zero map, so the injectivity hypothesis of proposition 5.2.5(1) fails for the truncations of I.

Corollary 5.2.13: Complexes of Torsion-Free Modules over a PID Are K-Flat

Let R be a principal ideal domain. Then every complex of torsion-free R-modules is K-flat. In particular every complex of torsion-free abelian groups is K-flat.

***Proof*:**
Over a principal ideal domain a module is flat if and only if it is torsion-free, by [8, Flat Equals Torsion-Free Over a PID], and a submodule of a torsion-free module is torsion-free by [8, Extensions And Submodules Of Torsion-Free Modules](1). So every submodule of a flat R-module is flat, a complex of torsion-free R-modules is a complex of flat R-modules, and theorem 5.2.12 makes it K-flat. Abelian groups are the modules over the principal ideal domain $\mathbb{Z}$, completing the proof.

Example 5.2.14: Reduction Modulo n of a Torsion-Free Complex
Let F be a complex of torsion-free abelian groups and $n \geq 1$. By corollary 5.2.13, F is K-flat, so proposition 5.2.11(2) gives

$$F \otimes^{L}_{\mathbb{Z}} \mathbb{Z}/n \cong \operatorname{Tot}(F \otimes_{\mathbb{Z}} \mathbb{Z}/n) \cong F/nF$$

where F/nF has the terms F^k/nF^k and the induced differentials. Each F^k is torsion-free, so multiplication by n is injective on it, and $0 \to F \xrightarrow{n} F \to F/nF \to 0$ is a short exact sequence of complexes. Its long exact sequence (theorem 2.1.20) yields the short exact sequences

$$0 \to H^k(F)/nH^k(F) \to H^k\big(F \otimes^{L}_{\mathbb{Z}} \mathbb{Z}/n\big) \to \{c \in H^{k+1}(F) : nc = 0\} \to 0$$

For a specific case, let $F^k = \mathbb{Z}$ for every k, with d^k multiplication by 2 for k odd and zero for k even. Then $d^{k+1}d^k = 0$, since one of the two factors is zero, and $H^k(F)$ is $\mathbb{Z}/2$ for k even and 0 for k odd. For $n = 2$, the complex $F/2F$ has $\mathbb{Z}/2$ in every degree and every differential zero, so $H^k(F \otimes^{L}_{\mathbb{Z}} \mathbb{Z}/2) \cong \mathbb{Z}/2$ for every k. The sequence above agrees: for k even the left term is $\mathbb{Z}/2$ and the right term is 0, and for k odd the left term is 0 and the right term is $\mathbb{Z}/2$.

The category R-Mod is a Grothendieck abelian category by example 5.1.5(1), so theorem 5.1.19 gives every complex of R-modules a K-injective resolution.

Definition 5.2.15: RHom of Unbounded Complexes

For a complex Z of R-modules let $\iota_Z \colon Z \to I_Z$ be the K-injective resolution fixed in corollary 5.1.22, with $\mathcal{A} = R$-Mod. For a complex Y of R-modules set

$$R\operatorname{Hom}_R(Y, Z) := \operatorname{Hom}^{\bullet}(Y, I_Z)$$

regarded as an object of $D(R)$.

The formula is that of definition 3.4.6 with the bounded-below injective resolution replaced by a K-injective one, and proposition 5.2.21 identifies the two constructions when the second variable is bounded below.

Example 5.2.16: $R\mathrm{Hom}$ out of R and into a K-Injective Complex

1. Let $Y = R$ in degree 0. By definition 3.4.1, $\mathrm{Hom}^n(R, I_Z) = \mathrm{Hom}_R(R, I_Z^n)$ with differential $\varphi \mapsto d_{I_Z}\varphi$, and evaluation at 1 identifies it with I_Z^n. So $R\mathrm{Hom}_R(R, Z)$ is the complex I_Z, which $Q(\iota_Z)$ makes isomorphic to Z in $D(R)$.

2. Let J be a K-injective complex. Then $\iota_J\colon J \to I_J$ is a quasi-isomorphism between K-injective complexes, hence a homotopy equivalence by corollary 5.1.13(1). For every complex Y the chain map $\mathrm{Hom}^\bullet(Y, \iota_J)$ is then a homotopy equivalence by lemma 3.4.3(2), hence a quasi-isomorphism by proposition 2.1.33, and $R\mathrm{Hom}_R(Y, J) \cong \mathrm{Hom}^\bullet(Y, J)$ in $D(R)$. For $R = \mathbb{Z}$, J may be the complex $\mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ of example 3.4.7, in degrees 0 and 1, a bounded-below complex of injective abelian groups and so K-injective by corollary 5.1.12. Then $R\mathrm{Hom}_\mathbb{Z}(Y, J) \cong \mathrm{Hom}^\bullet(Y, J)$ for every complex Y of abelian groups, bounded or not.

Proposition 5.2.17: $R\mathrm{Hom}$ of Unbounded Complexes Is a Bifunctor

Let R be a commutative ring.

1. $R\mathrm{Hom}_R$ is a bifunctor $D(R)^{\mathrm{op}} \times D(R) \to D(R)$, and another choice of the K-injective resolutions changes it by a natural isomorphism.

2. If $\pi\colon P \to Y$ is a K-projective resolution, then both chain maps in

$$R\mathrm{Hom}_R(Y, Z) = \mathrm{Hom}^\bullet(Y, I_Z) \xrightarrow{\pi^*} \mathrm{Hom}^\bullet(P, I_Z) \xleftarrow{(\iota_Z)_*} \mathrm{Hom}^\bullet(P, Z)$$

are quasi-isomorphisms. In particular $R\mathrm{Hom}_R(Y, Z) \cong \mathrm{Hom}^\bullet(P, Z)$ in $D(R)$.

***Proof*:**

1. *Step 1: the second variable.* For a morphism $\gamma\colon Z \to Z'$ of $D(R)$ let $I_\gamma\colon I_Z \to I_{Z'}$ be the morphism of $\mathrm{K}(R\text{-Mod})$ of corollary 5.1.22(1), the unique one with $Q(I_\gamma)\,Q(\iota_Z) = Q(\iota_{Z'})\,\gamma$, a functor of γ by that corollary. Set $R\mathrm{Hom}_R(Y, \gamma) := Q\big(\mathrm{Hom}^\bullet(Y, I_\gamma)\big)$, a functor of γ by lemma 3.4.3(2).

   Step 2: the first variable. Fix Z. By lemma 3.4.3(2), $Y \mapsto \mathrm{Hom}^\bullet(Y, I_Z)$ is a functor $\mathrm{K}(R\text{-Mod})^{\mathrm{op}} \to \mathrm{K}(R\text{-Mod})$, and by corollary 5.1.14 for the K-injective complex I_Z it carries quasi-isomorphisms to quasi-isomorphisms, so followed by Q it carries them to isomorphisms. By theorem 3.2.4, $D(R)^{\mathrm{op}}$ is the localization of $\mathrm{K}(R\text{-Mod})^{\mathrm{op}}$ at the quasi-isomorphisms, and the universal property of definition 1.2.1 factors the functor uniquely through $D(R)^{\mathrm{op}}$. The factorization is $R\mathrm{Hom}_R(-, Z)$.

   Step 3: both variables and other resolutions. Steps 3 and 4 of the proof of theorem 3.4.8 use its Steps 1 and 2, lemma 3.4.3(2), theorem 3.2.4, and the bijectivity of Q into the chosen resolutions, which there is lemma 3.4.4(2). With Steps 1 and 2 above and proposition 5.1.11(3) for that bijectivity, the same arguments show that the square formed by $R\mathrm{Hom}_R(\alpha, Z)$, $R\mathrm{Hom}_R(\alpha, Z')$, $R\mathrm{Hom}_R(Y, \gamma)$ and $R\mathrm{Hom}_R(Y', \gamma)$ commutes for all $\alpha\colon Y' \to Y$ and $\gamma\colon Z \to Z'$ in $D(R)$, so that $R\mathrm{Hom}_R$ is a bifunctor, and that another

choice $\iota'_Z\colon Z \to I'_Z$ of K-injective resolutions changes it by the natural isomorphism $\mathrm{Hom}^\bullet(Y, c_Z)$, where c_Z is the unique morphism of $\mathrm{K}(R\text{-Mod})$ with $Q(c_Z)Q(\iota_Z) = Q(\iota'_Z)$.

2. The map π^* is a quasi-isomorphism by corollary 5.1.14, applied to the quasi-isomorphism π and the K-injective complex I_Z. The map $(\iota_Z)_*$ is one by lemma 5.2.3(1), applied to the K-projective complex P and the quasi-isomorphism ι_Z, completing the proof.

Example 5.2.18: $R\mathrm{Hom}(\mathbb{Z}/2, V)$ for $\mathbb{Z}/2$ in Every Degree

Let $R = \mathbb{Z}$, let V be the complex of example 5.2.8, and let E be the complex $\mathbb{Z}s \xrightarrow{2} \mathbb{Z}t$ in degrees -1 and 0, a K-projective resolution of $\mathbb{Z}/2$ by example 5.2.2(1). By proposition 5.2.17(2), $R\mathrm{Hom}_\mathbb{Z}(\mathbb{Z}/2, V) \cong \mathrm{Hom}^\bullet(E, V)$, and

$$\mathrm{Hom}^m(E, V) = \mathrm{Hom}(\mathbb{Z}s, V^{m-1}) \times \mathrm{Hom}(\mathbb{Z}t, V^m) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2$$

For φ of degree m, the component of $D\varphi$ on s is $d_V\varphi(s) - (-1)^m\varphi(2t) = 0$, since $d_V = 0$ and $2\varphi(t) = 0$ in $\mathbb{Z}/2$, and its component on t is $d_V\varphi(t) = 0$. So every differential vanishes, and

$$H^m R\mathrm{Hom}_\mathbb{Z}(\mathbb{Z}/2, V) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2 \qquad \text{for every } m \in \mathbb{Z}$$

By lemma 5.2.3(2) and lemma 3.4.3(1) this group is also

$$\mathrm{Hom}_{D(\mathbb{Z})}(E, V[m]) \cong \mathrm{Hom}_{D(\mathbb{Z})}(\mathbb{Z}/2, V[m])$$

The complex V is bounded neither below nor above, so definition 3.4.6 does not apply to this pair.

Theorem 5.2.19: The Unbounded Tensor-Hom Adjunction

Let R be a commutative ring and let X, Y, Z be complexes of R-modules. There is an isomorphism

$$\mathrm{Hom}_{D(R)}\big(X \otimes^L_R Y,\ Z\big) \cong \mathrm{Hom}_{D(R)}\big(X,\ R\mathrm{Hom}_R(Y, Z)\big)$$

natural in X and Z.

***Proof*:**

Step 1: the adjunction for complexes. For complexes X, Y, Z, the map

$$\Phi\colon \mathrm{Hom}^\bullet\big(\mathrm{Tot}(X \otimes_R Y), Z\big) \to \mathrm{Hom}^\bullet\big(X, \mathrm{Hom}^\bullet(Y, Z)\big), \qquad \Phi(\varphi)(x)(y) := \varphi(x \otimes y)$$

is an isomorphism of complexes. By definition 3.4.1, the source in degree n is the product over all $m \in \mathbb{Z}$ of $\mathrm{Hom}_R\big(\mathrm{Tot}(X \otimes_R Y)^m, Z^{m+n}\big)$. A homomorphism out of the direct sum $\mathrm{Tot}(X \otimes_R Y)^m = \bigoplus_{i+j=m} X^i \otimes_R Y^j$ is a family of homomorphisms out of its summands, so the source is the product over all pairs (i, j) of $\mathrm{Hom}_R(X^i \otimes_R Y^j, Z^{i+j+n})$. The target in degree n is the product over all i of $\mathrm{Hom}_R\big(X^i, \mathrm{Hom}^{i+n}(Y, Z)\big)$, where $\mathrm{Hom}^{i+n}(Y, Z)$ is the product over all j of $\mathrm{Hom}_R(Y^j, Z^{i+j+n})$, and a homomorphism into a product is a family, so the target is the product over the same pairs of $\mathrm{Hom}_R\big(X^i, \mathrm{Hom}_R(Y^j, Z^{i+j+n})\big)$. On each factor Φ is the bijection of [8, Hom-Tensor Adjunction], with both rings equal to R. The comparison of differentials in Step 1 of the proof of theorem 3.4.19 is a computation on $x \otimes y$ for $x \in X^i$ and $y \in Y^j$, and it uses no bound.

Step 2: resolutions. Let $\pi_X\colon P_X \to X$ and $\pi_Y\colon P_Y \to Y$ be the K-projective resolutions of definition 5.2.9 and $\iota_Z\colon Z \to I_Z$ the K-injective resolution of definition 5.2.15. Set $T := \mathrm{Tot}(P_X \otimes_R P_Y)$ and $H := \mathrm{Hom}^\bullet(P_Y, I_Z)$. By proposition 5.2.11(2), with $F = P_X$ and $\varphi = \pi_X$, the chain map $\pi_X \otimes 1\colon T \to \mathrm{Tot}(X \otimes_R P_Y) = X \otimes^L_R Y$ is a quasi-isomorphism, and by proposition 5.2.17(2) so is $\pi_Y^*\colon R\mathrm{Hom}_R(Y,Z) \to H$.

Step 3: T is K-projective. Let A be acyclic. By Step 1, $\mathrm{Hom}^\bullet(T,A) \cong \mathrm{Hom}^\bullet\big(P_X, \mathrm{Hom}^\bullet(P_Y,A)\big)$. The complex $\mathrm{Hom}^\bullet(P_Y,A)$ is acyclic because P_Y is K-projective, so the right side is acyclic because P_X is K-projective. Hence $\mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(T,A) = H^0\mathrm{Hom}^\bullet(T,A) = 0$.

Step 4: the chain of isomorphisms. Composing with $Q(\pi_X \otimes 1)$ and $Q(\iota_Z)$, then using lemma 5.2.3(2) for T, lemma 3.4.3(1), Step 1, lemma 3.4.3(1) again, lemma 5.2.3(2) for P_X, and finally $Q(\pi_X)$ and $Q(\pi_Y^*)$,

$$\begin{aligned}\mathrm{Hom}_{D(R)}(X \otimes^L_R Y, Z) &\cong \mathrm{Hom}_{D(R)}(T, I_Z) \cong \mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(T, I_Z) = H^0\mathrm{Hom}^\bullet(T, I_Z)\\ &\cong H^0\mathrm{Hom}^\bullet(P_X, H) = \mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(P_X, H) \cong \mathrm{Hom}_{D(R)}(P_X, H)\\ &\cong \mathrm{Hom}_{D(R)}\big(X, R\mathrm{Hom}_R(Y,Z)\big)\end{aligned}$$

Step 5: naturality. A morphism $\alpha\colon X \to X'$ of $D(R)$ lifts, by lemma 5.2.3(2), to a unique $\tilde{\alpha}\colon P_X \to P_{X'}$ in $\mathrm{K}(R\text{-Mod})$ with $Q(\pi_{X'})Q(\tilde{\alpha}) = \alpha\, Q(\pi_X)$. The functor of Step 1 of the proof of proposition 5.2.11 sends $Q(g)$ to $Q(\mathrm{Tot}(g \otimes 1))$ for chain maps g, so applying it to this identity shows that $\alpha \otimes^L_R Y$ acts on T as $Q\big(\mathrm{Tot}(\tilde{\alpha} \otimes 1)\big)$. A morphism $\gamma\colon Z \to Z'$ acts on I_Z as $Q(I_\gamma)$, with I_γ as in Step 1 of the proof of proposition 5.2.17(1), and π_Y^* commutes with $(I_\gamma)_*$. Each isomorphism of Step 4 commutes with precomposition by these and postcomposition by I_γ: the uses of lemma 3.4.3(1) by its part (3), the uses of lemma 5.2.3(2) because Q is a functor, and Φ because $\Phi\big(\varphi \circ \mathrm{Tot}(\tilde{\alpha} \otimes 1)\big) = \Phi(\varphi) \circ \tilde{\alpha}$ and $\Phi(I_\gamma \circ \varphi) = (I_\gamma)_* \circ \Phi(\varphi)$ by the formula for Φ. So the composite isomorphism is natural in X and Z, completing the proof.

For fixed Y, the theorem says that $- \otimes^L_R Y$ is left adjoint to $R\mathrm{Hom}_R(Y,-)$ as functors $D(R) \to D(R)$. In the bounded case of theorem 3.4.19 the two functors have the different domains $D^-(R)$ and $D^+(R)$, which is also the form of Weibel [23] Theorem 10.8.7.

Example 5.2.20: Morphisms out of F/nF into Shifts of $\mathbb{Z}$

Let F be a complex of torsion-free abelian groups, $n \geq 1$ and $k \in \mathbb{Z}$. The complex $C_n = (\mathbb{Z} \xrightarrow{n} \mathbb{Z})$ in degrees -1 and 0 is a K-projective resolution of $\mathbb{Z}/n$ by example 5.2.2(1), so $R\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n, \mathbb{Z}[k]) \cong \mathrm{Hom}^\bullet(C_n, \mathbb{Z}[k])$ by proposition 5.2.17(2). As $\mathbb{Z}[k]$ is $\mathbb{Z}$ in degree $-k$, this complex is $\mathbb{Z} \to \mathbb{Z}$ in degrees $-k$ and $1-k$, with differential multiplication by $-(-1)^k n$, so its only cohomology is $\mathbb{Z}/n$ in degree $1-k$. A complex C whose only cohomology is in degree j is isomorphic in $D(\mathbb{Z})$ to $H^j(C)[-j]$, through $C \leftarrow \tau^{\leq j}C \to \tau^{\geq j}\tau^{\leq j}C$, both quasi-isomorphisms by proposition 2.1.14, the last complex being $Z^j(C)/B^j(C) = H^j(C)$ in degree j. So $R\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n, \mathbb{Z}[k]) \cong (\mathbb{Z}/n)[k-1]$, and with example 5.2.14 the theorem gives

$$\mathrm{Hom}_{D(\mathbb{Z})}\big(F/nF,\ \mathbb{Z}[k]\big) \cong \mathrm{Hom}_{D(\mathbb{Z})}\big(F,\ (\mathbb{Z}/n)[k-1]\big)$$

for every complex F of torsion-free abelian groups, bounded or not. For $F = \mathbb{Z}$ in degree 0 and $k = 1$, both sides are $\mathbb{Z}/n$: the left is $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n, \mathbb{Z}) \cong \mathbb{Z}/n$ by theorem 3.4.10(2) and example 3.4.7,

and the right is $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/n) \cong \mathbb{Z}/n$ by the last statement of theorem 3.4.10(2).

In the commutative case, as explained before theorem 3.4.19, definition 3.4.15 gives $\otimes^{\mathbf{L}}_{R}$ on $D^{-}(R) \times D^{-}(R)$ and definition 3.4.6 gives $R\mathrm{Hom}_{R}$ on $D(R)^{\mathrm{op}} \times D^{+}(R)$.

Proposition 5.2.21: The Unbounded Constructions Extend the Bounded Ones

Let R be a commutative ring.

1. On $D^{-}(R) \times D^{-}(R)$, there is a natural isomorphism from the derived tensor product of definition 3.4.15 to that of definition 5.2.9.
2. On $D(R)^{\mathrm{op}} \times D^{+}(R)$, there is a natural isomorphism from the $R\mathrm{Hom}$ of definition 3.4.6 to that of definition 5.2.15.

***Proof*:**

1. For a bounded-above complex Y, let $\pi^{-}_{Y} \colon P^{-}_{Y} \to Y$ be the projective resolution fixed in definition 3.4.15. It is K-projective by example 5.2.2(1), so the argument of Step 4 of the proof of proposition 5.2.11(1) applies to the two K-projective resolutions P^{-}_{Y} and P_{Y}. It gives an isomorphism $c_{Y} \colon P^{-}_{Y} \to P_{Y}$ of $\mathrm{K}(R\text{-Mod})$ with $Q(\pi_{Y})Q(c_{Y}) = Q(\pi^{-}_{Y})$, and $\mathrm{Tot}(X \otimes c_{Y})$ is an isomorphism from the bounded construction to the unbounded one. It is natural in X: for a chain map f the naturality square commutes because Tot is a bifunctor, and every morphism of $D^{-}(R)$ is a composite of morphisms $Q(f)$ and $Q(s)^{-1}$ of bounded-above complexes, as in Step 3 of the proof of theorem 3.4.17. For $\beta \colon Y \to Y'$ in $D^{-}(R)$, the morphism $\tilde{\beta}^{-} \colon P^{-}_{Y} \to P^{-}_{Y'}$ of Step 2 of the proof of theorem 3.4.17 satisfies $Q(\pi^{-}_{Y'})Q(\tilde{\beta}^{-}) = \beta\, Q(\pi^{-}_{Y})$, so $c_{Y'}\tilde{\beta}^{-}$ and $\tilde{\beta} c_{Y}$ both satisfy $Q(\pi_{Y'})\, Q(-) = \beta\, Q(\pi^{-}_{Y})$, and they are equal by the uniqueness in lemma 5.2.3(2) for the K-projective complex P^{-}_{Y}. This is naturality in Y.
2. For a bounded-below complex Z, let $\iota^{+}_{Z} \colon Z \to I^{+}_{Z}$ be the injective resolution fixed in definition 3.4.6. By proposition 5.1.11(3) for the K-injective complex I_{Z} there is a unique $c_{Z} \colon I^{+}_{Z} \to I_{Z}$ in $\mathrm{K}(R\text{-Mod})$ with $Q(c_{Z})Q(\iota^{+}_{Z}) = Q(\iota_{Z})$, and by lemma 3.4.4(2), whose source complex may be arbitrary, there is a unique $c'_{Z} \colon I_{Z} \to I^{+}_{Z}$ with $Q(c'_{Z})Q(\iota_{Z}) = Q(\iota^{+}_{Z})$. Both $c'_{Z}c_{Z}$ and id satisfy the condition $Q(-)Q(\iota^{+}_{Z}) = Q(\iota^{+}_{Z})$, so they are equal by the uniqueness in lemma 3.4.4(2), and $c_{Z}c'_{Z} = \mathrm{id}$ by the uniqueness for I_{Z}. So $\mathrm{Hom}^{\bullet}(Y, c_{Z})$ is an isomorphism of $\mathrm{K}(R\text{-Mod})$ from the bounded construction to the unbounded one. Naturality in Y holds for morphisms $Q(f)$, with f a chain map, because $\mathrm{Hom}^{\bullet}$ is a bifunctor (lemma 3.4.3(2)), and for morphisms $Q(s)^{-1}$ by inverting, every morphism of $D(R)$ being $Q(f)Q(s)^{-1}$ (theorem 3.2.4). For naturality in Z, let $\gamma \colon Z \to Z'$ be a morphism of $D^{+}(R)$ and $\tilde{\gamma}^{+} \colon I^{+}_{Z} \to I^{+}_{Z'}$ its lift from theorem 3.4.8(1), with $Q(\tilde{\gamma}^{+})Q(\iota^{+}_{Z}) = Q(\iota^{+}_{Z'})\gamma$. Then $c_{Z'}\tilde{\gamma}^{+}$ and $I_{\gamma}c_{Z}$, with I_{γ} as in Step 1 of the proof of proposition 5.2.17(1), both satisfy $Q(-)Q(\iota^{+}_{Z}) = Q(\iota_{Z'})\gamma$. They are equal, because proposition 5.1.11(3) for $I_{Z'}$ makes Q injective on $\mathrm{Hom}_{\mathrm{K}(R\text{-Mod})}(I^{+}_{Z}, I_{Z'})$ and $Q(\iota^{+}_{Z})$ is invertible. Applying $\mathrm{Hom}^{\bullet}(Y, -)$ gives naturality in Z, completing the proof.

By (1) and example 3.4.16, for $m, n \geq 1$ and $d = \gcd(m, n)$, the complex $\mathbb{Z}/m \otimes^{\mathbf{L}}_{\mathbb{Z}} \mathbb{Z}/n$ of definition 5.2.9 has cohomology $\mathbb{Z}/d$ in degrees -1 and 0 and 0 elsewhere.

Exercise 5.2.1

1. Let V be the complex of abelian groups of example 5.2.8, with $V^k = \mathbb{Z}/2$ for every k and every differential zero. Let A be the acyclic complex $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2$ in degrees -1, 0, 1. Compute the cohomology of $\mathrm{Tot}(V \otimes_{\mathbb{Z}} A)$ and conclude that V is not K-flat. Deduce from theorem 5.2.7(2) that V is not K-projective either.

2. Show that a direct sum of K-projective complexes is K-projective, and that a direct sum of K-flat complexes is K-flat. (Compute $\mathrm{Hom}^\bullet\big(\bigoplus_\alpha P_\alpha, A\big)$ and $\mathrm{Tot}\big((\bigoplus_\alpha X_\alpha) \otimes_R A\big)$ and use lemma 2.1.11.)

3. Let $0 \to X' \to X \to X'' \to 0$ be a short exact sequence of complexes of R-modules that is split in each degree. Show that if two of X', X, X'' are K-flat, so is the third. Show that $\mathrm{Tot}(X[1] \otimes_R A) = \mathrm{Tot}(X \otimes_R A)[1]$ as complexes, and deduce that the mapping cone of a chain map between K-flat complexes is K-flat.

4. Let k be a field, $\Lambda = k[x]/(x^2)$, and let I be the complex of Λ-modules of exercise 3.4.1 (1), with $I^m = \Lambda$ for every $m \in \mathbb{Z}$ and every differential multiplication by x. Using a bounded-above free resolution of $k = \Lambda/x\Lambda$, show that I is not K-flat, and deduce from theorem 5.2.7(2) that it is not K-projective. Show that the cycle module $x\Lambda \cong k$ of I is not a flat Λ-module, so that Λ does not satisfy the hypothesis of theorem 5.2.12.

5. Show that $R \otimes^L_R Y \cong Y$ in $D(R)$ for every complex Y, and that $\mathrm{Hom}_{D(R)}(R, W) \cong H^0(W)$ for every complex W. Applying theorem 5.2.19 with $X = R[-n]$, deduce that

$$H^n R\mathrm{Hom}_R(Y, Z) \cong \mathrm{Hom}_{D(R)}\big(Y, Z[n]\big)$$

   for all complexes Y and Z of R-modules and every $n \in \mathbb{Z}$.

6. Let F be a complex of torsion-free abelian groups and $n \geq 1$. Show that $F \otimes^L_{\mathbb{Z}} \mathbb{Z}/n \cong 0$ in $D(\mathbb{Z})$ if and only if multiplication by n is bijective on $H^k(F)$ for every $k \in \mathbb{Z}$. Check the criterion on $F = \mathbb{Q}$ in degree 0 and on $F = (\mathbb{Z} \xrightarrow{n} \mathbb{Z})$ in degrees -1 and 0.

7. With V and its resolution P as in example 5.2.8, show that for every $n \in \mathbb{Z}$

$$\mathrm{Hom}_{\mathrm{K}(\mathbb{Z}\text{-Mod})}\big(V, V[n]\big) \cong \prod_{k \in \mathbb{Z}} \mathbb{Z}/2, \qquad \mathrm{Hom}_{D(\mathbb{Z})}\big(V, V[n]\big) \cong \prod_{k \in \mathbb{Z}} \big(\mathbb{Z}/2 \oplus \mathbb{Z}/2\big)$$

   (For the second, compute $\mathrm{Hom}^\bullet(E_k, V)$ for each k.)

8. Let X and Y be K-flat complexes of R-modules. Show that $(x \otimes y) \otimes a \mapsto x \otimes (y \otimes a)$ is an isomorphism of complexes $\mathrm{Tot}\big(\mathrm{Tot}(X \otimes_R Y) \otimes_R A\big) \to \mathrm{Tot}\big(X \otimes_R \mathrm{Tot}(Y \otimes_R A)\big)$ for every complex A, with no sign, and conclude that $\mathrm{Tot}(X \otimes_R Y)$ is K-flat. Deduce that $X \otimes^L_R Y$ is represented by the K-flat complex $\mathrm{Tot}(P_X \otimes_R P_Y)$.

5.3 Truncation in the Derived Category

The canonical truncations of definition 2.1.12 cut a complex at a degree n and keep exactly its cohomology in degrees at most n, or in degrees at least n (proposition 2.1.14). In $D(\mathcal{A})$ they become functors onto the full subcategories of complexes whose cohomology vanishes above, or below, a

fixed degree. Every complex sits in a distinguished triangle between its two truncations, no nonzero morphism runs from the lower subcategory to the upper one, and each truncation is adjoint to the inclusion of its target. The complexes with cohomology only in degree 0 form a full subcategory equivalent to $\mathcal{A}$, so $\mathcal{A}$ is recovered from $D(\mathcal{A})$ together with the two subcategories cut out at degree 0, although distinct abelian categories can have equivalent derived categories (box 3.2.13).

Throughout, $\mathcal{A}$ is an arbitrary abelian category. The localization functor of corollary 2.1.17 is $Q\colon \mathrm{Ch}(\mathcal{A}) \to D(\mathcal{A})$. By theorem 3.2.4 every morphism $X \to Y$ of $D(\mathcal{A})$ is represented by a roof (N, s, f), which is the composite of (N, s, id_N), representing $Q(s)^{-1}$, with (N, id_N, f), representing $Q(f)$. Choosing chain maps in the two homotopy classes writes the morphism as $Q(f)Q(s)^{-1}$ with $f\colon N \to Y$ a chain map and $s\colon N \to X$ a quasi-isomorphism, any representative of the class s being a quasi-isomorphism because homotopic maps induce the same map on cohomology (proposition 2.1.31). An object M of $\mathcal{A}$ is regarded as a complex concentrated in degree 0, as in example 3.2.11, and $M[k]$ is then M in degree $-k$ by (3.1).

By proposition 2.1.14, $\tau^{\leq n}X$ has no cohomology above degree n and $\tau^{\geq n}X$ has none below it. The full subcategories of $D(\mathcal{A})$ cut out by these two vanishing conditions are the targets of the truncation functors.

Definition 5.3.1: The Subcategories $D^{\leq n}$ and $D^{\geq n}$

Let $\mathcal{A}$ be an abelian category and $n \in \mathbb{Z}$. Write $D^{\leq n}(\mathcal{A})$ for the full subcategory of $D(\mathcal{A})$ on the complexes X with $H^i(X) = 0$ for all $i > n$, and $D^{\geq n}(\mathcal{A})$ for the full subcategory on the complexes X with $H^i(X) = 0$ for all $i < n$. The full subcategory

$$D^{\leq 0}(\mathcal{A}) \cap D^{\geq 0}(\mathcal{A})$$

of complexes whose cohomology is concentrated in degree 0 is the *heart* of the pair $\big(D^{\leq 0}(\mathcal{A}), D^{\geq 0}(\mathcal{A})\big)$.

When $\mathcal{A}$ is fixed it is dropped from the notation. Membership depends only on the isomorphism class in $D(\mathcal{A})$, since H^i is a functor on $D(\mathcal{A})$ (proposition 3.2.8). The full subcategory $D^{+}(\mathcal{A})$ of box 3.2.5 consists of the complexes with cohomology bounded below, that is, with $H^i(X) = 0$ for all i below some n, so it is the union of the $D^{\geq n}$ over $n \in \mathbb{Z}$. Likewise $D^{-}(\mathcal{A})$, on the complexes with cohomology bounded above, is the union of the $D^{\leq n}$.

Example 5.3.2: The Complexes $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ and $\mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4$

Let X be $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ in degrees -1 and 0, as in example 2.1.13. Then $H^{-1}(X) = \ker(2) = 0$ and $H^0(X) = \mathbb{Z}/2$, so X lies in the heart $D^{\leq 0} \cap D^{\geq 0}$ although $X^{-1} \neq 0$. Example 2.1.13 computes $\tau^{\geq 0}X$ as $\mathbb{Z}/2$ in degree 0, and $p_0\colon X \to \tau^{\geq 0}X$ is a quasi-isomorphism by proposition 2.1.14, because $H^{-1}(X) = 0$. So $X \cong \mathbb{Z}/2$ in $D(\mathbf{Ab})$.

Let Y be $\mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4$ in degrees -1 and 0. Multiplication by 2 on $\mathbb{Z}/4$ has kernel $\{0, 2\}$ and image $\{0, 2\}$, so $H^{-1}(Y) \cong \mathbb{Z}/2$ and $H^0(Y) = (\mathbb{Z}/4)/\{0, 2\} \cong \mathbb{Z}/2$. Thus $Y \in D^{\leq 0} \cap D^{\geq -1}$ and $Y \notin D^{\geq 0}$.

The exact 2-periodic complex $\cdots \to \mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4 \to \cdots$ of box 3.2.5 is nonzero in every degree, and it lies in $D^{\leq n} \cap D^{\geq m}$ for every m and n, its cohomology being zero.

The functors $\tau^{\leq n}$ and $\tau^{\geq n}$ of definition 2.1.12 are defined on $\mathrm{Ch}(\mathcal{A})$. To act on $D(\mathcal{A})$ they must carry quasi-isomorphisms to quasi-isomorphisms, and they do, because each one preserves the cohomology in its range of degrees and has zero cohomology outside it.

Lemma 5.3.3: Truncation on the Derived Category

Let $n \in \mathbb{Z}$.

1. The functors $\tau^{\leq n}, \tau^{\geq n} \colon \mathrm{Ch}(\mathcal{A}) \to \mathrm{Ch}(\mathcal{A})$ carry quasi-isomorphisms to quasi-isomorphisms. Consequently there are unique functors $\tau^{\leq n}, \tau^{\geq n} \colon D(\mathcal{A}) \to D(\mathcal{A})$, written with the same symbols, such that $\tau^{\leq n} \circ Q = Q \circ \tau^{\leq n}$ and $\tau^{\geq n} \circ Q = Q \circ \tau^{\geq n}$.
2. The morphisms $Q(i_n) \colon \tau^{\leq n} X \to X$ and $Q(p_n) \colon X \to \tau^{\geq n} X$ are natural in $X \in D(\mathcal{A})$.
3. For every X, $\tau^{\leq n} X \in D^{\leq n}$ and $\tau^{\geq n} X \in D^{\geq n}$. Moreover i_n is a quasi-isomorphism, equivalently $Q(i_n)$ is an isomorphism, if and only if $X \in D^{\leq n}$, and p_n is a quasi-isomorphism, equivalently $Q(p_n)$ is an isomorphism, if and only if $X \in D^{\geq n}$.

***Proof*:**

1. Let $f \colon X \to Y$ be a quasi-isomorphism, and write i_n^X for the map i_n of the complex X. The chain map $\tau^{\leq n} f$ is f below degree n and the restriction of f^n to $Z^n(X)$ in degree n, so $i_n^Y \circ \tau^{\leq n} f = f \circ i_n^X$. For $i \leq n$, the maps $H^i(i_n^X)$ and $H^i(i_n^Y)$ are isomorphisms by proposition 2.1.14, and so is $H^i(f)$, hence so is
$$H^i(\tau^{\leq n} f) = H^i(i_n^Y)^{-1} \circ H^i(f) \circ H^i(i_n^X)$$
For $i > n$, proposition 2.1.14 gives $H^i(\tau^{\leq n} X) = 0 = H^i(\tau^{\leq n} Y)$, and the zero map between zero objects is an isomorphism. The chain map $\tau^{\geq n} f$ is the map $X^n/B^n(X) \to Y^n/B^n(Y)$ induced by f^n in degree n and f above, so $p_n^Y \circ f = \tau^{\geq n} f \circ p_n^X$, and the same argument with the range $i \geq n$ treats $\tau^{\geq n}$. So $Q \circ \tau^{\leq n}$ carries quasi-isomorphisms to isomorphisms, and the universal property of corollary 2.1.17 gives a unique functor $\tau^{\leq n}$ on $D(\mathcal{A})$ with $\tau^{\leq n} \circ Q = Q \circ \tau^{\leq n}$. The same holds for $\tau^{\geq n}$.
2. Write a morphism $\varphi \colon X \to Y$ of $D(\mathcal{A})$ as $Q(f)Q(s)^{-1}$ with $s \colon N \to X$ a quasi-isomorphism and $f \colon N \to Y$ a chain map. Applying Q to the naturality squares of i_n in $\mathrm{Ch}(\mathcal{A})$ for f and for s, and using (1), gives
$$Q(i_n^Y) \circ \tau^{\leq n} Q(f) = Q(f) \circ Q(i_n^N), \qquad Q(i_n^X) \circ \tau^{\leq n} Q(s) = Q(s) \circ Q(i_n^N)$$
In the second equation $Q(s)$ and $\tau^{\leq n} Q(s)$ are isomorphisms. Composing with $Q(s)^{-1}$ on the left and with $(\tau^{\leq n} Q(s))^{-1} = \tau^{\leq n}(Q(s)^{-1})$ on the right gives $Q(i_n^N) \circ \tau^{\leq n}(Q(s)^{-1}) = Q(s)^{-1} \circ Q(i_n^X)$, the naturality square for $Q(s)^{-1}$. Pasting it with the square for $Q(f)$ gives $Q(i_n^Y) \circ \tau^{\leq n} \varphi = \varphi \circ Q(i_n^X)$. The argument for p_n is the same.
3. The first sentence is proposition 2.1.14. A chain map f is a quasi-isomorphism exactly when $Q(f)$ is an isomorphism: Q inverts quasi-isomorphisms, and conversely if $Q(f)$ is an isomorphism then so is $H^i(Q(f)) = H^i(f)$ for every i, since H^i is a functor on $D(\mathcal{A})$ with $H^i \circ Q = H^i$ (proposition 3.2.8). By proposition 2.1.14, $H^i(i_n)$ is an isomorphism for every $i \leq n$, and for $i > n$ its source $H^i(\tau^{\leq n} X)$ is zero, so $H^i(i_n)$ is an isomorphism exactly when $H^i(X) = 0$. Hence i_n is a quasi-isomorphism if and only if $X \in D^{\leq n}$. The statement for p_n is proved the same way with $i < n$ in place of $i > n$, completing the proof.

By lemma 5.3.3(3), an object X of $D^{\leq n}$ is isomorphic in $D(\mathcal{A})$ to $\tau^{\leq n}X$, a complex that vanishes above degree n, and an object X of $D^{\geq n}$ is isomorphic to $\tau^{\geq n}X$, which vanishes below degree n. Conversely, a complex that vanishes above degree n has no cohomology there. So $D^{\leq n}$ consists of the objects isomorphic in $D(\mathcal{A})$ to a complex that vanishes above degree n, and likewise $D^{\geq n}$ consists of those isomorphic to a complex that vanishes below degree n.

The triangle relating a complex to its two truncations comes from theorem 3.2.10. The subcomplex $\tau^{\leq n}X$ of X has a quotient complex, and that quotient is quasi-isomorphic to $\tau^{\geq n+1}X$. The vanishing of morphisms from $D^{\leq n}$ to $D^{\geq n+1}$ is proved on a roof, after truncating its apex and its target so that the two have no degree in common.

Proposition 5.3.4: Orthogonality and the Truncation Triangle

Let $\mathcal{A}$ be an abelian category and $n \in \mathbb{Z}$.

1. $D^{\leq n} \subseteq D^{\leq n+1}$ and $D^{\geq n+1} \subseteq D^{\geq n}$. For every complex X, $X \in D^{\leq 0}$ if and only if $X[-n] \in D^{\leq n}$, and $X \in D^{\geq 0}$ if and only if $X[-n] \in D^{\geq n}$.
2. $\mathrm{Hom}_{D(\mathcal{A})}(X, Y) = 0$ for all $X \in D^{\leq n}$ and $Y \in D^{\geq n+1}$.
3. For every complex X there is a morphism $w\colon \tau^{\geq n+1}X \to (\tau^{\leq n}X)[1]$ of $D(\mathcal{A})$ for which

$$\tau^{\leq n}X \xrightarrow{Q(i_n)} X \xrightarrow{Q(p_{n+1})} \tau^{\geq n+1}X \xrightarrow{w} (\tau^{\leq n}X)[1]$$

   is a distinguished triangle of $D(\mathcal{A})$.

***Proof*:**

1. A complex with $H^i(X) = 0$ for all $i > n$ has $H^i(X) = 0$ for all $i > n+1$, and one with $H^i(X) = 0$ for all $i < n+1$ has $H^i(X) = 0$ for all $i < n$. For the second sentence, $H^i(X[-n]) = H^{i-n}(X)$ by (3.1), and $i - n > 0$ exactly when $i > n$, while $i - n < 0$ exactly when $i < n$.
2. Let $\varphi\colon X \to Y$ be a morphism of $D(\mathcal{A})$, and write $\varphi = Q(f)Q(s)^{-1}$ with $s\colon N \to X$ a quasi-isomorphism and $f\colon N \to Y$ a chain map. Since s is a quasi-isomorphism, $H^i(N) \cong H^i(X) = 0$ for $i > n$, so $N \in D^{\leq n}$, and $i := i_n^N\colon \tau^{\leq n}N \to N$ is a quasi-isomorphism by lemma 5.3.3(3). Likewise $p := p_{n+1}^Y\colon Y \to \tau^{\geq n+1}Y$ is a quasi-isomorphism, Y lying in $D^{\geq n+1}$. The chain map $p \circ f \circ i\colon \tau^{\leq n}N \to \tau^{\geq n+1}Y$ is zero in every degree k: for $k > n$ its source $(\tau^{\leq n}N)^k$ is zero, and for $k \leq n$ its target $(\tau^{\geq n+1}Y)^k$ is zero. Hence

$$Q(p) \circ \varphi \circ Q(s) \circ Q(i) = Q(p) \circ Q(f) \circ Q(i) = Q(pfi) = 0$$

   and since $Q(p)$, $Q(s)$ and $Q(i)$ are isomorphisms, $\varphi = 0$.
3. The chain map $i_n\colon \tau^{\leq n}X \to X$ is a monomorphism in every degree: it is the identity below degree n, the inclusion $Z^n \to X^n$ in degree n, and the zero map out of 0 above. Let C be its cokernel, computed degreewise, so that

$$0 \to \tau^{\leq n}X \xrightarrow{i_n} X \xrightarrow{\pi} C \to 0$$

   is a short exact sequence in $\mathrm{Ch}(\mathcal{A})$. Explicitly $C^k = 0$ for $k < n$, $C^n = X^n/Z^n$ and $C^k = X^k$ for $k > n$, with $d_C^n(x + Z^n) = d^n x$, well defined because $Z^n = \ker d^n$, and

$d_C^k = d^k$ for $k > n$. By theorem 3.2.10 there is a morphism $w' \colon C \to (\tau^{\leq n} X)[1]$ of $D(\mathcal{A})$ for which $\big(Q(i_n), Q(\pi), w'\big)$ is a distinguished triangle.

Step 1: the comparison map. Define $q \colon C \to \tau^{\geq n+1} X$ by $q^k = 0$ for $k \leq n$, $q^{n+1} \colon X^{n+1} \to X^{n+1}/B^{n+1}$ the quotient map, and $q^k = \mathrm{id}$ for $k \geq n+2$. Below degree n both complexes are zero, and above degree $n+1$ every component is an identity. In the square leaving degree n, $q^{n+1}(d_C^n(x + Z^n)) = d^n x + B^{n+1} = 0$ because $d^n x \in B^{n+1}$, matching $d^n_{\tau^{\geq n+1}X} \circ q^n = 0$. The square leaving degree $n+1$ commutes because the differential of $\tau^{\geq n+1} X$ in that degree is induced by d^{n+1}. Comparing degree by degree with definition 2.1.12, $q \circ \pi = p_{n+1}$.

Step 2: q is a quasi-isomorphism. For $i < n$, $C^i = 0$. In degree n the map d_C^n is injective, since $d^n x = 0$ forces $x \in Z^n$, and nothing arrives from $C^{n-1} = 0$, so $H^n(C) = 0$. Thus $H^i(C) = 0$ for $i \leq n$, and $H^i(\tau^{\geq n+1} X) = 0$ for $i \leq n$ by proposition 2.1.14, so $H^i(q)$ is an isomorphism for $i \leq n$. For $i \geq n+1$ the map $H^i(\pi)$ is the identity: π is the identity in degrees above n, the cycles of C in degree $n+1$ are Z^{n+1}, and the boundaries there are $d_C^n(C^n) = d^n(X^n) = B^{n+1}$. Since $H^i(q) \circ H^i(\pi) = H^i(p_{n+1})$, and $H^i(p_{n+1})$ is an isomorphism for $i \geq n+1$ by proposition 2.1.14, $H^i(q)$ is an isomorphism for $i \geq n+1$ as well.

Step 3: the triangle. Set $w := w' \circ Q(q)^{-1}$. The triple $\big(\mathrm{id}, \mathrm{id}, Q(q)\big)$ is a morphism of triangles from $\big(Q(i_n), Q(\pi), w'\big)$ to $\big(Q(i_n), Q(p_{n+1}), w\big)$: the middle square is $Q(q) \circ Q(\pi) = Q(p_{n+1})$ by Step 1, and the third is $w \circ Q(q) = w'$ by the definition of w. All three of its components are isomorphisms, so it is an isomorphism of triangles, and distinguished triangles are closed under isomorphism by (TR1) of definition 3.1.3, $D(\mathcal{A})$ being triangulated by theorem 3.2.7. This completes the proof.

Properties (1), (2) and (3), together with $\tau^{\leq n} X \in D^{\leq n}$ and $\tau^{\geq n+1} X \in D^{\geq n+1}$ from lemma 5.3.3(3), are the three axioms of a t-structure in the form of Gelfand-Manin [15, Def. IV.4.2], and section 5.4 develops them on an arbitrary triangulated category, with the pair $(D^{\leq 0}, D^{\geq 0})$ as the standard example. The proposition is Gelfand-Manin [15, Prop. IV.4.3], whose proof of (2) replaces the apex K of a roof by $\tau^{\leq 0} K$ and its target Y by the quasi-isomorphic quotient complex $Y/\tau^{\leq 0} Y$.

Orthogonality already constrains morphisms between objects of $\mathcal{A}$. For $M, N \in \mathcal{A}$ and $k < 0$, the complex $N[k]$ is N in degree $-k \geq 1$, so $N[k] \in D^{\geq 1}$, while $M \in D^{\leq 0}$, and (2) gives $\mathrm{Hom}_{D(\mathcal{A})}(M, N[k]) = 0$. When $\mathcal{A}$ has enough injectives this is the vanishing of negative degrees in theorem 3.4.10(2), and here it holds with no hypothesis on $\mathcal{A}$.

Example 5.3.5: The Truncation Triangle of $\mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4$

Let Y be $\mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4$ in degrees -1 and 0, as in example 5.3.2, and take $n = -1$. The cycles in degree -1 are $\{0, 2\}$, so $\tau^{\leq -1} Y$ is $\{0, 2\} \cong \mathbb{Z}/2$ in degree -1, which is the complex $\mathbb{Z}/2[1]$. The quotient complex C of the proof of proposition 5.3.4(3) is $(\mathbb{Z}/4)/\{0, 2\} \cong \mathbb{Z}/2$ in degree -1 followed by $\mathbb{Z}/4$ in degree 0, joined by the map $1 \mapsto 2$. The truncation $\tau^{\geq 0} Y$ is $(\mathbb{Z}/4)/\{0, 2\} \cong \mathbb{Z}/2$ in degree 0, and the comparison map $q \colon C \to \tau^{\geq 0} Y$ is the reduction $\mathbb{Z}/4 \to \mathbb{Z}/2$ in degree 0. It is surjective in each degree, and its kernel is the complex $\mathbb{Z}/2 \to \{0, 2\}$, $1 \mapsto 2$, whose differential is an isomorphism, so the kernel is exact and the long exact cohomology sequence of theorem 2.1.20

shows directly that q is a quasi-isomorphism. The triangle reads

$$\mathbb{Z}/2[1] \xrightarrow{Q(i_{-1})} Y \xrightarrow{Q(p_0)} \mathbb{Z}/2 \xrightarrow{w} \mathbb{Z}/2[2]$$

By proposition 2.1.14, $H^{-1}(i_{-1})$ is an isomorphism $H^{-1}(\mathbb{Z}/2[1]) = \mathbb{Z}/2 \to H^{-1}(Y)$ and $H^0(p_0)$ is an isomorphism $H^0(Y) \to H^0(\mathbb{Z}/2) = \mathbb{Z}/2$, while $H^0(\mathbb{Z}/2[1]) = 0$ and $H^{-1}(\mathbb{Z}/2) = 0$.

Combining proposition 5.3.4(2) with the triangle of (3) shows that every morphism from an object of $D^{\leq n}$ into X factors uniquely through $Q(i_n)$, and every morphism from X into an object of $D^{\geq n}$ factors uniquely through $Q(p_n)$.

Corollary 5.3.6: Truncation Is Adjoint to Inclusion

Let $n \in \mathbb{Z}$ and $X \in D(\mathcal{A})$.

1. For every $A \in D^{\leq n}$, composition with $Q(i_n)$ is a bijection
$$\mathrm{Hom}_{D(\mathcal{A})}(A, \tau^{\leq n}X) \xrightarrow{\sim} \mathrm{Hom}_{D(\mathcal{A})}(A, X)$$
natural in A and X. That is, $\tau^{\leq n}\colon D(\mathcal{A}) \to D^{\leq n}$ is right adjoint to the inclusion $D^{\leq n} \to D(\mathcal{A})$.

2. For every $B \in D^{\geq n}$, precomposition with $Q(p_n)$ is a bijection
$$\mathrm{Hom}_{D(\mathcal{A})}(\tau^{\geq n}X, B) \xrightarrow{\sim} \mathrm{Hom}_{D(\mathcal{A})}(X, B)$$
natural in X and B. That is, $\tau^{\geq n}\colon D(\mathcal{A}) \to D^{\geq n}$ is left adjoint to the inclusion $D^{\geq n} \to D(\mathcal{A})$.

***Proof*:**

1. Write $T := \tau^{\leq n}X$ and $S := \tau^{\geq n+1}X$. Since $D(\mathcal{A})$ is triangulated (theorem 3.2.7), $\mathrm{Hom}(A, -)$ is cohomological by proposition 3.1.8. Applied to the triangle of proposition 5.3.4(3) and to its rotation $\big(-w[-1], Q(i_n), Q(p_{n+1})\big)$ on $(S[-1], T, X)$, which is distinguished by (TR2), this gives exactness of
$$\mathrm{Hom}(A, S[-1]) \to \mathrm{Hom}(A, T) \xrightarrow{Q(i_n)\circ -} \mathrm{Hom}(A, X) \to \mathrm{Hom}(A, S)$$
at the two middle terms. Here $S \in D^{\geq n+1}$ by lemma 5.3.3(3), and $S[-1] \in D^{\geq n+2} \subseteq D^{\geq n+1}$ because $H^i(S[-1]) = H^{i-1}(S)$. Since $A \in D^{\leq n}$, both outer groups vanish by proposition 5.3.4(2), so composition with $Q(i_n)$ is injective and surjective. Naturality in A holds because composing on the left commutes with composing on the right. Naturality in X is the equation $Q(i_n^{X'}) \circ \tau^{\leq n}\varphi = \varphi \circ Q(i_n^X)$ of lemma 5.3.3(2), for every $\varphi\colon X \to X'$. Finally $\tau^{\leq n}X$ lies in $D^{\leq n}$ by lemma 5.3.3(3).

2. Write $T := \tau^{\leq n-1}X$ and $S := \tau^{\geq n}X$, and use the triangle of proposition 5.3.4(3) for $n-1$, which has vertices T, X, S. The functor $\mathrm{Hom}(-, B)$ is cohomological on $D(\mathcal{A})^{\mathrm{op}}$ by proposition 3.1.8. Applied to that triangle and to its rotation $\big(Q(p_n), w, -Q(i_{n-1})[1]\big)$

on $(X, S, T[1])$, it gives exactness of

$$\mathrm{Hom}(T[1], B) \to \mathrm{Hom}(S, B) \xrightarrow{-\circ Q(p_n)} \mathrm{Hom}(X, B) \to \mathrm{Hom}(T, B)$$

at the two middle terms. Here $T \in D^{\leq n-1}$, and $T[1] \in D^{\leq n-2} \subseteq D^{\leq n-1}$ because $H^i(T[1]) = H^{i+1}(T)$. Since $B \in D^{\geq n}$, both outer groups vanish by proposition 5.3.4(2) applied with $n-1$, so precomposition with $Q(p_n)$ is a bijection. Naturality is checked as in (1), and $\tau^{\geq n}X \in D^{\geq n}$ by lemma 5.3.3(3), completing the proof.

Parts (1) and (2) characterize the pairs $\left(\tau^{\leq n}X, Q(i_n)\right)$ and $\left(\tau^{\geq n}X, Q(p_n)\right)$ by properties of $D(\mathcal{A})$ and the subcategories $D^{\leq n}$ and $D^{\geq n}$, with no reference to the formulas of definition 2.1.12. For a distinguished triangle $A' \xrightarrow{a} X \xrightarrow{b} B' \xrightarrow{c} A'[1]$ with $A' \in D^{\leq n}$ and $B' \in D^{\geq n+1}$, part (1) gives a unique $f\colon A' \to \tau^{\leq n}X$ with $Q(i_n) \circ f = a$. There is also a unique $h\colon B' \to \tau^{\geq n+1}X$ with $h \circ b = Q(p_{n+1})$, both f and h are isomorphisms, and (f, id_X, h) is an isomorphism of triangles from (a, b, c) to the triangle of proposition 5.3.4(3) (exercise 5.3.1 (2)). Truncation functors exist in this form for an arbitrary t-structure, where no complex-level formula is available.

The objects of $\mathcal{A}$ are, up to isomorphism, the complexes with cohomology concentrated in degree 0, and the morphisms between two of them in $D(\mathcal{A})$ are exactly the morphisms of $\mathcal{A}$. A morphism of $D(\mathcal{A})$ between two objects of $\mathcal{A}$ is a roof whose apex is an arbitrary complex. Truncating the apex in degree 0 reduces it to a chain map out of a complex that stops in degree 0, and such a chain map into an object of $\mathcal{A}$ is determined by its effect on H^0.

Theorem 5.3.7: The Heart of the Standard Pair Is $\mathcal{A}$

Let $\mathcal{A}$ be an abelian category, and let $j\colon \mathcal{A} \to D(\mathcal{A})$ send an object M to the complex with M in degree 0 and zero elsewhere, and a morphism f to $Q(f)$.

1. j is fully faithful, and its essential image is the heart $D^{\leq 0} \cap D^{\geq 0}$.
2. $H^0 \circ j = \mathrm{id}_{\mathcal{A}}$, and there is a natural isomorphism $\vartheta\colon \mathrm{id} \to j \circ H^0$ of functors on $D^{\leq 0} \cap D^{\geq 0}$.

In particular $H^0\colon D^{\leq 0} \cap D^{\geq 0} \to \mathcal{A}$ and j are mutually quasi-inverse equivalences, and the heart is an abelian category.

***Proof*:**

Step 1: j lands in the heart, and $H^0 \circ j = \mathrm{id}_{\mathcal{A}}$. The complex $j(M)$ has $H^i(j(M)) = 0$ for $i \neq 0$ and $H^0(j(M)) = M$. For a morphism $f\colon M \to N$, $H^0(Q(f)) = H^0(f) = f$, because $H^0 \circ Q = H^0$ (proposition 3.2.8). In particular j is faithful.

Step 2: chain maps from a truncation into $j(N)$. Let L be a complex and $N \in \mathcal{A}$, and let $u\colon \tau^{\leq 0}L \to j(N)$ be a chain map. Its only possibly nonzero component is $u^0\colon Z^0(L) \to N$. The square leaving degree -1 reads $u^0 \circ d_L^{-1} = 0$, so u^0 vanishes on $B^0(L)$ and factors as $u^0 = \bar{u} \circ \pi_L$, where $\pi_L\colon Z^0(L) \to Z^0(L)/B^0(L)$ is the quotient map. In degree 0 the complex $\tau^{\leq 0}L$ has cycles $Z^0(L)$, since nothing leaves that degree, and boundaries $d_L^{-1}(L^{-1}) = B^0(L)$, so $H^0(\tau^{\leq 0}L) = Z^0(L)/B^0(L)$ and $H^0(u) = \bar{u}$. Thus $u^0 = H^0(u) \circ \pi_L$, and two chain maps $\tau^{\leq 0}L \to j(N)$ inducing the same map on H^0 are equal.

Step 3: j is full. Let $\varphi\colon j(M) \to j(N)$ be a morphism of $D(\mathcal{A})$, and write $\varphi = Q(g)Q(s)^{-1}$ with $s\colon L \to j(M)$ a quasi-isomorphism and $g\colon L \to j(N)$ a chain map. Then $H^k(L) \cong H^k(j(M)) = 0$ for $k > 0$, so $i := i_0^L\colon \tau^{\leq 0}L \to L$ is a quasi-isomorphism by lemma 5.3.3(3). The map $H^0(s)\colon H^0(L) \to M$ is an isomorphism, and we set

$$f := H^0(g) \circ H^0(s)^{-1}\colon M \to N$$

Regard f as a chain map $j(M) \to j(N)$. The chain maps $f \circ s \circ i$ and $g \circ i$ from $\tau^{\leq 0}L$ to $j(N)$ induce $f \circ H^0(s) \circ H^0(i)$ and $H^0(g) \circ H^0(i)$ on H^0, and these agree because $f \circ H^0(s) = H^0(g)$. By Step 2, $fsi = gi$ as chain maps. Applying Q and cancelling the isomorphisms $Q(i)$ and $Q(s)$ gives $Q(f) = Q(g)Q(s)^{-1} = \varphi$, so $\varphi = j(f)$. Together with Step 1, j is fully faithful.

Step 4: every object of the heart is isomorphic to one in the image. Let X lie in the heart and put $Y := \tau^{\leq 0}X$. The map $i := i_0^X\colon Y \to X$ is a quasi-isomorphism by lemma 5.3.3(3), since $X \in D^{\leq 0}$. For $k \leq 0$, $H^k(Y) \cong H^k(X)$ by proposition 2.1.14, and this vanishes for $k < 0$, so $Y \in D^{\geq 0}$ and $p := p_0^Y\colon Y \to \tau^{\geq 0}Y$ is a quasi-isomorphism as well. The complex $\tau^{\geq 0}Y$ is $Y^0/B^0(Y)$ in degree 0 and zero elsewhere, and $Y^0 = Z^0(X)$ and $B^0(Y) = d^{-1}(X^{-1}) = B^0(X)$, so $\tau^{\geq 0}\tau^{\leq 0}X = j\big(H^0(X)\big)$. Set

$$\vartheta_X := Q(p) \circ Q(i)^{-1}\colon X \to j\big(H^0(X)\big)$$

an isomorphism of $D(\mathcal{A})$. Both $H^0(i)$ and $H^0(p)$ are the identity of $Z^0(X)/B^0(X)$: the first is induced by the inclusion $Z^0(X) \to X^0$ and the second by the quotient $Z^0(X) \to Z^0(X)/B^0(X)$, each restricted to the cycles $Z^0(X)$ and taken modulo $B^0(X)$. Hence $H^0(\vartheta_X) = \mathrm{id}_{H^0(X)}$.

Step 5: naturality of ϑ. Let $\varphi\colon X \to X'$ be a morphism of the heart. The morphism $\psi := \vartheta_{X'} \circ \varphi \circ \vartheta_X^{-1}$ runs from $j(H^0(X))$ to $j(H^0(X'))$, so $\psi = j(a)$ for some morphism a of $\mathcal{A}$ by Step 3. By Step 1 and Step 4,

$$a = H^0(j(a)) = H^0(\vartheta_{X'}) \circ H^0(\varphi) \circ H^0(\vartheta_X)^{-1} = H^0(\varphi)$$

so $\vartheta_{X'} \circ \varphi = j(H^0(\varphi)) \circ \vartheta_X$, which is naturality.

Step 6: conclusion. Steps 1, 3 and 4 give (1), and Steps 1, 4 and 5 give (2), so j and the restriction of H^0 are quasi-inverse equivalences. The heart contains 0 and is closed under finite direct sums in the additive category $D(\mathcal{A})$, since each H^i is additive, so it is an additive category, and j is an additive functor because Q is. The conditions of definition 1.1.6 (kernels and cokernels exist, every monomorphism is a kernel and every epimorphism is a cokernel) are stated by universal properties, and an equivalence of categories carries each of them from $\mathcal{A}$ to the heart. For instance, the kernel of a morphism $\varphi\colon X \to X'$ of the heart is $\vartheta_X^{-1} \circ j(k)$, where k is a kernel of $H^0(\varphi)$ in $\mathcal{A}$. So the heart is abelian, completing the proof.

The theorem places $\mathcal{A}$ inside $D(\mathcal{A})$ with no hypothesis on $\mathcal{A}$. When $\mathcal{A}$ has enough injectives, full faithfulness of j is also the last clause of theorem 3.4.10(2), proved there through $R\mathrm{Hom}$. Short exact sequences of $\mathcal{A}$ become distinguished triangles of $D(\mathcal{A})$ with all three vertices in the heart, by theorem 3.2.10. The kernel of a morphism φ of the heart is computed in $\mathcal{A}$, as the kernel of $H^0(\varphi)$, and transported along j. Inside $D(\mathcal{A})$ it is produced by a cone followed by a truncation.

Example 5.3.8: The Cone of a Morphism of $\mathcal{A}$

Let $f\colon M \to N$ be a morphism of $\mathcal{A}$, regarded as a chain map of complexes concentrated in degree 0. In definition 2.1.23, $MC(f)^i = M^{i+1} \oplus N^i$ with differential $(l,m) \mapsto \big(-d_M(l),\ f(l) + d_N(m)\big)$, so $MC(f)$ is M in degree -1 and N in degree 0, joined by f:

$$MC(f)\colon \qquad \cdots \to 0 \to M \xrightarrow{f} N \to 0 \to \cdots$$

Hence $H^{-1}(MC(f)) = \ker f$ and $H^0(MC(f)) = \operatorname{coker} f$, and $MC(f) \in D^{\leq 0} \cap D^{\geq -1}$. The mapping cone triangle on f is distinguished in $\mathrm{K}(\mathcal{A})$ (theorem 3.1.12), so its image is distinguished in $D(\mathcal{A})$ (theorem 3.2.7), and by proposition 3.1.10 every distinguished triangle on $Q(f)$ has third vertex isomorphic to $MC(f)$.

Truncate at $n = -1$. The cycles of $MC(f)$ in degree -1 form $\ker f$, so $\tau^{\leq -1}MC(f)$ is $\ker f$ in degree -1, the complex $(\ker f)[1]$. The quotient of N by the boundaries $f(M)$ is $\operatorname{coker} f$, so $\tau^{\geq 0}MC(f)$ is $\operatorname{coker} f$ in degree 0. The triangle of proposition 5.3.4(3) is

$$(\ker f)[1] \longrightarrow MC(f) \longrightarrow \operatorname{coker} f \longrightarrow (\ker f)[2]$$

So the kernel and the cokernel of f are recovered from $D(\mathcal{A})$ and the pair $(D^{\leq 0}, D^{\geq 0})$: complete $Q(f)$ to a distinguished triangle, truncate its third vertex, and apply H^0 to $\big(\tau^{\leq -1}MC(f)\big)[-1]$ and to $\tau^{\geq 0}MC(f)$, both of which lie in the heart. When f is a monomorphism, $H^{-1}(MC(f)) = 0$, the cone itself lies in the heart, and it is isomorphic to $\operatorname{coker} f$ by the quasi-isomorphism of lemma 3.2.9 for the short exact sequence $0 \to M \to N \to \operatorname{coker} f \to 0$.

Exercise 5.3.1

1. Let $X \in D(\mathcal{A})$ and $n \in \mathbb{Z}$, and let w be the third morphism of the triangle of proposition 5.3.4(3). Show that $H^i(w) = 0$ for every i, and deduce from proposition 3.2.8 that

   $$0 \to H^i(\tau^{\leq n}X) \to H^i(X) \to H^i(\tau^{\geq n+1}X) \to 0$$

   is exact for every i. Compare with proposition 2.1.14.

2. Let $A' \xrightarrow{a} X \xrightarrow{b} B' \xrightarrow{c} A'[1]$ be a distinguished triangle of $D(\mathcal{A})$ with $A' \in D^{\leq n}$ and $B' \in D^{\geq n+1}$. Show that there are unique morphisms $f\colon A' \to \tau^{\leq n}X$ and $h\colon B' \to \tau^{\geq n+1}X$ with $Q(i_n) \circ f = a$ and $h \circ b = Q(p_{n+1})$, and that both are isomorphisms. (Apply $\operatorname{Hom}(T, -)$ for $T \in D^{\leq n}$ and $\operatorname{Hom}(-, T)$ for $T \in D^{\geq n+1}$, then argue as in the last paragraph of the proof of corollary 3.1.9.) Show further that $w \circ h = f[1] \circ c$, where w is the third morphism of the triangle of proposition 5.3.4(3), so that (f, id_X, h) is an isomorphism of triangles from (a, b, c) to $\big(Q(i_n), Q(p_{n+1}), w\big)$. (Complete (f, id_X) to a morphism of triangles by (TR3) of definition 3.1.3, and compare its third component with h.)

3. Let $X \in \mathrm{Ch}(\mathcal{A})$ and $m \leq n$. Show that the complexes $\tau^{\geq m}\tau^{\leq n}X$ and $\tau^{\leq n}\tau^{\geq m}X$ are canonically isomorphic, and that for $m = n$ both are $H^n(X)$ in degree n, that is $H^n(X)[-n]$.

4. Let $X' \to X \to X'' \to X'[1]$ be a distinguished triangle of $D(\mathcal{A})$. Show that if X' and X'' lie in $D^{\leq 0}$ then so does X, and that the same holds for $D^{\geq 0}$. Conclude that the heart is closed under extensions in $D(\mathcal{A})$.

5. Show that $X \in D^{\leq 0}$ if and only if $\mathrm{Hom}_{D(\mathcal{A})}(X, B) = 0$ for every $B \in D^{\geq 1}$, and that $X \in D^{\geq 1}$ if and only if $\mathrm{Hom}_{D(\mathcal{A})}(A, X) = 0$ for every $A \in D^{\leq 0}$. (Test against $\tau^{\geq 1}X$ and $\tau^{\leq 0}X$, and use corollary 5.3.6.)

6. Let $k \in \mathbb{Z}$. Show that H^k restricts to an equivalence from $D^{\leq k} \cap D^{\geq k}$ to $\mathcal{A}$, with quasi-inverse $M \mapsto j(M)[-k]$.

7. Let X be the complex $\mathbb{Z} \xrightarrow{d^{-1}} \mathbb{Z}^2 \xrightarrow{d^0} \mathbb{Z}$ of abelian groups in degrees -1, 0 and 1, with $d^{-1}(x) = (2x, 0)$ and $d^0(a, b) = 3b$. Compute $H^\bullet(X)$, $\tau^{\leq 0}X$ and $\tau^{\geq 1}X$, and show that $\tau^{\leq 0}X \cong \mathbb{Z}/2$ and $\tau^{\geq 1}X = \mathbb{Z}/3[-1]$ in $D(\mathbf{Ab})$. Then show that the third morphism w of the triangle of proposition 5.3.4(3) for $n = 0$ is zero. (The Hom-sets of $D(\mathbf{Ab})$ are abelian groups and composition is bilinear, by theorem 3.2.7. Compare $3w$ and $2w$.)

5.4 *t*-Structures and Their Hearts

The full subcategories $D^{\leq 0}$ and $D^{\geq 0}$ of $D(\mathcal{A})$ in section 5.3, cut out by the vanishing of cohomology above and below degree 0, have three properties (proposition 5.3.4). No nonzero morphism runs from an object of $D^{\leq 0}$ to an object of $D^{\geq 1}$, the subcategories nest under shifting, and every complex sits in a distinguished triangle whose outer vertices are its two canonical truncations. Once that triangle is replaced by the existence of some distinguished triangle with first vertex in $D^{\leq 0}$ and third vertex in $D^{\geq 1}$, the three properties involve only objects, morphisms and distinguished triangles, so they make sense in any triangulated category, and this section takes them as axioms. The objects lying in both halves form the *heart*. The main results are that the heart is an abelian category (theorem 5.4.13), and that a functor H^0_t built from two truncations carries every distinguished triangle to an exact sequence in the heart (corollary 5.4.15). For the standard t-structure on $D(\mathcal{A})$ the heart is equivalent to $\mathcal{A}$ and H^0_t corresponds to ordinary cohomology. One triangulated category can carry t-structures with inequivalent hearts, and example 5.4.3 puts one with zero heart on every triangulated category.

Definition 5.4.1: *t*-Structure

Let $\mathcal{K}$ be a triangulated category. A *t-structure* on $\mathcal{K}$ is a pair $(\mathcal{K}^{\leq 0}, \mathcal{K}^{\geq 0})$ of full subcategories, each closed under isomorphism, such that, writing for $n \in \mathbb{Z}$

$$\mathcal{K}^{\leq n} := \{X \in \mathcal{K} \big| X[n] \in \mathcal{K}^{\leq 0}\}, \qquad \mathcal{K}^{\geq n} := \{X \in \mathcal{K} \big| X[n] \in \mathcal{K}^{\geq 0}\}$$

the following hold.

(T1) $\mathrm{Hom}_{\mathcal{K}}(X, Y) = 0$ for all $X \in \mathcal{K}^{\leq 0}$ and $Y \in \mathcal{K}^{\geq 1}$.

(T2) $\mathcal{K}^{\leq 0} \subseteq \mathcal{K}^{\leq 1}$ and $\mathcal{K}^{\geq 1} \subseteq \mathcal{K}^{\geq 0}$.

(T3) Every $X \in \mathcal{K}$ sits in a distinguished triangle $A \to X \to B \to A[1]$ with $A \in \mathcal{K}^{\leq 0}$ and $B \in \mathcal{K}^{\geq 1}$.

The *heart* of the t-structure is the full subcategory $\mathcal{K}^\heartsuit := \mathcal{K}^{\leq 0} \cap \mathcal{K}^{\geq 0}$.

Since [1] is an automorphism, $X \in \mathcal{K}^{\leq n}$ if and only if $X[1] \in \mathcal{K}^{\leq n-1}$, both conditions saying that

$X[n] \in \mathcal{K}^{\leq 0}$, and likewise for $\mathcal{K}^{\geq n}$. This moves (T1) and (T2) to every degree: for all integers $m < n$,

$$\mathcal{K}^{\leq m} \subseteq \mathcal{K}^{\leq n}, \qquad \mathcal{K}^{\geq n} \subseteq \mathcal{K}^{\geq m}, \qquad \mathrm{Hom}_{\mathcal{K}}(X, Y) = 0 \ \text{ for } X \in \mathcal{K}^{\leq m} \text{ and } Y \in \mathcal{K}^{\geq n} \tag{5.2}$$

For the first, $X \in \mathcal{K}^{\leq m}$ gives $X[m] \in \mathcal{K}^{\leq 0} \subseteq \mathcal{K}^{\leq 1}$ by (T2), so $X[m+1] \in \mathcal{K}^{\leq 0}$ and $X \in \mathcal{K}^{\leq m+1}$, and induction reaches n. For the second, $X \in \mathcal{K}^{\geq n}$ gives $X[n-1] \in \mathcal{K}^{\geq 1} \subseteq \mathcal{K}^{\geq 0}$, so $X \in \mathcal{K}^{\geq n-1}$, and induction reaches m. For the third, $X \in \mathcal{K}^{\leq n-1}$ by the first, so $X[n-1] \in \mathcal{K}^{\leq 0}$ and $Y[n-1] \in \mathcal{K}^{\geq 1}$, and $\mathrm{Hom}_{\mathcal{K}}(X, Y) \cong \mathrm{Hom}_{\mathcal{K}}(X[n-1], Y[n-1])$ vanishes by (T1). In particular $X[1] \in \mathcal{K}^{\leq -1} \subseteq \mathcal{K}^{\leq 0}$ for $X \in \mathcal{K}^{\leq 0}$, so $\mathcal{K}^{\leq 0}$ is stable under $[1]$, and dually $\mathcal{K}^{\geq 0}$ is stable under $[-1]$.

The axioms say that every object is assembled, in a single distinguished triangle, from a piece in $\mathcal{K}^{\leq 0}$ and a piece in $\mathcal{K}^{\geq 1}$, and that every morphism from an object of the first kind to an object of the second is zero.[1]

Example 5.4.2: The Standard t-Structure on $D(\mathcal{A})$

Let $\mathcal{A}$ be an abelian category, and take the full subcategories $D^{\leq 0}$ and $D^{\geq 0}$ of $D(\mathcal{A})$ defined in definition 5.3.1. Each is closed under isomorphism, because H^i is a functor on $D(\mathcal{A})$ (proposition 3.2.8). By proposition 5.3.4(1), $X[n] \in D^{\leq 0}$ exactly when $X \in D^{\leq n}$, and likewise for $D^{\geq n}$, so the subcategories that definition 5.4.1 attaches to degree n are the $D^{\leq n}$ and $D^{\geq n}$ of definition 5.3.1. Axiom (T2) is the case $n = 0$ of proposition 5.3.4(1), and (T1) is proposition 5.3.4(2) with $n = 0$. For (T3), proposition 5.3.4(3) with $n = 0$ gives the distinguished triangle $\tau^{\leq 0}X \to X \to \tau^{\geq 1}X \to (\tau^{\leq 0}X)[1]$ on the canonical truncations of definition 2.1.12, and $\tau^{\leq 0}X \in D^{\leq 0}$ and $\tau^{\geq 1}X \in D^{\geq 1}$ by lemma 5.3.3(3). So $(D^{\leq 0}, D^{\geq 0})$ is a t-structure on $D(\mathcal{A})$, the *standard* one. Its heart is the heart of definition 5.3.1, the complexes with cohomology concentrated in degree 0, and theorem 5.3.7 shows that H^0 restricts to an equivalence from the heart to $\mathcal{A}$.

Example 5.4.3: A Degenerate t-Structure

Let $\mathcal{K}$ be any triangulated category, let $\mathcal{K}^{\leq 0} = \mathcal{K}$, and let $\mathcal{K}^{\geq 0}$ be the full subcategory of zero objects. Both are closed under isomorphism, and since $[n]$ carries zero objects to zero objects, $\mathcal{K}^{\leq n} = \mathcal{K}$ and $\mathcal{K}^{\geq n}$ is the subcategory of zero objects for every n. (T1) holds because every morphism into a zero object is zero, and (T2) holds with equalities. For (T3), the triangle $(\mathrm{id}_X, 0, 0)$ on $(X, X, 0)$ is distinguished by (TR1) and has $0 \in \mathcal{K}^{\geq 1}$. The heart is the subcategory of zero objects. On $\mathcal{K} = D(\mathcal{A})$ with $\mathcal{A}$ nonzero, this and the standard t-structure of example 5.4.2 are two t-structures on one triangulated category, one with heart equivalent to $\mathcal{A}$ and one with zero heart.

Exactness of Hom sequences converts the vanishing in (5.2) into statements about objects and morphisms. Let (u, v, w) be a distinguished triangle on (A, B, C). By (TR2) the rotations $(v, w, -u[1])$ on $(B, C, A[1])$, $(-w[-1], u, v)$ on $(C[-1], A, B)$ and $(-v[-1], -w[-1], u)$ on $(B[-1], C[-1], A)$ are distinguished, the last being the second rotation backwards. Applying proposition 3.1.8 to (u, v, w) and these three triangles gives, for every $T \in \mathcal{K}$, sequences

$$\begin{gathered}\mathrm{Hom}(T, B[-1]) \to \mathrm{Hom}(T, C[-1]) \to \mathrm{Hom}(T, A) \xrightarrow{u_*} \mathrm{Hom}(T, B) \xrightarrow{v_*} \mathrm{Hom}(T, C) \xrightarrow{w_*} \mathrm{Hom}(T, A[1]) \\ \mathrm{Hom}(A[1], T) \xrightarrow{w^*} \mathrm{Hom}(C, T) \xrightarrow{v^*} \mathrm{Hom}(B, T) \xrightarrow{u^*} \mathrm{Hom}(A, T) \to \mathrm{Hom}(C[-1], T)\end{gathered} \tag{5.3}$$

exact at every interior term, where the unlabelled maps are composition with $-v[-1]$, $-w[-1]$ and $-w[-1]$, in that order. A sign changes no kernel and no image, so it is dropped below whenever only exactness is used.

[1] The notion is due to Beilinson, Bernstein and Deligne [3, Def. 1.3.1], whose categories of perverse sheaves are hearts of t-structures. Gelfand and Manin [15, Def. IV.4.2] call the heart the *core*.

The axioms are stated in degree 0, and moving (T3) to another degree needs the translation to carry distinguished triangles to distinguished triangles. Rotating three times does this, up to a sign on the third edge.

Lemma 5.4.4: Translates of Distinguished Triangles

Let (u, v, w) be a distinguished triangle on (A, B, C) and let $n \in \mathbb{Z}$. There is a morphism $w_n \colon C[n] \to A[n+1]$ such that $(u[n], v[n], w_n)$ is a distinguished triangle on $(A[n], B[n], C[n])$. One may take $w_1 = -w[1]$ and $w_{-1} = -w[-1]$.

***Proof*:**
Three rotations by the first form of (TR2) give the distinguished triangles

$$(v, w, -u[1]), \qquad (w, -u[1], -v[1]), \qquad (-u[1], -v[1], -w[1])$$

on $(B, C, A[1])$, $(C, A[1], B[1])$ and $(A[1], B[1], C[1])$ respectively. From the last of these to $(u[1], v[1], -w[1])$ the triple $(\mathrm{id}_{A[1]}, -\mathrm{id}_{B[1]}, \mathrm{id}_{C[1]})$ is an isomorphism of triangles, because its three squares read

$$(-\mathrm{id})(-u[1]) = u[1]\,\mathrm{id}, \qquad \mathrm{id}\,(-v[1]) = v[1]\,(-\mathrm{id}), \qquad \mathrm{id}\,(-w[1]) = (-w[1])\,\mathrm{id}$$

by bilinearity of composition. So $(u[1], v[1], -w[1])$ is distinguished by (TR1). Three rotations by the second form of (TR2) give $(-w[-1], u, v)$ on $(C[-1], A, B)$, then $(-v[-1], -w[-1], u)$ on $(B[-1], C[-1], A)$, then $(-u[-1], -v[-1], -w[-1])$ on $(A[-1], B[-1], C[-1])$, and the same isomorphism $(\mathrm{id}, -\mathrm{id}, \mathrm{id})$ carries the last to $(u[-1], v[-1], -w[-1])$. For general n, induct on $|n|$: if $(u[n], v[n], w_n)$ is distinguished, the cases $n = \pm 1$, applied to it, give the distinguished triangles $(u[n+1], v[n+1], -w_n[1])$ and $(u[n-1], v[n-1], -w_n[-1])$, completing the proof.

Deciding whether an object lies in $\mathcal{K}^{\leq 0}$ will come down to deciding whether the first edge of a distinguished triangle is an isomorphism.

Lemma 5.4.5: A Triangle with Zero Third Vertex

Let (u, v, w) be a distinguished triangle on (A, B, C). Then u is an isomorphism if and only if $C \cong 0$.

***Proof*:**
Suppose $C \cong 0$. Then $C[-1] \cong 0$, since the automorphism $[-1]$ carries zero objects to zero objects. For every $T \in \mathcal{K}$ the first sequence of (5.3) contains

$$0 = \mathrm{Hom}(T, C[-1]) \to \mathrm{Hom}(T, A) \xrightarrow{u_*} \mathrm{Hom}(T, B) \to \mathrm{Hom}(T, C) = 0$$

exact at its two middle terms, so u_* is bijective. Taking $T = B$ gives $s \colon B \to A$ with $us = \mathrm{id}_B$. Taking $T = A$, both su and id_A are carried by u_* to u, since $u(su) = (us)u = u$, so $su = \mathrm{id}_A$ and u is an isomorphism with inverse s.

Conversely suppose u is an isomorphism. The triangle $(\mathrm{id}_A, 0, 0)$ on $(A, A, 0)$ is distinguished by (TR1), and $u \circ \mathrm{id}_A = u \circ \mathrm{id}_A$, so (TR3) gives $h \colon 0 \to C$ with (id_A, u, h) a morphism of triangles

from $(\mathrm{id}_A, 0, 0)$ to (u, v, w). Its first two components are isomorphisms, so corollary 3.1.9 makes h an isomorphism and $C \cong 0$, completing the proof.

(TR3) extends a commutative square to a morphism of triangles but does not make the extension unique, so the isomorphism of cones in proposition 3.1.10 need not be unique and the axioms give no functorial cone (box 3.1.4). An extension of a morphism of middle vertices exists and is unique when two Hom groups vanish, and in a t-structure (5.2) makes them vanish.

Lemma 5.4.6: Unique Extension to a Morphism of Triangles

Let (u, v, w) on (A, B, C) and (u', v', w') on (A', B', C') be distinguished triangles and let $g\colon B \to B'$.

1. If $\mathrm{Hom}(A, C') = 0$, there are $f\colon A \to A'$ and $h\colon C \to C'$ such that (f, g, h) is a morphism of triangles.
2. If $\mathrm{Hom}(A[1], C') = 0$, there is at most one f with $u'f = gu$ and at most one h with $hv = v'g$.

***Proof*:**

1. The first sequence of (5.3) for (u', v', w') and $T = A$ is exact at the middle term of
$$\mathrm{Hom}(A, A') \xrightarrow{u'_*} \mathrm{Hom}(A, B') \xrightarrow{v'_*} \mathrm{Hom}(A, C') = 0$$
so $gu = u'f$ for some $f\colon A \to A'$. (TR3), applied to f, g and the commutative square $gu = u'f$, gives h with (f, g, h) a morphism of triangles.
2. Applying [1] is a bijection $\mathrm{Hom}(A, C'[-1]) \to \mathrm{Hom}(A[1], C')$, so $\mathrm{Hom}(A, C'[-1]) = 0$. In the first sequence of (5.3) for (u', v', w') and $T = A$, the kernel of u'_* on $\mathrm{Hom}(A, A')$ is the image of $\mathrm{Hom}(A, C'[-1]) = 0$. So $u'f = u'f_1$ forces $f = f_1$. In the second sequence for (u, v, w) and $T = C'$, the kernel of v^* on $\mathrm{Hom}(C, C')$ is the image of $\mathrm{Hom}(A[1], C') = 0$. So $hv = h_1 v$ forces $h = h_1$, completing the proof.

Nothing in definition 5.4.1 singles out degree 0, and shifting the indexing gives another t-structure.

Lemma 5.4.7: Shifted t-Structures

Let $(\mathcal{K}^{\leq 0}, \mathcal{K}^{\geq 0})$ be a t-structure on $\mathcal{K}$ and let $m \in \mathbb{Z}$. Then $(\mathcal{K}^{\leq m}, \mathcal{K}^{\geq m})$ is a t-structure on $\mathcal{K}$, its subcategories in degree n are $\mathcal{K}^{\leq m+n}$ and $\mathcal{K}^{\geq m+n}$, and its heart $\mathcal{K}^{\leq m} \cap \mathcal{K}^{\geq m}$ consists of the objects X with $X[m] \in \mathcal{K}^{\heartsuit}$.

***Proof*:**
Both subcategories are closed under isomorphism, as $X \cong Y$ gives $X[m] \cong Y[m]$. An object X lies in the degree-n subcategory $\{X \,|\, X[n] \in \mathcal{K}^{\leq m}\}$ exactly when $X[n+m] \in \mathcal{K}^{\leq 0}$, that is when $X \in \mathcal{K}^{\leq m+n}$, and likewise for $\mathcal{K}^{\geq m+n}$. With this, (T1) and (T2) for the new pair say that $\mathrm{Hom}(X, Y) = 0$ for $X \in \mathcal{K}^{\leq m}$ and $Y \in \mathcal{K}^{\geq m+1}$, and that $\mathcal{K}^{\leq m} \subseteq \mathcal{K}^{\leq m+1}$ and $\mathcal{K}^{\geq m+1} \subseteq \mathcal{K}^{\geq m}$. All three are instances of (5.2). For (T3), let $X \in \mathcal{K}$ and take, by (T3) for the original pair,

a distinguished triangle (u, v, w) on $(A, X[m], B)$ with $A \in \mathcal{K}^{\leq 0}$ and $B \in \mathcal{K}^{\geq 1}$. Lemma 5.4.4 with $n = -m$ gives a distinguished triangle $(u[-m], v[-m], w_{-m})$ on $(A[-m], X, B[-m])$. Here $A[-m] \in \mathcal{K}^{\leq m}$ since $A[-m][m] = A$, and $B[-m] \in \mathcal{K}^{\geq m+1}$ since $B[-m][m+1] = B[1] \in \mathcal{K}^{\geq 0}$. The description of the heart is the definition of $\mathcal{K}^{\leq m}$ and $\mathcal{K}^{\geq m}$, completing the proof.

Example 5.4.8: Shifts of the Standard t-Structure

For the standard t-structure of example 5.4.2 and $m \in \mathbb{Z}$, the heart of $(D^{\leq m}, D^{\geq m})$ consists of the complexes X with $H^i(X) = 0$ for $i \neq m$. The functor $X \mapsto H^0(X[m]) = H^m(X)$ is an equivalence from it to $\mathcal{A}$, being the composite of the equivalence $[m]$ onto $D^{\leq 0} \cap D^{\geq 0}$ with the equivalence of theorem 5.3.7. So the shifted hearts are equivalent to one another. If $\mathcal{A}$ has a nonzero object M, they are also different subcategories of $D(\mathcal{A})$, since M, placed in degree 0, lies in the heart of $(D^{\leq m}, D^{\geq m})$ only for $m = 0$, while $M[-m]$ lies in it for every m.

Axiom (T3) asserts only that a triangle exists. Making it functorial in X needs a choice that is unique up to unique isomorphism, and (T1), through lemma 5.4.6, gives exactly that. The resulting functors and adjunctions extend lemma 5.3.3 and corollary 5.3.6 to an arbitrary t-structure.

Proposition 5.4.9: Truncation Functors of a t-Structure

Let $(\mathcal{K}^{\leq 0}, \mathcal{K}^{\geq 0})$ be a t-structure on $\mathcal{K}$ and let $n \in \mathbb{Z}$.

1. There are additive functors $\tau^{\leq n} \colon \mathcal{K} \to \mathcal{K}^{\leq n}$ and $\tau^{\geq n+1} \colon \mathcal{K} \to \mathcal{K}^{\geq n+1}$ and natural transformations $\alpha^n \colon \tau^{\leq n} \to \mathrm{id}$, $\beta^n \colon \mathrm{id} \to \tau^{\geq n+1}$ and $\partial^n \colon \tau^{\geq n+1} \to \tau^{\leq n}[1]$ such that for every $X \in \mathcal{K}$ the triangle

$$\tau^{\leq n} X \xrightarrow{\alpha^n_X} X \xrightarrow{\beta^n_X} \tau^{\geq n+1} X \xrightarrow{\partial^n_X} (\tau^{\leq n} X)[1]$$

   is distinguished.

2. For all $A \in \mathcal{K}^{\leq n}$, $B \in \mathcal{K}^{\geq n+1}$ and $X \in \mathcal{K}$, composition with α^n_X and with β^n_X gives bijections

$$\mathrm{Hom}(A, \tau^{\leq n} X) \xrightarrow{\sim} \mathrm{Hom}(A, X), \qquad \mathrm{Hom}(\tau^{\geq n+1} X, B) \xrightarrow{\sim} \mathrm{Hom}(X, B)$$

   natural in all three variables. So $\tau^{\leq n}$ is right adjoint to the inclusion of $\mathcal{K}^{\leq n}$ in $\mathcal{K}$, and $\tau^{\geq n+1}$ is left adjoint to the inclusion of $\mathcal{K}^{\geq n+1}$.

3. If (u, v, w) is a distinguished triangle on (A, X, B) with $A \in \mathcal{K}^{\leq n}$ and $B \in \mathcal{K}^{\geq n+1}$, there is exactly one isomorphism of triangles of the form (f, id_X, h) from it to the triangle of (1). In particular ∂^n_X is the only morphism completing (α^n_X, β^n_X) to a distinguished triangle.

***Proof*:**

By lemma 5.4.7, $(\mathcal{K}^{\leq n}, \mathcal{K}^{\geq n})$ is a t-structure whose subcategories in degrees 0 and 1 are $\mathcal{K}^{\leq n}$ and $\mathcal{K}^{\geq n+1}$, so the statement for n is the statement for $n = 0$ applied to that t-structure. Take $n = 0$, and write α, β, ∂ for α^0, β^0, ∂^0.

Step 1: The bijections for a chosen triangle. For each $X \in \mathcal{K}$ choose by (T3) a distinguished triangle $(\alpha_X, \beta_X, \partial_X)$ on $(\tau^{\leq 0}X, X, \tau^{\geq 1}X)$ with $\tau^{\leq 0}X \in \mathcal{K}^{\leq 0}$ and $\tau^{\geq 1}X \in \mathcal{K}^{\geq 1}$. For the standard t-structure on $D(\mathcal{A})$, where after the reduction $\mathcal{K}^{\leq 0}$ and $\mathcal{K}^{\geq 1}$ are the subcategories $D^{\leq n}$ and $D^{\geq n+1}$, we choose the triangle of proposition 5.3.4(3) for that n, whose outer vertices lie in them by lemma 5.3.3(3). Let $A \in \mathcal{K}^{\leq 0}$. The first sequence of (5.3) with $T = A$ contains

$$\mathrm{Hom}(A, (\tau^{\geq 1}X)[-1]) \to \mathrm{Hom}(A, \tau^{\leq 0}X) \xrightarrow{(\alpha_X)_*} \mathrm{Hom}(A, X) \to \mathrm{Hom}(A, \tau^{\geq 1}X)$$

exact at its middle terms. The last group vanishes by (T1). So does the first, since $(\tau^{\geq 1}X)[-1] \in \mathcal{K}^{\geq 2}$ and $\mathrm{Hom}(\mathcal{K}^{\leq 0}, \mathcal{K}^{\geq 2}) = 0$ by (5.2). So $(\alpha_X)_*$ is bijective. For $B \in \mathcal{K}^{\geq 1}$ the second sequence of (5.3) contains

$$\mathrm{Hom}((\tau^{\leq 0}X)[1], B) \to \mathrm{Hom}(\tau^{\geq 1}X, B) \xrightarrow{\beta_X^*} \mathrm{Hom}(X, B) \to \mathrm{Hom}(\tau^{\leq 0}X, B)$$

exact at its middle terms. The last group vanishes by (T1), and the first because $(\tau^{\leq 0}X)[1] \in \mathcal{K}^{\leq -1}$. So β_X^* is bijective.

Step 2: Functoriality. Let $g\colon X \to Y$ be a morphism of $\mathcal{K}$. The groups $\mathrm{Hom}(\tau^{\leq 0}X, \tau^{\geq 1}Y)$ and $\mathrm{Hom}((\tau^{\leq 0}X)[1], \tau^{\geq 1}Y)$ vanish by (5.2), since their sources lie in $\mathcal{K}^{\leq 0}$ and $\mathcal{K}^{\leq -1}$ and their target lies in $\mathcal{K}^{\geq 1}$. Hence lemma 5.4.6 extends g to a morphism of triangles $(\tau^{\leq 0}g, g, \tau^{\geq 1}g)$ from the chosen triangle on X to the chosen triangle on Y, where $\tau^{\leq 0}g$ is the unique morphism with $\alpha_Y \circ \tau^{\leq 0}g = g \circ \alpha_X$ and $\tau^{\geq 1}g$ the unique morphism with $\tau^{\geq 1}g \circ \beta_X = \beta_Y \circ g$. Uniqueness gives functoriality: $\mathrm{id}_{\tau^{\leq 0}X}$ satisfies the condition characterizing $\tau^{\leq 0}(\mathrm{id}_X)$, and for $g'\colon Y \to W$ the composite $\tau^{\leq 0}g' \circ \tau^{\leq 0}g$ satisfies

$$\alpha_W \circ \tau^{\leq 0}g' \circ \tau^{\leq 0}g = g' \circ \alpha_Y \circ \tau^{\leq 0}g = g'g \circ \alpha_X$$

the condition characterizing $\tau^{\leq 0}(g'g)$. Additivity is the same argument, as $\alpha_Y \circ (\tau^{\leq 0}g + \tau^{\leq 0}g_1) = (g + g_1) \circ \alpha_X$ by bilinearity of composition. The argument for $\tau^{\geq 1}$ is identical with β in place of α. The three squares of the morphism of triangles $(\tau^{\leq 0}g, g, \tau^{\geq 1}g)$ are the naturality of α, β and ∂. This proves (1).

Step 3: Naturality of the bijections. The bijection $(\alpha_X)_*$ is natural in A, being composition on the other side, and natural in X because $\alpha_Y \circ \tau^{\leq 0}g \circ \varphi = g \circ \alpha_X \circ \varphi$ for every $\varphi\colon A \to \tau^{\leq 0}X$. The same argument applies to β_X^*. This proves (2).

Step 4: Uniqueness of the triangle. Let (u, v, w) be distinguished on (A, X, B) with $A \in \mathcal{K}^{\leq 0}$ and $B \in \mathcal{K}^{\geq 1}$. As in Step 2, (5.2) gives the vanishing that lemma 5.4.6 needs for $g = \mathrm{id}_X$ in both directions between (u, v, w) and $(\alpha_X, \beta_X, \partial_X)$. So there are morphisms of triangles (f, id_X, h) from (u, v, w) to $(\alpha_X, \beta_X, \partial_X)$ and (f', id_X, h') back, each the only one with middle component id_X. A composite of morphisms of triangles is one, by functoriality of [1], so $(f'f, \mathrm{id}_X, h'h)$ and the identity are two morphisms from (u, v, w) to itself with middle component id_X, and the same lemma, applied to (u, v, w) and itself, makes them equal. Likewise $(ff', \mathrm{id}_X, hh')$ is the identity. So (f, id_X, h) is an isomorphism of triangles, and it is the only one with middle component id_X. Finally, if $(\alpha_X, \beta_X, \partial')$ is distinguished, the isomorphism (f, id_X, h) from it to $(\alpha_X, \beta_X, \partial_X)$ satisfies $\alpha_X f = \alpha_X$ and $h\beta_X = \beta_X$, so f and h are identities by the injectivity of $(\alpha_X)_*$ and β_X^* in Step 1. The third square then reads $\partial' = \partial_X$, completing the proof.

Proposition 5.4.9 is [3, Prop. 1.3.3] and [15, Lemma IV.4.5]. Part (2) says that every morphism into X from an object of $\mathcal{K}^{\leq n}$ factors uniquely through α^n_X, and every morphism out of X into an object of $\mathcal{K}^{\geq n+1}$ factors uniquely through β^n_X. Part (3) says that the triangle of (T3) is unique up to unique isomorphism fixing X, the uniqueness coming from (T1) through lemma 5.4.6. For the standard t-structure, Step 1 of the proof chooses the triangle of proposition 5.3.4(3), so $\alpha^n_X = Q(i_n)$ and $\beta^n_X = Q(p_{n+1})$. On morphisms, the naturality of $Q(i_n)$ and $Q(p_{n+1})$ in lemma 5.3.3(2) is the condition that characterizes $\tau^{\leq n}g$ and $\tau^{\geq n+1}g$ in Step 2. So the functors of (1) are those of lemma 5.3.3, (2) becomes corollary 5.3.6, and one symbol τ denotes both kinds of truncation.

The truncation functors convert membership in the two halves into a vanishing condition.

Lemma 5.4.10: The Two Halves Determine Each Other

Let $(\mathcal{K}^{\leq 0}, \mathcal{K}^{\geq 0})$ be a t-structure on $\mathcal{K}$ and let $n \in \mathbb{Z}$.

1. For $X \in \mathcal{K}$ the following are equivalent: (a) $X \in \mathcal{K}^{\leq n}$, (b) $\mathrm{Hom}(X, B) = 0$ for every $B \in \mathcal{K}^{\geq n+1}$, (c) $\tau^{\geq n+1}X \cong 0$, (d) α^n_X is an isomorphism.
2. For $Y \in \mathcal{K}$ the following are equivalent: (a) $Y \in \mathcal{K}^{\geq n+1}$, (b) $\mathrm{Hom}(A, Y) = 0$ for every $A \in \mathcal{K}^{\leq n}$, (c) $\tau^{\leq n}Y \cong 0$, (d) β^n_Y is an isomorphism.
3. $\mathcal{K}^{\leq n}$ and $\mathcal{K}^{\geq n}$ contain the zero objects and are closed under finite direct sums and under extensions: if (u, v, w) is distinguished on (A, X, C) and A and C lie in one of the two, then so does X.
4. $\tau^{\geq n}$ carries $\mathcal{K}^{\leq n}$ into $\mathcal{K}^{\leq n} \cap \mathcal{K}^{\geq n}$, and $\tau^{\leq n}$ carries $\mathcal{K}^{\geq n}$ into $\mathcal{K}^{\leq n} \cap \mathcal{K}^{\geq n}$.

***Proof*:**

1. (a) implies (b) by (5.2). If (b) holds, then $\mathrm{Hom}(\tau^{\geq n+1}X, B) = 0$ for every $B \in \mathcal{K}^{\geq n+1}$ by proposition 5.4.9(2), and taking $B = \tau^{\geq n+1}X$ gives $\mathrm{id}_{\tau^{\geq n+1}X} = 0$, so $\tau^{\geq n+1}X$ is a zero object, which is (c). If (c) holds, lemma 5.4.5 applied to the truncation triangle of proposition 5.4.9(1) makes α^n_X an isomorphism, which is (d). If (d) holds, $X \cong \tau^{\leq n}X \in \mathcal{K}^{\leq n}$, and $\mathcal{K}^{\leq n}$ is closed under isomorphism, which is (a).

2. The argument is that of (1) with the arrows turned around. (a) implies (b) by (5.2). If (b) holds, then $\mathrm{Hom}(A, \tau^{\leq n}Y) = 0$ for every $A \in \mathcal{K}^{\leq n}$ by proposition 5.4.9(2), and $A = \tau^{\leq n}Y$ gives $\tau^{\leq n}Y \cong 0$. If (c) holds, the rotated truncation triangle $(\beta^n_Y, \partial^n_Y, -\alpha^n_Y[1])$ on $(Y, \tau^{\geq n+1}Y, (\tau^{\leq n}Y)[1])$ has third vertex $(\tau^{\leq n}Y)[1] \cong 0$, and lemma 5.4.5 makes β^n_Y an isomorphism. If (d) holds, $Y \cong \tau^{\geq n+1}Y \in \mathcal{K}^{\geq n+1}$.

3. A zero object Z has $\mathrm{Hom}(Z, B) = 0 = \mathrm{Hom}(A, Z)$ for all A and B, so it lies in $\mathcal{K}^{\leq n}$ by (1) and in $\mathcal{K}^{\geq n}$ by (2) with $n-1$ in place of n. For $X, X' \in \mathcal{K}^{\leq n}$ and $B \in \mathcal{K}^{\geq n+1}$, $\mathrm{Hom}(X \oplus X', B) \cong \mathrm{Hom}(X, B) \times \mathrm{Hom}(X', B) = 0$ by the universal property of the coproduct, so $X \oplus X' \in \mathcal{K}^{\leq n}$ by (1). The same argument with the product and (2) treats $\mathcal{K}^{\geq n}$. For extensions, let $A, C \in \mathcal{K}^{\leq n}$ and $B \in \mathcal{K}^{\geq n+1}$. The second sequence of (5.3) contains $\mathrm{Hom}(C, B) \to \mathrm{Hom}(X, B) \to \mathrm{Hom}(A, B)$, exact in the middle with both ends zero, so $\mathrm{Hom}(X, B) = 0$ and $X \in \mathcal{K}^{\leq n}$ by (1). If instead $A, C \in \mathcal{K}^{\geq n}$, then for every $T \in \mathcal{K}^{\leq n-1}$ the first sequence contains $\mathrm{Hom}(T, A) \to \mathrm{Hom}(T, X) \to \mathrm{Hom}(T, C)$ with both ends zero, so $\mathrm{Hom}(T, X) = 0$, and $X \in \mathcal{K}^{\geq n}$ by (2) with $n-1$ in place of n.

4. Let $X \in \mathcal{K}^{\leq n}$. Rotating the truncation triangle for $n-1$ gives the distinguished triangle $(\beta_X^{n-1}, \partial_X^{n-1}, -\alpha_X^{n-1}[1])$ on $(X, \tau^{\geq n}X, (\tau^{\leq n-1}X)[1])$. Its outer vertices lie in $\mathcal{K}^{\leq n}$, the second because $(\tau^{\leq n-1}X)[1] \in \mathcal{K}^{\leq n-2} \subseteq \mathcal{K}^{\leq n}$, so $\tau^{\geq n}X \in \mathcal{K}^{\leq n}$ by (3), and $\tau^{\geq n}X \in \mathcal{K}^{\geq n}$ by construction. Let $X \in \mathcal{K}^{\geq n}$. Rotating the truncation triangle for n backwards gives the distinguished triangle $(-\partial_X^n[-1], \alpha_X^n, \beta_X^n)$ on $((\tau^{\geq n+1}X)[-1], \tau^{\leq n}X, X)$. Its outer vertices lie in $\mathcal{K}^{\geq n}$, the first because $(\tau^{\geq n+1}X)[-1] \in \mathcal{K}^{\geq n+2} \subseteq \mathcal{K}^{\geq n}$, so $\tau^{\leq n}X \in \mathcal{K}^{\geq n}$ by (3), completing the proof.

By (1) and (2), each half of a t-structure determines the other: $\mathcal{K}^{\geq 1}$ consists of the objects that receive only the zero morphism from objects of $\mathcal{K}^{\leq 0}$, and $\mathcal{K}^{\leq 0}$ of the objects that admit only the zero morphism to objects of $\mathcal{K}^{\geq 1}$. By (4), truncating an object of $\mathcal{K}^{\leq 0}$ from below lands in the heart, which gives a functor into the heart.

Definition 5.4.11: Cohomology Functors of a t-Structure

Let $(\mathcal{K}^{\leq 0}, \mathcal{K}^{\geq 0})$ be a t-structure on $\mathcal{K}$. Its *cohomology functors* are

$$H_t^0 := \tau^{\geq 0}\tau^{\leq 0} \colon \mathcal{K} \to \mathcal{K}^\heartsuit, \qquad H_t^n(X) := H_t^0(X[n]) \quad (n \in \mathbb{Z})$$

which take values in $\mathcal{K}^\heartsuit$ by lemma 5.4.10(4) with $n = 0$.

Example 5.4.12: The Cohomology Functors of the Standard t-Structure

Take the standard t-structure on $D(\mathcal{A})$ and let $X \in D(\mathcal{A})$ and $n \in \mathbb{Z}$. Ordinary cohomology is cohomological on $D(\mathcal{A})$ (proposition 3.2.8), so applying it to the truncation triangle of proposition 5.4.9(1) and its rotations gives the long exact sequence

$$\cdots \to H^{i-1}(\tau^{\geq n+1}X) \to H^i(\tau^{\leq n}X) \xrightarrow{H^i(\alpha_X^n)} H^i(X) \xrightarrow{H^i(\beta_X^n)} H^i(\tau^{\geq n+1}X) \to H^{i+1}(\tau^{\leq n}X) \to \cdots$$

By example 5.4.2, $H^i(\tau^{\leq n}X) = 0$ for $i > n$ and $H^i(\tau^{\geq n+1}X) = 0$ for $i \leq n$. For $i \leq n$ the terms on either side of $H^i(\alpha_X^n)$ are $H^{i-1}(\tau^{\geq n+1}X) = 0$ and $H^i(\tau^{\geq n+1}X) = 0$, and for $i \geq n+1$ the terms on either side of $H^i(\beta_X^n)$ are $H^i(\tau^{\leq n}X) = 0$ and $H^{i+1}(\tau^{\leq n}X) = 0$. Hence

$$H^i(\tau^{\leq n}X) \cong \begin{cases} H^i(X) & i \leq n \\ 0 & i > n \end{cases} \qquad H^i(\tau^{\geq n+1}X) \cong \begin{cases} 0 & i \leq n \\ H^i(X) & i \geq n+1 \end{cases}$$

the abstract counterpart of proposition 2.1.14. The first formula with $n = 0$, followed by the second with $n = -1$ applied to $\tau^{\leq 0}X$, gives $H^i(H_t^0 X) = H^i(\tau^{\geq 0}\tau^{\leq 0}X) \cong H^0(X)$ for $i = 0$ and 0 for $i \neq 0$, the isomorphism in degree 0 being $H^0(\alpha_X^0) \circ H^0(\beta^{-1}_{\tau^{\leq 0}X})^{-1}$, which is natural in X. So the equivalence H^0 of theorem 5.3.7 carries the functor H_t^0 to ordinary H^0, and H_t^n to $X \mapsto H^0(X[n]) = H^n(X)$. For the complex $X = (\mathbb{Z} \xrightarrow{2} \mathbb{Z})$ in degrees -1 and 0 of example 5.3.2, $H^{-1}(X) = 0$ and $H^0(X) = \mathbb{Z}/2$, so $H_t^0(X)$ corresponds to $\mathbb{Z}/2$ and $H_t^{-1}(X) \cong 0$.

By example 5.3.8, the cone of a morphism f of $\mathcal{A}$ lies in $D^{\leq 0} \cap D^{\geq -1}$, and under H^0 the objects $\big(\tau^{\leq -1}MC(f)\big)[-1]$ and $\tau^{\geq 0}MC(f)$ of the heart correspond to $\ker f$ and $\operatorname{coker} f$. For an arbitrary t-structure the cone of a morphism of the heart still lies in $\mathcal{K}^{\leq 0} \cap \mathcal{K}^{\geq -1}$, and the same two truncations give a kernel and a cokernel in the heart.

Theorem 5.4.13: The Heart of a t-Structure Is Abelian

Let $(\mathcal{K}^{\leq 0}, \mathcal{K}^{\geq 0})$ be a t-structure on a triangulated category $\mathcal{K}$.

1. The heart $\mathcal{K}^\heartsuit$ is an abelian category, and it is closed under extensions in $\mathcal{K}$.
2. Let $f\colon X \to Y$ be a morphism of $\mathcal{K}^\heartsuit$ and (f, g, h) a distinguished triangle on (X, Y, Z). Then $Z \in \mathcal{K}^{\leq 0} \cap \mathcal{K}^{\geq -1}$, the composite $\beta_Z^{-1} \circ g\colon Y \to \tau^{\geq 0} Z$ is a cokernel of f in $\mathcal{K}^\heartsuit$, and the composite $h[-1] \circ \alpha_Z^{-1}[-1]\colon (\tau^{\leq -1} Z)[-1] \to X$ is a kernel of f in $\mathcal{K}^\heartsuit$.
3. A sequence $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ of $\mathcal{K}^\heartsuit$ is exact if and only if there is a morphism $h\colon Z \to X[1]$ with (f, g, h) distinguished, and such an h is unique.

***Proof*:**

Step 1: $\mathcal{K}^\heartsuit$ is additive. By lemma 5.4.10(3) with $n = 0$, $\mathcal{K}^\heartsuit$ contains the zero objects and the direct sum of any two of its objects. A full subcategory of the additive category $\mathcal{K}$ with these two properties is additive (definition 1.1.1). It has the Hom groups of $\mathcal{K}$ with their bilinear composition, a zero object of $\mathcal{K}$ is a zero object of it, and the biproduct $X \oplus Y$ of $\mathcal{K}$, lying in it, has the universal properties of product and coproduct there because the subcategory is full. Closure under extensions is lemma 5.4.10(3) for $\mathcal{K}^{\leq 0}$ and $\mathcal{K}^{\geq 0}$ together.

Step 2: $Z \in \mathcal{K}^{\leq 0} \cap \mathcal{K}^{\geq -1}$. Rotating gives the distinguished triangle $(g, h, -f[1])$ on $(Y, Z, X[1])$. Here $Y \in \mathcal{K}^{\leq 0}$ and $X[1] \in \mathcal{K}^{\leq -1} \subseteq \mathcal{K}^{\leq 0}$, so $Z \in \mathcal{K}^{\leq 0}$ by lemma 5.4.10(3). Likewise $Y \in \mathcal{K}^{\geq 0} \subseteq \mathcal{K}^{\geq -1}$ and $X[1] \in \mathcal{K}^{\geq -1}$, so $Z \in \mathcal{K}^{\geq -1}$.

Step 3: The cokernel. Put $C := \tau^{\geq 0} Z$ and $c := \beta_Z^{-1} g$. Then $C \in \mathcal{K}^{\geq 0}$, and $C \in \mathcal{K}^{\leq 0}$ by lemma 5.4.10(4) because $Z \in \mathcal{K}^{\leq 0}$, so $C \in \mathcal{K}^\heartsuit$. Let $T \in \mathcal{K}^\heartsuit$. Precomposition with β_Z^{-1} is a bijection $\mathrm{Hom}(C, T) \to \mathrm{Hom}(Z, T)$ by proposition 5.4.9(2), as $T \in \mathcal{K}^{\geq 0}$. The second sequence of (5.3) for (f, g, h) gives the exact sequence

$$0 = \mathrm{Hom}(X[1], T) \to \mathrm{Hom}(Z, T) \xrightarrow{g^*} \mathrm{Hom}(Y, T) \xrightarrow{f^*} \mathrm{Hom}(X, T)$$

where $\mathrm{Hom}(X[1], T) = 0$ because $X[1] \in \mathcal{K}^{\leq -1}$ and $T \in \mathcal{K}^{\geq 0}$. Composing, $\varphi \mapsto \varphi c$ is a bijection from $\mathrm{Hom}(C, T)$ onto $\{\psi \in \mathrm{Hom}(Y, T) \big| \psi f = 0\}$. Taking $T = C$ and $\varphi = \mathrm{id}_C$ shows $cf = 0$, and the bijection for every $T \in \mathcal{K}^\heartsuit$ is the universal property of a cokernel of f in $\mathcal{K}^\heartsuit$.

Step 4: The kernel. Put $N := (\tau^{\leq -1} Z)[-1]$ and $k := h[-1] \circ \alpha_Z^{-1}[-1]$. Since $\tau^{\leq -1} Z \in \mathcal{K}^{\leq -1}$, and $\tau^{\leq -1} Z \in \mathcal{K}^{\geq -1}$ by lemma 5.4.10(4) because $Z \in \mathcal{K}^{\geq -1}$, the object N lies in $\mathcal{K}^{\leq 0} \cap \mathcal{K}^{\geq 0} = \mathcal{K}^\heartsuit$. Let $T \in \mathcal{K}^\heartsuit$. As $T[1] \in \mathcal{K}^{\leq -1}$, composition with α_Z^{-1} is a bijection $\mathrm{Hom}(T[1], \tau^{\leq -1} Z) \to \mathrm{Hom}(T[1], Z)$ by proposition 5.4.9(2), and applying the bijection $[-1]$ on Hom groups turns it into the bijection $\varphi \mapsto \alpha_Z^{-1}[-1] \circ \varphi$ from $\mathrm{Hom}(T, N)$ to $\mathrm{Hom}(T, Z[-1])$. The first sequence of (5.3) for (f, g, h) gives the exact sequence

$$0 = \mathrm{Hom}(T, Y[-1]) \to \mathrm{Hom}(T, Z[-1]) \xrightarrow{(-h[-1])_*} \mathrm{Hom}(T, X) \xrightarrow{f_*} \mathrm{Hom}(T, Y)$$

where $\mathrm{Hom}(T, Y[-1]) = 0$ because $T \in \mathcal{K}^{\leq 0}$ and $Y[-1] \in \mathcal{K}^{\geq 1}$. The map $(h[-1])_* = -(-h[-1])_*$ has the same kernel and image, so $\varphi \mapsto k\varphi$ is a bijection from $\mathrm{Hom}(T, N)$ onto

$\{\psi \in \mathrm{Hom}(T,X) \big| f\psi = 0\}$, the universal property of a kernel of f. So $\mathcal{K}^\heartsuit$ has kernels and cokernels.

Step 5: $\mathcal{K}^\heartsuit$ is abelian. Let f be a monomorphism of $\mathcal{K}^\heartsuit$. Its kernel N from Step 4 is a zero object by lemma 1.1.4, so $\tau^{\leq -1}Z \cong N[1] \cong 0$, and $Z \in \mathcal{K}^{\geq 0}$ by lemma 5.4.10(2) with $n = -1$. With Step 2, $Z \in \mathcal{K}^\heartsuit$. For $T \in \mathcal{K}^\heartsuit$ the first sequence of (5.3) gives the exact sequence

$$0 = \mathrm{Hom}(T, Z[-1]) \to \mathrm{Hom}(T,X) \xrightarrow{f_*} \mathrm{Hom}(T,Y) \xrightarrow{g_*} \mathrm{Hom}(T,Z)$$

the first group vanishing because $Z[-1] \in \mathcal{K}^{\geq 1}$. So f is a kernel of $g\colon Y \to Z$ in $\mathcal{K}^\heartsuit$. Now let f be an epimorphism. Its cokernel C from Step 3 is a zero object, so $Z \in \mathcal{K}^{\leq -1}$ by lemma 5.4.10(1) with $n = -1$, and with Step 2, $Z[-1] \in \mathcal{K}^{\leq 0} \cap \mathcal{K}^{\geq 0} = \mathcal{K}^\heartsuit$. For $T \in \mathcal{K}^\heartsuit$ the second sequence of (5.3) gives the exact sequence

$$0 = \mathrm{Hom}(Z,T) \to \mathrm{Hom}(Y,T) \xrightarrow{f^*} \mathrm{Hom}(X,T) \to \mathrm{Hom}(Z[-1],T)$$

whose last map is composition with $-h[-1]$, the first group vanishing because $Z \in \mathcal{K}^{\leq -1}$. So f is a cokernel of $-h[-1]\colon Z[-1] \to X$ in $\mathcal{K}^\heartsuit$. Every monomorphism of $\mathcal{K}^\heartsuit$ is therefore a kernel and every epimorphism a cokernel, and $\mathcal{K}^\heartsuit$ is abelian by definition 1.1.6. This proves (1) and (2).

Step 6: Exactness in $\mathcal{K}^\heartsuit$. In the abelian category $\mathcal{K}^\heartsuit$, the sequence $0 \to X \to Y \to Z \to 0$ is exact if and only if f is a kernel of g and g is a cokernel of f. If both hold, f is a monomorphism and g an epimorphism by lemma 1.1.3, which is exactness at X and at Z (example 1.1.14), and at Y the composite $\mathrm{coker}\, f \circ \ker g$ is $g \circ f = 0$. Conversely, suppose the sequence is exact. Then g is an epimorphism, hence a cokernel by definition 1.1.6, hence the cokernel of its kernel $\ker g\colon K \to Y$ by lemma 1.1.7. Exactness at Y gives $gf = 0$, so $f = (\ker g)a$ for some $a\colon X \to K$, and $(\mathrm{coker}\, f)(\ker g) = 0$, so $\ker g$ factors through the kernel of $\mathrm{coker}\, f$. That kernel is f, since f is a monomorphism, hence a kernel, hence the kernel of its cokernel by the same two results. Writing $\ker g = fb$, the identities $f = fba$ and $\ker g = (\ker g)ab$ give $ba = \mathrm{id}_X$ and $ab = \mathrm{id}_K$, as f and $\ker g$ are monomorphisms. So f is a kernel of g, and g, the cokernel of $\ker g$, is a cokernel of f.

Step 7: Exact sequences and triangles. Suppose (f,g,h) is distinguished with $X, Y, Z \in \mathcal{K}^\heartsuit$. The exact sequence of Step 5 for monomorphisms used only $Z \in \mathcal{K}^\heartsuit$, and it shows that f is a kernel of g. The exact sequence of Step 3 shows that g^* is a bijection from $\mathrm{Hom}(Z,T)$ onto $\{\psi | \psi f = 0\}$ for every $T \in \mathcal{K}^\heartsuit$, so g is a cokernel of f. By Step 6 the sequence is exact. Conversely, let the sequence be exact, and choose by (TR1) a distinguished triangle (f,g',h') on (X,Y,Z'). Since f is a monomorphism, Step 5 puts Z' in $\mathcal{K}^\heartsuit$, and the previous argument makes g' a cokernel of f. Cokernels are unique up to unique isomorphism, so $g = \varphi g'$ for an isomorphism $\varphi\colon Z' \to Z$, and $(\mathrm{id}_X, \mathrm{id}_Y, \varphi)$ is an isomorphism of triangles from (f,g',h') to $(f,g,h'\varphi^{-1})$, which is therefore distinguished by (TR1). For uniqueness, let (f,g,h) and (f,g,h_1) both be distinguished. (TR3) gives $\kappa\colon Z \to Z$ with $(\mathrm{id}_X, \mathrm{id}_Y, \kappa)$ a morphism from the first to the second, so $\kappa g = g$ and $h_1\kappa = h$. Then $(\kappa - \mathrm{id}_Z)g = 0$, and in the second sequence of (5.3) for (f,g,h) and $T = Z$, the kernel of g^* on $\mathrm{Hom}(Z,Z)$ is the image of $\mathrm{Hom}(X[1],Z)$, which is zero because $X[1] \in \mathcal{K}^{\leq -1}$ and $Z \in \mathcal{K}^{\geq 0}$. So $\kappa = \mathrm{id}_Z$ and $h_1 = h$, completing the proof.

Theorem 5.4.13 is [3, Thm. 1.3.6], and (1) is also [15, Thm. IV.4.4]. By (3), the short exact sequences of $\mathcal{K}^\heartsuit$ are the distinguished triangles with all three vertices in $\mathcal{K}^\heartsuit$, and the third edge of such a triangle is determined by the other two. The heart does not determine $\mathcal{K}$: in example 5.4.3 it is zero whatever $\mathcal{K}$ is.

Example 5.4.14: Kernel and Cokernel of Multiplication by 2 on $\mathbb{Z}/4$

In the standard t-structure on $D(\mathbf{Ab})$, let $f\colon \mathbb{Z}/4 \to \mathbb{Z}/4$ be multiplication by 2, both objects in degree 0. By example 5.3.8, its mapping cone triangle on $(\mathbb{Z}/4, \mathbb{Z}/4, MC(f))$ is distinguished in $D(\mathbf{Ab})$, and $MC(f)$ is the complex $\mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4$ in degrees -1 and 0, the complex Y of example 5.3.2. Both $H^{-1}(MC(f))$ and $H^0(MC(f))$ are isomorphic to $\mathbb{Z}/2$ there, so $MC(f) \in D^{\leq 0} \cap D^{\geq -1}$, as theorem 5.4.13(2) requires, and $MC(f)$ is not in the heart. For the standard t-structure the truncation functors $\tau^{\leq -1}$ and $\tau^{\geq 0}$ are the canonical truncations, by the choice in Step 1 of the proof of proposition 5.4.9 and the paragraph after that proof, and example 5.3.5 computes them for $MC(f)$: $\tau^{\leq -1}MC(f)$ is $\{0,2\}$ in degree -1, and $\tau^{\geq 0}MC(f)$ is $(\mathbb{Z}/4)/\{0,2\}$ in degree 0. So the kernel $\big(\tau^{\leq -1}MC(f)\big)[-1]$ and the cokernel $\tau^{\geq 0}MC(f)$ that theorem 5.4.13(2) gives are isomorphic to $\{0,2\}$ and $(\mathbb{Z}/4)/\{0,2\}$ in degree 0. The equivalence H^0 of theorem 5.3.7 carries them to the kernel and cokernel of multiplication by 2 computed in $\mathbf{Ab}$. Neither is zero, so f is neither a monomorphism nor an epimorphism of the heart.

For the standard t-structure, H^n_t is ordinary cohomology (example 5.4.12), which carries distinguished triangles to long exact sequences. For every t-structure, H^0_t is a cohomological functor $\mathcal{K} \to \mathcal{K}^\heartsuit$. A triangle whose first vertex lies in $\mathcal{K}^{\leq 0}$ goes to a right exact sequence in the heart, a triangle whose third vertex lies in $\mathcal{K}^{\geq 0}$ to a left exact one, and the octahedral axiom splits an arbitrary triangle into one of each.

Corollary 5.4.15: H^0_t Is a Cohomological Functor

Let $(\mathcal{K}^{\leq 0}, \mathcal{K}^{\geq 0})$ be a t-structure on $\mathcal{K}$. Then $H^0_t\colon \mathcal{K} \to \mathcal{K}^\heartsuit$ is a cohomological functor in the sense of definition 3.1.6. Consequently every distinguished triangle (u, v, w) on (X, Y, Z) gives a long exact sequence in $\mathcal{K}^\heartsuit$

$$\cdots \to H^n_t(X) \to H^n_t(Y) \to H^n_t(Z) \to H^{n+1}_t(X) \to H^{n+1}_t(Y) \to \cdots$$

***Proof*:**

Write H^0 for H^0_t, which is additive because $\tau^{\leq 0}$ and $\tau^{\geq 0}$ are (proposition 5.4.9). In the abelian category $\mathcal{K}^\heartsuit$ (theorem 5.4.13), a pair $P \xrightarrow{a} Q \xrightarrow{b} R$ with $ba = 0$ is exact at Q as soon as a is a kernel of b or b is a cokernel of a, since then $\operatorname{coker} a \circ \ker b$ is $\operatorname{coker} a \circ a = 0$, respectively $b \circ \ker b = 0$ (definition 1.1.13). If a and b are replaced by $\mu a \nu$ and $b\mu^{-1}$ for isomorphisms μ and ν, a kernel stays a kernel, and if they are replaced by $\mu a \nu$ and $\lambda b \mu^{-1}$, a cokernel stays a cokernel. The proof establishes one of the two conditions, case by case, using four facts.

(P) Maps out of $H^0(U)$ for $U \in \mathcal{K}^{\leq 0}$. By lemma 5.4.10(1), α^0_U is an isomorphism. Put $\theta_U := \beta^{-1}_{\tau^{\leq 0}U} \circ (\alpha^0_U)^{-1}\colon U \to H^0(U)$, which is natural in $U \in \mathcal{K}^{\leq 0}$ by the naturality of α^0 and β^{-1}. For $T \in \mathcal{K}^\heartsuit$, precomposition with θ_U is a bijection $\operatorname{Hom}(H^0(U), T) \to \operatorname{Hom}(U, T)$, by proposition 5.4.9(2) for β^{-1}, as $T \in \mathcal{K}^{\geq 0}$.

(P) Maps into $H^0(V)$ for $V \in \mathcal{K}^{\geq 0}$.* By lemma 5.4.10(4), $\tau^{\leq 0}V \in \mathcal{K}^{\geq 0}$, so $\beta^{-1}_{\tau^{\leq 0}V}$ is an isomorphism by lemma 5.4.10(2) with $n = -1$. Put $\kappa_V := \alpha^0_V \circ (\beta^{-1}_{\tau^{\leq 0}V})^{-1} \colon H^0(V) \to V$, natural in $V \in \mathcal{K}^{\geq 0}$. For $T \in \mathcal{K}^{\heartsuit}$, composition with κ_V is a bijection $\mathrm{Hom}(T, H^0(V)) \to \mathrm{Hom}(T, V)$, by proposition 5.4.9(2) for α^0, as $T \in \mathcal{K}^{\leq 0}$.

(E) $H^0(\alpha^0_X)$ is an isomorphism for every X. Naturality of α^0 at the morphism α^0_X gives $\alpha^0_X \circ \tau^{\leq 0}(\alpha^0_X) = \alpha^0_X \circ \alpha^0_{\tau^{\leq 0}X}$, so $\tau^{\leq 0}(\alpha^0_X) = \alpha^0_{\tau^{\leq 0}X}$ by the injectivity in proposition 5.4.9(2), and this is an isomorphism by lemma 5.4.10(1), as $\tau^{\leq 0}X \in \mathcal{K}^{\leq 0}$. Applying $\tau^{\geq 0}$ gives the claim.

Case 1: $X, Y, Z \in \mathcal{K}^{\leq 0}$. Then $H^0(v)$ is a cokernel of $H^0(u)$. For $T \in \mathcal{K}^{\heartsuit}$ the second sequence of (5.3) gives the exact sequence $0 = \mathrm{Hom}(X[1], T) \to \mathrm{Hom}(Z, T) \xrightarrow{v^*} \mathrm{Hom}(Y, T) \xrightarrow{u^*} \mathrm{Hom}(X, T)$, the first group vanishing because $X[1] \in \mathcal{K}^{\leq -1}$ and $T \in \mathcal{K}^{\geq 0}$. By (P) and the naturality of θ it is isomorphic to the sequence

$$0 \to \mathrm{Hom}(H^0(Z), T) \xrightarrow{H^0(v)^*} \mathrm{Hom}(H^0(Y), T) \xrightarrow{H^0(u)^*} \mathrm{Hom}(H^0(X), T)$$

which is therefore exact for every $T \in \mathcal{K}^{\heartsuit}$. That is the statement that $H^0(v)$ is a cokernel of $H^0(u)$.

Case 1: $X, Y, Z \in \mathcal{K}^{\geq 0}$. Then $H^0(u)$ is a kernel of $H^0(v)$.* For $T \in \mathcal{K}^{\heartsuit}$ the first sequence of (5.3) gives the exact sequence $0 = \mathrm{Hom}(T, Z[-1]) \to \mathrm{Hom}(T, X) \xrightarrow{u_*} \mathrm{Hom}(T, Y) \xrightarrow{v_*} \mathrm{Hom}(T, Z)$, the first group vanishing because $T \in \mathcal{K}^{\leq 0}$ and $Z[-1] \in \mathcal{K}^{\geq 1}$. By (P*) and the naturality of κ it is isomorphic to $0 \to \mathrm{Hom}(T, H^0(X)) \to \mathrm{Hom}(T, H^0(Y)) \to \mathrm{Hom}(T, H^0(Z))$ with maps $H^0(u)_*$ and $H^0(v)_*$, which says that $H^0(u)$ is a kernel of $H^0(v)$.

Case 2: $X \in \mathcal{K}^{\leq 0}$. Then $H^0(v)$ is a cokernel of $H^0(u)$. By proposition 5.4.9(2), $u = \alpha^0_Y u'$ for a unique $u' \colon X \to \tau^{\leq 0}Y$. Choose a distinguished triangle (u', r, e) on $(X, \tau^{\leq 0}Y, W)$ and apply (TR4) to the composable pair $X \xrightarrow{u'} \tau^{\leq 0}Y \xrightarrow{\alpha^0_Y} Y$, with (u', r, e) as the triangle on the first map, the truncation triangle $(\alpha^0_Y, \beta^0_Y, \partial^0_Y)$ on the second, and (u, v, w) on the composite. It gives a distinguished triangle $(f, g, r[1]\,\partial^0_Y)$ on $(W, Z, \tau^{\geq 1}Y)$ satisfying, among the four identities of (TR4), $fr = v\alpha^0_Y$. Rotating (u', r, e) gives the distinguished triangle $(r, e, -u'[1])$ on $(\tau^{\leq 0}Y, W, X[1])$, whose outer vertices lie in $\mathcal{K}^{\leq 0}$, so $W \in \mathcal{K}^{\leq 0}$ by lemma 5.4.10(3). The triangle $(f, g, r[1]\,\partial^0_Y)$ on $(W, Z, \tau^{\geq 1}Y)$ therefore has $W \in \mathcal{K}^{\leq 0}$ and $\tau^{\geq 1}Y \in \mathcal{K}^{\geq 1}$, and proposition 5.4.9(3) gives an isomorphism $\varphi \colon W \to \tau^{\leq 0}Z$ with $\alpha^0_Z \varphi = f$. Then $\alpha^0_Z \varphi r = fr = v\alpha^0_Y$. Case 1 applies to (u', r, e), whose three vertices lie in $\mathcal{K}^{\leq 0}$: $H^0(r)$ is a cokernel of $H^0(u')$. Finally $H^0(u) = H^0(\alpha^0_Y)H^0(u')$, and applying H^0 to $\alpha^0_Z \varphi r = v\alpha^0_Y$ gives $H^0(v) = H^0(\alpha^0_Z)H^0(\varphi)H^0(r)H^0(\alpha^0_Y)^{-1}$, where $H^0(\alpha^0_Y)$ and $H^0(\alpha^0_Z)$ are isomorphisms by (E), and $H^0(\varphi)$ is one because φ is. So $H^0(v)$ is a cokernel of $H^0(u)$.

(E) $H^0(\beta^{-1}_P)$ is an isomorphism for every P.* Case 2 applies to the truncation triangle $(\alpha^{-1}_P, \beta^{-1}_P, \partial^{-1}_P)$, whose first vertex lies in $\mathcal{K}^{\leq -1} \subseteq \mathcal{K}^{\leq 0}$, so $H^0(\beta^{-1}_P)$ is a cokernel of $H^0(\alpha^{-1}_P)$. The source of the latter is zero: $\tau^{\leq 0}\tau^{\leq -1}P \cong \tau^{\leq -1}P$ by lemma 5.4.10(1), and $\tau^{\geq 0}$ of an object of $\mathcal{K}^{\leq -1}$ is zero by lemma 5.4.10(1) with $n = -1$. The identity of $H^0(P)$ is a cokernel of the morphism from a zero object, and cokernels are unique up to isomorphism, so $H^0(\beta^{-1}_P)$ is an isomorphism.

Case 2: $Z \in \mathcal{K}^{\geq 0}$. Then $H^0(u)$ is a kernel of $H^0(v)$.* Choose a distinguished triangle $(u\alpha_X^{-1}, y, e)$ on $(\tau^{\leq -1}X, Y, V)$ and apply (TR4) to $\tau^{\leq -1}X \xrightarrow{\alpha_X^{-1}} X \xrightarrow{u} Y$, with the truncation triangle $(\alpha_X^{-1}, \beta_X^{-1}, \partial_X^{-1})$ on the first map, (u, v, w) on the second and $(u\alpha_X^{-1}, y, e)$ on the composite. It gives a distinguished triangle $(f, g, \beta_X^{-1}[1]\, w)$ on $(\tau^{\geq 0}X, V, Z)$ with $v = gy$ and $yu = f\beta_X^{-1}$. Its outer vertices lie in $\mathcal{K}^{\geq 0}$, so $V \in \mathcal{K}^{\geq 0}$ by lemma 5.4.10(3). Then $(u\alpha_X^{-1}, y, e)$ is distinguished on $(\tau^{\leq -1}X, Y, V)$ with $\tau^{\leq -1}X \in \mathcal{K}^{\leq -1}$ and $V \in \mathcal{K}^{\geq 0}$, and proposition 5.4.9(3) with $n = -1$ gives an isomorphism $\psi\colon V \to \tau^{\geq 0}Y$ with $\psi y = \beta_Y^{-1}$. Hence $\beta_Y^{-1}u = \psi yu = \psi f\beta_X^{-1}$. Case 1* applies to $(f, g, \beta_X^{-1}[1]\, w)$, whose three vertices lie in $\mathcal{K}^{\geq 0}$: $H^0(f)$ is a kernel of $H^0(g)$. Applying H^0 to $\beta_Y^{-1}u = \psi f\beta_X^{-1}$ and to $v = g\psi^{-1}\beta_Y^{-1}$ gives

$$H^0(u) = \mu\, H^0(f)\, H^0(\beta_X^{-1}), \qquad H^0(v) = H^0(g)\, \mu^{-1}, \qquad \mu := H^0(\beta_Y^{-1})^{-1} H^0(\psi)$$

with μ and $H^0(\beta_X^{-1})$ isomorphisms by (E*). So $H^0(u)$ is a kernel of $H^0(v)$.

The general case. Let (u, v, w) be any distinguished triangle on (X, Y, Z). Choose a distinguished triangle $(u\alpha_X^0, y, e)$ on $(\tau^{\leq 0}X, Y, U)$ and apply (TR4) to $\tau^{\leq 0}X \xrightarrow{\alpha_X^0} X \xrightarrow{u} Y$, with the truncation triangle $(\alpha_X^0, \beta_X^0, \partial_X^0)$ on the first map, (u, v, w) on the second and $(u\alpha_X^0, y, e)$ on the composite. It gives a distinguished triangle $(f, g, \beta_X^0[1]\, w)$ on $(\tau^{\geq 1}X, U, Z)$ with $v = gy$. Rotating it gives the distinguished triangle $(g, \beta_X^0[1]\, w, -f[1])$ on $(U, Z, (\tau^{\geq 1}X)[1])$, whose third vertex lies in $\mathcal{K}^{\geq 0}$. Case 2, applied to $(u\alpha_X^0, y, e)$, whose first vertex lies in $\mathcal{K}^{\leq 0}$, shows that $H^0(y)$ is a cokernel of $H^0(u)H^0(\alpha_X^0)$, hence of $H^0(u)$, as $H^0(\alpha_X^0)$ is an isomorphism by (E). Case 2*, applied to the rotated triangle, shows that $H^0(g)$ is a kernel, hence a monomorphism by lemma 1.1.3. So $\ker H^0(v) = \ker(H^0(g)H^0(y)) = \ker H^0(y)$, and $\operatorname{coker} H^0(u) \circ \ker H^0(v) = H^0(y) \circ \ker H^0(y) = 0$. Together with $H^0(v)H^0(u) = H^0(vu) = 0$ (lemma 3.1.7), this is exactness at $H^0(Y)$, so H^0 is cohomological. The long exact sequence follows by applying this to the rotations of (u, v, w) and splicing the resulting exact sequences, with $H^0(X[n]) = H_t^n(X)$, completing the proof.

The corollary is the last assertion of [3, Thm. 1.3.6] and is [15, Thm. IV.4.11(a)]. For the standard t-structure it returns the long exact cohomology sequence of a distinguished triangle of $D(\mathcal{A})$, by example 5.4.12.

Example 5.4.16: The Four-Term Sequence of Multiplication by 2

Apply corollary 5.4.15 to the mapping cone triangle of $f = 2\colon \mathbb{Z}/4 \to \mathbb{Z}/4$ from example 5.4.14. As $\mathbb{Z}/4$ in degree 0 has $H_t^n = 0$ for $n \neq 0$, and $MC(f)$ has $H_t^n = 0$ for $n \neq -1, 0$, the long exact sequence reduces, after applying the equivalence of theorem 5.3.7, which preserves kernels and cokernels, and example 5.4.12, to the exact sequence

$$0 \to H^{-1}(MC(f)) \to \mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4 \to H^0(MC(f)) \to 0$$

Exactness on the left makes the first map an injection onto $\ker(2) = \{0, 2\}$, and exactness on the right makes the last map a surjection with kernel $\{0, 2\}$. So the long exact sequence of the triangle is the four-term sequence $0 \to \ker f \to \mathbb{Z}/4 \to \mathbb{Z}/4 \to \operatorname{coker} f \to 0$, and it recovers both computations of example 5.4.14 at once.

In $D(\mathcal{A})$ a complex lying in every $D^{\leq n}$, or in every $D^{\geq n}$, has zero cohomology and so is zero (proposition 3.2.12). The axioms do not give this for an arbitrary t-structure: if X is a nonzero object of $\mathcal{K}$, then X lies in every $\mathcal{K}^{\leq n}$ for the t-structure of example 5.4.3. Nondegeneracy, the first

condition in definition 5.4.17, imposes this vanishing on a t-structure. Boundedness, the second, asks that every object lie in $\mathcal{K}^{\leq b} \cap \mathcal{K}^{\geq a}$ for some integers a and b. When $\mathcal{A}$ has a nonzero object it fails for the standard t-structure on $D(\mathcal{A})$, where a single complex can have nonzero cohomology in infinitely many degrees.

Definition 5.4.17: Nondegenerate and Bounded t-Structures

A t-structure $(\mathcal{K}^{\leq 0}, \mathcal{K}^{\geq 0})$ on $\mathcal{K}$ is *nondegenerate* if every object lying in $\mathcal{K}^{\leq n}$ for all $n \in \mathbb{Z}$ is a zero object, and every object lying in $\mathcal{K}^{\geq n}$ for all $n \in \mathbb{Z}$ is a zero object. It is *bounded* if every $X \in \mathcal{K}$ lies in $\mathcal{K}^{\leq b} \cap \mathcal{K}^{\geq a}$ for some integers a and b.

Example 5.4.18: The Standard t-Structure Is Nondegenerate and Not Bounded

An object X of $D^{\leq n}$ for every n, or of $D^{\geq n}$ for every n, has $H^i(X) = 0$ for all i by example 5.4.2, so X is exact and $X \cong 0$ in $D(\mathcal{A})$ by proposition 3.2.12. So the standard t-structure is nondegenerate. If $\mathcal{A}$ has a nonzero object M, the complex with M in every degree and all differentials zero has $H^i = M$ for every i, so it lies in no $D^{\leq b}$, and the standard t-structure is not bounded. The objects lying in some $D^{\leq b} \cap D^{\geq a}$ are the complexes with bounded cohomology, which form the subcategory $D^b(\mathcal{A})$ of box 3.2.5, and the truncation argument there shows that each such X is isomorphic in $D(\mathcal{A})$ to the bounded complex $\tau^{\geq a}\tau^{\leq b}X$. The t-structure of example 5.4.3 on a category with a nonzero object X is degenerate, since $X \in \mathcal{K}^{\leq n}$ for every n.

Besides its standard t-structure, the constructible derived category of a stratified space carries the perverse t-structure, whose heart, the category of perverse sheaves, is abelian by theorem 5.4.13. Its construction is the subject of the *Perverse Sheaves* volume [13]. A t-structure on a stable ∞-category is a t-structure in the sense of definition 5.4.1 on its homotopy category. There the enhancement provides functorial cones [12], which the axioms of a triangulated category do not give (box 3.1.4).

Exercise 5.4.1

1. Let $\mathcal{K}$ be a triangulated category, let $\mathcal{K}^{\leq 0}$ be the full subcategory of zero objects, and let $\mathcal{K}^{\geq 0} = \mathcal{K}$. Show that $(\mathcal{K}^{\leq 0}, \mathcal{K}^{\geq 0})$ is a t-structure, that its heart consists of the zero objects, and that it is nondegenerate if and only if every object of $\mathcal{K}$ is a zero object.
2. Show that a bounded t-structure is nondegenerate.
3. Let $X, Y \in \mathcal{K}^{\heartsuit}$ and $n < 0$. Show that $\mathrm{Hom}_{\mathcal{K}}(X, Y[n]) = 0$.
4. Let $T \in \mathcal{K}^{\heartsuit}$. Show that $H^0_t(T) \cong T$ and that $H^n_t(T) \cong 0$ for every $n \neq 0$.
5. Let $X \in \mathcal{K}^{\leq 0}$ and $Y \in \mathcal{K}^{\geq 0}$. Show that $f \mapsto H^0_t(f)$ is a bijection $\mathrm{Hom}_{\mathcal{K}}(X, Y) \to \mathrm{Hom}_{\mathcal{K}^{\heartsuit}}(H^0_t X, H^0_t Y)$.
6. In the standard t-structure on $D(\mathbf{Ab})$, let $f\colon \mathbb{Z}/4 \to \mathbb{Z}/8$ be the homomorphism $x \mapsto 4x$, both objects in degree 0. Compute the cohomology of $MC(f)$, decide whether f is a monomorphism or an epimorphism of the heart, and identify, through the equivalence of theorem 5.3.7, the kernel and cokernel that theorem 5.4.13(2) produces.
7. Let $X, Z \in \mathcal{K}^{\heartsuit}$, and let $i_X\colon X \to X \oplus Z$ and $p_Z\colon X \oplus Z \to Z$ be the inclusion and projection of the biproduct. Use theorem 5.4.13(3) and lemma 3.1.7 to show that $(i_X, p_Z, 0)$ is a distinguished triangle on $(X, X \oplus Z, Z)$.

8. Suppose the t-structure is nondegenerate.

    1. Show that if $X \in \mathcal{K}^{\geq 0}$ and $H^0_t(X) \cong 0$, then $X \in \mathcal{K}^{\geq 1}$, and that if $X \in \mathcal{K}^{\leq 0}$ and $H^0_t(X) \cong 0$, then $X \in \mathcal{K}^{\leq -1}$.
    2. Deduce that $X \cong 0$ whenever $H^n_t(X) \cong 0$ for every $n \in \mathbb{Z}$. (Treat $X \in \mathcal{K}^{\geq 0}$ and $X \in \mathcal{K}^{\leq 0}$ first, then apply the long exact sequence of corollary 5.4.15 to the truncation triangle of X.)
    3. Show that a morphism f of $\mathcal{K}$ is an isomorphism if and only if $H^n_t(f)$ is an isomorphism for every n.

6

Spectral Sequences

The derived functors of chapter 2 measure the failure of a single functor to be exact, and the long exact sequence of theorem 2.3.11 turns that measurement into something computable. Both tools carry the same limitation: each relates the cohomology of one object to the cohomology of two others, and no more. An object assembled from many pieces produces not one long exact sequence but a whole family of them at once, one for each stage of the assembly, and no finite amount of splicing turns that family into an answer. A module carrying a filtration $M = M_0 \supseteq M_1 \supseteq \cdots$ ([7, Filtrations]) is the basic case; a functor that factors through a third abelian category is another, and the double complexes of section 2.5 are a third.

A spectral sequence is the bookkeeping that assembles such a family. It produces a sequence of *pages* $E_1, E_2, E_3, \ldots$, each the cohomology of the one before it, beginning with a crude first approximation to the answer and correcting it repeatedly until the corrections run out. This chapter builds the machine and then gives the three standard sources of one. Section 6.1 isolates the algebraic mechanism, Massey's exact couple, from which every page and every differential is generated by a single operation. Section 6.2 constructs the spectral sequence of a filtered complex and proves the convergence and comparison theorems that make it usable. Section 6.3 treats the two spectral sequences of a double complex, completing the computations of section 2.5.1 that were deferred to this chapter. Section 6.4 derives a left exact functor on a bounded-below complex through a resolution of the complex by injectives, and gives the two spectral sequences of this hypercohomology. Section 6.5 obtains from them the spectral sequence of a composite of derived functors, the form in which the machine is most often applied. Section 6.6 treats complexes carrying two filtrations.

6.1 Exact Couples and Derived Couples

The obstacle a spectral sequence exists to overcome is visible already in the simplest case. A short exact sequence of complexes $0 \to A_\bullet \to C_\bullet \to C_\bullet/A_\bullet \to 0$ yields a long exact sequence relating

the three cohomologies, but that sequence does not determine $H_\bullet(C)$ from $H_\bullet(A)$ and $H_\bullet(C/A)$: it determines only the two ends of a short exact sequence

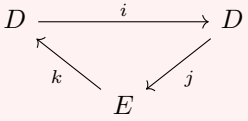
$$0 \to \operatorname{coker}\partial \to H_n(C) \to \ker \partial \to 0$$

and the extension itself is extra information the sequence does not carry. With one layer the ambiguity is a single extension problem. With a filtration of many layers there is one such problem per layer, and there are also correction terms of a different kind: a cohomology class of one layer may fail to extend over the next layer but succeed over the one after, and nothing in any individual long exact sequence detects that.

The device that organizes all of these at once packages every long exact sequence of the filtration into a single self-regenerating triangle. Its two vertices carry different roles: one is the answer being approximated, the other the auxiliary object that remembers how the layers are glued.

Definition 6.1.1: Exact Couple

Let $\mathcal{A}$ be an abelian category. An *exact couple* in $\mathcal{A}$ is a pair of objects D and E together with three morphisms



such that the triangle is exact at each of its three corners:

$$\operatorname{im} i = \ker j, \qquad \operatorname{im} j = \ker k, \qquad \operatorname{im} k = \ker i$$

The composite $d := j \circ k \colon E \to E$ is the *differential* of the couple, and the couple is written (D, E, i, j, k).

The data should be read as a compression of a long exact sequence, or rather of all of them simultaneously. In the filtered case treated in section 6.2, the object D collects the cohomologies of the stages of the filtration, the object E collects the cohomologies of the successive layers, the map i is induced by the inclusion of one stage in the next, the map j is the passage to the layer, and k is the connecting homomorphism. Exactness at the three corners is precisely the exactness of every one of those long exact sequences. Of the two objects only E is meant to be looked at, since it is the approximation to the answer; D is the auxiliary object, and it is the auxiliary object that makes the approximation improvable.

Two of the three exactness conditions are spent constructing the next stage, and one is spent immediately, on the claim that d deserves its name.

Lemma 6.1.2: The Differential of an Exact Couple Squares to Zero

Let (D, E, i, j, k) be an exact couple. Then $d \circ d = 0$, so (E, d) is a differential object and its cohomology

$$H(E, d) = \frac{\ker d}{\operatorname{im} d}$$

is defined.

***Proof*:**
Expand the composite and regroup:

$$d \circ d = (j \circ k) \circ (j \circ k) = j \circ (k \circ j) \circ k$$

Exactness at the corner E gives $\mathrm{im} j = \ker k$, so every morphism factoring through j is annihilated by k, that is $k \circ j = 0$. Hence $d \circ d = j \circ 0 \circ k = 0$, as we sought to show.

Only exactness at E was used. The remaining two conditions are exactly what is needed to promote $H(E, d)$ from a bare cohomology object to the E-term of a new exact couple, and it is that promotion which makes the construction repeat. The new auxiliary object is the image of i, and the new maps are induced by the old ones.

6.1.3 Convention: Elements in an Abelian Category The constructions below are phrased with elements, in the style of a diagram chase. This is legitimate in any abelian category: by the Freyd-Mitchell theorem (theorem 1.1.15), a small abelian category admits a fully faithful exact embedding into R-Mod for some ring R, and a diagram whose exactness is verified in R-Mod is exact in $\mathcal{A}$ because the embedding reflects exactness. Every statement in this section may therefore be proved for modules and read off for a general $\mathcal{A}$. The standard sources of exact couples, filtered complexes and double complexes of modules or of sheaves, are in any case module-like.

Definition 6.1.4: Derived Couple

Let (D, E, i, j, k) be an exact couple with differential $d = j \circ k$. Its *derived couple* (D', E', i', j', k') consists of the objects

$$D' := \mathrm{im} i, \qquad E' := H(E, d) = \frac{\ker d}{\mathrm{im} d}$$

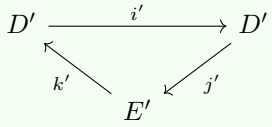
together with the morphisms

1. $i' := i|_{D'} \colon D' \to D'$, the restriction of i to its image;
2. $j' \colon D' \to E'$ defined by $j'(i(x)) := [\, j(x) \,]$ for $x \in D$;
3. $k' \colon E' \to D'$ defined by $k'([e]) := k(e)$ for $e \in \ker d$.

The name is older than, and unrelated to, the derived categories of chapter 3; it originates in Massey's work in algebraic topology. Three of the four items above require an argument before the definition is even meaningful: that i carries D' into itself, that j' is independent of the choice of preimage, and that k' both lands in D' and is independent of the representative cycle. Those checks and the exactness of the result are one theorem.

Theorem 6.1.5: The Derived Couple is an Exact Couple

Let (D, E, i, j, k) be an exact couple. Then the morphisms i', j' and k' of definition 6.1.4 are well defined, and the triangle



is an exact couple.

***Proof*:**

Step 1: The three maps are well defined. For i', note that $i(D') = i(i(D)) \subseteq i(D) = D'$, so restricting i to D' does indeed produce an endomorphism of D'.

For j', two things must be checked. First, $j(x)$ is a cycle for every $x \in D$: applying the differential gives $d(j(x)) = j(k(j(x))) = 0$, since $k \circ j = 0$ by exactness at E. So the class $[j(x)] \in E'$ makes sense. Second, the class does not depend on the choice of x. Suppose $i(x) = i(x')$. Then $x - x' \in \ker i = \mathrm{im}k$ by exactness at the target corner, say $x - x' = k(e)$ for some $e \in E$. Applying j gives

$$j(x) - j(x') = j(k(e)) = d(e) \in \mathrm{im}d$$

so $[j(x)] = [j(x')]$ in $E' = \ker d/\mathrm{im}d$.

For k', again two checks. First, k carries cycles into D': if $d(e) = 0$ then $j(k(e)) = 0$, so $k(e) \in \ker j = \mathrm{im}i = D'$ by exactness at the source corner. Second, the value is independent of the representative. If $e - e' = d(f) = j(k(f))$ for some $f \in E$, then

$$k(e) - k(e') = k(j(k(f))) = (k \circ j)(k(f)) = 0$$

again because $k \circ j = 0$. So k' is well defined.

Step 2: Exactness at the corner receiving i', that is $\mathrm{im}i' = \ker j'$. For the inclusion $\mathrm{im}i' \subseteq \ker j'$, take $y = i(x) \in D'$ and compute

$$j'(i'(y)) = j'(i(i(x))) = [\, j(i(x)) \,] = [0] = 0$$

the last equality because $\mathrm{im}i = \ker j$. For the reverse inclusion, let $y = i(x) \in D'$ satisfy $j'(y) = 0$, that is $[j(x)] = 0$, so $j(x) = d(e) = j(k(e))$ for some $e \in E$. Then $j(x - k(e)) = 0$, so $x - k(e) \in \ker j = \mathrm{im}i$, say $x - k(e) = i(z)$. Applying i and using $i \circ k = 0$ (exactness at the corner where k arrives) gives

$$y = i(x) = i(k(e)) + i(i(z)) = i(i(z)) = i'(i(z))$$

and $i(z) \in D'$, so $y \in \mathrm{im}i'$.

Step 3: Exactness at E', that is $\mathrm{im}j' = \ker k'$. For $\mathrm{im}j' \subseteq \ker k'$, take $y = i(x) \in D'$ and compute

$$k'(j'(y)) = k'([\, j(x) \,]) = k(j(x)) = 0$$

using $k \circ j = 0$. For the reverse inclusion, let $[e] \in E'$ satisfy $k'([e]) = k(e) = 0$. Then $e \in \ker k = \mathrm{im} j$, say $e = j(x)$ for some $x \in D$. The element $i(x)$ lies in D' and $j'(i(x)) = [j(x)] = [e]$, so $[e] \in \mathrm{im} j'$.

Step 4: Exactness at the corner receiving k', that is $\mathrm{im} k' = \ker i'$. For $\mathrm{im} k' \subseteq \ker i'$, take $[e] \in E'$; then $i'(k'([e])) = i(k(e)) = 0$ since $\mathrm{im} k = \ker i$. For the reverse inclusion, let $y = i(x) \in D'$ satisfy $i'(y) = 0$, that is $i(i(x)) = 0$. Then $i(x) \in \ker i = \mathrm{im} k$, say $i(x) = k(e)$ for some $e \in E$. This e is a cycle:

$$d(e) = j(k(e)) = j(i(x)) = 0$$

because $\mathrm{im} i = \ker j$. Hence $[e] \in E'$ and $k'([e]) = k(e) = i(x) = y$, so $y \in \mathrm{im} k'$.

All six inclusions hold, so the derived triangle is exact at each corner, completing the proof.

The theorem is the basis of the entire chapter. Because the derived couple is again an exact couple, the construction applies to it in turn, and iterating produces an infinite sequence of couples whose E-terms form a sequence of objects each of which is the cohomology of its predecessor. That sequence is the object the chapter is named for.

Definition 6.1.6: Spectral Sequence

Let $\mathcal{A}$ be an abelian category. A *spectral sequence* in $\mathcal{A}$ is a family $\{(E_r, d_r)\}_{r \geq r_0}$ of objects E_r of $\mathcal{A}$, each equipped with an endomorphism $d_r \colon E_r \to E_r$ satisfying $d_r \circ d_r = 0$, together with isomorphisms

$$E_{r+1} \cong H(E_r, d_r) = \frac{\ker d_r}{\mathrm{im} d_r}$$

for every $r \geq r_0$. The object E_r is the *r-th page*, d_r is the *r-th differential*, and r_0 is the *initial page*.

The spectral sequence is *bigraded* if each E_r carries a decomposition $E_r = \bigoplus_{p,q} E_r^{p,q}$ and each d_r is homogeneous of a fixed bidegree. Two conventions are in use. In the *homological* convention the pages are written $E^r_{p,q}$ and

$$d_r \colon E^r_{p,q} \to E^r_{p-r,\, q+r-1}$$

while in the *cohomological* convention they are written $E_r^{p,q}$ and

$$d_r \colon E_r^{p,q} \to E_r^{p+r,\, q-r+1}$$

Both differentials change the total degree $p+q$ by exactly one, downward in the homological convention and upward in the cohomological one. The two are exchanged by $E_r^{p,q} = E^r_{-p,-q}$.

The bidegrees are not a separate convention imposed on the definition; they are forced by the exact couple that generates the sequence, as the next result records. Every spectral sequence appearing in this book, and every one cited from it elsewhere in the series, arises from an exact couple in the manner described.

Theorem 6.1.7: The Spectral Sequence of an Exact Couple

Let $(D, E, i, j, k) = (D^1, E^1, i^{(1)}, j^{(1)}, k^{(1)})$ be an exact couple, and for $r \geq 1$ let $(D^{r+1}, E^{r+1}, i^{(r+1)}, j^{(r+1)}, k^{(r+1)})$ be the derived couple of the r-th one. Then $\{(E^r, d_r)\}_{r\geq 1}$, with $d_r = j^{(r)} \circ k^{(r)}$, is a spectral sequence, and $D^{r+1} = i^r(D)$ for every r.
Suppose in addition that D and E are bigraded and that i, j, k are homogeneous of bidegrees

$$\deg i = (1, -1), \qquad \deg j = (0, 0), \qquad \deg k = (-1, 0)$$

with the pages of the derived couples indexed by $D^{r+1}_{p,q} = i\big(D^r_{p-1,q+1}\big)$. Then

$$\deg i^{(r)} = (1, -1), \qquad \deg j^{(r)} = \big(-(r-1),\, r-1\big), \qquad \deg k^{(r)} = (-1, 0)$$

and consequently d_r is homogeneous of bidegree $(-r,\, r-1)$, so the sequence is bigraded in the homological convention of definition 6.1.6.

***Proof*:**
Step 1: The pages form a spectral sequence. By theorem 6.1.5 each derived couple is again an exact couple, so the construction may be iterated and every d_r satisfies $d_r \circ d_r = 0$ by lemma 6.1.2. By definition 6.1.4 the E-term of the derived couple is $E^{r+1} = H(E^r, d_r)$, which is the required isomorphism. The identity $D^{r+1} = i^r(D)$ follows by induction: it holds for $r = 0$ by convention, and if $D^r = i^{r-1}(D)$ then $D^{r+1} = i^{(r)}(D^r) = i(i^{r-1}(D)) = i^r(D)$, since $i^{(r)}$ is the restriction of i.

Step 2: The bidegree of $i^{(r)}$ and $k^{(r)}$. Each $i^{(r)}$ is a restriction of i to a subobject, so it is homogeneous of the same bidegree $(1, -1)$ for every r. Similarly $E^{r+1}_{p,q}$ is a subquotient of $E^r_{p,q}$ in the same bidegree, and $k^{(r+1)}$ is induced by $k^{(r)}$ on representatives, so $\deg k^{(r)} = (-1, 0)$ for every r by induction.

Step 3: The bidegree of $j^{(r)}$ shifts by $(-1, 1)$ at each stage. Fix r and suppose $\deg j^{(r)} = (a, b)$. By definition 6.1.4 the map $j^{(r+1)}$ is defined on an element of D^{r+1} written as $i(x)$ with $x \in D^r$, by $j^{(r+1)}(i(x)) = [\, j^{(r)}(x)\,]$. With the stated indexing, an element of $D^{r+1}_{p,q}$ is of the form $i(x)$ for $x \in D^r_{p-1,q+1}$, and $j^{(r)}(x)$ then lies in $E^r_{p-1+a,\, q+1+b}$. Hence $j^{(r+1)}$ carries $D^{r+1}_{p,q}$ into $E^{r+1}_{p+a-1,\, q+b+1}$, that is $\deg j^{(r+1)} = (a-1, b+1)$. Since $\deg j^{(1)} = (0, 0)$, induction gives $\deg j^{(r)} = (-(r-1), r-1)$.

Step 4: The bidegree of d_r. Combining Steps 2 and 3,

$$\deg d_r = \deg j^{(r)} + \deg k^{(r)} = \big(-(r-1),\, r-1\big) + (-1, 0) = (-r,\, r-1)$$

so $d_r \colon E^r_{p,q} \to E^r_{p-r,\, q+r-1}$, which is the homological convention of definition 6.1.6, completing the proof.

The bidegree computation is worth pausing over, because it explains what the higher pages are for. The differential d_1 moves one step in the filtration direction, d_2 moves two steps, and d_r moves r steps. Each derivation of the couple replaces D by $i(D)$, which is to say it discards the classes that

survive only one stage further and keeps those that survive r stages; the reach of the differential grows in step with that. A class killed by d_r and by no earlier differential encodes a gluing that spans exactly r layers of the filtration, information no single long exact sequence of the family can see. This is the precise sense in which the spectral sequence contains more than the collection of long exact sequences it was built from.

The simplest instance is a filtration with only two stages, where there is a single differential and the machine reduces to something already familiar.

Proposition 6.1.8: The Two-Step Filtration Recovers the Long Exact Sequence

Let $0 \to A_\bullet \xrightarrow{\iota} C_\bullet \xrightarrow{\pi} C_\bullet/A_\bullet \to 0$ be a short exact sequence of complexes in $\mathcal{A}$, and let $D = H_\bullet(A) \oplus H_\bullet(C)$ and $E = H_\bullet(A) \oplus H_\bullet(C/A)$ be the terms of the exact couple attached to the two-step filtration $0 \subseteq A_\bullet \subseteq C_\bullet$. Then:

1. the first page is $E^1 = H_\bullet(A) \oplus H_\bullet(C/A)$, with d_1 the connecting homomorphism $\partial\colon H_n(C/A) \to H_{n-1}(A)$ of theorem 2.1.20;
2. $E^2 = E^\infty$, that is $d_r = 0$ for all $r \geq 2$;
3. E^2 is the associated graded of $H_\bullet(C)$ for the induced two-step filtration, namely

$$E^2 \cong \operatorname{coker}\partial \oplus \ker \partial$$

with $\operatorname{coker}\partial$ the image of $H_n(A) \to H_n(C)$ and $\ker \partial$ its quotient.

***Proof*:**
Step 1: Identification of the first page. The filtration has stages $F_0 = A_\bullet$ and $F_1 = C_\bullet$, with $F_{-1} = 0$ and $F_p = C_\bullet$ for $p \geq 1$. The layers are $F_0/F_{-1} = A_\bullet$ and $F_1/F_0 = C_\bullet/A_\bullet$, and all other layers vanish. Since E^1 is by construction the sum of the cohomologies of the layers, $E^1 = H_\bullet(A) \oplus H_\bullet(C/A)$. The map k is the connecting homomorphism of the long exact sequence attached to the layer and j is the projection onto the layer, so on the summand $H_n(C/A)$ the composite $d_1 = j \circ k$ is the connecting homomorphism ∂ followed by the identification of $H_{n-1}(A)$ with the layer at filtration level zero, which is the identity. Hence $d_1 = \partial$ on that summand, and $d_1 = 0$ on $H_\bullet(A)$ because k vanishes there (its target is the cohomology of a layer at filtration level -1, which is zero).

Step 2: Degeneration after the second page. By theorem 6.1.7 the differential d_r has bidegree $(-r, r-1)$, so it decreases the filtration index by r. Only the filtration indices 0 and 1 carry nonzero terms, so for $r \geq 2$ the source and target of d_r cannot both be nonzero: a nonzero source at index 1 would have target at index $1 - r \leq -1$, and a source at index 0 would have target at index $-r < 0$. Hence $d_r = 0$ for $r \geq 2$ and the pages stabilize at E^2.

Step 3: Identification of the limit. By Step 1 the second page is the cohomology of $d_1 = \partial$, which is $\ker \partial$ in filtration index 1 and $\operatorname{coker}\partial$ in filtration index 0. The long exact sequence of theorem 2.1.20 reads

$$\cdots \to H_n(A) \xrightarrow{\iota_*} H_n(C) \xrightarrow{\pi_*} H_n(C/A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots$$

Exactness at $H_n(C)$ gives $\mathrm{im}\iota_* = \ker \pi_*$, and exactness at $H_n(A)$ identifies $\mathrm{im}\iota_* \cong H_n(A)/\mathrm{im}\partial = \mathrm{coker}\partial$ in degree n; exactness at $H_n(C/A)$ identifies the quotient $H_n(C)/\mathrm{im}\iota_* \cong \mathrm{im}\pi_* = \ker \partial$. These two are precisely the graded pieces of $H_n(C)$ for the filtration induced by $A_\bullet \subseteq C_\bullet$, and they agree with the two summands of E^2 computed above, as we sought to show.

The proposition is the sanity check for the whole apparatus: in the one case where the answer is already known, the spectral sequence returns exactly the long exact sequence and nothing more. It also isolates what the machine does *not* do. The second page records $\ker \partial$ and $\mathrm{coker}\partial$ as separate graded pieces, but the extension

$$0 \to \mathrm{coker}\partial \to H_n(C) \to \ker \partial \to 0$$

is not part of the output. A spectral sequence delivers the associated graded of the answer, never the answer itself; recovering the latter requires solving the extension problems left behind. This is a permanent feature, not a defect of the two-step case, and it is stated in general in theorem 6.2.6.

A worked instance shows the bookkeeping.

Example 6.1.9: The Spectral Sequence of a Filtered Complex of Abelian Groups
Let $C_\bullet$ be the complex of abelian groups concentrated in degrees 1 and 0,

$$C_1 = \mathbb{Z} \xrightarrow{\cdot 2} C_0 = \mathbb{Z}$$

and let $A_\bullet \subseteq C_\bullet$ be the subcomplex with $A_1 = 0$ and $A_0 = 2\mathbb{Z}$. This is a subcomplex because $\partial(A_1) = 0 \subseteq A_0$. Filter by $F_0 = A_\bullet$ and $F_1 = C_\bullet$.

The cohomology of $C_\bullet$ itself is immediate: $H_1(C) = \ker(\cdot 2) = 0$ and $H_0(C) = \mathbb{Z}/2\mathbb{Z}$. The spectral sequence recomputes this from the two layers. The layer at level 0 is $A_\bullet$, with $H_0(A) = 2\mathbb{Z} \cong \mathbb{Z}$ and $H_1(A) = 0$. The layer at level 1 is $C_\bullet/A_\bullet$, which is $\mathbb{Z}$ in degree 1 and $\mathbb{Z}/2\mathbb{Z}$ in degree 0, with induced differential $1 \mapsto 2 \bmod 2 = 0$; so $H_1(C/A) = \mathbb{Z}$ and $H_0(C/A) = \mathbb{Z}/2\mathbb{Z}$. Indexing by $E^1_{p,q} = H_{p+q}(F_p/F_{p-1})$, the nonzero entries of the first page are

$$E^1_{0,0} = H_0(A) \cong \mathbb{Z}, \qquad E^1_{1,0} = H_1(C/A) \cong \mathbb{Z}, \qquad E^1_{1,-1} = H_0(C/A) \cong \mathbb{Z}/2\mathbb{Z}$$

The only possibly nonzero differential on this page is $d_1 \colon E^1_{1,0} \to E^1_{0,0}$, the connecting homomorphism $H_1(C/A) \to H_0(A)$. Compute it on the generator: the class of $1 \in (C/A)_1$ lifts to $1 \in C_1$, whose boundary is $2 \in C_0$, and 2 is the generator of $A_0 = 2\mathbb{Z}$. So d_1 sends a generator to a generator and is an isomorphism $\mathbb{Z} \to \mathbb{Z}$.

The second page therefore has $E^2_{1,0} = E^2_{0,0} = 0$, while $E^2_{1,-1} = \mathbb{Z}/2\mathbb{Z}$ survives untouched (the differential out of the spot $(1,-1)$ lands in $E^1_{0,-1} = H_{-1}(A) = 0$, and the one into it comes from $E^1_{2,-1} = 0$). By proposition 6.1.8 the pages stabilize here. Reading off total degrees: the diagonal $p+q=1$ is entirely zero, giving $H_1(C) = 0$, and the diagonal $p+q=0$ contains the single group $E^\infty_{1,-1} = \mathbb{Z}/2\mathbb{Z}$, giving $H_0(C) = \mathbb{Z}/2\mathbb{Z}$. Both agree with the direct computation, and because each diagonal contains at most one nonzero group there is no extension problem to solve.

Not every exact couple comes from a filtration. A single short exact sequence of complexes, read the other way round, produces one whose iteration does not terminate and whose limit is a statement about torsion.

Example 6.1.10: The Bockstein Exact Couple

Let p be a prime and $C_\bullet$ a complex of free abelian groups. Multiplication by p is injective on each C_n, so there is a short exact sequence of complexes

$$0 \to C_\bullet \xrightarrow{\cdot p} C_\bullet \to C_\bullet/pC_\bullet \to 0$$

Its long exact sequence in cohomology is exactly an exact couple, with

$$D = H_\bullet(C), \qquad E = H_\bullet(C/pC), \qquad i = (\cdot p), \qquad j = \text{reduction}, \qquad k = \partial$$

The differential $d_1 = j \circ k$ is the *Bockstein homomorphism*. By theorem 6.1.7 the r-th auxiliary object is $D^{r+1} = i^r(D) = p^r H_\bullet(C)$, so passing to the derived couple repeatedly divides out by higher and higher powers of p. When $H_\bullet(C)$ is finitely generated, an element of p-power torsion is annihilated by p^r for some r and therefore disappears from D^{r+1}, while a free generator survives every stage; the limit page is $\big(H_\bullet(C)/\text{torsion}\big) \otimes \mathbb{F}_p$.

The smallest case exhibits the mechanism. Take $C_1 = \mathbb{Z} \xrightarrow{\cdot p} C_0 = \mathbb{Z}$, so that $H_1(C) = 0$ and $H_0(C) = \mathbb{Z}/p\mathbb{Z}$ is pure torsion. Then $C_\bullet/pC_\bullet$ has zero differential, so $E^1 = H_\bullet(C/pC)$ is $\mathbb{F}_p$ in degrees 1 and 0. The connecting map k sends the degree-one generator to the class of 1 in $H_0(C) = \mathbb{Z}/p\mathbb{Z}$ (lift $\bar{1}$ to $1 \in C_1$, take the boundary p, and divide by p), and j sends that class to a generator of $H_0(C/pC) = \mathbb{F}_p$. So d_1 is an isomorphism $E^1_1 \to E^1_0$, giving $E^2 = 0 = E^\infty$. This matches the general description: $H_\bullet(C)$ is entirely torsion, so $(H_\bullet(C)/\text{torsion}) \otimes \mathbb{F}_p = 0$.

Exercise 6.1.1

1. Let (D, E, i, j, k) be an exact couple in which i is an isomorphism. Show that $E = 0$. Deduce that every page of the associated spectral sequence vanishes.

2. Let (D, E, i, j, k) be an exact couple. Show that $d = j \circ k$ vanishes if and only if $\ker i \subseteq \text{im} i$. Then exhibit an exact couple with $d = 0$ but $k \neq 0$, so that a vanishing differential does not force the connecting map to vanish. (Take $D = \mathbb{Z}/4\mathbb{Z}$ and i multiplication by 2.)

3. Verify directly from definition 6.1.4 that the derived couple of the derived couple has $D'' = i^2(D)$, and that i'' is the restriction of i to $i^2(D)$, without appealing to theorem 6.1.7.

4. Let $C_\bullet$ be the complex $C_1 = \mathbb{Z} \xrightarrow{\cdot 6} C_0 = \mathbb{Z}$ filtered by $F_0 = A_\bullet$ with $A_1 = 0$, $A_0 = 3\mathbb{Z}$, and $F_1 = C_\bullet$. Compute the first page, the differential d_1, and the limit page, and check the answer against a direct computation of $H_\bullet(C)$.

5. In the Bockstein couple of example 6.1.10, take $C_\bullet$ to be $C_1 = \mathbb{Z} \xrightarrow{\cdot p^2} C_0 = \mathbb{Z}$. Show that $d_1 = 0$ but $d_2 \neq 0$, and that $E^3 = E^\infty = 0$. Explain how this illustrates the statement that d_r detects gluing spanning r stages.

6. Let $\mathcal{A}$ be an abelian category and (D, E, i, j, k) an exact couple with $E = 0$. Show that i is an isomorphism, so that the converse of exercise 6.1.1 (1) holds.

6.2 The Spectral Sequence of a Filtered Complex

The exact couples of section 6.1 were treated as abstract data. Their principal source is a complex equipped with a filtration, and this section carries out that construction, describes the pages explicitly enough to compute with, and proves the two theorems that make the output usable: a convergence theorem saying what the limit page computes, and a comparison theorem saying when two spectral sequences agree. These are the results cited from this chapter across the series, so both are proved in the generality in which they are used, for increasing filtrations of chain complexes and, in the restatement after them, for decreasing filtrations of cochain complexes.

Definition 6.2.1: Filtered Complex

Let $\mathcal{A}$ be an abelian category. A *filtered complex* is a chain complex $C_\bullet$ in $\mathcal{A}$ together with a family of subcomplexes $\{F_pC_\bullet\}_{p\in\mathbb{Z}}$ satisfying $F_pC_\bullet \subseteq F_{p+1}C_\bullet$ for every p. The filtration is called

1. *exhaustive* if $\bigcup_p F_pC_n = C_n$ for every n;
2. *bounded below* if for each n there is an $s(n)$ with $F_{s(n)}C_n = 0$;
3. *bounded* if it is bounded below and for each n there is a $t(n)$ with $F_{t(n)}C_n = C_n$.

The *associated graded* of the filtration is the complex $\mathrm{gr}_pC_\bullet = F_pC_\bullet/F_{p-1}C_\bullet$, and the filtration induced on homology is

$$F_pH_n(C) = \mathrm{im}\big(H_n(F_pC) \to H_n(C)\big)$$

Each inclusion $F_{p-1}C_\bullet \hookrightarrow F_pC_\bullet$ is a subcomplex inclusion in the sense of definition 2.1.7, so $\mathrm{gr}_pC_\bullet$ is again a complex.

Boundedness is the hypothesis that does the work later, and it is worth seeing why before it is used. In each fixed degree n a bounded filtration passes from 0 to all of C_n in finitely many steps, so any bookkeeping indexed by the filtration terminates in that degree after finitely many stages. Without some such condition the pages of the spectral sequence need not stabilize and the limit need not compute anything.

Every filtered complex produces an exact couple in one step: the long exact sequence of each pair $(F_pC_\bullet, F_{p-1}C_\bullet)$ gives the three maps, and the filtration index gives the bigrading.

Theorem 6.2.2: The Exact Couple of a Filtered Complex

Let $(C_\bullet, F_\bullet)$ be a filtered complex. Set

$$D^1_{p,q} = H_{p+q}(F_pC), \qquad E^1_{p,q} = H_{p+q}(\mathrm{gr}_pC)$$

and let i be induced by the inclusion $F_{p-1}C \hookrightarrow F_pC$, let j be induced by the projection $F_pC \to \mathrm{gr}_pC$, and let k be the connecting homomorphism of theorem 2.1.20 applied to

$$0 \to F_{p-1}C_\bullet \to F_pC_\bullet \to \mathrm{gr}_pC_\bullet \to 0$$

Then (D^1, E^1, i, j, k) is an exact couple with

$$\deg i = (1,-1), \qquad \deg j = (0,0), \qquad \deg k = (-1,0)$$

Consequently theorem 6.1.7 yields a spectral sequence with first page $E^1_{p,q} = H_{p+q}(\mathrm{gr}_pC)$ and differentials $d_r \colon E^r_{p,q} \to E^r_{p-r,\,q+r-1}$.
It is convenient to record one page earlier. Set

$$E^0_{p,q} := \mathrm{gr}_pC_{p+q}$$

with d_0 the differential induced by ∂ on the associated graded; then $E^1 = H(E^0, d_0)$, which is the identification just made, and the sequence is indexed from $r = 0$.

***Proof*:**
Step 1: Exactness. The displayed short exact sequence of complexes gives, by theorem 2.1.20, the long exact sequence

$$\cdots \to H_n(F_{p-1}C) \xrightarrow{i} H_n(F_pC) \xrightarrow{j} H_n(\mathrm{gr}_pC) \xrightarrow{k} H_{n-1}(F_{p-1}C) \to \cdots$$

Assembling these over all p and all n, the three maps i, j, k are defined on all of D^1 and E^1, and exactness of the triangle at each corner is exactness of the long exact sequences at the three types of term. This is exactly definition 6.1.1.

Step 2: Bidegrees. Write $n = p + q$. The map i carries $H_n(F_{p-1}C) = D^1_{p-1,q+1}$ to $H_n(F_pC) = D^1_{p,q}$, since $(p-1)+(q+1) = n$; hence $\deg i = (1,-1)$. The map j carries $H_n(F_pC) = D^1_{p,q}$ to $H_n(\mathrm{gr}_pC) = E^1_{p,q}$, so $\deg j = (0,0)$. The connecting map k carries $H_n(\mathrm{gr}_pC) = E^1_{p,q}$ to $H_{n-1}(F_{p-1}C) = D^1_{p-1,q}$, since $(p-1)+q = n-1$; hence $\deg k = (-1,0)$. These are the hypotheses of theorem 6.1.7, which then gives the spectral sequence and the stated bidegree of d_r, completing the proof.

The construction is quick, but the pages it produces are described only implicitly, as iterated subquotients of the first one. Computing with them, and proving anything about the limit, requires a description of E^r in terms of the original complex. The right notion is that of a chain which is a cycle only up to a given depth in the filtration.

Definition 6.2.3: Approximate Cycles

Let $(C_\bullet, F_\bullet)$ be a filtered complex and write $n = p + q$. For $r \geq 0$ set

$$Z^r_{p,q} = \{\, x \in F_pC_n \;:\; \partial x \in F_{p-r}C_{n-1} \,\}$$

the object of chains at filtration level p whose boundary drops r levels. By convention $Z^{-1}_{p,q} = F_pC_n$.

The objects $Z^r_{p,q}$ decrease in r, and their intersection over all r is, when the filtration is bounded below, the cycles of $C_\bullet$ lying in F_pC_n. An element of $Z^r_{p,q}$ is a class that survives to the r-th page: it is a cycle in the associated graded to accuracy r, and its failure to be a cycle of $C_\bullet$ is what d_r records.

Proposition 6.2.4: Explicit Description of the Pages

Let $(C_\bullet, F_\bullet)$ be a filtered complex and let $\{(E^r, d_r)\}$ be the spectral sequence of theorem 6.2.2. For every $r \geq 0$ there are isomorphisms

$$E^r_{p,q} \cong \frac{Z^r_{p,q}}{Z^{r-1}_{p-1,q+1} + \partial Z^{r-1}_{p+r-1,\,q-r+2}}$$

under which the differential d_r is induced by the boundary map ∂ of $C_\bullet$.

***Proof*:**
Write $\mathcal{E}^r_{p,q}$ for the right-hand side and $n = p + q$. The proof is an induction on r; the inductive statement is that $E^r \cong \mathcal{E}^r$ compatibly with the differentials, the differential on $\mathcal{E}^r$ being the one induced by ∂.

Step 1: The map induced by ∂ is a differential on $\mathcal{E}^r$. If $x \in Z^r_{p,q}$ then $\partial x \in F_{p-r}C_{n-1}$ and $\partial(\partial x) = 0 \in F_{p-2r}C_{n-2}$, so $\partial x \in Z^r_{p-r,\,q+r-1}$. Hence ∂ carries $Z^r_{p,q}$ into $Z^r_{p-r,q+r-1}$, and it carries the denominator at (p, q) into the denominator at $(p-r, q+r-1)$: the summand $Z^{r-1}_{p-1,q+1}$ is sent to $\partial Z^{r-1}_{p-1,q+1}$, which is literally the second summand of the target denominator, and the summand $\partial Z^{r-1}_{p+r-1,q-r+2}$ is sent to zero because $\partial\partial = 0$. So ∂ induces a map $\mathcal{E}^r_{p,q} \to \mathcal{E}^r_{p-r,q+r-1}$ squaring to zero.

Step 2: The base case $r = 0$. With the convention $Z^{-1}_{p,q} = F_pC_n$, the numerator is $Z^0_{p,q} = F_pC_n$ and the denominator is $F_{p-1}C_n + \partial F_{p-1}C_{n+1} = F_{p-1}C_n$, since $\partial F_{p-1}C_{n+1} \subseteq F_{p-1}C_n$. So $\mathcal{E}^0_{p,q} = F_pC_n / F_{p-1}C_n = \mathrm{gr}_pC_n$, which is $E^0_{p,q}$, and the induced differential is the one ∂ induces on the associated graded. Taking homology recovers $E^1_{p,q} = H_n(\mathrm{gr}_pC)$, matching theorem 6.2.2; the differential $d_1 = j \circ k$ is by construction the connecting homomorphism, which on representatives is ∂.

Step 3: The inductive step. Assume $E^r \cong \mathcal{E}^r$ compatibly with differentials. Then $E^{r+1} = H(E^r, d_r) \cong H(\mathcal{E}^r, \partial)$, so it suffices to identify the latter with $\mathcal{E}^{r+1}$. Abbreviate the denominator $D = Z^{r-1}_{p-1,q+1} + \partial Z^{r-1}_{p+r-1,q-r+2}$.

First the kernel. Let $x \in Z^r_{p,q}$ represent a class killed by d_r, so that ∂x lies in the denominator at $(p-r, q+r-1)$, namely $\partial x = z + \partial w$ with $z \in Z^{r-1}_{p-r-1,q+r}$ and $w \in Z^{r-1}_{p-1,q+1}$. Then $x - w$ still lies in F_pC_n, because $w \in F_{p-1}C_n$, and

$$\partial(x-w) = \partial x - \partial w = z \in F_{p-r-1}C_{n-1}$$

so $x - w \in Z^{r+1}_{p,q}$. Since $w \in D$, the classes of x and $x - w$ agree in $\mathcal{E}^r_{p,q}$. Hence $\ker d_r = (Z^{r+1}_{p,q} + D)/D$.

Next the image. A class in $\mathcal{E}^r_{p+r,q-r+1}$ is represented by some $y \in Z^r_{p+r,q-r+1}$, and d_r sends it to the class of ∂y. Hence $\mathrm{im} d_r = (\partial Z^r_{p+r,q-r+1} + D)/D$, and therefore

$$H(\mathcal{E}^r, \partial)_{p,q} = \frac{Z^{r+1}_{p,q} + D}{\partial Z^r_{p+r,q-r+1} + D} \cong \frac{Z^{r+1}_{p,q}}{Z^{r+1}_{p,q} \cap \left(\partial Z^r_{p+r,q-r+1} + D\right)}$$

the isomorphism being the second isomorphism theorem.

It remains to show that the intersection appearing there equals $Z^r_{p-1,q+1} + \partial Z^r_{p+r,q-r+1}$, which is exactly the denominator of $\mathcal{E}^{r+1}_{p,q}$. For the inclusion $\supseteq$, note $Z^r_{p-1,q+1} \subseteq Z^{r+1}_{p,q}$ (an element lies in $F_{p-1} \subseteq F_p$ and its boundary lies in F_{p-1-r}) and $Z^r_{p-1,q+1} \subseteq Z^{r-1}_{p-1,q+1} \subseteq D$; while $\partial Z^r_{p+r,q-r+1} \subseteq Z^{r+1}_{p,q}$ because such a boundary lies in F_pC_n and is itself a cycle. For the inclusion $\subseteq$, let $x \in Z^{r+1}_{p,q}$ be written as $x = \partial y + z + \partial w$ with $y \in Z^r_{p+r,q-r+1}$, $z \in Z^{r-1}_{p-1,q+1}$ and $w \in Z^{r-1}_{p+r-1,q-r+2}$. Then $z = x - \partial y - \partial w$ lies in $F_{p-1}C_n$ and $\partial z = \partial x \in F_{p-r-1}C_{n-1}$, so $z \in Z^r_{p-1,q+1}$. Moreover $w \in Z^{r-1}_{p+r-1,q-r+2}$ satisfies $w \in F_{p+r-1} \subseteq F_{p+r}$ and $\partial w \in F_{(p+r-1)-(r-1)} = F_p$, so $w \in Z^r_{p+r,q-r+1}$ and hence $\partial w \in \partial Z^r_{p+r,q-r+1}$. Therefore $x = z + \partial(y+w) \in Z^r_{p-1,q+1} + \partial Z^r_{p+r,q-r+1}$, as required.

Substituting gives $H(\mathcal{E}^r, \partial)_{p,q} \cong \mathcal{E}^{r+1}_{p,q}$. Write $\varphi_r \colon \mathcal{E}^r \to E^r$ for the isomorphism of the inductive statement. Composing with $E^{r+1} = H(E^r, d_r) \cong H(\mathcal{E}^r, \partial)$ gives φ_{r+1}, and since every isomorphism in this step is induced by the identity on representatives, φ_{r+1} sends the class of $x \in Z^{r+1}_{p,q}$ to the class of $\varphi_r[x]$ in $H(E^r, d_r)$.

Step 4: The isomorphism φ_r carries the map induced by ∂ to d_r. For $r = 0$ this is Step 2, so let $r \geq 1$. Each E^r is then a subquotient of E^1. For $e \in E^1$ write $\langle e \rangle_r \in E^r$ for its class whenever that class is defined: $\langle e \rangle_1 = e$, and $\langle e \rangle_{s+1}$ is the class of $\langle e \rangle_s$ in $H(E^s, d_s)$ when $d_s \langle e \rangle_s = 0$. For $x \in Z^r_{p,q}$ let $\bar{x}$ be its image in $\mathrm{gr}_p C_n$, a cycle because $\partial x \in F_{p-r}C_{n-1} \subseteq F_{p-1}C_{n-1}$. Step 3 at $r = 0$ gives $\varphi_1[x] = [\bar{x}]$, and the description of φ_{r+1} above then gives $\varphi_r[x] = \langle [\bar{x}] \rangle_r$ by induction. Iterating definition 6.1.4, whose maps are $k'([e]) = k(e)$ and $j'(i(y)) = [j(y)]$, gives

$$k^{(r)} \langle e \rangle_r = k(e), \qquad j^{(r)}\left(i^{r-1}(y)\right) = \langle j(y) \rangle_r \quad (y \in D^1)$$

where the class on the right is defined because at each stage $j^{(s)}$ takes values in $\ker k^{(s)} \subseteq \ker d_s$. The connecting homomorphism k of theorem 2.1.20 lifts the cycle $\bar{x}$ to x and applies ∂, so $k[\bar{x}]$ is the class of ∂x in $H_{n-1}(F_{p-1}C)$. Since ∂x is a cycle lying in $F_{p-r}C_{n-1}$, that class is $i^{r-1}(y)$ for y the class of ∂x in $H_{n-1}(F_{p-r}C)$, and $j(y)$ is the class of the image of ∂x in $\mathrm{gr}_{p-r}C_{n-1}$. Since $\partial x \in Z^r_{p-r,q+r-1}$ by Step 1, this gives

$$d_r \, \varphi_r[x] = j^{(r)} k^{(r)} \langle [\bar{x}] \rangle_r = j^{(r)}\left(i^{r-1}(y)\right) = \langle j(y) \rangle_r = \varphi_r[\partial x]$$

which closes the induction and completes the proof.

The formula repays reading slowly, because it makes the interpretive claim of section 6.1 precise. The numerator at page r consists of the chains whose boundary drops r filtration levels, and the denominator discards those already accounted for one level down together with the boundaries of chains that drop $r-1$ levels. Passing from page r to page $r+1$ tightens the requirement on the boundary by one level. A class that survives to page r and dies there is one that could be corrected to a cycle modulo F_{p-r} but not modulo F_{p-r-1}, which is the precise sense in which d_r detects gluing across r layers.

With the pages described, the limit can be identified. This is the theorem the rest of the series cites from this section.

Definition 6.2.5: Convergence of a Spectral Sequence

A bigraded spectral sequence $\{(E^r, d_r)\}$ *degenerates* at the spot (p,q) if the differentials into and out of that spot vanish for all sufficiently large r; the common value of $E^r_{p,q}$ for large r is then written $E^\infty_{p,q}$. The spectral sequence *converges* to a graded object $H_\bullet$ equipped with a filtration $F_\bullet H_\bullet$, written

$$E^1_{p,q} \implies H_{p+q}$$

if it degenerates at every spot, the filtration $F_\bullet H_n$ is bounded for each n (that is $F_pH_n = 0$ for p small and $F_pH_n = H_n$ for p large), and $E^\infty_{p,q} \cong F_pH_{p+q}/F_{p-1}H_{p+q}$ for all p,q. The boundedness clause is what lets a vanishing family of graded pieces be read back as a vanishing abutment; without it the graded pieces constrain only the successive quotients.

Theorem 6.2.6: Convergence for a Bounded Filtration

Let $(C_\bullet, F_\bullet)$ be a filtered complex whose filtration is bounded in the sense of definition 6.2.1. Then the spectral sequence of theorem 6.2.2 converges to $H_\bullet(C)$: it degenerates at every spot, and

$$E^\infty_{p,q} \cong \frac{F_pH_{p+q}(C)}{F_{p-1}H_{p+q}(C)}$$

for the filtration $F_pH_n(C) = \operatorname{im}(H_n(F_pC) \to H_n(C))$ of definition 6.2.1. Moreover that filtration of $H_n(C)$ is itself bounded, so $H_n(C)$ has a finite filtration whose graded pieces are the terms of the limit page on the diagonal $p+q=n$.

***Proof*:**
Fix n and let $s(m), t(m)$ be the bounds of definition 6.2.1 in degree m.

Step 1: Degeneration. Fix a spot (p,q) with $p+q=n$. The differential out of it lands in $E^r_{p-r,q+r-1}$, a subquotient of $E^1_{p-r,q+r-1} = H_{n-1}(\mathrm{gr}_{p-r}C)$. Once r is large enough that $p-r \le s(n-1)$, the complex $\mathrm{gr}_{p-r}C$ vanishes in degree $n-1$, so the target is zero. The differential into (p,q) comes from $E^r_{p+r,q-r+1}$, a subquotient of $H_{n+1}(\mathrm{gr}_{p+r}C)$; once $p+r-1 \ge t(n+1)$ one has $F_{p+r-1}C_{n+1} = F_{p+r}C_{n+1} = C_{n+1}$, so $\mathrm{gr}_{p+r}C$ vanishes in degree $n+1$ and the source is zero. Both hold for all large r, so $E^{r+1}_{p,q} = E^r_{p,q}$ eventually and $E^\infty_{p,q}$ is defined.

Step 2: The large-r value of the explicit description. Take r large enough that $F_{p-r}C_{n-1} = 0$ and $F_{p+r-1}C_{n+1} = C_{n+1}$, which is possible by boundedness. Write $Z_n = \ker(\partial\colon C_n \to C_{n-1})$

and $B_n = \mathrm{im}(\partial\colon C_{n+1} \to C_n)$. Then:

1. $Z^r_{p,q} = \{x \in F_pC_n : \partial x \in F_{p-r}C_{n-1} = 0\} = Z_n \cap F_pC_n$;
2. $Z^{r-1}_{p-1,q+1} = \{x \in F_{p-1}C_n : \partial x \in F_{p-r}C_{n-1} = 0\} = Z_n \cap F_{p-1}C_n$;
3. $\partial Z^{r-1}_{p+r-1,q-r+2} = \partial\{y \in F_{p+r-1}C_{n+1} : \partial y \in F_pC_n\} = \partial(C_{n+1}) \cap F_pC_n = B_n \cap F_pC_n$, using $F_{p+r-1}C_{n+1} = C_{n+1}$.

Substituting into proposition 6.2.4,

$$E^\infty_{p,q} \cong \frac{Z_n \cap F_pC_n}{(Z_n \cap F_{p-1}C_n) + (B_n \cap F_pC_n)}$$

Step 3: The same object is the graded piece of homology. By definition $F_pH_n(C)$ is the image of $H_n(F_pC)$ in $H_n(C)$, that is

$$F_pH_n(C) = \frac{(Z_n \cap F_pC_n) + B_n}{B_n} \cong \frac{Z_n \cap F_pC_n}{B_n \cap F_pC_n}$$

the isomorphism being the second isomorphism theorem. Taking the quotient by $F_{p-1}H_n(C)$, which under this identification is the image of $Z_n \cap F_{p-1}C_n$, gives

$$\frac{F_pH_n(C)}{F_{p-1}H_n(C)} \cong \frac{Z_n \cap F_pC_n}{(Z_n \cap F_{p-1}C_n) + (B_n \cap F_pC_n)}$$

which is the object computed in Step 2. Hence $E^\infty_{p,q} \cong \mathrm{gr}_pH_n(C)$.

Both sides were computed as subquotients of F_pC_n by the same formula, and the isomorphism between them is induced by the identity on representatives. A morphism of filtered complexes carries representatives to representatives and preserves every object appearing in that formula, so the identification is natural in $(C_\bullet, F_\bullet)$; this is what theorem 6.2.7 uses when it compares the two filtrations of homology.

Step 4: The induced filtration is bounded. If $p \le s(n)$ then $F_pC_n = 0$, so $H_n(F_pC) = 0$ and $F_pH_n(C) = 0$. If $p \ge t(n)$ then $F_pC_n = C_n$; taking p also at least $t(n+1)$ makes $F_pC_{n+1} = C_{n+1}$ as well, so every cycle and every boundary of degree n already lies at filtration level p and $F_pH_n(C) = H_n(C)$. Hence the filtration of $H_n(C)$ passes from 0 to everything in finitely many steps, completing the proof.

Two features of the statement deserve emphasis, both of them permanent. The limit page computes the *associated graded* of $H_n(C)$, not $H_n(C)$ itself; recovering the latter means solving the extension problems between successive graded pieces, exactly as in the two-step case of proposition 6.1.8. And convergence is a theorem about the filtration, not about the complex: boundedness is what forces the pages to stabilize, and without it the limit page can fail to compute anything.

The second theorem of the section makes a spectral sequence a tool for identification as well as calculation: a morphism of filtered complexes that is an isomorphism on any single page is an isomorphism on the abutment.

Theorem 6.2.7: Comparison Theorem

Let $f\colon (C_\bullet, F_\bullet) \to (C'_\bullet, F'_\bullet)$ be a morphism of filtered complexes, that is a chain map with $f(F_pC_\bullet) \subseteq F'_pC'_\bullet$, and suppose both filtrations are bounded. Then f induces a morphism of spectral sequences $f^r\colon E^r_{p,q} \to E'^r_{p,q}$ commuting with the differentials, and if f^{r_0} is an isomorphism for some $r_0 \geq 0$, then:

1. f^r is an isomorphism for every $r \geq r_0$, and hence so is f^∞;
2. $f_*\colon H_n(C) \to H_n(C')$ is an isomorphism for every n.

***Proof*:**

Step 1: Functoriality. On the zeroth page f induces f^0 on gr_pC_{p+q} directly, and the identification $E^1 = H(E^0, d_0)$ of theorem 6.2.2 is natural in the filtered complex, so the case $r_0 = 0$ reduces to $r_0 = 1$. For the remaining pages: because f preserves the filtration, it carries the short exact sequences of theorem 6.2.2 for C to those for C', and the connecting homomorphism of theorem 2.1.20 is natural. So f induces a morphism of exact couples, and the construction of the derived couple in definition 6.1.4 is built from i, j, k alone and is therefore natural in the couple. Hence f induces f^r on every page, commuting with d_r.

Step 2: Isomorphism on all later pages. Suppose f^r is an isomorphism. Since f^r commutes with the differentials, it carries $\ker d_r$ isomorphically onto $\ker d'_r$ and $\mathrm{im} d_r$ isomorphically onto $\mathrm{im} d'_r$, so the induced map on the quotients is an isomorphism. That induced map is f^{r+1}, so the claim follows by induction from r_0.

Step 3: Isomorphism on the limit page. Both filtrations are bounded, so by Step 1 of the proof of theorem 6.2.6 each spot degenerates: there is an R depending on (p, q) with $E^r_{p,q} = E^\infty_{p,q}$ and $E'^r_{p,q} = E'^\infty_{p,q}$ for $r \geq R$. Taking $r \geq \max(R, r_0)$ and applying Step 2 gives $f^\infty_{p,q}$ an isomorphism.

Step 4: Isomorphism on homology. Fix n. By theorem 6.2.6 the filtrations $F_\bullet H_n(C)$ and $F'_\bullet H_n(C')$ are finite, and f_* respects them. Proceed by induction up the filtration. The bottom stage is zero on both sides. If f_* is an isomorphism $F_{p-1}H_n(C) \to F'_{p-1}H_n(C')$, then in the commutative diagram of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & F_{p-1}H_n(C) & \longrightarrow & F_pH_n(C) & \longrightarrow & E^\infty_{p,n-p} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & F'_{p-1}H_n(C') & \longrightarrow & F'_pH_n(C') & \longrightarrow & E'^\infty_{p,n-p} & \longrightarrow & 0 \end{array}$$

the outer vertical maps are isomorphisms, the left one by the inductive hypothesis and the right one by Step 3. Each row is already a five-term exact sequence, so lemma 2.1.22 applies directly, with the outer maps $0 \to 0$ trivially epi and mono, and forces the middle map to be an isomorphism. Since the filtration is finite and exhausts $H_n(C)$, finitely many steps give that $f_*\colon H_n(C) \to H_n(C')$ is an isomorphism, completing the proof.

The applications elsewhere in the series are stated for cochain complexes with decreasing filtrations,

so the translation is recorded explicitly rather than left to the reader.

Corollary 6.2.8: Convergence for a Decreasing Filtration of a Cochain Complex

Let $(C^\bullet, d)$ be a cochain complex with a decreasing filtration $C^\bullet = F^0C^\bullet \supseteq F^1C^\bullet \supseteq \cdots$ compatible with d, that is $d(F^pC^n) \subseteq F^pC^{n+1}$, and suppose the filtration is exhaustive and bounded, so that for each n there is a P with $F^PC^n = 0$. Then there is a spectral sequence with

$$E_0^{p,q} = \frac{F^pC^{p+q}}{F^{p+1}C^{p+q}}, \qquad d_r \colon E_r^{p,q} \to E_r^{p+r,\,q-r+1}$$

each page the cohomology of the one before it, converging to $H^{p+q}(C)$:

$$E_1^{p,q} \implies H^{p+q}(C), \qquad E_\infty^{p,q} \cong \mathrm{gr}_F^p H^{p+q}(C) = \frac{F^pH^{p+q}(C)}{F^{p+1}H^{p+q}(C)}$$

where $F^pH^n(C) = \mathrm{im}\big(H^n(F^pC^\bullet) \to H^n(C^\bullet)\big)$.

***Proof*:**
Set $C_n = C^{-n}$ and $F_pC_\bullet = F^{-p}C^\bullet$. The differential d raises cochain degree by one and therefore lowers chain degree by one, so $(C_\bullet, F_\bullet)$ is a filtered chain complex in the sense of definition 6.2.1: the filtration is increasing because $F^\bullet$ is decreasing, it is exhaustive because $F^\bullet$ is, and it is bounded because for each n there is a P with $F^PC^n = 0$, giving $F_{-P}C_{-n} = 0$, while $F^0C^n = C^n$ gives $F_0C_{-n} = C_{-n}$.

Theorem 6.2.6 applies and gives a convergent spectral sequence with $E^0_{p,q} = \mathrm{gr}_pC_{p+q}$ and d_r of bidegree $(-r, r-1)$. Re-indexing by $E_r^{p,q} = E^r_{-p,-q}$, as in definition 6.1.6, turns the bidegree of d_r into $(r, -r+1)$ and turns the graded piece $\mathrm{gr}_{-p}H_{-p-q}(C_\bullet)$ into $\mathrm{gr}_F^pH^{p+q}(C^\bullet)$, which is the statement, as we sought to show.

The same re-indexing carries every other result of this section across, and it is worth saying so once rather than hedging at each use, since section 6.3 onward works exclusively with cochain complexes.

Corollary 6.2.9: The Cochain Form of the Section

Let $\mathcal{T}$ denote the translation $C_n = C^{-n}$, $F_p = F^{-p}$, $E_r^{p,q} = E^r_{-p,-q}$ of corollary 6.2.8. Under $\mathcal{T}$:

1. Theorem 6.2.7 holds for morphisms of cochain complexes with bounded decreasing filtrations: a morphism inducing an isomorphism on E_{r_0} for some r_0 induces one on E_∞ and on $H^\bullet$.
2. Definition 6.2.3, proposition 6.2.4 and definition 6.2.5 hold with the *bigrading* negated, the page index r unchanged. Explicitly $Z_r^{p,q} = \{x \in F^pC^{p+q} : dx \in F^{p+r}C^{p+q+1}\}$ and

$$E_r^{p,q} \cong \frac{Z_r^{p,q}}{Z_{r-1}^{p+1,q-1} + d\,Z_{r-1}^{p-r+1,\,q+r-2}}$$

***Proof*:**
$\mathcal{T}$ is an isomorphism between the category of cochain complexes with bounded decreasing filtrations and the category of chain complexes with bounded increasing filtrations, carrying morphisms to morphisms and the spectral sequence of one to the spectral sequence of the other, by the computation in the proof of corollary 6.2.8. A statement proved for one is therefore proved for the other; no hypothesis in theorem 6.2.7 or proposition 6.2.4 refers to the direction of the grading beyond boundedness, which $\mathcal{T}$ preserves, completing the proof.

A cohomological spectral sequence is *first-quadrant* if $E_r^{p,q} = 0$ for every $r \geq 2$ whenever $p < 0$ or $q < 0$. In such a sequence the corner spots $(p, 0)$ and $(0, q)$ each meet the abutment in only one way. The two results below use only that shape and convergence, so they apply to the first-quadrant spectral sequences among those of corollary 6.2.8, of a double complex, of section 6.4 and of section 6.5.

Proposition 6.2.10: Edge Morphisms

Let $E_2^{p,q} \Rightarrow H^{p+q}$ be a first-quadrant cohomological spectral sequence converging in the sense of definition 6.2.5, in the cochain form of corollary 6.2.9(2). Then for every $p, q \geq 0$ there are canonical morphisms

$$E_2^{p,0} \longrightarrow H^p, \qquad H^q \longrightarrow E_2^{0,q}$$

called the *edge morphisms*, given respectively by the composites

$$E_2^{p,0} \twoheadrightarrow E_\infty^{p,0} = F^pH^p \hookrightarrow H^p, \qquad H^q \twoheadrightarrow H^q/F^1H^q = E_\infty^{0,q} \hookrightarrow E_2^{0,q}$$

They are natural in the spectral sequence.

***Proof*:**
Step 1: The two composites are defined. Fix p. Every differential out of $(p, 0)$ lands in $(p+r, 1-r)$, which has negative second index for $r \geq 2$, and every differential into it comes from $(p-r, r-1)$, which has negative first index once $r > p$. So $E_\infty^{p,0}$ is a quotient of $E_2^{p,0}$, the successive pages being obtained by killing images. The filtration of H^p is bounded, so $F^aH^p = 0$ for large a, and its graded pieces $E_\infty^{a,p-a}$ vanish for $a > p$, the second index being negative. Hence $F^{p+1}H^p = 0$, and $E_\infty^{p,0} = F^pH^p/F^{p+1}H^p = F^pH^p$, a subobject of H^p. Composing gives the first morphism.

Dually, every differential into $(0, q)$ comes from $(-r, q + r - 1)$, which has negative first index, so $E_\infty^{0,q}$ is a subobject of $E_2^{0,q}$, obtained by taking successive kernels. The filtration of H^q is bounded, so $F^aH^q = H^q$ for small a, and its graded pieces $E_\infty^{a,q-a}$ vanish for $a < 0$. Hence $F^0H^q = H^q$, and $E_\infty^{0,q} = F^0H^q/F^1H^q = H^q/F^1H^q$. Composing gives the second morphism.

Step 2: Naturality. A morphism of spectral sequences compatible with the abutments respects the pages, hence the successive kernels and images defining E_∞, and respects the filtration of the abutment. Each edge morphism is built only from those data, so the two squares commute, completing the proof.

In total degree one the abutment has only the two graded pieces $E_\infty^{1,0}$ and $E_\infty^{0,1}$, and the only differential that changes either of them, or the bottom piece $E_\infty^{2,0}$ of H^2, is $d_2 \colon E_2^{0,1} \to E_2^{2,0}$.

Proposition 6.2.11: The Five-Term Exact Sequence of a First-Quadrant Spectral Sequence

Let $E_2^{p,q} \Rightarrow H^{p+q}$ be a first-quadrant cohomological spectral sequence, so that $E_r^{p,q} = 0$ for $r \geq 2$ whenever $p < 0$ or $q < 0$, converging to a filtered graded object $H^\bullet$ in the sense of definition 6.2.5, in the cochain form of corollary 6.2.9(2). Then there is an exact sequence

$$0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \to H^2$$

natural in the spectral sequence: a morphism of first-quadrant spectral sequences compatible with the abutment induces a morphism of these five-term sequences.

***Proof*:**
The spectral sequence is first-quadrant, so the filtration of H^n has graded pieces $E_\infty^{p,q}$ with $p+q=n$ and $0 \leq p \leq n$.

Step 1: Degree one. The diagonal $p+q=1$ contains the two spots $(1,0)$ and $(0,1)$. For every $r \geq 2$ the differentials touching $(1,0)$ run to $(1+r, 1-r)$ and from $(1-r, r-1)$, both outside the first quadrant, so $E_\infty^{1,0} = E_2^{1,0}$. The differentials arriving at $(0,1)$ come from $(-r, r)$ and are zero, and those leaving it run to $(r, 2-r)$, outside the first quadrant for $r \geq 3$. So $E_\infty^{0,1} = \ker\big(d_2 \colon E_2^{0,1} \to E_2^{2,0}\big)$. The filtration of H^1 therefore gives a short exact sequence

$$0 \to E_\infty^{1,0} \to H^1 \to E_\infty^{0,1} \to 0$$

that is $0 \to E_2^{1,0} \to H^1 \to \ker d_2 \to 0$. The first map here is the edge morphism $E_2^{1,0} \to H^1$ of proposition 6.2.10, and the second, followed by the inclusion of $\ker d_2$ into $E_2^{0,1}$, is the edge morphism $H^1 \to E_2^{0,1}$, both by construction.

Step 2: Splicing. Substituting $E_\infty^{0,1} = \ker d_2$ turns the short exact sequence into exactness of

$$0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \xrightarrow{d_2} E_2^{2,0}$$

the last map being d_2 and exactness at $E_2^{0,1}$ being the statement that the image of H^1 is exactly $\ker d_2$.

Step 3: The final term. It remains to show exactness at $E_2^{2,0}$, that is that the kernel of $E_2^{2,0} \to H^2$ is the image of d_2. The filtration of H^2 has $E_\infty^{2,0}$ as its bottom graded piece, and the map $E_2^{2,0} \to H^2$ is the composite $E_2^{2,0} \twoheadrightarrow E_\infty^{2,0} \hookrightarrow H^2$. The second map is a monomorphism, so the kernel of the composite is the kernel of the first, which is the image of the only differential arriving at $(2,0)$, namely $d_2 \colon E_2^{0,1} \to E_2^{2,0}$. Hence the sequence is exact there.

Step 4: Naturality. Let $\{E_r\} \to \{E'_r\}$ be a morphism of first-quadrant spectral sequences compatible with the abutments. Every object in the five-term sequence was constructed from data the morphism respects. The terms $E_2^{1,0}$, $E_2^{0,1}$, $E_2^{2,0}$ receive the induced map on the second page. The map d_2 is respected because a morphism of spectral sequences commutes with the differentials. Each $E_\infty^{p,q}$ receives the induced map because it is a subquotient of $E_2^{p,q}$ cut out by

those same differentials. And H^1, H^2 receive the given map on abutments, which by hypothesis is compatible with the filtrations and hence with the identifications $E_\infty^{p,q} \cong F^pH^{p+q}/F^{p+1}H^{p+q}$. Each of the four maps in the sequence was defined from exactly these, so all four squares commute, completing the proof.

A first application of the machinery, and the one that recurs in section 6.4, is the crudest filtration a complex carries: truncate it.

Example 6.2.12: The Stupid Filtration of a Cochain Complex

Let $(C^\bullet, d)$ be a cochain complex concentrated in degrees $0 \le n \le N$, and define the *stupid filtration* (the *filtration bête*) by

$$F^pC^n = \begin{cases} C^n & n \ge p \\ 0 & n < p \end{cases}$$

This is a decreasing filtration compatible with d, since d raises degree, and it is exhaustive and bounded because $F^0C^\bullet = C^\bullet$ and $F^{N+1}C^\bullet = 0$. The graded pieces are $\mathrm{gr}_F^pC^n = C^n$ if $n = p$ and 0 otherwise, so the zeroth page has a single nonzero entry in each column,

$$E_0^{p,q} = \begin{cases} C^p & q = 0 \\ 0 & q \ne 0 \end{cases}$$

The differential d_0 has bidegree $(0, 1)$ and so runs between entries with q differing by one; every such pair has a zero member, so $d_0 = 0$ and $E_1 = E_0$. The differential d_1 has bidegree $(1, 0)$ and is the map induced by d, namely $d\colon C^p \to C^{p+1}$. Hence

$$E_2^{p,q} = \begin{cases} H^p(C^\bullet) & q = 0 \\ 0 & q \ne 0 \end{cases}$$

and every later differential has target or source zero, so $E_2 = E_\infty$. By corollary 6.2.8 the sequence converges to $H^{p+q}(C^\bullet)$, and indeed the single surviving entry on the diagonal $p+q = n$ is $H^n(C^\bullet)$, with no extension problem. The stupid filtration therefore returns the cohomology of the complex and nothing more, which is the correct answer and a useful check on the indexing conventions. It becomes informative only when the entries C^p are themselves replaced by objects with cohomology of their own, which is what section 6.4 does.

Cochain complexes from topology and geometry often carry a product, as singular cochains do under the cup product and differential forms under the wedge product, and a filtration may be compatible with it.

Theorem 6.2.13: Multiplicative Structure on the Spectral Sequence of a Filtered DG Algebra

Let R be a commutative ring and (A, d) a *differential graded* (DG) R-algebra: a graded R-algebra $A = \bigoplus_n A^n$, associative with unit $1 \in A^0$, together with an R-linear map d of degree $+1$ satisfying $d \circ d = 0$ and

$$d(xy) = (dx)\, y + (-1)^m x\, (dy) \qquad (x \in A^m)$$

Let $A = F^0 A \supseteq F^1 A \supseteq \cdots$ be a decreasing filtration by subcomplexes which is multiplicative, $F^p A \cdot F^{p'} A \subseteq F^{p+p'} A$, and bounded in each degree: for each n there is a P with $F^P A^n = 0$, as happens when $F^p A^n = 0$ for all $p > n$. Let $\{(E_r, d_r)\}_{r \geq 0}$ be the spectral sequence of corollary 6.2.8. Then:

1. Each page carries an associative, unital, R-bilinear product
$$E_r^{p,q} \otimes_R E_r^{p',q'} \to E_r^{p+p',\, q+q'}$$
induced by the product of A on representatives, with unit the class of 1 in $E_r^{0,0}$.

2. Each differential is a derivation: for $x \in E_r^{p,q}$ and $y \in E_r$,
$$d_r(xy) = d_r(x)\, y + (-1)^{p+q}\, x\, d_r(y)$$

3. The isomorphism $E_{r+1} \cong H(E_r, d_r)$ is an isomorphism of bigraded algebras, where $H(E_r, d_r)$ carries the product induced by that of E_r.

4. The filtration $F^p H(A) = \operatorname{im}\big(H(F^p A) \to H(A)\big)$ makes $H(A)$ a filtered algebra, $F^p H(A) \cdot F^{p'} H(A) \subseteq F^{p+p'} H(A)$, and the isomorphism $E_\infty^{p,q} \cong F^p H^{p+q}(A) / F^{p+1} H^{p+q}(A)$ of corollary 6.2.8 is an isomorphism of bigraded algebras $E_\infty \cong \operatorname{gr} H(A)$, where E_∞ carries the product of E_r for large r.

5. If E_{r_0} is graded commutative, that is $xy = (-1)^{(p+q)(p'+q')}\, yx$ for $x \in E_{r_0}^{p,q}$ and $y \in E_{r_0}^{p',q'}$, then so are E_∞ and E_r for every $r \geq r_0$. This holds with $r_0 = 0$ if A is graded commutative, and with $r_0 = 1$ if A is graded commutative up to a homotopy preserving the filtration: an R-bilinear map h with $h(A^m \times A^{m'}) \subseteq A^{m+m'-1}$, $h(F^p A \times F^{p'} A) \subseteq F^{p+p'} A$ and
$$xy - (-1)^{mm'}\, yx = d\, h(x, y) + h(dx, y) + (-1)^m h(x, dy) \qquad (x \in A^m,\ y \in A^{m'})$$

***Proof*:**
Weibel [23] 5.4.8 gives (1) to (4) for chain complexes, checking the Leibniz rule on the zeroth page and appealing to 5.2.13 for the later pages.[a]

Set $F^p A = A$ for $p < 0$. With $p_+ = \max(p, 0)$, the inclusion $F^p A \cdot F^{p'} A \subseteq F^{p+p'} A$ then holds for all integers p and p', because $F^p A \cdot F^{p'} A = F^{p_+} A \cdot F^{p'_+} A \subseteq F^{p_+ + p'_+} A \subseteq F^{p+p'} A$. The same holds for h in (5). Write $n = p + q$ and $n' = p' + q'$, and for $s \geq -1$ set

$$Z_s^{p,q} = \{\, x \in F^p A^n \ :\ dx \in F^{p+s} A^{n+1} \,\}$$

which is all of F^pA^n for $s \le 0$ because d preserves the filtration, in agreement with the convention $Z_{-1}^{p,q} = F^pA^n$ of definition 6.2.3. Set

$$D_r^{p,q} = Z_{r-1}^{p+1,\,q-1} + d\,Z_{r-1}^{p-r+1,\,q+r-2}$$

whose second summand is $d\big(F^{p-r+1}A^{n-1}\big) \cap F^pA^n$. By proposition 6.2.4 in the cochain form of corollary 6.2.9, $E_r^{p,q} = Z_r^{p,q}/D_r^{p,q}$ for every $r \ge 0$, with $d_r[x] = [dx]$, and by Step 3 of the proof of proposition 6.2.4 the isomorphism $E_{r+1} \cong H(E_r, d_r)$ sends the class of $x \in Z_{r+1}^{p,q}$ to the class of $[x]$. Fix $x \in Z_r^{p,q}$ and $y \in Z_r^{p',q'}$.

Step 1: Products of approximate cycles. The product xy lies in $F^{p+p'}A^{n+n'}$, and

$$d(xy) = (dx)\,y + (-1)^n x\,(dy) \in F^{p+r}A \cdot F^{p'}A + F^pA \cdot F^{p'+r}A \subseteq F^{p+p'+r}A$$

so $xy \in Z_r^{p+p',\,q+q'}$.

Step 2: $D_r^{p,q} \cdot y$ and $x \cdot D_r^{p',q'}$ lie in $D_r^{p+p',q+q'}$. Let $z \in Z_{r-1}^{p+1,q-1}$. Then $zy \in F^{p+p'+1}A$, and $d(zy) = (dz)\,y + (-1)^n z\,(dy)$ lies in $F^{p+r}A \cdot F^{p'}A + F^{p+1}A \cdot F^{p'+r}A \subseteq F^{p+p'+r}A$, so $zy \in Z_{r-1}^{p+p'+1,\,q+q'-1}$. Next let $w \in F^{p-r+1}A^{n-1}$ with $dw \in F^pA^n$, so that dw is a general element of the second summand of $D_r^{p,q}$. The Leibniz rule gives

$$(dw)\,y = d(wy) + (-1)^n w\,(dy)$$

The element wy lies in $F^{p+p'-r+1}A^{n+n'-1}$, and $d(wy) = (dw)\,y - (-1)^n w\,(dy)$ lies in $F^{p+p'}A + F^{p+p'+1}A = F^{p+p'}A$, so $d(wy) \in d\,Z_{r-1}^{p+p'-r+1,\,q+q'+r-2}$. The element $w\,(dy)$ lies in $F^{p-r+1}A \cdot F^{p'+r}A \subseteq F^{p+p'+1}A$, and $d\big(w\,(dy)\big) = (dw)(dy)$ lies in $F^{p+p'+r}A$, so $w\,(dy) \in Z_{r-1}^{p+p'+1,\,q+q'-1}$. Hence $D_r^{p,q} \cdot y \subseteq D_r^{p+p',q+q'}$. The same estimates, applied to xz for $z \in Z_{r-1}^{p'+1,q'-1}$ and to the identity $x\,(dw) = (-1)^n\big(d(xw) - (dx)\,w\big)$ for $w \in F^{p'-r+1}A^{n'-1}$ with $dw \in F^{p'}A^{n'}$, give $x \cdot D_r^{p',q'} \subseteq D_r^{p+p',q+q'}$.

Step 3: Statements (1) and (2). By Steps 1 and 2, $(x + D_r^{p,q})(y + D_r^{p',q'}) \subseteq xy + D_r^{p+p',q+q'}$, so $[x][y] = [xy]$ is a well defined R-bilinear map $E_r^{p,q} \times E_r^{p',q'} \to E_r^{p+p',q+q'}$. It is associative because the product of A is. The unit 1 lies in $Z_r^{0,0}$, since $1 \in F^0A^0$ and $d1 = d(1 \cdot 1) = 2\,d1$ forces $d1 = 0$, and its class is a two-sided unit. This proves (1). For (2), $dx \in Z_r^{p+r,q-r+1}$ and $dy \in Z_r^{p'+r,q'-r+1}$, and

$$d_r\big([x][y]\big) = [d(xy)] = [(dx)\,y] + (-1)^n[x\,(dy)] = d_r[x]\,[y] + (-1)^{p+q}[x]\,d_r[y]$$

Step 4: Statement (3). By (2) the product of two d_r-cycles is a d_r-cycle, and for a d_r-cycle a of total degree m and any $c \in E_r$,

$$a\,d_r(c) = (-1)^m d_r(ac), \qquad d_r(c)\,a = d_r(ca)$$

So $\ker d_r$ is a bigraded subalgebra containing the unit, $\operatorname{im} d_r$ is a two-sided ideal of it, and $H(E_r, d_r)$ inherits the product $[a][b] = [ab]$. The isomorphism $E_{r+1} \to H(E_r, d_r)$ sends the class of $x \in Z_{r+1}^{p,q}$ to the class of $[x]$, so it sends the class of xy to the class of $[xy] = [x][y]$, which is

the product of the images. It also sends the unit to the unit, so it is an isomorphism of bigraded algebras.

Step 5: Statement (4). Write Z^n and B^n for the cocycles and coboundaries of A in degree n. The argument of Step 4, applied to d on A, shows that $Z = \bigoplus_n Z^n$ is a subalgebra and $B = \bigoplus_n B^n$ a two-sided ideal of it, so $H(A) = Z/B$ is a graded algebra. A class lies in $F^pH^n(A)$ exactly when it has a representative in $Z^n \cap F^pA^n$. Classes in $F^pH(A)$ and $F^{p'}H(A)$ therefore have representatives $\alpha \in Z \cap F^pA$ and $\beta \in Z \cap F^{p'}A$, and their product has the representative $\alpha\beta \in Z \cap F^{p+p'}A$. This proves $F^pH(A) \cdot F^{p'}H(A) \subseteq F^{p+p'}H(A)$, and $F^0H(A) = H(A)$ because $F^0A = A$. Consequently

$$\big(a + F^{p+1}H(A)\big)\big(b + F^{p'+1}H(A)\big) = ab + F^{p+p'+1}H(A) \qquad \big(a \in F^pH(A),\ b \in F^{p'}H(A)\big)$$

is a well defined bigraded product on $\mathrm{gr}H(A)$.

Fix (p, q) and take r with $F^{p+r}A^{n+1} = 0$ and $p - r + 1 \leq 0$, which holds for all large r by boundedness. Then $F^{p-r+1}A^{n-1} = A^{n-1}$, and the explicit description reads

$$Z_r^{p,q} = Z^n \cap F^pA^n, \qquad Z_{r-1}^{p+1,q-1} = Z^n \cap F^{p+1}A^n, \qquad d\,Z_{r-1}^{p-r+1,\,q+r-2} = B^n \cap F^pA^n$$

These do not depend on r, so for all such r the page $E_r^{p,q}$ is one fixed quotient of $Z^n \cap F^pA^n$, and that quotient is $E_\infty^{p,q}$. Take r this large for the three spots (p, q), (p', q') and $(p+p', q+q')$ at once. The product of E_r then sends the classes of cocycles $x \in Z^n \cap F^pA^n$ and $y \in Z^{n'} \cap F^{p'}A^{n'}$ to the class of xy, independently of r, and this is the product of E_∞. By Steps 2 and 3 of the proof of theorem 6.2.6, carried to cochains as in corollary 6.2.8, the isomorphism $E_\infty^{p,q} \cong \mathrm{gr}^pH^n(A)$ sends the class of x to $[x] + F^{p+1}H^n(A)$. It therefore sends the class of xy to $[xy] + F^{p+p'+1}H(A)$, which is the product of $[x] + F^{p+1}H(A)$ and $[y] + F^{p'+1}H(A)$ in $\mathrm{gr}H(A)$. It also sends 1 to 1, so $E_\infty \cong \mathrm{gr}H(A)$ as bigraded algebras.

Step 6: Statement (5). If E_r is graded commutative then so is $H(E_r, d_r)$, whose product is computed on representatives, and so is E_{r+1} by (3). Induction from r_0 covers every E_r with $r \geq r_0$, and E_∞ follows because its product is that of E_r for large r. If A is graded commutative then so is E_0, by (1). Suppose now that h is as in (5) and $r \geq 1$. Put $v = h(x, y)$ and $u = h(dx, y) + (-1)^n h(x, dy)$, so that $xy - (-1)^{nn'}yx = dv + u$. The element u lies in $F^{p+p'+r}A \subseteq F^{p+p'+1}A$, and $du = d(xy) - (-1)^{nn'}d(yx)$ lies in $F^{p+p'+r}A$ by Step 1, so $u \in Z_{r-1}^{p+p'+1,\,q+q'-1}$. The element v lies in $F^{p+p'}A^{n+n'-1} \subseteq F^{p+p'-r+1}A^{n+n'-1}$, and $dv = xy - (-1)^{nn'}yx - u$ lies in $F^{p+p'}A$, so $dv \in d\,Z_{r-1}^{p+p'-r+1,\,q+q'+r-2}$. Hence $xy - (-1)^{nn'}yx \in D_r^{p+p',q+q'}$, that is $[x][y] = (-1)^{nn'}[y][x]$ in E_r, and in particular E_1 is graded commutative, completing the proof.

[a] That appeal is not formal. The differential d_{r+1} is not determined by (E_r, d_r), so the Leibniz rule for d_r implies nothing about d_{r+1}. Step 3 below verifies it on every page from representatives in A.

Statement (2) is the one computations use: a derivation is determined by its values on a set of algebra generators, so on a page generated by a few classes the whole differential is fixed by its values on those classes.

Exercise 6.2.1

1. Let $(C_\bullet, F_\bullet)$ be a filtered complex whose filtration has exactly three nonzero stages, $0 = F_{-1} \subseteq F_0 \subseteq F_1 \subseteq F_2 = C_\bullet$. Show that $d_r = 0$ for $r \geq 3$, so that $E^3 = E^\infty$. Generalize to a filtration with N stages.

2. Show that the filtration induced on $H_n(C)$ in definition 6.2.1 is increasing, and that a morphism of filtered complexes $f\colon (C_\bullet, F_\bullet) \to (C'_\bullet, F'_\bullet)$ satisfies $f_*\big(F_pH_n(C)\big) \subseteq F'_pH_n(C')$. (This is the compatibility used in Step 4 of the proof of theorem 6.2.7.)

3. A spectral sequence *degenerates at* E^r if $d_s = 0$ for all $s \geq r$. Show that if the spectral sequence of a bounded filtered complex degenerates at E^1, then for every n the object $H_n(C)$ has a finite filtration with graded pieces $H_n(\mathrm{gr}_pC)$. Show by example that this does not force $H_n(C) \cong \bigoplus_p H_n(\mathrm{gr}_pC)$.

4. Verify proposition 6.2.4 directly at $r = 1$ by computing both sides for the filtered complex of example 6.1.9, and confirm that the differential induced by ∂ is the map computed there.

5. Let $f\colon (C_\bullet, F_\bullet) \to (C'_\bullet, F'_\bullet)$ be a morphism of bounded filtered complexes which is an isomorphism on E^2 but not on E^1. Show that f_* is still an isomorphism on homology, and construct such an f with $E^1 \neq E'^1$.

6.3 The Two Spectral Sequences of a Double Complex

A double complex carries two filtrations of its total complex, one by columns and one by rows, and section 6.2 converts each into a spectral sequence. Both converge to the homology of the same total complex, and that coincidence is the source of most of the identifications homological algebra makes: whenever one filtration degenerates for an easy reason and the other computes something of interest, the two are forced to agree. Section 2.5 promised this machinery twice, once for the cohomology of double complexes and once for the balancing of Tor and Ext, and this section delivers both.

Throughout, $A = \{A_{p,q}\}$ is a double complex in the sense of definition 2.5.1, with horizontal differential $d^h\colon A_{p,q} \to A_{p-1,q}$ and vertical differential $\delta^v\colon A_{p,q} \to A_{p,q-1}$ satisfying $d^hd^h = 0$, $\delta^v\delta^v = 0$ and $d^h\delta^v + \delta^vd^h = 0$. The total complex is $\mathrm{Tot}(A)_n = \bigoplus_{p+q=n} A_{p,q}$ with differential $d = d^h + \delta^v$, as in proposition 2.5.2; anticommutativity is exactly what makes $d \circ d = 0$. Write

$$H^h_p(A)_\bullet, \qquad H^v_q(A)_\bullet$$

for the homology of the rows and of the columns respectively, each of which is again a complex under the differential the other direction induces.

Definition 6.3.1: The Column and Row Filtrations

Let A be a double complex. The *column filtration* and the *row filtration* of $\mathrm{Tot}(A)$ are

$$^{\mathrm{I}}F_p\mathrm{Tot}(A)_n = \bigoplus_{\substack{p'+q'=n \\ p' \leq p}} A_{p',q'}, \qquad ^{\mathrm{II}}F_p\mathrm{Tot}(A)_n = \bigoplus_{\substack{p'+q'=n \\ q' \leq p}} A_{p',q'}$$

The double complex is *first-quadrant* if $A_{p,q} = 0$ whenever $p < 0$ or $q < 0$.

Both are filtrations by subcomplexes, and for the reason that each differential moves in only one of the two directions: d^h lowers p' and leaves q' unchanged, while δ^v lowers q' and leaves p' unchanged.

So d carries ${}^{\mathrm{I}}F_p$ into itself, and likewise for ${}^{\mathrm{II}}F_p$. The two differ only in which coordinate is being watched, which is why the same total complex acquires two different, and generally inequivalent, approximations to its homology.

Theorem 6.3.2: The Two Spectral Sequences of a Double Complex

Let A be a first-quadrant double complex. Then the column and row filtrations of $\mathrm{Tot}(A)$ give two spectral sequences, both converging to $H_\bullet(\mathrm{Tot}(A))$, with second pages

$$ {}^{\mathrm{I}}E^2_{p,q} = H^h_p\big(H^v_q(A)\big) \implies H_{p+q}(\mathrm{Tot}(A)) $$

$$ {}^{\mathrm{II}}E^2_{p,q} = H^v_p\big(H^h_q(A)\big) \implies H_{p+q}(\mathrm{Tot}(A)) $$

***Proof*:**

Step 1: Boundedness. Fix n. Since A is first-quadrant, $A_{p',q'} = 0$ unless $0 \le p' \le n$ on the diagonal $p' + q' = n$. Hence ${}^{\mathrm{I}}F_{-1}\mathrm{Tot}(A)_n = 0$ and ${}^{\mathrm{I}}F_n\mathrm{Tot}(A)_n = \mathrm{Tot}(A)_n$, so the column filtration is bounded in the sense of definition 6.2.1; the same argument applies to the row filtration. Theorem 6.2.6 therefore applies to each and gives convergence to $H_\bullet(\mathrm{Tot}(A))$ once the pages are identified.

Step 2: The zeroth and first pages of the column filtration. By definition 6.2.1 the graded piece is

$$ \mathrm{gr}_p\mathrm{Tot}(A)_n = \frac{{}^{\mathrm{I}}F_p\mathrm{Tot}(A)_n}{{}^{\mathrm{I}}F_{p-1}\mathrm{Tot}(A)_n} = A_{p,\,n-p} $$

so ${}^{\mathrm{I}}E^0_{p,q} = A_{p,q}$. The induced differential is the part of $d = d^h + \delta^v$ that preserves the filtration index p. The summand d^h lowers p, so it vanishes on the graded piece; the summand δ^v preserves p. Hence $d^0 = \delta^v$ and

$$ {}^{\mathrm{I}}E^1_{p,q} = H_q\big(A_{p,\bullet},\, \delta^v\big) = H^v_q(A)_p $$

the homology of the p-th column.

Step 3: The second page of the column filtration. By theorem 6.2.2 the differential d^1 has bidegree $(-1, 0)$, so it runs ${}^{\mathrm{I}}E^1_{p,q} \to {}^{\mathrm{I}}E^1_{p-1,q}$; by proposition 6.2.4 it is induced by the differential of $\mathrm{Tot}(A)$ on representatives. A representative of a class in ${}^{\mathrm{I}}E^1_{p,q}$ is an element of $A_{p,q}$ killed by δ^v, and applying $d = d^h + \delta^v$ to it gives d^h of it, an element of $A_{p-1,q}$. So d^1 is the map induced by d^h on vertical homology, and

$$ {}^{\mathrm{I}}E^2_{p,q} = H_p\big(H^v_q(A)_\bullet,\, d^h\big) = H^h_p\big(H^v_q(A)\big) $$

Step 4: The row filtration. The argument is identical with the two coordinates exchanged. The graded piece of the row filtration at index p in total degree n is $A_{n-p,\,p}$, so ${}^{\mathrm{II}}E^0_{p,q} = A_{q,p}$, and the differential preserving the index p is now d^h. Hence ${}^{\mathrm{II}}E^1_{p,q} = H_q(A_{\bullet,p}, d^h) = H^h_q(A)_p$, and the next differential is induced by δ^v, giving ${}^{\mathrm{II}}E^2_{p,q} = H^v_p(H^h_q(A))$. Both converge to the homology of the same total complex by Step 1, completing the proof.

The two sequences are not related by any map; they are two different routes to one answer, and the content of the theorem is that the answer is the same. The standard way to exploit this is to arrange for one of them to collapse. A double complex whose columns are exact except in one spot, for instance, has a first page concentrated in a single row, and then every differential from the second page onward has a zero source or a zero target.

Lemma 6.3.3: Collapse Along a Single Row

Let $\{(E^r, d_r)\}$ be a spectral sequence with $E^2_{p,q} = 0$ for all $q \neq 0$. Then $E^2 = E^\infty$, and if the sequence converges to $H_\bullet$ then $H_n \cong E^2_{n,0}$ for every n.

***Proof*:**
The differential d_r has bidegree $(-r, r-1)$ by theorem 6.1.7. For $r \geq 2$ the differential out of the spot $(p, 0)$ lands in $(p-r,\ r-1)$, which has second index $r-1 \geq 1$ and so vanishes; the differential into $(p, 0)$ comes from $(p+r,\ 1-r)$, which has negative second index and so vanishes. Every other spot is already zero. Hence $d_r = 0$ for all $r \geq 2$ and $E^2 = E^\infty$.

For the abutment, the diagonal $p + q = n$ of the limit page contains the single nonzero group $E^\infty_{n,0}$. By definition 6.2.5 the filtration of H_n is bounded and has these as graded pieces; a bounded filtration whose graded pieces vanish outside one step has exactly one nonzero step, so $H_n \cong E^\infty_{n,0} = E^2_{n,0}$, as we sought to show.

Both results are used in section 6.4 and section 6.5 for cochain double complexes, whose differentials raise indices. As with corollary 6.2.9, the translation is recorded once.

Corollary 6.3.4: The Cochain Form of the Double-Complex Sequences

Let $A^{\bullet,\bullet}$ be a cochain double complex concentrated in non-negative bidegrees, with differentials raising the two indices. Then the column and row filtrations of $\mathrm{Tot}(A)$ give two spectral sequences converging to $H^{p+q}(\mathrm{Tot}(A))$, with second pages

$$ {}^{\mathrm{I}}E_2^{p,q} = H_h^p\big(H_v^q(A)\big), \qquad {}^{\mathrm{II}}E_2^{p,q} = H_v^p\big(H_h^q(A)\big) $$

and lemma 6.3.3 holds in the form: if $E_2^{p,q} = 0$ for all $q \neq 0$ then $E_2 = E_\infty$ and $H^n \cong E_2^{n,0}$; symmetrically if $E_2^{p,q} = 0$ for all $p \neq 0$ then $H^n \cong E_2^{0,n}$.

***Proof*:**
Apply the translation $\mathcal{T}$ of corollary 6.2.9 in both indices, setting $A_{p,q} = A^{-p,-q}$. This turns a cochain double complex concentrated in non-negative bidegrees into a chain double complex concentrated in non-positive ones, which is bounded on diagonals in the sense of exercise 6.3.1 (5); theorem 6.3.2 and lemma 6.3.3 apply to it, and negating every index returns the displayed statements. The column form of the collapse is exercise 6.3.1 (4), translated the same way, completing the proof.

The lemma turns theorem 6.3.2 into an identification technique, and the standard application is the one section 2.5 deferred: that Tor may be computed by resolving either variable.

Corollary 6.3.5: Balancing Tor

Let R be a ring, M a right R-module and N a left R-module, and let $P_\bullet \to M$ and $Q_\bullet \to N$ be projective resolutions. Then

$$H_n\big(P_\bullet \otimes_R N\big) \cong H_n\big(\mathrm{Tot}(P_\bullet \otimes_R Q_\bullet)\big) \cong H_n\big(M \otimes_R Q_\bullet\big)$$

for every n. In particular $\mathrm{Tor}_n^R(M,N)$ may be computed by resolving either variable, recovering the Tor half of theorem 2.5.8 as the degenerate case of theorem 6.3.2.

***Proof*:**
Let $A_{p,q} = P_p \otimes_R Q_q$, with horizontal differential induced by that of $P_\bullet$ and vertical differential induced by that of $Q_\bullet$, the latter twisted by the sign $(-1)^p$ so that the two anticommute as definition 2.5.1 requires. Both resolutions are supported in non-negative degrees, so A is first-quadrant and theorem 6.3.2 applies.

Step 1: The column filtration. Each P_p is projective, hence a direct summand of a free module, hence flat, so the functor $P_p \otimes_R -$ is exact. Therefore it commutes with homology and

$${}^{\mathrm{I}}E^1_{p,q} = H_q\big(P_p \otimes_R Q_\bullet\big) \cong P_p \otimes_R H_q(Q_\bullet)$$

Since $Q_\bullet \to N$ is a resolution, $H_q(Q_\bullet)$ is N for $q = 0$ and zero otherwise. So the first page is concentrated in the row $q = 0$, where it is $P_p \otimes_R N$, and the second page is

$${}^{\mathrm{I}}E^2_{p,0} = H_p\big(P_\bullet \otimes_R N\big)$$

with ${}^{\mathrm{I}}E^2_{p,q} = 0$ for $q \neq 0$. By lemma 6.3.3, $H_n(\mathrm{Tot}(A)) \cong H_n(P_\bullet \otimes_R N)$.

Step 2: The row filtration. Symmetrically, each Q_q is flat, so ${}^{\mathrm{II}}E^1_{p,q} = H_q(P_\bullet \otimes_R Q_p) \cong H_q(P_\bullet) \otimes_R Q_p$, which is $M \otimes_R Q_p$ for $q = 0$ and zero otherwise. Hence ${}^{\mathrm{II}}E^2_{p,0} = H_p(M \otimes_R Q_\bullet)$ with the other rows zero, and lemma 6.3.3 gives $H_n(\mathrm{Tot}(A)) \cong H_n(M \otimes_R Q_\bullet)$.

Both computations describe the homology of the same total complex, so the two are isomorphic, completing the proof.

The argument is worth contrasting with the one given in section 2.5. There the comparison was made by hand, through an explicit quasi-isomorphism between the total complex and each of its edges. Here it is a two-line consequence of a general theorem, and the same two lines apply verbatim to Ext, to group cohomology, and to any other derived functor of a bifunctor. That is the characteristic economy of the method: the work moves once into theorem 6.2.6 and is then reused without repetition.

Exercise 6.3.1

1. Let A be a first-quadrant double complex all of whose columns are exact. Show that $\mathrm{Tot}(A)$ is acyclic. Deduce proposition 2.5.4 for first-quadrant double complexes.
2. State and prove the analogue of corollary 6.3.5 for Ext, using a projective resolution of M

and an injective resolution of N. Identify which functor plays the role of $P_p \otimes_R -$ in Step 1, and which exactness property replaces flatness.

3. Let A be the first-quadrant double complex with $A_{0,0} = A_{1,0} = \mathbb{Z}$, $A_{0,1} = A_{1,1} = \mathbb{Z}$, horizontal differentials the identity, and vertical differentials the identity, with the sign twist making it anticommute. Compute both spectral sequences and $H_\bullet(\mathrm{Tot}(A))$, and check that the two agree.

4. Show that a spectral sequence with $E^2_{p,q} = 0$ for all $p \neq 0$ also collapses, and that $H_n \cong E^2_{0,n}$. (No quadrant hypothesis is needed.) (This is the column analogue of lemma 6.3.3.)

5. The first-quadrant hypothesis in theorem 6.3.2 is used in exactly one place. Identify it, and show that the theorem still holds for any double complex that is *bounded on diagonals*, meaning that for each n only finitely many $A_{p,q}$ with $p + q = n$ are nonzero. Then show that a double complex with $A_{p,q} \neq 0$ for every $p, q \in \mathbb{Z}$ fails this condition, so that theorem 6.2.6 does not apply to either filtration.

6.4 Hypercohomology

The derived functors of section 2.3 take an object as input. Many constructions present a complex instead: a complex of sheaves in place of a single sheaf, a complex of modules in place of a module. For a left exact F and a bounded-below complex $C^\bullet$, this section resolves $C^\bullet$ by injectives in two directions at once. The resolution is a double complex, the Cartan-Eilenberg resolution, and the cohomology of the total complex of F applied to it is the *hypercohomology* $\mathbb{R}^n F(C^\bullet)$. The two filtrations of that double complex give two spectral sequences converging to $\mathbb{R}^n F(C^\bullet)$. The first starts from the derived functors $R^q F(C^p)$ of the terms of $C^\bullet$, and the second from the derived functors $R^p F(H^q(C^\bullet))$ of its cohomology objects. Hypercohomology is the cohomology of the total derived functor of section 3.3, by proposition 6.4.9.

Definition 6.4.1: Cartan-Eilenberg Resolution

Let $\mathcal{A}$ be an abelian category with enough injectives and $C^\bullet$ a cochain complex in $\mathcal{A}$ bounded below. A *Cartan-Eilenberg resolution* of $C^\bullet$ is a double complex $I^{\bullet,\bullet}$ of injective objects, with $I^{p,q} = 0$ for $q < 0$ and with the column $I^{p,\bullet}$ zero whenever $C^p = 0$, together with an augmentation $C^p \to I^{p,0}$ for each p, these forming a morphism of complexes $C^\bullet \to I^{\bullet,0}$, such that for every p the induced augmentations make each of

$$I^{p,\bullet}, \qquad B^p\big(I^{\bullet,\bullet}\big), \qquad Z^p\big(I^{\bullet,\bullet}\big), \qquad H^p\big(I^{\bullet,\bullet}\big)$$

an injective resolution of C^p, of $B^p(C^\bullet)$, of $Z^p(C^\bullet)$ and of $H^p(C^\bullet)$ respectively, where B^p, Z^p and H^p are taken in the first (horizontal) direction.

The conditions on B^p, Z^p and H^p tie the resolution to the cohomology of $C^\bullet$. A double complex whose columns are injective resolutions of the C^p need satisfy only the first condition, and its horizontal cohomology need not resolve the objects $H^p(C^\bullet)$. The vanishing conditions make a Cartan-Eilenberg resolution bounded below in both directions: if $C^p = 0$ for $p < p_0$, then $I^{p,q} = 0$ unless $p \geq p_0$ and $q \geq 0$ (the condition of Weibel [23] 5.7.1).

Theorem 6.4.2: Existence of Cartan-Eilenberg Resolutions

Every bounded-below cochain complex in an abelian category with enough injectives admits a Cartan-Eilenberg resolution.

***Proof*:**

Step 1: Choices. For each p, use that $\mathcal{A}$ has enough injectives (definition 2.2.3) to choose injective resolutions

$$B^p(C^\bullet) \to \mathcal{B}^{p,\bullet}, \qquad H^p(C^\bullet) \to \mathcal{H}^{p,\bullet}$$

concentrated in non-negative degrees, and taking the zero resolution of every zero object.

Step 2: Resolving the cycles. The definition of cohomology gives a short exact sequence

$$0 \to B^p(C^\bullet) \to Z^p(C^\bullet) \to H^p(C^\bullet) \to 0$$

Its two ends are already resolved by Step 1, so the horseshoe lemma (lemma 2.3.7), in its injective form, produces an injective resolution $Z^p(C^\bullet) \to \mathcal{Z}^{p,\bullet}$ fitting into a degreewise split short exact sequence of complexes $0 \to \mathcal{B}^{p,\bullet} \to \mathcal{Z}^{p,\bullet} \to \mathcal{H}^{p,\bullet} \to 0$.

Step 3: Resolving the terms. The complex $C^\bullet$ also gives, for each p, a short exact sequence

$$0 \to Z^p(C^\bullet) \to C^p \to B^{p+1}(C^\bullet) \to 0$$

whose outer terms are resolved by $\mathcal{Z}^{p,\bullet}$ from Step 2 and $\mathcal{B}^{p+1,\bullet}$ from Step 1. Applying lemma 2.3.7 again yields an injective resolution $C^p \to I^{p,\bullet}$ with $I^{p,q} \cong \mathcal{Z}^{p,q} \oplus \mathcal{B}^{p+1,q}$ in each degree, compatibly with the two inclusions.

Step 4: The horizontal differential. Define $I^{p,q} \to I^{p+1,q}$ as the composite

$$I^{p,q} \twoheadrightarrow \mathcal{B}^{p+1,q} \hookrightarrow \mathcal{Z}^{p+1,q} \hookrightarrow I^{p+1,q}$$

using the splitting of Step 3 for the first map and the inclusion of Step 2 for the second. The composite of two consecutive such maps is zero, because the image of the first lands in $\mathcal{Z}^{p+1,q}$, which the projection to $\mathcal{B}^{p+2,q}$ annihilates. So $I^{\bullet,\bullet}$ is a double complex of injectives, after the usual sign twist making the two differentials anticommute as definition 2.5.1 requires.

Step 5: The conditions hold. By construction $I^{p,\bullet}$ is an injective resolution of C^p, which is the first condition, and $I^{p,q} = 0$ for $q < 0$. If $C^p = 0$, then $B^p(C^\bullet)$, $H^p(C^\bullet)$ and $B^{p+1}(C^\bullet)$ are zero, so their chosen resolutions are zero, the horseshoe of Step 2 gives $\mathcal{Z}^{p,\bullet} = 0$, and $I^{p,\bullet} \cong \mathcal{Z}^{p,\bullet} \oplus \mathcal{B}^{p+1,\bullet}$ is zero. By Step 4 the horizontal cycles are $Z^p(I^{\bullet,q}) = \mathcal{Z}^{p,q}$ and the horizontal boundaries are $B^p(I^{\bullet,q}) = \mathcal{B}^{p,q}$, which are injective resolutions of $Z^p(C^\bullet)$ and $B^p(C^\bullet)$ by Steps 2 and 1. The quotient $\mathcal{Z}^{p,q}/\mathcal{B}^{p,q} \cong \mathcal{H}^{p,q}$ is an injective resolution of $H^p(C^\bullet)$ by Step 1 and the splitting of Step 2. The augmentations are compatible with the horizontal differential by construction, since d^h was defined through the factorization $C^p \twoheadrightarrow B^{p+1} \hookrightarrow Z^{p+1} \hookrightarrow C^{p+1}$ that the differential of $C^\bullet$ itself factors through. All four conditions of definition 6.4.1 hold, completing the proof.

Existence is only half of what the resolution is used for. Proposition 6.4.6 compares the resolutions of two different complexes, and the naturality of the spectral sequences built from them rests on the same comparison. Both need a morphism of complexes to lift to the resolutions. The object-level comparison theorem (lemma 2.2.13 and its dual) does not give this, since a Cartan-Eilenberg resolution is constrained not only degreewise but also on cycles, boundaries and cohomology. The lift has to be built through the same two horseshoes that produced the resolution, and the functoriality of that construction, lemma 2.3.8, is what the proof below uses at each of them.

Theorem 6.4.3: Comparison Theorem for Cartan-Eilenberg Resolutions

Let $f^\bullet\colon C^\bullet \to \widetilde{C}^\bullet$ be a morphism of bounded-below cochain complexes in an abelian category with enough injectives, and let $J^{\bullet,\bullet}$ and $\widetilde{J}^{\bullet,\bullet}$ be Cartan-Eilenberg resolutions of $C^\bullet$ and $\widetilde{C}^\bullet$. Then $f^\bullet$ lifts to a morphism of double complexes

$$\tilde{f}^{\bullet,\bullet}\colon J^{\bullet,\bullet} \to \widetilde{J}^{\bullet,\bullet}$$

compatible with the augmentations, and any two such lifts are homotopic. Consequently the induced maps on Z^p, B^p and H^p in the horizontal direction are the resolutions of $Z^p(f)$, $B^p(f)$ and $H^p(f)$.

***Proof*:**

Step 1: Lifts on the building blocks. The morphism $f^\bullet$ induces $B^p(f)\colon B^p(C^\bullet) \to B^p(\widetilde{C}^\bullet)$ and $H^p(f)\colon H^p(C^\bullet) \to H^p(\widetilde{C}^\bullet)$ for every p. By definition 6.4.1 the objects $\mathcal{B}^{p,\bullet}$ and $\mathcal{H}^{p,\bullet}$ occurring in J are injective resolutions of $B^p(C^\bullet)$ and $H^p(C^\bullet)$, and likewise for $\widetilde{J}$. The dual of lemma 2.2.13 therefore lifts $B^p(f)$ and $H^p(f)$ to morphisms of those resolutions, each unique up to homotopy.

Step 2: Lifts on cycles. Apply lemma 2.3.8 to the morphism of short exact sequences

$$\big(0 \to B^p(C) \to Z^p(C) \to H^p(C) \to 0\big) \longrightarrow \big(0 \to B^p(\widetilde{C}) \to Z^p(\widetilde{C}) \to H^p(\widetilde{C}) \to 0\big)$$

induced by $f^\bullet$, with the lifts of Step 1 on the two ends. This produces a lift $\mathcal{Z}^{p,\bullet} \to \widetilde{\mathcal{Z}}^{p,\bullet}$ of $Z^p(f)$.

Step 3: Lifts on the terms. Apply lemma 2.3.8 again, now to

$$\big(0 \to Z^p(C) \to C^p \to B^{p+1}(C) \to 0\big) \longrightarrow \big(0 \to Z^p(\widetilde{C}) \to \widetilde{C}^p \to B^{p+1}(\widetilde{C}) \to 0\big)$$

with the lift of Step 2 on the left end and that of Step 1 on the right. This produces $\tilde{f}^{p,\bullet}\colon J^{p,\bullet} \to \widetilde{J}^{p,\bullet}$ lifting f^p.

Step 4: Compatibility with the horizontal differential. For any Cartan-Eilenberg resolution the horizontal differential factors as $J^{p,q} \twoheadrightarrow \mathcal{B}^{p+1,q} \hookrightarrow \mathcal{Z}^{p+1,q} \hookrightarrow J^{p+1,q}$: it factors through its image $\mathcal{B}^{p+1,q}$ by definition, and that image lies in $\mathcal{Z}^{p+1,q}$ because $d^h \circ d^h = 0$. Each of the three maps is respected by the lifts of Steps 1 to 3, the projection and the two inclusions being the structure maps of the splittings. Hence $\tilde{f}^{\bullet,\bullet}$ commutes with d^h, and it commutes with d^v because each $\tilde{f}^{p,\bullet}$ is a morphism of resolutions. So it is a morphism of double complexes.

Step 5: Uniqueness up to homotopy. Let $\tilde{f}$ and $\tilde{g}$ both lift $f^\bullet$. In each column their difference lifts the zero morphism $C^p \to \widetilde{C}^p$, so by the dual of lemma 2.2.4 it is null-homotopic: there are $s^{p,\bullet}$ with $d^v s + s d^v = \tilde{f} - \tilde{g}$ in each column.

Producing a homotopy of *double* complexes, that is one which additionally satisfies $s d^h = d^h s$, requires choosing those column homotopies compatibly. This is a third induction of the same shape as lemma 2.3.8: choose null-homotopies on $\mathcal{B}^{p,\bullet}$ and $\mathcal{H}^{p,\bullet}$ by the object-level uniqueness of Step 1, extend over the first horseshoe to $\mathcal{Z}^{p,\bullet}$ by an obstruction computation of the same shape as the one for θ^q, and extend again over the second to $J^{p,\bullet}$. The bookkeeping is routine and carries no idea not already in lemma 2.3.8, so it is not repeated; it is carried out in *An Introduction to Homological Algebra* [23].

For the final clause, let $\tilde{f}$ be any lift. Its restriction to horizontal cocycles, coboundaries and cohomology is determined by its compatibility with the augmentations, since those subquotients are computed from d^h and $\tilde{f}$ commutes with d^h by Step 4; so on H^p it is a lift of $H^p(f)$, whether or not it is the one built in Steps 1 to 3. This completes the proof.

Applying F to a Cartan-Eilenberg resolution and totalizing gives a complex built from F of injective objects, as applying F to an injective resolution does for a single object.

Definition 6.4.4: Hypercohomology

Let $F\colon \mathcal{A} \to \mathcal{B}$ be a left-exact additive functor between abelian categories with $\mathcal{A}$ having enough injectives, and let $C^\bullet$ be a bounded-below cochain complex in $\mathcal{A}$. Choose a Cartan-Eilenberg resolution $J^{\bullet,\bullet}$ of $C^\bullet$ (theorem 6.4.2), where the first index is the degree in $C^\bullet$ and the second the resolution degree. The *hypercohomology* of $C^\bullet$ with respect to F is

$$\mathbb{R}^n F(C^\bullet) := H^n\Big(\mathrm{Tot}\big(F(J^{\bullet,\bullet})\big)\Big)$$

The two filtrations of the double complex $F(J^{\bullet,\bullet})$, by the degree in $C^\bullet$ and by resolution degree, give the two spectral sequences. Neither second page need determine the abutment, because differentials and extension problems intervene.

Theorem 6.4.5: The Two Hypercohomology Spectral Sequences

With the hypotheses of definition 6.4.4, there are two cohomological spectral sequences converging to $\mathbb{R}^{p+q}F(C^\bullet)$ in the sense of corollary 6.2.8, with filtrations bounded in each degree:

$$ {}^{\mathrm{I}}E_1^{p,q} = R^qF\big(C^p\big) \implies \mathbb{R}^{p+q}F(C^\bullet) $$

$$ {}^{\mathrm{II}}E_2^{p,q} = R^pF\big(H^q(C^\bullet)\big) \implies \mathbb{R}^{p+q}F(C^\bullet) $$

The first is filtered by the degree in $C^\bullet$, and its differential d_1 is R^qF of the differential of $C^\bullet$, so that ${}^{\mathrm{I}}E_2^{p,q} = H^p\big(R^qF(C^\bullet)\big)$. The second is filtered by resolution degree. Both are first-quadrant when $C^\bullet$ is concentrated in non-negative degrees. Both are natural:

1. A morphism $f\colon C^\bullet \to \widetilde{C}^\bullet$ of bounded-below complexes induces, through a lift of f to Cartan-Eilenberg resolutions, a morphism of each spectral sequence, compatible with the map the lift induces on the abutments, equal to $R^qF(f^p)$ on ${}^{\mathrm{I}}E_1^{p,q}$ and to $R^pF(H^q(f))$ on ${}^{\mathrm{II}}E_2^{p,q}$, and independent of the lift from those pages onwards.
2. A natural transformation $\tau\colon F \to F'$ of left-exact additive functors $\mathcal{A} \to \mathcal{B}$ induces a morphism from each spectral sequence for F to the corresponding one for F', compatible with the map τ induces on the abutments, and equal on ${}^{\mathrm{I}}E_1$ and ${}^{\mathrm{II}}E_2$ to the maps $H^k(\tau_I)\colon R^kF(A) \to R^kF'(A)$, for I an injective resolution of the object A in question.

***Proof*:**
Both come from corollary 6.3.4 applied to the double complex $F(J^{\bullet,\bullet})$, and the work is in identifying the pages.

Step 1: The two spectral sequences. The complex $C^\bullet$ is bounded below, so there is an integer p_0 with $C^p = 0$ for $p < p_0$, and then $J^{p,q} = 0$ unless $p \geq p_0$ and $q \geq 0$, by definition 6.4.1. The cochain double complex $D^{p,q} := F(J^{p+p_0,q})$, with the differentials of $F(J^{\bullet,\bullet})$, is concentrated in non-negative bidegrees, and $\mathrm{Tot}(D)^n = \mathrm{Tot}(F(J))^{n+p_0}$ with the same differential. Corollary 6.3.4 therefore applies to D. Adding p_0 back to the index that records the degree in $C^\bullet$ turns its column filtration into the filtration of $\mathrm{Tot}(F(J))$ by the degree in $C^\bullet$, and its row filtration into the filtration by resolution degree. This gives two spectral sequences converging to $H^{p+q}(\mathrm{Tot}(F(J))) = \mathbb{R}^{p+q}F(C^\bullet)$, and their filtrations are bounded in each degree because each diagonal $p+q=n$ of $J^{\bullet,\bullet}$ carries only the finitely many terms with $p_0 \leq p \leq n$.

Step 2: The first sequence. For the filtration by the degree in $C^\bullet$ the graded piece at level p is the column $F(J^{p,\bullet})$ with the vertical differential, since the horizontal differential raises the filtration index. By corollary 6.2.8 the first page is its cohomology,

$$ {}^{\mathrm{I}}E_1^{p,q} = H^q\big(F(J^{p,\bullet})\big) = R^qF(C^p) $$

because $J^{p,\bullet}$ is an injective resolution of C^p by definition 6.4.1. The sign twist of the vertical differential does not change the cohomology. By proposition 6.2.4, in the cochain form of corollary 6.2.9(2), the differential d_1 is induced by the total differential on representatives, and on a vertical cocycle in $F(J^{p,q})$ the total differential is the horizontal one. The horizontal differential $J^{p,\bullet} \to J^{p+1,\bullet}$ is a morphism of injective resolutions lying over the differential $C^p \to C^{p+1}$, the augmentations forming a morphism of complexes, so the map it induces on cohomology after

applying F is R^qF of that differential. This gives the stated d_1 and ${}^{\mathrm{I}}E_2^{p,q} = H^p(R^qF(C^\bullet))$.

Step 3: The second sequence. By corollary 6.3.4 the row filtration has ${}^{\mathrm{II}}E_2^{p,q} = H_v^p\big(H_h^q(F(J))\big)$: horizontal cohomology at the degree q in $C^\bullet$, followed by vertical cohomology at the resolution degree p. Fix a resolution degree s. By definition 6.4.1 the cycles, coboundaries and cohomology objects of the row $J^{\bullet,s}$ are terms of injective resolutions, hence injective, so the short exact sequences

$$0 \to Z^q(J^{\bullet,s}) \to J^{q,s} \to B^{q+1}(J^{\bullet,s}) \to 0$$
$$0 \to B^q(J^{\bullet,s}) \to Z^q(J^{\bullet,s}) \to H^q(J^{\bullet,s}) \to 0$$

have injective first terms and are split. An additive functor preserves split exact sequences, so F commutes with forming Z^q, B^q and H^q along the row:

$$H^q\big(F(J^{\bullet,s})\big) \cong F\big(H^q(J^{\bullet,s})\big)$$

By definition 6.4.1 the complex $H^q(J^{\bullet,\bullet})$, as s varies, is an injective resolution of $H^q(C^\bullet)$, so the vertical cohomology of $F(H^q(J^{\bullet,\bullet}))$ in degree p is $R^pF(H^q(C^\bullet))$. Hence ${}^{\mathrm{II}}E_2^{p,q} = R^pF(H^q(C^\bullet))$.

When $C^\bullet$ is concentrated in non-negative degrees, $R^qF(C^p)$ and $R^pF(H^q(C^\bullet))$ vanish unless $p, q \geq 0$, since derived functors vanish in negative degrees, so both sequences are first-quadrant.

Step 4: Naturality. For (1), let $\widetilde{J}^{\bullet,\bullet}$ be the chosen Cartan-Eilenberg resolution of $\widetilde{C}^\bullet$ and $\tilde{f}\colon J \to \widetilde{J}$ a lift of f, which exists by theorem 6.4.3. Then $F(\tilde{f})$ is a morphism of double complexes, so its totalization preserves both filtrations, and theorem 6.2.7, in the cochain form of corollary 6.2.9, makes it induce a morphism of each spectral sequence compatible with the induced map on the abutments. Each column $\tilde{f}^{p,\bullet}$ is a morphism of injective resolutions lying over f^p, so the map it induces on ${}^{\mathrm{I}}E_1^{p,q} = H^q(F(J^{p,\bullet}))$ is $R^qF(f^p)$. By the final clause of theorem 6.4.3, $\tilde{f}$ restricts on horizontal cohomology to a morphism of injective resolutions lying over $H^q(f)$, and the isomorphism $H^q(F(J^{\bullet,s})) \cong F(H^q(J^{\bullet,s}))$ of Step 3 commutes with it, so the map on ${}^{\mathrm{II}}E_2^{p,q}$ is $R^pF(H^q(f))$. Neither depends on the lift. A morphism of spectral sequences is determined from a page onwards by its value on that page, its value on each later page being the map induced on cohomology, so the two morphisms are independent of the lift from those pages onwards. For (2), naturality of τ makes the maps $\tau_{J^{p,q}}$ a morphism of double complexes $F(J) \to F'(J)$, and the same argument applies to it. On ${}^{\mathrm{II}}E_2$ it uses that κ is natural in the functor, $\kappa^{F'} \circ H^q(\tau_X) = \tau_{H^qX} \circ \kappa^F$, which holds because κ is the functor applied to $Z^q \to H^q$ and τ is natural, completing the proof.

The definition depends on a choice of resolution, and the second spectral sequence, through the comparison theorem of section 6.2, shows that the answer does not.

Proposition 6.4.6: Hypercohomology is Well Defined

The objects $\mathbb{R}^nF(C^\bullet)$ of definition 6.4.4 are independent of the choice of Cartan-Eilenberg resolution, up to canonical isomorphism, and a quasi-isomorphism $C^\bullet \to \widetilde{C}^\bullet$ of bounded-below complexes induces an isomorphism $\mathbb{R}^nF(C^\bullet) \cong \mathbb{R}^nF(\widetilde{C}^\bullet)$ for every n.

***Proof*:**
Let $J^{\bullet,\bullet}$ and $\widetilde{J}^{\bullet,\bullet}$ be Cartan-Eilenberg resolutions of $C^\bullet$ and $\widetilde{C}^\bullet$ respectively, and let $f\colon C^\bullet \to \widetilde{C}^\bullet$ be a quasi-isomorphism. Taking f to be the identity handles the first claim.

Step 1: A comparison map exists. By theorem 6.4.3, f lifts to a morphism of double complexes $J^{\bullet,\bullet} \to \widetilde{J}^{\bullet,\bullet}$, uniquely up to homotopy, and the lift induces $H^q(f)$ on the horizontal cohomology resolutions. Applying F and totalizing gives a map of filtered complexes $\mathrm{Tot}(F(J)) \to \mathrm{Tot}(F(\widetilde{J}))$ compatible with the filtration by resolution degree.

Step 2: The comparison is an isomorphism on a page. By theorem 6.4.5, the filtration by resolution degree has second page ${}^{\mathrm{II}}E_2^{p,q} = R^pF(H^q(C^\bullet))$. By the final clause of theorem 6.4.3 the lift of Step 1 restricts on horizontal cohomology to a resolution of $H^q(f)$, so the map it induces on that second page is exactly $R^pF(H^q(f))$. Since f is a quasi-isomorphism, $H^q(f)$ is an isomorphism for every q, and hence so is $R^pF(H^q(f))$ by functoriality of the derived functors. So the induced map is an isomorphism on the second page.

Step 3: Conclusion. Both filtrations by resolution degree are bounded in each degree, by theorem 6.4.5, so theorem 6.2.7 applies with $r_0 = 2$, in the cochain form of corollary 6.2.9. It gives that the map is an isomorphism on the abutment, that is $\mathbb{R}^nF(C^\bullet) \cong \mathbb{R}^nF(\widetilde{C}^\bullet)$ for every n. Independence of the resolution is the case $f = \mathrm{id}$, and canonicity follows because the lift of Step 1 is unique up to homotopy and homotopic maps agree on cohomology, completing the proof.

Two special cases recover the ordinary theory and explain the name. Both follow by collapsing one of the two sequences, and together they say that hypercohomology interpolates between the two computations one might have attempted directly.

Corollary 6.4.7: Hypercohomology Recovers the Ordinary Derived Functors

With the hypotheses of definition 6.4.4:

1. If $C^\bullet$ is concentrated in degree 0, with $C^0 = A$, then $\mathbb{R}^nF(C^\bullet) \cong R^nF(A)$ for every n.
2. If every C^p is F-acyclic, then $\mathbb{R}^nF(C^\bullet) \cong H^n\big(F(C^\bullet)\big)$ for every n.

***Proof*:**

1. If $C^\bullet$ is concentrated in degree 0 then $H^q(C^\bullet) = 0$ for $q \neq 0$ and $H^0(C^\bullet) = A$. So ${}^{\mathrm{II}}E_2^{p,q} = R^pF(H^q(C^\bullet))$ vanishes unless $q = 0$, where it is $R^pF(A)$. The second page is concentrated in the row $q = 0$, so by lemma 6.3.3, in the cohomological indexing of definition 6.1.6, the sequence collapses and the diagonal $p + q = n$ carries the single group $R^nF(A)$. Hence $\mathbb{R}^nF(C^\bullet) \cong R^nF(A)$.
2. If every C^p is F-acyclic then $R^qF(C^p) = 0$ for $q \geq 1$ by definition 2.3.15, so ${}^{\mathrm{I}}E_1^{p,q} = R^qF(C^p)$ vanishes unless $q = 0$, where it is $R^0F(C^p) = F(C^p)$ by left exactness. The first page is concentrated in the row $q = 0$, and by theorem 6.4.5 its differential d_1 is F of the differential of $C^\bullet$, so the second page is $H^p(F(C^\bullet))$ in that row and zero elsewhere.

Collapsing by lemma 6.3.3, as in (1), gives $\mathbb{R}^n F(C^\bullet) \cong H^n(F(C^\bullet))$.

This completes the proof.

The second case is what makes hypercohomology a practical tool: a complex of acyclic objects may be substituted for the original complex, and the functor is then applied termwise. It is also the precise sense in which the definition generalizes the ordinary theory, since a resolution of a single object is a complex of acyclics quasi-isomorphic to that object, and proposition 6.4.6 says the answer depends only on the quasi-isomorphism class.

Example 6.4.8: Hypercohomology of a Complex with Zero Differential

Let $C^\bullet$ be a bounded-below complex with zero differential. Then $H^q(C^\bullet) = C^q$, and the second spectral sequence reads

$$ {}^{\mathrm{II}}E_2^{p,q} = R^p F(C^q) \implies \mathbb{R}^{p+q} F(C^\bullet) $$

The first has ${}^{\mathrm{I}}E_1^{p,q} = R^q F(C^p)$ with d_1 induced by the zero differential, hence zero, so ${}^{\mathrm{I}}E_2 = {}^{\mathrm{I}}E_1$. A Cartan-Eilenberg resolution may be taken to be the direct sum of injective resolutions of the individual C^p, the zero resolution where $C^p = 0$, since with zero differential the conditions of definition 6.4.1 on Z^p, B^p and H^p reduce to the condition on C^p itself. The resulting double complex splits as a direct sum, so every differential from the second page onward vanishes and there are no extension problems:

$$ \mathbb{R}^n F(C^\bullet) \cong \bigoplus_{p+q=n} R^p F(C^q) $$

The example is the sanity check for the definition: when the complex carries no gluing, hypercohomology is the direct sum of the ordinary derived functors of its terms, and the content of the general case is exactly the failure of that splitting.

The total right derived functor $RF\colon D^+(\mathcal{A}) \to D^+(\mathcal{B})$ of definition 3.3.5, written $R\mathscr{F}$ there, is also computed by replacing a complex with a quasi-isomorphic bounded-below complex of injectives, and the total complex of a Cartan-Eilenberg resolution is one.

Proposition 6.4.9: Hypercohomology Is the Cohomology of the Total Derived Functor

With the hypotheses of definition 6.4.4, let $RF\colon D^+(\mathcal{A}) \to D^+(\mathcal{B})$ be the total right derived functor of F. Then for every n

$$ \mathbb{R}^n F(C^\bullet) \cong H^n\big(RF(C^\bullet)\big) $$

***Proof*:**

Let $J^{\bullet,\bullet}$ be a Cartan-Eilenberg resolution of $C^\bullet$, and choose p_0 with $J^{p,q} = 0$ unless $p \geq p_0$ and $q \geq 0$, as in Step 1 of the proof of theorem 6.4.5.

Step 1: $\mathrm{Tot}(J)$ *is a bounded-below complex of injectives.* The diagonal $p + q = n$ carries only the terms with $p_0 \leq p \leq n$, so $\mathrm{Tot}(J)^n$ is a finite direct sum of injective objects, and $\mathrm{Tot}(J)^n = 0$ for $n < p_0$. A finite direct sum of injectives is injective, since $\mathrm{Hom}(-, I_1 \oplus I_2) \cong$

$\mathrm{Hom}(-, I_1) \oplus \mathrm{Hom}(-, I_2)$ is exact when both summands are.

Step 2: The augmentation is a quasi-isomorphism. The augmentations $C^p \to J^{p,0}$ define a morphism $\varepsilon\colon C^\bullet \to \mathrm{Tot}(J)$. It commutes with the differentials: the vertical differential vanishes on the image of an augmentation, which is the kernel of the first map of the resolution $J^{p,\bullet}$, and the horizontal differential commutes with the augmentations because they form a morphism of complexes $C^\bullet \to J^{\bullet,0}$ (definition 6.4.1). Filter $C^\bullet$ by the subcomplexes $\sigma^{\geq p}C^\bullet$, equal to C^n in degrees $n \geq p$ and to 0 below, as in example 6.2.12, and $\mathrm{Tot}(J)$ by the degree in $C^\bullet$, through the subcomplexes with nth term $\bigoplus_{p' \geq p} J^{p',\, n-p'}$. Both filtrations are bounded in each degree, and ε preserves them, since $\varepsilon(C^n)$ lies in $J^{n,0}$. After degrees and filtration indices are shifted by p_0, both filtrations begin at the whole complex in index 0, so corollary 6.2.8 applies, and the first page of each spectral sequence is the cohomology of its graded pieces. The graded piece of $C^\bullet$ at p is C^p in degree p, so its first page is C^p at $(p, 0)$ and zero elsewhere. The graded piece of $\mathrm{Tot}(J)$ at p is the column $J^{p,\bullet}$, placed in total degrees $p + q$ with the vertical differential, so its first page is $H^q(J^{p,\bullet})$ at (p, q). That is C^p for $q = 0$, through the augmentation, and zero for $q \neq 0$, because $J^{p,\bullet}$ is a resolution of C^p. So ε induces an isomorphism on the first pages, and theorem 6.2.7, in the cochain form of corollary 6.2.9(1), makes ε a quasi-isomorphism.

Step 3: Comparison with RF. Construct RF from the class of injective objects, which is adapted to F by example 3.3.3(1). By theorem 3.3.6(3) another choice changes $RF(C^\bullet)$ only up to isomorphism. By Step 1, $\mathrm{Tot}(J)$ lies in K^+ of that class, and by Step 2, ε is a quasi-isomorphism. So in the notation of theorem 3.3.6, with $X = C^\bullet$, $s = \varepsilon$ and $Z = \mathrm{Tot}(J)$, the morphism $\gamma\colon \Phi(C^\bullet) \to \mathrm{Tot}(J)$ is an isomorphism, and applying the functor induced by F term by term gives an isomorphism $RF(C^\bullet) \cong F(\mathrm{Tot}(J))$ in $D^+(\mathcal{B})$.

Step 4: Conclusion. The functor F is additive and each $\mathrm{Tot}(J)^n$ is a finite direct sum, so $F(\mathrm{Tot}(J)) = \mathrm{Tot}(F(J))$ as complexes. Taking cohomology, $H^n(RF(C^\bullet)) \cong H^n(\mathrm{Tot}(F(J))) = \mathbb{R}^n F(C^\bullet)$ by definition 6.4.4, as we sought to show.

Exercise 6.4.1

1. Show that if F is exact then $\mathbb{R}^n F(C^\bullet) \cong F(H^n(C^\bullet))$ for every n, using whichever of the two spectral sequences of theorem 6.4.5 collapses.
2. Let $C^\bullet$ be a bounded-below complex with $H^q(C^\bullet) = 0$ for all q. Show that $\mathbb{R}^n F(C^\bullet) = 0$ for every n, and explain why this is the expected answer given proposition 6.4.6.
3. Let $C^\bullet$ be a bounded-below complex with $H^q(C^\bullet) = 0$ for $q < 0$. Write down the five-term exact sequence that proposition 6.2.11 attaches to the second spectral sequence of theorem 6.4.5. Then show that the hypothesis on $C^\bullet$ cannot be omitted, using $F = \mathrm{Hom}_{\mathbb{Z}/4}(\mathbb{Z}/2, -)$ on $\mathbb{Z}/4$-modules and the complex $C^\bullet$ with $\mathbb{Z}/2$ in degree -1 and zero elsewhere.
4. Let $A \to K^\bullet$ be a resolution of an object A of $\mathcal{A}$ by F-acyclic objects. Deduce from proposition 6.4.6 and corollary 6.4.7 that $R^n F(A) \cong H^n(F(K^\bullet))$ for every n, which is the conclusion of theorem 2.3.18.
5. Let $C^\bullet$ be the two-term complex $A \xrightarrow{f} B$ in degrees 0 and 1. Show that $\mathbb{R}^0 F(C^\bullet) \cong F(\ker f)$, by both spectral sequences of theorem 6.4.5.

6.5 The Grothendieck Spectral Sequence

Derived functors were built in section 2.3 one functor at a time. A functor of interest is often a composite: global sections of a sheaf factor through pushforward along a map, invariants under a group factor through invariants under a normal subgroup, and global Hom factors through sheaf Hom. The derived functors of the two factors are then understood, and the derived functors of the composite are wanted. They are related by a spectral sequence, which is an instance of theorem 6.4.5: for $G \circ F$ and an injective resolution $I^\bullet$ of an object, apply hypercohomology of G to the complex $F(I^\bullet)$. What makes the instance work is a hypothesis relating the two functors, that F carry injective objects to G-acyclic ones in the sense of definition 2.3.15.

Theorem 6.5.1: The Grothendieck Spectral Sequence

Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ be abelian categories with $\mathcal{A}$ and $\mathcal{B}$ having enough injectives, and let

$$F\colon \mathcal{A} \to \mathcal{B}, \qquad G\colon \mathcal{B} \to \mathcal{C}$$

be left-exact additive functors. Suppose F carries injective objects of $\mathcal{A}$ to G-acyclic objects of $\mathcal{B}$. Then for every object A of $\mathcal{A}$ there is a first-quadrant cohomological spectral sequence, natural in A,

$$E_2^{p,q} = R^pG\big(R^qF(A)\big) \implies R^{p+q}(G \circ F)(A)$$

***Proof*:**

Step 1: The complex. Choose an injective resolution $A \to I^\bullet$ in $\mathcal{A}$. The complex $F(I^\bullet)$ in $\mathcal{B}$ is concentrated in non-negative degrees, and $H^q(F(I^\bullet)) = R^qF(A)$ by definition of the right-derived functors. Each I^p is injective, so each $F(I^p)$ is G-acyclic by hypothesis. As $\mathcal{B}$ has enough injectives, definition 6.4.4 applies to G and the complex $F(I^\bullet)$.

Step 2: The abutment. Every term of $F(I^\bullet)$ is G-acyclic, so corollary 6.4.7(2) gives

$$\mathbb{R}^nG\big(F(I^\bullet)\big) \cong H^n\big(G(F(I^\bullet))\big) = R^n(G \circ F)(A)$$

the equality because $G \circ F$ is left exact and $I^\bullet$ is an injective resolution of A.

Step 3: The second page. The second spectral sequence of theorem 6.4.5, for G and $F(I^\bullet)$, reads

$${}^{\mathrm{II}}E_2^{p,q} = R^pG\big(H^q(F(I^\bullet))\big) = R^pG\big(R^qF(A)\big) \implies \mathbb{R}^{p+q}G\big(F(I^\bullet)\big)$$

It is first-quadrant, since $R^pG = 0$ for $p < 0$ and $F(I^\bullet)$ has no cohomology in negative degrees. Composed with the isomorphism of Step 2, its abutment is $R^{p+q}(G \circ F)(A)$, which gives the spectral sequence of the statement.

Step 4: Naturality. Let $a\colon A \to A'$ be a morphism and $A' \to I'^\bullet$ an injective resolution. The dual of lemma 2.2.13 lifts a to a morphism $\tilde{a}\colon I^\bullet \to I'^\bullet$, unique up to homotopy. By the naturality clause (1) of theorem 6.4.5, applied to G and $F(\tilde{a})$, there are morphisms of both spectral sequences, compatible with the map on the common abutments. On the second page of the second sequence the morphism is $R^pG(H^q(F(\tilde{a}))) = R^pG(R^qF(a))$. On the first page of the

first sequence it is $R^qG(F(\tilde{a}^p))$, so on the row $q=0$ of its second page, which is $H^p(G(F(I^\bullet)))$, it is $H^p(G(F(\tilde{a}))) = R^p(G\circ F)(a)$. The isomorphism of Step 2 is the collapse of the first sequence onto that row, so under it the map on abutments is $R^n(G\circ F)(a)$. These maps depend only on a, since homotopic lifts induce the same maps on cohomology, and by the same clause the morphism of the second sequence is determined from its second page onwards by its value there. This is the asserted naturality, completing the proof.

The spectral sequence gives $R^n(G\circ F)(A)$ a finite filtration whose graded pieces are subquotients of the groups $R^pG(R^qF(A))$ with $p+q=n$, cut out by the differentials. In low degrees the corners of the second page give exact information, through the edge morphisms and the five-term sequence of section 6.2. For the Grothendieck spectral sequence both edge morphisms are maps defined without a spectral sequence. The first uses functoriality of hypercohomology: a morphism u of bounded-below complexes lifts to their Cartan-Eilenberg resolutions by theorem 6.4.3, uniquely up to homotopy, and the map the lift induces on the cohomology of the total complexes after applying G, which does not depend on the lift by the argument of proposition 6.4.6, is written $\mathbb{R}^pG(u)$. The second uses, for a complex X in $\mathcal{B}$, the morphism

$$\kappa_X \colon H^q\big(G(X)\big) \longrightarrow G\big(H^q(X)\big)$$

induced by the projection $Z^q(X)\to H^q(X)$. By left exactness the cocycles of $G(X)$ in degree q are $G(Z^q(X))$, and G of the projection vanishes on the coboundaries of $G(X)$, which are the image of $G(X^{q-1})\to G(Z^q(X))$, since $X^{q-1}\to Z^q(X)\to H^q(X)$ is zero. The morphism κ_X is natural in X.

Corollary 6.5.2: Edge Morphisms of the Grothendieck Spectral Sequence

Under the hypotheses of theorem 6.5.1, let $A\to I^\bullet$ be an injective resolution. The two edge morphisms of the spectral sequence of theorem 6.5.1 (proposition 6.2.10) are maps

$$R^pG\big(F(A)\big) \longrightarrow R^p(G\circ F)(A), \qquad R^q(G\circ F)(A) \longrightarrow G\big(R^qF(A)\big)$$

natural in A. The first is $\mathbb{R}^pG(\iota)$ for the inclusion $\iota\colon F(A)\to F(I^\bullet)$ of $F(A)=H^0(F(I^\bullet))$ as a complex concentrated in degree 0, composed with the isomorphisms $\mathbb{R}^pG(F(A))\cong R^pG(F(A))$ and $\mathbb{R}^pG(F(I^\bullet))\cong R^p(G\circ F)(A)$ of corollary 6.4.7. The second is $\kappa_{F(I^\bullet)}\colon H^q(G(F(I^\bullet)))\to G(H^q(F(I^\bullet)))$.

***Proof*:**

Step 1: The shape. By proposition 6.2.10 the edge morphisms run $E_2^{p,0}\to H^p$ and $H^q\to E_2^{0,q}$. Here $E_2^{p,0}=R^pG(R^0F(A))=R^pG(F(A))$ and $E_2^{0,q}=R^0G(R^qF(A))=G(R^qF(A))$, since $R^0F=F$ and $R^0G=G$ by left exactness, and $H^n=R^n(G\circ F)(A)$. The edge morphisms are natural in the spectral sequence by proposition 6.2.10, and the spectral sequence is natural in A by theorem 6.5.1, so the edge morphisms are natural in A. Write $X=F(I^\bullet)$, so that the spectral sequence is the second spectral sequence of theorem 6.4.5 for G and X, and let $J^{\bullet,\bullet}$ be the chosen Cartan-Eilenberg resolution of X and $\varepsilon\colon X\to J^{\bullet,0}$ its augmentation.

Step 2: The first edge morphism. The inclusion ι is a morphism of complexes, since $F(A)$ is the kernel of $X^0\to X^1$. By the naturality clause (1) of theorem 6.4.5, it induces a morphism from the second spectral sequence for G and $F(A)$ to that for X, compatible with $\mathbb{R}^pG(\iota)$ on the abutments and equal to $R^pG(H^q(\iota))$ on the second page, which on $E_2^{p,0}$ is the identity of

$R^pG(F(A))$. The edge morphisms commute with this morphism by proposition 6.2.10. For the complex $F(A)$, concentrated in degree 0, the second page lies in the row $q = 0$, so the sequence collapses, and its edge morphism $E_2^{p,0} \to \mathbb{R}^pG(F(A))$ is the isomorphism of corollary 6.4.7(1). Hence the first edge morphism of X is $\mathbb{R}^pG(\iota)$ composed with that isomorphism.

Step 3: The second edge morphism. Write $T = \mathrm{Tot}(G(J))$ and $T_{\geq s}$ for its subcomplex of resolution degree at least s, so that $T/T_{\geq 1}$ is the row $G(J^{\bullet,0})$, and $H^q(T)_{\geq s}$ for the image of $H^q(T_{\geq s})$. By proposition 6.2.10 the second edge morphism is $H^q(T) \to H^q(T)/H^q(T)_{\geq 1} = E_\infty^{0,q} \subseteq E_2^{0,q} \subseteq E_1^{0,q} = H^q(G(J^{\bullet,0}))$, the inclusions holding because no differential arrives at a spot with first index 0. The identification of E_∞ with the graded pieces is induced by the identity on representatives, by its construction in the proof of theorem 6.2.6, and so are the pages (proposition 6.2.4). So this is the map $H^q(T) \to H^q(G(J^{\bullet,0}))$ induced by the projection $T \to T/T_{\geq 1}$. The isomorphism $H^q(G(X)) \cong H^q(T)$ of corollary 6.4.7(2) is the collapse of the first spectral sequence, whose row $q = 0$ is represented by the cocycles of $G(X)$ placed in $G(J^{\bullet,0})$ through $G(\varepsilon)$, so it is induced by the morphism of complexes $G(X) \to T$ given by $G(\varepsilon)$. The composite $H^q(G(X)) \to H^q(G(J^{\bullet,0}))$ is therefore $H^q(G(\varepsilon))$, and naturality of κ gives

$$\kappa_{J^{\bullet,0}} \circ H^q\big(G(\varepsilon)\big) = G\big(H^q(\varepsilon)\big) \circ \kappa_X$$

The map $\kappa_{J^{\bullet,0}}$ is an isomorphism: by definition 6.4.1 the cycles and coboundaries of the row $J^{\bullet,0}$ are injective, so the sequences $0 \to Z^q \to J^{q,0} \to B^{q+1} \to 0$ and $0 \to B^q \to Z^q \to H^q \to 0$ of the row split, and G preserves split exact sequences. By left exactness of G, the morphism $G(H^q(\varepsilon))$ identifies $G(H^q(X))$ with the kernel of $G(H^q(J^{\bullet,0})) \to G(H^q(J^{\bullet,1}))$, which is $E_2^{0,q}$ under $\kappa_{J^{\bullet,0}}$, and this is the identification $E_2^{0,q} = G(H^q(X))$ of theorem 6.4.5. Under it the second edge morphism is κ_X, completing the proof.

Specializing proposition 6.2.11 to theorem 6.5.1 recovers the form in which the five-term sequence is usually quoted, and it is the one example 6.5.4 and the group-cohomology literature use.

Corollary 6.5.3: The Five-Term Exact Sequence of a Composite

Under the hypotheses of theorem 6.5.1, there is an exact sequence

$$0 \to R^1G\big(F(A)\big) \to R^1(GF)(A) \to G\big(R^1F(A)\big) \to R^2G\big(F(A)\big) \to R^2(GF)(A)$$

natural in A.

***Proof*:**
The Grothendieck spectral sequence is first-quadrant and converges to $R^\bullet(GF)(A)$, so proposition 6.2.11 applies. Substituting its second page, $E_2^{p,q} = R^pG(R^qF(A))$, gives $E_2^{1,0} = R^1G(F(A))$ since $R^0F = F$, then $E_2^{0,1} = G(R^1F(A))$ since $R^0G = G$, and $E_2^{2,0} = R^2G(F(A))$; and H^1, H^2 are $R^1(GF)(A)$, $R^2(GF)(A)$. Naturality in A follows from naturality of the spectral sequence in A, as we sought to show.

The machine can now be put to work. The instance closest to hand uses the group cohomology of section 4.1: invariants under a group factor through invariants under a normal subgroup.

Example 6.5.4: The Lyndon-Hochschild-Serre Spectral Sequence

Let G be a group, $N \trianglelefteq G$ a normal subgroup, and M a G-module. Taking N-invariants produces a G/N-module, because N acts trivially on M^N and the residual action of G factors through G/N, and taking G/N-invariants of that returns M^G. So the functor $(-)^G$ factors as

$$\mathrm{Mod}_G \xrightarrow{(-)^N} \mathrm{Mod}_{G/N} \xrightarrow{(-)^{G/N}} \mathbf{Ab}$$

Both factors are left exact: for $(-)^{G/N}$ this is proposition 4.1.6, and for $(-)^N \colon \mathrm{Mod}_G \to \mathrm{Mod}_{G/N}$ the same argument applies, exactness being tested on the underlying abelian groups. That $R^p(-)^{G/N} = H^p(G/N, -)$ is section 4.1 by definition.

The identification $R^q(-)^N = H^q(N, -)$ needs one step more than it looks. The left-hand side is computed from an injective resolution in Mod_G, while the right-hand side is computed from one in Mod_N, so the two agree only because *restriction to a subgroup preserves injectives*. That holds here because $\mathbb{Z}[G]$ is free as a $\mathbb{Z}[N]$-module, on any set of coset representatives, so the relative induction functor $\mathrm{Ind}_N^G = \mathbb{Z}[G] \otimes_{\mathbb{Z}[N]} -$, which is the evident generalization of the absolute one in definition 4.1.16, is exact and is left adjoint to restriction; a right adjoint to an exact functor preserves injectives, by the adjunction $\mathrm{Hom}_N(-, \mathrm{Res} I) \cong \mathrm{Hom}_G(\mathrm{Ind}(-), I)$, which is then a composite of two exact functors. Hence restricting an injective G-module gives an injective N-module, and an injective resolution in Mod_G restricts to one in Mod_N, computing $H^q(N, -)$.

The hypothesis of theorem 6.5.1 is that $(-)^N$ carries injective G-modules to $(-)^{G/N}$-acyclic G/N-modules, and in fact it carries them to injectives. Let I be an injective G-module, let $0 \to A \to B$ be a monomorphism of G/N-modules, and let $\varphi \colon A \to I^N$ be G/N-equivariant. Regard A and B as G-modules through $G \twoheadrightarrow G/N$; then $A \to I^N \hookrightarrow I$ is G-equivariant, so injectivity of I extends it to $\psi \colon B \to I$. For $b \in B$ and $n \in N$ one has $n \cdot \psi(b) = \psi(n \cdot b) = \psi(b)$, since N acts trivially on B; so ψ lands in I^N and is the required extension. Hence I^N is injective, and injectives are acyclic for any left-exact functor by definition 2.3.15.

Theorem 6.5.1 therefore gives a first-quadrant spectral sequence

$$E_2^{p,q} = H^p\big(G/N,\, H^q(N, M)\big) \implies H^{p+q}(G, M)$$

the *Lyndon-Hochschild-Serre spectral sequence*, and corollary 6.5.3 specializes to the inflation-restriction sequence

$$0 \to H^1(G/N, M^N) \to H^1(G, M) \to H^1(N, M)^{G/N} \to H^2(G/N, M^N) \to H^2(G, M)$$

using $H^0(G/N, H^1(N, M)) = H^1(N, M)^{G/N}$. The same argument runs verbatim for a profinite group acting on discrete modules, once the category of discrete modules is known to have enough injectives; that is the setting in which the sequence is used in Galois cohomology.

Exercise 6.5.1

1. Show that if $R^q F(A) = 0$ for all $q \geq 1$, that is if A is F-acyclic, then the Grothendieck spectral sequence collapses and $R^n(GF)(A) \cong R^n G(F(A))$ for every n.
2. Show that if G is exact, then $R^n(GF)(A) \cong G(R^n F(A))$ for every n. Identify which page of the spectral sequence collapses and why.

3. In the situation of example 6.5.4 take $N = G$, so that G/N is trivial. Verify directly that the spectral sequence degenerates and returns $H^n(G, M) = H^n(G, M)$, and explain which hypothesis makes the E_2 page concentrated in one column.

4. Prove the dual statement of theorem 6.5.1 for right-exact functors and left-derived functors: if $F\colon \mathcal{A} \to \mathcal{B}$ and $G\colon \mathcal{B} \to \mathcal{C}$ are right exact, $\mathcal{A}$ and $\mathcal{B}$ have enough projectives, and F carries projectives to G-acyclics, then $E^2_{p,q} = L_pG(L_qF(A)) \implies L_{p+q}(G \circ F)(A)$.

5. Show that the hypothesis that F carries injectives to G-acyclics cannot simply be dropped, by explaining exactly which step of the proof of theorem 6.5.1 fails without it and what the first filtration computes in its absence.

6.6 Filtered Complexes and the Filtered Derived Category

Section 6.2 extracted a spectral sequence from a filtered complex and stopped there. That is enough when one filtration is present. It is not enough when two are, and two is the case that arises: a complex carrying both a weight filtration and a Hodge filtration is the object mixed Hodge theory is built from, and the question there is whether the second filtration passes to the pages of the spectral sequence of the first without ambiguity. Answering it needs the filtered complexes themselves organised into a category, with the right notion of equivalence.

Definition 6.6.1: Filtered Object and Strict Morphism

Let $\mathcal{A}$ be an abelian category. A *filtered object* is a pair (A, F) with F a decreasing chain of subobjects $\cdots \supseteq F^pA \supseteq F^{p+1}A \supseteq \cdots$. A morphism $f\colon (A, F) \to (B, F)$ is a morphism $f\colon A \to B$ with $f(F^pA) \subseteq F^pB$ for all p; it is *strict* if

$$f(A) \cap F^pB = f(F^pA)$$

for all p, that is, if every element of the image that lies in F^pB is the image of something already in F^pA.

Strictness is the condition that makes filtered algebra behave. Without it the category of filtered objects is not abelian: a morphism can be a monomorphism and an epimorphism without being an isomorphism, because the filtration on the image can be finer than the filtration inherited from the target.

Proposition 6.6.2: Strictness and Exactness

A morphism $f\colon (A, F) \to (B, F)$ of filtered objects is strict if and only if the induced sequence

$$0 \to F^p(\ker f) \to F^pA \to F^pB$$

has gr_F exact, equivalently if and only if gr_F applied to $0 \to \ker f \to A \to B$ stays exact. A composite of strict morphisms need not be strict.

***Proof*:**
If f is strict and $b = f(a) \in F^pB$, then $b = f(a')$ with $a' \in F^pA$, so $\mathrm{gr}_F^p f$ is surjective onto the image, and its kernel is $\mathrm{gr}_F^p(\ker f)$ because $a - a' \in \ker f$ lies in F^pA when both do. Conversely if gr_F preserves exactness, an element of $f(A) \cap F^pB$ lifts step by step through the graded pieces from a large p downwards, the filtration being exhaustive and separated in each degree.

For the last claim, filter each of the $\mathbb{Q}$-vector spaces below with F^p equal to the whole space for $p \leq 0$ and to 0 for $p \geq 2$. Let $B = \mathbb{Q}^2$ with $F^1B = \mathbb{Q}(e_1 + e_2)$, let $A = \mathbb{Q}e_1$ with $F^1A = 0$, and let $C = B/\mathbb{Q}e_2$ with F^1C the image of F^1B. Let $f\colon A \to B$ be the inclusion and $g\colon B \to C$ the projection. Then $f(A) \cap F^1B = 0 = f(F^1A)$, so f is strict, and g is strict because it is surjective and $F^pC = g(F^pB)$ for every p. But $e_1 + e_2$ maps to a generator of the line C, so $F^1C = C$, and $(g \circ f)(A) \cap F^1C = C$ while $(g \circ f)(F^1A) = 0$. So $g \circ f$ is not strict.

Definition 6.6.3: Filtered Quasi-Isomorphism

A morphism $f\colon (K^\bullet, F) \to (L^\bullet, F)$ of filtered complexes is a *filtered quasi-isomorphism* if $\mathrm{gr}_F^p f$ is a quasi-isomorphism of complexes for every p.

When the filtrations are finite in each degree, a filtered quasi-isomorphism is a quasi-isomorphism (theorem 6.6.4), and the converse fails: a map can induce an isomorphism on cohomology while changing the filtration, and such maps must not become invertible, or the spectral sequence would not be an invariant of the object. Without the finiteness the first implication fails as well. Give $\mathbb{Q}$ in degree 0 the filtration with $F^p\mathbb{Q} = \mathbb{Q}$ for every p. Every graded piece is zero, so $\mathbb{Q} \to 0$ is a filtered quasi-isomorphism, and it is not a quasi-isomorphism.

Theorem 6.6.4: The Filtered Derived Category

Let $\mathcal{A}$ be an abelian category with enough injectives. Consider the bounded-below filtered complexes $(K^\bullet, F)$ over $\mathcal{A}$ whose filtration is *finite in each degree*: for each n there are integers $a \leq b$ with $F^aK^n = K^n$ and $F^bK^n = 0$. Their *filtered homotopy category* has the filtered morphisms of complexes as morphisms, taken modulo the homotopies h with $h(F^pK^n) \subseteq F^pL^{n-1}$ for all p and n. Then:

1. The localization of the filtered homotopy category at the filtered quasi-isomorphisms exists. It is called the *filtered derived category* $D^+F(\mathcal{A})$.
2. Every filtered quasi-isomorphism is a quasi-isomorphism, and forgetting the filtration induces a functor $D^+F(\mathcal{A}) \to D^+(\mathcal{A})$.
3. For each $r \geq 1$ the assignment $(K, F) \mapsto E_r(K, F)$ of theorem 6.2.2, in the cochain form of corollary 6.2.9, factors through $D^+F(\mathcal{A})$: filtered-homotopic morphisms induce the same morphism of spectral sequences from the E_1-page onwards, and a filtered quasi-isomorphism induces an isomorphism of spectral sequences from the E_1-page onwards.

***Proof*:**
Through the translation $\mathcal{T}$ of corollary 6.2.9, a filtration finite in each degree is bounded in the sense of definition 6.2.1. So the spectral sequence of (K, F) is defined, with $E_1^{p,q} = H^{p+q}(\mathrm{gr}_F^p K)$

by theorem 6.2.2, and by theorem 6.2.7 a filtered morphism f induces a morphism of spectral sequences, which on E_1 is $H^{p+q}(\mathrm{gr}_F^p f)$. On each later page it is the map induced on the cohomology of the page before, since E_{r+1} is the cohomology of (E_r, d_r). So a morphism of spectral sequences is determined from E_1 onwards by its value on E_1.

1. This is theorem 1.2.2, which needs no hypothesis on the class of morphisms inverted.

2. Let f be a filtered quasi-isomorphism. Each $H^{p+q}(\mathrm{gr}_F^p f)$ is an isomorphism by definition 6.6.3, so f induces an isomorphism on E_1, and theorem 6.2.7, in the cochain form of corollary 6.2.9(1) with $r_0 = 1$, makes $H^\bullet(f)$ an isomorphism. So f is a quasi-isomorphism. Let $Q\colon \mathrm{Ch}(\mathcal{A}) \to D(\mathcal{A})$ be the localization of corollary 2.1.17. It identifies homotopic morphisms, by proposition 2.1.40, and a filtered homotopy is a homotopy, so Q composed with forgetting the filtration is a functor on the filtered homotopy category. It takes values in $D^+(\mathcal{A})$, the complexes being bounded below, and it carries filtered quasi-isomorphisms to isomorphisms. Clause (2) of definition 1.2.1 gives the factorization through $D^+F(\mathcal{A})$.

3. Let h be a filtered homotopy from f to g, so $f - g = dh + hd$. Since h preserves the filtration, it induces maps $\mathrm{gr}_F^p K^n \to \mathrm{gr}_F^p L^{n-1}$ forming a homotopy from $\mathrm{gr}_F^p f$ to $\mathrm{gr}_F^p g$, and these two induce the same map on cohomology (proposition 2.1.31). So f and g induce the same morphism on E_1, hence on every E_r with $r \geq 1$, and E_r is a functor on the filtered homotopy category. By (2) a filtered quasi-isomorphism induces an isomorphism on E_1, hence on every later page by theorem 6.2.7, so clause (2) of definition 1.2.1 gives the factorization through $D^+F(\mathcal{A})$. The case $r = 0$ is excluded: $E_0 = \mathrm{gr}_F K$, on which a filtered quasi-isomorphism need not be an isomorphism.

This completes the proof.

The lemma below is the reason this section exists. It is the tool that lets a complex carrying two filtrations be handled at all, and it is what mixed Hodge theory uses: there W is a weight filtration, F is a Hodge filtration, and the conclusion is that F passes unambiguously to the pages of the spectral sequence of W.

Theorem 6.6.5: The Lemma on Two Filtrations

Let $K^\bullet$ be a bounded-below complex in an abelian category $\mathcal{A}$, carrying an increasing filtration W by subcomplexes that is biregular (in each degree, finite) and a decreasing filtration F by subcomplexes that is biregular as well. Give the subcomplex $F^pK^\bullet$ the filtration $W_n(F^pK) = W_nK \cap F^pK$ and the quotient $K^\bullet/F^pK^\bullet$ the filtration $W_n(K/F^pK) = \mathrm{im}(W_nK)$, and write ${}_WE_r(-)$ for the spectral sequences of theorem 6.2.2. The three filtrations F induces on ${}_WE_r(K)$ are

$$F_d^p = \mathrm{im}\big({}_WE_r(F^pK) \to {}_WE_r(K)\big), \qquad F_{d^*}^p = \ker\big({}_WE_r(K) \to {}_WE_r(K/F^pK)\big)$$

on ${}_WE_r(K)$ for r finite, and on ${}_WE_\infty(K)$ by the same two formulas with $r = \infty$, the maps being the stabilized ones; and the *recursive* filtration F_{rec}: on ${}_WE_0$ it is the filtration F induces on $\mathrm{gr}_n^W K^\bullet$, and on ${}_WE_{r+1} = H({}_WE_r, d_r)$ it is the filtration a subquotient inherits from F_{rec} on ${}_WE_r$. Then:

1. $F_d^p \subseteq F_{\mathrm{rec}}^p \subseteq F_{d^*}^p$ on ${}_WE_r$ for every $r \geq 0$ and on ${}_WE_\infty$, and the three coincide on ${}_WE_0$ and on ${}_WE_1$, with no hypothesis beyond the above;
2. let $r_0 \geq 0$ and suppose that d_s is strictly compatible with F_{rec} on ${}_WE_s$ for every $s \leq r_0$. Then for every p the sequence
$$0 \to {}_WE_r(F^pK) \to {}_WE_r(K) \to {}_WE_r(K/F^pK) \to 0$$
is exact for every $r \leq r_0 + 1$, and the three filtrations coincide on ${}_WE_r$ for every $r \leq r_0 + 2$;
3. if d_s is strictly compatible with F_{rec} for every $s \geq 0$, then the three filtrations coincide on every page and on ${}_WE_\infty$, the differential of $K^\bullet$ is strictly compatible with F in the sense of definition 6.6.1, and the spectral sequence of $(K^\bullet, F)$ degenerates at E_1.

The hypothesis of (2) at $s = 0$ says exactly that the differential of $\mathrm{gr}_n^W K^\bullet$ is strictly compatible with the filtration induced by F, for every n; on its own it therefore gives coincidence on ${}_WE_1$ and ${}_WE_2$, and example 6.6.6 shows that it gives nothing more.

***Proof*:**

Conventions. By theorem 1.1.15 the arguments below may be run with elements. Index the spectral sequence of W by the weight n and the cohomological degree m: setting $C_m = K^{-m}$ and $F_pC_\bullet = W_pK^{-\bullet}$ presents $(K^\bullet, W)$ as a filtered chain complex in the sense of definition 6.2.1, and theorem 6.2.2 produces

$${}_WE_0^{n,m} = \mathrm{gr}_n^W K^m, \qquad {}_WE_1^{n,m} = H^m(\mathrm{gr}_n^W K^\bullet), \qquad d_r \colon {}_WE_r^{n,m} \to {}_WE_r^{n-r,\,m+1}$$

Biregularity of W and boundedness below of $K^\bullet$ make that filtration bounded in each degree, so theorem 6.2.6 applies: the pages stabilize at every spot and ${}_WE_\infty^{n,m} \cong \mathrm{gr}_n^W H^m(K^\bullet)$ for the filtration $W_nH^m(K) = \mathrm{im}\big(H^m(W_nK) \to H^m(K)\big)$, itself finite in each degree. The same holds for $F^pK^\bullet$ and for $K^\bullet/F^pK^\bullet$: a biregular filtration restricts and corestricts to biregular filtrations, and both complexes are bounded below.

Fix p for the whole argument except where stated, and abbreviate

$$A_r = {}_W E_r(F^p K^\bullet), \qquad B_r = {}_W E_r(K^\bullet), \qquad C_r = {}_W E_r(K^\bullet/F^p K^\bullet)$$

writing $\alpha_r \colon A_r \to B_r$ and $\beta_r \colon B_r \to C_r$ for the maps induced by the inclusion and the projection. The construction of theorem 6.2.2 is functorial in the filtered complex, so α_r and β_r commute with d_r, and α_{r+1}, β_{r+1} are the maps they induce on d_r-cohomology. By definition $F^p_d B_r = \mathrm{im}\alpha_r$ and $F^p_{d^*} B_r = \ker \beta_r$.

Step 1: the sequence at $r = 0$, and the three filtrations there. For every n and m the sequence

$$0 \to W_n K^m \cap F^p K^m \to W_n K^m \to W_n(K/F^p K)^m \to 0$$

is exact, by the definition of the two induced filtrations. These sequences are compatible with the inclusions $W_{n-1} \subseteq W_n$, and the vertical maps from the sequence at $n-1$ to the sequence at n are monomorphisms. Applying theorem 2.1.20 to that map of short exact sequences, all three kernels vanish and the six-term sequence collapses to

$$0 \to \mathrm{gr}^W_n(F^p K)^m \to \mathrm{gr}^W_n K^m \to \mathrm{gr}^W_n(K/F^p K)^m \to 0$$

which is therefore exact. This is $0 \to A_0 \to B_0 \to C_0 \to 0$ at the spot (n, m), and the maps commute with d_0, so it is a short exact sequence of complexes in the cohomological degree m, one for each n.

For the filtrations, $F^p_d B_0 = \mathrm{im}\alpha_0$ is $\big(W_n K^m \cap F^p K^m + W_{n-1} K^m\big)/W_{n-1} K^m$, which is the filtration F induces on $\mathrm{gr}^W_n K^m$, that is $F^p_{\mathrm{rec}} B_0$ by definition; and $F^p_{d^*} B_0 = \ker \beta_0$ is the same subobject by the exactness just proved. So all three agree on ${}_W E_0$, and in particular the phrase "d_0 is strict" is unambiguous and says exactly that the differential of each $\mathrm{gr}^W_n K^\bullet$ is strict for the induced F.

Step 2: the two inclusions, and equality on ${}_W E_1$. We prove $F^p_d \subseteq F^p_{\mathrm{rec}} \subseteq F^p_{d^*}$ on B_r by induction on r, the case $r = 0$ being the equality of Step 1.

For the first inclusion let $\xi \in F^p_d B_{r+1}$, say $\xi = \alpha_{r+1}(\eta)$ with $\eta \in A_{r+1} = H(A_r, d_r)$. Write $\eta = [a]$ with $a \in A_r$, $d_r a = 0$. Then $\alpha_r(a)$ is killed by d_r and lies in $\mathrm{im}\alpha_r = F^p_d B_r$, which by induction lies in $F^p_{\mathrm{rec}} B_r$; and $\xi = [\alpha_r(a)]$. Thus ξ is the class of an element of $\ker d_r \cap F^p_{\mathrm{rec}} B_r$, which is precisely what $F^p_{\mathrm{rec}} B_{r+1}$ consists of.

For the second let $\xi \in F^p_{\mathrm{rec}} B_{r+1}$, say $\xi = [b]$ with $b \in \ker d_r \cap F^p_{\mathrm{rec}} B_r$. By induction $b \in F^p_{d^*} B_r = \ker \beta_r$, so $\beta_{r+1}(\xi) = [\beta_r(b)] = 0$ and $\xi \in \ker \beta_{r+1} = F^p_{d^*} B_{r+1}$. Since each spot of each page equals its limit value for large r, and the three filtrations are defined at $r = \infty$ as those common values, the inclusions persist on ${}_W E_\infty$.

Equality on ${}_W E_1$ needs the short exact sequence of Step 1. For fixed n the objects $B_0^{n,\bullet}$ with d_0 form a complex with $H^m = B_1^{n,m}$, and likewise for A and C; theorem 2.1.20 gives the long exact sequence

$$\cdots \to A_1^{n,m} \xrightarrow{\alpha_1} B_1^{n,m} \xrightarrow{\beta_1} C_1^{n,m} \to A_1^{n,m+1} \to \cdots$$

Exactness at $B_1^{n,m}$ reads $\mathrm{im}\alpha_1 = \ker\beta_1$, that is $F_d^p = F_{d^*}^p$ on ${}_W E_1$, and F_{rec} is squeezed between them. This proves (1).

Step 3: two equivalences. (Throughout, theorem 2.1.20 is applied to the E_r-pages with differential d_r, which for each fixed residue class of the weight index form ordinary complexes, so the lemma applies as stated.) For $r \geq 0$ write

$$(\star_r)\colon \qquad 0 \to A_r \to B_r \to C_r \to 0 \quad \text{is exact}$$

a statement about complexes, since α_r and β_r commute with d_r. Step 1 is $(\star_0)$.

(a) $(\star_r)$ implies $F_d^p = F_{d^*}^p$ on ${}_W E_{r+1}$. By theorem 2.1.20 the sequence

$$\cdots \to A_{r+1} \xrightarrow{\alpha_{r+1}} B_{r+1} \xrightarrow{\beta_{r+1}} C_{r+1} \xrightarrow{\delta_r} A_{r+1} \to \cdots$$

is exact; exactness at B_{r+1} is $\mathrm{im}\alpha_{r+1} = \ker\beta_{r+1}$. Combined with Step 2 this forces all three filtrations to agree on ${}_W E_{r+1}$.

(b) Assume $(\star_r)$. Then $(\star_{r+1})$ holds if and only if d_r is strictly compatible with F_d^p on ${}_W E_r$, that is, if and only if $d_r(B_r) \cap F_d^p B_r = d_r(F_d^p B_r)$ at every spot. By the long exact sequence of (a), α_{r+1} is a monomorphism exactly when the δ_r landing in A_{r+1} vanishes and β_{r+1} is an epimorphism exactly when the δ_r leaving C_{r+1} vanishes; so $(\star_{r+1})$ is equivalent to $\delta_r = 0$. By $(\star_r)$ the map α_r is a monomorphism, so δ_r is described as follows: for $c \in C_r$ with $d_r c = 0$, choose $b \in B_r$ with $\beta_r(b) = c$; then $\beta_r(d_r b) = d_r c = 0$, so $d_r b \in \ker\beta_r = \mathrm{im}\alpha_r$ and there is a unique $a \in A_r$ with $\alpha_r(a) = d_r b$; it satisfies $d_r a = 0$, and $\delta_r[c] = [a]$.

Suppose d_r is strict for F_d^p, and let c, b, a be as above. Then $d_r b \in \mathrm{im}\alpha_r = F_d^p B_r$, so strictness gives $d_r b = d_r b'$ with $b' \in F_d^p B_r$, say $b' = \alpha_r(a')$. Then $\alpha_r(a) = d_r\alpha_r(a') = \alpha_r(d_r a')$, so $a = d_r a'$ and $\delta_r[c] = 0$. Conversely suppose $\delta_r = 0$ and let $b \in B_r$ with $d_r b \in F_d^p B_r = \ker\beta_r$. Then $d_r\beta_r(b) = \beta_r(d_r b) = 0$, so $c := \beta_r(b)$ is a d_r-cocycle and $\delta_r[c] = [a]$ with $\alpha_r(a) = d_r b$. Since $\delta_r = 0$ we get $a = d_r a'$ for some $a' \in A_r$, whence $d_r b = \alpha_r(d_r a') = d_r\big(\alpha_r(a')\big)$ with $\alpha_r(a') \in F_d^p B_r$. Thus $d_r(B_r) \cap F_d^p B_r \subseteq d_r(F_d^p B_r)$, and the reverse inclusion is automatic.

Step 4: the induction, giving (2). Assume d_s is strictly compatible with F_{rec} on ${}_W E_s$ for every $s \leq r_0$. We show by induction on $s \leq r_0 + 1$, simultaneously for every p, that

$$(\mathrm{I}_s)\colon (\star_s) \text{ holds}, \qquad (\mathrm{II}_s)\colon F_d = F_{\mathrm{rec}} = F_{d^*} \text{ on } {}_W E_s$$

Step 1 gives (I_0) and (II_0). Let $s \leq r_0$ and assume both at stage s. By (II_s) the hypothesis at stage s says that d_s is strict for F_d^p, so Step 3(b) upgrades $(\star_s)$ to $(\star_{s+1})$, which is (I_{s+1}); and (I_s) with Step 3(a) gives (II_{s+1}). This delivers $(\star_r)$ for every $r \leq r_0 + 1$ and coincidence on ${}_W E_r$ for every $r \leq r_0 + 1$. Applying Step 3(a) once more to $(\star_{r_0+1})$ gives coincidence on ${}_W E_{r_0+2}$ as well. Each stage uses the strictness produced by the previous one and produces the exactness that makes the next stage's strictness hypothesis unambiguous, which is why the two conclusions cannot be separated.

Step 5: part (3), first the limit page. Assume now that d_s is strict for every $s \geq 0$. Then $(\star_r)$

holds for every r and every p, and the three filtrations agree on every page. Fix p and a spot (n,m). By convergence there is an R with $A_r^{n,m} = A_\infty^{n,m}$, $B_r^{n,m} = B_\infty^{n,m}$ and $C_r^{n,m} = C_\infty^{n,m}$ for $r \geq R$, compatibly with α and β, so

$$0 \to {}_W E_\infty^{n,m}(F^pK) \to {}_W E_\infty^{n,m}(K) \to {}_W E_\infty^{n,m}(K/F^pK) \to 0$$

is exact and the three filtrations agree on ${}_W E_\infty$.

Step 6: from the limit page to the complex. By theorem 6.2.6 and the naturality recorded in its proof, $\alpha_\infty^{n,m}$ is gr_n^W of the map $H^m(F^pK) \to H^m(K)$ induced by the inclusion, for the filtrations $W_nH^m(-) = \mathrm{im}\big(H^m(W_n -) \to H^m(-)\big)$, both finite in each degree. Apply theorem 2.1.20 to the map of short exact sequences

$$\begin{array}{ccccccccc} 0 & \to & W_{n-1}H^m(F^pK) & \to & W_nH^m(F^pK) & \to & \mathrm{gr}_n^W H^m(F^pK) & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & W_{n-1}H^m(K) & \to & W_nH^m(K) & \to & \mathrm{gr}_n^W H^m(K) & \to & 0 \end{array}$$

The sequence $0 \to \ker(W_{n-1}) \to \ker(W_n) \to \ker(\mathrm{gr}_n^W)$ is exact, the third kernel vanishes by Step 5, and $W_nH^m(F^pK) = 0$ for n small; so induction upwards on n gives $\ker\big(W_nH^m(F^pK) \to W_nH^m(K)\big) = 0$ for every n, and since the filtration is exhaustive in each degree, $H^m(F^pK) \to H^m(K)$ is a monomorphism for every p and m.

Strictness of d follows at once. Let $z \in d(K^{m-1}) \cap F^pK^m$, say $z = dx$ with $x \in K^{m-1}$. Then z is a cocycle of $F^pK^\bullet$ whose class maps to $[z] = 0$ in $H^m(K)$, so by injectivity $z = dy$ for some $y \in F^pK^{m-1}$. Hence $d(K^{m-1}) \cap F^pK^m = d(F^pK^{m-1})$, which is strict compatibility with F in the sense of definition 6.6.1.

Step 7: degeneration of the F-sequence. The filtration F is decreasing and biregular and $K^\bullet$ is bounded below, so after a shift of the index corollary 6.2.8 applies, and by corollary 6.2.9(2)

$$E_r^{p,q}(K,F) \cong \frac{Z_r^{p,q}}{Z_{r-1}^{p+1,q-1} + d\,Z_{r-1}^{p-r+1,\,q+r-2}}, \qquad Z_r^{p,q} = \{\, x \in F^pK^{p+q} : dx \in F^{p+r}K^{p+q+1} \,\}$$

Write $Z_\infty^{p,q} = \{x \in F^pK^{p+q} : dx = 0\}$. For $r \geq 1$ we claim $Z_r^{p,q} = Z_\infty^{p,q} + Z_{r-1}^{p+1,q-1}$. Given $x \in Z_r^{p,q}$, strictness at level $p+r$ gives $dx \in d(K^{p+q}) \cap F^{p+r}K^{p+q+1} = d(F^{p+r}K^{p+q})$, say $dx = dy$ with $y \in F^{p+r}K^{p+q}$. Then $x - y \in Z_\infty^{p,q}$, while $y \in F^{p+1}K^{p+q}$ has $dy \in F^{p+r}K^{p+q+1} = F^{(p+1)+(r-1)}K^{p+q+1}$, so $y \in Z_{r-1}^{p+1,q-1}$, which is a summand of the denominator. Hence every class of $E_r^{p,q}(K,F)$ is represented by a cocycle of $K^\bullet$, on which d vanishes, so $d_r = 0$ for every $r \geq 1$ and the spectral sequence of $(K^\bullet, F)$ degenerates at E_1. This proves (3), completing the proof.

The proof never manipulated the objects Z_r of definition 6.2.3 on the W-side, and it is worth saying what those would show. By proposition 6.2.4, with $Z_r^{n,m} = \{x \in W_nK^m : dx \in W_{n-r}K^{m+1}\}$ and $D_r^{n,m} = Z_{r-1}^{n-1,m} + d\,Z_{r-1}^{n+r-1,m-1}$, one has ${}_W E_r^{n,m} = Z_r^{n,m}/D_r^{n,m}$, and F_d^p is the image of $F^pK^m \cap Z_r^{n,m}$, while $F_{d^*}^p$ is the image of $Z_r^{n,m} \cap \big(F^pK^m + \widehat{D}_r^{n,m}\big)$, where $\widehat{D}$ is built from the objects $\{y \in W_aK^b : dy \in W_{a-s}K^{b+1} + F^pK^{b+1}\}$ in place of $Z_s^{a,b}$. The gap between the two is exactly the failure of the boundary condition to be detectable modulo F^p, and closing it by hand is the several-page computation the argument above replaces with a long exact sequence.

Example 6.6.6: Sharpness of the Hypothesis

Take $\mathcal{A}$ the category of $\mathbb{Q}$-vector spaces and let $K^\bullet$ be $\mathbb{Q}e$ in degree 0 and $\mathbb{Q}f$ in degree 1, with $de = f$. Filter it by

$$W_{-1}K = 0, \quad W_0K^0 = 0, \; W_0K^1 = \mathbb{Q}f, \quad W_1K = K$$

$$F^0K = K, \quad F^1K^0 = 0, \; F^1K^1 = \mathbb{Q}f, \quad F^2K = 0$$

Both are biregular filtrations by subcomplexes. The graded pieces are $\mathrm{gr}_0^W K = \mathbb{Q}f$ in degree 1 and $\mathrm{gr}_1^W K = \mathbb{Q}e$ in degree 0, each with zero differential, so the differential of every $\mathrm{gr}_n^W K^\bullet$ is trivially strict for F: the hypothesis of theorem 6.6.5(2) holds with $r_0 = 0$ and no more.

It holds with no more. The differential $d_1 \colon {}_WE_1^{1,0} = \mathbb{Q}e \to {}_WE_1^{0,1} = \mathbb{Q}f$ is the connecting map of $0 \to W_0K \to K \to \mathrm{gr}_1^W K \to 0$, hence $e \mapsto f$, an isomorphism; its source has $F^1 = 0$ and its target has F^1 equal to everything, so d_1 is not strict. Likewise $H^\bullet(K) = 0$ while $\mathrm{gr}_F^0 K = \mathbb{Q}e$ in degree 0 and $\mathrm{gr}_F^1 K = \mathbb{Q}f$ in degree 1, so the spectral sequence of (K, F) has $E_1^{0,0} = \mathbb{Q}e$ and $E_1^{1,0} = \mathbb{Q}f$ with $E_\infty = 0$ and does not degenerate at E_1; equivalently $d(K^0) \cap F^1K^1 = \mathbb{Q}f$ while $d(F^1K^0) = 0$, so d is not strict for F. Conclusion (2) of theorem 6.6.5 with $r_0 = 0$ is therefore exactly right: here ${}_WE_2 = 0$, so the three filtrations do agree on ${}_WE_1$ and ${}_WE_2$, and nothing beyond that is true.

The example is not pathological, and it should be read as a warning about the shape of the input. The strictness of d_0 is a statement about the graded pieces of W alone and can never see how those pieces are glued; the strictness of d_1 is a statement about the gluing, and it has to be given. In mixed Hodge theory it is given by weights: the terms of ${}_WE_1$ are cohomologies of smooth projective varieties, d_1 is a morphism of pure Hodge structures of a single weight, and such a morphism is strict, while d_r for $r \geq 2$ lowers the weight and vanishes. With all d_r strict, part (3) applies and the Hodge filtration degenerates at E_1.

The last of the three exercises below is used often enough elsewhere to be worth stating as a result in its own right.

Lemma 6.6.7: Degeneration at E_1 and Strictness

Let $(K^\bullet, F)$ be a complex with F a biregular decreasing filtration by subcomplexes. The spectral sequence of $(K^\bullet, F)$ degenerates at E_1 if and only if the differential of $K^\bullet$ is strictly compatible with F.

***Proof*:**

This is exercise 6.6.1 (3), proved there in both directions. No finiteness or field hypothesis is needed: biregularity alone makes the correction argument in the reverse direction terminate.

Exercise 6.6.1

1. Let $f \colon A \to B$ and $g \colon B \to C$ be strict morphisms of filtered vector spaces. Show that $g \circ f$ is strict if f is surjective or g is injective, and identify which of the two conditions each of f and g fails in the example in the proof of proposition 6.6.2.

2. Give a morphism of filtered complexes that is a quasi-isomorphism but not a filtered quasi-isomorphism.

3. Show that the spectral sequence of a filtered complex (K, F) degenerates at E_1 if and only if the differential of K is strictly compatible with F.